Probabilistic Data Science for Psychology

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Welcome

Welcome to the working version of *Probabilistic Data Science for Psychology (PDSP)*, a textbook on data-scientific methodology at the Institute of Psychology at Otto von Guericke University Magdeburg.

Figure 1 Probabilistic Data Science for Psychology

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Part I

Mathematical foundations

As the language of scientific modeling, mathematics is also the language of probabilistic data science in psychology. In this role, mathematics mediates between intuitive scientific language and the variety of programming languages used to implement data-scientific analyses. Mathematical concept formation makes it possible to grasp concepts very precisely and in a generally understandable way, and to communicate them independently of the imprecision of everyday scientific language or the idiosyncrasies of different programming languages. Of course, mathematical language cannot assume this role detached from scientific-language intuition or practical application. In probabilistic data analysis, mathematics is therefore never an end in itself, but must always be understood as applied mathematics.

In this part, after some thoughts on the fundamental concepts of mathematics (Chapter 1), we present sets and functions (Chapter 2 and Chapter 4), the two pillars of modern mathematics, in compressed form. We supplement this presentation with some notational aspects in Chapter 3. Closely interwoven with the concept of a function are differential calculus and integral calculus. Here, too, the aim is not an exhaustive presentation, but in particular an explanation of some foundations that are central to the data-scientific concepts of optimization and to working with probability functions. We supplement these sections with some introductory thoughts on the analytical foundations of differential and integral calculus in Chapter 5. In dealing with the large data sets of contemporary science, matrix notation has proved useful. We first consider this subfield of linear algebra for single-column matrices and then for matrices in detail, again with the aim of providing notational and computational foundations for later sections.

Overall, this part therefore primarily serves as a brief introduction to foundations that will be taken up again in other parts, and in particular also as the introduction of a unified notation. Depending on need and interest, more in-depth self-study of the various contents is of course advisable. In the following bibliographic remarks, we provide an overview of sources and reading recommendations for the contents considered here.

Bibliographic remarks

Bärwolff (2017) and Arens et al. (2018) form the primary basis for many of the topics presented here and offer an excellent and comprehensive entry point into the material covered. At the international level, Spivak (2008) and Strang (2009) provide very readable introductions to differential calculus and linear algebra, respectively. A deeper understanding, especially of analytical concepts, is provided, for example, by Abbott (2015) and Chiossi (2021). S. Searle (1982) offers a compact presentation of those aspects of matrix calculus and linear algebra that are used mainly in probabilistic data analysis. Deisenroth et al. (2020) provides an overview of many of the topics treated here under the moniker of machine learning.

1 Language and logic

1.1 Mathematics is a language

Mathematics is the language of scientific model building. For example, the expression

$$F = ma \tag{1.1}$$

corresponds, in the sense of Newton's second axiom, to a theory of the motion of objects under the action of forces (Newton (1687)). Likewise, the expression

$$\max_{q(z)} \int q(z) \ln \left(\frac{p(y,z)}{q(z)} \right) dz \tag{1.2}$$

corresponds, in the sense of variational inference, to contemporary theory about how the brain works (Friston (2005)). Mathematical symbolism serves in particular to communicate scientific insights precisely and aims to describe complex matters exactly and efficiently. As in reflective engagement with any form of language, the question "What is this supposed to mean?" therefore always remains the guiding question when working with mathematical content and symbolism.

As a language system, mathematics has a number of special features. First, its contents are often abstract. This is because mathematics strives for the broadest possible general comprehensibility and applicability. Mathematical approaches to the phenomena of the world are interested in making insights as easy as possible to transfer to other contexts. To make this possible, mathematics tries to work as precisely and understandably as possible, that is, in terms of precise concepts. It proceeds in a strictly hierarchical way, so that concepts introduced later often presuppose a good understanding of the underlying concepts introduced earlier.

The precision of mathematical language implies a high density of information. It is therefore rather sober and leaves out what is superfluous, so that, in the best case, *everything* in a mathematical text is relevant for communicating an idea. As recipients of mathematical texts, we perceive their information density through the high cognitive energy required when reading such a text. This high energy expenditure calls in particular for calm and slowness when reading with the aim of genuine understanding. The following quotation may serve as a guiding principle for working with mathematical texts: "A mathematical text cannot be read like a novel; one has to work through it" (Unger (2000)). After reading a short mathematical text, one should always ask critically whether one has really understood what has been read or whether further sources should be consulted to clarify the matter. It is also helpful, in the spirit of Richard Feynman's famous quotation "What I cannot create, I do not understand", to make one's own notes and to reconstruct mathematical language systems oneself.

If one wants to gain access to the world of scientific model building, then it is helpful, when dealing with its mathematical expressions and symbolism, to use the same strategies as when learning a foreign language. In addition to immersing oneself in the corresponding language space, that is, constant exposure to mathematical modes of expression, this certainly includes first memorizing concepts and translating texts into everyday language. A deep and secure understanding of mathematical model building then arises in particular through the application of mathematical ways of thinking in written and spoken form.

1.2 Basic building blocks

In the following, we introduce three basic building blocks of mathematical communication that will accompany us throughout: the concepts of a *definition*, a *theorem*, and a *proof*.

Definition

A *definition* is a cornerstone of a mathematical system that is neither justified nor deductively derived within that system. Definitions can only be evaluated according to their usefulness within a mathematical system. A definition is best memorized first and questioned only once one has understood its use in application or is not convinced by it. Some relaxation and calm are generally helpful when dealing with definitions that appear complex at first sight. To indicate that we define a symbol as something, we use the notation "$:=$". For example, the expression "$a := 2$" defines the symbol a as the number two. In this text, definitions always end with the symbol $\bullet$.

Theorem

A *theorem* is a mathematical statement that can be recognized as correct (true) by means of a proof. That is, a theorem is always derived from definitions and/or other theorems. In this sense, theorems are the empirical results of mathematics. In applied, data-analytic mathematics, theorems are often helpful for calculations. It is therefore worth memorizing them, because they usually form the basis for data analysis and data interpretation. Equations often appear in theorems. These equations arise from the assumptions of the theorem. To mark equations, we use the equals sign "$=$". For example, in a given context, the expression "$a = 2$" says that, because of certain assumptions, the variable a has the value two. In this text, theorems always end with the symbol $\circ$.

Proof

A *proof* is a chain of argumentation that draws on known definitions and theorems to establish the correctness (truth) of a theorem. Short proofs often contribute to understanding a theorem; long proofs tend to do so less. Proofs are therefore, in particular, answers to the question why a mathematical statement holds ("Why is this so?"). Proofs are not memorized. When proofs are short, it is sensible to work through them, because they usually repeat content that is assumed to be known. When they are long, it is more sensible to skip them at first so as not to get lost in details and lose the main path through the mathematical structure. In this text, proofs always end with the symbol $\square$.

In addition to definitions, theorems, and proofs, mathematical texts contain other typical building blocks, such as *axioms*, *lemmas*, *corollaries*, and *conjectures*. We will not use these terms and therefore give only a brief overview of them.

- *Axioms* are unproved fundamental assumptions in the sense that they serve as starting points for building mathematical systems. The transition between definitions and axioms is often fluid. Because we will not work at a particularly deep mathematical level, we prefer the term definition in almost all cases.
- A *lemma* is an "auxiliary theorem", that is, a mathematical statement that is proved but is not as important as a theorem. Because, on the one hand, we focus on important content and, on the other hand, we do not want to discriminate among mathematical statements, we avoid this term and instead use the term theorem throughout.
- A *corollary* is a mathematical statement that follows from a theorem by a simple proof. Because the "simplicity" of mathematical proofs is a relative property, we avoid this term and instead also use the term theorem throughout.
- *Conjectures* are mathematical statements for which it is unknown whether they are provable or refutable. Because we work in the area of applied mathematics, we will not encounter conjectures.

1.3 Propositional logic

Having now become acquainted with some basic building blocks of mathematical language systems, we turn to *propositional logic*, a simple system that allows relationships between mathematical statements to be established and formalized. Propositional logic plays a central role, for example, in the definition of set operations, in optimization conditions for functions, and in many proofs. In data-analytic applications, propositional logic is the basis of Boolean logic in programming. In mathematical psychology, propositional logic is the basis of the representational theory of measurement.

We begin with the definition of the concept of a mathematical *statement*.

Definition 1.1 (Statement)**.** A *statement* is a sentence to which the property *true* or *false* can be assigned unambiguously.

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The adjective *true* can also be understood as *correct*. We abbreviate true by "t" and false by "f". In the field of real numbers, for example, the statement $1 + 1 = 2$ is true and the statement $1+1 = 3$ is false. Note that the binarity of the truth value of statements is a basic assumption of propositional logic and therefore to be understood formally and scientifically, not empirically. Truth values do not refer to definitions; definitions fix meanings.

A first way of working with statements is to negate them. This leads to the following definition.

Definition 1.2 (Negation)**.** Let A be a statement. Then the *negation of A* is the statement that is false when A is true and true when A is false. The negation of A is denoted by $\neg A$, read as "not A".

•

For example, the negation of the statement "The sun is shining" is the statement "The sun is not shining". The negation of the statement $1 + 1 = 2$ is the statement $1 + 1 \neq 2$, and the negation of the statement $x > 1$ is the statement $x \leq 1$. The definition of the negation of a statement A is represented in tabular form as follows:

Table 1.1 Truth table of negation

A	$\neg A$
t	f
f	t

Tables of this form are called *truth tables*. They are a popular tool in propositional logic, and we will use them often in the following.

If one wants to connect two statements logically, the concepts of *conjunction* and *disjunction* are available.

Definition 1.3 (Conjunction)**.** Let A and B be statements. Then the *conjunction of A and B* is the statement that is true if and only if both A and B are true. The conjunction of A and B is denoted by $A \wedge B$, read as "A and B".

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The definition of conjunction implies the following truth table:

Table 1.2 Truth table of conjunction

A	B	$A \wedge B$
t	t	t
t	f	f
f	t	f
f	f	f

As an example, let A be the statement $2 \geq 1$ and let B be the statement $2 > 1$. Since both A and B are true, the statement $(2 \geq 1) \wedge (2 > 1)$ is also true. As another example, let A be the statement $1 \geq 1$ and let B be the statement $1 > 1$. Here, only A is true and B is false. Thus the statement $(1 \geq 1) \wedge (1 > 1)$ is false.

Definition 1.4 (Disjunction)**.** Let A and B be statements. Then the *disjunction of A and B* is the statement that is true if and only if at least one of the two statements A and B is true. The disjunction of A and B is denoted by $A \vee B$, read as "A or B".

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The definition of disjunction implies the following truth table:

Table 1.3 Truth table of disjunction

A	B	$A \vee B$
t	t	t
t	f	t
f	t	t
f	f	f

$A \vee B$ is therefore also true when both A and B are true. The "or" considered here is thus more precisely an "and/or". For this reason, disjunction is also called a "non-exclusive or". As an example, let A be the statement $2 \geq 1$ and let B be the statement $2 > 1$. A is true and B is true. Thus the statement $(2 \geq 1) \vee (2 > 1)$ is true. Now again let A be the statement $1 \geq 1$ and let B be the statement $1 > 1$. Then A is true and B is false. Thus the statement $(1 \geq 1) \vee (1 > 1)$ is true.

One way of placing statements in a mechanical logical relationship is *implication*. It is defined as follows.

Definition 1.5 (Implication)**.** Let A and B be statements. Then the *implication*, denoted by $A \Rightarrow B$, is the statement that is false if and only if A is true and B is false. A is called the *assumption (premise)* and B the *conclusion* of the implication. $A \Rightarrow B$ is read as "from A follows B", "A implies B", or "if A, then B".

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The definition of implication can be represented by the following truth table:

Table 1.4 Truth table of implication

A	B	$A \Rightarrow B$
t	t	t
t	f	f
f	t	t
f	f	t

An intuitive understanding of the definition of implication in the sense of the truth table above is obtained most readily by reading it as an attempt to capture and formalize the intuitive idea of a consequence in the context of propositional logic. To understand this, it is best to read the truth states in Table 1.4 in the order truth state of A, truth state of $A \Rightarrow B$, and finally truth state of B. Read in this way, the truth table shows that if A is true and $A \Rightarrow B$ is true, then B is true. Thus, if one constructs a true implication based on a true statement, for example by transforming equations, it follows that B is also true. If this is not possible, that is, if A is true and $A \Rightarrow B$ is always false, then B is also false. This is how statements may be refuted. Finally, one sees that if A is false and $A \Rightarrow B$ is true, then B can be either true or false. Only from a true premise does a true implication always yield a true conclusion. In particular, the definition of implication therefore satisfies the principle "anything follows from falsehood (ex falso sequitur quodlibet)". Thus one cannot infer anything meaningful from false statements by means of implication.

In the context of implication, the concepts of *sufficient* and *necessary* conditions arise: If $A \Rightarrow B$ is true, one says that "A is *sufficient* for B" and that "B is *necessary* for A". This terminology is explained in the context of implication as follows: If $A \Rightarrow B$ is true, then if A is true, B is also true. The truth of A is therefore enough for the truth of B. Thus A is sufficient for B. Furthermore, if $A \Rightarrow B$ is true, then if B is false, A is also false. The truth of B is therefore necessary for the truth of A.

Finally, to illustrate Table 1.4, we give a concrete everyday example. Let A be the statement "The bridge is damaged", let B be the statement "The bridge is closed", and let $A \Rightarrow B$ be the implication "If the bridge is damaged, then the bridge is closed". We discuss the four cases of Table 1.4.

- A true, B true, $A \Rightarrow B$ true. If A is true, then the bridge is damaged, and if B is also true, then the bridge is also closed. In this case, the implication "If the bridge is damaged, then the bridge is closed" is therefore obviously true.
- A true, B false, $A \Rightarrow B$ false. If A is true, then the bridge is damaged, and if B is false, then the bridge is not closed. In this case, the implication "If the bridge is damaged, then the bridge is closed" is therefore obviously false.
- A false, B true, $A \Rightarrow B$ true. If A is false, then the bridge is not damaged, and if B is true, then the bridge is closed. Since the bridge is not damaged, however, no conclusion can be drawn regarding the implication "If the bridge is damaged, then the bridge is closed"; in particular, one cannot show that it would be false. Thus one assumes that the statement "If the bridge is damaged, then the bridge is closed" is true.
- A false, B false, $A \Rightarrow B$ true. If A is false, then the bridge is not damaged, and if B is false, then the bridge is not closed. Since the bridge is not damaged, however, no conclusion can be drawn regarding the implication "If the bridge is damaged, then the bridge is closed"; in particular, one cannot show that it would be false. Thus one again assumes that the statement "If the bridge is damaged, then the bridge is closed" is true.

A very frequently occurring relation between two statements is their *equivalence*.

Definition 1.6 (Equivalence)**.** Let A and B be statements. The *equivalence of A and B* is the statement that is true if and only if A and B are both true or if A and B are both false. The equivalence of A and B is denoted by $A \Leftrightarrow B$ and read as "A if and only if B" or "A is equivalent to B".

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The definition of equivalence implies the following truth table:

Table 1.5 Truth table of equivalence

A	B	$A \Leftrightarrow B$
t	t	t
t	f	f
f	t	f
f	f	t

The definition of the concept of *logical equivalence* allows, among other things, the equivalence of two statements to be established by means of implications.

Definition 1.7 (Logical equivalence)**.** Two statements are called *logically equivalent* if their truth tables are the same.

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As examples of logical equivalences that are frequently used in proof arguments, we show the statements of the following theorem.

Theorem 1.1 (Logical equivalences)**.** *Let A and B be two statements. Then the following statements are logically equivalent:*

(1) $A \Leftrightarrow B$ *and* $(A \Rightarrow B) \wedge (B \Rightarrow A)$
(2) $A \Rightarrow B$ *and* $(\neg B) \Rightarrow (\neg A)$

∘

Proof. By the definition of logical equivalence, we have to show that the truth tables of the statements under consideration are the same. We show first (1), then (2).

(1) We recall the truth table of $A \Leftrightarrow B$:

A	B	$A \Leftrightarrow B$
t	t	t
t	f	f
f	t	f
f	f	t

We further consider the truth table of $(A \Rightarrow B) \wedge (B \Rightarrow A)$:

A	B	$A \Rightarrow B$	$B \Rightarrow A$	$(A \Rightarrow B) \wedge (B \Rightarrow A)$
t	t	t	t	t
t	f	f	t	f
f	t	t	f	f
f	f	t	t	t

Comparing the truth table of $A \Leftrightarrow B$ with the first two and the last column of the truth table of $(A \Rightarrow B) \wedge (B \Rightarrow A)$ shows their equality.

(2) We recall the truth table of $A \Rightarrow B$:

A	B	$A \Rightarrow B$
t	t	t
t	f	f
f	t	t
f	f	t

We further consider the truth table of $(\neg B) \Rightarrow (\neg A)$:

A	B	$\neg B$	$\neg A$	$(\neg B) \Rightarrow (\neg A)$
t	t	f	f	t
t	f	t	f	f
f	t	f	t	t
f	f	t	t	t

Comparing the truth table of $A \Rightarrow B$ with the first two and the last column of the truth table of $(\neg B) \Rightarrow (\neg A)$ shows their equality. □

The first statement of Theorem 1.1 says that the statement "A and B are equivalent" is logically equivalent to the statement "from A follows B and from B follows A". This is the basis for many so-called *direct proofs* using equivalence transformations. The second statement of Theorem 1.1 says that the statement "from A follows B" is logically equivalent to the statement "from not B follows not A". This is the basis for the technique of *indirect proof*. We consider these proof techniques more closely in the following section. First, we summarize the meanings of the symbols introduced in this section once more in the table below.

Table 1.10 Symbol overview. A and B are statements, that is, A and B are either true or false.

Symbol	Meaning	Note
$\neg A$	Not A	True when A is false, and vice versa
$A \wedge B$	A and B	True only when both A and B are true
$A \vee B$	A and/or B	True when at least one of the statements is true
$A \Rightarrow B$	From A follows B	B is necessary for A, A is sufficient for B
$A \Leftrightarrow B$	A is equivalent to B	Both $A \Rightarrow B$ and $B \Rightarrow A$ hold

1.4 Equivalence transformations

Mathematical problems often lead to the case in which information about an unknown variable is represented implicitly by means of an equation or an inequality. To represent the information about the corresponding variable explicitly, that is, to determine its value in the case of equations or to determine the range of numbers in which the value of the variable lies in the case of inequalities, one uses *equivalence transformations*. Here, we consider only the equivalence transformations of equations and inequalities with respect to real variables that are familiar from school mathematics. Equivalence transformations of equations and inequalities have the property that the truth value of the statement formulated by an equation or inequality does not change when the corresponding transformation is applied. Both sides of the equation or inequality are always transformed. For the application of a mathematical operation that transforms an equation or inequality statement A into an equation or inequality statement B to be an equivalence transformation of the form $A \Leftrightarrow B$, both $A \Rightarrow B$ and $B \Rightarrow A$ must hold, as is well known. This implies that equivalence transformations are reversible (invertible) operations.

For equations, admissible equivalence transformations include in particular

- adding a number to both sides of the equation,
- subtracting a number from both sides of the equation,
- multiplying both sides of the equation by a nonzero number,
- dividing both sides of the equation by a nonzero number, and

- applying an invertible function to both sides of the equation.

Example

Anticipating Section 4.3, we consider the statement

$$2\exp(x) - 2 = 0. \tag{1.3}$$

Then

$$\begin{aligned}
& 2\exp(x) - 2 = 0 \\
\Leftrightarrow\; & 2\exp(x) - 2 + 2 = +2 \\
\Leftrightarrow\; & 2\exp(x) = 2 \\
\Leftrightarrow\; & \frac{1}{2}\cdot 2\exp(x) = \frac{1}{2}\cdot 2 \\
\Leftrightarrow\; & \exp(x) = 1 \\
\Leftrightarrow\; & \ln(\exp(x)) = \ln(1) \\
\Leftrightarrow\; & x = 0.
\end{aligned} \tag{1.4}$$

In summary, therefore,

$$2\exp(x) - 2 = 0 \Leftrightarrow x = 0. \tag{1.5}$$

For inequalities, admissible equivalence transformations include in particular

- adding a number to both sides of the inequality,
- subtracting a number from both sides of the inequality,
- multiplying both sides of the inequality by a nonzero number, with multiplication by a negative number reversing the inequality sign,
- dividing both sides of the inequality by a nonzero number, with division by a negative number reversing the inequality sign, and
- applying an invertible monotonically increasing function to both sides of the inequality; for monotonically decreasing functions, the inequality sign is reversed.

Example

We consider the statement

$$-5x - 2 \geq 8. \tag{1.6}$$

Then

$$\begin{aligned}
& -5x - 2 \geq 8 \\
\Leftrightarrow\; & -5x - 2 + 2 \geq 8 + 2 \\
\Leftrightarrow\; & -5x \geq 10 \\
\Leftrightarrow\; & -\frac{1}{5}\cdot 5x \geq \frac{1}{5}\cdot 10 \\
\Leftrightarrow\; & -x \geq 2 \\
\Leftrightarrow\; & x \leq -2.
\end{aligned} \tag{1.7}$$

In summary, therefore,

$$-5x - 2 \geq 8 \Leftrightarrow x \leq -2. \tag{1.8}$$

1.5 Proof techniques

In this section, we outline three fundamental proof techniques: *direct* and *indirect proofs*, and *proof by contradiction.* In what follows, especially the first technique will be used repeatedly to justify theorems.

We have:

- *Direct proofs* use equivalence transformations to show $A \Rightarrow B$.
- *Indirect proofs* use the logical equivalence of $A \Rightarrow B$ and $(\neg B) \Rightarrow (\neg A)$.
- *Proofs by contradiction* show that $(\neg B) \wedge A$ is false.

Equipped with this, we now prove the following theorem by means of a direct proof, an indirect proof, and a proof by contradiction (cf. Arens et al. (2018)).

Theorem 1.2 (Squares of positive numbers)**.** *Let a and b be two positive numbers. Then $a^2 < b^2 \Rightarrow a < b$.*

∘

Proof. We first give a *direct proof.* Let $a^2 < b^2$ be the statement A and let $a < b$ be the statement B. Then

$$a^2 < b^2 \Leftrightarrow 0 < b^2 - a^2 \Leftrightarrow 0 < (b+a)(b-a) \Leftrightarrow 0 < (b-a) \Leftrightarrow a < b. \tag{1.9}$$

We now give an *indirect proof.* Let $a^2 \geq b^2$ be the statement $\neg A$. Furthermore, let $a \geq b$ be the statement $\neg B$. Then

$$a \geq b \Leftrightarrow a^2 \geq ab \wedge ab \geq b^2 \Leftrightarrow a^2 \geq b^2. \tag{1.10}$$

Finally, we give a *proof by contradiction.* To do so, we show that the assumption $(\neg B) \wedge A$ leads to a false statement. We have

$$a \geq b \wedge a^2 < b^2 \Leftrightarrow a^2 \geq ab \wedge a^2 < b^2 \Leftrightarrow ab \leq a^2 < b^2. \tag{1.11}$$

Furthermore,

$$a \geq b \wedge a^2 < b^2 \Leftrightarrow ab \geq b^2 \wedge a^2 < b^2 \Leftrightarrow a^2 < b^2 \leq ab. \tag{1.12}$$

Overall, this yields the false statement

$$ab \leq a^2 < b^2 \leq ab \Leftrightarrow ab < ab. \tag{1.13}$$

□

2 Sets

2.1 Basic definitions

Sets collect mathematical objects and form the foundation of modern mathematics. We begin with the following definition.

Definition 2.1 (Sets)**.** Following Cantor (1895), a *set* M is defined as “a collection into a whole of definite, distinct objects m of our intuition or of our thought (which are called the elements of the set)”. We write

$$m \in M \text{ or } m \notin M \tag{2.1}$$

to express that m is an element or not an element of M, respectively.

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There are at least the following ways of defining sets:

- Listing the elements in curly braces, for example $M := \{1, 2, 3\}$.
- Specifying the properties of the elements, for example $M := \{x \in \mathbb{N} | x < 4\}$.
- Equating the set with another uniquely defined set, for example $M := \mathbb{N}_3$.

The notation $\{x \in \mathbb{N} | x < 4\}$ is read as “$x \in \mathbb{N}$ such that $x < 4$”, where the meaning of $\mathbb{N}$ will be explained below. It is important to recognize that sets are *unordered* mathematical objects, that is, the order in which the elements of a set are listed does not matter. For example, $\{1, 2, 3\}$, $\{1, 3, 2\}$, and $\{2, 3, 1\}$ denote the same set, namely the set of the first three natural numbers.

Basic relations between several sets are fixed in the next definition.

Definition 2.2 (Subsets and set equality)**.** Let M and N be two sets.

- A set M is called a *subset* of a set N if, for every element $m \in M$, we also have $m \in N$. If M is a subset of N, we write
$$M \subseteq N \tag{2.2}$$
and call M a *subset* of N and N a *superset* of M.
- A set M is called a *proper subset* of a set N if, for every element $m \in M$, we also have $m \in N$, but there is at least one element $n \in N$ for which $n \notin M$. If M is a proper subset of N, we write
$$M \subset N. \tag{2.3}$$

- Two sets M and N are called *equal* if, for every element $m \in M$, we also have $m \in N$, and if, for every element $n \in N$, we also have $n \in M$. If the sets M and N are equal, we write
$$M = N. \tag{2.4}$$

•

Example

For example, consider the sets $M := \{1\}$, $N := \{1, 2\}$, and $O := \{1, 2\}$. Then, by the definitions above, $M \subset N$, because $1 \in M$ and $1 \in N$, but $2 \in N$ and $2 \notin M$. Furthermore, $N \subseteq O$, because $1 \in N$ and $1 \in O$, as well as $2 \in N$ and $2 \in O$, and there is no element of O that is not in N. Likewise, $O \subseteq N$, because $1 \in O$ and $1 \in N$, as well as $2 \in O$ and $2 \in N$, and there is no element of N that is not in O. Finally, we even have $N = O$, because, for every element $n \in N$, we also have $n \in O$, and at the same time, for every element $o \in O$, we also have $o \in N$. We represent these relations schematically by means of *Venn-Euler diagrams* in Figure 2.1.

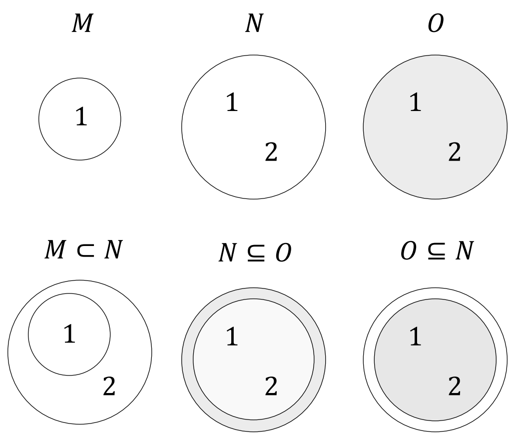

Figure 2.1 Venn-Euler diagrams of the subset properties of the sets $M := \{1\}$, $N := \{1, 2\}$, and $O := \{1, 2\}$.

An important property of a set is the number of elements it contains. This is called the *cardinality* of the set.

Definition 2.3 (Cardinality)**.** The number of elements of a set M is called its *cardinality* and is denoted by $|M|$.

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A special set is the set without elements.

Definition 2.4. A set with cardinality zero is called the *empty set* and is denoted by $\emptyset$.

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As examples, let $M := \{1, 2, 3\}$, $N = \{a, b, c, d\}$, and $O := \emptyset$. Then $|M| = 3$, $|N| = 4$, and $|O| = 0$.

For every set, one can consider the set of all subsets of this set. This leads to the important concept of the *power set*.

Definition 2.5 (Power set)**.** The set of all subsets of a set M is called the *power set of* M and is denoted by $\mathcal{P}(M)$.

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Note that the empty subset of M and M itself are also always elements of $\mathcal{P}(M)$. We first consider four examples of the concept of a power set.

- Let $M_0 := \emptyset$ be the empty set. Then

$$\mathcal{P}(M_0) = \{\emptyset\}. \tag{2.5}$$

- Let M_1 be the one-element set $M_1 := \{a\}$. Then

$$\mathcal{P}(M_1) = \{\emptyset, \{a\}\}. \tag{2.6}$$

- Let $M_2 := \{a, b\}$. Then M_2 has both one-element and two-element subsets, and

$$\mathcal{P}(M_2) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}. \tag{2.7}$$

- Finally, let $M_3 := \{a, b, c\}$. Then M_3 has, among others, one-element, two-element, and three-element subsets, and

$$\mathcal{P}(M_3) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}. \tag{2.8}$$

Theorem 2.1 (Cardinality of the power set)**.** *Given a set M with cardinality $|M| = n$, and let $\mathcal{P}(M)$ be its power set. Then $|\mathcal{P}(M)| = 2^n$.*

∘

Proof. To prove the statement of the theorem, we associate each element P of the power set of M uniquely with a binary sequence of length n, where the entry at the i-th position represents whether the i-th element of M is an element of P or not. For example, let $M := \{m_1, m_2, m_3\}$ and $P := \{m_2, m_3\}$. Then P corresponds to the binary sequence 011. The empty set $P := \emptyset$ corresponds to the binary sequence 000, and the original set $P = M$ corresponds to the binary sequence 111. Thus the question is how many unique binary sequences of length n there are. Since each element of the sequence has two possible states, there are n factors $2 \cdot 2 \cdots 2$, that is, 2^n. □

In the examples above, we have the cases

- $|M_0| = 0 \Rightarrow |\mathcal{P}(M_0)| = 2^0 = 1$,
- $|M_1| = 1 \Rightarrow |\mathcal{P}(M_1)| = 2^1 = 2$,
- $|M_2| = 2 \Rightarrow |\mathcal{P}(M_2)| = 2^2 = 4$,
- $|M_3| = 3 \Rightarrow |\mathcal{P}(M_3)| = 2^3 = 8$,

as can be verified by counting the elements of the corresponding power sets.

2.2 Operations

Two sets can be combined with one another in different ways. The result of such a combination is another set. We call the combination of two sets a *set operation* and give the following definitions.

Definition 2.6 (Set operations)**.** Let M and N be two sets.

- The *union of M and N* is defined as the set

$$M \cup N := \{x | x \in M \vee x \in N\}, \tag{2.9}$$

  where $\vee$ is to be understood according to Definition 1.4 as a *non-exclusive or*, that is, as and/or.

- The *intersection of M and N* is defined as the set

$$M \cap N := \{x | x \in M \wedge x \in N\}. \tag{2.10}$$

  If M and N satisfy $M \cap N = \emptyset$, then M and N are called *disjoint*.

- The *difference of M and N* is defined as the set

$$M \setminus N := \{x | x \in M \wedge x \notin N\}. \tag{2.11}$$

  The difference of M and N is also called, especially when $N \subseteq M$, the *complement of N with respect to M* and is denoted by N^c. In this context, M is also called the *basic set* or the *universe*.

- The *symmetric difference of M and N* is defined as the set

$$M \Delta N := \{x | (x \in M \vee x \in N) \wedge x \notin M \cap N\}. \tag{2.12}$$

  The symmetric difference can therefore be understood as an *exclusive or*.

•

Example

As an example, consider the sets $M := \{1, 2, 3\}$ and $N := \{2, 3, 4, 5\}$. Then

- $M \cup N = \{1, 2, 3, 4, 5\}$, because $1 \in M$, $2 \in M$, $3 \in M$, $4 \in N$, and $5 \in N$.
- $M \cap N = \{2, 3\}$, because only for 2 and 3 do we have $2 \in M, 3 \in M$ and also $2 \in N, 3 \in N$. For 1, we only have $1 \in M$, and for 4 and 5, we only have $4 \in N$ and $5 \in N$.
- $M \setminus N = \{1\}$, because $1 \in M$, but $1 \notin N$, and $2 \in M$, but also $2 \in N$.
- $N \setminus M = \{4, 5\}$, because $4 \in N$ and $5 \in N$, but $4 \notin M$ and $5 \notin M$. This shows in particular that the difference of M and N is *not* symmetric, that is, it is *not* necessarily the case that $M \setminus N$ equals $N \setminus M$.
- $M \Delta N = \{1, 4, 5\}$, because $1 \in M$, but $1 \notin \{2, 3\}$, $2 \in M$, but $2 \in \{2, 3\}$, $3 \in M$, but $3 \in \{2, 3\}$, $4 \in N$, but $4 \notin \{2, 3\}$, and $5 \in N$, but $5 \notin \{2, 3\}$.

Figure 2.2 visualizes the set operations considered in this example.

Finally, we introduce the concept of a partition of a set.

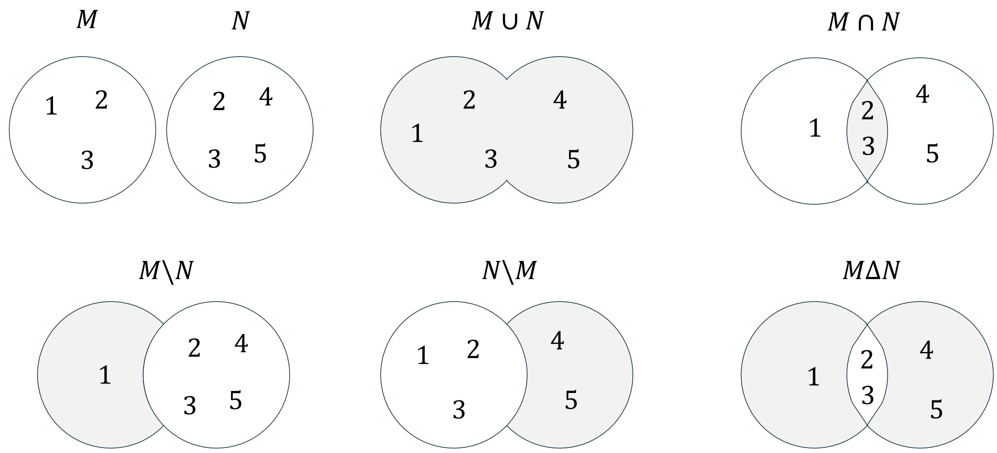

Figure 2.2 Venn-Euler diagrams of the set operations considered in the example for $M := \{1, 2, 3\}$ and $N := \{2, 3, 4, 5\}$. The gray shaded regions correspond to the resulting sets.

Definition 2.7 (Partition)**.** Let M be a set, and let $P := \{N_i\}$ be a set of sets N_i with $i = 1, ..., n$, such that

$$\left(M = \cup_{i=1}^{n} N_i\right) \wedge \left(N_i \cap N_j = \emptyset \text{ for } i, j = 1, ..., n \text{ and } i \neq j\right). \tag{2.13}$$

Then P is called a *partition of* M.

•

The partition of a set therefore corresponds to splitting the set into disjoint subsets. Partitions are generally not unique, that is, there are usually several ways to partition a given set.

As an example, consider the set $M := \{1, 2, 3, 4, 5, 6\}$. Then $P_1 := \{\{1\}, \{2, 3, 4, 5, 6\}\}$, $P_2 := \{\{1, 2, 3\}, \{4, 5, 6\}\}$, and $P_3 := \{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$ are three possible partitions of M. Figure 2.3 visualizes the partitions considered in this example.

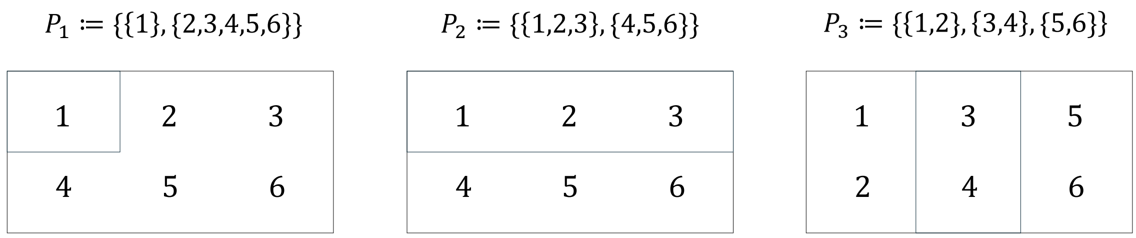

Figure 2.3 Diagrams of the partitions considered in the example for $M := \{1, 2, 3, 4, 5, 6\}$.

2.3 Special sets

Number sets

In natural science, one attempts to describe phenomena of the world that are intuitively identified as discrete or continuous by means of numbers. Depending on the type of phenomenon, different number sets are suitable for this purpose. Mathematics provides, among others, the number sets given in the following definition.

Definition 2.8 (Number sets)**.** The following denote:

- $\mathbb{N} \;:= \{1, 2, 3, ...\}$ the *natural numbers*,
- $\mathbb{N}_n := \{1, 2, 3, ..., n\}$ the *natural numbers of order n*,
- $\mathbb{N}^0 := \mathbb{N} \cup \{0\}$ the *natural numbers* and zero,
- $\mathbb{Z} \;:= \{..., -3, -2, -1, 0, 1, 2, 3...\}$ the *integers*,
- $\mathbb{Q} \;:= \{\frac{p}{q} | p \in \mathbb{Z}, q \in \mathbb{N}\}$ the *rational numbers*,
- $\mathbb{R}$ the *real numbers*, and
- $\mathbb{C} \;:= \{a + ib | a, b \in \mathbb{R}, i := \sqrt{-1}\}$ the *complex numbers*.

•

The natural numbers and the integers are suitable for quantifying discrete phenomena. The rational numbers and especially the real numbers are suitable for quantifying continuous phenomena. $\mathbb{R}$ includes the rational numbers and the so-called *irrational numbers* $\mathbb{R} \setminus \mathbb{Q}$. Rational numbers are numbers that can be expressed as fractions of integers and natural numbers. These include all integers as well as negative and positive decimal numbers such as $-\frac{9}{10} = -0.9$, $\frac{1}{3} = 0.\bar{3}$, and $\frac{196}{100} = 1.96$. Irrational numbers are numbers that cannot be expressed as rational numbers. Examples of irrational numbers are *Euler's number* $e \approx 2.71$, the *circle constant* $\pi \approx 3.14$, and the square root of 2, $\sqrt{2} \approx 1.41$.

The real numbers contain the natural numbers, the integers, and the rational numbers as subsets. Thus there are very many real numbers. Indeed, Cantor (1874) proved that there are more real numbers than natural numbers, even though there are infinitely many real numbers and infinitely many natural numbers. This property of the real numbers is called their *uncountability*. We will deepen this concept in Chapter 4. In particular,

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}. \tag{2.14}$$

Between any two real numbers there are infinitely many further real numbers. Positive infinity ∞ and negative infinity $-\infty$ are not numbers with which one could calculate in standard mathematics. They also do not belong to the number sets given in the definition above; thus $\infty \notin \mathbb{R}$ and $-\infty \notin \mathbb{R}$. Complex numbers are suitable for describing two-dimensional continuous phenomena. The values of the first dimension are represented in the real part a, and the values of the second dimension in the complex part b of a complex number. Complex numbers are used, for example, in the modeling of physical phenomena and in Fourier analysis.

Important subsets of the real numbers are the so-called *intervals*. We give the following definitions.

Intervals

Definition 2.9. Connected subsets of the real numbers are called *intervals*. For $a, b \in \mathbb{R}$, one distinguishes

- the *closed interval*

$$[a, b] := \{x \in \mathbb{R} | a \le x \le b\}, \tag{2.15}$$

- the *open interval*
$$]a, b[:= \{x \in \mathbb{R} | a < x < b\}, \tag{2.16}$$
- and the *half-open intervals*
$$]a, b] := \{x \in \mathbb{R} | a < x \leq b\} \text{ and } [a, b[:= \{x \in \mathbb{R} | a \leq x < b\}. \tag{2.17}$$

•

As examples, Figure 2.4 graphically represents the intervals $[1, 2]$, $]1, 2]$, $[1, 2[$, and $]1, 2[$. Here one imagines the real numbers as a continuous number line and must in each case note whether the left and right endpoints are part of the interval or not. As mentioned above, positive infinity (∞) and negative infinity $-\infty$ are not elements of $\mathbb{R}$. Thus one always writes $] - \infty, b]$ or $] - \infty, b[$ and $]a, \infty[$ or $[a, \infty[$, as well as $\mathbb{R} =] - \infty, \infty[$.

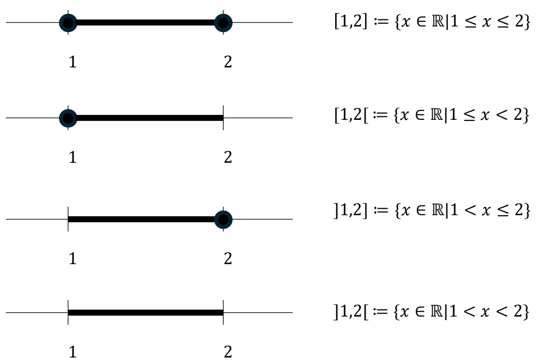

Figure 2.4 Representation of intervals on the number line. The black dot indicates that the corresponding number is part of the interval.

Cartesian products

Often one wants to quantitatively describe several independent properties of a phenomenon at the same time. For this purpose, the one-dimensional number sets defined above can be extended to multidimensional number sets by forming *Cartesian products*. The elements of Cartesian products are called *ordered tuples* or *vectors*.

Definition 2.10 (Cartesian products)**.** Let M and N be two sets. Then the *Cartesian product of the sets M and N* is the set of all ordered tuples (m, n) with $m \in M$ and $n \in N$, formally
$$M \times N := \{(m, n) | m \in M, n \in N\}. \tag{2.18}$$

The Cartesian product of a set M with itself is denoted by
$$M^2 := M \times M. \tag{2.19}$$

Furthermore, let $M_1, M_2, ..., M_n$ be sets. Then the *Cartesian product of the sets $M_1, ..., M_n$* is the set of all ordered n-tuples $(m_1, ..., m_n)$ with $m_i \in M_i$ for $i = 1, ..., n$, formally
$$\prod_{i=1}^{n} M_i := M_1 \times \cdots \times M_n := \{(m_1, ..., m_n) | m_i \in M_i \text{ for } i = 1, ..., n\}. \tag{2.20}$$

The n-fold Cartesian product of a set M with itself is denoted by

$$M^n := \prod_{i=1}^{n} M := \{(m_1, ..., m_n)|m_i \in M \text{ for } i = 1, ..., n\}. \tag{2.21}$$

•

In contrast to sets, the tuples introduced in Definition 2.10 are *ordered*. This means, for example, that for sets we have $\{1, 2\} = \{2, 1\}$, but for tuples we have $(1, 2) \neq (2, 1)$.

Example

Let $M := \{1, 2\}$ and $N := \{1, 2, 3\}$. Then the Cartesian product $M \times N$ is given by

$$M \times N := \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\} \tag{2.22}$$

and the Cartesian product $N \times M$ is given by

$$N \times M := \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}. \tag{2.23}$$

The Cartesian product is therefore generally not commutative, that is, it is not necessarily the case that $M \times N = N \times M$. One may construct the sets $M \times N \neq N \times M$ in this example from Table 2.1 and Table 2.2.

Table 2.1 Cartesian product $M \times N$

(m, n)	$n = 1$	$n = 2$	$n = 3$
$m = 1$	$(1, 1)$	$(1, 2)$	$(1, 3)$
$m = 2$	$(2, 1)$	$(2, 2)$	$(2, 3)$

Table 2.2 Cartesian product $N \times M$

(n, m)	$m = 1$	$m = 2$
$n = 1$	$(1, 1)$	$(1, 2)$
$n = 2$	$(2, 1)$	$(2, 2)$
$n = 3$	$(3, 1)$	$(3, 2)$

$\mathbb{R}$ to the Power of n

As described above, the real numbers are particularly suitable for describing continuous phenomena. For the simultaneous description of several aspects of a continuous phenomenon, the *set of real tuples of n-th order*, or $\mathbb{R}$ to the power of n for short, is correspondingly useful.

Definition 2.11 (Set of real tuples of n-th order)**.** The n-fold Cartesian product of the real numbers with themselves is denoted by

$$\mathbb{R}^n := \prod_{i=1}^{n} \mathbb{R} := \{x := (x_1, ..., x_n)|x_i \in \mathbb{R}\} \tag{2.24}$$

and is read as "$\mathbb{R}$ to the power of n". We write the elements of $\mathbb{R}^n$ as columns

$$x := \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \tag{2.25}$$

and call them *n-dimensional vectors*. For distinction, we also call the elements of $\mathbb{R}^1 = \mathbb{R}$ *scalars*.

•

Examples

Familiar examples of $\mathbb{R}^n$ are $\mathbb{R}^1$ as the set of real numbers, $\mathbb{R}^2$ as the set of real tuples in the model of the two-dimensional plane, and $\mathbb{R}^3$ as the set of real triples in the model of three-dimensional space, as visualized in Figure 2.5.

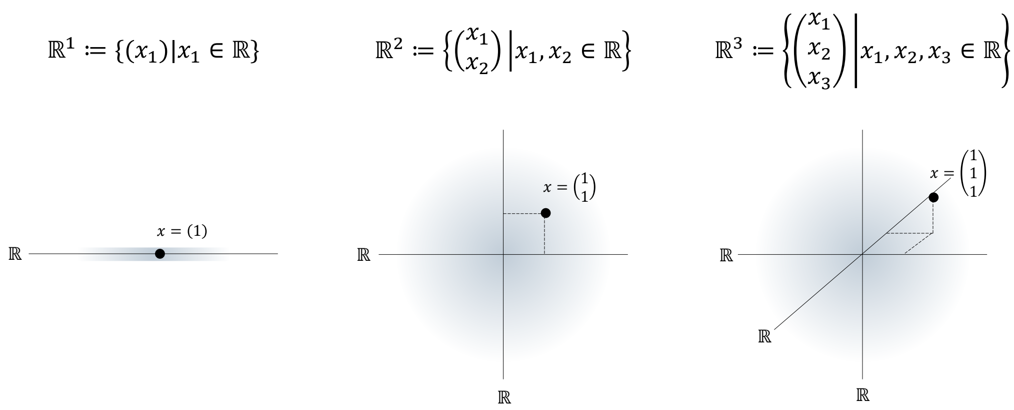

Figure 2.5 $\mathbb{R}^n$ for $n = 1, n = 2$, and $n = 3$. The black dots each represent an element of the corresponding set.

An example of an $x \in \mathbb{R}^4$ is

$$x = \begin{pmatrix} 0.16 \\ 1.76 \\ 0.23 \\ 7.11 \end{pmatrix}. \tag{2.26}$$

3 Sums, products, powers

This unit introduces notation for the elementary arithmetic operations.

3.1 Sums

Definition 3.1 (Summation symbol)**.** It denotes

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n. \tag{3.1}$$

Here,

- Σ denotes the Greek letter *S*igma, mnemonically standing for *s*um,
- the subscript $i = 1$ denotes the *running index* and the *starting index*,
- the superscript n denotes the *terminal index*, and
- $x_1, x_2, ..., x_n$ denote the *summands*.

•

For meaningful use of the summation symbol, it is essential to specify the beginning and the end of the summation by means of the subscript and the superscript. The precise name of the running index, by contrast, is irrelevant for the value of the sum; thus,

$$\sum_{i=1}^{n} x_i = \sum_{j=1}^{n} x_j. \tag{3.2}$$

Sometimes the running index is also given as an element of an *index set*. If, for example, the index set $I := \{1, 5, 7\}$ is defined, then

$$\sum_{i \in I} x_i := x_1 + x_5 + x_7. \tag{3.3}$$

In the following, we briefly consider a few examples of how the summation symbol is used.

- *Summation of predefined summands.* Let $x_1 := 2$, $x_2 := 10$, and $x_3 := -4$. Then

$$\sum_{i=1}^{3} x_i = x_1 + x_2 + x_3 = 2 + 10 - 4 = 8. \tag{3.4}$$

- *Summation of weighted predefined summands.* Again let $x_1 := 2$, $x_2 := 10$, and $x_3 := -4$. In addition, let the *weighting coefficients* $a_1 := \frac{1}{2}$, $a_2 := \frac{1}{5}$, and $a_3 := 2$ be defined. Then

$$\sum_{i=1}^{3} a_i x_i = a_1 x_1 + a_2 x_2 + a_3 x_3 = \frac{1}{2} \cdot 2 + \frac{1}{5} \cdot 10 + 2 \cdot (-4) = 1 + 2 - 8 = -5. \quad (3.5)$$

  Expressions of the form $\sum_{i=1}^{n} a_i x_i$ are also called *linear combinations* of $x_1, ..., x_n$ with the *coefficients* or *weighting parameters* $a_1, ..., a_n$.
- *Summation of natural numbers.* We have

$$\sum_{i=1}^{5} i = 1 + 2 + 3 + 4 + 5 = 15. \quad (3.6)$$

- *Summation of even natural numbers.* We have

$$\sum_{i=1}^{5} 2i = 2 \cdot 1 + 2 \cdot 2 + 2 \cdot 3 + 2 \cdot 4 + 2 \cdot 5 = 2 + 4 + 6 + 8 + 10 = 30. \quad (3.7)$$

- *Summation of odd natural numbers.* We have

$$\sum_{i=1}^{5} (2i-1) = 2\cdot1-1+2\cdot2-1+2\cdot3-1+2\cdot4-1+2\cdot5-1 = 1+3+5+7+9 = 25. \quad (3.8)$$

Working with the summation symbol can often be simplified by applying the following calculation rules.

Theorem 3.1 (Calculation rules for sums)**.**

(1) Sums of identical summands

$$\sum_{i=1}^{n} x = nx \quad (3.9)$$

(2) Associativity for sums of equal length

$$\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i = \sum_{i=1}^{n} (x_i + y_i) \quad (3.10)$$

(3) Distributivity under multiplication by a constant

$$\sum_{i=1}^{n} a x_i = a \sum_{i=1}^{n} x_i \quad (3.11)$$

(4) Splitting sums for $1 < m < n$

$$\sum_{i=1}^{n} x_i = \sum_{i=1}^{m} x_i + \sum_{i=m+1}^{n} x_i \quad (3.12)$$

(5) Reindexing

$$\sum_{i=0}^{n} x_i = \sum_{j=m}^{n+m} x_{j-m} \quad (3.13)$$

∘

Proof. These calculation rules can be verified by writing out the sums and applying the rules of addition and multiplication. Here, we show associativity for sums of equal length and distributivity under multiplication by a constant as examples. For the former, we have

$$
\begin{aligned}
\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i &= x_1 + x_2 + \cdots + x_n + y_1 + y_2 + \cdots + y_n \\
&= x_1 + y_1 + x_2 + y_2 + \cdots + x_n + y_n \\
&= \sum_{i=1}^{n} (x_i + y_i).
\end{aligned} \tag{3.14}
$$

For the latter, we have

$$
\begin{aligned}
\sum_{i=1}^{n} a x_i &= a x_1 + a x_2 + \cdots + a x_n \\
&= a(x_1 + x_2 + \cdots + x_n) \\
&= a \sum_{i=1}^{n} x_i.
\end{aligned} \tag{3.15}
$$

□

Examples

As a first example of applying the calculation rules recorded in Theorem 3.1, we consider the evaluation of a *mean* (sometimes also called an *average*). Let $x_1, x_2, ..., x_n$ be real numbers. The mean of these numbers is the sum of $x_1, x_2, ..., x_n$ divided by the number of numbers n. By statement (3) of Theorem 3.1, it is irrelevant whether the numbers are first added up and the resulting sum is then divided by n, or whether the individual numbers are each divided by n and the corresponding results are then added up. More precisely, by applying Theorem 3.1 (3) with $a = 1/n$, we obtain

$$
\frac{1}{n} \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} \frac{x_i}{n}. \tag{3.16}
$$

For example, the mean of $x_1 := 1, x_2 := 4, x_3 := 2$, and $x_4 := 1$ is given by

$$
\frac{1}{4} \sum_{i=1}^{4} x_i = \frac{1}{4}(1 + 4 + 2 + 1) = \frac{8}{4} = 2 = \frac{8}{4} = \frac{1}{4} + \frac{4}{4} + \frac{2}{4} + \frac{1}{4} = \sum_{i=1}^{4} \frac{x_i}{4}. \tag{3.17}
$$

As a second example, we consider the reindexing rule recorded in Theorem 3.1 (5). Let $n := 3$ and $m := 2$, and let $x_0 := 2$, $x_1 := 3$, $x_2 := 5$, and $x_3 := 10$. Then, evidently,

$$
\begin{aligned}
\sum_{i=0}^{3} x_i &= x_0 + x_1 + x_2 + x_3 \\
&= 2 + 3 + 5 + 10 \\
&= 20.
\end{aligned} \tag{3.18}
$$

But we also have

$$
\begin{aligned}
\sum_{j=m}^{n+m} x_{j-m} &= \sum_{j=2}^{3+2} x_{j-2} \\
&= \sum_{j=2}^{5} x_{j-2} \\
&= x_{2-2} + x_{3-2} + x_{4-2} + x_{5-2} \\
&= x_0 + x_1 + x_2 + x_3 \\
&= 2 + 3 + 5 + 10 \\
&= 20.
\end{aligned} \tag{3.19}
$$

Double Sums

In applications, one often encounters expressions that carry out several summations one after another. For each of these sums, the definitions and calculation rules listed above apply. The meaning of double sums is best made clear by writing them out from the inside to the outside, while noting that the running index of the outer sum remains constant during each iteration of the inner sum.

The following examples may illustrate this.

(1)

$$
\begin{aligned}
\sum_{i=1}^{2}\sum_{j=1}^{3}(i+j) &= \sum_{i=1}^{2}(i+1+i+2+i+3) \\
&= (1+1+1+2+1+3) + (2+1+2+2+2+3) \\
&= 9 + 12 \\
&= 21
\end{aligned} \tag{3.20}
$$

(2)

$$
\begin{aligned}
\sum_{i=1}^{2}\sum_{j=1}^{3}(x_i+y_j) &= \sum_{i=1}^{2}(x_i+y_1+x_i+y_2+x_i+y_3) \\
&= (x_1+y_1+x_1+y_2+x_1+y_3) + (x_2+y_1+x_2+y_2+x_2+y_3)
\end{aligned} \tag{3.21}
$$

(3)

$$
\begin{aligned}
\sum_{i=1}^{3}\sum_{j=1}^{2}(x_iy_j) &= \sum_{i=1}^{3}(x_iy_1+x_iy_2) \\
&= (x_1y_1+x_1y_2) + (x_2y_1+x_2y_2) + (x_3y_1+x_3y_2)
\end{aligned} \tag{3.22}
$$

3.2 Products

The product symbol provides notation for products that is analogous to the summation symbol.

Definition 3.2 (Product symbol)**.** It denotes

$$\prod_{i=1}^{n} x_i = x_1 \cdot x_2 \cdot \dots \cdot x_n. \tag{3.23}$$

Here,

- $\prod$ denotes the Greek letter *P*i, mnemonically standing for *p*roduct,
- the subscript $i = 1$ denotes the *running index* and the *starting index*,
- the superscript n denotes the *terminal index*, and
- $x_1, x_2, ..., x_n$ denote the *product terms.*

•

Analogously to the summation symbol, the product symbol is meaningful only with a subscript and a superscript specifying the running and terminal index. The precise name of the running index is again irrelevant; thus,

$$\prod_{i=1}^{n} x_i = \prod_{j=1}^{n} x_j. \tag{3.24}$$

Here, too, the running index is in rare cases given as an element of an index set. If, for example, the index set $J := \mathbb{N}_2^0$ is defined, then

$$\prod_{j \in J} x_j := x_0 \cdot x_1 \cdot x_2. \tag{3.25}$$

One example of the use of the product symbol is the definition of the *factorial* of a natural number n by

$$n! := \prod_{i=1}^{n} i. \tag{3.26}$$

For example,

$$3! := \prod_{i=1}^{3} i = 1 \cdot 2 \cdot 3 = 6. \tag{3.27}$$

There are also a number of calculation rules for products that often simplify working with them. We list some in the following theorem. In doing so, we make anticipatory use of the notation of *powers* defined in Section 3.3.

Theorem 3.2 (Calculation rules for products)**.**

(1) Products of identical factors

$$\prod_{i=1}^{n} x = x^n \tag{3.28}$$

(2) Raising constants to powers

$$\prod_{i=1}^{n} a x_i = a^n \prod_{i=1}^{n} x_i \tag{3.29}$$

(3) Splitting products for $1 < m < n$

$$\prod_{i=1}^{n} x_i = \prod_{i=1}^{m} x_i \prod_{j=m+1}^{n} x_j \tag{3.30}$$

(4) Product of products

$$\prod_{i=1}^{n} x_i y_i = \prod_{i=1}^{n} x_i \prod_{i=1}^{n} y_i \tag{3.31}$$

∘

3.3 Powers

Products of numbers with themselves can be abbreviated using power notation.

Definition 3.3 (Power)**.** For $a \in \mathbb{R}$ and $n \in \mathbb{N}^0$, the *n-th power of* a is defined by

$$a^0 := 1 \text{ and } a^{n+1} := a^n \cdot a. \tag{3.32}$$

Furthermore, for $a \in \mathbb{R} \setminus \{0\}$ and $n \in \mathbb{N}^0$, the *negative n-th power of* a is defined by

$$a^{-n} := (a^n)^{-1} := \frac{1}{a^n}. \tag{3.33}$$

a is called the *base*, and n is called the *exponent*.

•

The way a^{n+1} is defined by referring back to the power a^n in the above definition is called *recursive*. The definition $a^0 := 1$ is called the *initial case* of the recursion; it is what makes the recursive definition of a^{n+1} possible in the first place. The definition $a^{n+1} := a^n \cdot a$ is also called the *recursion step*. The following calculation rules simplify computations with powers.

Theorem 3.3 (Calculation rules for powers)**.** *For* $a, b \in \mathbb{R}$ *and* $n, m \in \mathbb{Z}$ *with* $a \neq 0$ *for negative exponents, the following calculation rules hold:*

$$a^n a^m = a^{n+m} \tag{3.34}$$
$$(a^n)^m = a^{nm} \tag{3.35}$$
$$(ab)^n = a^n b^n \tag{3.36}$$

∘

Instead of a proof, we consider the following three examples.

(1)

$$2^2 \cdot 2^3 = (2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2^5 = 2^{2+3} \tag{3.37}$$

(2)

$$(3^2)^3 = (3 \cdot 3)^3 = (3 \cdot 3) \cdot (3 \cdot 3) \cdot (3 \cdot 3) = 3^6 = 3^{2 \cdot 3} \tag{3.38}$$

(3)
$$(2\cdot 4)^2 = (2\cdot 4)\cdot(2\cdot 4) = (2\cdot 2)\cdot(4\cdot 4) = 2^2\cdot 4^2 \tag{3.39}$$

Closely related to powers is the definition of the n-th root:

Definition 3.4 (n-th root)**.** For $a \in \mathbb{R}_{\geq 0}$ and $n \in \mathbb{N}$, the *n-th root of a* is defined as the nonnegative real number r with
$$r^n = a. \tag{3.40}$$

•

When computing with roots, power notation for roots is often helpful because it allows the calculation rules for powers to be applied directly.

Theorem 3.4 (Power notation for the n-th root)**.** *Let $a \in \mathbb{R}_{\geq 0}$, let $n \in \mathbb{N}$, and let r be the n-th root of a. Then*
$$r = a^{\frac{1}{n}} \tag{3.41}$$

∘

Proof. We have
$$\left(a^{\frac{1}{n}}\right)^n = a^{\frac{1}{n}}\cdot a^{\frac{1}{n}}\cdot \cdots \cdot a^{\frac{1}{n}} = a^{\sum_{i=1}^n \frac{1}{n}} = a^1 = a. \tag{3.42}$$
Thus, by Definition 3.4, $r = a^{\frac{1}{n}}$. □

Computing with square roots is greatly facilitated by the power notation
$$\sqrt{x} = x^{\frac{1}{2}} \tag{3.43}$$

For example,
$$\frac{2\pi}{\sqrt{2\pi}} = \frac{2\pi}{(2\pi)^{\frac{1}{2}}} = (2\pi)^1\cdot(2\pi)^{-\frac{1}{2}} = (2\pi)^{1-\frac{1}{2}} = (2\pi)^{\frac{1}{2}} = \sqrt{2\pi}. \tag{3.44}$$

The following relation between square, root, and absolute value is used quite often.

Theorem 3.5 (Root and absolute value)**.** *Let $x \in \mathbb{R}$, and let*
$$|x| := \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0 \end{cases} \tag{3.45}$$
denote the absolute value of x. *Then*
$$\sqrt{x^2} = |x|. \tag{3.46}$$

∘

Proof. We consider the cases $x \geq 0$ and $x < 0$. First, let $x \geq 0$. Then
$$x \geq 0 \Rightarrow x^2 \geq 0 \Rightarrow \sqrt{x^2} = x = |x|. \tag{3.47}$$
Now let $x < 0$. Then
$$x < 0 \Rightarrow x^2 > 0 \Rightarrow \sqrt{x^2} = -x = |x|. \tag{3.48}$$
□

4 Functions

Alongside sets, functions are the second foundational pillar of modern mathematics. In this unit, we define the concept of a function, introduce first properties of functions, and give an overview of several elementary functions.

4.1 Definition and properties

Definition 4.1 (Function)**.** A *function* or *mapping* f is an assignment rule that assigns exactly one element of a set Z to each element of a set D. D is called the *domain of* f, and Z is called the *codomain of* f. We write

$$f : D \to Z, x \mapsto f(x), \tag{4.1}$$

where $f : D \to Z$ is read as "the function f maps all elements of the set D uniquely to elements in Z" and $x \mapsto f(x)$ is read as "x, which is an element of D, is mapped by the function f to $f(x)$, where $f(x)$ is an element of Z." The arrow $\to$ denotes the mapping between the sets D and Z, while the arrow $\mapsto$ denotes the mapping between an element of D and an element of Z.

•

It is essential to distinguish between the *function* f as an assignment rule and a *value of the function* $f(x)$ as an element of Z. x is the *argument* of the function (the function's *input*), and $f(x)$ is the value that the function f takes for the argument x (the function's *output*). Usually, the definition of a function specifies, after $f(x)$, the *functional form of* f, that is, a rule for forming the value $f(x)$ from x. For example, in the following definition of a function

$$f : \mathbb{R} \to \mathbb{R}_{\geq 0}, x \mapsto f(x) := x^2 \tag{4.2}$$

the definition of a power is used. Note that functions are always *unique* in the sense that, whenever the function is applied, they always assign one and the same $f(x) \in Z$ to each $x \in D$.

Figure 4.1 visualizes several aspects of the definition of a function. Figure 4.1 A first illustrates the central terms of the definition. Figure 4.1 B shows a first example of a function in which the function values assigned to the arguments are determined not by a computational rule but by direct definition. Figure 4.1 C and Figure 4.1 D use the same pictorial language to emphasize two aspects of the definition by means of counterexamples: the drawing in Figure 4.1 C is not the representation of a function, because by Definition 4.1 a function assigns *each element* of a set D exactly one element of a codomain Z. The sketched assignment rule does not assign any element in Z to the element $2 \in D$, and therefore it is not a function. Similarly, according to Definition 4.1, a function assigns exactly one element of a codomain Z to each element of a set D. The assignment rule

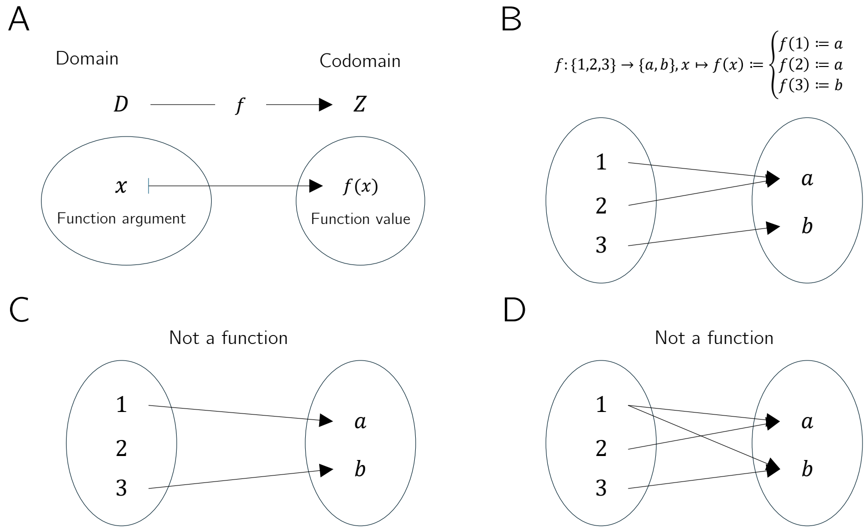

Figure 4.1 Aspects of the definition of a function

sketched in Figure 4.1 D assigns both the element $a \in Z$ and the element $b \in Z$ to $1 \in D$, and therefore it is incompatible with Definition 4.1.

Thus, functions put elements of sets into relation with one another. The sets of these elements receive special names.

Definition 4.2 (Image, range, preimage set, preimage)**.** Let $f : D \to Z, x \mapsto f(x)$ be a function and let $D' \subseteq D$ and $Z' \subseteq Z$. The set

$$f(D') := \{z \in Z | \text{there exists an } x \in D' \text{ with } z = f(x)\} \tag{4.3}$$

is called the *image of* D', and $f(D) \subseteq Z$ is called the *range* of f. Furthermore, the set

$$f^{-1}(Z') := \{x \in D | f(x) \in Z'\} \tag{4.4}$$

is called the *preimage set of* Z'. An $x \in D$ with $z = f(x) \in Z$ is called a *preimage of* z.

•

Note that the range $f(D)$ of f and the codomain Z of f need not be identical.

Example

To clarify the concepts introduced in Definition 4.2, we consider the function shown in Figure 4.2 A,

$$f : \{1,2,3,4,5\} \to \{a,b,c,d\}, x \mapsto f(x) := \begin{cases} f(1) := b \\ f(2) := d \\ f(3) := c \\ f(4) := c \\ f(5) := d \end{cases}. \tag{4.5}$$

According to Definition 4.2, an image is always defined with respect to a subset D' of the domain D. Thus, let $D' := \{2,3\} \subset D$, as shown in Figure 4.2 B. Then the image of D' is the set of those $z \in Z$ for which there exists an $x \in D'$ such that $z = f(x)$. Here, the relevant $z \in Z$ for which such an $x \in D'$ exists are precisely $c, d \in Z$, because $f(2) = d$ and $f(3) = c$. Beyond these, there is no $z \in Z$ with $f(x) = z$ and $x \in \{2, 3\}$. By Definition 4.2, the range $f(D)$ is the subset of those $z \in Z$ for which there exists an $x \in D$ such that $z = f(x)$. This is true for all elements of Z except for $a \in Z$, because there is no $x \in D$ with $a = f(x)$. For the function considered here, the range of f is therefore given by $f(D) := \{b, c, d\}$.

Conversely, according to Definition 4.2, a preimage set is always defined with respect to a subset Z' of the codomain Z. Thus, let $Z' := \{c, d\} \subset Z$, as shown in Figure 4.2 C. Then the preimage set of Z' is the set of those $x \in D$ for which $f(x) \in Z'$. The elements of D whose function values under f are elements of Z' are precisely $\{2, 3, 4, 5\}$. By contrast, $1 \in D$ is not an element of this preimage set, because f maps $1 \in D$ to $b \in Z$ and $b \notin Z'$. Nevertheless, $1 \in D$ is of course a preimage of b.

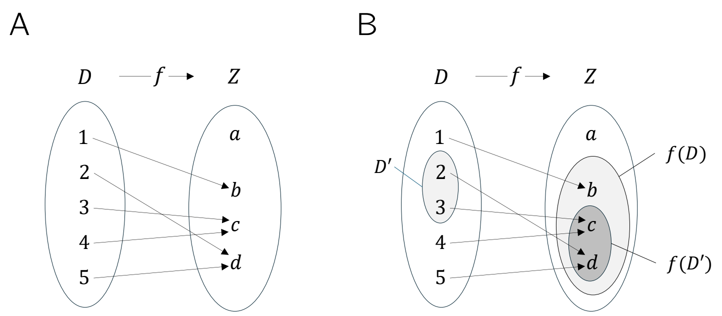

Figure 4.2 (A) Visualization of the function Equation 4.5. (B) Range and an image of f. (C) Preimage set of a set Z' under f

Basic properties of functions are named in the following definition.

Definition 4.3 (Injectivity, surjectivity, bijectivity)**.** Let $f : D \to Z, x \mapsto f(x)$ be a function. f is called *injective* if, for every image element $z \in f(D)$, there is exactly one preimage $x \in D$. Equivalently, f is injective if $x_1, x_2 \in D$ and $x_1 \neq x_2$ imply $f(x_1) \neq f(x_2)$. f is called *surjective* if $f(D) = Z$, that is, if every element of the codomain Z has a preimage in the domain D. Finally, f is called *bijective* if it is both injective and surjective. Bijective functions are also called *one-to-one correspondences* or *one-to-one mappings*.

•

Examples

We first illustrate Definition 4.3 using three (counter)examples in Figure 4.3. Figure 4.3 A shows the *non-injective* function

$$f : \{1, 2, 3\} \to \{a, b\}, x \mapsto f(x) := \begin{cases} f(1) := a \\ f(2) := a \\ f(3) := b \end{cases} . \tag{4.6}$$

The function is non-injective because the element a in the range of f has more than one preimage in the domain of f, namely the elements 1 and 2. Figure 4.3 B shows the *non-surjective* function

$$g : \{1, 2, 3\} \to \{a, b, c, d\}, x \mapsto g(x) := \begin{cases} g(1) := a \\ g(2) := b \\ g(3) := d \end{cases} . \tag{4.7}$$

The function is non-surjective because the element c in the codomain of g has no preimage in the domain of g. Finally, Figure 4.3 C shows the bijective function

$$h : \{1, 2, 3\} \to \{a, b, c\}, x \mapsto h(x) := \begin{cases} h(1) := a \\ h(2) := b \\ h(3) := c \end{cases} . \tag{4.8}$$

For *every* element in the codomain of h there is *exactly one* preimage, so the function is injective and surjective and therefore bijective.

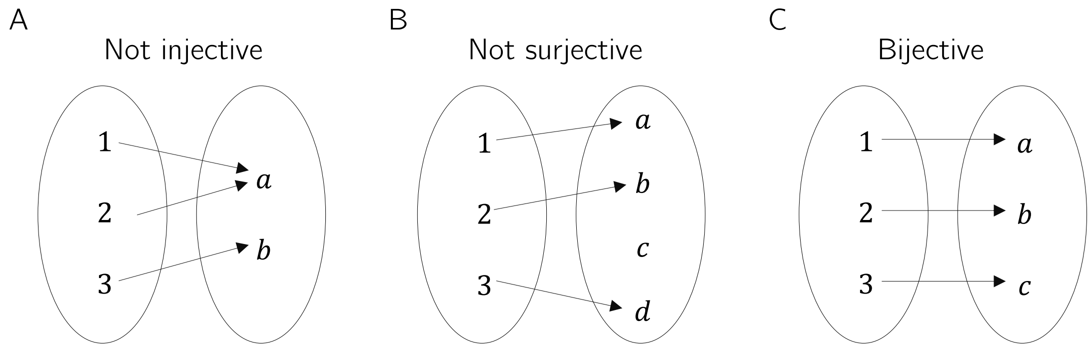

Figure 4.3 Injectivity, surjectivity, bijectivity.

As a further example, consider the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2 \tag{4.9}$$

This function is not injective because, for example, with $x_1 = 2 \neq -2 = x_2$ we have $f(x_1) = 2^2 = 4 = (-2)^2 = f(x_2)$. Moreover, f is not surjective because, for example, $-1 \in \mathbb{R}$ has no preimage under f. However, if the domain of f is restricted to the non-negative real numbers, that is, if we define the function

$$\tilde{f} : [0, \infty[\to [0, \infty[, x \mapsto \tilde{f}(x) := x^2, \tag{4.10}$$

then, in contrast to f, $\tilde{f}$ is injective and surjective, hence bijective.

On the uncountability of the real numbers

At this point, we use the concepts of surjectivity and bijectivity of functions to deepen the idea of the *uncountability of the real numbers* a little. This idea says that there are different kinds of infinities, which is certainly not easy to grasp intuitively at first. Moreover, some aspects of probability theory are formulated in ways that accommodate the uncountability of the real numbers, so even a rough understanding of this property can help motivate some of the subtleties of probability theory. Specifically, we briefly sketch *Cantor's diagonal argument* (Cantor (1891), Gray (1994)) for the uncountability of the real numbers. We begin by recording what, against the background of bijective mappings, is meant by a *finite* and a *countably infinite* set.

Definition 4.4. A set M is called *finite* if there exists a bijective mapping of the form

$$f : \mathbb{N}_n \to M \tag{4.11}$$

onto the elements of M. Furthermore, a set M is called *countably infinite* if there exists a bijective mapping of the form

$$f : \mathbb{N} \to M \tag{4.12}$$

onto the elements of M.

•

In the case of a finite set, each element of the set can thus be assigned a natural number with maximum value $n < \infty$ in the sense of a bijective mapping. In the case of a countably infinite set, each element of the set can at least be assigned a natural number in the sense of a bijective mapping. The elements of these sets can therefore be counted.

Cantor (1891) showed that this is not true in general for the real numbers: no bijective mapping between the real numbers and the natural numbers can be constructed, and hence there must be more real numbers than natural numbers. More specifically, one can argue that even the interval $[0, 1]$ already contains more real numbers than there are natural numbers, that is, no surjective function from $\mathbb{N}$ to $[0, 1]$ can be constructed.

Cantor's argument has the following form. Suppose that $f : \mathbb{N} \to [0, 1]$ is an injective function represented in tabular form, for example as in Table 4.1.

Table 4.1 Mapping $f : \mathbb{N} \to [0, 1]$. The dots $\cdots$ and $\vdots$ indicate that the table continues indefinitely to the right and downward.

n	$f(n)$										
1	0.	**3**	1	4	1	5	9	2	6	5	$\cdots$
2	0.	8	**9**	4	5	9	7	8	1	0	$\cdots$
3	0.	9	6	**2**	3	2	1	5	8	7	$\cdots$
4	0.	5	3	6	**8**	8	8	9	7	3	$\cdots$
5	0.	7	4	3	8	**1**	8	0	9	7	$\cdots$
$\vdots$	$\vdots$										

Now construct another number in $[0, 1]$ by adding 1 to each of the bold diagonal entries in Table 4.1 whenever the corresponding value is less than 9, and otherwise setting the value to 0. This is shown in Table 4.2.

Table 4.2 Construction of another real number in $[0, 1]$.

n	$f(n)$										
1	0.	**4**	1	4	1	5	9	2	6	5	$\cdots$
2	0.	8	**0**	4	5	9	7	8	1	0	$\cdots$
3	0.	9	6	**3**	3	2	1	5	8	7	$\cdots$
4	0.	5	3	6	**9**	8	8	9	7	3	$\cdots$
5	0.	7	4	3	8	**2**	8	0	9	7	$\cdots$
$\vdots$	$\vdots$										

The number thus constructed,

$$0.40392... \tag{4.13}$$

cannot occur in Table 4.1, because it differs from $f(1)$ in its first decimal place, from $f(2)$ in its second decimal place, from $f(3)$ in its third decimal place, ..., from $f(n)$ in its nth decimal place, and so on indefinitely. Thus there is at least one number in $[0,1]$ to which no natural number can be assigned. The injective mapping f is not surjective and therefore not bijective. Consequently, $[0,1]$ is not countably infinite and is accordingly called *uncountable*.

4.2 Types of functions

By composition, further functions can be formed from given functions.

Definition 4.5 (Composition of functions)**.** Let $f : D \to Z$ and $g : Z \to S$ be two functions, where the range of f is assumed to agree with the domain of g. Then

$$g \circ f : D \to S, x \mapsto (g \circ f)(x) := g(f(x)) \tag{4.14}$$

defines a function called the *composition of f and g*.

•

The notation for composed functions takes a little getting used to. It is important to recognize that $g \circ f$ denotes the composed function and $(g \circ f)(x)$ denotes an element in the codomain of the composed function. Intuitively, when evaluating $(g \circ f)(x)$, one first applies the function f to x and then applies the function g to the element $f(x)$ of Z. This is captured by the functional form $g(f(x))$. For simplicity, the composition of two functions is often denoted by a single letter; for example, one writes $h := g \circ f$ with $h(x) = g(f(x))$. A mild source of confusion can arise when elements in the codomain of f are denoted by y, so that the notation $y = f(x)$ and $h(x) = g(y)$ is used. However, this notation is sometimes needed for notational simplicity. We visualize Definition 4.5 in Figure 4.4.

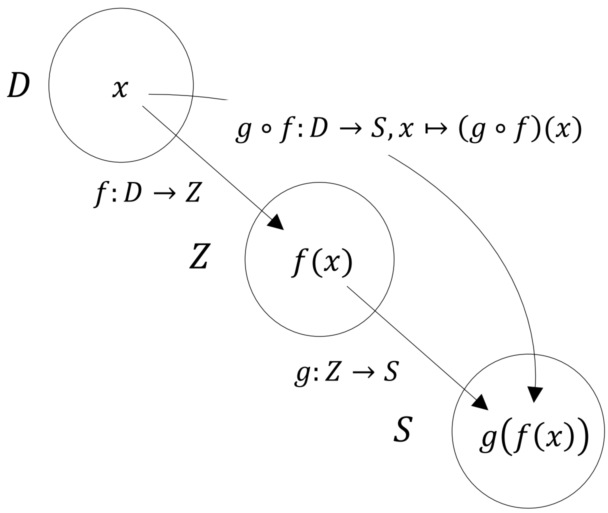

Figure 4.4 Composition of functions.

As an example of the composition of two functions, consider

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := -x^2 \tag{4.15}$$

and

$$g : \mathbb{R} \to \mathbb{R}, x \mapsto g(x) := \exp(x). \tag{4.16}$$

In this case, the composition of f and g is

$$g \circ f : \mathbb{R} \to \mathbb{R}, x \mapsto (g \circ f)(x) := g(f(x)) = \exp\left(-x^2\right). \tag{4.17}$$

A first application of function composition appears in the following definition.

Definition 4.6 (Inverse function)**.** Let $f : D \to Z, x \mapsto f(x)$ be a bijective function. Then the function f^{-1} with

$$f^{-1} \circ f : D \to D, x \mapsto (f^{-1} \circ f)(x) := f^{-1}(f(x)) = x \tag{4.18}$$

is called the *inverse function*, *inverse mapping*, or simply the *inverse of f*.

•

Inverse functions are always bijective. This follows because f is bijective and therefore each $x \in D$ is assigned exactly one $f(x) = z \in Z$. Hence, each $z \in Z$ is also assigned exactly one $x \in D$, namely $f^{-1}(f(x)) = x$. Intuitively, the inverse function of f undoes the effect of f on an element x. We visualize Definition 4.6 in Figure 4.5 A.

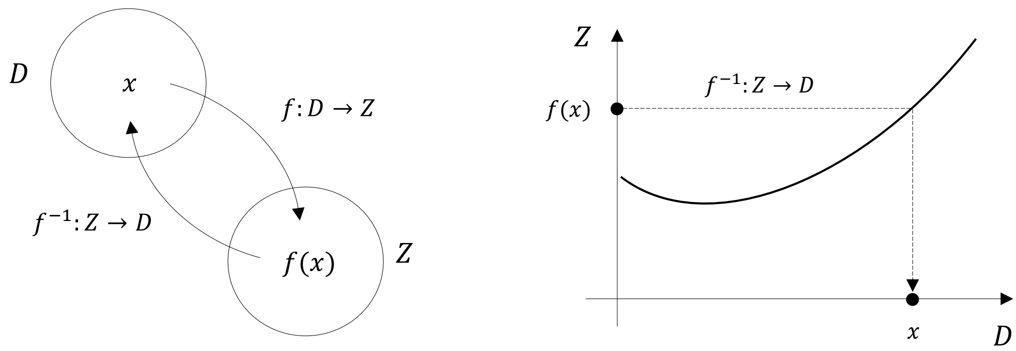

Figure 4.5 Inverse function.

If one considers the graph of a function in a Cartesian coordinate system, then applying a function to a value on the x-axis leads to a value on the y-axis. Applying the inverse function correspondingly leads from a value on the y-axis to a value on the x-axis. We visualize this in Figure 4.5 B. For example, consider the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := 2x =: y. \tag{4.19}$$

Then the inverse function of f is given by

$$f^{-1} : \mathbb{R} \to \mathbb{R}, y \mapsto f^{-1}(y) := \frac{1}{2}y, \tag{4.20}$$

because for every $x \in \mathbb{R}$,

$$(f^{-1} \circ f)(x) := f^{-1}(f(x)) = f^{-1}(2x) = \frac{1}{2} \cdot 2x = x. \tag{4.21}$$

An important class of functions consists of *linear maps*.

Definition 4.7 (Linear map)**.** A map $f : D \to Z, x \mapsto f(x)$ is called a *linear map* if, for $x, y \in D$ and a scalar c,

$$f(x+y) = f(x) + f(y) \qquad \text{(additivity)}$$

and

$$f(cx) = cf(x) \qquad \text{(homogeneity)}$$

hold. A map for which the above properties do not hold is called a *nonlinear map*.

•

Linear maps are often known as "straight lines." The general definition of linear maps is not completely congruent with this intuition. In particular, linear maps are only those functions that map zero to zero. We show this in the following theorem.

Theorem 4.1 (Linear map of zero)**.** *Let $f : D \to Z$ be a linear map. Then*

$$f(0) = 0. \tag{4.22}$$

∘

Proof. First note that by additivity of f,

$$f(0) = f(0+0) = f(0) + f(0). \tag{4.23}$$

Adding $-f(0)$ to both sides of the above equation gives

$$f(0) - f(0) = f(0) + f(0) - f(0) \Leftrightarrow f(0) = 0 \tag{4.24}$$

and this proves the claim. □

We illustrate the concept of a linear map with two examples.

- For $a \in \mathbb{R}$, the map
$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := ax \tag{4.25}$$
is a linear map because
$$f(x+y) = a(x+y) = ax+ay = f(x)+f(y) \text{ and } f(cx) = acx = cax = cf(x). \tag{4.26}$$
- For $a, b \in \mathbb{R}$, by contrast, the map
$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := ax + b \tag{4.27}$$
is nonlinear because, for example, for $a := b := 1$,
$$f(x+y) = 1(x+y) + 1 = x + y + 1 \neq x + 1 + y + 1 = f(x) + f(y). \tag{4.28}$$

A map of the form $f(x) := ax + b$ is called a *linear-affine map* or *linear-affine function*. Somewhat imprecisely, functions of the form $f(x) := ax + b$ are sometimes also called *linear functions*.

Besides the types of functions discussed so far, there are many further classes of functions. In the following definition, we classify functions by the dimensionality of their domains and codomains. This type of function classification is often helpful for gaining an initial overview of a mathematical model.

Definition 4.8 (Function types)**.** We distinguish

- *univariate real-valued functions* of the form

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x), \tag{4.29}$$

- *multivariate real-valued functions* of the form

$$f : \mathbb{R}^n \to \mathbb{R}, x \mapsto f(x) = f(x_1, ..., x_n), \tag{4.30}$$

- and *multivariate vector-valued functions* of the form

$$f : \mathbb{R}^n \to \mathbb{R}^m, x \mapsto f(x) = \begin{pmatrix} f_1(x_1, ..., x_n) \\ \vdots \\ f_m(x_1, ..., x_n) \end{pmatrix}, \tag{4.31}$$

  where $f_i, i = 1, ..., m$ are called the *components* or *component functions* of f.

•

In physics, multivariate real-valued functions are called *scalar fields*, and multivariate vector-valued functions are called *vector fields*. In some applications, *matrix-variate matrix-valued functions* also occur.

4.3 Elementary functions

By *elementary functions* we mean a small set of univariate real-valued functions that frequently appear as building blocks of more complex functions. These are the *polynomial functions*, the *exponential function*, the *logarithm function*, and the *gamma function*. In the following, we give the definitions of these functions and state their main properties as theorems, and then present their graphs. For proofs of the properties introduced here, we refer to the advanced literature.

Definition 4.9 (Polynomial functions)**.** A function of the form

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \sum_{i=0}^{k} a_i x^i = a_0 + a_1 x^1 + a_2 x^2 + \cdots + a_k x^k \tag{4.32}$$

is called a *polynomial function* of degree k with coefficients $a_0, a_1, ..., a_k \in \mathbb{R}$. Typical polynomial functions are listed in Table 4.3.

Table 4.3 Polynomial functions.

Name	Form	Coefficients
Constant function	$f(x) = a$	$a_0 := a,\ a_i := 0, i > 0$
Identity function	$f(x) = x$	$a_0 := 0,\ a_1 := 1,\ a_i := 0, i > 1$
Linear-affine function	$f(x) = ax + b$	$a_0 := b,\ a_1 := a,\ a_i := 0, i > 1$
Square function	$f(x) = x^2$	$a_0 := 0,\ a_1 := 0,\ a_2 := 1,\ a_i := 0, i > 2$

•

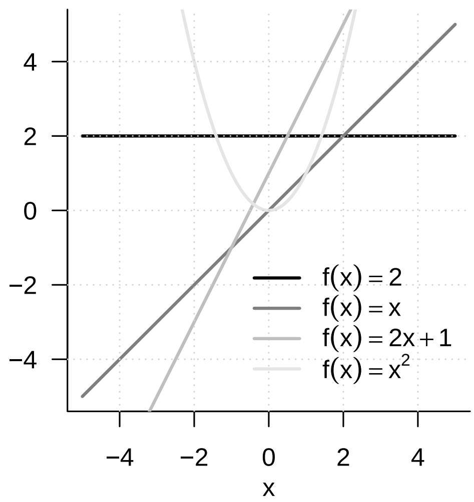

Figure 4.6 Selected polynomial functions

Figure 4.6 shows the graphs of the polynomial functions listed in Definition 4.9.

An important pair of functions consists of the exponential function and the logarithm function.

Theorem 4.2 (Exponential function and its properties)**.** *The* exponential function *is defined as*

$$\exp : \mathbb{R} \to \mathbb{R}, x \mapsto \exp(x) := e^x := \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots \tag{4.33}$$

Among others, the exponential function has the properties listed in Table 4.4.

Table 4.4 Properties of the exponential function.

Property	*Meaning*
Range	$x \in]-\infty, 0[\Rightarrow \exp(x) \in]0, 1[$
	$x \in]0, \infty[\quad \Rightarrow \exp(x) \in]1, \infty[$
Monotonicity	$x < y \Rightarrow \exp(x) < \exp(y)$
Special values	$\exp(0) = 1$ *and* $\exp(1) = e$
Summation property	$\exp(x + y) = \exp(x)\exp(y)$
Subtraction property	$\exp(x - y) = \frac{\exp(x)}{\exp(y)}$

∘

In particular, the exponential function only takes positive values and intersects the y-axis at $x = 0$. The number $\exp(1) := e \approx 2.71...$ is called *Euler's number.* Finally, the special values of the exponential function imply

$$\exp(x)\exp(-x) = \exp(x - x) = \exp(0) = 1. \tag{4.34}$$

Theorem 4.3 (Logarithm function and its properties)**.** *The* logarithm function *is defined as the inverse function of the exponential function,*

$$\ln :]0, \infty[\to \mathbb{R}, x \mapsto \ln(x) \textit{ with } \ln(\exp(x)) = x \textit{ for all } x \in \mathbb{R}. \tag{4.35}$$

Among others, the logarithm function has the properties listed in Table 4.5.

Table 4.5 Properties of the logarithm function.

Property	*Meaning*
Range	$x \in]0, 1[\;\Rightarrow \ln(x) \in]-\infty, 0[$
	$x \in]1, \infty[\Rightarrow \ln(x) \in]0, \infty[$
Monotonicity	$x < y \Rightarrow \ln(x) < \ln(y)$
Special values	$\ln(1) = 0$ *and* $\ln(e) = 1$
Product property	$\ln(xy) = \ln(x) + \ln(y)$
Power property	$\ln(x^c) = c\ln(x)$
Division property	$\ln\left(\frac{1}{x}\right) = -\ln(x)$

∘

In contrast to the exponential function, the logarithm function takes both negative and positive values. The logarithm function intersects the x-axis at $x = 1$. The product property and the power property are central when calculating with the logarithm function. Intuitively, they can be remembered as: "The logarithm function turns products into sums and powers into products." The graphs of the exponential and logarithm functions are shown in Figure 4.7.

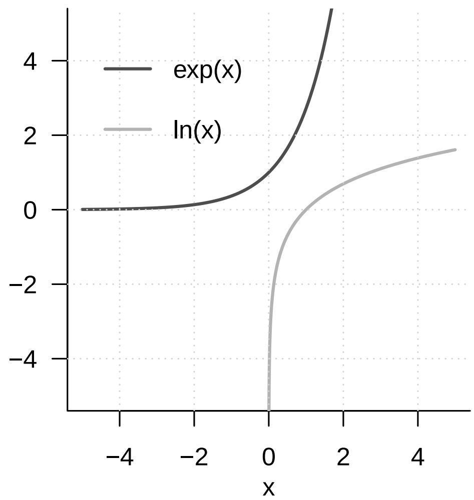

Figure 4.7 Exponential function and logarithm function

A frequent companion in probability theory is the *gamma function*.

Definition 4.10 (Gamma function)**.** The *gamma function* is defined by

$$\Gamma : \mathbb{R} \to \mathbb{R}, x \mapsto \Gamma(x) := \int_0^\infty \xi^{x-1} \exp(-\xi)\, d\xi \tag{4.36}$$

Among others, the gamma function has the properties listed in Table 4.6.

Table 4.6 Properties of the gamma function.

Property	Meaning
Special values	$\Gamma(1) = 1$ $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$
Recursion property	For $x > 0$, $\Gamma(x+1) = x\Gamma(x)$

•

A section of the graph of the gamma function is shown in Figure 4.8.

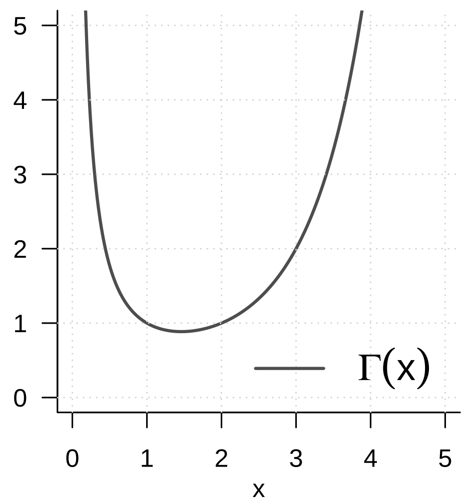

Figure 4.8 Gamma function

5 Sequences, limits, continuity

The topics treated in this chapter are not central to probabilistic data science but form basic building blocks of real analysis. Because modern probability theory is closely intertwined with analytic approaches, however, they help us understand certain results of probability theory. One example of such a result is the *central limit theorem*. The central limit theorem, in turn, forms the basis for the widely used normal-distribution assumption in probabilistic data science. The foundations considered here therefore indirectly increase our understanding of data-scientific principles. In brief, the central limit theorem is a statement about the *limit function* of a *sequence of functions*, more precisely about a sequence of random variables. Knowledge of the nature of *sequences*, *sequences of functions*, and their *limits* therefore allows an informed entry into the study of the central limit theorem. Furthermore, the topics treated in this chapter provide at least a first entry point for understanding continuity and smoothness of functions, which are important basic concepts in nonlinear optimization, for example when determining parameter estimators of probabilistic models.

5.1 Sequences

We begin with the definition of the concept of a *real sequence*.

Definition 5.1 (Real sequence)**.** A *real sequence* is a function of the form

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n). \tag{5.1}$$

The function values $f(n)$ of a real sequence are usually denoted by x_n and called *sequence terms*. Common notations for sequences are

$$(x_1, x_2, ...) \text{ or } (x_n)_{n=1}^{\infty} \text{ or } (x_n)_{n\in\mathbb{N}} \text{ or } (x_n). \tag{5.2}$$

•

Note that a real sequence, because there are infinitely many natural numbers, always has infinitely many sequence terms. This should be kept in mind especially when using the notation $(x_1, x_2, ...)$. We consider two standard examples of real sequences.

Examples of Real Sequences

(1) Real sequences of the form

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n) := \left(\frac{1}{n}\right)^{\frac{p}{q}} \text{ with } p, q \in \mathbb{N} \tag{5.3}$$

are called *harmonic sequences*. For $p := q := 1$, a harmonic sequence has the sequence-term form

$$\left(\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, ...\right). \tag{5.4}$$

(2) Real sequences of the form

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n) := q^n \text{ with } q \in]-1, 1[\tag{5.5}$$

are called *geometric sequences*. For $q := \frac{1}{2}$, a geometric sequence has the sequence-term form

$$\begin{aligned} \left(\left(\frac{1}{2}\right)^1, \left(\frac{1}{2}\right)^2, \left(\frac{1}{2}\right)^3, ...\right) &= \left(\frac{1^1}{2^1}, \frac{1^2}{2^2}, \frac{1^3}{2^3}, ...\right) \\ &= \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, ...\right). \end{aligned} \tag{5.6}$$

In addition to real sequences, that is, sequences of real numbers, one can also consider sequences of other mathematical objects. An important type of sequence is the *sequence of functions*.

Definition 5.2 (Sequence of functions)**.** Let ϕ be a set of univariate real-valued functions with domain $D \subseteq \mathbb{R}$. Then a sequence of functions is a function of the form

$$F : \mathbb{N} \to \phi, n \mapsto F(n). \tag{5.7}$$

The function values $F(n)$ of a sequence of functions are usually denoted by f_n and called *sequence terms*. Common notations for sequences of functions are

$$(f_1, f_2, ...) \text{ or } (f_n)_{n=1}^{\infty} \text{ or } (f_n)_{n \in \mathbb{N}} \text{ or } (f_n). \tag{5.8}$$

•

The definition of a sequence of functions is evidently analogous to the definition of a real sequence. The difference between a real sequence and a sequence of functions is that the sequence terms of a real sequence are real numbers, whereas the sequence terms of a sequence of functions are univariate real-valued functions. Here, too, we discuss two standard examples.

Examples of Sequences of Functions

(1) We consider the set ϕ of univariate real-valued functions of the form

$$\phi := \{f_n | f_n : [0, 1] \to \mathbb{R}, x \mapsto f_n(x) := x^n \text{ for } n \in \mathbb{N}\}. \tag{5.9}$$

Then

$$F : \mathbb{N} \to \phi, n \mapsto F(n) := f_n \tag{5.10}$$

defines a sequence of functions. For the function values of the sequence terms of F, we have

$$f_1(x) := x^1, f_2(x) := x^2, f_3(x) := x^3, ... \tag{5.11}$$

(2) We consider the set ϕ of univariate real-valued functions of the form

$$\phi := \{f_n | f_n : [-a, a] \to \mathbb{R}, x \mapsto f_n(x) := \sum_{k=0}^{n} \frac{x^k}{k!} \text{ for } n \in \mathbb{N}\}. \tag{5.12}$$

Then

$$F : \mathbb{N} \to \phi, n \mapsto F(n) := f_n \tag{5.13}$$

defines a sequence of functions. For the function values of the sequence terms of F, we have

$$f_1(x) := \sum_{k=0}^{1} \frac{x^k}{k!}, f_2(x) := \sum_{k=0}^{2} \frac{x^k}{k!}, f_3(x) := \sum_{k=0}^{3} \frac{x^k}{k!}, ... \tag{5.14}$$

5.2 Limits

When considering the terms of a sequence, one can ask which values a sequence might take when the sequence index n becomes very large, that is, tends to infinity. If, in this case, the sequence terms take very similar values (and do not themselves become infinitely large), one is led to the concept of a *limit* for real sequences and a *limit function* for sequences of functions.

Definition 5.3 (Limit of a real sequence). $x \in \mathbb{R}$ is called the limit of a real sequence $(x_n)_{n=1}^{\infty}$ if, for every $\epsilon > 0$, there exists an $m \in \mathbb{N}$ such that

$$|x_n - x| < \epsilon \text{ for all } n \geq m. \tag{5.15}$$

A sequence that has a limit is called a *convergent sequence*; a sequence that has no limit is called a *divergent sequence*. To express that $x \in \mathbb{R}$ is the limit of the sequence $(x_n)_{n=1}^{\infty}$, one also writes

$$\lim_{n\to\infty} x_n = x \text{ or } x_n \to x \text{ for } n \to \infty \text{ or } x_n \xrightarrow{n\to\infty} x. \tag{5.16}$$

•

According to Definition 5.3, the limit of a sequence may exist, but it need not. For example, the sequence

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n) := n \tag{5.17}$$

has no limit, because here both n and $f(n)$ become infinitely large. For a convergent sequence, that is, a sequence whose limit exists, Definition 5.3 says that, for an arbitrarily small but positive ϵ, an $m \in \mathbb{N}$ can be specified such that, for all sequence terms x_n with $n \geq m$, the distance from the limit in the positive or negative direction is smaller than ϵ. Thus the sequence terms with $n \geq m$ lie arbitrarily close to the limit x. It may often be the case that a smaller ϵ requires a larger m.

Examples

The examples of real sequences considered above, by contrast, have limits.

(1) For the generalized harmonic sequences, with $p, q \in \mathbb{N}$,

$$\lim_{n\to\infty} \left(\frac{1}{n}\right)^{\frac{p}{q}} = 0. \tag{5.18}$$

(2) For the geometric sequences, with $q \in]-1, 1[$,

$$\lim_{n\to\infty} q^n = 0. \tag{5.19}$$

The harmonic and geometric sequences are therefore also called *null sequences*. For general proofs of Equation 5.18 and Equation 5.19, we refer to the advanced literature. In fact, these proofs are not trivial and touch on the basic assumptions about the nature of the real numbers. In Figure 5.1, we visualize the first ten sequence terms as well as the limits of the harmonic sequence for $p := q := 1$ and of the geometric sequence for $q := 1/2$. The sequence terms shown as points evidently lie increasingly closer to the limit shown by a gray line as n increases.

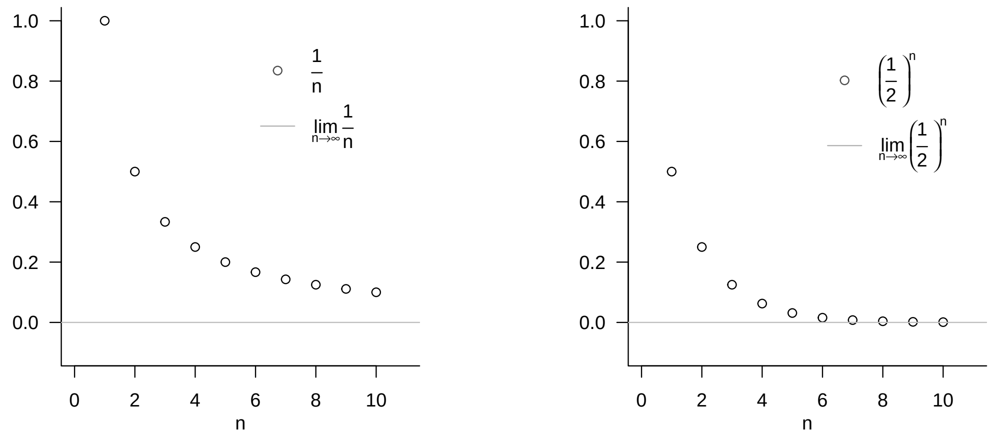

Figure 5.1 Examples of limits of real sequences.

For the special case of the harmonic sequence with $p = q = 1$, we briefly present the argument for the limit

$$\lim_{n\to\infty} \frac{1}{n} = 0. \tag{5.20}$$

According to Definition 5.3, Equation 5.20 holds if and only if, for an arbitrary $\epsilon > 0$ (and in particular for an arbitrarily small one), an $m \in \mathbb{N}$ can be specified such that, for all $n \geq m$,

$$\left|\frac{1}{n} - 0\right| < \epsilon \Leftrightarrow \frac{1}{n} < \epsilon.$$

Thus, using the ceiling function $\lceil \cdot \rceil$, if for an arbitrary $\epsilon > 0$ we set

$$m := \left\lceil \frac{1}{\epsilon} \right\rceil + 1,$$

for example, for $\epsilon := 0.01$ correspondingly $m := \lceil \frac{1}{0.01} \rceil + 1 = 101$, then for all $n \geq m$ we have $\frac{1}{n} < \epsilon$. Since $\epsilon > 0$ was chosen arbitrarily, this construction is possible for all $\epsilon > 0$.

For sequences of functions, one possible extension of the concepts of convergence and limit is the following.

Definition 5.4 (Pointwise convergence and limit function of a sequence of functions)**.** Let $F = (f_n)_{n\in\mathbb{N}}$ be a sequence of functions of univariate real-valued functions with domain D. F is called *pointwise convergent* if the real sequence $(f_n(x))_{n\in\mathbb{N}}$ is a convergent sequence for every $x \in D$, that is, if it has a limit. The function that assigns to each $x \in D$ this limit of $(f_n(x))_{n\in\mathbb{N}}$ is then called the *limit function of the sequence of functions* F and has the form

$$f : D \to \mathbb{R}, x \mapsto f(x) := \lim_{n\to\infty} f_n(x). \tag{5.21}$$

•

Note that the limits of convergent real sequences are real numbers, whereas the limit functions of pointwise convergent sequences of functions are functions. In addition to pointwise convergence of sequences of functions, there is the more powerful concept of *uniform convergence* of sequences of functions, for which we refer to the advanced literature. As examples, we consider the limit functions of the sequences of functions discussed above; for proofs, we again refer to the advanced literature.

Examples

(1) We consider the sequence of functions

$$F : \mathbb{N} \to \phi, n \mapsto F(n) \tag{5.22}$$

with

$$\phi := \{f_n | f_n : [0, 1] \to \mathbb{R}, x \mapsto f_n(x) := x^n \text{ for } n \in \mathbb{N}\}. \tag{5.23}$$

Then F is pointwise convergent with limit function

$$f : [0, 1] \to \mathbb{R}, x \mapsto f(x) := \begin{cases} 0, & \text{for } x \in [0, 1[\\ 1, & \text{for } x = 1 \end{cases} \tag{5.24}$$

because $f_n(x) := x^n$ is a geometric sequence, and thus a null sequence, for $x \in [0, 1[$, while $f_n(x) := x^n$ is a constant sequence for $x = 1$, all of whose sequence terms have distance 0 from 1. The sequence of functions F therefore converges to a function that is equal to zero on the entire interval $[0, 1]$, except at the point 1. This function evidently has a jump.

(2) We consider the sequence of functions

$$F : \mathbb{N} \to \phi, n \mapsto F(n) \tag{5.25}$$

with

$$\phi := \{f_n | f_n : [-a, a] \to \mathbb{R}, x \mapsto f_n(x) := \sum_{k=0}^{n} \frac{x^k}{k!} \text{ for } n \in \mathbb{N}\}. \tag{5.26}$$

Then F is pointwise convergent with limit function

$$f : [-a, a] \to \mathbb{R}, x \mapsto f(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!} =: \exp(x). \tag{5.27}$$

The sequence of functions F therefore converges to the exponential function on $[-a, a]$. Conversely, the exponential function is defined precisely by

$$\exp(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!}. \tag{5.28}$$

5.3 Continuity

In this section, we approach the concept of the *continuity* of a function. Intuitively, a function is continuous if it has no jumps or, equivalently, if small changes in its arguments always lead only to small changes in its function values (and therefore precisely to no jumps). To define continuity, we first need the concept of the *limit of a function*.

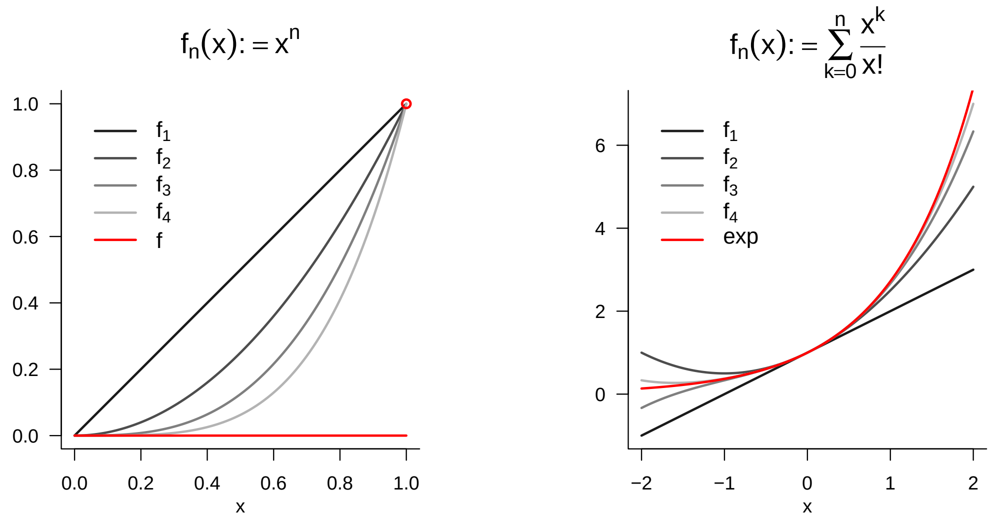

Figure 5.2 Examples of limits of sequences of functions.

Definition 5.5 (Limit of a function)**.**

For $D \subseteq \mathbb{R}$ and $Z \subseteq \mathbb{R}$, let $f : D \to Z, x \mapsto f(x)$ be a function, and let $a, b \in \mathbb{R}$. b is called the *limit of the function f as x tends to a* if

(1) there exists a real sequence $(x_n)_{n=1}^{\infty}$ with sequence terms in D and limit a, that is, $\lim_{n\to\infty} x_n = a$, and

(2) for every such sequence, b is the limit of the sequence of function values $f(x_n)$ of the sequence terms of $(x_n)_{n=1}^{\infty}$, that is, $\lim_{n\to\infty} f(x_n) = b$.

If b is the limit of the function f as x tends to a, one also writes $\lim_{x\to a} f(x) = b$.

•

In Figure 5.3, we visualize the limit of the exponential function at $a = 1$ by representing sequence terms $x_n \to 1$ as points parallel to the x-axis and the corresponding sequence terms $f(x_n)$ on the function graph. Evidently, $\lim_{x\to 1} \exp(x) = e \approx 2.71$.

We can now define the concept of continuity of a function.

Definition 5.6 (Continuity of a function)**.** A function $f : D \to Z$ with $D \subseteq \mathbb{R}$ and $Z \subseteq \mathbb{R}$ is called *continuous at $a \in D$* if

$$\lim_{x\to a} f(x) = f(a). \tag{5.29}$$

If f is continuous at every $x \in D$, then f *is continuous on D.*

•

Note that, for a function continuous at a, it follows that

$$\lim_{x\to a} f(x) = f\left(\lim_{x\to a} x\right). \tag{5.30}$$

For continuous functions, taking limits and evaluating the function can therefore be interchanged.

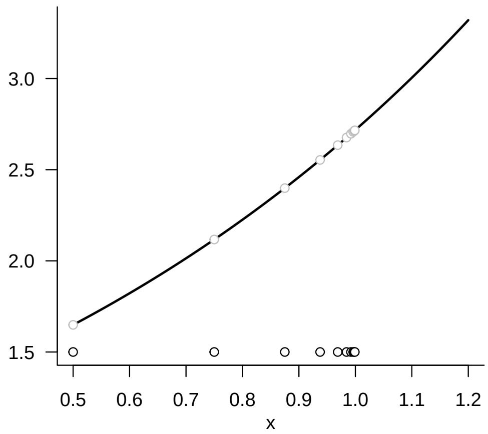

Figure 5.3 Example of the limit of a function. The sequence of x-values is defined here as $x_n := 1-(1/2)^n$ for $n = 1, 2, \ldots$.

6 Differential calculus

Differential calculus is concerned with the change of functions. On the one hand, it provides the basis for the mathematical modelling of dynamic systems by means of *differential equations*, that is, the description of functions in terms of their rates of change. On the other hand, differential calculus provides the basis of *optimization*, that is, of determining extrema of functions. In Section 6.1 we first introduce the concept of the derivative and elementary calculation rules associated with it. In Section 6.2 we then turn to the question of how derivatives can be used to determine extrema of functions.

6.1 Definitions and calculation rules

We begin with the following definition.

Definition 6.1 (Differentiability and derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval and let

$$f : I \to \mathbb{R}, x \mapsto f(x) \tag{6.1}$$

be a univariate real-valued function. f is called *differentiable* at $a \in I$ if the limit

$$f'(a) := \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{6.2}$$

exists. $f'(a)$ is then called the *derivative of f at the point a*. If f is differentiable for all $x \in I$, then f is called *differentiable*, and the function

$$f' : I \to \mathbb{R}, x \mapsto f'(x) \tag{6.3}$$

is called the *derivative of f*.

•

For $h > 0$, the expression

$$\frac{f(a+h) - f(a)}{h} \tag{6.4}$$

is called the *Newton difference quotient*. As shown in Figure 6.1, the Newton difference quotient measures the change $f(a+h) - f(a)$ of f on the y-axis per distance h on the x-axis. If, for example, $f(a)$ and $f(a+h)$ represent the position of an object at a time a and at a later time $a+h$, then $f(a+h) - f(a)$ is the distance travelled by this object in the time h, that is, its average velocity over the time interval h. For $h \to 0$, the Newton difference quotient then measures the instantaneous rate of change of f at a, in this example the velocity of the object at time a.

From a mathematical point of view, it is important to distinguish between the symbols $f'(a)$ and f' in the definition of the derivative. As usual, $f'(a)$ denotes the value of a

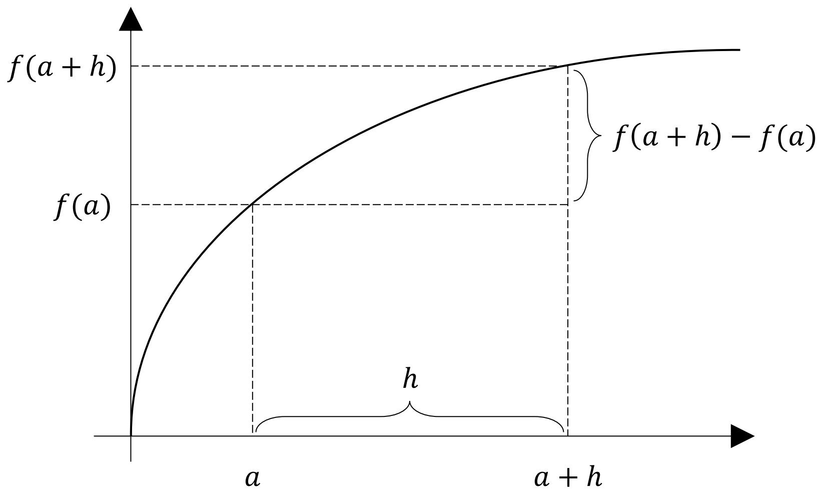

Figure 6.1 Elements of the Newton difference quotient.

function, that is, a number. In contrast, f' denotes a function, namely the function whose values are given by $f'(a)$ for all $a \in \mathbb{R}$.

Several historically established notations for derivatives exist in the literature; all of them represent the same concept of the derivative.

Definition 6.2 (Notation for derivatives of univariate real-valued functions)**.** Let f be a univariate real-valued function. Equivalent notations for the derivative of f and the derivative of f at a point x are

- the *Lagrange notation* f' and $f'(x)$,
- the *Leibniz notation* $\frac{df}{dx}$ and $\frac{df(x)}{dx}$,
- the *Newton notation* $\dot{f}$ and $\dot{f}(x)$,
- the *Euler notation* Df and $Df(x)$.

•

In the following, for univariate real-valued functions we will mainly use the Lagrange notation f' and $f'(x)$ as labels. In calculations, we also use an adapted form of Leibniz notation and understand the expression $\frac{d}{dx}f(x)$ as the instruction to compute the derivative of f. Newton notation is used mainly when the function argument represents time and is then usually denoted by t for *time*. Accordingly, $\dot{f}(t)$ denotes the rate of change of f at time t. Euler notation is particularly useful in the context of multivariate real-valued or vector-valued functions.

We assume a number of derivatives of elementary functions to be known; these are summarized in Table 6.1. For proofs, we refer to the advanced literature.

Table 6.1 Derivatives of elementary functions

Name	Definition	Derivative
Polynomial function	$f(x) := \sum_{i=0}^{k} a_i x^i$	$f'(x) = \sum_{i=1}^{k} i a_i x^{i-1}$
Constant function	$f(x) := a$	$f'(x) = 0$
Identity function	$f(x) := x$	$f'(x) = 1$
Linear-affine function	$f(x) := ax + b$	$f'(x) = a$
Square function	$f(x) := x^2$	$f'(x) = 2x$
Exponential function	$f(x) := \exp(x)$	$f'(x) = \exp(x)$
Logarithm function	$f(x) := \ln(x)$	$f'(x) = \frac{1}{x}$

In Figure 6.2 we visualize the identity function, a linear function, and the square function together with their respective derivatives. In Figure 6.3 we visualize the exponential and logarithm functions together with their respective derivatives.

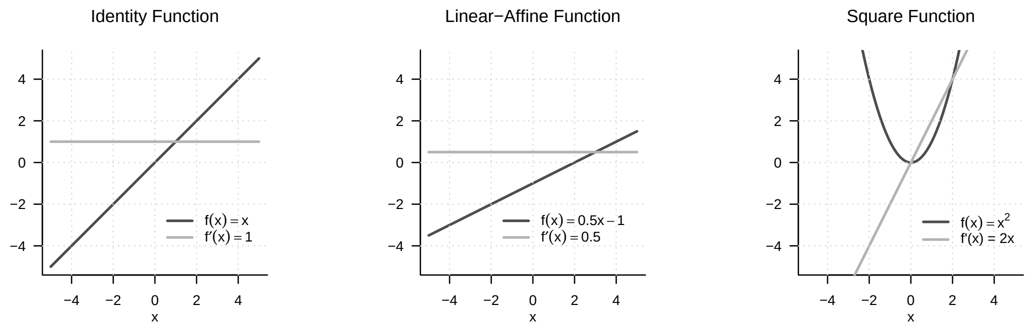

Figure 6.2 Derivatives of three elementary functions. The derivative of the identity function is the constant function with value 1, the derivative of the linear-affine function considered here is the constant function with value 0.5, and the derivative of the square function is the linear-affine function with $a = 2$ and $b = 0$.

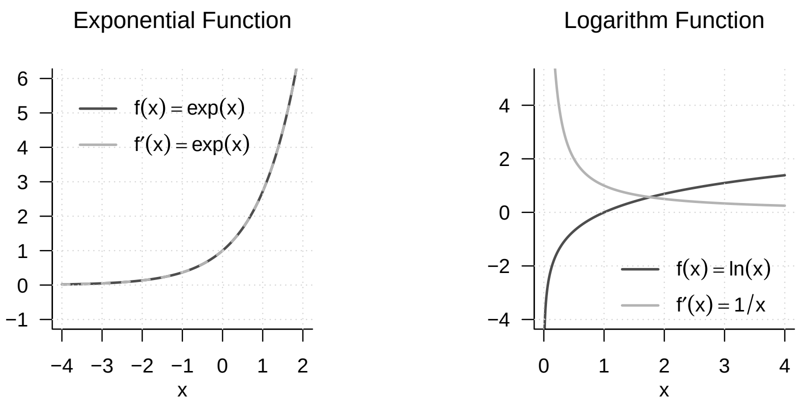

Figure 6.3 Derivatives of the exponential function and the logarithm function. The derivative of the exponential function is the exponential function itself; the derivative of the logarithm function has function values $1/x$.

Based on the definition of the derivative of a univariate real-valued function, further derivatives of such a function can be defined easily.

Definition 6.3 (Higher derivatives)**.** Let f be a univariate real-valued function and let

$$f^{(1)} := f' \tag{6.5}$$

be the derivative of f. The kth derivative of f is defined recursively by

$$f^{(k)} := \left(f^{(k-1)}\right)' \text{ for } k > 1, \tag{6.6}$$

assuming that $f^{(k-1)}$ is differentiable. In particular, the *second derivative of* f is defined as the derivative of f', that is,

$$f'' := (f')'. \tag{6.7}$$

•

In analogy to the above, in calculations we also write $\frac{d^2}{dx^2}f(x)$ for the instruction to determine the second derivative of a function f. The zeroth derivative $f^{(0)}$ of f is f itself. By tradition and for simplicity, for $k < 4$ one usually writes f', f'', and f''' according to Lagrange notation instead of $f^{(1)}, f^{(2)}$, and $f^{(3)}$.

Example

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2. \tag{6.8}$$

Then

$$f^{(1)}(x) = f'(x) = \frac{d}{dx}\left(x^2\right) = 2x. \tag{6.9}$$

Furthermore,

$$\begin{aligned}
f^{(2)}(x) &= \left(f^{(2-1)}\right)'(x) = \left(f^{(1)}\right)'(x) = \frac{d}{dx}(2x) = 2,\\
f^{(3)}(x) &= \left(f^{(3-1)}\right)'(x) = \left(f^{(2)}\right)'(x) = \frac{d}{dx}(2) = 0,\\
f^{(4)}(x) &= \left(f^{(4-1)}\right)'(x) = \left(f^{(3)}\right)'(x) = \frac{d}{dx}(0) = 0.
\end{aligned} \tag{6.10}$$

Thus, throughout this calculation, one computes only first derivatives.

A number of calculation rules are helpful for determining the derivative of a function because they allow the derivative of a function to be derived from the derivatives of its subfunctions. For proofs of the calculation rules introduced in the following theorem, we refer to the advanced literature.

Theorem 6.1 (Calculation rules for derivatives)**.** *For $i = 1, ..., n$, let g_i be real-valued univariate differentiable functions. Then the following calculation rules hold:*

(1) Sum rule

$$\text{For } f(x) := \sum_{i=1}^{n} g_i(x),\ f'(x) = \sum_{i=1}^{n} g_i'(x). \tag{6.11}$$

(2) Product rule

$$\text{For } f(x) := g_1(x)g_2(x),\ f'(x) = g_1'(x)g_2(x) + g_1(x)g_2'(x). \tag{6.12}$$

(3) Quotient rule

$$\text{For } f(x) := \frac{g_1(x)}{g_2(x)},\ f'(x) = \frac{g_1'(x)g_2(x) - g_1(x)g_2'(x)}{g_2^2(x)}. \tag{6.13}$$

(4) Chain rule

$$\text{For } f(x) := g_1(g_2(x)),\ f'(x) = g_1'(g_2(x))g_2'(x). \tag{6.14}$$

∘

Examples

(1) Sum rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := 4x^3 + 3x^2. \tag{6.15}$$

Then f has the form

$$f(x) = \sum_{i=1}^{2} g_i(x) = g_1(x) + g_2(x) \text{ with } g_1(x) := 4x^3 \text{ and } g_2(x) := 3x^2. \tag{6.16}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(4x^3\right) = 12x^2 \text{ and } g_2'(x) = \frac{d}{dx}\left(3x^2\right) = 6x. \tag{6.17}$$

Thus, by the sum rule,

$$f'(x) = \sum_{i=1}^{2} g_i'(x) = g_1'(x) + g_2'(x) = 12x^2 + 6x. \tag{6.18}$$

(2) Product rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2 \sin(x). \tag{6.19}$$

Then f has the form

$$f(x) = g_1(x)g_2(x) \text{ with } g_1(x) := x^2 \text{ and } g_2(x) := \sin(x). \tag{6.20}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(x^2\right) = 2x \text{ and } g_2'(x) = \frac{d}{dx}\left(\sin x\right) = \cos(x). \tag{6.21}$$

Thus, by the product rule,

$$f'(x) = g_1'(x)g_2(x) + g_1(x)g_2'(x) = 2x\sin(x) + x^2\cos(x). \tag{6.22}$$

(3) Quotient rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \frac{x^2}{x+1}. \tag{6.23}$$

Then f has the form

$$f(x) = \frac{g_1(x)}{g_2(x)} \text{ with } g_1(x) := x^2 \text{ and } g_2(x) := x + 1. \tag{6.24}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(x^2\right) = 2x \text{ and } g_2'(x) = \frac{d}{dx}\left(1 + x\right) = 1. \tag{6.25}$$

Thus, by the quotient rule,

$$f'(x) = \frac{g_1'(x)g_2(x) - g_1(x)g_2'(x)}{g_2^2(x)} = \frac{2x(x+1) - x^2}{(x+1)^2} = \frac{2x^2 + 2x - x^2}{(x+1)^2} = \frac{x^2 + 2x}{(x+1)^2}. \tag{6.26}$$

(4) Chain rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \exp\left(-x^2\right). \tag{6.27}$$

Then f has the form

$$f(x) = g_1(g_2(x)) \text{ with } g_1(x) := \exp(x) \text{ and } g_2(x) := -x^2. \tag{6.28}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(\exp(x)\right) = \exp(x) \text{ and } g_2'(x) = \frac{d}{dx}\left(-x^2\right) = -2x. \tag{6.29}$$

Thus, by the chain rule,

$$f'(x) = g_1'(g_2(x))g_2'(x) = \exp\left(-x^2\right)(-2x) = -2x\exp\left(-x^2\right). \tag{6.30}$$

One can therefore remember the derivative resulting from the chain rule as: "The derivative of the outer function at the value of the inner function times the derivative of the inner function."

6.2 Analytic optimization

An important application of differential calculus is the determination of extrema of functions. At its core, this concerns the question for which values in its domain a function assumes a maximum or a minimum. For simple functions this is possible analytically. The general procedure is often also known under the keyword "curve sketching". In applications, an analytic approach to optimizing functions is usually not possible, and numerical algorithms are used to determine extrema. Understanding these algorithms, however, presupposes an understanding of the principles of analytic optimization. In this section, we give an introduction to analytic optimization of univariate real-valued functions. We begin by making precise the concepts of maxima and minima of univariate real-valued functions mentioned above.

Definition 6.4 (Extreme points and extreme values)**.** Let $U \subseteq \mathbb{R}$ and let $f : U \to \mathbb{R}$ be a univariate real-valued function. f has at the point $x_0 \in U$

- a *local minimum* if there is an interval $I :=]a, b[$ with $x_0 \in]a, b[$ and
$$f(x_0) \leq f(x) \text{ for all } x \in I \cap U, \tag{6.31}$$
- a *global minimum* if
$$f(x_0) \leq f(x) \text{ for all } x \in U, \tag{6.32}$$
- a *local maximum* if there is an interval $I :=]a, b[$ with $x_0 \in]a, b[$ and
$$f(x_0) \geq f(x) \text{ for all } x \in I \cap U, \tag{6.33}$$
- a *global maximum* if
$$f(x_0) \geq f(x) \text{ for all } x \in U. \tag{6.34}$$

The value $x_0 \in U$ of the domain of f is called, respectively, a *local* or *global minimizer* or *maximizer*, and the function value $f(x_0) \in \mathbb{R}$ is called, respectively, a *local* or *global minimum* or *maximum*. In general, the value $x_0 \in U$ is called an *extreme point*, and the function value $f(x_0) \in \mathbb{R}$ is called an *extreme value*.

•

Extreme points of functions are often denoted by
$$\underset{x \in I \cap U}{\operatorname{argmin}} f(x) \text{ or } \underset{x \in I \cap U}{\operatorname{argmax}} f(x), \tag{6.35}$$
and extreme values of functions are often denoted by
$$\min_{x \in I \cap U} f(x) \text{ or } \max_{x \in I \cap U} f(x). \tag{6.36}$$

Figure 6.4 visualizes the different types of extreme values according to Definition 6.4.

Analytic optimization of univariate real-valued functions is based on the so-called *necessary* and *sufficient conditions for extrema*. The former makes a statement about the behavior of the first derivative of a function at an extreme point; the latter makes a statement about the behavior of a function at a point that satisfies certain requirements on its first and second derivatives.

Theorem 6.2 (Necessary condition for extrema)**.** *Let f be a univariate real-valued function that is differentiable in a neighborhood of x_0. If x_0 is an interior extreme point of f, then*
$$x_0 \textit{ is an interior extreme point of } f \Rightarrow f'(x_0) = 0. \tag{6.37}$$

∘

If x_0 is an interior extreme point of f, then the first derivative of f at x_0 is therefore equal to zero. Instead of giving a proof, let us reason that, for example, at a local maximizer x_0 of f, f increases to the left of x_0 and decreases to the right of x_0. At x_0, however, f neither increases nor decreases, so it is plausible that $f'(x_0) = 0$.

Theorem 6.3 (Sufficient conditions for local extrema)**.** *Let f be a twice differentiable univariate real-valued function.*

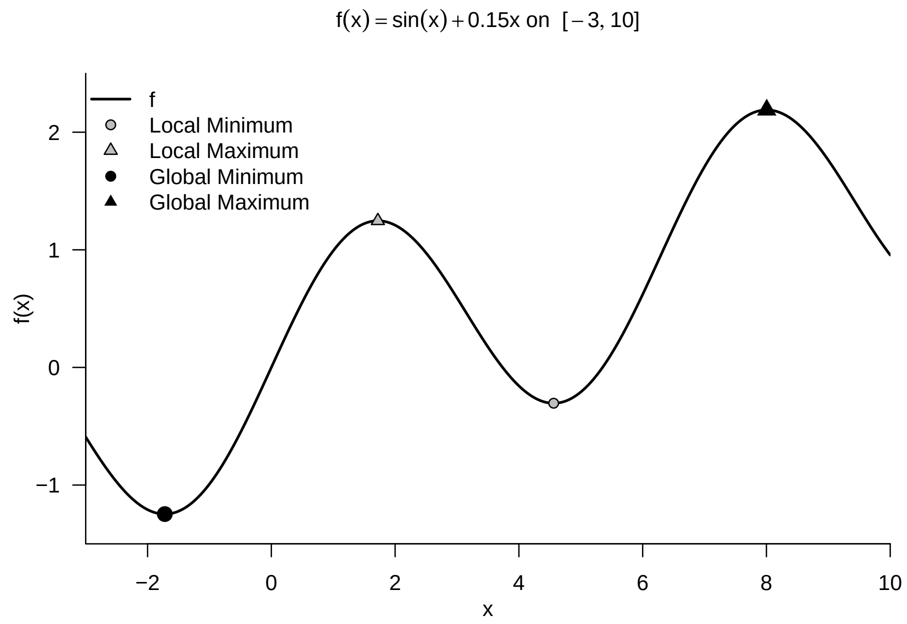

Figure 6.4 Extreme points of a function.

- *If, for $x_0 \in U \subseteq \mathbb{R}$,*
$$f'(x_0) = 0 \text{ and } f''(x_0) > 0 \tag{6.38}$$
holds, then f has a local minimum at the point x_0.
- *If, for $x_0 \in U \subseteq \mathbb{R}$,*
$$f'(x_0) = 0 \text{ and } f''(x_0) < 0 \tag{6.39}$$
holds, then f has a local maximum at the point x_0.

∘

Again we omit a proof and illustrate the condition with the example shown in Figure 6.5. Here, clearly, $x_0 = 1$ is a local minimizer of $f(x) = (x-1)^2$. We can see that to the left of x_0, f decreases, and to the right of x_0, f increases. At x_0, f neither increases nor decreases, so $f'(x_0) = 0$. Moreover, to the left and right of x_0 and at x_0, the change f'' of f' is positive: to the left of x_0, the negativity of f' weakens to 0, and to the right of x_0, the positivity of f' increases.

In particular, the sufficient conditions for local extrema suggest the following *standard procedure* for determining local extreme points.

Theorem 6.4 (Standard procedure of analytic optimization)**.** *Let f be a univariate real-valued function. Local extreme points of f can be identified by the following* standard procedure of analytic optimization*:*

1. *Compute the first and second derivatives of f.*
2. *Determine zeros x^* of f' by solving $f'(x^*) = 0$ for x^*. The zeros of f' are then candidates for extreme points of f.*

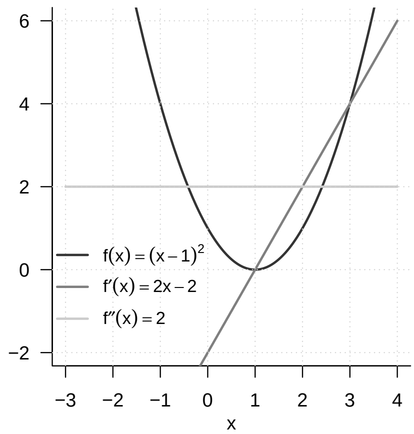

Figure 6.5 Analytic optimization of $f(x) := (x-1)^2$.

3. *Evaluate $f''(x^*)$: If $f''(x^*) > 0$, then x^* is a local minimizer of f; if $f''(x^*) < 0$, then x^* is a local maximizer of f; if $f''(x^*) = 0$, then this criterion does not allow a decision.*

∘

Example

Instead of giving a proof, we consider the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := (x-1)^2 \tag{6.40}$$

as an example. The first derivative of f from Figure 6.5 is obtained by the chain rule as

$$f'(x) = \frac{d}{dx}\left((x-1)^2\right) = 2(x-1)\cdot\frac{d}{dx}(x-1) = 2x-2. \tag{6.41}$$

The second derivative of f is

$$f''(x) = \frac{d}{dx}f'(x) = \frac{d}{dx}(2x-2) = 2 > 0 \text{ for all } x \in \mathbb{R}. \tag{6.42}$$

Solving $f'(x^*) = 0$ for x^* yields

$$f'(x^*) = 0 \Leftrightarrow 2x^* - 2 = 0 \Leftrightarrow 2x^* = 2 \Leftrightarrow x^* = 1. \tag{6.43}$$

Consequently, $x^* = 1$ is a minimizer of f with associated minimum value $f(1) = 0$.

6.3 Differential calculus of multivariate real-valued functions

We first recall the concept of a multivariate real-valued function.

Definition 6.5 (Multivariate real-valued function)**.** A function of the form

$$f : \mathbb{R}^n \to \mathbb{R}, x \mapsto f(x) = f(x_1, ..., x_n) \tag{6.44}$$

is called a *multivariate real-valued* function.

•

The arguments of multivariate real-valued functions are thus real n-tuples of the form $x := (x_1, ..., x_n)$, whereas their function values are real numbers. An example of a multivariate real-valued function with $n := 2$ is

$$f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_2^2. \tag{6.45}$$

We visualize this function in Figure 6.6. The right-hand panel shows a representation by means of so-called *isocontours*, that is, lines in the domain of the function for which the function assumes identical values. The corresponding values are marked for selected isocontours in the figure.

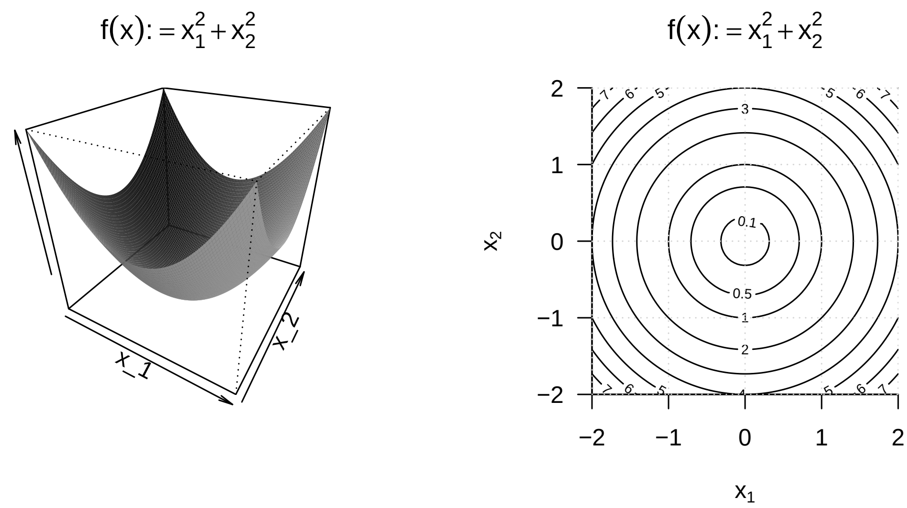

Figure 6.6 Visualizations of a bivariate function. The left-hand panel visualizes the function values as a surface above the two-dimensional domain of the function. Although this representation has a certain three-dimensional character, it is a bivariate function with a correspondingly two-dimensional domain. The right-hand panel visualizes lines with equal function values in the two-dimensional domain of the function.

We now begin to extend the concepts of differentiability and the derivative of univariate real-valued functions to the case of multivariate real-valued functions. To this end, we first introduce the concepts of *partial differentiability* and the *partial derivative*.

Definition 6.6 (Partial differentiability and partial derivative)**.** Let $D \subseteq \mathbb{R}^n$ be a set and let

$$f : D \to \mathbb{R}, x \mapsto f(x) \tag{6.46}$$

be a multivariate real-valued function. f is called *partially differentiable with respect to* x_i at $a \in D$ if the limit

$$\frac{\partial}{\partial x_i} f(a) := \lim_{h \to 0} \frac{f(a + he_i) - f(a)}{h} \tag{6.47}$$

exists. $\frac{\partial}{\partial x_i} f(a)$ is then called the *partial derivative of f with respect to x_i at the point a*. If f is partially differentiable with respect to x_i for all $x \in D$, then f is called *partially differentiable with respect to x_i*, and the function

$$\frac{\partial}{\partial x_i} f : D \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_i} f(x) \tag{6.48}$$

is called the *partial derivative of f with respect to x_i*. f is called *partially differentiable at $x \in D$* if f is partially differentiable with respect to x_i at $x \in D$ for all $i = 1, ..., n$, and f is called *partially differentiable* if f is partially differentiable with respect to x_i at all $x \in D$ for all $i = 1, ..., n$.

•

In Definition 6.6, $e_i \in \mathbb{R}^n$ denotes the ith canonical unit vector, for which $(e_i)_j = 1$ for $i = j$ and $(e_i)_j = 0$ for $i \neq j$ with $j = 1, ..., n$. In analogy and generalization to the Newton difference quotient, the difference quotient occurring here,

$$\frac{f(x + he_i) - f(x)}{h}, \tag{6.49}$$

therefore measures the change $f(x + he_i) - f(x)$ of f per distance h in the direction e_i. We visualize the components of this quotient for the case of a bivariate function in Figure 6.7.

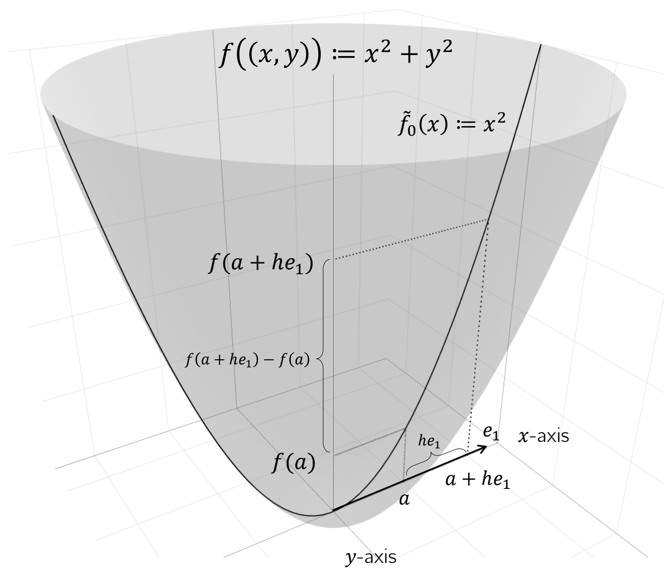

Figure 6.7 Components of the partial Newton difference quotient.

For $h \to 0$, the difference quotient correspondingly measures the *rate of change* of f at x in the direction e_i. As in the discussion of derivatives, $\frac{\partial}{\partial x_i} f(x)$ is a number, whereas $\frac{\partial}{\partial x_i} f$ is a function. In practice, one computes $\frac{\partial}{\partial x_i} f$ as the (ordinary) derivative

$$\frac{d}{dx_i} \tilde{f}_{x_1,...x_{i-1},x_{i+1},...,x_n}(x_i) \tag{6.50}$$

of the univariate real-valued function

$$\tilde{f} : \mathbb{R} \to \mathbb{R}, x_i \mapsto \tilde{f}_{x_1,...x_{i-1},x_{i+1},...,x_n}(x_i) := f(x_1, ..., x_i, ..., x_n). \tag{6.51}$$

Thus, for the ith partial derivative, all x_j with $j \neq i$ are regarded as constants, and one is led back to the familiar computation of derivatives of univariate real-valued functions. We illustrate the procedure for computing partial derivatives with a first example.

Example (1)

We consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_2^2. \tag{6.52}$$

Because the domain of this function is two-dimensional, two partial derivatives can be computed:

$$\frac{\partial}{\partial x_1} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_1} f(x) \text{ and } \frac{\partial}{\partial x_2} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_2} f(x). \tag{6.53}$$

To compute the first of these partial derivatives, one considers the function

$$f_{x_2} : \mathbb{R} \to \mathbb{R}, x_1 \mapsto f_{x_2}(x_1) := x_1^2 + x_2^2, \tag{6.54}$$

where x_2 takes the role of a constant. To make explicit that x_2 is not an argument of the function, while the function still depends on x_2, we have used the subscript notation $f_{x_2}(x_1)$. To compute the partial derivative, we now compute the (ordinary) derivative of f_{x_2},

$$f'_{x_2}(x_1) = 2x_1. \tag{6.55}$$

Thus,

$$\frac{\partial}{\partial x_1} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_1} f(x) = \frac{\partial}{\partial x_1}(x_1^2 + x_2^2) = f'_{x_2}(x_1) = 2x_1. \tag{6.56}$$

Analogously, with the corresponding formulation of f_{x_1},

$$\frac{\partial}{\partial x_2} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_2} f(x) = \frac{\partial}{\partial x_2}(x_1^2 + x_2^2) = f'_{x_1}(x_2) = 2x_2. \tag{6.57}$$

As for the derivative of a univariate real-valued function, it is also possible for a multivariate real-valued function to define a higher derivative recursively.

Definition 6.7 (Second partial derivatives)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function, and let $\frac{\partial}{\partial x_i} f$ be the partial derivative of f with respect to x_i. Then the second partial derivative of f with respect to x_i and x_j is defined as

$$\frac{\partial^2}{\partial x_j \partial x_i} f(x) := \frac{\partial}{\partial x_j}\left(\frac{\partial}{\partial x_i} f\right). \tag{6.58}$$

•

Note that for each partial derivative $\frac{\partial}{\partial x_i} f$ for $i = 1, ..., n$, there are in total n second partial derivatives $\frac{\partial^2}{\partial x_j \partial x_i} f$ for $j = 1, ..., n$. The resulting n^2 second partial derivatives, however, are not all distinct. This is an essential statement of *Schwarz's theorem*.

Theorem 6.5 (Schwarz's theorem)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a partially differentiable multivariate real-valued function. Then*

$$\frac{\partial^2}{\partial x_j \partial x_i} f(x) = \frac{\partial^2}{\partial x_i \partial x_j} f(x) \text{ for all } 1 \le i, j \le n. \tag{6.59}$$

$\circ$

For a proof, we refer to the advanced literature. Schwarz's theorem states in particular that, when forming second partial derivatives, the order of partial differentiation is irrelevant. In this way, the theorem facilitates the calculation of second partial derivatives and also helps detect analytic errors when computing them. We illustrate this by continuing the example above.

Example (1)

We want to compute the second-order partial derivatives of the function

$$f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_2^2 \tag{6.60}$$

With the results for the first-order partial derivatives of this function, we obtain

$$\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_1} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_1}(2x_1) = 2 \\
\frac{\partial^2}{\partial x_1 \partial x_2} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_1}(2x_2) = 0 \\
\frac{\partial^2}{\partial x_2 \partial x_1} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_2}(2x_1) = 0 \\
\frac{\partial^2}{\partial x_2 \partial x_2} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_2}(2x_2) = 2.
\end{aligned} \tag{6.61}$$

Obviously,

$$\frac{\partial^2}{\partial x_1 x_2} f(x) = \frac{\partial^2}{\partial x_2 x_1} f(x). \tag{6.62}$$

Example (2)

As a further example, we want to compute the first- and second-order partial derivatives of the function

$$f : \mathbb{R}^3 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_1 x_2 + x_2 \sqrt{x_3} \tag{6.63}$$

With the calculation rules for derivatives, the first-order partial derivatives are

$$\begin{aligned}
\frac{\partial}{\partial x_1} f(x) &= \frac{\partial}{\partial x_1}\left(x_1^2 + x_1 x_2 + x_2 \sqrt{x_3}\right) = 2x_1 + x_2, \\
\frac{\partial}{\partial x_2} f(x) &= \frac{\partial}{\partial x_2}\left(x_1^2 + x_1 x_2 + x_2 \sqrt{x_3}\right) = x_1 + \sqrt{x_3}, \\
\frac{\partial}{\partial x_3} f(x) &= \frac{\partial}{\partial x_3}\left(x_1^2 + x_1 x_2 + x_2 \sqrt{x_3}\right) = \frac{x_2}{2\sqrt{x_3}}.
\end{aligned} \tag{6.64}$$

For the second partial derivatives with respect to x_1, we obtain

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_1} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_1}(2x_1 + x_2) = 2, \\
\frac{\partial^2}{\partial x_2 \partial x_1} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_2}(2x_1 + x_2) = 1, \\
\frac{\partial^2}{\partial x_3 \partial x_1} f(x) &= \frac{\partial}{\partial x_3}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_3}(2x_1 + x_2) = 0.
\end{aligned}
\tag{6.65}
$$

For the second partial derivatives with respect to x_2, we obtain

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_2} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_1}(x_1 + \sqrt{x_3}) = 1, \\
\frac{\partial^2}{\partial x_2 \partial x_2} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_2}(x_1 + \sqrt{x_3}) = 0, \\
\frac{\partial^2}{\partial x_3 \partial x_2} f(x) &= \frac{\partial}{\partial x_3}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_3}(x_1 + \sqrt{x_3}) = \frac{1}{2\sqrt{x_3}}.
\end{aligned}
\tag{6.66}
$$

For the second partial derivatives with respect to x_3, we obtain

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_3} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_3} f(x)\right) = \frac{\partial}{\partial x_1}\left(\frac{x_2}{2\sqrt{x_3}}\right) = 0, \\
\frac{\partial^2}{\partial x_2 \partial x_3} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_3} f(x)\right) = \frac{\partial}{\partial x_2}\left(\frac{x_2}{2\sqrt{x_3}}\right) = \frac{1}{2\sqrt{x_3}}, \\
\frac{\partial^2}{\partial x_3 \partial x_3} f(x) &= \frac{\partial}{\partial x_3}\left(\frac{\partial}{\partial x_3} f(x)\right) = \frac{\partial}{\partial x_3}\left(x_2 \frac{1}{2} x_3^{-\frac{1}{2}}\right) = -\frac{1}{4} x_2 x_3^{-\frac{3}{2}}.
\end{aligned}
\tag{6.67}
$$

Furthermore, one sees that the order of partial differentiation is irrelevant, because

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_2} f(x) &= \frac{\partial^2}{\partial x_2 \partial x_1} f(x) = 1, \\
\frac{\partial^2}{\partial x_1 \partial x_3} f(x) &= \frac{\partial^2}{\partial x_3 \partial x_1} f(x) = 0, \\
\frac{\partial^2}{\partial x_2 \partial x_3} f(x) &= \frac{\partial^2}{\partial x_3 \partial x_2} f(x) = \frac{1}{2\sqrt{x_3}}.
\end{aligned}
\tag{6.68}
$$

As seen above, for a multivariate real-valued function $f : \mathbb{R}^n \to \mathbb{R}$ there are in total n first partial derivatives and n^2 second partial derivatives. These are collected in the *gradient* and the *Hessian matrix* of a multivariate real-valued function.

Definition 6.8 (Gradient)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function. The *gradient* $\nabla f(x)$ of f at the point $x \in \mathbb{R}^n$ is defined as

$$
\nabla f(x) := \begin{pmatrix} \frac{\partial}{\partial x_1} f(x) \\ \frac{\partial}{\partial x_2} f(x) \\ \vdots \\ \frac{\partial}{\partial x_n} f(x) \end{pmatrix} \in \mathbb{R}^n.
\tag{6.69}
$$

•

Note that gradients are multivariate vector-valued functions of the form

$$\nabla f : \mathbb{R}^n \to \mathbb{R}^n, x \mapsto \nabla f(x). \tag{6.70}$$

For $n = 1$, $\nabla f(x) = f'(x)$. An important property of the gradient is that $-\nabla f(x)$ indicates the direction of steepest descent of f in $\mathbb{R}^n$. This insight is not trivial, however, and will be deepened at a later point. As examples, we consider the gradients of the functions analyzed above.

Example (1)

For the function $f : \mathbb{R}^2 \to \mathbb{R}$ considered in Equation 6.60,

$$\nabla f(x) := \begin{pmatrix} \frac{\partial}{\partial x_1} f(x) \\ \frac{\partial}{\partial x_2} f(x) \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix} \in \mathbb{R}^2. \tag{6.71}$$

In Figure 6.8 we visualize color-coded selected values of this gradient for $(0.7, 0.7)^T$, $(-0.3, 0.1)^T$, $(-0.5, -0.4)^T$, and $(0.1, -1.0)^T$.

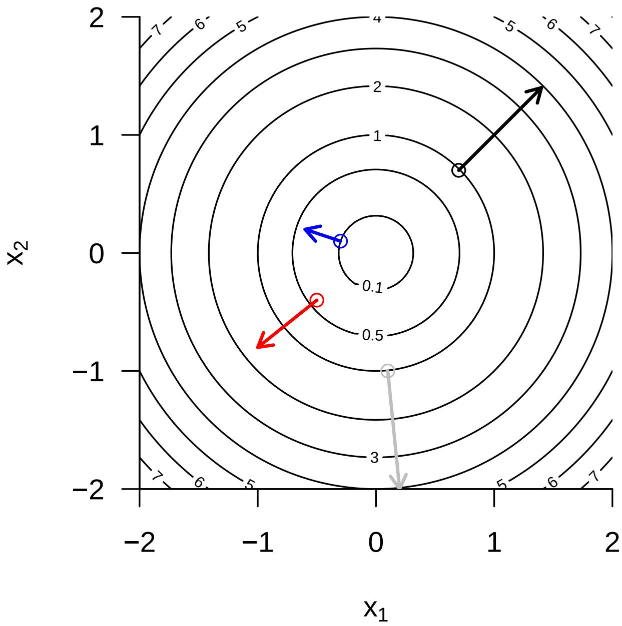

Figure 6.8 Example gradient values of the bivariate function $f(x) = x_1^2 + x_2^2$.

Example (2)

For the function $f : \mathbb{R}^3 \to \mathbb{R}$ considered in Equation 6.63,

$$\nabla f(x) := \begin{pmatrix} \frac{\partial}{\partial x_1} f(x) \\ \frac{\partial}{\partial x_2} f(x) \\ \frac{\partial}{\partial x_3} f(x) \end{pmatrix} = \begin{pmatrix} 2x_1 + x_2 \\ x_1 + \sqrt{x_3} \\ \frac{x_2}{2\sqrt{x_3}} \end{pmatrix} \in \mathbb{R}^3. \tag{6.72}$$

Finally, we turn to collecting the second partial derivatives of a multivariate real-valued function in the *Hessian matrix*.

Definition 6.9 (Hessian matrix)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function. The *Hessian matrix* $\nabla^2 f(x)$ of f at the point $x \in \mathbb{R}^n$ is defined as

$$\nabla^2 f(x) := \begin{pmatrix} \frac{\partial^2}{\partial x_1 \partial x_1} f(x) & \frac{\partial^2}{\partial x_1 \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_1 \partial x_n} f(x) \\ \frac{\partial^2}{\partial x_2 \partial x_1} f(x) & \frac{\partial^2}{\partial x_2 \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_2 \partial x_n} f(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_n \partial x_1} f(x) & \frac{\partial^2}{\partial x_n \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_n \partial x_n} f(x) \end{pmatrix} \in \mathbb{R}^{n \times n}. \tag{6.73}$$

•

Note that Hessian matrices are multivariate matrix-valued mappings of the form

$$\nabla^2 f : \mathbb{R}^n \to \mathbb{R}^{n \times n}, x \mapsto \nabla^2 f(x). \tag{6.74}$$

For $n = 1$, $\nabla^2 f(x) = f''(x)$. Furthermore, from

$$\frac{\partial^2}{\partial x_i \partial x_j} f(x) = \frac{\partial^2}{\partial x_j \partial x_i} f(x) \text{ for } 1 \le i, j \le n \tag{6.75}$$

it follows that the Hessian matrix is symmetric, that is,

$$\left(\nabla^2 f(x)\right)^T = \nabla^2 f(x). \tag{6.76}$$

Example (1)

For the function $f : \mathbb{R}^2 \to \mathbb{R}$ considered in Equation 6.60,

$$\nabla^2 f(x) := \begin{pmatrix} \frac{\partial^2}{\partial x_1 x_1} f(x) & \frac{\partial^2}{\partial x_1 x_2} f(x) \\ \frac{\partial^2}{\partial x_2 x_1} f(x) & \frac{\partial^2}{\partial x_2 x_2} f(x) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}. \tag{6.77}$$

The Hessian matrix of this function is therefore a constant function that does not depend on x.

Example (2)

For the function $f : \mathbb{R}^3 \to \mathbb{R}$ considered in Equation 6.63,

$$\nabla^2 f(x) := \begin{pmatrix} \frac{\partial^2}{\partial x_1 x_1} f(x) & \frac{\partial^2}{\partial x_1 x_2} f(x) & \frac{\partial^2}{\partial x_1 x_3} f(x) \\ \frac{\partial^2}{\partial x_2 x_1} f(x) & \frac{\partial^2}{\partial x_2 x_2} f(x) & \frac{\partial^2}{\partial x_2 x_3} f(x) \\ \frac{\partial^2}{\partial x_3 x_1} f(x) & \frac{\partial^2}{\partial x_3 x_2} f(x) & \frac{\partial^2}{\partial x_3 x_3} f(x) \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & \frac{1}{2\sqrt{x_3}} \\ 0 & \frac{1}{2\sqrt{x_3}} & -\frac{1}{4} x_2 x_3^{-3/2} \end{pmatrix}. \tag{6.78}$$

In contrast to Example (1), the Hessian matrix of the function considered here is not a constant function, and its value depends on the value of the function argument $x \in \mathbb{R}^3$.

7 Integral calculus

This chapter gives an overview of central concepts of integral calculus. The main focus throughout is on clarifying terminology, mathematical symbolism, and the intuition conveyed by it, rather than on the concrete computation of integrals.

7.1 Indefinite integrals

We begin with the definition of the indefinite integral and the concept of an antiderivative.

Definition 7.1 (Indefinite integral and antiderivative)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a univariate real-valued function. Then a differentiable function $F : I \to \mathbb{R}$ with the property

$$F' = f \tag{7.1}$$

is called an *antiderivative of* f. If F is an antiderivative of f, then

$$\int f(x)\,dx := F + c \text{ with } c \in \mathbb{R} \tag{7.2}$$

is called the *indefinite integral of the function* f. The indefinite integral of a function thus denotes the set of all antiderivatives of a function.

•

The above definition states that the derivative of an antiderivative of a function f is precisely f. In addition, the indefinite integral of a function f is the set of *all* antiderivatives of f obtained by adding different constants $c \in \mathbb{R}$. Such a constant $c \in \mathbb{R}$ is also called an *integration constant*; of course, $\frac{d}{dx}c = 0$. The symbol $\int f(x)\,dx$ is defined as $F + c$. In this expression, $f(x)$ is called the *integrand*. $\int$ and dx have no meaning of their own; they are merely symbols.

For the elementary functions introduced in the previous sections, the antiderivatives listed in Table 7.1 result. This can be verified by differentiating the respective antiderivative with the calculation rules of differential calculus. The indefinite integrals of these elementary functions then follow directly from these antiderivatives by adding an integration constant.

Table 7.1 Antiderivatives of elementary functions

Name	Definition	Antiderivative
Polynomial function	$f(x) := \sum_{i=0}^{k} a_i x^i$	$F(x) = \sum_{i=0}^{k} \frac{a_i}{i+1} x^{i+1}$
Constant function	$f(x) := a$	$F(x) = ax$
Identity function	$f(x) := x$	$F(x) = \frac{1}{2}x^2$

Table 7.1 Antiderivatives of elementary functions

Name	Definition	Antiderivative
Linear-affine function	$f(x) := ax + b$	$F(x) = \frac{1}{2}ax^2 + bx$
Square function	$f(x) := x^2$	$F(x) = \frac{1}{3}x^3$
Exponential function	$f(x) := \exp(x)$	$F(x) = \exp(x)$
Logarithm function	$f(x) := \ln(x)$	$F(x) = x\ln x - x$

The calculation rules collected in the following theorem are often helpful for determining antiderivatives of functions that are composed of functions with known antiderivatives.

Theorem 7.1 (Calculation rules for antiderivatives)**.** *Let f and g be univariate real-valued functions that possess antiderivatives, and let g be invertible. Then the following calculation rules hold for determining antiderivatives.*

(1) Sum rule

$$\int af(x) + bg(x)\,dx = a\int f(x)\,dx + b\int g(x)\,dx \text{ for } a, b \in \mathbb{R} \tag{7.3}$$

(2) Integration by parts

$$\int f'(x)g(x)\,dx = f(x)g(x) - \int f(x)g'(x)\,dx \tag{7.4}$$

(3) Substitution rule

$$\int f(g(x))g'(x)\,dx = \int f(t)\,dt \text{ with } t = g(x) \tag{7.5}$$

$\circ$

Proof. For a proof of the sum rule, we refer to the advanced literature. The calculation rule for integration by parts follows by integrating the product rule of differentiation. Recall that

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x). \tag{7.6}$$

Integrating both sides of the equation and taking into account the sum rule for antiderivatives then yields

$$\begin{aligned} \textstyle\int (f(x)g(x))'\,dx &= \textstyle\int f'(x)g(x) + f(x)g'(x)\,dx \\ \Leftrightarrow f(x)g(x) &= \textstyle\int f'(x)g(x)\,dx + \int f(x)g'(x)\,dx \\ \Leftrightarrow \textstyle\int f'(x)g(x)\,dx &= f(x)g(x) - \textstyle\int f(x)g'(x)\,dx. \end{aligned} \tag{7.7}$$

For $F' = f$, the substitution rule follows by applying the chain rule of differentiation to the composite function $F(g)$. Specifically, first

$$(F(g(x)))' = F'(g(x))g'(x) = f(g(x))g'(x). \tag{7.8}$$

Integrating both sides of the equation

$$(F(g(x)))' = f(g(x))g'(x) \tag{7.9}$$

then yields

$$\begin{aligned} \textstyle\int (F(g(x)))'\,dx &= \textstyle\int f(g(x))g'(x)\,dx \\ \Leftrightarrow F(g(x)) + c &= \textstyle\int f(g(x))g'(x)\,dx \\ \Leftrightarrow \textstyle\int f(g(x))g'(x)\,dx &= \textstyle\int f(t)\,dt \text{ with } t := g(x). \end{aligned} \tag{7.10}$$

Here, the right-hand side of the last equation above is to be understood as $F(g(x)) + c$, that is, as the antiderivative of f evaluated at the point $t := g(x)$. The dt is not to be replaced by $dg(x)$; it is purely notational in nature. $\square$

Indefinite integrals occupy a central place in the solution of differential equations. More immediate, however, is the use of indefinite integrals in the context of evaluating *definite integrals*, as introduced in the next section.

7.2 Definite integrals

Intuitively, a definite integral corresponds to the signed area between the graph of a function f and the x-axis, restricted to an interval $[a, b]$ (cf. Figure 7.1). *Signed* means that areas between the x-axis and positive values of f contribute positively to the area, whereas areas between the x-axis and negative values of f contribute negatively. For example, the value of the definite integral shown in Figure 7.1 A is 0.68, the value of the definite integral shown in Figure 7.1 B is 0.95 (the shaded area is clearly larger than in Figure 7.1 A), and the value of the definite integral shown in Figure 7.1 C is 0 (the shaded positive and negative areas cancel exactly). The latter example also suggests interpreting the integral as the average value of a function f over an interval $[a, b]$.

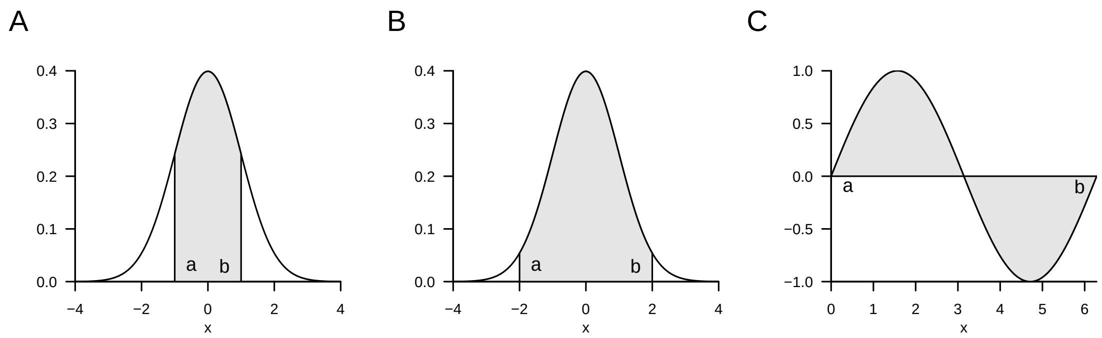

Figure 7.1 Examples of definite integrals

To introduce the concept of the *definite integral* in the sense of the *Riemann integral*, we first need to do some preliminary work. We begin by introducing a term for the subdivision of an interval into smaller sections.

Definition 7.2 (Partition of an interval and mesh)**.** Let $[a, b] \subset \mathbb{R}$ be an interval and let $x_0, x_1, x_2, ..., x_n \in [a, b]$ be a set of points with

$$a =: x_0 < x_1 < x_2 < \cdots < x_n := b \tag{7.11}$$

and

$$\Delta x_i := x_i - x_{i-1} \text{ for } i = 1, ..., n. \tag{7.12}$$

Then the set

$$Z := \{[x_0, x_1], [x_1, x_2], ..., [x_{n-1}, x_n]\} \tag{7.13}$$

of subintervals of $[a, b]$ defined by $x_0, x_1, x_2, ..., x_n$ is called a *partition of* $[a, b]$. Furthermore,

$$Z_{\max} := \max_{i \in \mathbb{N}_n} \Delta x_i, \tag{7.14}$$

that is, the largest of the subinterval lengths Δx_i, is called the *mesh of* Z.

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Intuitively, Δx_i is the width of the rectangles in Figure 7.2, as we will see in what follows. With the concepts of the partition of an interval, we can now introduce the concept of *Riemann sums*.

Definition 7.3 (Riemann sums)**.** Let $f : [a,b] \to \mathbb{R}$ be a bounded function on $[a,b]$, that is, $|f(x)| < c$ for $0 < c < \infty$ and all $x \in [a,b]$. Let Z be a partition of $[a,b]$ with subinterval lengths Δx_i for $i = 1, ..., n$. Furthermore, let ξ_i for $i = 1, ..., n$ be an arbitrary point in the subinterval $[x_{i-1}, x_i]$ of the partition Z. Then

$$R(Z) := \sum_{i=1}^{n} f(\xi_i)\Delta x_i \tag{7.15}$$

is called the *Riemann sum of* f *on* $[a,b]$ *with respect to the partition* Z.

•

If, for example, one chooses the maximum of f in each subinterval in the Riemann sum, the so-called *upper Riemann sum* results,

$$R_o(Z) := \sum_{i=1}^{n} \left(\max_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.16}$$

If, by contrast, one chooses the minimum of f in each subinterval, the so-called *lower Riemann sum* results,

$$R_u(Z) := \sum_{i=1}^{n} \left(\min_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.17}$$

Figure 7.2 illustrates the definition of these Riemann sums: the dark gray rectangles each have area $\Delta x_i \cdot \min_{[x_{i-1},x_i]} f(\xi)$ and thus form the summands in the lower Riemann sum

$$R_u(Z) := \sum_{i=1}^{4} \left(\min_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.18}$$

The vertical combination of dark gray and light gray rectangles each has area $\Delta x_i \cdot \max_{[x_{i-1},x_i]} f(\xi)$ and thus forms the summands in the upper Riemann sum

$$R_o(Z) := \sum_{i=1}^{4} \left(\max_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.19}$$

Now imagine letting Δx_i tend to zero for all $i = 1, ..., n$, thereby making the mesh of the partition Z smaller and smaller. The values of $\min_{[x_{i-1},x_i]} f(\xi_i)$ and $\max_{[x_{i-1},x_i]} f(\xi_i)$, and hence also the values of $R_u(Z)$ and $R_o(Z)$, will then approach each other more and more. This limiting process is used in the definition of the Riemann integral.

Definition 7.4 (Definite riemann integral)**.** Let $f : [a,b] \to \mathbb{R}$ be a bounded real-valued function on $[a,b]$. Furthermore, for Z_k with $k = 1, 2, 3...$, let there be a sequence of partitions of $[a,b]$ with corresponding mesh $Z_{\max,k}$. If, for every sequence of partitions $Z_1, Z_2, ...$ with $|Z_{\max,k}| \to 0$ as $k \to \infty$ and for arbitrarily chosen points ξ_{ki} with $i = 1, ..., n$ in the subinterval $[x_{k,i-1}, x_{k,i}]$ of the partition Z_k, the sequence of corresponding Riemann sums $R(Z_1), R(Z_2), ...$ tends to the same limit, then f is called *integrable* on $[a,b]$. The

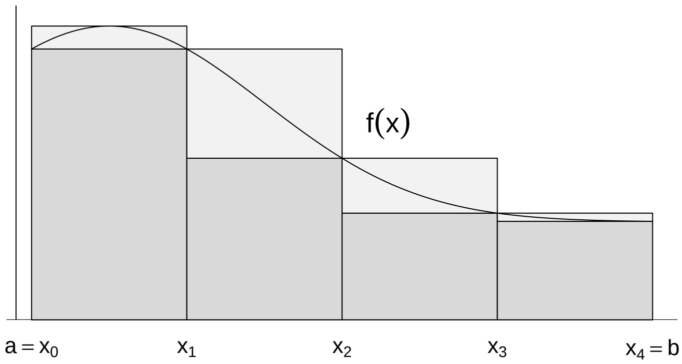

Figure 7.2 Riemann sums

corresponding limit of the sequence of Riemann sums is called the *definite Riemann integral* and denoted by

$$\int_a^b f(x)\,dx := \lim_{k\to\infty} R(Z_k) \text{ for } |Z_{\max,k}| \to 0. \tag{7.20}$$

In this context, the values a and b are called the *lower* and *upper* limits of integration, respectively, $f(x)$ is called the *integrand*, and x is called the *variable of integration*.

•

The Riemann integrability of a function and the value of a definite Riemann integral are thus defined in terms of a limiting process. However, the theory of Riemann integrals can be extended by the fundamental theorems of differential and integral calculus, so that the concrete computation of a definite integral rarely requires forming partitions and determining a limit. For simplicity, in what follows we omit the designation *Riemann* and simply speak of *definite integrals*.

A first step toward simplifying the computation of definite integrals is to record the following calculation rules, for whose proof we refer to the advanced literature.

Theorem 7.2 (Calculation rules for definite integrals)**.** *Let f and g be integrable functions on $[a,b]$. Then the following calculation rules hold.*

(1) Linearity. For $c_1, c_2 \in \mathbb{R}$,

$$\int_a^b (c_1 f(x) + c_2 g(x))\,dx = c_1 \int_a^b f(x)\,dx + c_2 \int_a^b g(x)\,dx. \tag{7.21}$$

(2) Additivity. For $a < c < b$,

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx. \tag{7.22}$$

(3) Change of sign when reversing the limits of integration

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx. \tag{7.23}$$

(4) Independence from the choice of the variable of integration

$$\int_a^b f(x)\,dx = \int_a^b f(y)\,dy. \tag{7.24}$$

(5) Independence of the integral from the type of interval

$$\int_a^b f(x)\,dx = \int_{]a,b[} f(x)\,dx = \int_{[a,b[} f(x)\,dx = \int_{]a,b]} f(x)\,dx = \int_{[a,b]} f(x)\,dx, \tag{7.25}$$

where $\int_I$ *denotes the definite integral of* f *on the interval* $I \subseteq \mathbb{R}$.

∘

Note that the linearity of the definite integral is analogous to associativity for sums of equal length and distributivity when multiplying by a constant (cf. Theorem 3.1),

$$\sum_{i=1}^{n}(c_1 x_i + c_2 y_i) = c_1 \sum_{i=1}^{n} x_i + c_2 \sum_{i=1}^{n} y_i,$$

that the additivity of definite integrals is analogous to splitting sums, and that the independence of the value of an integral from the choice of the variable of integration has its counterpart in the independence of the value of a sum from its summation index. A graphical representation of the additivity calculation rule for integrals is shown in Figure 7.3. The sum of the areas given by the definite integrals $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ with $a < c < b$ is the area of $\int_a^b f(x)\,dx$.

With the help of additivity and the zero-interval integral

$$\int_a^a f(x)\,dx = 0 \tag{7.26}$$

the change of sign when reversing the limits of integration, which will be important below, can be justified. To this end, as in Figure 7.3, let

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx. \text{ and } \int_b^a f(x)\,dx = \int_b^c f(x)\,dx + \int_c^a f(x)\,dx. \tag{7.27}$$

Adding both equations gives

$$\begin{aligned}
\int_a^b f(x)\,dx + \int_b^a f(x)\,dx &= \int_a^c f(x)\,dx + \int_c^b f(x)\,dx + \int_b^c f(x)\,dx + \int_c^a f(x)\,dx \\
\Leftrightarrow \int_a^a f(x)\,dx &= \int_a^b f(x)\,dx + \int_b^a f(x)\,dx \\
\Leftrightarrow 0 &= \int_a^b f(x)\,dx + \int_b^a f(x)\,dx \\
\Leftrightarrow \int_a^b f(x)\,dx &= -\int_b^a f(x)\,dx.
\end{aligned} \tag{7.28}$$

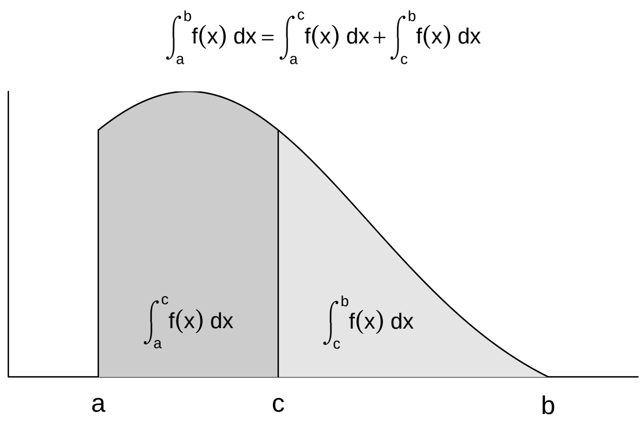

Figure 7.3 Additivity of definite integrals

The *fundamental theorems of differential and integral calculus* finally make it possible to compute definite integrals of a function f directly with the help of the antiderivative F of f. For the proof of the first fundamental theorem of differential and integral calculus, we need the *mean value theorem of integral calculus*, which we state here without proof and illustrate in Figure 7.4.

Theorem 7.3 (Mean value theorem of integral calculus). *For a continuous function* $f : [a,b] \to \mathbb{R}$*, there exists a* $\xi \in]a,b[$ *such that*

$$\int_a^b f(x)\,dx = f(\xi)(b-a) \tag{7.29}$$

∘

The mean value theorem of integral calculus guarantees the existence of a $\xi \in [a,b]$ such that the definite integral $\int_a^b f(x)\,dx$ equals the product of the "rectangle height" $f(\xi)$ and the "rectangle width" $(b-a)$. In Figure 7.4, this ξ lies exactly halfway between a and b. That the resulting gray rectangle area equals $\int_a^b f(x)\,dx$ follows from the visually at least plausible fact that the areas between $f(x)$ and $f(\xi)$ on the interval $[a,\xi]$ and between $f(\xi)$ and $f(x)$ on the interval $[\xi,b]$ have the same magnitude, but the former has a negative sign. In general, however, the mean value theorem of integral calculus only guarantees the existence of a $\xi \in [a,b]$ with the discussed property; it does not provide a formula for determining ξ.

With this preliminary work, we can now formulate and prove the first fundamental theorem of differential and integral calculus.

Theorem 7.4 (First fundamental theorem of calculus). *If* $f : I \to \mathbb{R}$ *is a continuous function on the interval* $I \subset \mathbb{R}$*, then the function*

$$F : I \to \mathbb{R}, x \mapsto F(x) := \int_a^x f(t)\,dt \text{ with } x, a \in I \tag{7.30}$$

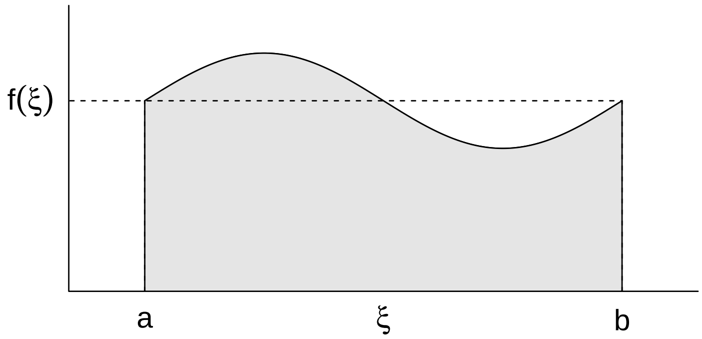

Figure 7.4 On the mean value theorem of integral calculus

is an antiderivative of f.

∘

Proof. We consider the difference quotient

$$\frac{1}{h}(F(x+h) - F(x)) \tag{7.31}$$

With the definition $F(x) := \int\limits_a^x f(t)\,dt$ and the additivity of the definite integral, it follows that

$$\frac{1}{h}(F(x+h) - F(x)) = \frac{1}{h}\left(\int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt\right) = \frac{1}{h}\int_x^{x+h} f(t)\,dt \tag{7.32}$$

By the mean value theorem of integral calculus, there is therefore a $\xi \in]x, x+h[$ such that

$$\frac{1}{h}(F(x+h) - F(x)) = f(\xi) \tag{7.33}$$

Taking the limit then yields

$$\lim_{h\to 0}\frac{1}{h}(F(x+h) - F(x)) = \lim_{h\to 0} f(\xi) \text{ for } \xi \in]x, x+h[\Leftrightarrow F'(x) = f(x). \tag{7.34}$$

□

The second fundamental theorem of differential and integral calculus states how to compute a definite integral with the help of the antiderivative.

Theorem 7.5 (Second fundamental theorem of calculus). *If F is an antiderivative of a continuous function $f : I \to \mathbb{R}$ on an interval I, then, for $a, b \in I$ with $a \le b$,*

$$\int_a^b f(x)\,dx = F(b) - F(a) =: F(x)|_a^b. \tag{7.35}$$

∘

Proof. Using the calculation rules for definite integrals and the first fundamental theorem of differential and integral calculus, we obtain

$$F(b) - F(a) = \int_{\alpha}^{b} f(t)\,dt - \int_{\alpha}^{a} f(t)\,dt = \int_{a}^{\alpha} f(t)\,dt + \int_{\alpha}^{b} f(t)\,dt = \int_{a}^{b} f(x)\,dx. \tag{7.36}$$

□

We want to apply the second fundamental theorem of differential and integral calculus in three examples (cf. Figure 7.5).

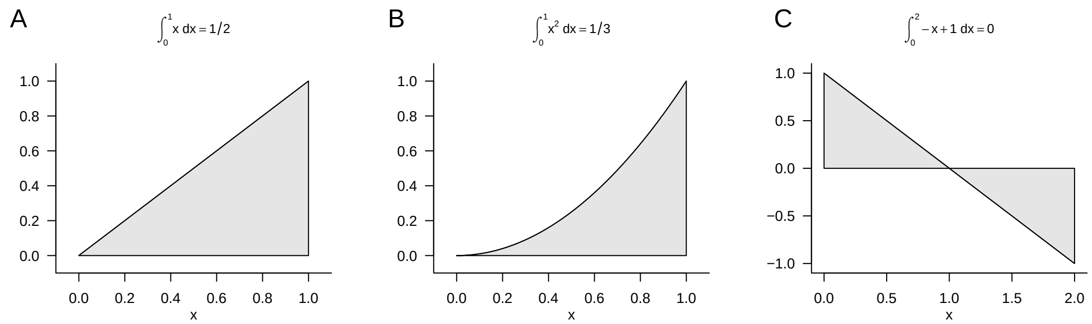

Figure 7.5 Examples of the second fundamental theorem of differential and integral calculus

Example (1)

We consider the identity function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x \tag{7.37}$$

and want to compute the definite integral of this function on the interval $[0, 1]$, that is,

$$\int_0^1 f(x)\,dx = \int_0^1 x\,dx. \tag{7.38}$$

To do so, recall that an antiderivative of f is given by

$$F : \mathbb{R} \to \mathbb{R}, x \mapsto F(x) := \frac{1}{2}x^2 \tag{7.39}$$

because

$$F'(x) = \frac{d}{dx}\left(\frac{1}{2}x^2\right) = 2 \cdot \frac{1}{2}x^{2-1} = x. \tag{7.40}$$

Substitution into the second fundamental theorem of differential and integral calculus then gives

$$\int_0^1 x\,dx = \frac{1}{2}1^2 - \frac{1}{2}0^2 = \frac{1}{2}. \tag{7.41}$$

This result agrees with the intuition suggested by the gray area in Figure 7.5 A.

Example (2)

Next, we consider the square function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2 \tag{7.42}$$

and want to compute the definite integral of this function on the interval $[0,1]$, that is,

$$\int_0^1 f(x)\,dx = \int_0^1 x^2\,dx. \tag{7.43}$$

To do so, recall that an antiderivative of f is given by

$$F : \mathbb{R} \to \mathbb{R}, x \mapsto F(x) := \frac{1}{3}x^3 \tag{7.44}$$

because

$$F'(x) = \frac{d}{dx}\left(\frac{1}{3}x^3\right) = 3 \cdot \frac{1}{3}x^{3-1} = x^2. \tag{7.45}$$

Substitution into the second fundamental theorem of differential and integral calculus then gives

$$\int_0^1 x^2\,dx = \frac{1}{3}1^3 - \frac{1}{3}0^3 = \frac{1}{3}. \tag{7.46}$$

This result agrees with the intuition that follows from comparing the gray areas in Figure 7.5 A and Figure 7.5 B.

Example (3)

Finally, we consider the linear-affine function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := -x + 1 \tag{7.47}$$

and want to compute the definite integral of this function on the interval $[0,2]$, that is,

$$\int_0^2 f(x)\,dx = \int_0^2 -x + 1\,dx. \tag{7.48}$$

To do so, recall that an antiderivative of the linear function with $a = -1$ and $b = 1$ (cf. Table 7.1) is given by

$$F : \mathbb{R} \to \mathbb{R}, x \mapsto F(x) := -\frac{1}{2}x^2 + x \tag{7.49}$$

because

$$F'(x) = \frac{d}{dx}\left(-\frac{1}{2}x^2 + x\right) = -2 \cdot \frac{1}{2}x^{2-1} + 1 \cdot x^{1-1} = -x + 1. \tag{7.50}$$

Substitution into the second fundamental theorem of differential and integral calculus then gives

$$\int_0^2 -x + 1\,dx = \left(-\frac{1}{2}2^2 + 2\right) - \left(-\frac{1}{2}0^2 + 0\right) = -2 + 2 - 0 = 0. \tag{7.51}$$

This result agrees with the intuition that the positive and negative gray areas in Figure 7.5 C cancel each other.

7.3 Improper integrals

Improper integrals are definite integrals in which at least one limit of integration is not a real number, but rather $-\infty$ or ∞. We illuminate the nature of improper integrals with the following definition and an example.

Definition 7.5 (Improper integrals). Let $f : \mathbb{R} \to \mathbb{R}$ be a univariate real-valued function. With the definitions

$$\int_{-\infty}^{b} f(x)\,dx := \lim_{a\to-\infty}\int_{a}^{b} f(x)\,dx \text{ and } \int_{a}^{\infty} f(x)\,dx := \lim_{b\to\infty}\int_{a}^{b} f(x)\,dx \tag{7.52}$$

and the additivity of integrals

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{b} f(x)\,dx + \int_{b}^{\infty} f(x)\,dx \tag{7.53}$$

the concept of the definite integral is extended to the unbounded intervals of integration $]-\infty, b]$, $[a, \infty[$, and $]-\infty, \infty[$. Integrals with unbounded intervals of integration are called *improper integrals*. If the corresponding limits exist, one says that the improper integrals *converge*.

•

As an example, we consider the improper integral of the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \frac{1}{x^2} \tag{7.54}$$

on the interval $[1, \infty[$, that is,

$$\int_{1}^{\infty} \frac{1}{x^2}\,dx. \tag{7.55}$$

According to the conventions in the definition of improper integrals,

$$\int_{1}^{\infty} \frac{1}{x^2}\,dx = \lim_{b\to\infty}\int_{1}^{b} \frac{1}{x^2}\,dx. \tag{7.56}$$

With the antiderivative $F(x) = -x^{-1}$ of $f(x) = x^{-2}$, the definite integral in the above equation becomes

$$\int_{1}^{b} \frac{1}{x^2}\,dx = F(b) - F(1) = -\frac{1}{b} - \left(-\frac{1}{1}\right) = -\frac{1}{b} + 1. \tag{7.57}$$

Thus,

$$\int_{1}^{\infty} \frac{1}{x^2}\,dx = \lim_{b\to\infty}\int_{1}^{b} \frac{1}{x^2}\,dx = \lim_{b\to\infty}\left(-\frac{1}{b}+1\right) = -\lim_{b\to\infty}\frac{1}{b} + \lim_{b\to\infty} 1 = 0 + 1 = 1. \tag{7.58}$$

7.4 Multidimensional integrals

So far, we have only considered integrals of univariate real-valued functions. The concept of the integral can also be extended to multivariate real-valued functions. In that case, however, the integration domain of the function is not necessarily as easy to describe as an interval. In particular, arbitrarily shaped integration domains are already possible for bivariate functions, for example. Here, we now want to consider the simplest case of a *hyperrectangle*. In this case, we can define multidimensional definite integrals as follows.

Definition 7.6 (Multidimensional definite integrals on hyperrectangles)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function. Then integrals of the form

$$\int_{[a_1,b_1]\times\cdots\times[a_n,b_n]} f(x)\,dx = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1,...,x_n)\,dx_1...\,dx_n \tag{7.59}$$

are called *multidimensional definite integrals on hyperrectangles.* Furthermore, integrals of the form

$$\int_{\mathbb{R}^n} f(x)\,dx = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1,...,x_n)\,dx_1...\,dx_n \tag{7.60}$$

are called *multidimensional improper integrals.*

•

As an example of Definition 7.6, we consider the two-dimensional definite integral of the function

$$f : \mathbb{R}^2 \to \mathbb{R}, (x_1,x_2) \mapsto f(x_1,x_2) := x_1^2 + 4x_2 \tag{7.61}$$

on the rectangle $[0,1] \times [0,1]$. Fubini's theorem from the theory of multidimensional integrals states that multidimensional integrals can be evaluated in any coordinate order. Thus, for example,

$$\int_{a_1}^{b_1} \left(\int_{a_2}^{b_2} f(x_1,x_2)\,dx_2 \right) dx_1 = \int_{a_2}^{b_2} \left(\int_{a_1}^{b_1} f(x_1,x_2)\,dx_1 \right) dx_2. \tag{7.62}$$

In this sense, for the example we consider

$$\int_0^1 \int_0^1 x_1^2 + 4x_2\,dx_1\,dx_2 = \int_0^1 \left(\int_0^1 x_1^2 + 4x_2\,dx_1 \right) dx_2 \tag{7.63}$$

and therefore first the inner integral. Here, x_2 takes on the role of a constant. An antiderivative of $g(x_1) := x_1^2 + 4x_2$ is $G(x_1) = \frac{1}{3}x_1^3 + 4x_2x_1$, which can be verified by differentiating G. Thus, for the inner integral,

$$\begin{aligned}\int_0^1 x_1^2 + 4x_2\,dx_1 &= G(1) - G(0)\\ &= \frac{1}{3}\cdot 1^3 + 4x_2 \cdot 1 - \frac{1}{3}\cdot 0^3 - 4x_2 \cdot 0\\ &= \frac{1}{3} + 4x_2.\end{aligned} \tag{7.64}$$

Considering the outer integral

$$\int_0^1 4x_2 + \frac{1}{3}\,dx_2$$

then gives, with the antiderivative

$$H(x_2) = 2x_2^2 + \frac{1}{3}x_2 \tag{7.65}$$

of

$$h(x_2) := 4x_2 + \frac{1}{3}, \tag{7.66}$$

that

$$
\begin{aligned}
\int_0^1 \int_0^1 x_1^2 + 4x_2 \, dx_1 \, dx_2 &= \int_0^1 4x_2 + \frac{1}{3} \, dx_2 \\
&= H(1) - H(0) \\
&= 2 \cdot 1^2 + \frac{1}{3} \cdot 1 - 2 \cdot 0^2 - \frac{1}{3} \cdot 0 \\
&= \frac{7}{3}.
\end{aligned}
\tag{7.67}
$$

8 Vectors

In scientific modeling, one often considers phenomena characterized by several quantitative features. For example, the position of an object in three-dimensional space is determined by three coordinates with respect to the three axes of a Cartesian coordinate system. Similarly, a person's health state may be characterized by three measured values, for example a self-report score, a biomarker, and an expert rating. For modeling and analyzing such multidimensional quantitative phenomena, mathematics provides a versatile tool in the form of the real vector space. In this chapter, we first introduce the concept of a real vector space and the basic arithmetic of vectors (Section 8.1). A vector space structure that is strongly guided by three-dimensional spatial intuition is then provided by the Euclidean vector space (Section 8.2). With vector arithmetic, all vectors of a vector space can be formed from a small collection of distinguished vectors. We discuss the concepts underlying this principle in Section 8.3 and Section 8.4.

8.1 Real vector space

We begin with the general definition of a vector space, which specifies basic rules for computing with vectors.

Definition 8.1 (Vector space)**.** Let V be a nonempty set and let S be a set of scalars. Furthermore, let a mapping

$$+ : V \times V \to V, (v_1, v_2) \mapsto +(v_1, v_2) =: v_1 + v_2, \tag{8.1}$$

called *vector addition*, be defined. Finally, let a mapping

$$\cdot : S \times V \to V, (s, v) \mapsto \cdot(s, v) =: sv, \tag{8.2}$$

called *scalar multiplication*, be defined. Then the tuple $(V, S, +, \cdot)$ is called a *vector space* if and only if, for arbitrary elements $v, w, u \in V$ and $a, b \in S$, the following conditions hold:

(1) *Commutativity of vector addition.*

$$v + w = w + v.$$

(2) *Associativity of vector addition.*

$$(v + w) + u = v + (w + u)$$

(3) *Existence of an identity element for vector addition.*

$$\text{There exists a vector } 0 \in V \text{ with } v + 0 = 0 + v = v.$$

(4) *Existence of inverse elements for vector addition.*

$$\text{For all vectors } v \in V \text{ there exists a vector } -v \in V \text{ with } v + (-v) = 0.$$

(5) *Existence of an identity element for scalar multiplication.*

$$\text{There exists a scalar } 1 \in S \text{ with } 1 \cdot v = v.$$

(6) *Associativity of scalar multiplication.*

$$a \cdot (b \cdot v) = (a \cdot b) \cdot v.$$

(7) *Distributivity with respect to vector addition.*

$$a \cdot (v + w) = a \cdot v + a \cdot w.$$

(8) *Distributivity with respect to scalar addition.*

$$(a + b) \cdot v = a \cdot v + b \cdot v.$$

•

It is striking that Definition 8.1 specifies how one should compute with vectors, but does not state what a vector actually is beyond being an element of a set. This is due to the fact that there are many different mathematical objects for which vector space structures can be defined. Examples include the set of real m-tuples, the set of matrices, the set of polynomials, the set of solutions of a linear system of equations, the set of real sequences, the set of continuous functions, and many others.

Here we are initially interested only in the vector space of the set of real m-tuples. We recall that we denote the real m-tuples by

$$\mathbb{R}^m := \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} | x_i \in \mathbb{R} \text{ for all } 1 \le i \le m \right\} \tag{8.3}$$

and pronounce $\mathbb{R}^m$ as "$\mathbb{R}$ to the power of m". The elements $x \in \mathbb{R}^m$ are called *real vectors* or simply *vectors*. We now want to use the set $\mathbb{R}^m$ as the basis for the definition of a vector space. To this end, we first define vector addition for elements of $\mathbb{R}^m$ and scalar multiplication for elements of $\mathbb{R}$ and $\mathbb{R}^m$.

Definition 8.2 (Vector addition and scalar multiplication in $\mathbb{R}^m$)**.** For all $x, y \in \mathbb{R}^m$ and $a \in \mathbb{R}$, let *vector addition* be defined by

$$+ : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}^m, (x, y) \mapsto x + y = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} := \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_m + y_m \end{pmatrix} \tag{8.4}$$

and let *scalar multiplication* be defined by

$$\cdot : \mathbb{R} \times \mathbb{R}^m \to \mathbb{R}^m, (a, x) \mapsto ax = a \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} := \begin{pmatrix} ax_1 \\ \vdots \\ ax_m \end{pmatrix} \tag{8.5}$$

•

This yields the following result.

Theorem 8.1 (Real vector space)**.** $(\mathbb{R}^m, \mathbb{R}, +, \cdot)$ *with the arithmetic rules of addition and multiplication in* $\mathbb{R}$ *is a vector space.*

∘

For a proof, which we omit here, one has to verify conditions (1) to (8) from Definition 8.1 for the set considered here and the forms of vector addition and scalar multiplication specified here. These follow easily from the arithmetic rules of addition and multiplication in $\mathbb{R}$ and from the fact that vector addition and scalar multiplication for elements of $\mathbb{R}^m$ were defined componentwise in Definition 8.2. We thus define the concept of the *real vector space*.

Definition 8.3 (Real vector space)**.** For $\mathbb{R}^m$, let $+$ and $\cdot$ be the vector addition and scalar multiplication defined in Definition 8.2. Then, based on Theorem 8.1, we call the vector space $(\mathbb{R}^m, \mathbb{R}, +, \cdot)$ the *real vector space*.

•

On the basis of Definition 8.3, we now illustrate computing with real vectors by means of a few examples.

Examples

(1) For

$$x := \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \in \mathbb{R}^4 \text{ and } y := \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

we have

$$x + y = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+2 \\ 2+1 \\ 3+0 \\ 4+1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \\ 5 \end{pmatrix} \in \mathbb{R}^4.$$

In **R**, one implements this example as follows.

```
x = matrix(c(1,2,3,4), nrow = 4)       # vector definition
y = matrix(c(2,1,0,1), nrow = 4)       # vector definition
x + y                                  # vector addition
```

```
     [,1]
[1,]    3
[2,]    3
[3,]    3
[4,]    5
```

(2) For

$$x := \begin{pmatrix} 2 \\ 3 \end{pmatrix} \in \mathbb{R}^2 \text{ and } y := \begin{pmatrix} 1 \\ 3 \end{pmatrix} \in \mathbb{R}^2$$

we have

$$x - y = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2-1 \\ 3-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathbb{R}^2.$$

In **R**, one implements this example as follows.

```
x = matrix(c(2,3), nrow = 2)          # vector definition
y = matrix(c(1,3), nrow = 2)          # vector definition
x - y                                 # vector subtraction
```

```
     [,1]
[1,]    1
[2,]    0
```

(3) For

$$x := \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \in \mathbb{R}^3 \text{ and } a := 3 \in \mathbb{R}$$

we have

$$ax = 3 \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \cdot 2 \\ 3 \cdot 1 \\ 3 \cdot 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 9 \end{pmatrix} \in \mathbb{R}^3.$$

In **R**, one implements this example as follows.

```
x = matrix(c(2,1,3), nrow = 3)        # vector definition
a = 3                                 # scalar definition
a*x                                   # scalar multiplication
```

```
     [,1]
[1,]    6
[2,]    3
[3,]    9
```

For $m \in \{1, 2, 3\}$, real vectors and computations with them can be visualized. For $m > 3$, for example when more than three quantitative features of a person's health state are available, as is regularly the case in applications, this is not possible. Nevertheless, visual intuition for $m \leq 3$ may facilitate an initial understanding of vector spaces. We focus here on the case $m := 2$. In this case, the real vectors under consideration lie in the two-dimensional plane and are usually visualized as points or arrows (Figure 8.1).

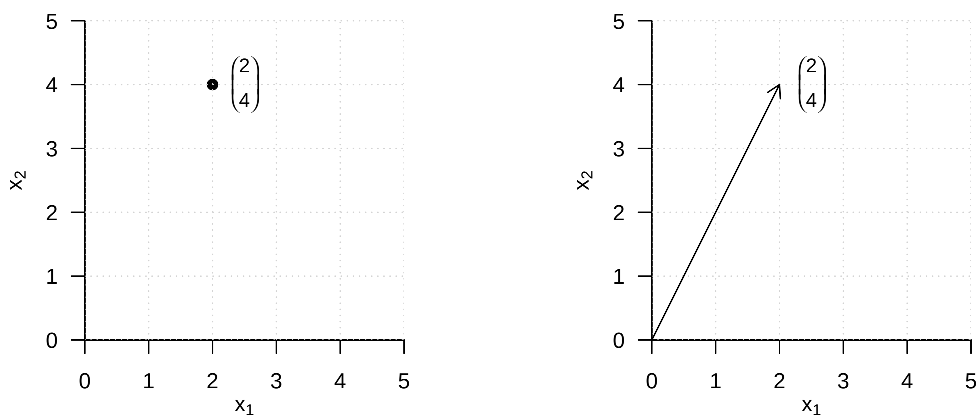

Figure 8.1 Visualization of vectors in $\mathbb{R}^2$

Figure 8.2 visualizes the vector addition

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}. \tag{8.6}$$

The sum vector corresponds to the diagonal of the parallelogram spanned by the two summands.

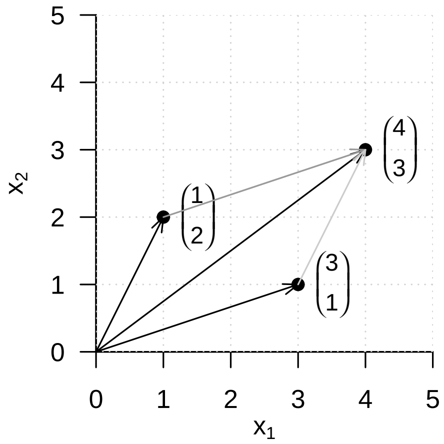

Figure 8.2 Vector addition in $\mathbb{R}^2$

Figure 8.3 visualizes the vector subtraction

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \tag{8.7}$$

The resulting vector corresponds to the diagonal of the parallelogram spanned by the first vector and the opposite vector of the second vector.

Finally, Figure 8.4 visualizes the scalar multiplication

$$3\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \tag{8.8}$$

Multiplication of a vector by a positive scalar changes only its length, but not its direction.

8.2 Euclidean vector space

By defining the *inner product*, the real vector space can be endowed with spatial-geometric intuition in the sense of a *Euclidean vector space*. In particular, this makes it possible to define and compute concepts such as the *length of a vector*, the *distance between two vectors*, and, not least, the *angle between two vectors*. We first introduce the *inner product*.

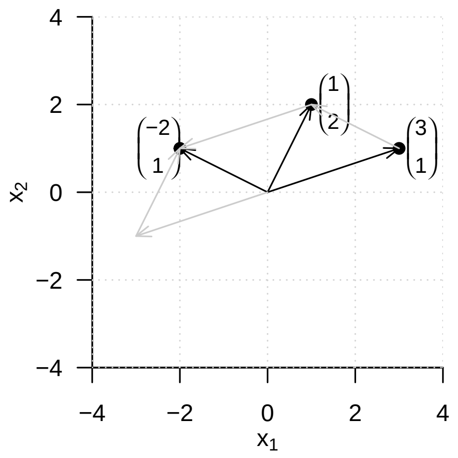

Figure 8.3 Vector subtraction in $\mathbb{R}^2$

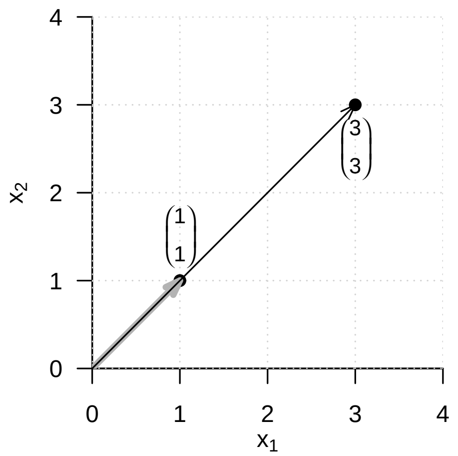

Figure 8.4 Scalar multiplication in $\mathbb{R}^2$

Definition 8.4 (Inner product on $\mathbb{R}^m$)**.** The *inner product on* $\mathbb{R}^m$ is defined as the mapping

$$\langle\rangle : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}, (x,y) \mapsto \langle(x,y)\rangle := \langle x,y\rangle := \sum_{i=1}^{m} x_i y_i. \tag{8.9}$$

•

The inner product is called an inner product because it yields a scalar, not because it involves multiplication by scalars. The inner product is closely related to the matrix product, as we will see later. We first consider an example and its implementation in **R**.

Example

Let

$$x := \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \text{ and } y := \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \tag{8.10}$$

Then

$$\langle x,y\rangle = x_1y_1 + x_2y_2 + x_3y_3 = 1\cdot 2 + 2\cdot 0 + 3\cdot 1 = 2 + 0 + 3 = 5. \tag{8.11}$$

In **R**, there are several ways to evaluate an inner product. We list two of them for the given example below.

```
# vector definitions
x = matrix(c(1,2,3), nrow = 3)
y = matrix(c(2,0,1), nrow = 3)

# inner product using R's componentwise multiplication and sum() function
sum(x*y)
```

```
[1] 5
```

```
# inner product using R's matrix transposition and multiplication
t(x) %*% y
```

```
     [,1]
[1,]    5
```

With the help of the inner product, the concept of the real vector space can be extended to the concept of the *real canonical Euclidean vector space.*

Definition 8.5 (Euclidean vector space)**.** The tuple $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ consisting of the real vector space $(\mathbb{R}^m, +, \cdot)$ and the inner product $\langle\rangle$ on $\mathbb{R}^m$ is called the *real canonical Euclidean vector space.*

•

In general, every tuple consisting of a vector space and an inner product is called a "Euclidean vector space". Informally, however, we often simply speak of $\mathbb{R}^m$ as a "Euclidean vector space" and, in particular, of $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ as the "Euclidean vector space". A Euclidean vector space is a vector space with geometric structure induced by the inner product. In particular, the geometric concepts of *length*, *distance*, and *angle* now acquire meaning in the Euclidean vector space. We define them as follows.

Definition 8.6. $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ is the Euclidean vector space.

(1) The *length* of a vector $x \in \mathbb{R}^m$ is defined as

$$\|x\| := \sqrt{\langle x, x\rangle}. \tag{8.12}$$

(2) The *distance* between two vectors $x, y \in \mathbb{R}^m$ is defined as

$$d(x, y) := \|x - y\|. \tag{8.13}$$

(3) The *angle* α between two vectors $x, y \in \mathbb{R}^m$ with $x, y \neq 0$ is defined by

$$0 \leq \alpha \leq \pi \text{ and } \cos\alpha := \frac{\langle x, y\rangle}{\|x\|\|y\|} \tag{8.14}$$

•

The length $\|x\|$ of a vector $x \in \mathbb{R}^m$ is also called the *Euclidean norm of* x, the ℓ_2*-norm of* x, or simply the *norm of* x. It is often denoted by $\|x\|_2$. We consider three examples for determining the length of a vector and their corresponding **R** implementation. We illustrate these examples in Figure 8.5.

Example (1)

$$\left\|\begin{pmatrix}2\\0\end{pmatrix}\right\| = \sqrt{\left\langle\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}\right\rangle} = \sqrt{2^2 + 0^2} = \sqrt{4} = 2.00 \tag{8.15}$$

```
norm(matrix(c(2,0),nrow = 2), type = "2")            # vector length = l_2 norm
```

```
[1] 2
```

Example (2)

$$\left\|\begin{pmatrix}2\\2\end{pmatrix}\right\| = \sqrt{\left\langle\begin{pmatrix}2\\2\end{pmatrix}, \begin{pmatrix}2\\2\end{pmatrix}\right\rangle} = \sqrt{2^2 + 2^2} = \sqrt{8} \approx 2.83 \tag{8.16}$$

```
norm(matrix(c(2,2),nrow = 2), type = "2")            # vector length = l_2 norm
```

```
[1] 2.828427
```

Example (3)

$$\left\|\begin{pmatrix}2\\4\end{pmatrix}\right\| = \sqrt{\left\langle\begin{pmatrix}2\\4\end{pmatrix}, \begin{pmatrix}2\\4\end{pmatrix}\right\rangle} = \sqrt{2^2 + 4^2} = \sqrt{20} \approx 4.47 \tag{8.17}$$

```
norm(matrix(c(2,4),nrow = 2), type = "2")            # vector length = l_2 norm
```

```
[1] 4.472136
```

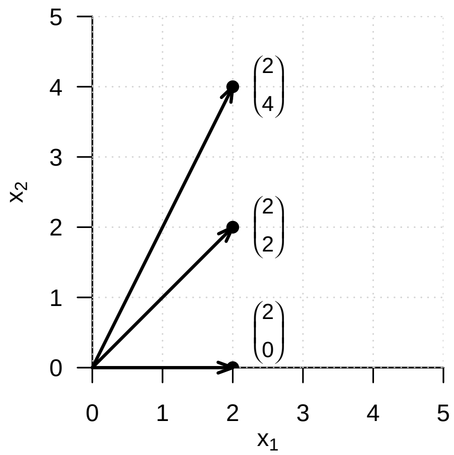

Figure 8.5 Vector length in $\mathbb{R}^2$

For the distance $d(x,y)$ between two vectors $x,y \in \mathbb{R}^m$, we note without proof that it is nonnegative and symmetric, that is,

$$d(x,y) \geq 0, d(x,x) = 0 \text{ and } d(x,y) = d(y,x) \tag{8.18}$$

hold. Moreover, $d(x,y)$ satisfies the so-called *triangle inequality*, which states that the direct path between two points in space is always shorter than an indirect path through a third point,

$$d(x,y) \leq d(x,z) + d(z,y). \tag{8.19}$$

Thus, $d(x,y)$ satisfies important aspects of spatial intuition. We give two examples for determining distances between vectors in $\mathbb{R}^2$, which we visualize in Figure 8.6.

Example (1)

$$d\left(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}2\\2\end{pmatrix}\right) = \left\|\begin{pmatrix}1\\1\end{pmatrix} - \begin{pmatrix}2\\2\end{pmatrix}\right\| = \left\|\begin{pmatrix}-1\\-1\end{pmatrix}\right\| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2} \approx 1.41 \tag{8.20}$$

```
norm(matrix(c(1,1),nrow = 2) - matrix(c(2,2),nrow = 2), type = "2")
```

```
[1] 1.414214
```

Example (2)

$$d\left(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}4\\1\end{pmatrix}\right) = \left\|\begin{pmatrix}1\\1\end{pmatrix} - \begin{pmatrix}4\\1\end{pmatrix}\right\| = \left\|\begin{pmatrix}-3\\0\end{pmatrix}\right\| = \sqrt{(-3)^2 + 0^2} = \sqrt{9} = 3 \tag{8.21}$$

```
norm(matrix(c(1,1),nrow = 2) - matrix(c(1,4),nrow = 2), type = "2")
```

```
[1] 3
```

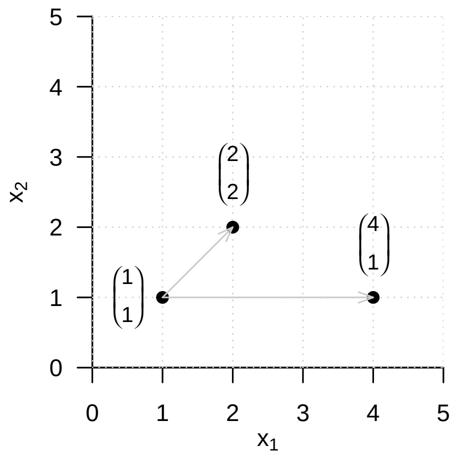

Figure 8.6 Vector distances in $\mathbb{R}^2$

Finally, we note that for the calculation of the angle between two vectors based on the above definition, the cosine function cos is bijective on $[0, \pi]$ and hence invertible with inverse function *acos*, the arccosine function. We also consider two examples for the concept of angle. Note in particular that Definition 8.6 gives the angle in radians. For a specification in degrees, a corresponding conversion is required.

Example (1)

$$\operatorname{acos}\left(\frac{\left\langle\begin{pmatrix}3\\0\end{pmatrix},\begin{pmatrix}3\\3\end{pmatrix}\right\rangle}{\left\|\begin{pmatrix}3\\0\end{pmatrix}\right\|\left\|\begin{pmatrix}3\\3\end{pmatrix}\right\|}\right) = \operatorname{acos}\left(\frac{3\cdot 3+3\cdot 0}{\sqrt{3^2+0^2}\cdot\sqrt{3^2+3^2}}\right) = \operatorname{acos}\left(\frac{9}{3\cdot\sqrt{18}}\right) = \frac{\pi}{4} \approx 0.785 \tag{8.22}$$

The conversion to degrees then gives

$$0.785 \cdot \frac{180^\circ}{\pi} = 45^\circ \tag{8.23}$$

In **R**, one implements this as follows.

```
x = matrix(c(3,0), nrow = 2)                                 # vector 1
y = matrix(c(3,3), nrow = 2)                                 # vector 2
w = acos(sum(x*y)/(sqrt(sum(x*x))*sqrt(sum(y*y)))) * 180/pi  # angle in degrees
print(w)
```

```
[1] 45
```

Example (2)

$$\alpha = \operatorname{acos}\left(\frac{\left\langle\begin{pmatrix}3\\0\end{pmatrix},\begin{pmatrix}0\\3\end{pmatrix}\right\rangle}{\left\|\begin{pmatrix}3\\0\end{pmatrix}\right\|\left\|\begin{pmatrix}0\\3\end{pmatrix}\right\|}\right) = \operatorname{acos}\left(\frac{3\cdot 0+0\cdot 3}{\sqrt{3^2+0^2}\cdot\sqrt{0^2+3^2}}\right) = \operatorname{acos}\left(\frac{0}{3\cdot 3}\right) = \frac{\pi}{2} \approx 1.57 \tag{8.24}$$

The conversion to degrees then gives

$$\frac{\pi}{2} \cdot \frac{180^\circ}{\pi} = 90^\circ \tag{8.25}$$

The corresponding **R** implementation is as follows.

```
x = matrix(c(3,0), nrow = 2)                               # vector 1
y = matrix(c(0,3), nrow = 2)                               # vector 2
w = acos(sum(x*y)/(sqrt(sum(x*x))*sqrt(sum(y*y)))) * 180/pi     # angle in degrees
print(w)
```

```
[1] 90
```

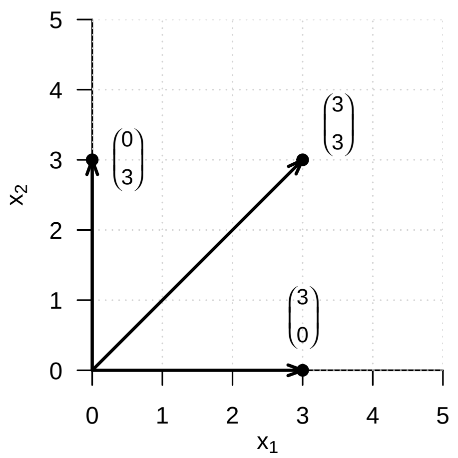

Figure 8.7 Angles in $\mathbb{R}^2$

The fact that two vectors can form a right angle, that is, can in a sense be maximally non-parallel, is an important geometric principle and is therefore singled out by the following definition.

Definition 8.7 (Orthogonality and orthonormality of vectors)**.** Let $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ be the Euclidean vector space.

(1) Two vectors $x, y \in \mathbb{R}^m$ are called *orthogonal* if

$$\langle x, y\rangle = 0 \tag{8.26}$$

(2) Two vectors $x, y \in \mathbb{R}^m$ are called *orthonormal* if

$$\langle x, y\rangle = 0 \text{ and } \|x\| = \|y\| = 1. \tag{8.27}$$

•

For orthogonal and orthonormal vectors, in particular, we therefore also have

$$\cos\alpha = \frac{\langle x, y\rangle}{\|x\|\|y\|} = \frac{0}{\|x\|\|y\|} = 0, \tag{8.28}$$

and hence

$$\alpha = \frac{\pi}{2} = 90^\circ. \tag{8.29}$$

8.3 Linear independence

In this section, we introduce the concept of *linear independence* of vectors. To this end, we first define the concept of a *linear combination* of vectors.

Definition 8.8 (Linear combination)**.** Let $\{v_1, v_2, ..., v_k\}$ be a set of k vectors of a vector space V, and let $a_1, a_2, ..., a_k$ be scalars. Then the *linear combination* of the vectors in $\{v_1, v_2, ..., v_k\}$ with the *coefficients* $a_1, a_2, ..., a_k$ is defined as the vector

$$w := \sum_{i=1}^{k} a_i v_i \in V. \tag{8.30}$$

•

Example

Let

$$v_1 := \begin{pmatrix} 2 \\ 1 \end{pmatrix}, v_2 := \begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_3 := \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } a_1 := 2, a_2 := 3, a_3 := 0. \tag{8.31}$$

Then the linear combination of v_1, v_2, v_3 with coefficients a_1, a_2, a_3 is

$$\begin{aligned} w &= a_1 v_1 + a_2 v_2 + a_3 v_3 \\ &= 2 \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ 5 \end{pmatrix}. \end{aligned} \tag{8.32}$$

Based on the concept of a linear combination, one can now define the concept of *linear independence* of vectors.

Definition 8.9 (Linear independence)**.** Let V be a vector space. A set $W := \{w_1, w_2, ..., w_k\}$ of vectors in V is called *linearly independent* if the only representation of the zero element $0 \in V$ by a linear combination of the $w \in W$ is the so-called *trivial representation*

$$0 = a_1 w_1 + a_2 w_2 + \cdots + a_k w_k \text{ with } a_1 = a_2 = \cdots = a_k = 0 \tag{8.33}$$

If the set W is not linearly independent, it is called *linearly dependent*.

•

To check whether a given set of vectors is linearly dependent or independent, in principle one has to check whether a possible linear combination of the given vectors is zero. Theorem 8.2 and Theorem 8.3 show how this can be achieved with less effort for two and for finitely many vectors, respectively.

Theorem 8.2 (Linear dependence of two vectors)**.** *Let V be a vector space. Two vectors $v_1, v_2 \in V$ are linearly dependent if one of the vectors is a scalar multiple of the other vector.*

∘

Proof. Let v_1 be a scalar multiple of v_2, that is,

$$v_1 = \lambda v_2 \text{ with } \lambda \neq 0. \tag{8.34}$$

Then

$$v_1 - \lambda v_2 = 0. \tag{8.35}$$

But this corresponds to the linear combination

$$a_1 v_1 + a_2 v_2 = 0 \tag{8.36}$$

with $a_1 = 1 \neq 0$ and $a_2 = -\lambda \neq 0$. Thus, there exists a linear combination of the zero element that is not the trivial representation, and hence v_1 and v_2 are not linearly independent. □

Theorem 8.3 (Linear dependence of a set of vectors)**.** *Let V be a vector space, and let $w_1, ..., w_k \in V$ be a set of vectors in V. If one of the vectors w_i with $i = 1, ..., k$ is a linear combination of the other vectors, then the set of vectors is linearly dependent.*

∘

Proof. The vectors $w_1, ..., w_k$ are linearly dependent if and only if $\sum_{i=1}^k a_i w_i = 0$ with at least one $a_i \neq 0$. Thus, for example, let $a_j \neq 0$. Then

$$0 = \sum_{i=1}^k a_i w_i = \sum_{i=1, i\neq j}^k a_i w_i + a_j w_j \tag{8.37}$$

Therefore,

$$a_j w_j = -\sum_{i=1, i\neq j}^k a_i w_i \tag{8.38}$$

and thus

$$w_j = -a_j^{-1} \sum_{i=1, i\neq j}^k a_i w_i = -\sum_{i=1, i\neq j}^k (a_j^{-1} a_i) w_i \tag{8.39}$$

Thus, w_j is a linear combination of the w_i for $i = 1, ..., k$ with $i \neq j$. □

8.4 Vector space bases

In this section, we introduce the concept of a *vector space basis.* A basis of a vector space is a subset of vectors of the vector space that can be used to represent all vectors of the vector space. In the sense of linear combinations of vectors, a vector space basis therefore contains all information needed to construct the corresponding vector space. However, a vector space basis is usually not unique, and many vector spaces indeed have infinitely many bases. The following definition first states how infinitely many vectors can be formed from a limited number of vectors by means of linear combinations.

Definition 8.10 (Span and spanning)**.** Let V be a vector space, and let $W := \{w_1, ..., w_k\} \subset V$. Then the *span* of W is defined as the set of all linear combinations of the elements of W,

$$\operatorname{Span}(W) := \left\{ \sum_{i=1}^{k} a_i w_i | a_1, ..., a_k \text{ are scalar coefficients} \right\} \tag{8.40}$$

A set of vectors $W \subseteq V$ is said to *span a vector space* V if every $v \in V$ can be written as a linear combination of vectors in W.

•

We now define the concept of a *basis* of a vector space.

Definition 8.11 (Basis)**.** Let V be a vector space, and let $B \subseteq V$. B is called a *basis of* V if

(1) the vectors in B are linearly independent, and
(2) the vectors in B span the vector space V.

•

Bases of vector spaces have the following important properties.

Theorem 8.4 (Properties of bases)**.**

(1) All bases of a vector space contain the same number of vectors.
(2) Every set of m linearly independent vectors is a basis of an m-dimensional vector space.

∘

For a proof of this very deep theorem, we refer to the advanced literature. The unique number of vectors in a basis of a vector space named by the above theorem is called the *dimension of the vector space.* Since there are usually infinitely many sets of m linearly independent vectors in a vector space, vector spaces usually have infinitely many bases.

If one considers a single vector in a vector space, one may ask how this vector can be represented with the help of a vector space basis. This leads to the following concepts.

Definition 8.12 (Basis representation and coordinates)**.** Let $B := \{b_1, ..., b_m\}$ be a basis of an m-dimensional vector space V, and let $v \in V$. Then the linear combination

$$v = \sum_{i=1}^{m} c_i b_i \tag{8.41}$$

is called the *representation of v with respect to the basis B*, and the coefficients $c_1, ..., c_m$ are called the *coordinates of v with respect to the basis B.*

•

For a fixed basis, the coordinates of a vector with respect to this basis are also fixed and unique. This is the statement of the following theorem.

Theorem 8.5 (Uniqueness of basis representation). *The basis representation of a $v \in V$ with respect to a basis B is unique.*

∘

Proof. Without loss of generality, assume that the vector space has dimension m. Assume that two representations of v with respect to the basis B exist, that is,

$$\begin{aligned} v &= a_1 b_1 + \cdots + a_m b_m \\ v &= c_1 b_1 + \cdots + c_m b_m \end{aligned} \tag{8.42}$$

Subtracting the lower equation from the upper equation gives

$$0 = (a_1 - c_1) b_1 + \cdots + (a_m - c_m) b_m \tag{8.43}$$

Because $b_1, ..., b_m$ are linearly independent, however, $(a_i - c_i) = 0$ for all $i = 1, ..., m$, and thus the two representations of v with respect to the basis B are identical. □

At the end of this section, we consider a special basis of the real vector space.

Definition 8.13 (Orthonormal basis of $\mathbb{R}^m$). A set of m vectors $v_1, ..., v_m \in \mathbb{R}^m$ is called an *orthonormal basis* of $\mathbb{R}^m$ if $v_1, ..., v_m$ each have length 1 and are mutually orthogonal, that is, if

$$\langle v_i, v_j \rangle = \begin{cases} 1 & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases}. \tag{8.44}$$

•

We first consider an example of an orthonormal basis.

Example (1)

$$B_1 := \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \tag{8.45}$$

is an orthonormal basis of $\mathbb{R}^2$, because B_1 consists of two vectors and

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = 1 \cdot 1 + 0 \cdot 0 = 1 + 0 = 1 \tag{8.46}$$

as well as

$$\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle = 0 \cdot 0 + 1 \cdot 1 = 0 + 1 = 1 \tag{8.47}$$

and

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle = 1 \cdot 0 + 0 \cdot 1 = 0 + 0 = 0 \tag{8.48}$$

hold.

For general real vector spaces, bases of the form of B_1 are singled out by the concept of the *canonical basis*.

Definition 8.14 (Canonical basis and standard unit vectors)**.** The orthonormal basis

$$B := \left\{e_1, ..., e_m | (e_i)_j = 1 \text{ for } i = j \text{ and } (e_i)_j = 0 \text{ for } i \neq j,\ i, j = 1, ..., m\right\} \subset \mathbb{R}^m \tag{8.49}$$

is called the *canonical basis* of $\mathbb{R}^m$, and the e_i are called *standard unit vectors.*

•

B_1 from Example (1) is therefore the canonical basis of $\mathbb{R}^2$.

The canonical basis of $\mathbb{R}^3$ is

$$B := \left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}. \tag{8.50}$$

However, there are also non-canonical orthonormal bases. To see this, we consider another example.

Example (2)

$$B_2 := \left\{\begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}, \begin{pmatrix}-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}\right\} \tag{8.51}$$

is also an orthonormal basis of $\mathbb{R}^2$, because B_2 consists of two vectors and

$$\left\langle\begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}, \begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}\right\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} + \frac{1}{2} = 1, \tag{8.52}$$

as well as

$$\left\langle\begin{pmatrix}-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}, \begin{pmatrix}-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}\right\rangle = \left(-\frac{1}{\sqrt{2}}\right) \cdot \left(-\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} + \frac{1}{2} = 1 \tag{8.53}$$

and

$$\left\langle\begin{pmatrix}-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}, \begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}\right\rangle = -\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = -\frac{1}{2} + \frac{1}{2} = 0 \tag{8.54}$$

hold.

We visualize the two orthonormal bases B_1 and B_2 of $\mathbb{R}^2$ in Figure 8.8.

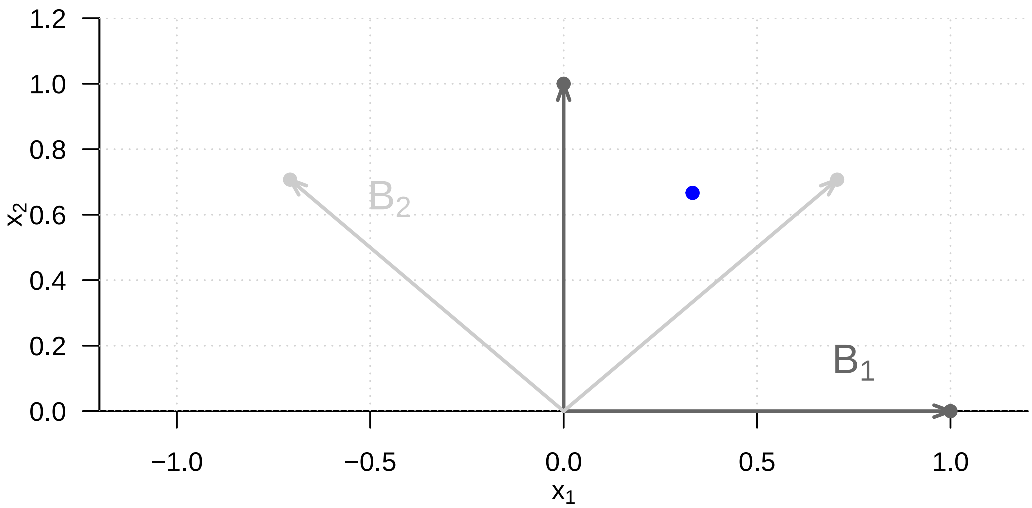

Figure 8.8 Two bases of $\mathbb{R}^2$

9 Matrices

Matrices are the words of the language of modern data analysis. An understanding of modern data-analytic methods and their implementation is not possible without a basic understanding of the concept of a matrix and knowledge of the fundamental matrix operations. Matrices can play very different roles. For example, matrices can represent data, experimental designs, and model parameters. In the context of linear algebra, matrices are used to represent linear mappings and vector spaces; here vectors are then understood as special matrices.

In this chapter, we introduce working with matrices while largely avoiding abstract terminology from linear algebra. We first introduce the concept of a matrix and then discuss the first fundamental matrix operations, namely matrix addition, matrix subtraction, scalar multiplication, and matrix transposition (Section 9.1 and Section 9.2). We then introduce the central concepts of matrix multiplication and matrix inversion (Section 9.3 and Section 9.4). With the matrix determinant, we discuss a first measure for describing matrices in Section 9.5. We close in Section 9.6 with an overview of particularly common matrices.

9.1 Definition

We begin with the definition of a matrix.

Definition 9.1. A matrix is a rectangular arrangement of numbers denoted as follows:

$$A := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} := (a_{ij})_{1\le i\le n,\, 1\le j\le m}. \tag{9.1}$$

•

Matrices consist of *rows* and *columns*. The matrix entries a_{ij} are indexed with a *row index* i and a *column index* j. For example, for

$$A := \begin{pmatrix} 2 & 7 & 5 & 2 \\ 8 & 2 & 5 & 6 \\ 6 & 4 & 0 & 9 \\ 9 & 2 & 1 & 2 \end{pmatrix}, \tag{9.2}$$

we have $a_{32} = 4$. The *size* or *dimension* of a matrix is determined by the number of its rows $n \in \mathbb{N}$ and columns $m \in \mathbb{N}$. Matrices with $n = m$ are called *square matrices*.

In what follows, we only need matrices with real entries, that is, $a_{ij} \in \mathbb{R}$ for all $i = 1, ..., n$ and $j = 1, ..., m$. We call matrices with real entries *real matrices* and denote the set of real matrices with n rows and m columns by $\mathbb{R}^{n \times m}$. From the expression

$$A \in \mathbb{R}^{n \times m} \tag{9.3}$$

we can therefore see that A is a real matrix with n rows and m columns. We identify the set $\mathbb{R}^{1 \times 1}$ with the set $\mathbb{R}$ and the set $\mathbb{R}^{n \times 1}$ with the set $\mathbb{R}^n$. Thus, real matrices with one column and n rows correspond to n-dimensional real vectors, and real matrices with one column and one row correspond to real numbers.

Defining matrices in R

In **R**, matrices are defined by transforming **R** vectors into the representation of a mathematical matrix with the `matrix()` function. The entries of an **R** vector are distributed over the matrix according to their total number and the specified number of rows `nrow`. For example, if we want to define the matrix

$$A := \begin{pmatrix} 2 & 3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \tag{9.4}$$

in **R**, we obtain

```r
# columnwise definition of A (R default)
A = matrix(c(2,1,3,6,0,5), nrow = 2)
print(A)
```

```
     [,1] [,2] [,3]
[1,]    2    3    0
[2,]    1    6    5
```

By default, **R** follows a so-called *column-major order*, that is, the elements of the **R** vector `c(2,1,3,6,0,5)` are transferred one after another from top to bottom into the columns of the matrix from left to right. A somewhat clearer connection between the visual layout of the **R** code and the resulting matrix is obtained by formatting the **R** vector with line breaks according to the intended matrix layout and then changing the column-major order to *row-major order* with the argument `byrow = TRUE`. Then the first row of the matrix is filled from left to right with the elements of the **R** vector, then the second row, and so on until all elements of the vector are distributed over the matrix.

```r
# rowwise definition of A (byrow = TRUE)
A = matrix(c(2,3,0,
             1,6,5),
             nrow = 2,
             byrow = TRUE)
print(A)
```

```
     [,1] [,2] [,3]
[1,]    2    3    0
[2,]    1    6    5
```

```r
# rowwise definition of B
B = matrix(c(4,1,0,
            -4,2,0),
            nrow = 2,
            byrow = TRUE)
print(B)
```

```
     [,1] [,2] [,3]
[1,]    4    1    0
[2,]   -4    2    0
```

9.2 Basic matrix operations

One can compute with matrices. The following matrix operations are fundamental:

- the addition of matrices of the same size, called *matrix addition*,
- the subtraction of matrices of the same size, called *matrix subtraction*,
- the multiplication of a matrix by a scalar, called *scalar multiplication*,
- the interchange of the rows and columns of a matrix, called *matrix transposition*.

In what follows, we introduce these operations in operator form, that is, as functions. This serves in particular to emphasize, for each operation, by means of its domain what kind of objects the respective operation takes, and by means of its codomain what kind of object the result of the respective operation is.

9.2.1 Matrix addition

Definition 9.2. Let $A, B \in \mathbb{R}^{n\times m}$. Then the *addition* of A and B is defined as the mapping

$$+ : \mathbb{R}^{n\times m} \times \mathbb{R}^{n\times m} \to \mathbb{R}^{n\times m},\ (A,B) \mapsto +(A,B) := A + B \tag{9.5}$$

with

$$\begin{aligned} A + B &= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{pmatrix} \\ &:= \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1m}+b_{1m} \\ a_{21}+b_{21} & a_{22}+b_{22} & \cdots & a_{2m}+b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}+b_{n1} & a_{n2}+b_{n2} & \cdots & a_{nm}+b_{nm} \end{pmatrix}. \end{aligned} \tag{9.6}$$

•

In particular, the definition of matrix addition specifies that only matrices of the same size can be added and that the operation of matrix addition is defined elementwise.

Example

Let $A, B \in \mathbb{R}^{2\times 3}$ be defined as

$$A := \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \text{ and } B := \begin{pmatrix} 4 & 1 & 0 \\ -4 & 2 & 0 \end{pmatrix}. \tag{9.7}$$

Because A and B have the same size, we can add them:

$$\begin{aligned} C = A + B &= \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} + \begin{pmatrix} 4 & 1 & 0 \\ -4 & 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 2+4 & -3+1 & 0+0 \\ 1-4 & 6+2 & 5+0 \end{pmatrix} \\ &= \begin{pmatrix} 6 & -2 & 0 \\ -3 & 8 & 5 \end{pmatrix}. \end{aligned} \tag{9.8}$$

In **R**, one carries out the above calculation as follows.

```
# definition
A = matrix(c(2, -3, 0,
             1,  6, 5),
             nrow  = 2,
             byrow = TRUE)
B = matrix(c( 4, 1, 0,
             -4, 2, 0),
              nrow = 2,
             byrow = TRUE)

# addition
C = A + B
print(C)
```

```
     [,1] [,2] [,3]
[1,]    6   -2    0
[2,]   -3    8    5
```

9.2.2 Matrix subtraction

The subtraction of matrices of the same size is defined analogously to addition.

Definition 9.3 (Matrix subtraction)**.** Let $A, B \in \mathbb{R}^{n\times m}$. Then the *subtraction* of A and B is defined as the mapping

$$-: \mathbb{R}^{n\times m} \times \mathbb{R}^{n\times m} \to \mathbb{R}^{n\times m},\ (A,B) \mapsto -(A,B) := A - B \tag{9.9}$$

with

$$\begin{aligned} A - B &= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{pmatrix} \\ &:= \begin{pmatrix} a_{11} - b_{11} & a_{12} - b_{12} & \cdots & a_{1m} - b_{1m} \\ a_{21} - b_{21} & a_{22} - b_{22} & \cdots & a_{2m} - b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} - b_{n1} & a_{n2} - b_{n2} & \cdots & a_{nm} - b_{nm} \end{pmatrix}. \end{aligned} \tag{9.10}$$

•

As with matrix addition, the definition of matrix subtraction specifies that only matrices of the same size can be subtracted from one another and that the subtraction of two equally sized matrices is defined elementwise.

Example

We can also subtract the matrices A and B defined in the example on matrix addition from one another:

$$\begin{aligned} D = A - B &= \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} - \begin{pmatrix} 4 & 1 & 0 \\ -4 & 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 2-4 & -3-1 & 0-0 \\ 1+4 & 6-2 & 5-0 \end{pmatrix} \\ &= \begin{pmatrix} -2 & -4 & 0 \\ 5 & 4 & 5 \end{pmatrix}. \end{aligned} \tag{9.11}$$

In **R**, one carries out this calculation as follows.

```
# subtraction
D = A - B
print(D)
```

```
     [,1] [,2] [,3]
[1,]   -2   -4    0
[2,]    5    4    5
```

9.2.3 Scalar multiplication

The *scalar multiplication* of a matrix denotes the multiplication of a scalar by a matrix.

Definition 9.4 (Scalar multiplication)**.** Let $c \in \mathbb{R}$ be a scalar and let $A \in \mathbb{R}^{n\times m}$. Then the *scalar multiplication* of c and A is defined as the mapping

$$\cdot : \mathbb{R} \times \mathbb{R}^{n\times m} \to \mathbb{R}^{n\times m},\ (c, A) \mapsto \cdot(c, A) := cA \tag{9.12}$$

with

$$cA = c\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} := \begin{pmatrix} ca_{11} & ca_{12} & \cdots & ca_{1m} \\ ca_{21} & ca_{22} & \cdots & ca_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{nm} \end{pmatrix}. \tag{9.13}$$

•

With this definition, scalar multiplication is therefore defined elementwise.

Example

Let $c := -3$ and let $A \in \mathbb{R}^{4\times 3}$ be defined as

$$A := \begin{pmatrix} 3 & 1 & 1 \\ 5 & 2 & 5 \\ 2 & 7 & 1 \\ 3 & 4 & 2 \end{pmatrix}. \tag{9.14}$$

Then

$$B := cA = -3\begin{pmatrix} 3 & 1 & 1 \\ 5 & 2 & 5 \\ 2 & 7 & 1 \\ 3 & 4 & 2 \end{pmatrix} = \begin{pmatrix} -3\cdot 3 & -3\cdot 1 & -3\cdot 1 \\ -3\cdot 5 & -3\cdot 2 & -3\cdot 5 \\ -3\cdot 2 & -3\cdot 7 & -3\cdot 1 \\ -3\cdot 3 & -3\cdot 4 & -3\cdot 2 \end{pmatrix} = \begin{pmatrix} -9 & -3 & -3 \\ -15 & -6 & -15 \\ -6 & -21 & -3 \\ -9 & -12 & -6 \end{pmatrix}. \tag{9.15}$$

In **R**, one carries out this scalar multiplication as follows.

```
# definitions
A = matrix(c(3,1,1,
             5,2,5,
             2,7,1,
             3,4,2),
           nrow = 4,
           byrow = TRUE)
c = -3

# scalar multiplication
B = c*A
print(B)
```

```
     [,1] [,2] [,3]
[1,]   -9   -3   -3
[2,]  -15   -6  -15
[3,]   -6  -21   -3
[4,]   -9  -12   -6
```

With the help of the definition of matrix addition and scalar multiplication, it is possible to define a vector space whose elements are the real matrices. In particular, this definition also specifies the arithmetic rules for working with matrix addition and scalar multiplication.

Theorem 9.1 (Vector space of real-valued matrices). *The triple* $(\mathbb{R}^{n\times m}, +, \cdot)$ *with the matrix addition and scalar multiplication defined above is a vector space. In particular, for* $A, B, C \in \mathbb{R}^{n\times m}$ *and* $r, s, t \in \mathbb{R}$, *the following arithmetic rules therefore hold:*

(1) Commutativity of addition: $A + B = B + A$.
(2) Associativity of addition: $(A + B) + C = A + (B + C)$.
(3) Existence of an identity element for addition: $\exists\, 0 \in \mathbb{R}^{n\times m}$ *with* $A + 0 = 0 + A = A$.
(4) Existence of inverse elements for addition: $\forall A\, \exists - A \in \mathbb{R}^{n\times m}$ *with* $A + (-A) = 0$.
(5) Existence of an identity element for scalar multiplication: $\exists\, 1 \in \mathbb{R}$ *with* $1 \cdot A = A$.
(6) Associativity of scalar multiplication: $r \cdot (s \cdot A) = (r \cdot s) \cdot A$.
(7) Distributivity with respect to matrix addition: $r \cdot (A + B) = r \cdot A + r \cdot B$.
(8) Distributivity with respect to scalar addition: $(r + s) \cdot A = r \cdot A + s \cdot A$.

∘

We omit a proof, which follows directly, with some notational effort, from the elementwise character of matrix addition and scalar multiplication as well as the arithmetic rules known from working with real numbers. The identity element of addition mentioned in the theorem is called the *zero matrix*; we will introduce a general notation for it later. The inverse elements of addition are given by

$$-A := (-a_{ij})_{1\le i\le n, 1\le j\le m} \tag{9.16}$$

and make it possible to view matrix subtraction as a special case of matrix addition.

9.2.4 Matrix transposition

Another frequently occurring basic matrix operation is the interchange of the row and column arrangement of a matrix, called *matrix transposition.*

Definition 9.5 (Matrix transposition). Let $A \in \mathbb{R}^{n\times m}$. Then the *transposition* of A is defined as the mapping

$$\cdot^T : \mathbb{R}^{n\times m} \to \mathbb{R}^{m\times n},\ A \mapsto \cdot^T(A) := A^T \tag{9.17}$$

with

$$A^T = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}^T := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \cdots & a_{nm} \end{pmatrix}. \tag{9.18}$$

•

Thus, for $A \in \mathbb{R}^{n\times m}$, we always have $A^T \in \mathbb{R}^{m\times n}$. Furthermore, the following arithmetic rules of matrix transposition hold, as one can see from examples:

(1) For $A \in \mathbb{R}^{1\times 1}$,

$$A^T = A. \tag{9.19}$$

(2) We have

$$\left(A^T\right)^T = A. \tag{9.20}$$

(3) We have

$$(a_{ii})_{1\le i\le \min(n,m)} = (a_{ii})^T_{1\le i\le \min(n,m)}. \tag{9.21}$$

The latter property of transposition states that the elements on the main diagonal of a matrix remain unchanged under transposition.

Example

Let $A \in \mathbb{R}^{2\times 3}$ be defined by

$$A := \begin{pmatrix} 2 & 3 & 0 \\ 1 & 6 & 5 \end{pmatrix}, \tag{9.22}$$

Then $A^T \in \mathbb{R}^{3\times 2}$ and, specifically,

$$A^T := \begin{pmatrix} 2 & 1 \\ 3 & 6 \\ 0 & 5 \end{pmatrix}. \tag{9.23}$$

Furthermore, apparently $\min(m,n) = 2$, and consequently

$$(a_{11}) = (a_{11})^T \text{ and } (a_{22}) = (a_{22})^T. \tag{9.24}$$

In **R**, one transposes a matrix as follows.

```
# definition
A = matrix(c(2,3,0,
             1,6,5),
           nrow = 2,
           byrow = TRUE)
print(A)
```

```
     [,1] [,2] [,3]
[1,]    2    3    0
[2,]    1    6    5
```

```
# transposition
AT = t(A)
print(AT)
```

```
     [,1] [,2]
[1,]    2    1
[2,]    3    6
[3,]    0    5
```

Finally, in connection with matrix addition, matrix subtraction, and scalar multiplication, the following arithmetic rules hold, as one can see from examples:

(1) For $A, B \in \mathbb{R}^{n\times m}$,

$$(A+B)^T = A^T + B^T. \tag{9.25}$$

(2) For $A, B \in \mathbb{R}^{n\times m}$,

$$(A-B)^T = A^T - B^T. \tag{9.26}$$

(3) For $c \in \mathbb{R}$ and $A \in \mathbb{R}^{n\times m}$,

$$(cA)^T = cA^T. \tag{9.27}$$

9.3 Matrix multiplication

Matrix multiplication is the central operation when computing with matrices. It is defined as follows.

Definition 9.6 (Matrix multiplication)**.** Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times k}$. Then the *matrix multiplication* of A and B is defined as the mapping

$$\cdot : \mathbb{R}^{n\times m} \times \mathbb{R}^{m\times k} \to \mathbb{R}^{n\times k},\ (A,B) \mapsto \cdot(A,B) := AB \tag{9.28}$$

with

$$\begin{aligned} AB &= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1k} \\ b_{21} & b_{22} & \cdots & b_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mk} \end{pmatrix} \\ &:= \begin{pmatrix} \sum_{i=1}^m a_{1i}b_{i1} & \sum_{i=1}^m a_{1i}b_{i2} & \cdots & \sum_{i=1}^m a_{1i}b_{ik} \\ \sum_{i=1}^m a_{2i}b_{i1} & \sum_{i=1}^m a_{2i}b_{i2} & \cdots & \sum_{i=1}^m a_{2i}b_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^m a_{ni}b_{i1} & \sum_{i=1}^m a_{ni}b_{i2} & \cdots & \sum_{i=1}^m a_{ni}b_{ik} \end{pmatrix} \\ &= \left(\sum_{i=1}^m a_{ji}b_{il}\right)_{1\le j\le n, 1\le l\le k} \end{aligned} \tag{9.29}$$

•

The matrix product AB is therefore defined only if A has exactly as many columns as B has rows. Informally, the following mnemonic applies to the involved matrix sizes:

$$(n \times m)(m \times k) = (n \times k). \tag{9.30}$$

The entry $(AB)_{ij}$ in AB corresponds to the sum of the multiplied ith row of A and jth column of B. To compute $(AB)_{ij}$ for $i = 1, ..., n$ and $j = 1, ..., k$, one can therefore proceed mentally as follows:

(1) Place the transpose of the ith row of A over the jth column of B.
(2) Because A has exactly m columns and B has exactly m rows, each element of the row from A has a corresponding element in the column of B.
(3) Multiply the corresponding elements with one another.
(4) The sum of these products is then the entry with index ij in AB.

Example

Let $A \in \mathbb{R}^{2\times 3}$ and $B \in \mathbb{R}^{3\times 2}$ be defined as

$$A := \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \text{ and } B := \begin{pmatrix} 4 & 2 \\ -1 & 0 \\ 1 & 3 \end{pmatrix}. \tag{9.31}$$

We want to compute $C := AB$ and $D := BA$. With $n = 2, m = 3$, and $k = 2$, we already know that $C \in \mathbb{R}^{2\times 2}$ and $D \in \mathbb{R}^{3\times 3}$, because

$$(2 \times 3)(3 \times 2) = (2 \times 2) \tag{9.32}$$

and

$$(3 \times 2)(2 \times 3) = (3 \times 3). \tag{9.33}$$

Thus here surely $AB \neq BA$. For C, we obtain

$$\begin{aligned}
C &= AB \\
&= \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ -1 & 0 \\ 1 & 3 \end{pmatrix} \\
&= \begin{pmatrix} 2 \cdot 4 + (-3) \cdot (-1) + 0 \cdot 1 & 2 \cdot 2 + (-3) \cdot 0 + 0 \cdot 3 \\ 1 \cdot 4 + 6 \cdot (-1) + 5 \cdot 1 & 1 \cdot 2 + 6 \cdot 0 + 5 \cdot 3 \end{pmatrix} \\
&= \begin{pmatrix} 8 + 3 + 0 & 4 + 0 + 0 \\ 4 - 6 + 5 & 2 + 0 + 15 \end{pmatrix} \\
&= \begin{pmatrix} 11 & 4 \\ 3 & 17 \end{pmatrix}.
\end{aligned} \tag{9.34}$$

In **R**, the `%*%` operator is used for matrix multiplication.

```
# definitions
A = matrix(c(2,-3,0,
             1, 6,5),
           nrow  = 2,
           byrow = TRUE)
B = matrix(c( 4,2,
             -1,0,
              1,3),
           nrow  = 3,
           byrow = TRUE)

# matrix multiplication
C = A %*% B
print(C)
```

```
     [,1] [,2]
[1,]   11    4
[2,]    3   17
```

For D, we further obtain

$$\begin{aligned}
D &= BA \\
&= \begin{pmatrix} 4 & 2 \\ -1 & 0 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \\
&= \begin{pmatrix} 4 \cdot 2 + 2 \cdot 1 & 4 \cdot (-3) + 2 \cdot 6 & 4 \cdot 0 + 2 \cdot 5 \\ (-1) \cdot 2 + 0 \cdot 1 & (-1) \cdot (-3) + 0 \cdot 6 & (-1) \cdot 0 + 0 \cdot 5 \\ 1 \cdot 2 + 3 \cdot 1 & 1 \cdot (-3) + 3 \cdot 6 & 1 \cdot 0 + 3 \cdot 5 \end{pmatrix} \\
&= \begin{pmatrix} 8 + 2 & -12 + 12 & 0 + 5 \\ -2 + 0 & 3 + 0 & 0 + 0 \\ 2 + 3 & -3 + 18 & 0 + 15 \end{pmatrix} \\
&= \begin{pmatrix} 10 & 0 & 10 \\ -2 & 3 & 0 \\ 5 & 15 & 15 \end{pmatrix}
\end{aligned} \tag{9.35}$$

In **R**, one checks this calculation as follows.

```
# definitions
A = matrix(c(2,-3,0,
             1, 6,5),
           nrow  = 2,
           byrow = TRUE)
B = matrix(c( 4,2,
             -1,0,
              1,3),
           nrow  = 3,
           byrow = TRUE)

# matrix multiplication
D = B %*% A
print(D)
```

```
     [,1] [,2] [,3]
[1,]   10    0   10
[2,]   -2    3    0
[3,]    5   15   15
```

However, if a matrix multiplication is not defined because of incompatible matrix sizes, then it also cannot be evaluated numerically.

```
# example of an undefined matrix multiplication
E = t(A) %*% B      # (3 x 2)(3 x 2)
```

```
Error in `t(A) %*% B`:
! nicht passende Argumente
```

The following theorem, which we do not prove, establishes the relation between the inner product of two vectors and the multiplication of two matrices. This essentially follows from the identification of $\mathbb{R}^n$ and $\mathbb{R}^{n\times 1}$ and from the fact that, by definition, the entry $(AB)_{ij}$ in the product of $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times k}$ corresponds to the vector inner product of the ith column of A^T and the jth column of B.

Theorem 9.2 (Matrix multiplication and vector inner product)**.** *Let $x, y \in \mathbb{R}^n$. Then*

$$\langle x, y\rangle = x^T y. \tag{9.36}$$

Furthermore, for $A \in \mathbb{R}^{n\times m}$ and $i = 1, ..., n$, let

$$\bar{a}_i := (a_{ij})_{1\le j\le m} \in \mathbb{R}^m \tag{9.37}$$

be the columns of A^T, and for $B \in \mathbb{R}^{m\times k}$ and $j = 1, ..., k$, let

$$\bar{b}_j := (b_{ij})_{1\le i\le m} \in \mathbb{R}^m \tag{9.38}$$

be the columns of B, that is,

$$A^T = \begin{pmatrix} \bar{a}_1 & \bar{a}_2 & \cdots & \bar{a}_n \end{pmatrix} \in \mathbb{R}^{m\times n} \text{ and } B = \begin{pmatrix} \bar{b}_1 & \bar{b}_2 & \cdots & \bar{b}_k \end{pmatrix} \in \mathbb{R}^{m\times k}. \tag{9.39}$$

Then

$$AB = \left(\langle \bar{a}_i, \bar{b}_j\rangle\right)_{1\le i\le n, 1\le j\le k}. \tag{9.40}$$

∘

9.3.1 Arithmetic rules for matrix multiplication

In the following, we collect a few basic arithmetic rules for matrix multiplication, especially also in combination with other matrix operations.

For proofs of the following two theorems on associativity and distributivity, which essentially follow from the corresponding arithmetic rules for sums and products of real numbers, we refer to the advanced literature.

Theorem 9.3 (Associativity)**.** *Let $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{m\times k}$, $C \in \mathbb{R}^{k\times p}$, and $c \in \mathbb{R}$. Then:*

(1) Matrix multiplication is associative, that is,

$$A(BC) = (AB)C. \tag{9.41}$$

(2) The combination of matrix multiplication and scalar multiplication is associative,

$$c(AB) = (cA)B = A(cB). \tag{9.42}$$

∘

The associativity of matrix multiplication and scalar multiplication can be seen easily by considering the j,lth element of $c(AB)$, $(cA)B$, and $A(cB)$:

$$c\left(\sum_{i=1}^{m} a_{ji}b_{il}\right) = \sum_{i=1}^{m}\left(ca_{ji}\right)b_{il} = \sum_{i=1}^{m} a_{ji}\left(cb_{il}\right). \tag{9.43}$$

Theorem 9.4 (Distributivity)**.** *Let $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{n\times m}$, and $C \in \mathbb{R}^{m\times p}$. Then*

$$(A + B)C = AC + BC \tag{9.44}$$

and

$$C^T(A + B) = C^T A + C^T B \tag{9.45}$$

∘

In contrast to the commutativity of multiplication of real numbers, matrix multiplication is in general not commutative.

Theorem 9.5 (Noncommutativity)**.** *Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times p}$. In general,*

$$AB \neq BA. \tag{9.46}$$

∘

Proof. If $p \neq n$, then BA is not defined, so we consider only the case $p = n$. We show by giving a counterexample with $A, B \in \mathbb{R}^{2\times 2}$ that in general $AB = BA$ does *not* hold. Let

$$A := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \text{ and } B := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{9.47}$$

Then

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = BA. \tag{9.48}$$

□

Theorem 9.6 (Combination of matrix multiplication and transposition). *Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times k}$. Then*

$$(AB)^T = B^T A^T. \tag{9.49}$$

∘

Proof. A proof is obtained as follows:

$$\begin{aligned}
(AB)^T &= \left(\left(\sum_{i=1}^n a_{ji}b_{il}\right)_{1\le j\le m, 1\le l\le k}\right)^T \\
&= \left(\sum_{i=1}^n a_{li}b_{ij}\right)_{1\le j\le k, 1\le l\le m} \\
&= \left(\sum_{i=1}^n b_{ij}a_{li}\right)_{1\le j\le k, 1\le l\le m} \\
&= B^T A^T.
\end{aligned} \tag{9.50}$$

□

9.4 Matrix inversion

To motivate the concept of the inverse matrix, we first consider the problem of *solving a system of linear equations*. To this end, let $A \in \mathbb{R}^{n\times n}$, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}^n$, and suppose that

$$Ax = b. \tag{9.51}$$

Let A and b be known, and let x be unknown. Specifically, for example, let

$$A := \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } b := \begin{pmatrix} 5 \\ 11 \end{pmatrix}. \tag{9.52}$$

Then the following system of linear equations with two equations and two unknowns is given:

$$Ax = b \Leftrightarrow \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix} \Leftrightarrow \begin{matrix} 1x_1 + 2x_2 & = 5 \\ 3x_1 + 4x_2 & = 11 \end{matrix}. \tag{9.53}$$

The goal of solving systems of linear equations is, as is well known, to find those x for which the system is satisfied. To introduce the concept of the inverse matrix of A in this context, we simplify the situation further. We assume that $A = a$ is a 1×1 matrix, that is, a scalar, and likewise that x and b are scalars, so that for $a, x, b \in \mathbb{R}$ we have the equation

$$ax = b \tag{9.54}$$

To solve this equation for x, one would naturally multiply both sides of the equation by the *multiplicative inverse* of a, where the *multiplicative inverse* of a denotes the value which, when multiplied by a, yields 1. As is well known, this is given by

$$a^{-1} = \frac{1}{a} \tag{9.55}$$

Then

$$ax = b \Leftrightarrow a^{-1}ax = a^{-1}b \Leftrightarrow 1 \cdot x = a^{-1}b \Leftrightarrow x = \frac{b}{a}. \tag{9.56}$$

For example, quite concretely,

$$2x = 6 \Leftrightarrow 2^{-1}2x = 2^{-1}6 \Leftrightarrow \frac{1}{2}2x = \frac{1}{2}6 \Leftrightarrow x = 3. \tag{9.57}$$

Analogously to the case in which all matrices in $Ax = b$ are scalars, in the case of a system of linear equations one would like to be able to multiply both sides of the equation by the *multiplicative inverse* A^{-1} of A, so that an equation of the form

$$A^{-1}A = \text{"1"}. \tag{9.58}$$

results. Then one would have

$$Ax = b \Leftrightarrow A^{-1}Ax = A^{-1}b \Leftrightarrow x = A^{-1}b. \tag{9.59}$$

This intuitive idea of the multiplicative inverse of a matrix A is formalized below under the concept of the *inverse matrix.* To this end, we first need the concept of the *identity matrix.*

Definition 9.7 (Identity matrix)**.** The matrix

$$I_n := (a_{ij})_{1\le i\le n, 1\le j\le n} \in \mathbb{R}^{n\times n} := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \tag{9.60}$$

with $a_{ij} = 1$ for $i = j$ and $a_{ij} = 0$ for $i \neq j$ is called the *n-dimensional identity matrix.*

•

In **R**, I_n is generated with the command `diag(n)`. For matrix multiplication, the identity matrix is the analogue of the 1 in the multiplication of real numbers. This is the statement of the following theorem.

Theorem 9.7 (Identity element of matrix multiplication)**.** *I_n is the identity element of matrix multiplication, that is, for $A \in \mathbb{R}^{n\times m}$ we have*

$$I_nA = A \text{ and } AI_m = A. \tag{9.61}$$

∘

Proof. Let $B = (b_{ij}) = I_nA \in \mathbb{R}^{n\times m}$. Then, for all $1 \le i \le n$ and all $1 \le j \le m$,

$$b_{ij} = 0 \cdot a_{1j} + 0 \cdot a_{2j} + \cdots + 0 \cdot a_{i-1,j} + 1 \cdot a_{ij} + \cdots + 0 \cdot a_{i+1,j} + 0 \cdot a_{nj} = a_{ij}. \tag{9.62}$$

The proof for AI_m is analogous. □

With the concept of the identity matrix, we can now define the concepts of the inverse matrix and the invertible matrix:

Definition 9.8 (Invertible matrix and inverse matrix)**.** A square matrix $A \in \mathbb{R}^{n\times n}$ is called *invertible* if there exists a square matrix $A^{-1} \in \mathbb{R}^{n\times n}$ such that

$$A^{-1}A = AA^{-1} = I_n \tag{9.63}$$

The matrix A^{-1} is called the *inverse matrix of* A.

•

Note that the concepts of the inverse matrix and invertibility refer *only* to square matrices. In particular, square matrices can be invertible, but need not be (systems of linear equations can therefore have solutions, but need not). Non-invertible matrices are also called *singular* matrices, and invertible matrices are sometimes called *nonsingular* matrices. Finally, note that Definition 9.8 merely states what an inverse matrix is, but not how to compute it.

Example of an invertible matrix

The matrix

$$A := \begin{pmatrix} 2.0 & 1.0 \\ 3.0 & 4.0 \end{pmatrix} \tag{9.64}$$

is invertible, and its inverse matrix is given by

$$A^{-1} = \begin{pmatrix} 0.8 & -0.2 \\ -0.6 & 0.4 \end{pmatrix}, \tag{9.65}$$

because

$$\begin{pmatrix} 2.0 & 1.0 \\ 3.0 & 4.0 \end{pmatrix}\begin{pmatrix} 0.8 & -0.2 \\ -0.6 & 0.4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.8 & -0.2 \\ -0.6 & 0.4 \end{pmatrix}\begin{pmatrix} 2.0 & 1.0 \\ 3.0 & 4.0 \end{pmatrix}, \tag{9.66}$$

as can be verified by direct calculation.

Example of a non-invertible matrix

The matrix

$$B := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{9.67}$$

is not invertible, because if B were invertible, then there would exist

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{9.68}$$

with

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{9.69}$$

But this would mean that $0 = 1$ in $\mathbb{R}$, which is a contradiction. Therefore, B cannot be invertible.

On computing inverse matrices

2×2 up to about 5×5 matrices can in principle be inverted by hand; linear algebra provides various methods for this. We omit an introduction to matrix inversion by hand here, since matrices are inverted numerically by default in applications. Numerical matrix inversion is also a large field of research in numerical mathematics, which provides a variety of algorithms for this purpose. In **R**, matrices are inverted by default with the function `solve()`, in analogy to solving systems of linear equations. For the above example of an invertible matrix, this yields the following **R** code.

```
# definition
A = matrix(c(2,1,
             3,4),
           nrow  = 2,
           byrow = TRUE)

# computing A^{-1}
print(solve(A))
```

```
     [,1] [,2]
[1,]  0.8 -0.2
[2,] -0.6  0.4
```

```
# checking the properties of an inverse matrix
print(solve(A) %*% A)
```

```
     [,1] [,2]
[1,]    1    0
[2,]    0    1
```

```
# the reverse calculation yields small rounding errors
print(A %*% solve(A))
```

```
     [,1]          [,2]
[1,]    1 -5.551115e-17
[2,]    0  1.000000e+00
```

Non-invertible matrices are, of course, also numerically non-invertible, as the following error message in **R** demonstrates for the above example of a non-invertible matrix.

```
B = matrix(c(1,0,
             0,0),
           nrow  = 2,
           byrow = TRUE)
solve(B)
```

```
Error in `solve.default()`:
! Lapackroutine dgesv: System ist genau singulär: U[2,2] = 0
```

9.5 Determinants

The determinant is a versatile measure of a square matrix. For understanding *eigenanalysis* and *matrix decomposition*, the concept of the determinant is fundamental in the context of the *characteristic polynomial*.

In general, a determinant is a nonlinear mapping of the form

$$|\cdot| : \mathbb{R}^{n\times n} \to \mathbb{R}, A \mapsto |A|, \tag{9.70}$$

that is, a determinant assigns the real number $|A|$ to a square matrix A. The number $|A|$ is determined recursively by the following definition.

Definition 9.9 (Determinant)**.** For $A = (a_{ij})_{1 \le i,j \le n} \in \mathbb{R}^{n \times n}$ with $n > 1$, let $A_{ij} \in \mathbb{R}^{n-1 \times n-1}$ be the matrix obtained from A by removing the ith row and the jth column. Then the number

$$|A| := a_{11} \qquad \text{for } n = 1 \tag{9.71}$$

$$|A| := \sum_{j=1}^{n} a_{1j}(-1)^{1+j} \det\left(A_{1j}\right) \text{ for } n > 1 \tag{9.72}$$

is called the *determinant of* A.

•

The definition thus successively reduces the determination of the determinant of a square matrix, by deleting rows and columns, to the determinant of a 1×1 matrix, which is given by its only element. For

$$A := \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \in \mathbb{R}^{3 \times 3} \tag{9.73}$$

we obtain, for example, the following matrices of the form $A_{ij} \in \mathbb{R}^{3-1 \times 3-1}$:

$$A_{11} = \begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix}, A_{12} = \begin{pmatrix} 4 & 6 \\ 7 & 9 \end{pmatrix}, A_{21} = \begin{pmatrix} 2 & 3 \\ 8 & 9 \end{pmatrix}, A_{22} = \begin{pmatrix} 1 & 3 \\ 7 & 9 \end{pmatrix}. \tag{9.74}$$

For computing determinants of two- and three-dimensional square matrices, there are direct, non-recursive arithmetic rules, which are recorded in the following theorem.

Theorem 9.8 (Determinants of two- and three-dimensional matrices)**.**
Let $A = (a_{ij})_{1 \le i,j \le 2} \in \mathbb{R}^{2 \times 2}$. *Then*

$$|A| = a_{11}a_{22} - a_{12}a_{21}. \tag{9.75}$$

Let $A = (a_{ij})_{1 \le i,j \le 3} \in \mathbb{R}^{3 \times 3}$. *Then*

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31}. \tag{9.76}$$

∘

Proof. For $A \in \mathbb{R}^{2 \times 2}$, by definition

$$\begin{aligned} |A| &= \sum_{j=1}^{n} a_{1j}(-1)^{1+j}|A_{1j}| \\ &= a_{11}(-1)^{1+1}|A_{11}| + a_{12}(-1)^{1+2}|A_{12}| \\ &= a_{11}|(a_{22})| - a_{12}|(a_{21})| \\ &= a_{11}a_{22} - a_{12}a_{21}. \end{aligned} \tag{9.77}$$

For $A \in \mathbb{R}^{3\times 3}$, by definition and with the formula for determinants of 2×2 matrices,

$$
\begin{aligned}
|A| &= \sum_{j=1}^{n} a_{1j}(-1)^{1+j}|A_{1j}| \\
&= a_{11}(-1)^{1+1}|A_{11}| + a_{12}(-1)^{1+2}|A_{12}| + a_{13}(-1)^{1+3}|A_{13}| \\
&= a_{11}|A_{11}| - a_{12}|A_{12}| + a_{13}|A_{13}| \\
&= a_{11}\left|\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}\right| - a_{12}\left|\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix}\right| + a_{13}\left|\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}\right| \\
&= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\
&= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} \\
&= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31}.
\end{aligned}
\tag{9.78}
$$

□

For determining the determinants of 2×2 and 3×3 matrices, the so-called *rule of Sarrus* therefore applies:

> "sum of the products on the diagonals minus sum of the products on the counterdiagonals."

For 3×3 matrices, the mnemonic refers to the scheme

$$
\left(\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}\right). \tag{9.79}
$$

Examples for determinants of 2×2 and 3×3 matrices

Let

$$
A := \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}, B := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } C := \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \tag{9.80}
$$

Then

$$
|A| = 2\cdot 4 - 1\cdot 3 = 8 - 3 = 5 \tag{9.81}
$$

and

$$
|B| = 1\cdot 0 - 0\cdot 0 = 0 - 0 = 0 \tag{9.82}
$$

and

$$
|C| = 2\cdot 1\cdot 3 + 0\cdot 0\cdot 0 + 0\cdot 0\cdot 0 - 0\cdot 0\cdot 3 - 0\cdot 0\cdot 0 - 0\cdot 1\cdot 0 = 2\cdot 1\cdot 3 = 6. \tag{9.83}
$$

In **R**, one checks this with the `det()` function as follows.

```
# matrix definition and determinant calculation
A = matrix(c(2,1,
             3,4),
           nrow = 2,
           byrow = TRUE)
det(A)
```

```
[1] 5
```

```
# matrix definition and determinant calculation
B = matrix(c(1,0,
             0,0),
           nrow = 2,
           byrow = TRUE)
det(B)
```

```
[1] 0
```

```
# matrix definition and determinant calculation
C = matrix(c(2,0,0,
             0,1,0,
             0,0,3),
           nrow = 3,
           byrow = TRUE)
det(C)
```

```
[1] 6
```

There are numerous arithmetic rules for determinants in conjunction with matrix multiplication and matrix inversion. Without proof, we collect these in the following theorem.

Theorem 9.9 (Arithmetic rules for determinants)**.**

(Multiplication theorem for determinants.) For $A, B \in \mathbb{R}^{n\times n}$*,*

$$|AB| = |A||B|. \tag{9.84}$$

(Transposition.) For $A \in \mathbb{R}^{n\times n}$*,*

$$|A| = \left|A^T\right|. \tag{9.85}$$

(Inversion.) For an invertible matrix $A \in \mathbb{R}^{n\times n}$*,*

$$\left|A^{-1}\right| = \frac{1}{|A|}. \tag{9.86}$$

(Triangular matrices.) For matrices $A = (a_{ij})_{1\le i,j\le n} \in \mathbb{R}^{n\times n}$ *with* $a_{ij} = 0$ *for* $i > j$ *or* $a_{ij} = 0$ *for* $j > i$*,*

$$|A| = \prod_{i=1}^{n} a_{ii}. \tag{9.87}$$

∘

The following very deep theorem, which we do not want to prove completely, provides a way to determine from the determinant of a square matrix whether it is invertible.

Theorem 9.10. $A \in \mathbb{R}^{n\times n}$ *is invertible if and only if* $|A| \neq 0$*. Thus,*

$$A \text{ is invertible} \Leftrightarrow |A| \neq 0 \text{ and } A \text{ is not invertible} \Leftrightarrow |A| = 0. \tag{9.88}$$

∘

Proof. We only indicate a proof and show that invertibility of A implies that $|A|$ cannot be zero. Suppose, then, that A is invertible. Then there exists a matrix B with $AB = I_n$, and by the multiplication theorem for determinants it follows that

$$|AB| = |A||B| = |I_n| = 1. \tag{9.89}$$

Thus, $|A| = 0$ cannot hold, because otherwise $0 = 1$. □

Visual Intuition

The abstract concept of the determinant of a square matrix can be illustrated somewhat with the concept of a vector space. To this end, let $a_1, ..., a_n \in \mathbb{R}^n$ be the columns of $A \in \mathbb{R}^{n \times n}$. Then (as we do not prove) $|A|$ corresponds to the signed volume of the parallelotope spanned by $a_1, ..., a_n \in \mathbb{R}^n$. To illustrate this visually, we consider the matrices

$$A_1 = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, A_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, A_3 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \tag{9.90}$$

with the respective determinants

$$|A_1| = 3 \cdot 2 - 1 \cdot 1 = 5, \quad |A_2| = 2 \cdot 2 - 0 \cdot 0 = 4, \quad |A_3| = 2 \cdot 2 - 2 \cdot 2 = 0. \tag{9.91}$$

Figure 9.1 visualizes the corresponding intuition.

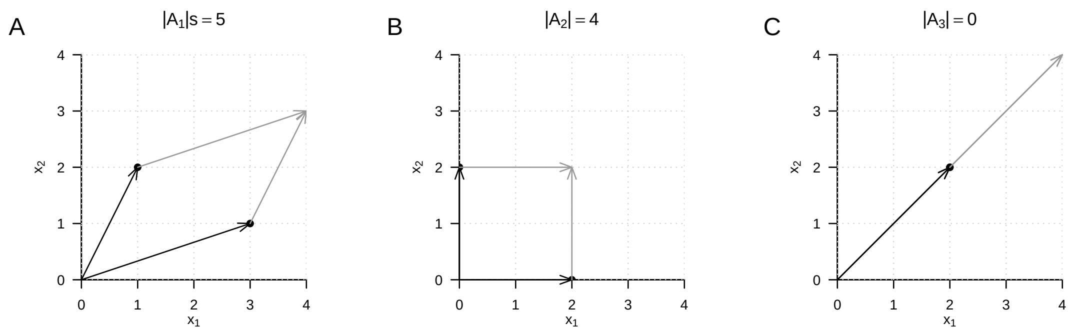

Figure 9.1 Determinants as parallelotope volumes.

9.6 Special matrices

In this section, we collect several frequently occurring types of matrices and their properties. For proofs of the vast majority of properties, we refer to the advanced literature.

9.6.1 Identity matrices

We have already encountered the identity matrix and the unit vectors. We collect them here once more in a joint definition.

Definition 9.10 (Identity matrix and unit vectors)**.** We denote the *identity matrix* by

$$I_n := (i_{jk})_{1 \le j \le n, 1 \le k \le n} \in \mathbb{R}^{n \times n} \text{ with } i_{jk} = 1 \text{ for } j = k \text{ and } i_{jk} = 0 \text{ for } j \neq k. \tag{9.92}$$

We denote the *unit vectors* $e_i, i = 1, ..., n$ by

$$e_i := (e_{i_j})_{1 \le j \le n} \in \mathbb{R}^n \text{ with } e_{i_j} = 1 \text{ for } i = j \text{ and } e_{i_j} = 0 \text{ for } i \neq j. \tag{9.93}$$

•

The identity matrix I_n consists only of zeros and diagonal elements equal to one; the unit vectors consist only of zeros and one single 1 in the respectively indexed component. We have

$$I_n = \begin{pmatrix} e_1 & \cdots & e_n \end{pmatrix} \in \mathbb{R}^{n \times n} \tag{9.94}$$

For $n = 3$, for example,

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{9.95}$$

Furthermore, as is well known, for the unit vectors and $1 \le i, j \le n$,

$$e_i^T e_j = 0 \text{ for } i \neq j, e_i^T e_i = 1 \text{ and } e_i^T v = v^T e_i = v_i \text{ for } v \in \mathbb{R}^n. \tag{9.96}$$

9.6.2 One matrices and zero matrices

Definition 9.11 (Zero matrices, zero vectors, one matrices, one vectors)**.** We denote *zero matrices* and *zero vectors* by

$$0_{nm} := (0)_{1 \le i \le n, 1 \le j \le m} \in \mathbb{R}^{n \times m} \text{ and } 0_n := (0)_{1 \le i \le n} \in \mathbb{R}^n. \tag{9.97}$$

We denote *one matrices* and *one vectors* by

$$1_{nm} := (1)_{1 \le i \le n, 1 \le j \le m} \in \mathbb{R}^{n \times m} \text{ and } 1_n := (1)_{1 \le i \le n} \in \mathbb{R}^n. \tag{9.98}$$

•

0_{nm} and 0_n therefore consist only of zeros, and 1_{nm} and 1_n consist only of ones. For example,

$$0_{32} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, 0_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, 1_{32} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \text{ and } 1_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \tag{9.99}$$

Furthermore, for example,

$$0_n 0_n^T = 0_{nn} \text{ and } 1_n 1_n^T = 1_{nn}, \tag{9.100}$$

as can be verified by direct calculation.

9.6.3 Diagonal matrices

Definition 9.12 (Diagonal matrix)**.** A matrix $D \in \mathbb{R}^{n \times m}$ is called a *diagonal matrix* if $d_{ij} = 0$ for $1 \le i \le n, 1 \le j \le m$ with $i \neq j$.

•

A square diagonal matrix $D \in \mathbb{R}^{n\times n}$ with diagonal elements $d_1, ..., d_n \in \mathbb{R}$ is also written as

$$D = \text{diag}(d_1, ..., d_n). \tag{9.101}$$

For example,

$$D := \text{diag}(1, 2, 3) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \tag{9.102}$$

and for $\sigma^2 \in \mathbb{R}$,

$$\Sigma = \text{diag}(\sigma^2, \sigma^2, \sigma^2) = \begin{pmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{pmatrix} = \sigma^2 I_3. \tag{9.103}$$

In the following theorem, we collect a few important properties of square diagonal matrices.

Theorem 9.11 (Properties of square diagonal matrices)**.**

(Determinant.) Let $D := diag(d_1, ..., d_n) \in \mathbb{R}^{n\times n}$ be a square diagonal matrix. Then

$$|D| = \prod_{i=1}^{n} d_i. \tag{9.104}$$

∘

9.6.4 Symmetric matrices

Symmetric matrices are square matrices that remain unchanged under transposition:

Definition 9.13. A matrix $S \in \mathbb{R}^{n\times n}$ is called *symmetric* if $S^T = S$.

•

An example of a symmetric matrix is

$$S := \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix}. \tag{9.105}$$

In the following theorem, we collect a few important properties of symmetric matrices.

Theorem 9.12 (Properties of symmetric matrices)**.**

(Summation.) Let $S_1 \in \mathbb{R}^{n\times n}$ and $S_2 \in \mathbb{R}^{n\times n}$ be symmetric matrices. Then

$$S_1 + S_2 = (S_1 + S_2)^T. \tag{9.106}$$

(Inverse.) Let S be an invertible symmetric matrix and let S^{-1} be its inverse. Then S^{-1} is also a symmetric matrix, that is,

$$\left(S^{-1}\right)^T = S^{-1}. \tag{9.107}$$

∘

9.6.5 Orthogonal matrices

Definition 9.14. A matrix $Q \in \mathbb{R}^{n\times n}$ is called *orthogonal* if $Q^TQ = I_n$.

•

The columns of an orthogonal matrix are therefore pairwise orthogonal. For

$$Q = \begin{pmatrix} q_1 & \cdots & q_n \end{pmatrix} \text{ with } q_i \in \mathbb{R}^n \text{ for } 1 \le i \le n, \tag{9.108}$$

we have

$$q_i^T q_j = 0 \text{ for } i \neq j \text{ and } q_i^T q_j = 1 \text{ for } i = j \text{ with } 1 \le i, j \le n. \tag{9.109}$$

Theorem 9.13 (Properties of orthogonal matrices)**.** *Let $Q \in \mathbb{R}^{n\times n}$ be an orthogonal matrix. Then the following properties of Q hold.*

(Inverse.) The inverse of Q is Q^T, that is,

$$Q^{-1} = Q^T. \tag{9.110}$$

(Transposition.) The rows of Q are orthonormal, that is,

$$QQ^T = I_n \tag{9.111}$$

∘

Proof. (Inverse.) Under the assumption that Q^{-1} exists, we have

$$Q^TQ = I_n \Leftrightarrow Q^TQQ^{-1} = I_nQ^{-1} \Leftrightarrow Q^{-1} = Q^T. \tag{9.112}$$

(Transposition.) We have

$$Q^TQ = I_n \Leftrightarrow QQ^TQ = QI_n \Leftrightarrow QQ^TQQ^T = QQ^T \Leftrightarrow QQ^T = I_n. \tag{9.113}$$

□

9.6.6 Positive-definite matrices

Positive-definite matrices are central to probabilistic modeling using multivariate normal distributions.

Definition 9.15. A square matrix $C \in \mathbb{R}^{n\times n}$ is called positive-definite (p.d.) if

- C is a symmetric matrix and
- for all $x \in \mathbb{R}^n, x \neq 0_n$, $x^TCx > 0$.

•

In the following theorem, we collect a few important properties of positive-definite matrices.

Theorem 9.14 (Properties of positive-definite matrices)**.**

(Inverse.) Let $C \in \mathbb{R}^{n\times n}$ be a positive-definite matrix. Then C^{-1} exists and is also positive-definite.

∘

9.7 Eigenanalysis

With the *eigenanalysis of a square matrix*, the *orthonormal decomposition of a symmetric matrix*, and the *singular value decomposition of an arbitrary matrix*, we discuss in this section three closely related concepts of matrix theory that play central roles in many areas of data-analytic applications. However, the meaning of these concepts becomes apparent above all in the respective application context, so this section may necessarily seem somewhat abstract.

Eigenvectors and eigenvalues

The *eigenanalysis* of a square matrix is understood as determining its *eigenvectors* and *eigenvalues*. These are defined for a square matrix as follows.

Definition 9.16 (Eigenvector and eigenvalue)**.** Let $A \in \mathbb{R}^{m \times m}$ be a square matrix. Then every vector $v \in \mathbb{R}^m$ different from the zero vector 0_m for which, with a scalar $\lambda \in \mathbb{R}$,

$$Av = \lambda v \tag{9.114}$$

holds is called an *eigenvector of* A, and λ is then called an *eigenvalue of* A.

•

By definition, every eigenvector therefore has an associated eigenvalue, although the eigenvalues of different eigenvectors can certainly be identical. Intuitively, the definition of eigenvector and eigenvalue means that an eigenvector of a matrix is changed in its length, but not in its direction, by multiplication with this very matrix. The associated eigenvalue of the eigenvector corresponds to the factor of the change in length. However, the assignment of eigenvectors and eigenvalues is not unique, as the following theorem shows.

Theorem 9.15 (Multiplicativity of eigenvectors)**.** *Let $A \in \mathbb{R}^{m \times m}$ be a square matrix. If $v \in \mathbb{R}^m$ is an eigenvector of A with eigenvalue $\lambda \in \mathbb{R}$, then for $c \in \mathbb{R} \setminus \{0\}$, $cv \in \mathbb{R}^m$ is also an eigenvector of A, again with eigenvalue $\lambda \in \mathbb{R}$.*

∘

Proof. This follows from

$$A(cv) = cAv = c\lambda v = \lambda(cv). \tag{9.115}$$

Thus, cv is an eigenvector of A with eigenvalue λ. □

To resolve the non-uniqueness in the definition of the eigenvector associated with an eigenvalue, we use the convention of considering only those vectors as eigenvectors for an eigenvalue λ that have length 1 and hence are elements of the unit sphere in $\mathbb{R}^m$:

$$\mathbb{S}^{m-1} := \{x \in \mathbb{R}^m : \|x\|_2 = 1\}. \tag{9.116}$$

Thus, if we should have found an eigenvector v for an eigenvalue λ of a matrix A with length other than 1, then

$$\frac{v}{\|v\|_2} \tag{9.117}$$

is of length 1 and, by Theorem 9.15, is likewise an eigenvector of A with eigenvalue λ. Before turning to the determination of eigenvalues and eigenvectors, we illustrate the concepts of eigenvalue and eigenvector for the case of a 2×2 matrix using an example.

Example

Consider the matrix

$$A := \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{9.118}$$

and the vector

$$v := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{9.119}$$

Then v is an eigenvector of A with eigenvalue $\lambda = 3$, because

$$\begin{aligned} Av &= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \lambda v. \end{aligned} \tag{9.120}$$

Inspection of Figure 9.2 accordingly shows that for the matrix A defined here, the vectors v and Av point in the same direction, but that Av is three times as long as v. Now consider the vector

$$w := \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{9.121}$$

This vector also has length 1, but in contrast to v it is not an eigenvector of A. We have

$$\begin{aligned} Aw &= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \end{aligned} \tag{9.122}$$

Because the second entry of w is 0, there can be no scalar λ that would transform the second entry of w into the second entry of Aw by multiplication. Figure 9.2 therefore shows accordingly that the vector resulting from multiplication of w by A no longer points in the same direction as w.

Determining eigenvalues and eigenvectors

The following theorem states how the eigenvalues and eigenvectors of a square matrix can be computed.

Theorem 9.16 (Determining eigenvalues and eigenvectors)**.** *Let $A \in \mathbb{R}^{m\times m}$ be a square matrix. Then the eigenvalues of A are obtained as the roots of the characteristic polynomial*

$$\chi_A(\lambda) := |A - \lambda I_m| \tag{9.123}$$

of A. Furthermore, let $\lambda_i^*, i = 1, 2, \ldots$ be the eigenvalues of A determined in this way. The corresponding eigenvectors $v_i, i = 1, 2, \ldots$ of A can then be determined by solving the systems of linear equations

$$(A - \lambda_i^* I_m)v_i = 0_m \textit{ for } i = 1, 2, \ldots \tag{9.124}$$

∘

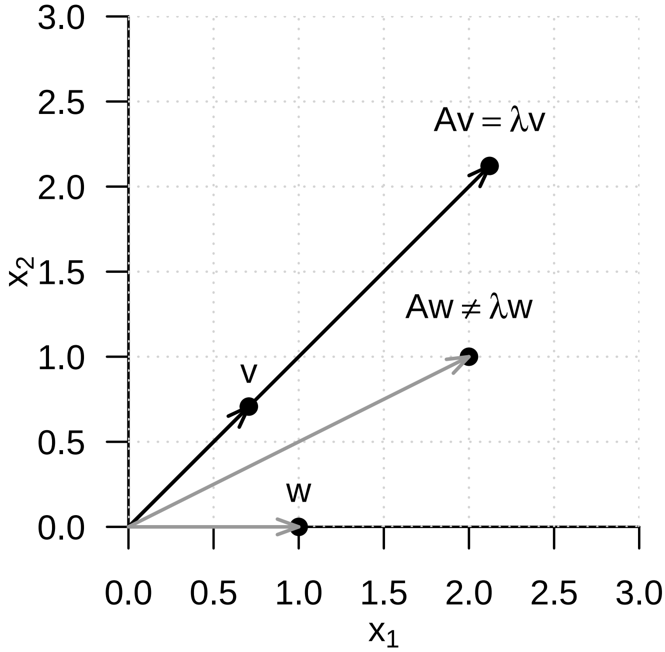

Figure 9.2 Eigenvector of a 2×2 matrix. For the matrix A defined in the text, v is an eigenvector, but w is not.

Proof. (1) Determining eigenvalues

We first note that, by the definition of eigenvectors and eigenvalues,

$$Av = \lambda v \Leftrightarrow Av - \lambda v = 0_m \Leftrightarrow (A - \lambda I_m)v = 0_m. \tag{9.125}$$

For the eigenvalue λ, the eigenvector v is therefore mapped to the zero vector 0_m by multiplication with $(A - \lambda I_m)$. But because by definition $v \neq 0_m$, the matrix $(A - \lambda I_m)$ is not invertible: both the zero vector and v are mapped to 0_m by $(A - \lambda I_m)$, so the mapping

$$f : \mathbb{R}^m \to \mathbb{R}^m, x \mapsto (A - \lambda I_m)x \tag{9.126}$$

is not bijective, and $(A - \lambda I_m)^{-1}$ cannot exist. The fact that $(A - \lambda I_m)$ is not invertible, however, is equivalent to the determinant of $(A - \lambda I_m)$ being equal to zero. Thus,

$$\chi_A(\lambda) = |A - \lambda I_m| = 0 \tag{9.127}$$

is a necessary and sufficient condition for λ to be an eigenvalue of A.

(2) Determining eigenvectors

Let λ_i^* be an eigenvalue of A. Then the above considerations imply that solving

$$(A - \lambda_i^* I_m)v_i^* = 0_m \tag{9.128}$$

for v_i^* yields an eigenvector for the eigenvalue λ_i^*. □

In general, determining eigenvalues and eigenvectors therefore requires determining polynomial roots and solving systems of linear equations. For small matrices with $m \leq 4$, this can indeed be done manually. The matrices that occur in applications, however, are usually much larger, so numerical methods for root finding and for solving systems of linear equations are used for eigenanalysis, for example in functions such as **R**'s `eigen()`, **SciPy's** `linalg.eig()`, or **Julia**'s `eigvals()` and `eigvecs()`. For details on these methods, we refer to the advanced literature, for example Burden et al. (2016) and Richter & Wick (2017). Here we merely illustrate Theorem 9.16 by means of an example.

Example

To this end, let again

$$A := \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{9.129}$$

We first want to compute the eigenvalues of A. By Theorem 9.16, these are the roots of the characteristic polynomial of A. We therefore first compute the characteristic polynomial of A by

$$\chi_A(\lambda) = \left| \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = \left| \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} \right| = (2-\lambda)^2 - 1. \tag{9.130}$$

Using the quadratic formula to solve quadratic equations, one then finds

$$(2 - \lambda_{1/2}^*)^2 - 1 = 0 \Leftrightarrow \lambda_1^* = 3 \text{ or } \lambda_2^* = 1. \tag{9.131}$$

The eigenvalues of A are therefore $\lambda_1 = 3$ and $\lambda_2 = 1$. The associated eigenvectors are then obtained for $i = 1, 2$ by solving the system of linear equations

$$(A - \lambda_i I_2)v_i = 0_2. \tag{9.132}$$

Specifically, for $\lambda_1 = 3$,

$$(A - 3I_2)v_1 = 0_2 \Leftrightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_{1_1} \\ v_{1_2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{9.133}$$

implies that

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{9.134}$$

is an eigenvector for the eigenvalue λ_1, and for $\lambda_2 = 1$,

$$(A - I_2)v_2 = 0_2 \Leftrightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_{2_1} \\ v_{2_2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{9.135}$$

implies that

$$v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{9.136}$$

is an eigenvector for the eigenvalue $\lambda_2 = 1$. Furthermore, here obviously

$$v_1^T v_2 = 0 \text{ and } \|v_1\| = \|v_2\| = 1. \tag{9.137}$$

The following **R** code demonstrates the determination of the eigenvalues and eigenvectors of the matrix considered here with the help of the `eigen()` function.

```
# matrix definition
A = matrix(c(2,1,
             1,2),
           nrow  = 2,
           byrow = TRUE)

# eigenanalysis
eigen(A)
```

```
eigen() decomposition
$values
[1] 3 1

$vectors
          [,1]       [,2]
[1,] 0.7071068 -0.7071068
[2,] 0.7071068  0.7071068
```

At the end of this section, we consider two technical theorems that make statements about the relation between special matrix products and their eigenvalues and eigenvectors. We need these theorems in the context of canonical correlation analysis.

Theorem 9.17 (Eigenvalues and eigenvectors of matrix products)**.** *For $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times n}$, the eigenvalues of $AB \in \mathbb{R}^{n \times n}$ and $BA \in \mathbb{R}^{m \times m}$ are equal. Furthermore, if v is an eigenvector for a nonzero eigenvalue λ of AB, then $w := Bv$ is an eigenvector of BA for the eigenvalue λ.*

∘

For a proof, we refer to Mardia et al. (1979), p. 468. We demonstrate the statement of this theorem using the **R** code below.

```
A   = matrix(1:6, nrow = 2,  byrow = T)          # matrix A \in \mathbb{R}^{2 x 3}
B   = matrix(1:6, ncol = 2,  byrow = T)          # matrix B \in \mathbb{R}^{3 x 2}
EAB = eigen(A %*% B)                             # eigenanalysis of AB \in \mathbb{R}^{2 \times 2}
EBA = eigen(B %*% A)                             # eigenanalysis of BA \in \mathbb{R}^{3 \times 3}
w   = B %*% EAB$vectors[,1]                      # eigenvector of BA
cat("Eigenvalues of AB :"  , EAB$values[1:2],
    "\nEigenvalues of BA :", EBA$values[1:2],
    "\nBAw with w = Bv   :", B %*% A %*% w,
    "\nlw with w = Bv    :", EBA$values[1] * w)
```

```
Eigenvalues of AB : 85.57934 0.4206623
Eigenvalues of BA : 85.57934 0.4206623
BAw with w = Bv   : -191.1333 -416.7586 -642.3839
lw with w = Bv    : -191.1333 -416.7586 -642.3839
```

Theorem 9.18. *For $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{p\times n}$, $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^p$, the only nonzero eigenvalue of $Aab^TB \in \mathbb{R}^{n\times n}$ is equal to b^TBAa, with associated eigenvector Aa.*

∘

For a proof, we refer to Mardia et al. (1979), p. 468. We demonstrate the statement of this theorem using the **R** code below.

```
A      = matrix(1:6, nrow = 2,  byrow = T)      # matrix A \in \mathbb{R}^{2 x 3}
B      = matrix(1:8, ncol = 2,  byrow = T)      # matrix B \in \mathbb{R}^{4 x 2}
a      = matrix(1:3, nrow = 3,  byrow = T)      # vector a \in \mathbb{R}^{3 x 1}
b      = matrix(1:4, nrow = 4,  byrow = T)      # vector b \in \mathbb{R}^{4 x 1}
EAabTB = eigen(A %*% a %*% t(b) %*% B)          # eigenanalysis of Aab^TB \in \mathbb{R}^{2 x 2}
cat("Eigenvalues of AabTB :", EAabTB$values,
    "\nbTBAa                :", t(b) %*% B %*% A %*% a,
    "\nAa                   :", A %*% a,
    "\n(AabTB)Aa            :",(A %*% a %*% t(b) %*% B) %*% A %*% a,            # mv
    "\n(bTBAa)Aa            :",as.vector((t(b) %*% B %*% A %*% a)) * (A %*% a)) # = \lambda v
```

```
Eigenvalues of AabTB : 2620 0
bTBAa                : 2620
Aa                   : 14 32
(AabTB)Aa            : 36680 83840
(bTBAa)Aa            : 36680 83840
```

9.8 Orthonormal decomposition

The term *decomposition* of a matrix denotes the splitting of a given matrix into the matrix product of several matrices. A wide variety of matrix decompositions play an important role in many mathematical applications; for an overview, see for example Golub & Van Loan (2013). In this section, we introduce a special matrix decomposition, the *orthonormal decomposition of a symmetric matrix*, which builds directly on eigenanalysis. We first record the following fundamental theorem on the eigenvalues and eigenvectors of symmetric matrices.

Theorem 9.19 (Eigenvalues and eigenvectors of symmetric matrices)**.**

Let $S \in \mathbb{R}^{m\times m}$ be a symmetric matrix. Then:

(1) The eigenvalues of S are real.

(2) The eigenvectors corresponding to any two distinct eigenvalues of S are orthogonal.

∘

Proof. We take the fact that a symmetric matrix has m real eigenvalues as given and show only that the eigenvectors corresponding to any two distinct eigenvalues of a symmetric matrix are orthogonal. Without loss of generality, let $\lambda_i, \lambda_j \in \mathbb{R}$ with $1 \le i, j \le m$ and $\lambda_i \ne \lambda_j$ be two distinct eigenvalues of S with associated eigenvectors q_i and q_j, respectively. Then, as shown below,

$$\lambda_i q_i^T q_j = \lambda_j q_i^T q_j. \tag{9.138}$$

With $q_i \ne 0_m, q_j \ne 0_m$ and $\lambda_i \ne \lambda_j$, it follows that $q_i^T q_j = 0$, because there is no number c other than zero for which, with $a, b \in \mathbb{R}$ and $a \ne b$,

$$ac = bc \tag{9.139}$$

holds. To finally show

$$\lambda_i q_i^T q_j = \lambda_j q_i^T q_j, \tag{9.140}$$

we first note that

$$Sq_i = \lambda_i q_i \Leftrightarrow (Sq_i)^T = (\lambda_i q_i)^T \Leftrightarrow q_i^T S^T = \lambda_i q_i^T \Leftrightarrow q_i^T S = \lambda_i q_i^T \Leftrightarrow q_i^T S q_j = \lambda_i q_i^T q_j \tag{9.141}$$

and

$$Sq_j = \lambda_j q_j \Leftrightarrow q_j^T S = \lambda_j q_j^T \Leftrightarrow q_j^T S q_i = \lambda_j q_j^T q_i \Leftrightarrow (q_j^T S q_i)^T = (\lambda_j q_j^T q_i)^T \Leftrightarrow q_i^T S q_j = \lambda_j q_i^T q_j \tag{9.142}$$

hold. Thus, both $\lambda_i q_i^T q_j$ and $\lambda_j q_i^T q_j$ are identical with $q_i^T S q_j$ and therefore also with each other. □

Obviously, we have proven only statement (2) of Theorem 9.19. A complete proof of the theorem can be found, for example, in Strang (2009). We also note that, because by convention we consider eigenvectors of length 1, the orthogonal eigenvectors addressed in Theorem 9.19 are in particular also orthonormal. With the help of Theorem 9.19, we can now formulate the orthonormal decomposition of a symmetric matrix and prove its existence.

Theorem 9.20 (Orthonormal decomposition of a symmetric matrix)**.** *Let $S \in \mathbb{R}^{m \times m}$ be a symmetric matrix with m distinct eigenvalues. Then S can be written as*

$$S = Q\Lambda Q^T, \tag{9.143}$$

where $Q \in \mathbb{R}^{m \times m}$ is an orthogonal matrix and $\Lambda \in \mathbb{R}^{m \times m}$ is a diagonal matrix.

∘

Proof. Let $\lambda_1 > \lambda_2 > \ldots > \lambda_m$ be the eigenvalues of S ordered by size, and let $q_1, \ldots, q_m$ be the associated orthonormal eigenvectors. With

$$Q := \begin{pmatrix} q_1 & q_2 & \cdots & q_m \end{pmatrix} \in \mathbb{R}^{m \times m} \text{ and } \Lambda := \operatorname{diag}\left(\lambda_1, \lambda_2, \ldots, \lambda_m\right) \in \mathbb{R}^{m \times m}, \tag{9.144}$$

the definitions of eigenvalues and eigenvectors first imply that

$$Sq_i = \lambda_i q_i \text{ for } i = 1, \ldots, m \Leftrightarrow SQ = Q\Lambda. \tag{9.145}$$

Right multiplication by Q^T then gives, with $QQ^T = I_m$,

$$SQQ^T = Q\Lambda Q^T \Leftrightarrow SI_m = Q\Lambda Q^T \Leftrightarrow S = Q\Lambda Q^T. \tag{9.146}$$

□

Because of the diagonality of Λ, the splitting of S into the matrix product $Q\Lambda Q^T$ is also called a *diagonalization of S*. As shown in the proof, to represent S in diagonal form, the diagonal elements of Λ are chosen as the eigenvalues of S ordered by size, and the columns of Q are chosen as the corresponding eigenvectors of S. We illustrate this with an example.

Example

For the symmetric matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{9.147}$$

with the eigenvalues $\lambda_1 = 3$ and $\lambda_2 = 1$ determined above and the associated orthonormal eigenvectors

$$v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{9.148}$$

let

$$Q := \begin{pmatrix} v_1 & v_2 \end{pmatrix} \text{ and } \Lambda = \operatorname{diag}(\lambda_1, \lambda_2). \tag{9.149}$$

Then obviously

$$\begin{aligned}
Q\Lambda Q^T &= \begin{pmatrix} v_1 & v_2 \end{pmatrix} \operatorname{diag}(\lambda_1, \lambda_2) \begin{pmatrix} v_1 & v_2 \end{pmatrix}^T \\
&= \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \right) \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \right) \\
&= \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix} \right) \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \right) \\
&= \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \\
&= A
\end{aligned}$$

and we have verified Theorem 9.20 for this example.

Symmetric square root of a matrix

The definition of the orthonormal decomposition of a symmetric matrix allows us to introduce the concept of the *symmetric square root of a matrix*.

Definition 9.17 (Symmetric square root of a matrix)**.** Let $S \in \mathbb{R}^{m \times m}$ be an invertible symmetric matrix with positive eigenvalues. Then for $r \in \mathbb{N}^0$ and $s \in \mathbb{N}$, the rational powers of S are defined, with the orthonormal matrix $Q \in \mathbb{R}^{m \times m}$ of the eigenvectors of S and the diagonal matrix $\Lambda = \operatorname{diag}(\lambda_i) \in \mathbb{R}^{m \times m}$ of the associated eigenvalues $\lambda_1, ..., \lambda_m$ of S, as

$$S^{r/s} = Q\Lambda^{r/s}Q^T \text{ with } \Lambda^{r/s} = \operatorname{diag}\left(\lambda_i^{r/s}\right). \tag{9.150}$$

The special case $r := 1, s := 2$ is called the *symmetric square root of* S and has the form

$$S^{1/2} = Q\Lambda^{1/2}Q^T \text{ with } \Lambda^{1/2} = \operatorname{diag}\left(\lambda_i^{1/2}\right). \tag{9.151}$$

•

We note that Definition 9.17 immediately implies

$$\left(S^{1/2}\right)^2 = Q\Lambda^{1/2}Q^TQ\Lambda^{1/2}Q^T = Q\Lambda^{1/2}\Lambda^{1/2}Q^T = Q\Lambda Q^T = S. \tag{9.152}$$

Furthermore,

$$\left(S^{-1/2}\right)^2 = Q\Lambda^{-1/2}Q^TQ\Lambda^{-1/2}Q^T = Q\Lambda^{-1/2}\Lambda^{-1/2}Q^T = Q\Lambda^{-1}Q^T = S^{-1}. \tag{9.153}$$

Finally,

$$\begin{aligned}
S^{-1/2}SS^{-1/2} &= Q\Lambda^{-1/2}Q^TQ\Lambda Q^TQ\Lambda^{-1/2}Q^T \\
&= Q\Lambda^{-1/2}\Lambda\Lambda^{-1/2}Q^T \\
&= Q\Lambda\Lambda^{-1}Q^T \\
&= I_m
\end{aligned} \tag{9.154}$$

9.9 Singular value decomposition

A versatile matrix decomposition of an arbitrary matrix is the *singular value decomposition*. At this point, we are interested only in the relation between singular value decomposition and eigenanalysis and refer to the advanced literature, for example Strang (2009), for a detailed discussion of the singular value decomposition. The concept of singular value decomposition is defined as follows.

Definition 9.18 (Singular value decomposition)**.** Let $Y \in \mathbb{R}^{m\times n}$ be a matrix. Then the decomposition

$$Y = USV^T, \tag{9.155}$$

where $U \in \mathbb{R}^{m\times m}$ is an orthogonal matrix, $S \in \mathbb{R}^{m\times n}$ is a diagonal matrix, and $V \in \mathbb{R}^{n\times n}$ is an orthogonal matrix, is called the *singular value decomposition* of Y. The diagonal elements of S are called the *singular values* of Y.

•

Singular value decompositions are abbreviated as *SVD*. We omit a discussion of the computation of a singular value decomposition and merely note that singular value decompositions can be computed, for example, in **R** with the function `svd()`, in **SciPy** with `scipy.linalg.svd()`, and in **Julia** with `svd()`. The following theorem describes the relation between singular value decomposition and eigenanalysis and is used in many places.

Theorem 9.21 (Singular value decomposition and eigenanalysis)**.**

Let $Y \in \mathbb{R}^{m\times n}$ be a matrix and

$$Y = USV^T \tag{9.156}$$

be its singular value decomposition. Then:

- *the columns of U are the eigenvectors of YY^T,*
- *the columns of V are the eigenvectors of Y^TY, and*
- *the singular values are the square roots of the associated eigenvalues.*

∘

Proof. We first note that, with

$$\left(YY^T\right)^T = YY^T \text{ and } \left(Y^TY\right)^T = Y^TY, \tag{9.157}$$

YY^T and Y^TY are symmetric matrices and thus have orthonormal decompositions. We further note that, with $V^TV = I_n$ and $U^TU = I_m$,

$$YY^T = USV^T\left(USV^T\right)^T = USV^TVS^TU^T = USS^TU^T =: U\Lambda_U U^T \tag{9.158}$$

and

$$Y^TY = \left(USV^T\right)^T USV^T = VS^TU^TUSV^T = VS^TSV^T =: V\Lambda_V V^T \tag{9.159}$$

where we have defined $\Lambda_U := SS^T$ and $\Lambda_V := S^TS$. Because the product of diagonal matrices is again a diagonal matrix, Λ_U and Λ_V are diagonal matrices, and by definition U and V are orthogonal matrices. Thus, we have written YY^T and Y^TY in the form of the orthonormal decompositions

$$YY^T = U\Lambda_U U^T \text{ and } Y^TY = V\Lambda_V V^T \tag{9.160}$$

where, for the nonzero diagonal elements of Λ_U and Λ_V, these elements are the squared singular values, that is, the squared diagonal elements of S. □

10 Descriptive statistics

The aim of evaluating descriptive statistics is to make data sets accessible to human observation as efficiently as possible. As soon as the number of data points in a data set exceeds a certain number, pure numerical representations are usually difficult for human cognition to grasp. This is where descriptive statistics comes in and, in addition to graphical representations of larger data sets, also offers measures that describe certain characteristic aspects of a data set, such as its "average" value or its variability, in the sense of data reduction. In this section we want to use univariate descriptive statistics to get to know the first procedures and measures that make it possible to summarize data sets in which there is one number per study unit. We look in particular at *frequency distributions* and *histograms*, *distribution functions* and *quantiles*, as well as *measures of central tendency*, which allow statements about "average" data values, and *measures of data variability*, which allow statements about the spread of data values.

In contrast to probabilistic modeling in the context of frequentist and Bayesian inference, descriptive statistics is not based on probability theory and can therefore be viewed independently of it. However, many aspects of descriptive statistics ultimately only make sense against the background of probability theory considerations and, conversely, many concepts in probability theory are motivated by descriptive statistical concepts. If the present section does not have a background in probability theory, its full meaning will certainly only become apparent in the context of the concepts of probability theory.

In this section, we focus on descriptive statistics for describing a data set of the form

$$y := (y_1, ..., y_n) \text{ with } y_i \in \mathbb{R} \text{ for } i = 1, ..., n. \tag{10.1}$$

It is initially irrelevant whether such a data set is in the form of a column vector $y \in \mathbb{R}^{n \times 1}$ or a row vector $y \in \mathbb{R}^{1 \times n}$. Furthermore, we focus on the practical evaluation of the descriptive statistics considered, so we provide many examples for their calculation with **R**.

Application example

As an application example, we consider a data set for the evidence-based evaluation of psychotherapy for depression. To do this, we assume that $n = 100$ patients were randomly divided into a treatment and a control study condition and were asked about their depression severity using the BDI-II questionnaire at the beginning of the respective intervention. We do not want to consider the BDI-II scores collected after the interventions in this chapter. So we assume that we have a data set as shown by the following **R** code in Table 10.1.

```
D = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
knitr::kable(D[,1:2], digits = 2, align = "c")          # r markdown table output
```

Table 10.1 BDI-II scores (PRE) of n = 100 patients in a treatment (T) and a control study group (C) before the start of a psychological intervention

GROUP	PRE
T	17
T	20
T	16
T	18
T	21
T	17
T	17
T	17
T	18
T	18
T	20
T	17
T	16
T	18
T	16
T	18
T	17
T	14
T	18
T	18
T	20
T	20
T	21
T	19
T	19
T	17
T	20
T	21
T	17
T	17
T	19
T	20
T	22
T	20
T	21
T	20
T	16
T	15
T	15
T	18
T	21
T	17
T	18
T	21
T	17
T	19
T	16
T	17
T	17
T	18
C	22
C	19
C	21
C	18
C	19
C	17
C	20
C	19
C	19
C	19
C	17
C	20
C	18
C	20
C	18
C	18
C	15
C	18
C	17
C	19
C	19
C	19
C	20
C	19
C	20
C	19
C	21
C	19
C	19
C	18
C	20
C	16
C	21
C	19
C	20
C	18
C	23
C	18
C	19
C	18

Table 10.1 BDI-II scores (PRE) of n = 100 patients in a treatment (T) and a control study group (C) before the start of a psychological intervention

GROUP	PRE
C	21
C	17
C	20
C	21
C	21
C	20
C	18
C	18
C	19
C	19

10.1 Frequency distributions

We start with the concepts of *absolute* and *relative frequency distribution*. For data sets with values in which the same value occurs multiple times, frequency distributions can provide a first impression of the nature of the data set.

Definition 10.1 (Absolute and relative frequency distributions)**.** $y := (y_1, ..., y_n)$ is a data set and $W := \{w_1, ..., w_k\}$ with $k \leq n$ are the values occurring in the data set. Then the function is called

$$h : W \to \mathbb{N}, w \mapsto h(w) := \text{number of } y_i \text{ in } y \text{ with } y_i = w \tag{10.2}$$

the *absolute frequency distribution* of the values of y and the function

$$r : W \to [0, 1], w \mapsto r(w) := \frac{h(w)}{n} \tag{10.3}$$

is called the *relative frequency distribution* of the values of y.

•

Frequency distributions can be evaluated and displayed in **R** by combining the `table()` and `barplot()` functions. The following **R** code first generates the absolute and relative frequency distributions of the pre-intervention BDI-II data of the example data set.

```
D        = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y        = D$PRE                                               # double vector of the PRE values
n        = length(y)                                           # number of data values (100)
H        = as.data.frame(table(y))                             # absolute frequency distribution (data frame)
names(H) = c("w", "ah")                                        # column naming
H$rh     = H$ah/n                                              # relative frequency distribution
print(H, digits = 1)                                           # output
```

```
    w ah   rh
1  14  1 0.01
2  15  3 0.03
3  16  6 0.06
4  17 17 0.17
5  18 21 0.21
6  19 20 0.20
7  20 17 0.17
8  21 12 0.12
9  22  2 0.02
10 23  1 0.01
```

From the output of the data frame `H` we can see that in the data set 10 under consideration, different values `w` occur between 14 and 23, whereby, for example, the values 18 and 22 occur 21 times and twice respectively. Both the absolute and the relative frequency distribution show that values around 18 occur more frequently in the present data set, whereas extreme values such as 13 or 23 occur rather rarely. Note that the relative frequency distribution is simply a scaled form of the absolute frequency distribution. Qualitatively, i.e. with regard to the frequent or rare occurrence of certain values, the two frequency distributions do not differ. This becomes particularly clear when both frequency distributions are visualized, as in Figure 10.1. The following **R** code uses the `barplot()` function for this purpose. The height of the bar plotted above a value w corresponds to the function values $h(w)$ or $r(w)$. The visual insight naturally corresponds to the inspection of the `H` data frame. Both frequency distributions show that values around 18 occur more frequently in the present data set, whereas lower or higher values such as 13 or 23 are rather rare.

```
pdf(                                                          # pdf copy function
file      = "./_figures/110-frequency-distributions.pdf",     # filename
width     = 7,                                                # width (inches)
height    = 4)                                                # height (inches)
par(                                                          # figure parameters
mfcol     = c(1,2),                                           # rows and columns layout
las       = 1,                                                # x-tick orientation
cex       = 0.7,                                              # annotation enlargement
cex.main  = 1.3)                                              # title enlargement
h         = H$ah                                              # h(w) values
r         = H$rh                                              # r(w) values
names(h)  = H$w                                               # barplot needs w values as names
names(r)  = H$w                                               # barplot needs w values as names
barplot(                                                      # bar chart
h,                                                            # absolute frequencies
col       = "gray90",                                         # bar color
xlab      = "w",                                              # x axis label
ylab      = "h(w)",                                           # y axis label
ylim      = c(0,25),                                          # y axis limits
main      = "Absolute frequency distribution")                # title
barplot(                                                      # bar chart
r,                                                            # absolute frequencies
col       = "gray90",                                         # bar color
xlab      = "w",                                              # x axis label
ylab      = "r(w)",                                           # y axis label
ylim      = c(0,.25),                                         # y axis limits
main      = "Relative frequency distribution")                # title
dev.off()                                                     # end image editing
```

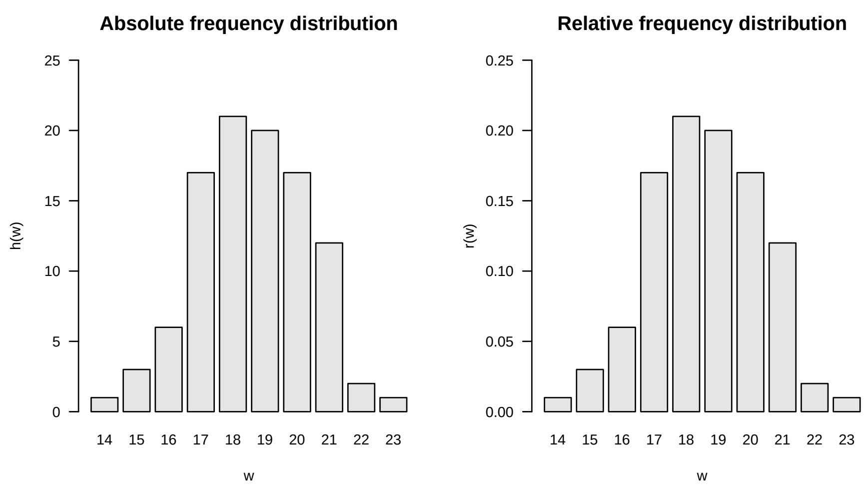

Figure 10.1 Absolute and relative frequency distributions of pre-intervention BDI-II scores.

10.2 Histograms

If no values appear multiple times in a data set, the absolute and relative frequency distributions for each of these values take the value 1 and $1/n$, respectively, and then provide no value beyond visual inspection of the data set. However, the histograms presented in this section offer a way to generate visual representations of frequency distributions in this case too. Different values that are "close to each other" in a well-defined sense are combined into classes and the absolute or relative frequencies of these classes are determined. Since in data sets in which the same values occur multiple times, these values are naturally in the same class, histograms can also be usefully used in the case considered in Section 10.1. Histograms generalize the frequency distributions considered in the previous section for data sets with multiple values.

We first give a definition of the concept of histogram. Note that in this definition the process of class formation, the determination of their absolute and relative frequencies, and the visualization of these frequencies are closely interlinked. This is due to the convention that the term histogram describes both a specific form of data analysis and its visualization.

Definition 10.2 (Histogram)**.** A *histogram* is a diagram in which, for a data set $y = (y_1, ..., y_n)$ with values $W := \{w_1, ..., w_m\}$ with $m \le n$ for $j = 1, ..., k$ over adjacent intervals $[b_{j-1}, b_j[$ with $b_0 := \min W$ and $b_k := \max W$, rectangles with the *width* $d_j := b_j - b_{j-1}$ and the *height* $h(w)$ or $r(w)$ for $w \in [b_{j-1}, b_j[$ are shown. The intervals $[b_{j-1}, b_j[$ are called *classes* (*bins*).

•

Of course, the visual appearance of a histogram depends heavily on how precisely the values of a data set are grouped into classes. According to Definition 10.2, the decisive parameter for this is the number of classes k that are selected between the minimum value b_0 and the maximum value b_k of the data set. Below we present some conventional values for the number of classes k and calculate and visualize them using the example data set as an example. We always assume that the neighboring intervals $[b_{j-1}, b_j[$ with $j = 1, ..., k$ are chosen for the number of classes k with a constant width $d = b_j - b_{j-1}$.

To calculate and visualize histograms we use the **R** function `hist()`. In this case, the classes $[b_{j-1}, b_j[$ are specified for $j = 1, ..., k$ as the argument `breaks`. Specifically, `breaks` is the **R** vector `c(b_0,b_1,...,b_k)` with length $k+1$. To illustrate the effect of different choices of the number of classes, we first read the data set and determine its minimum and maximum values b_0 and b_k.

```
# dataset
D   = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y   = D$PRE                                                # pre-intervention BDI-II data
b_0 = min(y)                                               # b_0
b_k = max(y)                                               # b_k
```

```
Minimum b_0              :  14
Maximum b_k              :  23
```

A first possibility for choosing the classes k is to achieve a desired constant width d as precisely as possible. This is what you rely on

$$k := \left\lceil \frac{b_k - b_0}{d} \right\rceil. \tag{10.4}$$

Note that the application of the rounding function implies that the resulting class width is at most the desired value, but can also be smaller. The following **R** code illustrates this effect.

```
# histogram with desired class width d = 2
d   = 2                                # desired class width
b_0 = min(y)                           # b_0
b_k = max(y)                           # b_k
k   = ceiling((b_k - b_0)/d)           # number of classes
b   = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Desired class width d      :  2
Number of classes k        :  5
Intervals [b_j-1, b_j]    :  14 15.8 17.6 19.4 21.2 23
Achieved class width d     :  1.8
```

Choosing a desired class width of $d = 1.5$, on the other hand, leads to a resulting class width for the present data set that corresponds to the desired one.

```
# histogram with desired class width d = 1.5
d = 1.5                              # desired class width
k = ceiling((b_k - b_0)/d)           # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Desired class width d      :  1.5
Number of classes k        :  6
Intervals [b_j-1, b_j]    :  14 15.5 17 18.5 20 21.5 23
Achieved class width d     :  1.5
```

Another option is to make the number of classes dependent on the number of data points in the data set. The Excel standard choice is included

$$k := \lceil \sqrt{n} \rceil. \tag{10.5}$$

The following **R** code implements this choice for the present data set.

```
# histogram with Excel standard
n = length(y)                        # number of data points
k = ceiling(sqrt(n))                 # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Number of data points n    :  100
Number of classes k        :  10
Intervals [b_j-1, b_j]    :  14 14.9 15.8 16.7 17.6 18.5 19.4 20.3 21.2 22.1 23
Achieved class width d     :  0.9
```

The number of classes suggested by Sturges (1926) is similar to the Excel standard, which optimizes the representation under an implicit normal distribution assumption

$$k := \lceil \log_2 n + 1 \rceil \tag{10.6}$$

is given. In this case the result is:

```
# class number according to Sturges (1926)
n = length(y)                          # number of data points
k = ceiling(log2(n)+1)                 # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Number of data points n    :  100
Number of classes k        :  8
Intervals [b_j-1, b_j]    :  14 15.125 16.25 17.375 18.5 19.625 20.75 21.875 23
Achieved class width d     :  1.125
```

Another possibility is to incorporate descriptive statistics of the data set into the choice of the number of classes. As a measure of the dispersion of the data in the data set, the number of classes according to Scott (1979) uses the empirical standard deviation (cf. Section 10.6) in order to use the histogram to achieve a density estimate for a normal distribution. We do not want to delve into the non-parametric background of this approach here, but simply note that the choice of the number of classes according to Scott (1979) by determining the class width

$$d := 3.49S/\sqrt[3]{n} \tag{10.7}$$

and the subsequent choice of the number of classes

$$k := \left\lceil \frac{b_k - b_0}{d} \right\rceil \tag{10.8}$$

is given, where S denotes the standard deviation of the data set. In this case the result is:

```
# number of classes according to Scott (1979)
n = length(y)                          # number of data values
S = sd(y)                              # empirical standard deviation
d = ceiling(3.49*S/(n^(1/3)))          # class width
k = ceiling((b_k - b_0)/d)             # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Number of data points n    :  100
Standard deviation         :  1.740167
Class width by Scott d     :  2
Number of classes k        :  5
Intervals [b_j-1, b_j]    :  14 15.8 17.6 19.4 21.2 23
Achieved class width d     :  1.8
```

By default, the `hist()` function implemented in **R** uses a modification of the number of classes suggested by Sturges (1926) and offers a number of additional options for selecting classes. `?hist` provides an overview of this. The following **R** code visualizes the absolute frequencies of the values of the example data set using the classes specified above.

```
# figure parameters
pdf(                                                                      # pdf copy function
file     = "./_figures/110-histograms.pdf",                               # filename
width    = 6,                                                             # width (inches)
height   = 9)                                                             # height (inches)
par(                                                                      # figure parameters
mfcol    = c(3,2),                                                        # rows and columns layout
las      = 1,                                                             # x-tick orientation
cex      = .7,                                                            # annotation enlargement
cex.main = .9)                                                            # title enlargement
x_min    = 12                                                             # x-axis limit
x_max    = 25                                                             # x-axis limit
y_min    = 0                                                              # y-axis limit
y_max    = 45                                                             # y-axis limit
label    = c("", "", "Excel", "Sturges", "Scott")                         # title

# histograms with predefined number of classes and class width
for(i in 1:5){
    hist(                                                                 # histogram
    y,                                                                    # dataset
    breaks   = bs[[i]],                                                   # breaks argument
    xlim     = c(x_min, x_max),                                           # x-axis limits
```

```
    ylim     = c(y_min, y_max),                                                # y-axis limits
    ylab     = "frequency",                                                    # y-axis label
    xlab     = "BDI",                                                          # x-axis label
    main     = sprintf(paste(label[i], "k = %.0f, d = %.2f"), ks[i],ds[i]))   # title
}

# r default histogram
hist(                                                                          # histogram
y,                                                                             # dataset
xlim     = c(x_min, x_max),                                                    # x-axis limits
ylim     = c(y_min, y_max),                                                    # y-axis limits
ylab     = "frequency",                                                        # y-axis label
xlab     = "",                                                                 # x-axis label
main     = "R Default")                                                        # title
mtext(LETTERS[6], adj=0, line=2, cex = 1.2, at = 8)                            # subplot letter
dev.off()                                                                      # end image editing
```

10.3 Empirical distribution functions

In addition to the question of how frequently certain values or classes of values occur in a data set, it can be interesting to examine how frequently values that are smaller than a certain value occur. Although the answer to this question is of course implicit in the absolute and relative frequencies or the corresponding histograms, it is sometimes advantageous to have this information directly available. The concepts of *cumulative absolute and relative frequency distribution* and *empirical distribution function* introduced in this section denote the corresponding analogues to the concepts discussed in Section 10.1 and Section 10.2. We begin by defining the *cumulative absolute* and *cumulative relative frequency distributions*.

Definition 10.3 (Cumulative absolute and relative frequency distributions)**.** Let $y = (y_1, ..., y_n)$ be a data set, $W := \{w_1, ..., w_k\}$ with $k \le n$ the numerical values occurring in the data set and h and r the absolute and relative frequency distributions of the values of y, respectively. Then the function is called

$$H : W \to \mathbb{N}, w \mapsto H(w) := \sum_{w' \le w} h(w') \tag{10.9}$$

the *cumulative absolute frequency distribution* of y and the function

$$R : W \to [0, 1], w \mapsto R(w) := \sum_{w' \le w} r(w') \tag{10.10}$$

the *cumulative relative frequency distribution* of the numerical values of y.

•

In the sense of the definition Definition 10.1 applies to the cumulative frequency distributions

$$H(w) = \text{number of } y_i \text{ in } y \text{ with } y_i \le w \tag{10.11}$$

and

$$R(w) = \text{number of } y_i \text{ in } y \text{ with } y_i \le w \text{ divided by } n. \tag{10.12}$$

In **R**, the function `cumsum()` allows the calculation of cumulative sums and thus enables the evaluation of cumulative frequency distributions, as the following **R** code demonstrates using the example data set.

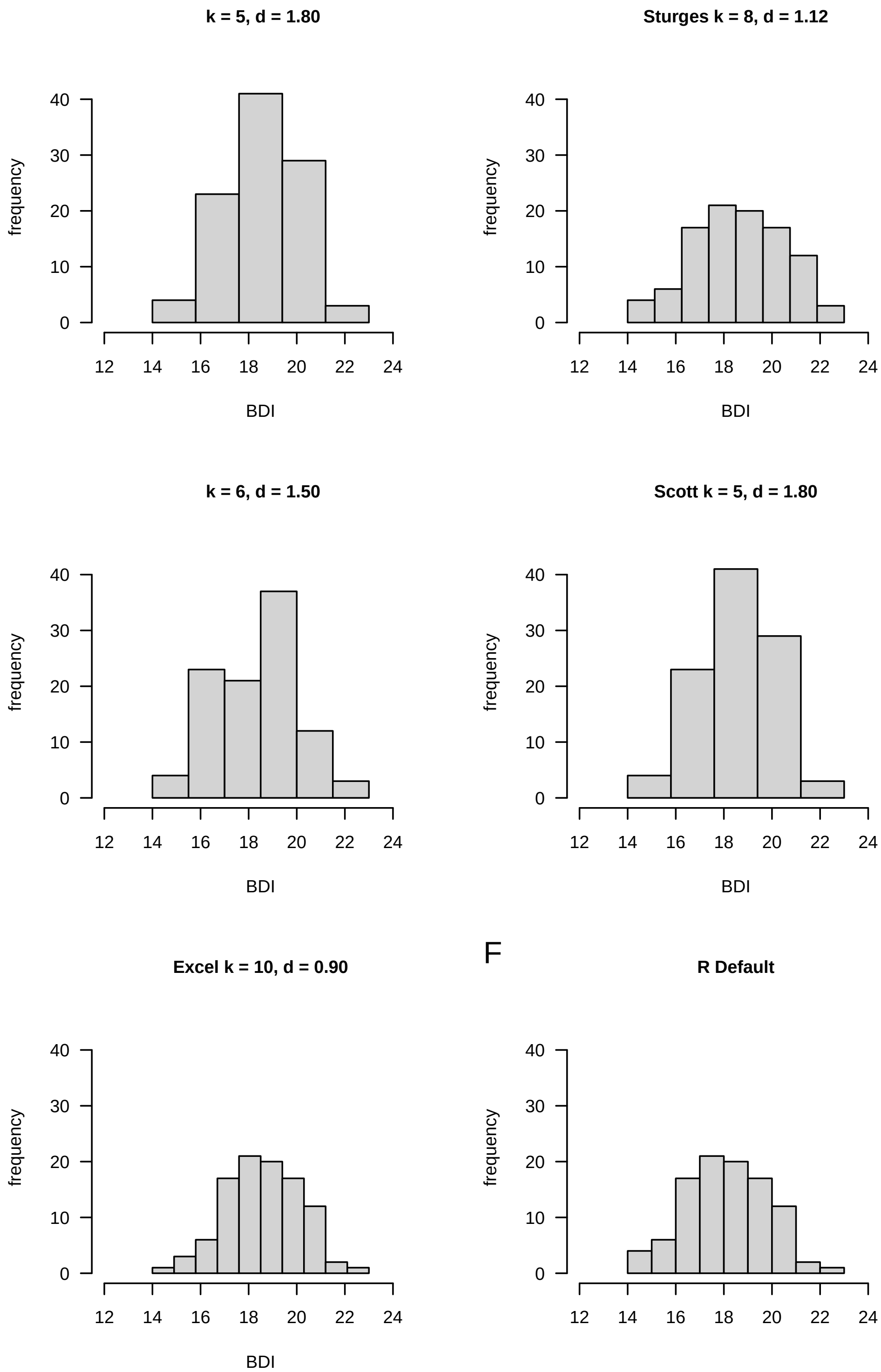

Figure 10.2 Histograms of the absolute frequencies of the pre-intervention BDI-II scores. Note that the qualitative impression that values around 18 occur frequently in the data set and that lower and higher values are rarer is constant across all choices of the number of classes. The exact quantitative appearance, however, depends on the exact choice of the number of classes and the class width.

```
D        = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y        = D$PRE                                               # pre-intervention BDI-II data
n        = length(y)                                           # number of data values
H        = as.data.frame(table(y))                             # absolute frequency distribution as a data frame
names(H) = c("w", "h")                                         # column naming
H$H      = cumsum(H$h)                                         # cumulative absolute frequency distribution
H$r      = H$h/n                                               # relative frequency distribution
H$R      = cumsum(H$r)                                         # cumulative relative frequency distribution
print(H, digits = 1)
```

```
    w  h   H    r    R
1  14  1   1 0.01 0.01
2  15  3   4 0.03 0.04
3  16  6  10 0.06 0.10
4  17 17  27 0.17 0.27
5  18 21  48 0.21 0.48
6  19 20  68 0.20 0.68
7  20 17  85 0.17 0.85
8  21 12  97 0.12 0.97
9  22  2  99 0.02 0.99
10 23  1 100 0.01 1.00
```

Pay particular attention to the comparison of `h` and `H` as well as `r` and `R`, which illustrate the cumulative totals in Definition 10.3. The following **R** code enables the representation of the cumulative frequency distributions determined as shown in Figure 10.3. Obviously, the absolute and relative frequency of values less than the smallest value in the data set is zero. On the other hand, the absolute and relative frequency of values smaller than the largest value in the data set are n and $n/n = 1$, respectively.

```
pdf(                                                                 # pdf copy function
file       = "./_figures/110-cumulative-frequency-distributions.pdf",  # filename
width      = 10,                                                     # width (inches)
height     = 5)                                                      # height (inches)
Hw         = H$H                                                     # h(w) values
Rw         = H$R                                                     # r(w) values
names(Hw)  = H$w                                                     # barplot needs w values as names
names(Rw)  = H$w                                                     # barplot needs w values as names
par(                                                                 # figure parameters
mfcol      = c(1,2),                                                 # rows and columns layout
las        = 1,                                                      # x-tick orientation
cex        = 1)                                                      # annotation enlargement
barplot(                                                             # bar chart
Hw,                                                                  # h(w) values
col        = "gray90",                                               # bar color
xlab       = "w",                                                    # x-axis label
ylab       = "H(w)",                                                 # y-axis label
ylim       = c(0,110),                                               # y-axis limits
las        = 1,                                                      # axis tick label orientation
main       = "Absolute cumulative frequency PRE")                    # title
barplot(                                                             # bar chart
Rw,                                                                  # r(w) values
col        = "gray90",                                               # bar color
xlab       = "w",                                                    # x-axis label
ylab       = "R(w)",                                                 # y-axis label
ylim       = c(0,1),                                                 # y-axis limits
las        = 1,                                                      # axis tick label orientation
main       = "Relative cumulative frequency PRE")                    # title
dev.off()                                                            # end image editing
```

The cumulative analogue of a histogram is the *empirical distribution function*. In contrast to a histogram, no definition of classes is required to determine an empirical distribution function. Instead of the interval division of the real numbers between the minimum and the maximum of a data set, the empirical distribution function simply uses the real numbers, as the following definition makes clear.

Definition 10.4 (Empirical distribution function)**.** Let $y = (y_1, ..., y_n)$ be a data set. Then the function is called

$$F : \mathbb{R} \to [0, 1], x \mapsto F(x) := \frac{\text{number of } y_i \text{ in } y \text{ with } y_i \le x}{n} \tag{10.13}$$

the empirical distribution function (EVF) of y.

•

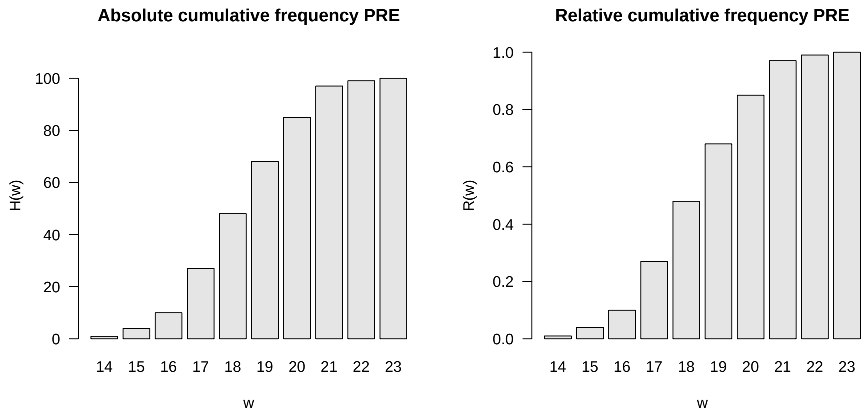

Figure 10.3 Absolute and relative cumulative frequency distributions of pre-intervention BDI-II scores.

The empirical distribution function is sometimes called the *empirical cumulative distribution function*. In general, the empirical distribution function as in Definition 10.4 is limited to specifying the relative frequencies. Of course, not all real numbers appear in data sets. This implies that the relative frequency assigned to real numbers that lie between two adjacent values in the data set is equal to that of the smaller of the two values. Typically, empirical distribution functions are step functions. In **R**, the function `ecdf()` can be used to evaluate and visualize the empirical distribution function, as the following **R** code demonstrates using the example data set and is visualized in Figure 10.4. For example, you can read from Figure 10.4 that the relative frequency of values smaller than 17.9 is approximately 0.3. Values smaller than 17.8 have the same relative frequency.

```
D      = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y      = D$PRE                                               # pre-intervention BDI-II data
evf    = ecdf(y)                                             # evaluation of the EVF
pdf(                                                         # pdf copy function
file   = "./_figures/110-ecdf.pdf",                          # filename
width  = 8,                                                  # width (inches)
height = 5)                                                  # height (inches)
plot(                                                        # plot() knows how to handle ecdf object
evf,                                                         # ecdf object
xlab   = "BDI",                                              # x-axis label
ylab   = "Relative frequency",                               # y-axis label
main   = "Empirical distribution function")                  # title
dev.off()                                                    # end image editing
```

10.4 Quantiles and boxplots

The empirical distribution function provides an answer, at least graphically represented, to the question of how large the relative proportion of values in a data set is that are smaller than a given value. The quantiles introduced in this section can be understood as an inversion of this data set interrogation. Specifically, with quantiles you specify a relative proportion $0 \le p \le 1$ of the values of a data set and ask which value of the data set is such that $p \cdot 100\%$ of the values of the data set are less than or equal to this value. We use the following definition of the term p *quantile*.

Definition 10.5 (p quantile)**.** Let $y = (y_1, ..., y_n)$ be a data set and

$$y_s = \left(y_{(1)}, y_{(2)}, ..., y_{(n)}\right) \text{ with } \min_{1 \le i \le n} y_i = y_{(1)} \le y_{(2)} \le \cdots \le y_{(n)} = \max_{1 \le i \le n} y_i \tag{10.14}$$

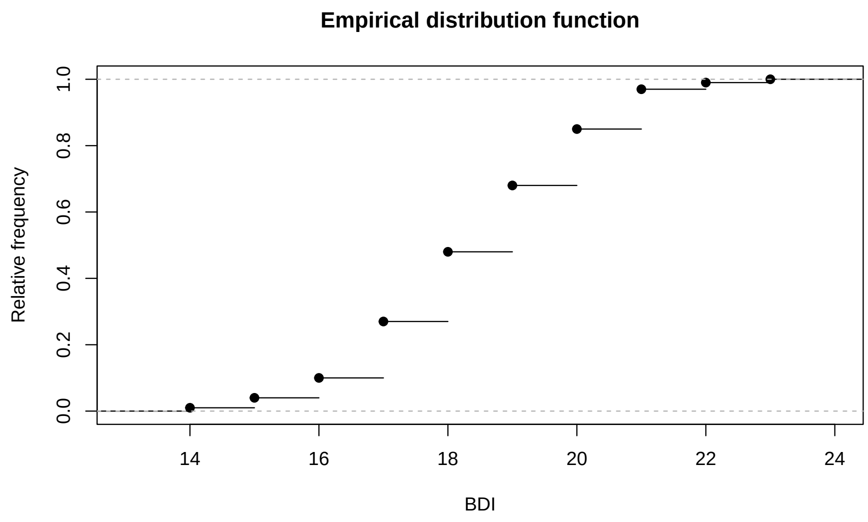

Figure 10.4 Empirical distribution function of the PRE values.

be the corresponding data set sorted in ascending order. Furthermore, let $\lfloor \cdot \rceil$ denote the rounding function. Then for a $p \in [0, 1]$ is the number

$$y_p := \begin{cases} y_{(\lfloor np+1 \rceil)} & \text{if } np \notin \mathbb{N} \\ \frac{1}{2}\left(y_{(np)} + y_{(np+1)}\right) & \text{if } np \in \mathbb{N} \end{cases} \tag{10.15}$$

the p *quantile* of the data set y.

•

Note that the p quantile y_p to Definition 10.5 is either a value of the data set or a value between two values of the data set. After constructing y_p, it applies in particular that at least $p \cdot 100\%$ of all values in the data set are less than or equal to y_p and at least $(1-p) \cdot 100\%$ of all values in the data set are greater than or equal to y_p. The p quantile therefore divides the ordered data set y_s in the ratio p to $(1-p)$. Depending on the choice of p, the p quantiles receive special names:

- $y_{0.25}, y_{0.50}, y_{0.75}$ are called *lower quartile*, *median* and *upper quartile*, respectively,
- $y_{j \cdot 0.10}$ for $j = 1, ..., 9$ are called *deciles* and
- $y_{j \cdot 0.01}$ for $j = 1, ..., 99$ are called *percentiles*.

To clarify the concept of the p quantile, let us consider an example based on Henze (2018), chapter 5. First of all, the data set shown in the table below and its version sorted in ascending order are given.

i	1	2	3	4	5	6	7	8	9	10
y_i	8.5	1.5	75	4.5	6.0	3.0	3.0	2.5	6.0	9.0
$y_{(i)}$	1.5	2.5	3.0	3.0	4.5	6.0	6.0	8.5	9.0	75

We first want to determine the lower quartile, i.e. the 0.25 quantile, of this data set. It's apparently $n = 10$ and we chose $p := 0.25$. Based on Definition 10.5 we find that

$$np = 10 \cdot 0.25 = 2.5 \notin \mathbb{N}. \tag{10.16}$$

So according to Definition 10.5 it applies that

$$y_{0.25} = y_{(\lfloor 2.5+1 \rfloor)} = y_{(3)} = 3.0. \tag{10.17}$$

The lower quartile of the data set under consideration is therefore the value 3.0 of the data set.

Next we want to determine the 0.80 quantile of the data set. Apparently it is still $n = 10$ and we have now chosen $p := 0.80$. In this case we find that

$$np = 10 \cdot 0.80 = 8 \in \mathbb{N}. \tag{10.18}$$

So follows after Definition 10.5

$$y_{0.80} = \frac{1}{2}\left(y_{(8)} + y_{(8+1)}\right) = \frac{1}{2}\left(y_{(8)} + y_{(9)}\right) = \frac{8.5 + 9.0}{2} = 8.75. \tag{10.19}$$

The $y_{0.80}$ quantile of the data set under consideration is with 8.75 the value between the values 8.5 and 9 of the data set. In **R** the quantiles determined in this way are evaluated as follows.

```
y   = c(8.5,1.5,75,4.5,6.0,3.0,3.0,2.5,6.0,9.0)  # sample data
n   = length(y)                                   # number of data values
y_s = sort(y)                                     # sorted data set
p   = 0.25                                        # np \notin \mathbb{N}
y_p = y_s[floor(n*p + 1)]                         # 0.25 quantile
cat("0.25 quantile: ", round(y_p,2))              # output
```

```
0.25 quantile:  3
```

```
p   = 0.80                                        # np \in \mathbb{N}
y_p = (1/2)*(y_s[n*p] + y_s[n*p + 1])             # 0.80 quantile
cat("0.80 quantile: ", round(y_p,2))              # output
```

```
0.80 quantile:  8.75
```

With `quantile()`, **R** also provides a function that automates the quantile determination of a data set. However, it should be noted that the definition of the quantile term given in Definition 10.5 is only one of at least nine different quantile definitions and the exact values can differ from definition to definition. The specific quantile definition used by `quantile()` is selected via the function argument `type`. Hyndman & Fan (1996) provide an overview on the theoretical level, and `?quantile` on the implementation level. As an example, we look at determining the 0.25 quantile of the pre-intervention BDI-II sample data set using `type = 8`.

```
D   = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y   = D$PRE                                               # pre-intervention BDI-II data
y_p = quantile(y, 0.25, type = 8)                         # 0.25 quantile, definition 1
```

```
0.25 quantile (type = 8):  17
```

Finally, a *boxplot* visualizes a quantile-based summary of a data set, typically visualizing the minimum and maximum of a data set as well as the lower quartile, median and upper quartile. In **R** you create a boxplot using the `boxplot()` function, as the following **R** code demonstrates using the pre-intervention BDI-II data set.

```
D          = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
pdf(                                                            # pdf copy function
file       = "./_figures/110-boxplot.pdf",                      # filename
width      = 8,                                                 # width (inches)
height     = 5)                                                 # height (inches)
boxplot(                                                        # box plot
D$PRE,                                                          # dataset
horizontal = T,                                                 # horizontal representation
range      = 0,                                                 # whiskers up to min y and max y
xlab       = "BDI",                                             # x axis label
main       = "Box plot")                                        # title
dev.off()                                                       # end image editing
```

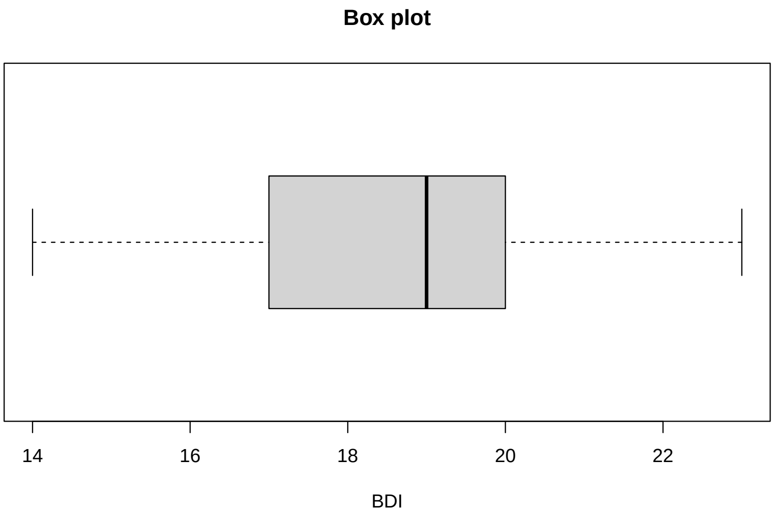

Figure 10.5 Boxplot of the pre-intervention BDI-II scores of the example data set.

In Figure 10.5, $\min y$ and $\max y$ are represented as so-called "whisker endpoints", $y_{0.25}$ and $y_{0.75}$ form the lower and upper boundaries of the central gray box and the median $y_{0.50}$ is represented as a line in the central gray box. Here too, there are a lot of box plot variations, see McGill et al. (1978). A precise explanation of the data set properties displayed graphically is always necessary. Finally, it should be noted that the representation of data sets using boxplots has become somewhat out of fashion in the last twenty years and is only very rarely found in current scientific publications.

10.5 Measures of central tendency

In this section we look at some basic data set metrics that are central to descriptive statistics and provide information about the "average" value of a data set. We start by defining the mean.

10.5.1 Mean

Definition 10.6 (Mean)**.** Let $y = (y_1, ..., y_n)$ be a data set. Then that means

$$\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i \tag{10.20}$$

the *mean* of y.

•

The mean of a data set is the sum of the data values divided by the number of data values. The mean defined here is also referred to as the *arithmetic mean* or *average*. In the context of frequentist inference, we will refer to the mean as the *sample mean* in the context of the random sampling model. In particular, the models of frequentist inference also allow the functional form of the mean to be more deeply substantiated, as we will see at a given point. Using the sum function, you can calculate the mean of a data set in **R** as follows.

```
D     = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y     = D$PRE                                               # pre-intervention BDI-II data
n     = length(y)                                           # number of values
y_bar = (1/n)*sum(y)                                        # average calculation
```

```
Mean of PRE-BDI-II data 18.61
```

With the `mean()` function, **R** of course also provides a function for automated calculation of the mean value.

```
cat("Mean of PRE-BDI-II data", mean(y))  # output
```

```
Mean of PRE-BDI-II data 18.61
```

In the following theorem we first note two properties of the mean value of a data set that make its classification as a measure of the central tendency of a data set plausible.

Theorem 10.1 (Centrality properties of the mean)**.** *Let $y = (y_1, ..., y_n)$ be a data set and $\bar{y}$ be the mean of y. Then apply*

(1) The sum of the deviations from the mean is zero,

$$\sum_{i=1}^{n}(y_i - \bar{y}) = 0. \tag{10.21}$$

(2) The absolute sums of negative and positive deviations from the mean are equal, i.e. if $j = 1, ..., n_j$ denotes the data point indices with $(y_j - \bar{y}) < 0$ and $k = 1, ..., n_k$ denotes the data point indices with $(y_k - \bar{y}) \geq 0$, then with $n = n_j + n_k$ it applies that

$$|\sum_{j=1}^{n_j}(y_j - \bar{y})| = |\sum_{k=1}^{n_k}(y_k - \bar{y})|. \tag{10.22}$$

∘

Proof. (1) It applies

$$\begin{aligned}
\sum_{i=1}^{n}(y_i - \bar{y}) &= \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \bar{y} \\
&= \sum_{i=1}^{n} y_i - n\bar{y} \\
&= \sum_{i=1}^{n} y_i - \frac{n}{n}\sum_{i=1}^{n} y_i \\
&= \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} y_i \\
&= 0.
\end{aligned} \tag{10.23}$$

(2) Let $j = 1, ..., n_j$ be the indices with $(y_j - \bar{y}) < 0$ and $k = 1, ..., n_k$ the indices with $(y_k - \bar{y}) \geq 0$, such that $n = n_j + n_k$. Then applies

$$
\begin{aligned}
& \sum_{i=1}^{n}(y_i - \bar{y}) = 0 \\
\Leftrightarrow & \sum_{j=1}^{n_j}(y_j - \bar{y}) + \sum_{k=1}^{n_k}(y_k - \bar{y}) = 0 \\
\Leftrightarrow & \sum_{k=1}^{n_k}(y_k - \bar{y}) = -\sum_{j=1}^{n_j}(y_j - \bar{y}) \\
\Leftrightarrow & |\sum_{j=1}^{n_j}(y_j - \bar{y})| = |\sum_{k=1}^{n_k}(y_k - \bar{y})|.
\end{aligned}
\tag{10.24}
$$

□

According to Theorem 10.1, the mean value of a data set is just such that the total deviations of the data points from this value equalize to zero and that positive and negative deviations from the mean balance each other out. In **R** you can verify Theorem 10.1 using the example data set as follows.

```
# dataset
D   = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y   = D$PRE                                                 # pre-intervention BDI-II data
s   = sum(y - mean(y))                                      # sum of deviations from the mean
s_1     = sum(y[y <= mean(y)] - mean(y))                    # sum of all negative deviations
s_2 = sum(y[y > mean(y)]  - mean(y))                        # sum of all positive deviations
```

```
Sum of deviations from the mean: 0
Sums of positive and negative deviations: -71.28 71.28
```

Theorem 10.2 (Summation properties of the mean). *Given two data sets $y = (y_1, ..., y_n)$ and $z = (z_1, ..., z_n)$ of the same size and let it be by*

$$
y + z := (y_1 + z_1, ..., y_n + z_n) \tag{10.25}
$$

the sum of these data sets is defined. Furthermore, let $\bar{y}$ and $\bar{z}$ be the mean values of the data sets y and z, respectively. Then

$$
\overline{y + z} = \bar{y} + \bar{z} \tag{10.26}
$$

∘

Proof. It applies

$$
\begin{aligned}
\overline{y+z} & := \frac{1}{n}\sum_{i=1}^{n}(y_i + z_i) \\
& = \frac{1}{n}\sum_{i=1}^{n} y_i + \frac{1}{n}\sum_{i=1}^{n} z_i \\
& =: \bar{y} + \bar{z}.
\end{aligned}
\tag{10.27}
$$

□

If you look at the sum of two data sets, it is irrelevant to determining their mean value whether you first add up the data sets and then determine the mean value of the summed data set or whether you determine the mean value for each of the two data sets and add them up. We verify this result using the pre- and post-intervention BDI-II data from the example data set in **R** as follows.

```
D         = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y         = D$PRE                                               # pre-intervention BDI-II data
y_bar     = mean(y)                                             # mean of pre-intervention BDI-II data
z         = D$POS                                               # post-intervention BDI-II data
z_bar     = mean(z)                                             # mean of post-intervention BDI-II data
yz_bar_1  = mean(y + z)                                         # mean of summed pre and post data
yz_bar_2  = y_bar + z_bar                                       # sum of the means of the pre- and post-data
```

```
Mean of z: 31.68
Sum of the means of y and z :  31.68
```

Theorem 10.3 (Mean under linear-affine transformation). *Let $y = (y_1, ..., y_n)$ be a data set and $\bar{y}$ be the mean of y. Furthermore, be for $a, b \in \mathbb{R}$*

$$ay + b := (ay_1 + b, ..., ay_n + b) \tag{10.28}$$

a linear-affine transformed data set of y is defined. Then that applies

$$\overline{ay + b} = a\bar{y} + b \tag{10.29}$$

∘

Proof. It applies

$$\begin{aligned}
\overline{ay + b} &:= \frac{1}{n}\sum_{i=1}^{n}(ay_i + b) \\
&= \sum_{i=1}^{n}\left(\frac{1}{n}ay_i + \frac{1}{n}b\right) \\
&= \left(\sum_{i=1}^{n}\frac{1}{n}ay_i\right) + \left(\sum_{i=1}^{n}\frac{1}{n}b\right) \\
&= a\left(\frac{1}{n}\sum_{i=1}^{n}y_i\right) + \frac{1}{n}\sum_{i=1}^{n}b \\
&= a\bar{y} + \frac{1}{n}nb \\
&= a\bar{y} + b.
\end{aligned} \tag{10.30}$$

□

A linear-affine transformation of a data set transforms the mean value of the data set in a linear-affine manner. You can take advantage of this if you think about what the mean value should be when rescaling a data set, for example converting from grams to kilograms. Of course, to be on the safe side, you can also simply determine the average of the rescaled data set. We demonstrate this property as an example by rescaling the pre-intervention BDI-II data set with the value $a = 2$ while simultaneously adding the constant $b := 5$.

```
D         = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y         = D$PRE                                               # pre-intervention BDI-II data
y_bar     = mean(y)                                             # mean of pre-intervention BDI-II data
a         = 2                                                   # multiplication constant
b         = 5                                                   # addition constant
z         = a*y + b                                             # linear-affine transformation of the PRE data
z_bar     = mean(z)                                             # mean value of transformed PRE data
ay_bar_b  = a*y_bar + b                                         # pre data mean transformation
```

```
Mean value of the transformed data set: 42.22
Transformation of the mean       : 42.22
```

10.5.2 Median

In the mean value defined above, each value of a data set is included summatively with the same weight $1/n$. This particularly applies to values that are very atypical compared to the other values in the data set. One way to determine the “center” of a data set independent of such atypical values is to determine its median, which we define below. We already learned about the median in section Section 10.4 as the 0.5 quantile, with which it is identical.

Definition 10.7 (Median)**.** Let $y = (y_1, ..., y_n)$ be a data set and $y_s = (y_{(1)}, ..., y_{(n)})$ the corresponding data set sorted in ascending order. Then the median of y is defined as

$$\tilde{y} := \begin{cases} y_{((n+1)/2)} & \text{if } n \text{ ungerade,} \\ \frac{1}{2}\left(y_{(n/2)} + y_{(n/2+1)}\right) & \text{if } n \text{ gerade.} \end{cases} \tag{10.31}$$

•

Definition 10.7 implies that at least 50% of all data values y_i of the data set are less than or equal to $\tilde{y}$ and at the same time at least 50% of all data values y_i of the data set are greater than or equal to $\tilde{y}$. Instead of a formal proof of these statements, we refer to Figure 10.6. As in the definition of the median, we differentiate between the cases where the number of data points is odd (here $n = 5$, Figure 10.6 A) or even ($n = 6$, Figure 10.6 B). In the case of an odd number of data points, the median is the data value with the index number that lies in the middle of the index numbers of the sorted data set, in the example Figure 10.6 A, i.e. the value $y_{(3)}$. In this case, $3/5$, i.e. 60% of the data values (specifically $y_{(1)}$, $y_{(2)}$ and $y_{(3)}$) are less than or equal to the median, but at the same time $3/5$, i.e. 60% of the data values (specifically $y_{(3)}$, $y_{(4)}$ and $y_{(5)}$) greater than or equal to the median. In the case of an even number of data points, the location of the median is somewhat more intuitive: the median lies in the middle of the two middle values of the sorted data set (in Figure 10.6 B specifically in the middle of the values $y_{(3)}$ and $y_{(4)}$). This means that $3/6$, i.e. 50% of the data values (specifically $y_{(1)}$, $y_{(2)}$ and $y_{(3)}$) are smaller than the median and $3/6$, i.e. 50% of the data values (specifically $y_{(4)}$, $y_{(5)}$ and $y_{(6)}$) are larger than the median.

The following **R** code determines the median of the example data set using Definition 10.7.

```
D       = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y       = D$PRE                                               # pre-intervention BDI-II data
n       = length(y)                                           # number of values
y_s     = sort(y)                                             # ascending sorted vector
if(n %% 2 == 1){                                              # n odd, n mod 2 == 1
   y_tilde = y_s[(n+1)/2]                                     # median formula, case n odd
} else {                                                      # n even, n mod 2 == 0
   y_tilde = (y_s[n/2] + y_s[n/2 + 1])/2}                     # median formula, case n even
```

```
Median of PRE-BDI-II data: 19
```

Of course, **R** also provides a function with `median()` for directly determining the median, as the following **R** code demonstrates.

```
cat("Median of PRE-BDI-II data:", median(y))  # output
```

```
Median of PRE-BDI-II data: 19
```

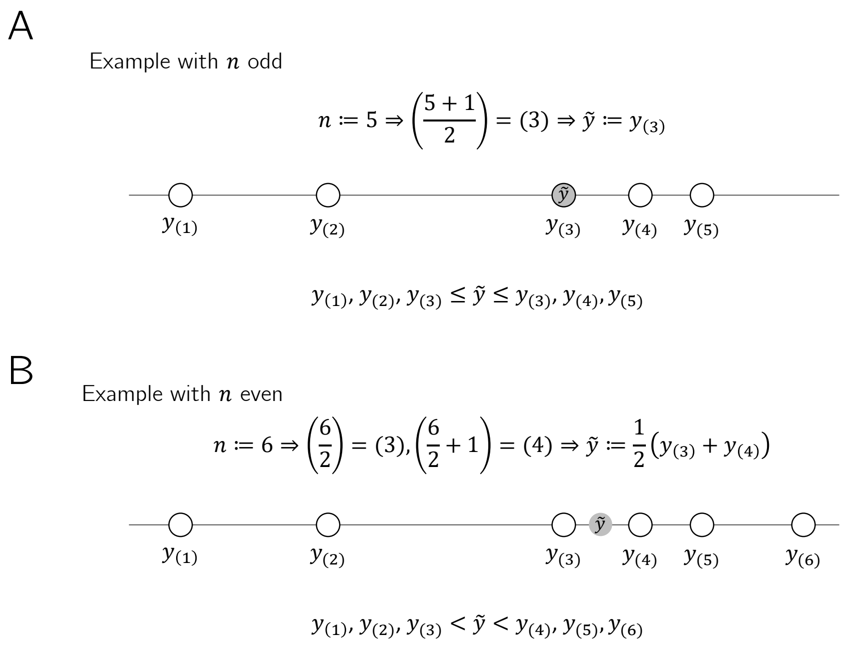

Figure 10.6 Determination and properties of the median.

The following simulation shows as an example that the median, as a measure of the middle of a data set, is less susceptible to outliers than the mean.

```
D = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y = D$PRE                                               # pre-intervention BDI-II data
```

```
Mean and median for data set without outliers: 18.61 19
```

```
z    = y      # new simulated data set
z[1] = 10000  # ... with an extreme value
```

```
Mean and median for data set with outliers: 118.44 19
```

So, although the median appears to be a more robust measure of the central tendency of a data set than the mean, determining averages is certainly the more common approach. On the one hand, this is due to the fact that the mean value is easier to use in an analytical context for designing probabilistic models due to its mathematically less complex definition. On the other hand, the occurrence of extreme values is usually a sign of data collection errors (e.g. a BDI-II value of 10.000 simply should not occur) or large differences between the experimental units under consideration, which are usually noticed by visual inspection of the data set before determining measures of central tendency. However, there is a whole subfield of statistics that deals with - in the broadest sense - measures that are independent of outliers, see e.g. Huber (1981).

10.5.3 Mode

If some values occur repeatedly in a data set, a third measure of central tendency is given with the *mode* as the value that occurs most frequently in a data set. We use the following definition.

Definition 10.8 (Mode). $y := (y_1, ..., y_n)$ with $y_i \in \mathbb{R}$ is a data set, $W := \{w_1, ..., w_k\}$ with $k \leq n$ are the different numerical values occurring in the data set and $h : W \to \mathbb{N}$ is the absolute frequency distribution of the numerical values of y. Then the *mode (or mode)* of y is defined as

$$\text{argmax}_{w \in W} h(w), \tag{10.32}$$

i.e. the value that occurs most frequently in the data set.

•

In **R** you can determine the mode of a corresponding data set as follows.

```
D        = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y        = D$PRE                                               # pre-intervention BDI-II data
n        = length(y)                                           # number of data values (100)
H        = as.data.frame(table(y))                             # absolute frequency distribution (data frame)
names(H) = c("a", "w")                                         # consistent naming
mode     = H$w[which.max(H$w)]                                 # mode
```

```
cat("Mode of PRE-BDI-II data:", as.numeric(as.vector(mode)))  # output as numeric vector, not factor
```

```
Mode of PRE-BDI-II data: 21
```

10.5.4 Data distribution shapes and measures of central tendency

How useful it is to use the mean, median or mode to provide information about the central tendency of a data set depends largely on the nature of the data set. For example, if you have a data set of high-resolution data in which no data value appears multiple times, the mode is certainly not suitable for making a quantitative statement about the central tendency of the data set. The specification of a measure of central tendency should always be viewed in the context of the overall nature of the data set and, if applicable, the inferential statistical model that is used to interrogate the data set. In Figure 10.7 we give four examples of how the measures of central tendency considered above can behave in different data value distribution scenarios. Of course, these four scenarios are not exhaustive and each data set should be viewed critically for a meaningful measure of central tendency.

Figure 10.7 A visualizes the position of the mean, median and mode in a data set that is symmetrically distributed around a value of approximately $y = 50$ and for which values close to this central value occur frequently and values far from this central value occur rarely. As we see later, this corresponds to the typical characteristics of normally distributed data values. In this case, the mean, median and mode are quite close to each other and each of these measures in fact describes the central tendency of the data set.

Figure 10.7 B visualizes the position of the mean, median and mode in a data set in which the data values occur clustered around two values of approximately $y = 25$ and $y = 75$ and rarely occur in the positive and negative directions from these values. For example, there are very few data points in this data set around $y = 50$. The mean and median now take on values around $y = 50$. As such, they do not provide any information about which values occur particularly frequently in this data set, but only about where the average "physical center of gravity" of the data set is. The mode, on the other hand, is at the larger of the two data accumulation points due to the slightly more frequent occurrence of values around $y = 75$. Overall, none of the three measures really satisfactorily describes the central tendency of the data values for such a data set, which is also called *bimodal*.

Figure 10.7 C visualizes the position of the mean, median and mode in a data set in which the data values in the width under consideration occur with approximately the same frequency throughout. Here the mode is relatively arbitrary at around $y = 15$, as there is a minimal accumulation of data values there. In the sense of the "physical center of gravity" of the data set, the mean and median are around $y = 50$, even if the values of this data set have no tendency to occur more frequently there than elsewhere. Here too, the description of the central tendency of the data set by the measures considered is not really satisfactory. Taken together, using Figure 10.7 B and Figure 10.7 C, one might be able to conclude that if a data set does not have a clear central tendency, the measures of central tendency cannot reflect it either.

Finally, Figure 10.7 D shows the location of the mean, median and mode for a data set that is skew-symmetrically distributed around values of approximately $y = 0.5$. The data in this data set only takes positive values, many data values are in fact around $y = 0.5$, but there are also higher data values. We will see later that this appearance is typical for squared normally distributed data values. In a psychological context, reaction times in particular often have a similar distribution, as they cannot be negative, in most cases being in the range of 0.3 to 0.6 seconds, but in some cases can be much longer due to inattention or stimulus variability. In this case, the median describes the actual central tendency of the data values somewhat better than the mean, as this is slightly shifted towards higher values due to the rarely occurring outlier values.

Overall, the specification of one or more measures of central tendency must always be viewed critically against the background of the distribution of the values of a data set and classified qualitatively.

10.6 Measures of variability

In addition to the question of what "average value" a data set assumes, it is often of interest to quantify how much the values in the data set are scattered or, in other words, how "variable" they are. Here we consider the *range*, the *empirical variance* and the *empirical standard deviation* as quantitative measures of the variability of a data set. As with the mean, the full meaning of the latter measures becomes apparent primarily in the context of inferential statistical methods.

10.6.1 Range

The range of a data set is the difference between its largest and smallest values.

Definition 10.9 (Range)**.** Let $y = (y_1, ..., y_n)$ be a data set. Then the *range* of $y_1, ..., y_n$ is defined as

$$r := \max(y_1, ..., y_n) - \min(y_1, ..., y_n). \tag{10.33}$$

•

In **R** you determine the range either by evaluating the maximum and minimum of the data set using the `max()` and `min()` functions or directly using the `range()` function. The following **R** code demonstrates the use of the `max()` and `min()` functions.

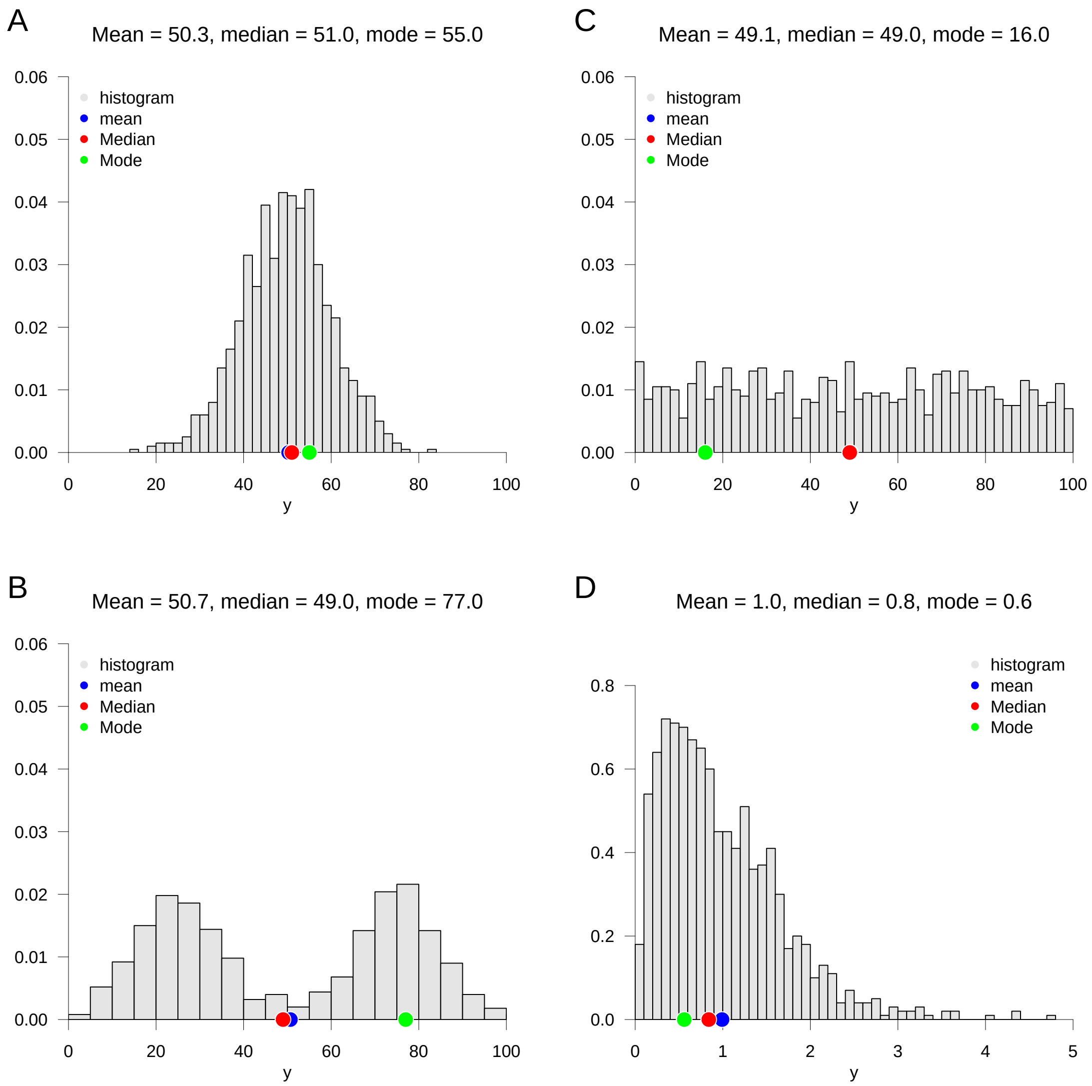

Figure 10.7 Visual intuition on measures of central tendency in different data distribution forms

```
D     = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y     = D$PRE                                                # pre-intervention BDI-II data
y_max = max(y)                                               # maximum of PRE values
y_min = min(y)                                               # minimum of PRE values
r     = y_max - y_min                                        # range
```

```
Range of PRE-BDI-II data: 9
```

The following **R** code demonstrates the use of the `range()` function.

```
MinMax = range(y)              # "automatic" calculation of min(x), max(x)
r      = MinMax[2] - MinMax[1]  # range
```

```
Range of PRE-BDI-II data: 9
```

10.6.2 Empirical variance

A typical statistical measure of the variability of data sets is the standardized sum of the squared differences between the individual data values and their mean. The squared differences between the individual data values and their mean are also referred to as *difference squares*. Depending on the form of standardization, i.e. the formation of an average of the squared deviations, a distinction is made between *uncorrected empirical variance* and *empirical variance*, as the following definition states.

Definition 10.10 (Uncorrected empirical variance and empirical variance)**.** Let $y = (y_1, ..., y_n)$ be a data set with $n > 1$ and $\bar{y}$ be its mean. Then the *uncorrected empirical variance* of y is defined as

$$\tilde{s}^2 := \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 \tag{10.34}$$

and the *empirical variance* of y is defined as

$$s^2 := \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2. \tag{10.35}$$

•

We saw in Theorem 10.1 that the sum of the differences between the data values and their mean is always zero. Squaring the differences

$$y_i - \bar{y} \text{ for } i = 1, ..., n \tag{10.36}$$

in Definition 10.10 now leads to the fact that negative and positive differences between data points do not balance each other out and, depending on the level of the positive and negative differences, with

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 \tag{10.37}$$

a higher or lower measure of the overall deviation of the data values from their mean can be determined. The sum of the squares of the deviations is zero if all data values are identical and therefore identical to their mean. Alternatively, for example, the determination of the absolute amounts $|y_i - \bar{y}|$ would also be conceivable, but the resulting measure of variability has different properties than the measure of the squared deviations in Definition 10.10, which is why this is usually preferred, also in view of its proximity to probabilistic normal distribution models. The uncorrected empirical variance $\tilde{s}^2$ and the (corrected) empirical variance s^2 then differ in terms of the form of the averaging of the squared deviations. For $\tilde{s}^2$ the sum of the squares of the deviations is divided by n, for s^2 by $n-1$. So $\tilde{s}^2$ is always slightly smaller than s^2. However, this difference will be rather small for large n, think for example of the numerical difference between $\frac{1}{3}$ and $\frac{1}{2}$ for small n and between $\frac{1}{1001}$ and $\frac{1}{1000}$ for large n. Note that the empirical variance for $n = 1$ is not defined, but the uncorrected empirical variance can in principle be determined even in this special case. The term *uncorrected* (and implicitly *corrected*) cannot be understood at this point, but only makes sense against the background of a frequentist inference model and will be explained in the later section on parameter estimation.

We summarize the above discussion in the following theorem.

Theorem 10.4 (Ratio of uncorrected empirical variance and empirical variance)**.** *Let $y = (y_1, ..., y_n)$ be a data set with $n > 1$ and let $\tilde{s}^2$ and s^2 be its uncorrected empirical variance and empirical variance, respectively. Then apply*

(1) (conversion)

$$\tilde{s}^2 = \frac{n-1}{n}s^2 \text{ and } s^2 = \frac{n}{n-1}\tilde{s}^2. \tag{10.38}$$

(2) (inequality)

$$0 \le \tilde{s}^2 \le s^2. \tag{10.39}$$

∘

Proof. (1) After Definition 10.10 applies on the one hand

$$\begin{aligned}
\tilde{s}^2 &= \frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2 \\
\Leftrightarrow \frac{n-1}{n-1}\tilde{s}^2 &= \frac{n-1}{n-1}\frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2 \\
\Leftrightarrow \tilde{s}^2 &= \frac{n-1}{n}\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \\
\Leftrightarrow \tilde{s}^2 &= \frac{n-1}{n}s^2
\end{aligned} \tag{10.40}$$

and on the other

$$\begin{aligned}
s^2 &= \frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \\
\Leftrightarrow \frac{n}{n}s^2 &= \frac{n}{n}\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \\
\Leftrightarrow s^2 &= \frac{n}{n-1}\frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2 \\
\Leftrightarrow s^2 &= \frac{n}{n-1}\tilde{s}^2.
\end{aligned} \tag{10.41}$$

(2) With the non-negativity of the square function and $\frac{1}{n} > 0$ and $\frac{1}{n-1} > 0$ for $n > 1$, $\tilde{s}^2 \ge 0$ and $s^2 \ge 0$ result directly. Finally applies to $\sum_{i=1}^n (y_i - \bar{y})^2 \neq 0$

$$\frac{1}{n} < \frac{1}{n-1} \Leftrightarrow \frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2 < \frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \Leftrightarrow \tilde{s}^2 < s^2. \tag{10.42}$$

and for $\sum_{i=1}^n (y_i - \bar{y})^2 = 0$ apparently $\tilde{s}^2 = s^2$. So overall:

$$0 \le \tilde{s}^2 \le s^2. \tag{10.43}$$

□

The following **R** code demonstrates the determination of the uncorrected empirical variance and the empirical variance using the formulas specified in Definition 10.10.

```
D        = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y        = D$PRE                                               # pre-intervention BDI-II data
n        = length(y)                                           # number of values
s2_tilde = (1/n)*sum((y - mean(y))^2)                          # uncorrected empirical variance
s2       = (1/(n-1))*sum((y - mean(y))^2)                      # empirical variance
```

```
Uncorrected empirical variance of PRE-BDI-II data: 2.998
Empirical variance of PRE-BDI-II data: 3.028
```

The **R** function `var()` determines the empirical variance by default. Using statement (1) in Theorem 10.4, the uncorrected empirical variance can also be determined based on this function.

```
s2       = var(y)            # empirical variance
s2_tilde = ((n-1)/n)*var(y)  # uncorrected empirical variance
```

```
Uncorrected empirical variance of PRE-BDI-II data: 2.998
Empirical variance of PRE-BDI-II data: 3.028
```

The following theorem states how the empirical variance behaves when a data set is transformed with linear affinity.

Theorem 10.5 (Variance in linear-affine transformations)**.** *Let $y = (y_1, ..., y_n)$ be a data set with empirical variance S_y^2 and $z = (ay_1 + b, ..., ay_n + b)$ be the data set with empirical variance S_z^2 transformed linearly with $a, b \in \mathbb{R}$. Then applies*

$$s_z^2 = a^2 s_y^2. \tag{10.44}$$

∘

Proof.

$$\begin{aligned}
s_z^2 &= \frac{1}{n-1}\sum_{i=1}^n (z_i - \bar{z})^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n (ay_i + b - (a\bar{y} + b))^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n (ay_i + b - a\bar{y} - b)^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n (a(y_i - \bar{y}))^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n a^2(y_i - \bar{y})^2 \\
&= a^2\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \\
&= a^2 S_y^2.
\end{aligned} \tag{10.45}$$

□

For example, converting a data set from meters to centimeters changes the variance of the original data set by the factor 100^2. We reproduce Theorem 10.5 using the following **R** codes as an example.

```
y   = D$PRE                                                             # pre-intervention BDI-II data
s2y = var(y)                                                            # empirical variance of y_1,...,y_n
a   = 2                                                                 # multiplication constant
b   = 5                                                                 # addition constant
z   = a*y + b                                                           # z_i = ay_i + b
s2z = var(z)                                                            # empirical variance of z_1,...,z_n
cat("Empirical variance of z_1,...,z_n by direct calculation:", round(s2z,3))  # output
```

```
Empirical variance of z_1,...,z_n by direct calculation: 12.113
```

```
s2z = a^2*s2y                                                           # empirical variance of z_1,...,z_n
cat("Empirical variance of z_1,...,z_n according to theorem:", round(s2z,3))   # output
```

```
Empirical variance of z_1,...,z_n according to theorem: 12.113
```

The following theorem shows a way to determine the uncorrected empirical variance of a data set solely by determining mean values. The theorem finds its analytical equivalent in the variance shift theorem.

Theorem 10.6 (Shift theorem for uncorrected empirical variance). *Let $y = (y_1, ..., y_n)$ be a data set, $y^2 := (y_1^2, ..., y_n^2)$ be its element-wise square, and $\bar{y}$ and $\overline{y^2}$ be the mean values, respectively. Then applies*

$$\tilde{s}^2 = \overline{y^2} - \bar{y}^2 \tag{10.46}$$

∘

Proof.

$$\begin{aligned}
\tilde{s}^2 &:= \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}\left(y_i^2 - 2y_i\bar{y} + \bar{y}^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n} y_i^2 - 2\bar{y}\frac{1}{n}\sum_{i=1}^{n} y_i + \frac{1}{n}\sum_{i=1}^{n}\bar{y}^2 \\
&= \overline{y^2} - 2\bar{y}\bar{y} + \frac{1}{n}n\bar{y}^2 \\
&= \overline{y^2} - 2\bar{y}^2 + \bar{y}^2 \\
&= \overline{y^2} - \bar{y}^2.
\end{aligned} \tag{10.47}$$

□

We reproduce Theorem 10.6 using the following **R** codes as an example.

```
y        = D$PRE                          # pre-intervention BDI-II data
s2_tilde = mean(y^2) - (mean(y))^2        # \bar{y^2} - \bar{y}^2
cat("Uncorrected empirical variance according to the shift theorem:", round(s2_tilde,3))  # output
```

```
Uncorrected empirical variance according to the shift theorem: 2.998
```

Note that the shift theorem of the uncorrected empirical variance does not apply to the empirical variance, since in this case the multiplication factor $\frac{1}{n-1}$ does not lead to the corresponding mean values. In particular, we have already seen above that for the example data set in question, $s^2 = 3.028$.

10.6.3 Empirical standard deviation

As seen above, the empirical variance is the average squared deviation of a data set value from its mean. If, for example, the unit of the data set in a reaction time task is the millisecond, then subtracting and then squaring results in the unit millisecond^2, a not very intuitive unit. In order to obtain a measure of dispersion whose unit corresponds to the unit of the original data set, the square root function is often applied to the empirical variance. This leads to the concept of *empirical standard deviation*.

Definition 10.11 (Empirical standard deviation)**.** Let $y = (y_1, ..., y_n)$ be a data set. The *empirical standard deviation* of y is defined as

$$s := \sqrt{s^2} \tag{10.48}$$

and the *uncorrected empirical standard deviation* of y is defined as

$$\tilde{s} := \sqrt{\tilde{s}^2}. \tag{10.49}$$

•

In analogy to the properties of the empirical variance and its uncorrected version, the uncorrected empirical standard deviation can also easily be converted into the empirical standard deviation and vice versa, apparently the following apply here

$$\tilde{s} = \sqrt{\frac{n-1}{n}}s \text{ and } s = \sqrt{\frac{n}{n-1}}\tilde{s}. \tag{10.50}$$

The following **R** code demonstrates the determination of the uncorrected empirical standard deviation and the empirical standard deviation using the formulas specified in Definition 10.11.

```
y = D$PRE                              # pre-intervention BDI-II data
n = length(y)                          # number of values
s = sqrt((1/(n-1))*sum((y - mean(y))^2))  # standard deviation
```

```
Empirical standard deviation: 1.74
```

The **R** function `sd()` determines the empirical standard deviation by default.

```
s = sd(y)
```

```
Empirical standard deviation: 1.74
```

If you transform a data set with linear affinity, the multiplication constant of the transformation is only included in the transformation of the empirical standard deviation in terms of its amount. This is the statement of the following theorem.

Theorem 10.7 (Empirical standard deviation for linear-affine transformations)**.** *Let $y = (y_1, ..., y_n)$ be a data set with empirical standard deviation s_y and $z = (ay_1 + b, ..., ay_n + b)$ be the data set linearly-affinically transformed with $a, b \in \mathbb{R}$ with empirical standard deviation s_z. Then applies*

$$s_z = |a| s_y. \tag{10.51}$$

∘

Proof.

$$
\begin{aligned}
s_z &:= \left(\frac{1}{n-1}\sum_{i=1}^n (z_i - \bar{z})^2\right)^{1/2} \\
&= \left(\frac{1}{n-1}\sum_{i=1}^n \left(ay_i + b - (a\bar{y} + b)\right)^2\right)^{1/2} \\
&= \left(\frac{1}{n-1}\sum_{i=1}^n \left(a(y_i - \bar{y})\right)^2\right)^{1/2} \\
&= \left(\frac{1}{n-1}\sum_{i=1}^n a^2(y_i - \bar{y})^2\right)^{1/2} \\
&= \left(a^2\right)^{1/2}\left(\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2\right)^{1/2}.
\end{aligned}
\tag{10.52}
$$

So $s_z = as_y$ applies when $a \geq 0$ and $s_z = -as_y$ applies when $a < 0$. But this corresponds to $s_z = |a|s_y$. □

We reproduce Theorem 10.7 as an example using the following **R** codes once for the $a \geq 0$ case and once for the $a < 0$ case

```
# a >= 0
D  = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y  = D$PRE                                                # pre-intervention BDI-II data
a  = 2                                                    # multiplication constant
b  = 5                                                    # addition constant
z  = a*y + b                                              # z_i = ay_i + b
sz = sd(z)                                                # empirical standard deviation of z
```

```
Empirical standard deviation after transformation: 3.48
```

```
sz = a*sd(y)  # empirical standard deviation according to theorem
```

```
Empirical standard deviation after transformation: 3.48
```

```
# a < 0
D  = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y  = D$PRE                                                # pre-intervention BDI-II data
a  = -3                                                   # multiplication constant
b  = 10                                                   # addition constant
z  = a*y + b                                              # z_i = ay_i + b
sz = sd(z)                                                # empirical standard deviation of z
```

```
Empirical standard deviation after transformation: 5.221
```

```
sz = -a*sd(y)  # empirical standard deviation according to theorem
```

```
Empirical standard deviation after transformation: 5.221
```

Part II

Imperative programming

On the importance of imperative programming in psychology

Today, scientific data are usually available in digital form. To evaluate these digital data, they are analyzed with the help of a computer. For this purpose, one writes special computer programs that are tailored precisely to the research question at hand. Such programs are called data analysis scripts.

Computer-assisted data analysis follows a typical structure. First, a digital data set is imported and cleaned in order to correct erroneous or incomplete data and prepare them for analysis. Descriptive statistics are then usually computed and visualized in order to obtain an initial overview of the data. This is often followed by a step of probabilistic modeling and inference, in which models based on probability theory are used to test hypotheses or make predictions. Finally, the results are documented and presented clearly, again with the help of the computer.

The central element of computer-assisted data analysis is the *data analysis script*, a text file that documents all steps from the raw data to the final data visualization and thus makes data analysis processes transparent and reproducible. A data analysis script enables third parties to reproduce scientific results, which is a cornerstone of good scientific practice. Data analysis scripts are therefore indispensable tools of everyday scientific work and essential components of scientific publications. In the following sections, starting from a discussion of some basic concepts from computer science and structure-forming elements of imperative programming, we will become familiar with the development of data analysis scripts in the programming environment R.

Various digital tools are used to analyze psychological data. Particularly common are freely available programming languages and platforms such as R, which is widely used in data science, statistics, and psychology, and Python, with its broad range of applications in data science, machine learning, and artificial intelligence. The free programming language Julia, which is in fact optimized for data analysis, still occupies more of a niche and currently has not yet found wide adoption, especially in industrial applications. Free programming languages have the advantage that they are often further developed by a broad community of users and can in this way develop a versatile infrastructure.

In addition, commercial programming environments whose source code cannot be inspected continue to be used in psychology. In neuroimaging and engineering applications, for example, the commercial programming environment MATLAB remains popular even today. Some programs that now seem rather old-fashioned also continue to be used, including SPSS, which was very widespread in the social sciences and psychology around the turn of the millennium, JMP, with areas of application in biology and psychology, and STATA, which was popular in medicine and economics, among other fields.

The popularity of digital tools is subject to constant change. One way to measure the current popularity of different programming languages and environments is to quantify Google searches for programming language tutorials. An overview of current trends is provided by the PYPL Index, which is based on the analysis of such search queries.

In summary, digitalization has a lasting impact on science in particular. Effective research data management poses an acute challenge that researchers must face. Programming is increasingly becoming a central craft of scientific work, and basic knowledge of computer science has become indispensable in the modern working world. This applies not least to psychotherapists, for example in the context of online interventions, digitally supported treatment services, or psychotherapy research.

11 Basic concepts of computer science

In ordinary usage (cf. Wikipedia), computer science is the science of the systematic representation, storage, processing, and transmission of information, with particular emphasis on the automatic processing of information by computers. Computer science is both a basic and formal science and an engineering discipline.

Central components of computer science are, on the one hand, *computers* and, on the other hand, *algorithms and programs*. *Computers* are machines that can store data and carry out simple data operations. They perform these simple operations at extremely high speed. The universality of computers rests on the fact that they can store both data and programs.

Programs are *algorithms* formulated in a programming language. *Algorithms*, in turn, are sequences of *instructions* that prescribe certain *operations*. In algorithms, one distinguishes different levels: the *description level*, as found, for example, in a recipe, assembly instructions, or a data analysis script; the *instructions* of an algorithm, such as "mix flour and water", the pictorial depiction of screwing two parts together, or the instruction `x = c(1, 2, 3)` in **R**; and finally the *execution level*, that is, the actual cooking process, the assembly of a piece of furniture, or the execution of the data analysis script.

Computer science is divided into several subfields that may also be relevant to psychology. *Theoretical computer science* deals with the foundations of computation, such as automata theory, computability theory, and complexity theory. *Applied computer science* concerns the development and use of application software, questions of human-computer interaction, and the social effects of computer science. *Computer engineering* examines the hardware side of computer science, including microprocessor technology, computer architecture, and network technology. Finally, *practical computer science* deals with concrete implementation in the form of programming, the development and analysis of algorithms, and the construction and use of databases. In psychological basic research, practical computer science is the subfield that is applied most often. However, questions from complexity theory, and thus from theoretical computer science, are also of interest in the development of quantitative psychological theories (cf. Van Rooij & Wareham (2012)), and human-computer interaction is one of the topic areas of modern work psychology (cf. Dahm (2005)).

In addition, there are many special areas of computer science that are closely connected with psychology. These include, in particular, the areas of *machine learning* and *artificial intelligence*, which have their origins in quantitative cognitive psychology and today make it possible to analyze large amounts of data automatically and to make predictions. Another important field is *computer graphics and visual computing*, which deals with image recognition and image synthesis as well as the design of virtual reality (VR) and augmented reality (AR). These technologies use findings from perceptual psychology on the one hand and, on the other hand, open up new possibilities for experiments, therapies, and training in psychology. *Computational linguistics* contributes methods of speech recognition and speech synthesis that help analyze, model, and make human communication usable in applications such as chatbots or assistance systems. Finally, *bioinformatics* also plays a major role for psychology, for example by developing methods for evaluating medical imaging procedures that are also used in neuropsychological research and diagnostics.

11.1 Computer architecture

Even though we do not want to go into the technical aspects of computer science in depth here, for imperative programming it is helpful to have at least a rudimentary understanding of the typical physical structure, that is, the hardware, of a computer. A computer consists of several essential hardware components that work together to handle all computation and storage tasks. At the center is the main circuit board, also called the motherboard or mainboard. It is the central printed circuit board on which important components such as the processor, memory, and connectors are linked to one another.

The heart of a computer is the CPU (central processing unit, also called the microprocessor), which combines the arithmetic unit, the control unit, and the registers of the system. It performs all basic computations, controls the flow of data, and coordinates the processes. In addition, it has a small but very fast *cache*, a volatile memory that temporarily stores frequently needed data in order to speed up processing. Typical examples of current CPUs are Intel Core i processors, AMD Ryzen processors, or Apple Silicon M processors.

The CPU is complemented by RAM (random access memory), the temporary, volatile working memory of the system. During the computer's runtime, the programs and data currently needed are stored here to enable fast access. RAM is limited in its capacity, for example to 32 to 64 GB in many modern systems.

Mass storage, the computer's stationary storage, is used for the permanent storage of data and programs. Today, SSDs (solid state drives) are usually used for this purpose; they are much faster and more robust than conventional magnetic hard drives. Cloud storage is also increasingly used as a complement to, or replacement for, local mass storage.

Another important component is the GPU (graphics processing unit), a powerful processor optimized specifically for visualization tasks. The GPU relieves the CPU in graphics-intensive applications such as 3D rendering or video editing and also plays a growing role in computationally intensive tasks such as training neural networks, where it can effectively support the CPU.

The basic functional architecture of modern computers goes back to Von Neumann (1945) and is therefore referred to as the *Von Neumann architecture*. The decisive characteristic of the Von Neumann architecture is that both instructions (program instructions) and data are stored in the same memory, namely in the form of binary numbers. The central processing unit (CPU) reads both the instructions and the data from this shared memory. As a result, programs can be changed dynamically during execution or even generated anew, since in memory they are present merely as data.

Specifically, Von Neumann (1945) describes the following computer structure: the computer is composed of the basic functional units control unit, arithmetic unit, memory, input unit, and output unit. Programs and data are entered into memory together, so that data, programs, and intermediate and final results are stored in the same memory area. The memory itself is divided into equally sized, numbered (addressed) cells whose contents can be retrieved or changed selectively via the respective address. The instructions of a program typically lie in consecutive memory cells, so that the control unit can call the next instruction simply by increasing the instruction address by one. This principle of sequentially processing instructions is the basic principle of imperative programming and is therefore very closely connected with the Von Neumann architecture. Special jump instructions also make it possible to deviate from this fixed order and jump to another point in the program. The basic instructions include arithmetic operations such as addition and multiplication, logical comparisons such as logical AND or OR, and transport instructions by which data are transferred, for example, from the input unit to memory or from memory to the arithmetic unit. All data, that is, both instructions and addresses, are encoded in binary. A suitable circuit technology in the computer handles the binary encoding and decoding of the information.

11.2 Algorithms and programs

A *real-world problem* denotes the problem that is to be solved with the help of a computer. A typical example is the evaluation of questionnaire data from a psychological study. To work on such a problem, one first creates a *problem specification*, that is, the precise verbal formulation of the real-world problem, as one might find it in the methods section of a scientific publication. On this basis, an *algorithm* is designed, that is, a sequence of instructions for solving the problem, for example reading in the data, computing descriptive statistics, and carrying out a statistical test. Finally, the algorithm is implemented in the form of a *program*, that is, as a text file written in a programming language that can be executed by a computer.

Definition 11.1 (Algorithm)**.** An algorithm is a sequence of instructions for deriving certain output data from certain input data, where the following conditions must be satisfied:

- Finiteness. The sequence of instructions is completely described in a finite text.
- Effectiveness. Each instruction must actually be executable.
- Termination. The algorithm ends after finitely many instructions.
- Determinacy. The course of the algorithm is prescribed at every point in time.

If E denotes the set of admissible input data and A denotes the set of admissible output data, then an algorithm is therefore a function:

$$f : E \to A, \quad e \mapsto f(e). \tag{11.1}$$

Conversely, functions that can be described by an algorithm are called *computable functions*.

•

11.3 Programming languages

Programming languages specify the rules that a program must obey. They define a *syntax*, that is, the vocabulary and structure of a program, as well as the *semantics*, that is, the meaning of the permitted instructions. In the following, we give some terminological clarifications for categorizing different forms of programming languages. Here one distinguishes *machine language* and so-called *higher-level programming languages*, different *generations of programming languages*, the *imperative programming* that is the focus here from *declarative programming*, and *compiled* from *interpreted* programming languages.

Machine language and higher-level programming languages

Machine language consists of elementary operation instructions such as storing, comparing, or adding, which are encoded as binary numbers. Table 11.1 shows some examples of such instructions:

Table 11.1 Examples of machine language

Instruction	Binary code
Add contents of R1 to contents of R2	1001 0010
Increase contents of R by 1	1001 0110
Transfer contents of R1 to R3	0010 0011

Programs written in machine language are called machine programs. De facto, a computer executes only machine programs, because these are understood directly by the hardware. For humans, however, programming in machine language is very laborious, which is why higher-level programming languages are predominantly used in practice.

Higher-level programming languages use words and sentences modeled on human language to make programming more understandable and efficient. Programs in higher-level programming languages are then translated into machine language by a *compiler* or an *interpreter* so that they can be executed by the computer. Well-known examples of higher-level programming languages are FORTRAN, COBOL, C++, Python, and **R**.

Generations of programming languages

The *first-generation programming languages (1GL)* are the machine languages. Programs are written directly in the form of binary numbers, for example `10110000 01100001`, which corresponds to `B0 61` in hexadecimal notation. Such machine languages were used mainly on the first vacuum-tube and relay computers such as ENIAC or early IBM mainframes of the 1940s and early 1950s.

With the *second generation (2GL)*, assembly languages emerged from the 1950s on as the first form of symbolic programming. Here, human-readable abbreviations are used, although they are still processor-specific. An example of such a processor-specific Intel instruction is `MOV AL, 61H`. This generation of programming languages was used mainly on mainframe computers such as the IBM 704 or early mainframes of the 1950s and 1960s, which already used transistors instead of vacuum tubes.

From the 1970s onward, the *third-generation programming languages (3GL)* became established. These include higher-level programming languages such as FORTRAN, C, C++, and Java. They are much more programming-friendly and, above all, processor-independent, which considerably improves the portability of programs. Such languages were first used widely on minicomputers such as the DEC PDP-11 and later on workstations, PCs, and servers.

Finally, from the 1980s onward, the *fourth-generation programming languages (4GL)* developed; they include languages such as Python, MATLAB, and **R**. These are characterized by minimizing code overhead, high flexibility, support for multiparadigmatic approaches, and a high degree of automation. They are used mainly on modern PCs, high-performance computing clusters (HPC clusters), and cloud-based systems, which became increasingly available from the 1990s onward.

Imperative and declarative programming

In *imperative programming* (from *imperare*, Latin for "to command"), the path to a solution is specified as a sequence of instructions or commands. These commands process data that are addressed via variables. Within imperative programming, two different approaches are distinguished: *procedural* imperative programming and *object-oriented* imperative programming.

In *procedural imperative programming*, data and the commands that manipulate them are treated separately. The central structural concept here is procedures or functions, which encapsulate reusable units of instructions. By contrast, in *object-oriented imperative programming*, data and the commands that act on them are combined as objects. These objects are at the same time data carriers and providers of functions and form the central structural concept of this approach.

In practice, mixed forms of the two paradigms are often used, because the approaches can be combined well depending on the problem. An example of such a mixed form is the newer programming language Julia, which does not offer a classical object-oriented approach but instead a *multiple-dispatch* approach: functions can have different implementations depending on the types of their arguments and can thus combine aspects of object-oriented and procedural programming.

In *declarative programming*, the focus is not on the path to the solution but on the description of the problem itself. The programmer formulates what is to be achieved, and not in detail how the computer is to reach this goal step by step. The path to the solution is determined automatically by the system on the basis of given rules and mechanisms.

A typical feature of declarative approaches is that the focus is on defining relations, conditions, or goals rather than on an explicit sequence of commands. The programmer thus describes the desired result in a formal language, and the computer independently searches for a solution that satisfies these conditions.

Well-known approaches to declarative programming are, for example, logic programming, as implemented in the language Prolog, or functional programming, as in Haskell. In logic programming, facts and rules are formulated from which the system draws conclusions by inference. In functional programming, computations are described by the composition of functions, while the state of the data is not changed explicitly. Declarative programming is particularly suitable for problems in which the path to the solution can be complex and varied, but the desired properties of the result can be clearly specified. Typical application areas include database queries (for example SQL), knowledge bases, constraint solving, and formal proofs.

A modern example of the use of declarative programming is the system Lean, a so-called *theorem prover*. Lean is used to formally prove mathematical theorems. Here the user formulates the axioms, definitions, and statements to be proved in a mixture of declarative and interactive commands. The system automatically checks whether the specified steps are valid and completes the missing parts of the proof according to fixed rules. In this way, Lean combines declarative programming with automated proof and makes it possible to develop large mathematical theories in a formally correct way. For an introduction to Lean in the context of mathematical proof, see, for example, The Mechanics of Proof.

Compiled and interpreted programming languages

Compiled programming languages are characterized by the fact that the entire source code is translated into machine language before execution. A special program, the so-called *compiler*, is used for this translation. The machine code generated by the compiler is executed directly by the processor, which enables the program to run very quickly. Since the executable program is already available in machine language, it no longer has to be translated at runtime. However, whenever the source code is changed, it is necessary to compile the program again. Typical examples of compiled programming languages are C, C++, and Rust.

By contrast, in *interpreted programming languages*, the source code is translated into a machine-like language during execution. This is done by a special program, the so-called *interpreter*. Since translation and execution take place at the same time, execution is usually slower than in compiled programs. One advantage, however, is that after changes no separate compilation step is required; the program can be executed immediately. Typical examples of interpreted programming languages are Python and **R**.

In the sense of the concepts introduced here, **R** is an interpreted multiparadigmatic fourth-generation programming language that supports object-oriented concepts but is often used procedurally and functionally and is tailored to statistical data analysis.

11.4 R

R is a programming language and a software package originally developed by Ross Ihaka (Ihaka & Gentleman (1996)). It is a free dialect of the proprietary software S, introduced by Becker et al. (Becker et al. (1988)). Today, **R** is developed and maintained by the R Core Team and the R Foundation. **R** has a large and active community that has so far contributed more than 20,000 **R** packages, which extend the system with numerous additions. The core of **R** deliberately develops in an evolutionary and conservative way, while the large number of R packages enables consistent and progressive further development. **R** can be easily downloaded and installed from CRAN.

With **R**, one can load, manipulate, and save data sets in different formats. The language makes it possible to carry out a wide variety of computations on different data structures. In addition, **R** offers a broad spectrum of statistical analysis methods that can be applied directly to the data. It is also typical to write reproducible data analysis scripts with which analyses can be automated and results can be documented transparently. Finally, **R** is very well suited for creating figures and visualizations in order to present data clearly.

However, some things that are important in psychological data science can be implemented only to a limited extent with **R**. For example, **R** per se lacks a modern and appealing development environment of its own; embedding **R** in VS Code remedies this. In the area of scientific computing, **R** reaches its limits. Here, languages such as Python, MATLAB, or Julia often offer more powerful and flexible tools. Finally, **R** is hardly suitable for programming psychological experiments, for which Python or MATLAB are also more commonly used.

There are many ways to get help with **R**. Quite classically, a quick search on Google, a request to ChatGPT, or an interaction with Copilot in VS Code often already helps. During programming, and when one already knows the name of a function, one can call help directly in **R**:

```
?mean
help(mean)
```

For more comprehensive, longer tutorials and background information, it is worth taking a look at the so-called *vignettes* of installed packages:

```
browseVignettes()
```

In addition, there are numerous helpful online resources, for example the specialized search engine rseek or the many articles and guides on R-Bloggers, although most information on these platforms is certainly also part of the databases accessible through chatbots such as ChatGPT, Gemini, Claude, or DeepSeek.

11.5 Visual studio code

Visual Studio Code (VS Code) is a free source code editor from Microsoft that has been available for Windows, macOS, and Linux since 2015. Roughly speaking, VS Code is something like the perhaps better-known RStudio environment, except that it is for all programming languages. VS Code can be flexibly extended through extensions and used as an integrated development environment (IDE) for almost any language, for example **R**, Python, Julia, Shell, Quarto, and many more. Since 2018, VS Code has been regarded as the most popular editor overall, as annual surveys show. Remarkably, this means that a Microsoft product in particular is also the most-used editor in the Linux world. VS Code is highly configurable, strongly shaped by the community, and enables the synchronization of settings and extensions across multiple devices via Microsoft's GitHub. Detailed documentation with instructions for use can be found at www.code.visualstudio.com/docs. The corresponding documentation R in Visual Studio Code provides an introduction to working with **R** in VS Code.

12 Arithmetic and variables

12.1 Arithmetic

If one opens an **R** terminal, one can first use it as a calculator, as the following example shows.

```
1+1                                         # addition
```

```
[1] 2
```

In the terminal, this shows, on the one hand, the result 2 and, on the other hand, with `[1]`, an indication that the result is the first entry in the result vector generated by **R**. We will discuss vectors in detail later. Further examples of using an **R** terminal as a calculator are the following.

```
2*3                                         # multiplication
```

```
[1] 6
```

```
sqrt(2)                                     # square root
```

```
[1] 1.414214
```

```
exp(0)                                      # exponential function
```

```
[1] 1
```

```
log(1)                                      # logarithm function
```

```
[1] 0
```

Table 12.1 gives an initial overview of the arithmetic operators available in Base **R**. In addition to the familiar operations of addition, subtraction, multiplication, division, and exponentiation, Base **R** also offers special operators for more advanced arithmetic computations. These include, for example, *matrix multiplication*, *integer division* (`5 %/% 2 = 2`), and the *modulo operation*, which returns the remainder of an integer division (`5 %% 2 = 1`).

Table 12.1 Examples of arithmetic operators in Base **R**.

Operation	Operator
Addition	`+`
Subtraction	`-`
Multiplication	`*`
Division	`/`
Power	`^`
Matrix multiplication	`%*%`
Integer division	`%/%`
Modulo operation	`%%`

In addition to the arithmetic operators for combining two numbers, Base **R** naturally also offers the possibility of evaluating standard mathematical functions. Table 12.2 gives an initial overview.

Table 12.2 Examples of mathematical functions in Base **R**.

Function	Call
Exponential function	`exp(x)`
Logarithm function	`log(x)`
Absolute value	`abs(x)`
Square root	`sqrt(x)`
Round up	`ceiling(x)`
Round down	`floor(x)`
Mathematical rounding	`round(x)`

Although the operators listed in Table 12.1 and the functions listed in Table 12.2 seem to have a somewhat different character at first glance, Base **R** does not formally distinguish between operators and functions. In particular, operators can also be used and understood as functions of several arguments by means of so-called *infix notation*. The **R** code below shows how the arithmetic combination $2 + 3$ can be understood as the execution of a function of the form $+ : \mathbb{R}^2 \to \mathbb{R}$ with the name "+".

```
`+`(2,3)                                              # infix notation for 2 + 3
```

```
[1] 5
```

Finally, like every programming language, Base **R** also offers the possibility of evaluating expressions with respect to their logical content. Here, logic is to be understood in the sense of the propositional logic from Chapter 1, in which statements can take one of two values, true (TRUE) or false (FALSE). The relational operators `<`, `<=`, `>`, and `>=` listed in Table 12.3 are usually applied to numerical values. The equality operators `==` and `!=` can be applied to arbitrary data structures. Finally, the operators for combining logical values, `&` and `|`, correspond to the concepts of logical conjunction ("and") and disjunction ("non-exclusive or") known from Chapter 1.

Table 12.3 Examples of logical operators in Base **R**.

Logical connection	Operator
Equal	`==`
Unequal	`!=`
Conjunction	`&`
Disjunction	`|`
Less than	`<`
Less than or equal	`<=`

The operators and functions listed in Table 12.1, Table 12.2, and Table 12.3 represent only a small excerpt of the operators and functions provided by Base **R**. A complete overview is provided by the call `names(methods:::.BasicFunsList)`, which lists the names of all functions provided by Base **R**. We will get to know many of these functions later.

```r
names(methods:::.BasicFunsList)
```

```
  [1] "$"                  "$<-"                "["                 
  [4] "[<-"                "[["                 "[[<-"              
  [7] "crossprod"          "tcrossprod"         "xtfrm"             
 [10] "c"                  "all"                "any"               
 [13] "sum"                "prod"               "max"               
 [16] "min"                "range"              "cummax"            
 [19] "rep"                "^"                  "cos"               
 [22] "levels<-"           "cumsum"             "asin"              
 [25] "anyNA"              "<="                 "Conj"              
 [28] "exp"                "is.matrix"          "log10"             
 [31] "as.environment"     "ceiling"            "asinh"             
 [34] "abs"                "as.raw"             "is.infinite"       
 [37] "is.array"           "floor"              "=="                
 [40] "sign"               "cummin"             "|"                 
 [43] "round"              "Re"                 "!"                 
 [46] "is.numeric"         "acosh"              "!="                
 [49] "names<-"            "&"                  "atanh"             
 [52] "sinh"               ">="                 "sin"               
 [55] "*"                  "atan"               "+"                 
 [58] "length"             "length<-"           "sinpi"             
 [61] "-"                  "is.nan"             "/"                 
 [64] "sqrt"               "%/%"                "as.numeric"        
 [67] "seq.int"            "trunc"              "digamma"           
 [70] "acos"               "<"                  "as.logical"        
 [73] "cosh"               "Mod"                "tanpi"             
 [76] "%*%"                "dimnames<-"         "log2"              
 [79] ">"                  "tanh"               "is.na"             
 [82] "dim"                "signif"             "gamma"             
 [85] "is.finite"          "as.integer"         "dim<-"             
 [88] "as.double"          "lgamma"             "Arg"               
 [91] "log1p"              "trigamma"           "%%"                
 [94] "tan"                "cumprod"            "Im"                
 [97] "expm1"              "as.call"            "log"               
[100] "as.complex"         "cospi"              "as.character"      
[103] "dimnames"           "names"              "("                 
[106] ":"                  "="                  "@"                 
[109] "{"                  "~"                  "&&"                
[112] ".C"                 "baseenv"            "quote"             
[115] "::"                 "<-"                 "is.name"           
[118] "if"                 "||"                 "attr<-"            
[121] "untracemem"         ".cache_class"       "substitute"        
[124] "interactive"        "is.call"            "switch"            
[127] "function"           "is.single"          "is.null"           
[130] "is.language"        "is.pairlist"        ".External.graphics"
[133] "declare"            "globalenv"          "class<-"           
[136] ".Primitive"         "is.logical"         "enc2utf8"          
[139] "UseMethod"          ".subset"            "proc.time"         
[142] "enc2native"         "repeat"             ":::"               
[145] "<<-"                "@<-"                "missing"           
[148] "nargs"              "isS4"               ".isMethodsDispatchOn"
[151] "forceAndCall"       "Exec"               ".primTrace"        
[154] "storage.mode<-"     ".Call"              "unclass"           
[157] "gc.time"            ".subset2"           "environment<-"     
[160] "emptyenv"           "seq_len"            ".External2"        
[163] "is.symbol"          "class"              "on.exit"           
[166] "is.raw"             "for"                "is.complex"        
[169] "list"               "invisible"          "is.character"      
[172] "oldClass<-"         "is.environment"     "attributes"        
[175] "break"              "return"             "attr"              
[178] "tracemem"           "next"               ".Call.graphics"    
[181] "standardGeneric"    "is.atomic"          "retracemem"        
```

```
[184] "expression"          "is.expression"        "call"
[187] "is.object"           "pos.to.env"           "attributes<-"
[190] ".primUntrace"        "...length"            "Tailcall"
[193] ".External"           "oldClass"             ".Internal"
[196] ".Fortran"            "browser"              "is.double"
[199] ".class2"             "while"                "nzchar"
[202] "is.list"             "lazyLoadDBfetch"      "unCfillPOSIXlt"
[205] "...elt"              "...names"             "is.integer"
[208] "is.function"         "is.recursive"         "seq_along"
[211] "unlist"              "as.vector"            "lengths"
```

12.2 Precedence

An important property when using operators is their *precedence*, as defined by the respective programming language. These are rules specified by the developers of programming languages that determine the order of operations. From mathematics, one is especially familiar with the precedence rule "multiplication and division before addition and subtraction". This rule says that in expressions containing both products or divisions and sums or differences, the products or divisions are carried out first, and their results then enter the formation of sums or differences second. Thus, for example,

$$2 \cdot 5 - 3 = 10 - 3 = 7 \tag{12.1}$$

and not, for instance,

$$2 \cdot 5 - 3 \neq 2 \cdot 2 = 4. \tag{12.2}$$

We assume as known that precedence rules can be emphasized or overridden by parentheses. For example, the expression in Equation 12.1 is equivalent to

$$(2 \cdot 5) - 3 = 10 - 3 = 7 \tag{12.3}$$

and the calculation intended in Equation 12.2 can be made correct by placing parentheses as

$$2 \cdot (5 - 3) = 2 \cdot 2 = 4 \tag{12.4}$$

These familiar calculation rules and their overriding by parentheses are implemented correspondingly in **R**:

```r
2*5 - 3
```

```
[1] 7
```

```r
2*(5 - 3)
```

```
[1] 4
```

Another precedence rule assumed to be known is that powers have higher precedence than products or divisions and sums or differences. Thus, for example,

$$2^3 + 3 = (2 \cdot 2 \cdot 2) + 3 = 8 + 3 = 11 \tag{12.5}$$

and not, for instance,

$$2^3 + 3 \neq 2^{3+3} = 2^6 = 64. \tag{12.6}$$

Correspondingly, in **R** we have

```r
2^3 + 3
```

```
[1] 11
```

```r
2^(3 + 3)
```

```
[1] 64
```

An important special feature of the precedence rules in **R** is that *unary signs*, that is, signs that identify a number as a negative number, also fall within the domain of operator precedence. Although one might be inclined to interpret an expression of the form -1^2 as $(-1)^2 = 1$, **R** nevertheless follows the rule that in -1^2 the power is evaluated first and the sign afterward, so that $-1^2 = -(1^2) = -1$ holds:

```r
-1^2                        # -(1^2)   = -1
```

```
[1] -1
```

```r
(-1)^2                      # (-1)^2   =  1
```

```
[1] 1
```

Furthermore, in **R**, powers are processed from right to left, whereas products, divisions, sums, and differences are processed from left to right. For example,

```r
2^2^3                       # 2^(2^3)  = 2^8 = 256
```

```
[1] 256
```

```r
(2^2)^3                     # (2^2)^3  = 4^3 = 64
```

```
[1] 64
```

but

```r
2+3/4*5                     # 2+(3/4)*5 = 2+(0.75*5) = 2+3.75 = 5.75
```

```
[1] 5.75
```

```r
2+3/(4*5)                   # 2+3/(4*5) = 2+3/20 = 2+0.15 = 2.15
```

```
[1] 2.15
```

In general, in programming one should not guess precedence rules or try to derive them logically. Nor should one trust that the developers of a programming language will probably have implemented the precedence rules according to one's own state of knowledge. Instead, one should always check computations critically and, to be safe, emphasize the intended calculation with parentheses once too often rather than once too little.

In addition to the precedence rules considered here for the basic arithmetic operations, **R** has a whole series of further rules for dealing with many other operators, of which we have so far become familiar with only very few. When becoming familiar with a new operator while learning a programming language, one should therefore also make sure to clarify the precedence of this operator. The text from the **R** documentation shown in Figure 12.1, which can be called with the command `?Syntax`, discusses the precedence rules that apply in **R** in full and should be consulted frequently when working with **R**.

Operator Syntax and Precedence

Description

Outlines R syntax and gives the precedence of operators.

Details

The following unary and binary operators are defined. They are listed in precedence groups, from highest to lowest.

<table>
<tr><td><code>:: :::</code></td><td>access variables in a namespace</td></tr>
<tr><td><code>$ @</code></td><td>component / slot extraction</td></tr>
<tr><td><code>[[[</code></td><td>indexing</td></tr>
<tr><td><code>^</code></td><td>exponentiation (right to left)</td></tr>
<tr><td><code>- +</code></td><td>unary minus and plus</td></tr>
<tr><td><code>:</code></td><td>sequence operator</td></tr>
<tr><td><code>%any% |></code></td><td>special operators (including <code>%%</code> and <code>%/%</code>)</td></tr>
<tr><td><code>* /</code></td><td>multiply, divide</td></tr>
<tr><td><code>+ -</code></td><td>(binary) add, subtract</td></tr>
<tr><td><code>< > <= >= == !=</code></td><td>ordering and comparison</td></tr>
<tr><td><code>!</code></td><td>negation</td></tr>
<tr><td><code>& &&</code></td><td>and</td></tr>
<tr><td><code>| ||</code></td><td>or</td></tr>
<tr><td><code>~</code></td><td>as in formulae</td></tr>
<tr><td><code>-> ->></code></td><td>rightwards assignment</td></tr>
<tr><td><code><- <<-</code></td><td>assignment (right to left)</td></tr>
<tr><td><code>=</code></td><td>assignment (right to left)</td></tr>
<tr><td><code>?</code></td><td>help (unary and binary)</td></tr>
</table>

Within an expression operators of equal precedence are evaluated from left to right except where indicated. (Note that `=` is not necessarily an operator.)

Figure 12.1 Precedence rules in **R**.

12.3 Variables

In programming, *variables* are abstract containers for data whose contents can be changed during the execution of a program. Variables are denoted by names in program code and, during program execution, have an address in working memory. Variables can be of different *types*. This means that variables represent different kinds of data, and only those kinds. In traditional programming languages, the type of a variable is explicitly *declared*. At the beginning of program execution, it is specified which kind of data a particular variable can represent. The declaration of a variable `A`, which, for example, can represent integers and only integers, has roughly the form

```
VAR A : INTEGER     # a is a variable of type integer
```

Subsequently, the variable `A` can be assigned a value of type integer, for example in the form

```
A := 1              # variable A is assigned the numerical value 1
```

In a 4GL language such as **R**, the variable type is usually specified directly by an initialization statement. Thus, the following **R** code simultaneously declares the variable `a` as being of type `double` (decimal number) with the value `1`,

```
a = 1               # a is a variable of type double; its value is 1
```

In the vast majority of programming languages, the assignment command is `=`. **R** is special in this respect because it allows the additional assignment commands `<-` and `->`: `<-` assigns from right to left, `->` from left to right. The usual assignment from right to left is also achieved by `=`. The officially recommended assignment command for **R** is `<-`. This is, however, highly idiosyncratic, and we mostly use `=`, as in every other programming language. A first example should illustrate how to work with variables.

Example

We consider the following task:

"Lina goes to the stationery store and buys four notebooks and two pens. One notebook costs 1 euro and one pen costs 3 euros. How many items does Lina buy in total, and how many euros does Lina have to pay in total?"

To solve the task, we first define the variables `number_notebooks` and `number_pens` and assign to them their respective values from the task.

```
number_notebooks = 4
number_pens      = 2
```

After assignment, the variables with their assigned values now exist in working memory, the so-called *workspace*. The command `ls()` displays the existing usable variables in working memory.

```
ls()
```

```
[1] "number_notebooks" "number_pens"
```

The crucial point is that the variable names can now be used like numbers in computations. `print()` outputs variable values in an **R** terminal:

```
total_items = number_notebooks + number_pens
print(total_items)
```

```
[1] 6
```

To calculate the total price, we next define the variables `price_notebook` and `price_pen` according to the task specification.

```
price_notebook = 1
price_pen      = 2
```

The total price is then calculated as follows.

```
total_price = number_notebooks*price_notebook + number_pens*price_pen
print(total_price)
```

```
[1] 8
```

Variables in the workspace can also be deleted again. `rm()` (remove) allows individual variables to be deleted.

```
rm(total_price)
ls()
```

```
[1] "number_notebooks" "number_pens"      "price_notebook"   "price_pen"
[5] "total_items"
```

`rm(list=ls())` deletes all variables.

```
rm(list = ls())
ls()
```

```
character(0)
```

Variable names

In principle, one is free to choose names for variables, with minor restrictions. Admissible variable names in **R**

- consist of letters, numbers, periods (.), and underscores (_),
- begin with a letter or a period, but then not followed by a number,
- must not be expressions already reserved in **R**, such as `for`, `if`, `NaN`, …

The help page `?reserved` provides assistance with the restrictions on variable names. A function for generating a syntactically admissible variable name from arbitrary suggestions is `make.names()`. Its help page also describes the rules for variable naming sketched here in further detail. In general, useful variables are short, to save space in the code, and meaningful, to increase human understanding of the code. Variables usually consist only of lowercase letters and underscores.

Variable representation

We now want to examine the relationship between variable names (identifiers) and variable contents (objects in working memory) in a little more detail. The following concepts and relationships are especially important when computations reach the limits of the available working memory and must be optimized.

We begin with the concept of *binding*. To do so, we first consider the following variable initialization:

```
x = 1
```

Intuitively, this statement creates a variable named `x` with the value `1`. At the technical level, two things happen (cf. Figure 12.2):

1. **R** creates an object (a vector with value 1) with working-memory address `lobstr::obj_addr(x)`.
2. **R** connects this object with the name `x` (*binding*), which references the object in working memory.

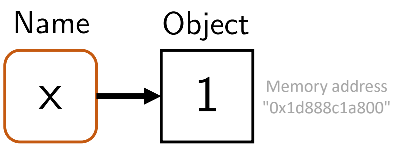

Figure 12.2 Binding of a name to an object in working memory.

Now consider the following statement:

```
y = x
```

Intuitively, this creates a variable named `y` with a value equal to the value of `x`. At the technical level, a new name `y` is created that references the same object in working memory as `x` (cf. Figure 12.3).

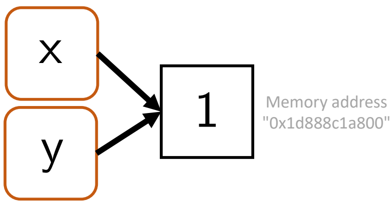

Figure 12.3 Binding of two names to one object in working memory.

One can use `lobstr::obj_addr(x)` and `lobstr::obj_addr(y)` to verify that the addresses of the object in working memory are identical:

```
print(lobstr::obj_addr(x))
```

```
[1] "0x24114314820"
```

```
print(lobstr::obj_addr(y))
```

```
[1] "0x24114314820"
```

Thus, the relevant object is *not* copied when assigned to `y`. This saves working memory.

Furthermore, **R** uses the so-called *copy-on-modify* principle. This principle states that an object in working memory is copied and receives a new address as soon as it is modified. The original object in working memory remains unchanged. Figure 12.4 and the following **R** code first illustrate the *copy-on-modify* principle:

```
x = 1        # object (e.g., 0x74b) is created; x references the object's memory address
y = x        # y references the same memory address as x (0x74b)
y = 3        # y is modified; a modified copy (e.g., 0xcd2) is stored
y            # y now references (0xcd2)
```

```
[1] 3
```

```
x            # x continues to reference (0x74b)
```

```
[1] 1
```

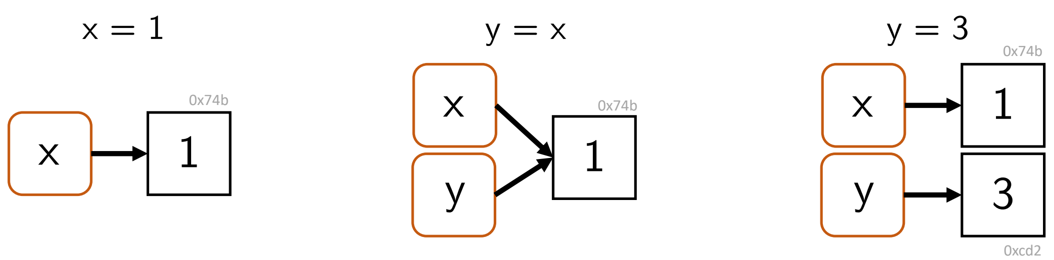

Figure 12.4 Copy-on-modify. Only after changing the value assigned to the identifier `y` is the value copied in working memory.

The fact that objects are copied in working memory when they are changed, while the original objects are in principle preserved unchanged, is also referred to as the *immutability* of objects in **R**. However, the immutability of objects in **R** does not go very far, as the following example shows. First, an object of a `1` is created in working memory and can be referenced with the variable name `x`. By *indexing* (cf. Chapter 13), the object is then readdressed in working memory when an intended modification takes place.

```
x   = 1                   # object is created; x references the object's memory address
print(lobstr::obj_addr(x))  # object's memory address
```

```
[1] "0x24117606f58"
```

```
x[1] = 2                      # object is changed
print(lobstr::obj_addr(x))   # new memory address
```

```
[1] "0x241178119a0"
```

The copy-on-modify principle also applies in the reverse order. That is, if an object to which another variable name points is changed, the originally referenced object does not change. The following **R** code may illustrate this.

```
x = 1       # object (e.g., 0x74b) is created; x references the object's memory address
y = x       # y references the same memory address as x (0x74b)
x = 3       # a new object (e.g., 0x2a08) is created; x now references 0x2a08
y           # y continues to reference the original object (0x74b)
```

```
[1] 1
```

Finally, *unbinding* of objects in working memory is also possible, for example as a consequence of the following **R** commands (cf. Figure 12.5):

```
x = 1       # x references object (0x74b)
x = "a"     # x references object (0x2a08); object (0x74b) is now unreferenced
```

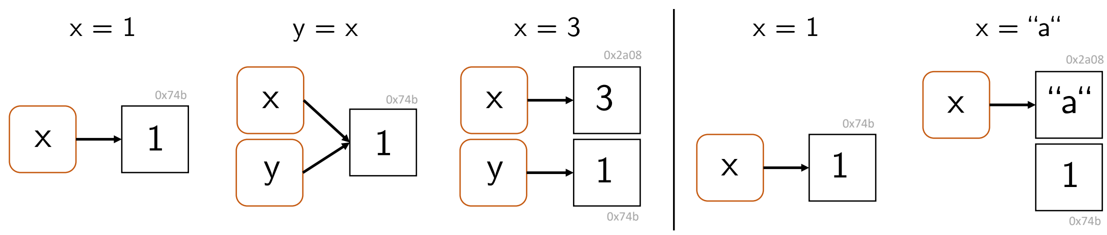

Figure 12.5 Unbinding. Allocated working memory remains that is not referenced by any identifier.

The deletion of unreferenced objects in working memory is called *garbage collection*. **R** automatically deletes unreferenced objects in working memory, although usually only when the memory is needed. In general, however, this works well, and it is usually not necessary to actively use the function `gc()` provided by **R** for garbage collection.

Representation of missing values

In data-analytic programming, one often has to deal with missing values, for example when an experimental measurement for a participant had to be aborted. To represent missing values, **R** uses `NA` (*not available*). If one attempts to calculate with `NA`, the result is usually again `NA`:

```
3 * NA       # multiplication of an NA value yields NA
```

```
[1] NA
```

```
NA < 2        # relational comparison of an NA value yields NA
```

```
[1] NA
```

```
NA^0          # missing value to the power of 0 yields 1 because every value to the power of 0 is one (?)
```

```
[1] 1
```

```
NA & FALSE    # logical conjunction with FALSE yields FALSE
```

```
[1] FALSE
```

One tests for `NA` with `is.na()`:

```
x = c(NA, 5, NA, 10)  # vector with NAs
x == NA               # no testing for NAs: 5 == NA is NA, not FALSE
```

```
[1] NA NA NA NA
```

```
is.na(x)              # logical test for NA
```

```
[1]  TRUE FALSE  TRUE FALSE
```

In addition to `NA`, **R** also has `NaN` as a way to represent mathematically undefined values. In contrast to `NA`, `NaN` is always of numerical type. The results when calculating with `NaN` are similar to those with `NA`.

```
3 * NaN       # multiplication of a NaN value yields NaN
```

```
[1] NaN
```

```
NaN < 2       # relational comparison of a NaN value yields NA
```

```
[1] NA
```

```
NaN^0         # undefined value to the power of 0 yields 1 because every value to the power of 0 is one (?)
```

```
[1] 1
```

```
NaN & FALSE   # logical conjunction with FALSE yields FALSE
```

```
[1] FALSE
```

One tests for `NaN` with `is.nan()`.

```
x = c(NaN, 5, NaN, 10)  # vector with NaNs
is.nan(x)               # logical test for NaN
```

13 Data structures

In every programming language, there are abstract containers for representing different kinds of data and working with them. In classical 3GL programming languages, one distinguishes *fundamental*, *composite*, and *user-defined* data structures.

Fundamental data structures are predefined in the programming language. Typical examples are containers for logical values (`TRUE`, `FALSE`), integers (`int8`), which, for example, represent the 256 values from -128 to 127, floating-point numbers (`single`, `double`) with different precision, or characters (`a`, `b`), which are often placed in quotation marks. These basic data types are typically associated with suitable operations, such as `AND`/`OR` for logical values, `+` and `-` for integers, `+`, `-`, `*`, and `/` for floating-point numbers, or concatenation for characters.

Composite data structures are also predefined and serve to combine several variables of the same type. In **R**, these include, for example, vectors, lists, arrays, matrices, and data frames. Specific operations also exist for composite structures, such as vector indexing or matrix multiplication.

Finally, in 3GL programming languages, *user-defined data structures* can also be created. They allow the programming environment to be adapted to specific application cases and are therefore especially relevant in practice. As a rule, they are assembled from predefined structures and, in the context of object-oriented programming, can also be associated with their own operations.

When one learns a programming language, an important first step is to become at least rudimentarily familiar with its data structures. One then gets to know the subtleties of data-structure representation in daily practical work with the programming language. To become familiar with the data structures of a programming language, the following guiding questions are therefore useful:

Fundamental data structures

- Which fundamental data structures does the language provide?
- Which operations on them are already defined?
- What is the syntax for defining a variable of a fundamental data type?
- What is the syntax for calling predefined operations?

Composite data structures

- Which containers and associated operations does the programming language provide?
- What is the syntax for working with a container?

User-defined data structures

- How does one create user-defined data structures and associated operations?
- What is the syntax for working with a user-defined data structure?

13.1 Data structures in R

As a 4GL language, **R** goes beyond the classical data structures of 3GL languages in its data structures. In **R**, first of all, everything is an *object*. Regardless of whether it is a single value, a vector, a function, or a complex data structure, everything is represented by **R** as an object. In general, the concept of an *object* denotes the possibility of representing both data and functionality in a single unit of meaning. Here, however, we are primarily interested only in data representation.

In general, one distinguishes in **R** between *atomic* objects and *recursive* objects. *Atomic objects* consist of components of the same data type. Typical examples are numerical vectors, logical vectors, or character strings. *Recursive objects*, by contrast, can contain components of different types. These include especially lists and data frames, which make it possible to combine heterogeneous data.

Every **R** object can be described uniquely by three central properties: its *mode*, its *length*, and, optionally, further *attributes*. The *mode* of an object specifies the basic data type of an object and is closely related to the *type* and *class* of an object, as explained below. In addition to the mode, every object has a *length*, which indicates how many elements it contains. For a vector, for example, the length is the number of values it contains; for a list, it is the number of its components. In addition, objects in **R** can be provided with additional *attributes*. Attributes are metadata that provide additional information about the object, such as the names of elements (`names`), dimensions (`dim`) for matrices and arrays, or class information (`class`), which is relevant for object-oriented programming in **R**.

Mode, type, and class

The classification of data structures in **R** is somewhat complex, because the concepts of the *mode*, the *type*, and the *class* of an object exist side by side. In practical application, the subtleties of this classification are usually not important, and the mode, type, and class of an object are also often identical.

Roughly speaking, the *mode* corresponds to the basic nature of an object and is chosen especially so that there is high compatibility with the data-structure classification system of **S** (Becker et al. (1988)). Examples of modes are `numeric`, `character`, `logical`, or `list`. The mode of an object can be retrieved with the function `mode()`, as the following example shows.

```
mode(42)                          # numeric
```

```
[1] "numeric"
```

```
mode(1L)                          # numeric
```

```
[1] "numeric"
```

```
mode("hello")                     # character
```

```
[1] "character"
```

```r
mode(TRUE)                      # logical
```

```
[1] "logical"
```

```r
mode(list(1, 2, 3))             # list
```

```
[1] "list"
```

```r
mode(factor(c("a", "b")))       # r factor => numeric
```

```
[1] "numeric"
```

```r
mode(Sys.Date())                # r date => numeric
```

```
[1] "numeric"
```

```r
mode(lm(mpg ~ wt, data=mtcars)) # r linear model => list
```

```
[1] "list"
```

Somewhat closer to the internal representation of an object in memory, and therefore somewhat more technical in nature, is the *type* of an **R** object. This becomes particularly clear with respect to objects of mode `numeric`. Here, the classical 3GL type distinctions `integer` or `double` exist. Another example is the mode *factor*. As we will see later, factors in **R** look to the user like categorical data, often of character type. Their mode, however, is `numeric` and their type is `integer`. This is related to the fact that factors in **R** are often used at the user level to specify experimental conditions that usually have meaningful names such as "Treatment" or "Control", but in data analysis are represented by mathematical matrices with integer entries. The type of an object can be retrieved with the function `typeof()`, as the following example shows. Note the similarities and differences with the modes of the same objects called above.

```r
typeof(42)                      # double
```

```
[1] "double"
```

```r
typeof(1L)                      # integer
```

```
[1] "integer"
```

```r
typeof("hello")                 # character
```

```
[1] "character"
```

```
typeof(TRUE)                    # logical
```

```
[1] "logical"
```

```
typeof(list(1,2,3))             # list
```

```
[1] "list"
```

```
typeof(factor(c("a", "b")))     # r factor => integer
```

```
[1] "integer"
```

```
typeof(Sys.Date())              # r date => double
```

```
[1] "double"
```

```
typeof(lm(mpg ~ wt, data=mtcars))   # r linear model => list
```

```
[1] "list"
```

Finally, the class of an object specifies the kind of object from the perspective of object-oriented programming in **R**. For an object, it is especially relevant how generic functions such as `print()` or `summary()` deal with it. In contrast to modes and types, programmers can specify the class of objects themselves and thus develop generic functions that are applicable to many kinds of objects and, despite having the same name, carry out different operations with different objects. The class of an object can be retrieved with the function `class()`, as the following example shows. Note the similarities and differences with the modes and types of the same objects called above.

```
class(42)                       # numeric
```

```
[1] "numeric"
```

```
class(1L)                       # integer
```

```
[1] "integer"
```

```
class("hello")                  # character
```

```
[1] "character"
```

```
class(TRUE)                     # logical
```

```
[1] "logical"
```

```r
class(list(1,2,3))                 # list
```

```
[1] "list"
```

```r
class(factor(c("a", "b")))         # r factor => "factor"
```

```
[1] "factor"
```

```r
class(Sys.Date())                  # r date => "Date"
```

```
[1] "Date"
```

```r
class(lm(mpg ~ wt, data=mtcars))   # r linear model => "lm"
```

```
[1] "lm"
```

In summary, the mode describes the general kind of an object, the type describes its internal, technical storage, and the class defines **R**'s perspective for the functional processing of the object. For many objects, these three concepts agree to a large extent, but for special objects such as factors, date values, or user-defined objects, they differ.

14 Vectors

In this section, we now want to get to know **R** vectors in more detail as a first fundamental data structure. Many principles that can be explained using vectors carry over analogously to more complex data structures in **R**, so that competent handling of the basic data structure considered here is indispensable in application contexts as well. It should be noted that the term *vector* in this section refers strictly to the corresponding **R** data structure, not to the mathematical object of the same name. Although **R** vectors resemble their mathematical counterpart in some properties and certain operations between **R** vectors can represent operations between the corresponding mathematical objects, vectors in imperative programming often have many more and different properties than the corresponding mathematical objects.

In general, when discussing a data structure, we first consider how it is *generated* using **R** code, by which properties it is *characterized*, how individual elements of the data structure can be *indexed* and edited, and which *arithmetic operations* are admissible for the data structure. Finally, we consider possible further special features of the data structure. In the context of **R** vectors, these include, for example, the concepts of *data type coercion*, *recycling*, or the representation of missing values in data sets.

R vectors are ordered sequences of data values. The individual data values of a vector are called the *elements* of the vector. Vectors whose elements are all of the same data type are called *atomic*. Here, the term *vector* always means an atomic vector. The central data types for vectors are the previously introduced *numeric*, (*double*, *integer*), *logical*, and *character* types. Figure 14.1 shows graphical representations of corresponding vectors.

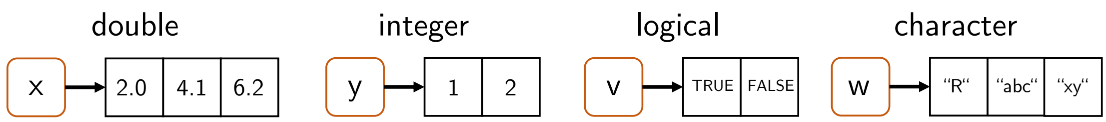

Figure 14.1 Vectors in **R**.

14.1 Generation

We first consider the generation of one-element vectors, which are also called *elementary values*. Like all data structures, elementary values are generated by assignments of the form

```
elementary_value = value
```

Here, the specification of `value` determines what kind of elementary value is generated.

Elementary values of type `numeric` can be generated by assigning a number. By default, this creates an elementary value of type `double`. The numbers can be specified without decimal places, in decimal notation, or in scientific notation. Further possible values for generating an elementary value of type `double` are `Inf` and `-Inf` for positive and negative infinity and `NaN` (Not-a-Number) as a placeholder for numerical values.

```
x = 1       # one-element vector
typeof(x)   # of type double
```

```
[1] "double"
```

```
y = 2.1e2   # one-element vector
typeof(y)   # of type double
```

```
[1] "double"
```

```
z = Inf     # one-element vector
typeof(z)   # of type double
```

```
[1] "double"
```

Numeric elementary values of type `integer` are generated like `double` values without decimal places, followed by an `L` for `long integer`.

```
x = 1L      # one-element vector
typeof(x)   # of type integer
```

```
[1] "integer"
```

```
y = 200L    # one-element vector
typeof(y)   # of type integer
```

```
[1] "integer"
```

Elementary values of type `logical` can be generated by assigning the logical values `TRUE` or `FALSE`, or abbreviated as `T` or `F`.

```
x = TRUE    # one-element vector
typeof(x)   # of type logical
```

```
[1] "logical"
```

```
y = F       # one-element vector
typeof(y)   # of type logical
```

```
[1] "logical"
```

Finally, elementary values of type `character` are generated by assigning values in single quotes (`'a'`) or double quotes (`"a"`).

```
x = "a"     # one-element vector
typeof(x)   # of type character
```

```
[1] "character"
```

```
y = 'test'  # one-element vector
typeof(y)   # of type character
```

```
[1] "character"
```

To combine several elementary values into one vector, one concatenates them with the function `c()`. The term concatenation is derived from the Latin *catena* ("the chain").

```
x = c(1,2,3)          # numeric vector [1,2,3]
s = c("a", "b", "c")  # character vector ["a", "b", "c"]
l = c(TRUE, FALSE)    # logical vector [TRUE, FALSE]
```

```
x
```

```
[1] 1 2 3
```

```
s
```

```
[1] "a" "b" "c"
```

```
l
```

```
[1]  TRUE FALSE
```

Not only elementary values, but also vectors with elementary values or with other vectors can be concatenated. With the vector `x` above, for example, one generates the following vectors `y` and `z`:

```
y = c(0,x,4)  # numeric vector [0,1,2,3,4]
z = c(x,y)    # numeric vector [1,2,3,0,1,2,3,4]
```

```
y
```

```
[1] 0 1 2 3 4
```

```
z
```

```
[1] 1 2 3 0 1 2 3 4
```

Vectors do not have to be generated from already existing values; "empty" vectors of a desired data type can also be generated. **R** provides the functions `vector()` as well as `double()`, `integer()`, `logical()`, and `character()` for this purpose. This is potentially helpful for already occupying computer memory at the beginning of a longer computation, so that no memory has to be searched for during program execution, which can increase program runtime under certain circumstances.

```
# generate "empty" vectors with vector()
v = vector("double",3)      # double vector [0,0,0]
w = vector("integer",3)     # integer vector [0,0,0]
l = vector("logical",2)     # logical vector [FALSE, FALSE]
s = vector("character",4)   # character vector ["", "", "", ""]

# alternative generation of "empty" vectors
v = double(3)               # double vector [0,0,0]
w = integer(3)              # integer vector [0,0,0]
l = logical(2)              # logical vector [FALSE, FALSE]
s = character(4)            # character vector ["", "", "", ""]
```

However, the "empty" vectors generated in this way are not actually empty: `v` and `w` consist of zeros, `l` consists of the logical values `FALSE`, and `s` consists of the characters `""`. For initializing vectors as containers and then filling them in the context of a program, `vector()`, `double()`, `integer()`, `logical()`, and `character()` are therefore suitable only to a limited extent: if the code overlooks the filling of individual entries, computations are performed with the values generated by these functions, which is not necessarily intended. In the chapter on control structures, we will learn a method for initializing data structures while generating reasonably safe code.

Sequences

A frequent tool in programming is vectors that represent sequences of numerical values, for example as representations of the domains of univariate functions. **R** provides the so-called *colon operator* `:` and the function `seq()` and its variants for generating them.

The colon operator in the form `a:b` generates sequences of integer steps between a and b according to

$$(s_i)_{i=0,1,2...} \text{ with } s_i = a + i \text{ for } i = 0, 1, 2, ... \text{ and } s_i \leq b. \tag{14.1}$$

Here, "between `a` and `b`" specifically means that `a` is the first number of the generated sequence and `b` is the largest possible value. The following examples illustrate this.

```
x = 0:5       # 0 is the start of the sequence, 5 can be included
y = 1.5:6.1   # 1.5 is the start, the sequence ends at 5.5 < 6.1
```

```
x
```

```
[1] 0 1 2 3 4 5
```

```
y
```

```
[1] 1.5 2.5 3.5 4.5 5.5
```

The function `seq()` with the syntax `seq(from, to, by)`, where `from` denotes the start value, `to` the maximal end value, and `by` the step size, offers further possibilities for generating sequences with non-integer step sizes as well. The following examples illustrate this.

```
x_1 = seq(0,5)           # like 0:5
x_2 = seq(0,1,len = 5)   # 5 numbers between 0 (incl.) and 1 (incl.)
x_3 = seq(0,2,by = .15)  # 0.15 steps between 0 (incl.) and 2 (max.)
x_4 = seq(1,0,by = -.1)  # -0.1 steps between 1 (incl.) and 0 (min.)
```

```
x_1
```

```
[1] 0 1 2 3 4 5
```

```
x_2
```

```
[1] 0.00 0.25 0.50 0.75 1.00
```

```
x_3
```

```
 [1] 0.00 0.15 0.30 0.45 0.60 0.75 0.90 1.05 1.20 1.35 1.50 1.65 1.80 1.95
```

```
x_4
```

```
 [1] 1.0 0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2 0.1 0.0
```

The `seq()` variant `seq.int()` can be used to generate integer sequences and may be faster than `seq()` under certain circumstances. `seq_len()` generates integer sequences of a given length, and `seq_along()` generates integer sequences based on the entries of a vector.

```
x_1 = seq.int(0,5)          # like 0:5, [0,1,2,3,4,5]
x_2 = seq_len(5)            # natural numbers up to 5, [1,2,3,4,5]
x_3 = seq_along(c("a","b")) # like seq_len(length(c("a", "b")))
```

```
x_1
```

```
[1] 0 1 2 3 4 5
```

```
x_2
```

```
[1] 1 2 3 4 5
```

```
x_3
```

```
[1] 1 2
```

14.2 Characterization

For characterizing vectors, the functions `length()` and `typeof()` are available; they output the number of entries of a vector and the data type of its entries, as the following examples illustrate.

```
x = 0:10   # vector
length(x)  # number of elements of the vector
```

```
[1] 11
```

```
x = 1:3L    # vector
typeof(x)  # data type of the atomic vector
```

```
[1] "integer"
```

```
y = c(T,F,T)  # vector
typeof(y)     # data type of the atomic vector
```

```
[1] "logical"
```

`is.logical()`, `is.double()`, `is.integer()`, and `is.character()` test the data type of a vector and return a logical binary value.

```
is.double(x)  # test the vector above
```

```
[1] FALSE
```

```
is.logical(y)  # test the vector above
```

```
[1] TRUE
```

Data type coercion

If one tries to concatenate different data types in **R** in one vector, for example elementary values of type `numeric`, `character`, and `logical`, then **R** enforces a uniform data type for the resulting vector. The following ranking applies:

$$\text{character} > \text{double} > \text{integer} > \text{logical}$$

Thus, for example, if one of the elementary values is of type `character`, then the resulting vector is of type `character` and numerical or logical values are converted to this type.

```
x = c(1, "a", TRUE)  # concatenation of different data types
typeof(x)            # coerced data type
```

```
[1] "character"
```

```
x                    # character vector
```

```
[1] "1"    "a"    "TRUE"
```

If, by contrast, one of the elementary values is of type `integer` and another is of type `logical`, then the resulting vector is of type `integer` and the logical values are converted to integers.

```
x = c(1L, TRUE, FALSE)  # combination of mixed data types (integer beats logical)
typeof(x)               # generated vector is of type integer
```

```
[1] "integer"
```

```
x
```

```
[1] 1 1 0
```

Since **R** performs this data type coercion implicitly, one should be careful to concatenate only values of the same data type with the function `c()`. To be as flexible as possible, **R** also performs more subtle data coercions. Thus, for example, it is possible to compute the sum of logical values, where the logical values `TRUE` and `FALSE` are implicitly treated as the numerical values `0` and `1`.

```
x = c(TRUE, FALSE, TRUE, TRUE)  # logical vector
s = sum(x)                      # sum of logical elements converted to integer
s
```

```
[1] 3
```

Explicit data type coercion is allowed by the functions `as.logical()`, `as.integer()`, `as.double()`, and `as.character()`. If one applies these to a vector, the resulting vector is of the desired type.

```
x = c(0,1,1,0)      # double vector
typeof(x)
```

```
[1] "double"
```

```
x
```

```
[1] 0 1 1 0
```

```
y = as.logical(x)  # converted to logical
typeof(y)
```

```
[1] "logical"
```

```
y
```

```
[1] FALSE  TRUE  TRUE FALSE
```

```
z = as.integer(x)
typeof(z)
```

```
[1] "integer"
```

```
z
```

```
[1] 0 1 1 0
```

14.3 Indexing

In **R** and many other programming languages, individual or several vector entries are addressed by *indexing*. Indexing is also called *subsetting* or *slicing*. In **R**, square brackets of the form [ ] are used for indexing. **R** uses *one-based indexing*, which means that the index of the first element is `1` (and not, for example, `0` as in Python).

Indexing of vector entries can be used to copy a specific entry of a vector into another data structure:

```
x = c("a", "b", "c")  # character vector ["a", "b", "c"]
y = x[2]              # copy of "b" in y
y                     # output
```

```
[1] "b"
```

Likewise, indexing of vector entries can be used to manipulate a specific entry of the indexed vector:

```
x    = c("a", "b", "c")  # character vector ["a", "b", "c"]
x[3] = "d"               # change of x to x = ["a", "b", "d"]
x                        # output
```

```
[1] "a" "b" "d"
```

In principle, in **R**, indexing

(1) with a vector of positive numbers addresses the corresponding entries,
(2) with a vector of negative numbers addresses the corresponding complementary entries,
(3) with a logical vector addresses the entries with logical value `TRUE`, and
(4) with a `character` vector addresses the correspondingly named entries.

We want to illustrate these principles using the following examples.

(1) Indexing with a vector of positive numbers

```
x = c(1,4,9,16,25)  # [1,4,9,16,25] = [1^2, 2^2, 3^2, 4^2, 5^2]
y = x[1:3]          # 1:3 generates vector [1,2,3], x[1:3] = [1,4,9]
z = x[c(1,3,5)]     # c(1,3,5) generates vector [1,3,5], x[c(1,3,5)] = [1,9,25]
x
```

```
[1]  1  4  9 16 25
```

```
y
```

```
[1] 1 4 9
```

```
z
```

```
[1]  1  9 25
```

(2) Indexing with a vector of negative numbers

```
x = c(1,4,9,16,25)  # [1,4,9,16,25] = [1^2, 2^2, 3^2, 4^2, 5^2]
y = x[c(-2,-4)]     # all components except 2 and 4, x[c(-2,-4)] = [1,9,25]
x
```

```
[1]  1  4  9 16 25
```

```
y
```

```
[1]  1  9 25
```

```
z = x[c(-1,2)]      # mixed indexing is not allowed (error message)
```

```
Error in `x[c(-1, 2)]`:
! nur Nullen dürfen mit negativen Indizes gemischt werden
```

(3) Indexing with a logical vector

```
x = c(1,4,9,16,25)   # [1,4,9,16,25] = [1^2, 2^2, 3^2, 4^2, 5^2]
y = x[c(T,T,F,F,T)]  # components with TRUE, x[c(T,T,F,F,T)] = [1,4,25]
z = x[x > 5]         # x > 5 = [F,F,T,T,T], x[x > 5] = [9,16,25]
x
```

```
[1]  1  4  9 16 25
```

```
y
```

```
[1]  1  4 25
```

```
z
```

```
[1]  9 16 25
```

(4) Indexing with a `character` vector

```
x = c(1,4,9,16,25)     # [1,4,9,16,25] = [1^2, 2^2, 3^2, 4^2, 5^2]
names(x) = c("a","b")  # naming of components as [a  b <NA> <NA> <NA>]
y = x["a"]             # x["a"] = 1
```

It remains to be noted that **R** exhibits high flexibility in indexing, that is, it often does not issue errors even though an indexing operation makes no sense per se. For example, out-of-range indices, that is, indices that exceed the number of entries of a vector, do not cause errors, but instead return `NA` (not available).

```
x = c(1,4,9,16,25)  # [1,4,9,16,25] = [1^2, 2^2, 3^2, 4^2, 5^2]
y = x[10]           # x[10] = NA (not available)
y
```

```
[1] NA
```

Furthermore, non-integer indices do not cause errors, but are rounded down to the next integer.

```
y = x[4.9]   # x[4.9]  = x[4]  = 16
y
```

```
[1] 16
```

```
z = x[-4.9]  # x[-4.9] = x[-4] = [1,4,9,25]
z
```

```
[1]  1  4  9 25
```

Finally, empty indices index the entire vector.

```
y = x[]  # y = x
```

The flexibility of **R** in nonsensical indexing has the advantage that programs are less often interrupted with an error. On the other hand, it has the disadvantage that unintended misindexing remains unnoticed and data analysis programs can thus produce unintended results. When working with indexing in **R**, a high degree of attention is therefore required, and ideally the validation of created data analysis programs using simulated data with known results. A similar degree of caution is advisable because of **R**'s *recycling* functionality, which we will get to know in the next section.

14.4 Arithmetic of numeric vectors

With **R** vectors, one can carry out arithmetic operations, that is, one can compute with them. We distinguish between *unary arithmetic operations*, that is, arithmetic operations applied to one vector, and *binary arithmetic operations*, in which two vectors are linked with each other. In general, arithmetic vector operators in **R** are evaluated elementwise, but some special features must be observed for binary operators.

Let us first consider some unary arithmetic operators and the application of functions to a vector. In the following example, the unary operators `-` and `^2`, as well as the function `log()`, are applied to a vector `a`. The evaluation of the operators or the function occurs *elementwise*, which means that the operator or function is applied to each element of the vector independently of its other elements.

```
a = seq(0,1,len=11)  # a = [ 0.0   ,  0.1   , ...,  0.9    , 1.0    ]
a
```

```
 [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
```

```
b = -a                # b = [-0.0   , -0.1   , ..., -0.9   ,-1.0   ]
b
```

```
 [1]  0.0 -0.1 -0.2 -0.3 -0.4 -0.5 -0.6 -0.7 -0.8 -0.9 -1.0
```

```
v = a^2               # v = [ 0.0^2 ,  0.1^2 , ...,  0.9^2  , 1.0^2  ]
v
```

```
 [1] 0.00 0.01 0.04 0.09 0.16 0.25 0.36 0.49 0.64 0.81 1.00
```

```
w = log(a)            # w = [ln(0.0), ln(0.1), ...,  ln(0.9), ln(1.0)]
w
```

```
 [1]       -Inf -2.3025851 -1.6094379 -1.2039728 -0.9162907 -0.6931472
 [7] -0.5108256 -0.3566749 -0.2231436 -0.1053605  0.0000000
```

Binary arithmetic operators, that is, the linking of two vectors, are likewise evaluated elementwise. When linking two vectors of the same length, the vector entries with the same indices are linked using the binary arithmetic operator. The result is a vector with the same length as the two initial vectors.

```
a = c(1,2,3)  # a = [1,2,3]
b = c(2,1,4)  # b = [2,1,4]
v = a + b     # v = [1,2,3] + [2,1,4] = [1+2,2+1,3+4] = [   3,   3,    7]
v
```

```
[1] 3 3 7
```

```
w = a - b     # w = [1,2,3] - [2,1,4] = [1-2,2-1,3-4] = [  -1,   1,   -1]
w
```

```
[1] -1  1 -1
```

```
x = a * b     # x = [1,2,3] * [2,1,4] = [1*2,2*1,3*4] = [   2,   2,   12]
x
```

```
[1]  2  2 12
```

```
y = a / b     # y = [1,2,3] / [2,1,4] = [1/2,2/1,3/4] = [0.50,   2, 0.75]
y
```

```
[1] 0.50 2.00 0.75
```

If one links a vector and a *scalar*, that is, a vector of length 1, in **R**, then **R** expands the scalar to a vector of the same length and performs the elementwise linking of these vectors. The following examples illustrate this.

```
a = c(1,2,3)  # a = [1,2,3]
b = 2         # b = [2]
v = a + b     # v = [1,2,3] + [2,2,2] = [1+2,2+2,3+2] = [  3,  4,   5]
v
```

```
[1] 3 4 5
```

```
w = a - b     # w = [1,2,3] - [2,2,2] = [1-2,2-2,3-2] = [ -1,  2,   1]
w
```

```
[1] -1  0  1
```

```
x = a * b     # x = [1,2,3] * [2,2,2] = [1*2,2*2,3*2] = [  2,  4,   6]
x
```

```
[1] 2 4 6
```

```
y = a / b     # y = [1,2,3] / [2,2,2] = [1/2,2/2,3/2] = [0.5,  1, 1.5]
y
```

```
[1] 0.5 1.0 1.5
```

Under the term *recycling*, **R** allows, in addition to elementwise arithmetic for vectors of the same length or for scalars and vectors, arithmetic with vectors of different lengths. For this purpose, elements of the shorter of two vectors are used to extend it to the length of the longer vector, that is, they are *recycled* in this sense. This property of vector arithmetic in **R** again means that programs abort with an error less often and that recycling can, under certain circumstances, offer elegant solutions for computations in informed code. However, it has the clear disadvantage that unintended misallocations of vectors, for example due to the absence of data points, can remain undetected. Here, too, a high degree of attention and, in particular, extensive testing are therefore indicated when creating programs. Specifically, one can distinguish two cases in the recycling of vector arithmetic in **R**: If the length of the longer of the two vectors is an integer multiple of the length of the shorter vector, then the shorter vector is expanded by repeating its elements in their original order the corresponding number of times. The following example illustrates this.

```
x = 1:2    # x = [1,2], length(x) = 2
x
```

```
[1] 1 2
```

```
y = 3:6    # y = [3,4,5,6], length(y) = 4
y
```

```
[1] 3 4 5 6
```

```
v = x + y  # v = [1,2,1,2] + [3,4,5,6] = [4,6,6,8]
v
```

```
[1] 4 6 6 8
```

In this sense, the arithmetic of vectors and scalars, that is, vectors of length 1, is a special case of this principle. If the length of the longer of the two vectors is not an integer multiple of the length of the shorter vector, then elements of the shorter vector are used in their original order to fill up the shorter vector until both vectors have the same length. The following example illustrates this.

```
x = c(1,3,5)       # x = [1,3,5], length(x) = 3
x
y = c(2,4,6,8,10)  # y = [2,4,6,8,10], length(y) = 5
y
v = x + y          # v = [1,3,5,1,3] + [2,4,6,8,10] = [3,7,11,9,13]
v
```

14.5 Attributes

Attributes in **R** are metadata of objects in the form of key-value pairs. In general, the function `attributes()` calls all attributes of an object. Vectors do not have attributes per se.

```
a = 1:3        # a numeric vector
attributes(a)  # call all attributes of a
```

```
NULL
```

The function `attr()` can be used to call and define attributes. The following code defines an attribute with the key (name) `"S"` and the value `"W"` for the vector `a`.

```
attr(a, "S") = "W"  # a receives attribute with key S and value W
attr(a, "S")        # the attribute with key S has value W
```

```
[1] "W"
```

However, attributes are often removed during operations.

```
b = a[1]       # copy of the first element of a in vector b
attributes(b)  # call all attributes of b
```

```
NULL
```

The specification of the attribute `names` gives names to the elements of a vector that can be used for indexing and are not removed during operations.

```
v = c(x=1,y=2,z=3)  # element-name generation during vector creation
v                   # vector output
```

```
x y z
1 2 3
```

Indexing using names has the following form:

```
v["y"]  # indexing by name
```

```
y
2
```

After the definition of a vector, the function **names()** can be used to define names for the vector entries and to look them up.

```
y = 4:6                   # generation of a vector
names(y) = c("a","b","c")  # definition of names
names(y)                  # call element names
```

```
[1] "a" "b" "c"
```

Named vector entries can be helpful if the vector forms a unit of meaning and, for example, represents properties of a participant such as age, height, and weight.

```
p = c(age    = 31,   # age (years)
      height = 198,  # height (cm)
      weight = 75)   # weight (kg)
p                    # vector output
```

```
   age height weight
    31    198     75
```

However, working with named vector entries is rather rare. Nevertheless, the principle of providing entries of a data structure with a meaningful label instead of a numerical index is common practice in **R**, as we will see when working with the central data structure of data frames in the chapter on lists, data frames, and tibbles.

15 Matrices and arrays

R matrices and arrays can be understood as higher-dimensional generalizations of **R** vectors. As with vectors, it is important in programming to distinguish matrices from the mathematical objects of the same name. Thus, in programming, matrices can simply serve for the tabular storage of numerical data, without the primary goal necessarily being the application of mathematical matrix calculus. Conversely, elementwise arithmetic operations with matrices in programming represent rather special mathematical operations such as the Hadamard product in mathematics. In this section, we primarily want to examine matrices and arrays as data containers in imperative programming with **R**. An introduction to matrix calculus with the same data structures is given in Chapter 9.

15.1 Matrices

Abstractly speaking, **R** matrices are two-dimensional, rectangular data structures of the form

$$M = \begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1n_c} \\ m_{21} & m_{22} & \cdots & m_{2n_c} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n_r1} & m_{n_r2} & \cdots & m_{n_rn_c} \end{pmatrix} \tag{15.1}$$

The elements $m_{ij}, i = 1, ..., n_r, j = 1, ..., n_c$ of the matrix are necessarily of the same data type. In the above scheme, n_r denotes the number of rows and n_c denotes the number of columns of the matrix M. Each element of a matrix therefore has a row index i and a column index j. Intuitively, **R** matrices are thus numerically indexed tables. Formally, matrices in **R** are atomic vectors interpreted two-dimensionally.

15.1.1 Generation

The **R** function `matrix()` is used to generate a matrix; it fills matrices with the elements of a vector. The general syntax of this function is `M = matrix(v, nrow, ncol, byrow)`, where

- `v` denotes a vector of length `nrow`$\cdot$ `ncol`,
- `nrow` denotes the desired number of rows of `M`,
- `ncol` denotes the desired number of columns of `M`,
- the logical variable `byrow` denotes the filling order of `M`.

Since, for a given length of `v` and a given number of rows or columns, the corresponding number of columns or rows of the matrix follows, it is only necessary to specify either `nrow` or `ncol`, as the following examples show.

```
matrix(c(1:12), nrow = 3)                # 3 x 4 matrix of the numbers 1,...,12, byrow = FALSE
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12
```

```
matrix(c(1:12), ncol = 4)                # 3 x 4 matrix of the numbers 1,...,12, byrow = FALSE
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12
```

```
matrix(c(1:12), nrow = 3, byrow = TRUE)  # 3 x 4 matrix of the numbers 1,...,12, byrow = TRUE
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    2    3    4
[2,]    5    6    7    8
[3,]    9   10   11   12
```

Existing matrices with the same number of rows can be concatenated columnwise with the function `cbind()`.

```
A = matrix(c(1:4) , nrow = 2)  # 2 x 2 matrix of the numbers 1,...,4
A
```

```
     [,1] [,2]
[1,]    1    3
[2,]    2    4
```

```
B = matrix(c(5:10), nrow = 2)  # 2 x 3 matrix of the numbers 5,...,10
B
```

```
     [,1] [,2] [,3]
[1,]    5    7    9
[2,]    6    8   10
```

```
C = cbind(A,B)                 # columnwise concatenation of A and B
C
```

```
     [,1] [,2] [,3] [,4] [,5]
[1,]    1    3    5    7    9
[2,]    2    4    6    8   10
```

Equivalently, existing matrices with the same number of columns can be concatenated rowwise with the function `rbind()`.

```
A = matrix(c(1:6) , nrow = 2, byrow = TRUE)  # 2 x 3 matrix of the numbers 1,...,6
print(A)
```

```
     [,1] [,2] [,3]
[1,]    1    2    3
[2,]    4    5    6
```

```
B = matrix(c(7:9), nrow = 1)                # 1 x 3 matrix of the numbers 7,...,9
print(B)
```

```
     [,1] [,2] [,3]
[1,]    7    8    9
```

```
C = rbind(A,B)                              # rowwise concatenation of A and B
print(C)
```

```
     [,1] [,2] [,3]
[1,]    1    2    3
[2,]    4    5    6
[3,]    7    8    9
```

15.1.2 Characterization

The familiar function `typeof()` can be used to output the elementary data type of a matrix:

```
typeof(matrix(c(1,0,0,1)    , nrow = 2))  # 2 x 2 double matrix
```

```
[1] "double"
```

```
typeof(matrix(c(1L,0L,0L,1L), nrow = 2))  # 2 x 2 integer matrix
```

```
[1] "integer"
```

```
typeof(matrix(c(T,T,F,F)    , nrow = 2))  # 2 x 2 logical matrix
```

```
[1] "logical"
```

```
typeof(matrix(c("a","b","c"), nrow = 1))  # 1 x 3 character matrix
```

```
[1] "character"
```

The number of rows or columns of a matrix can be output with the functions `nrow()` (number of rows) and `ncol()` (number of columns).

```
C = matrix(1:12, nrow = 3)  # 3 x 4 matrix
nrow(C)                     # number of rows
```

```
[1] 3
```

```
ncol(C)                     # number of columns
```

```
[1] 4
```

15.1.3 Indexing

In general, matrix elements are indexed with a row index and a column index. The first of two matrix indices always indexes the row, and the second always indexes the column. The remaining principles of matrix indexing correspond essentially to the principles of vector indexing. One special feature is that a missing row or column index indexes all elements of the corresponding dimension.

```
A = matrix(c(2:7)^2, nrow = 2)  # 2 x 3 matrix of the numbers 2^2,...,7^2
A
```

```
     [,1] [,2] [,3]
[1,]    4   16   36
[2,]    9   25   49
```

```
A[1,3]                          # element in row 1, column 3 of A [36]
```

```
[1] 36
```

```
A[2,2]                          # element in row 2, column 2 of A [25]
```

```
[1] 25
```

```
A[2,]                           # all elements of row 2 [9,25,49]
```

```
[1]  9 25 49
```

```
A[,3]                           # all elements of column 3 [36,49]
```

```
[1] 36 49
```

```
A[1:2,1:2]                      # submatrix of the first two rows and columns
```

```
     [,1] [,2]
[1,]    4   16
[2,]    9   25
```

```
A[A>10]                         # elements of A greater than 10 [16,25,36,49]
```

```
[1] 16 25 36 49
```

```
A[1,c(F,F,T)]                   # element in row 1, column 3 of A [36]
```

```
[1] 36
```

15.1.4 Arithmetic

If one applies unary arithmetic operators and functions to matrices, these are evaluated elementwise, as with vectors.

```
A = matrix(c(1:4), nrow = 2)  # 2 x 2 matrix of the numbers 1,2,3,4
A
```

```
     [,1] [,2]
[1,]    1    3
[2,]    2    4
```

```
B = A^2                       # result B[i,j] = A[i,j]^2, 1 <= i,j <= 2
B
```

```
     [,1] [,2]
[1,]    1    9
[2,]    4   16
```

```
C = sqrt(B)                   # result C[i,j] = sqrt(A[i,j]^2), 1 <= i,j <= 2
C
```

```
     [,1] [,2]
[1,]    1    3
[2,]    2    4
```

```
D = exp(A)                    # result D[i,j] = exp(A[i,j]), 1 <= i,j <= 2
D
```

```
         [,1]     [,2]
[1,] 2.718282 20.08554
[2,] 7.389056 54.59815
```

Like vectors, matrices with identical numbers of rows and columns can be linked element-wise with the binary arithmetic operators `+`, `-`, `*`, and `/`. The following examples illustrate this:

```
A = matrix(c(1:4), nrow = 2)  # 2 x 2 matrix of the numbers 1,2,3,4
A
```

```
     [,1] [,2]
[1,]    1    3
[2,]    2    4
```

```
B = matrix(c(5:8), nrow = 2)  # 2 x 2 matrix of the numbers 5,6,7,8
B
```

```
     [,1] [,2]
[1,]    5    7
[2,]    6    8
```

```
C = A + B                     # result C[i,j] = A[i,j] + B[i,j], 1 <= i,j <= 2
C
```

```
     [,1] [,2]
[1,]    6   10
[2,]    8   12
```

```
D = A * B                    # result D[i,j] = A[i,j] * B[i,j], 1 <= i,j <= 2
D
```

```
     [,1] [,2]
[1,]    5   21
[2,]   12   32
```

In addition, **R** matrices can be used for computations in the sense of matrix calculus from linear algebra. We show only a few examples here; a detailed presentation of the mathematical principles and their **R** implementation can be found in Chapter 9.

```
C = A %*% B       # 2 x 2 matrix product
C
```

```
     [,1] [,2]
[1,]   23   31
[2,]   34   46
```

```
A_T = t(A)        # transposition of A
A_T
```

```
     [,1] [,2]
[1,]    1    2
[2,]    3    4
```

```
A_inv = solve(A)  # inverse of A
A_inv
```

```
     [,1] [,2]
[1,]   -2  1.5
[2,]    1 -0.5
```

```
A_det = det(A)    # determinant of A
A_det
```

```
[1] -2
```

15.1.5 Attributes

Formally, matrices are atomic vectors with a `dim` attribute.

```
A = matrix(1:12, nrow = 4 )  # 4 x 3 matrix
attributes(A)                # call the attributes of A
```

```
$dim
[1] 4 3
```

As with vectors, one can also give names to matrix elements, although this is rather unusual for matrices as well. Specifically, the functions `rownames()` and `colnames()` define the matrix attribute `dimnames`.

```
rownames(A) = c("P1","P2","P3","P4")  # naming the rows of A
colnames(A) = c("Age", "Hgt", "Wgt")  # naming the columns of A
A                                     # matrix with attribute dimnames
```

```
   Age Hgt Wgt
P1   1   5   9
P2   2   6  10
P3   3   7  11
P4   4   8  12
```

```
attr(A, "dimnames")                   # call the attribute dimnames
```

```
[[1]]
[1] "P1" "P2" "P3" "P4"

[[2]]
[1] "Age" "Hgt" "Wgt"
```

15.2 Arrays

R arrays are d-dimensional, hyperrectangular data structures. Intuitively, arrays can be understood as the generalization of matrices to more than two dimensions. For $d = 1$, an array corresponds to a vector; for $d = 2$, an array corresponds to a matrix. As with vectors and matrices, the elements of an array must be of the same data type. Arrays are characterized by how many elements $d_i, i = 1, ..., d$ the i-th dimension can represent. Each element of an array is indexed by a d-dimensional index $i_1, i_2, ..., i_d$.

15.2.1 Generation

Analogously to the function `matrix()`, the function `array()` is used to fill an array with vector elements. The general syntax of this function is `A = array(v, dim)`, where `v` is a vector of array data and `dim` is a vector of integer values that represents the maximal indices in each of the `length(dim)` dimensions of the array. For example, the following command generates a $2 \times 2 \times 3$ array.

```
A = array(1:12, dim = c(2,2,3))  # 2 x 2 x 3 array of the numbers 1,...,12
A
```

```
, , 1

     [,1] [,2]
[1,]    1    3
[2,]    2    4

, , 2

     [,1] [,2]
[1,]    5    7
[2,]    6    8

, , 3
```

```
     [,1] [,2]
[1,]    9   11
[2,]   10   12
```

Three-dimensional arrays can be imagined intuitively as matrices placed one behind another in space. The third dimension then represents the "pages" of the array. However, when working with higher-dimensional arrays, it is useful to try to detach oneself from spatial conceptions of arrays and instead understand arrays as data containers whose elements can be addressed uniquely by their indices.

15.2.2 Characterization

The `length()` function outputs the number of elements of an array.

```
A = array(1:12, dim = c(2,2,3))  # 2 x 2 x 3 array of the numbers 1,...,12
length(A)                        # number of elements of the array
```

```
[1] 12
```

The function `typeof()` outputs the elementary data type of an array:

```
typeof(A)  # type of the array
```

```
[1] "integer"
```

The function `dim()` outputs the dimensions of an array:

```
dim(A)  # dimension of the array
```

```
[1] 2 2 3
```

15.2.3 Indexing

Indexing of arrays is analogous to vector and matrix indexing, with the difference that three or more indices are needed to index an array element.

```
A     = array(1:12, dim = c(2,2,3))  # 2 x 2 x 3 array of the numbers 1,...,12
a_223 = A[2,2,3]                     # array element with index address [2,2,3] (12)
```

15.2.4 Arithmetic

As with vectors and matrices, unary arithmetic operators are evaluated elementwise for arrays. Likewise, binary arithmetic operators are evaluated elementwise for arrays of identical dimensionality.

```r
A = array(1:4, dim = c(1,2,2))  # 1 x 2 x 2 array of the numbers 1,...,4
A
```

```
, , 1

     [,1] [,2]
[1,]    1    2

, , 2

     [,1] [,2]
[1,]    3    4
```

```r
B = array(5:8, dim = c(1,2,2))  # 1 x 2 x 2 array of the numbers 5,...,8
B
```

```
, , 1

     [,1] [,2]
[1,]    5    6

, , 2

     [,1] [,2]
[1,]    7    8
```

```r
C = A^2                         # elementwise exponentiation
C
```

```
, , 1

     [,1] [,2]
[1,]    1    4

, , 2

     [,1] [,2]
[1,]    9   16
```

```r
D = A + B                       # elementwise addition
D
```

```
, , 1

     [,1] [,2]
[1,]    6    8

, , 2

     [,1] [,2]
[1,]   10   12
```

15.2.5 Attributes

Formally, arrays are atomic vectors with a `dim` attribute.

```
A = array(1:12, dim = c(2,2,3))  # 2 x 2 x 3 array of the numbers 1,...,12
attributes(A)                    # call the attributes of A
```

```
$dim
[1] 2 2 3
```

As with matrices, the function `dimnames()` can be used to name the array dimensions, but here too the naming of array dimensions is rather unusual.

16 Lists, dataframes, and tibbles

16.1 Lists

R lists are ordered sequences of **R** objects. Lists are called a recursive data type because they can bundle objects of different data types, including lists themselves. Although the intuitive view that lists “contain” objects is natural, lists de facto do not “contain” objects, but only the references to the corresponding objects in memory (Figure 16.1). Lists are an essential building block of **R** dataframes.

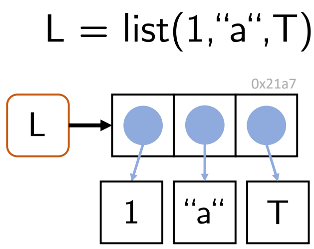

Figure 16.1 List representation. A list in **R** is an indexable sequence of references to **R** objects in memory.

Generation

Lists can be generated by directly concatenating the list elements with the function `list()`.

```
# generation of a list with a vector, matrix, and function element
L = list(c(1,4,5), matrix(1:8, nrow = 2), exp)
print(L)
```

```
[[1]]
[1] 1 4 5

[[2]]
     [,1] [,2] [,3] [,4]
[1,]    1    3    5    7
[2,]    2    4    6    8

[[3]]
function (x)  .Primitive("exp")
```

In particular, lists themselves can also be elements of lists, which motivates their designation as a *recursive* data type.

```
L = list(list(1))  # list with element 1 in a list
print(L)           # output
```

The function `c()`, familiar from **R** vectors, can also be used to concatenate several lists.

```
L = c(list(pi), list("a"))  # concatenation of two lists
print(L)                    # output
```

Characterization

The data type of lists is `list`:

```
L = list(1:2, "a", log)  # generation of a list
typeof(L)                # type determination
```

```
[1] "list"
```

`length()` returns the number of list elements at the top list level; the element contents are therefore ignored:

```
L = list(1:2, list("a", pi), exp)  # list with three top-level elements
length(L)                          # length() ignores element contents
```

```
[1] 3
```

The dimension, number of rows, and number of columns of lists are `NULL`:

```
L = list(1:2, "a", sin)  # a list
dim(L)                   # the dimension of lists is NULL
```

```
NULL
```

```
nrow(L)                  # the number of rows of lists is NULL
```

```
NULL
```

```
ncol(L)                  # the number of columns of lists is NULL
```

```
NULL
```

Indexing

When indexing lists, one must note that lists can be indexed in two different ways. Single square brackets `[ ]` index list elements as lists:

```
L  = list(1:3, "a", exp)  # a list
l1 = L[1]                 # indexing a list element
print(l1)                 # output
```

```
[[1]]
[1] 1 2 3
```

```
typeof(l1)                # type determination of l1
```

```
[1] "list"
```

Double square brackets `[[ ]]` index the contents of list elements:

```
L  = list(1:3, "a", exp)  # a list
i2 = L[[2]]               # indexing the list-element content
print(i2)                 # output of the content of list element L[2]
typeof(i2)                # type determination of i2
```

These conventions must also be observed when replacing list contents. For example, it is not possible to change the first list element with the expression `L[1] = c(1,2,3)`, because the vector `c(1,2,3)` is not a list:

```
L      = list(1:3, "a", exp)  # a list
L[1]   = 4:6                  # no type conversion, error message
```

```
Warning in L[1] = 4:6: Anzahl der zu ersetzenden Elemente ist kein Vielfaches
der Ersetzungslänge
```

```
L[1]   = list(4:6)            # replacement of the 1st list element
L[[3]] = "c"                  # replacement of the 3rd list-element content
```

Indexing

The principles of list indexing are analogous to those of vector indexing. Vectors of positive numbers address the corresponding elements:

```
L = list(1:3, "a", pi)  # a list
l = L[c(1,3)]           # 1st and 3rd list element
l
```

```
[[1]]
[1] 1 2 3

[[2]]
[1] 3.141593
```

Vectors of negative numbers address the elements complementary to their values:

```
L = list(1:3, "a", pi)  # a list
l = L[-c(1,3)]          # 2nd list element
l
```

```
[[1]]
[1] "a"
```

Logical vectors address elements with index `TRUE`:

```
L = list(1:3, "a", pi)  # a list
l = L[c(T,T,F)]         # 1st and 2nd list element
l
```

```
[[1]]
[1] 1 2 3

[[2]]
[1] "a"
```

For non-numerical addressing, list elements can also be given names. This is possible after the generation of a list with the `names()` function:

```
K        = list(1:2, TRUE)    # an unnamed list
names(K) = c("Frodo", "Sam")  # naming with names()
K                             # output
```

```
$Frodo
[1] 1 2

$Sam
[1] TRUE
```

It is likewise possible to assign the names of the list elements directly during the generation of the list:

```
L = list(frodo = 1:3, sam = "a", eowyn = "b")  # list generation with names
```

Both list elements and list-element contents can then be addressed with their names:

```
L = list(frodo = 1:3, sam = "a", eowyn = "b")  # list generation with names
L["frodo"]                                     # list-element output
```

```
$frodo
[1] 1 2 3
```

```
typeof(L["frodo"])                             # list-element type
```

```
[1] "list"
```

```
L[["frodo"]]                                # list-element content output
```

```
[1] 1 2 3
```

```
typeof(L[["sam"]])                          # list-element content type
```

```
[1] "character"
```

In particular, list-element contents can also be indexed with the $ operator and their name. In the context of dataframes, this is the most common kind of list indexing:

```
L = list(frodo = 1:3, sam = "a", eowyn = "b")  # list generation with names
L$frodo                                     # list-element content output
```

```
[1] 1 2 3
```

```
typeof(L$frodo)                             # list-element content type
```

```
[1] "integer"
```

```
L$sam                                       # list-element content output
```

```
[1] "a"
```

```
typeof(L$sam)                               # list-element content type
```

```
[1] "character"
```

```
L$eowyn                                     # list-element content output
```

```
[1] "b"
```

```
typeof(L$eowyn)                             # list-element content type
```

```
[1] "character"
```

Arithmetic

General list arithmetic is not defined, because list-element contents can be of different data types for which unary or binary operations are not defined:

```
L1 = list(1:3, "a" )  # a list
L2 = list(T, exp )    # a list
L1+L2                 # attempt at list addition
```

```
Error in `L1 + L2`:
! nicht-numerisches Argument für binären Operator
```

If, however, the contents of list elements are of a common data type for which certain binary operations are defined, then they can also be linked, for example arithmetically:

```
L1 = list(1:3, pi )   # a list
L2 = list(4:6, exp )  # a list
L1[[1]] + L2[[1]]     # addition of the 1st list-element contents, [1+4, 2+5,3+6]
```

```
[1] 5 7 9
```

```
L2[[2]](1)            # application of the 2nd list-element content, exp(1)
```

```
[1] 2.718282
```

Copy-on-modify

As with vectors, the *copy-on-modify* principle applies to lists. Specifically, for lists, so-called *shallow copies* are generated when a list is modified. In this process, only the list object itself is copied and receives a new address in memory, but not the objects bound by the list, which remain at the same addresses in memory. `lobstr::ref()` makes it possible to trace this behavior.

```
L1 = list(1,2,3)  # generation of a list as an object
lobstr::ref(L1)   # output of the references
```

```
 [1:0x272aa1c7dc8] <list>
[2:0x272a7394318] <dbl>
[3:0x272a7394120] <dbl>
[4:0x272a7419e38] <dbl>
```

```
L2 = L1           # same object referenced by L1 and L2
lobstr::ref(L2)   # output of the references
```

```
 [1:0x272aa1c7dc8] <list>
[2:0x272a7394318] <dbl>
[3:0x272a7394120] <dbl>
[4:0x272a7419e38] <dbl>
```

```
L2[[3]] = 4       # copy-on-modify with shallow copy
lobstr::ref(L2)   # output of the references
```

```
 [1:0x272aa2399a8] <list>
[2:0x272a7394318] <dbl>
[3:0x272a7394120] <dbl>
[4:0x272a740b1c0] <dbl>
```

Apparently, the assignment `L2 = L1` initially changes nothing about the memory addresses to which `L1` and `L2` as well as their elements refer. The modification of the content of the third list element of `L2` by `L2[[3]] = 4`, however, then changes the address of the list object in memory because a copy of the list object is generated. At the same time, according to the shallow-copy principle, the addresses of the first two contents of `L2` do not change. Figure 16.2 visualizes these principles with an example.

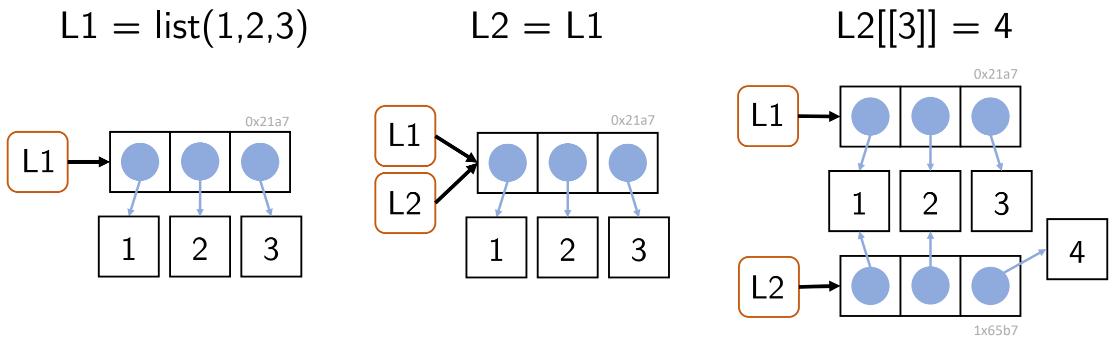

Figure 16.2 Effects of the assignment `L2 = L1` for a list `L1` and of the modification of an element of `L2`.

16.2 Dataframes

R dataframes are the central data structure in **R**. Dataframes are best imagined as tables whose columns have names. Technically, a dataframe is a list whose elements are vectors of the same length. The list elements correspond to the columns of a table, and the vector elements at the same position correspond to the rows of a table.

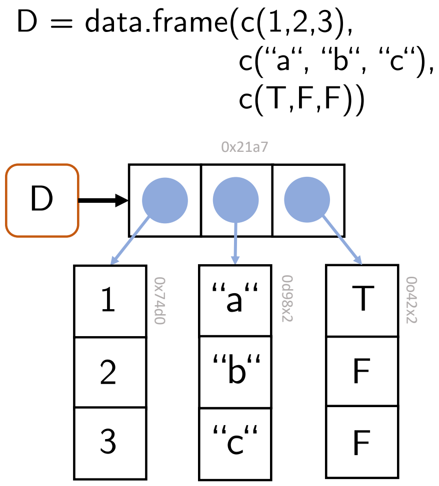

Figure 16.3 Dataframe representation. An **R** dataframe is an **R** list whose elements are **R** vectors of the same length.

Generation

Dataframes are generated with the function `data.frame()`. The column names and their vector-valued contents are specified as arguments of the function:

```
D = data.frame(x = letters[1:4],   # 1st column with name x
               y = 1:4,            # 2nd column with name y
               z = c(T,T,F,T))     # 3rd column with name z
print(D)                           # output of the dataframe as a table
```

```
  x y     z
1 a 1  TRUE
2 b 2  TRUE
3 c 3 FALSE
4 d 4  TRUE
```

The columns of the dataframe must have the same length; otherwise a dataframe is not defined:

```
D = data.frame(x = letters[1:4],  # 1st column with name x
               y = 1:4,           # 2nd column with name y
               z = c(T,T,F))      # 3rd column with name z
```

```
Error in `data.frame()`:
! Argumente implizieren unterschiedliche Anzahl Zeilen: 4, 3
```

The columns of a dataframe can evidently be of different data types; this is, of course, congruent with the character of **R** lists.

Characterization

The functions `names()` and `colnames()` can be used to output the names of the columns of a dataframe, and the function `rownames()` can be used to output its row names. By default, the row names of a dataframe are the integers 1,2,...,n:

```
D = data.frame(age    = c(30,35,40,45),        # 1st column
               height = c(178,189,165,171),    # 2nd column
               weight = c(67, 76, 81, 92))     # 3rd column
names(D)                                        # names outputs the column names
```

```
[1] "age"    "height" "weight"
```

```
colnames(D)                                     # colnames corresponds to names
```

```
[1] "age"    "height" "weight"
```

```
rownames(D)                                     # default rownames are 1,2,...
```

```
[1] "1" "2" "3" "4"
```

Furthermore, a dataframe is characterized by its number of rows and columns. These can be output with the functions `nrow()` and `ncol()`, respectively. For dataframes, as for lists, the function `length()` returns the number of list entries; for dataframes this correspondingly means the number of columns.

```
nrow(D)    # number of rows
```

```
[1] 4
```

```
ncol(D)    # number of columns
```

```
[1] 3
```

```
length(D)  # length is the number of columns
```

```
[1] 3
```

A helpful function for characterizing a dataframe is the function `str()`, which displays essential aspects of a dataframe in compact form. `str()` stands for *structure* and denotes the columns of the dataframe as *variables* and row entries as *observations*:

```
str(D)
```

```
'data.frame':	4 obs. of  3 variables:
 $ age   : num  30 35 40 45
 $ height: num  178 189 165 171
 $ weight: num  67 76 81 92
```

Attributes

Technically, dataframes are lists with attributes for (column) `names` and `row.names`. Dataframes are of `class` "data.frame":

```
typeof(D)
```

```
[1] "list"
```

```
attributes(D)
```

```
$names
[1] "age"    "height" "weight"

$class
[1] "data.frame"

$row.names
[1] 1 2 3 4
```

Indexing

Dataframes can be indexed both like matrices and like lists. The indexing principles correspond to those of vectors. The following example first shows how a dataframe is indexed with single brackets in the sense of a list. The corresponding list element is then itself a dataframe for dataframes:

```
D = data.frame(x = letters[1:4],  # 1st column with name x
               y = 1:4,           # 2nd column with name y
               z = c(T,T,F,T))    # 3rd column with name z
class(D)                          # dataframe D
```

```
[1] "data.frame"
```

```
v = D[1]                        # 1st list element as dataframe
v
```

```
  x
1 a
2 b
3 c
4 d
```

```
class(v)                        # v is a dataframe
```

```
[1] "data.frame"
```

Furthermore, dataframes can be indexed like lists with double brackets. This addresses the contents of a list element:

```
w = D[[1]]   # content of the 1st list element
w
```

```
[1] "a" "b" "c" "d"
```

```
class(w)     # w is a character vector
```

```
[1] "character"
```

The typical indexing of a dataframe uses the $ indexing familiar from lists, which addresses the content of the corresponding dataframe column:

```
y = D$y    # $ for indexing the y column
y
```

```
[1] 1 2 3 4
```

```
class(y)  # y is a vector of type "integer" (!)
```

```
[1] "integer"
```

If one is interested in specific contents of the dataframe in specific rows and columns, the principles of matrix indexing are useful, as the following example shows:

```
D[2:3,-2]        # 1st index for rows, 2nd index for columns
```

```
  x     z
2 b  TRUE
3 c FALSE
```

```
D[c(T,F,T,F),]   # 1st index for rows, 2nd index for columns
```

```
  x y     z
1 a 1  TRUE
3 c 3 FALSE
```

```
D[,c("x", "z")]  # 1st index for rows, 2nd index for columns
```

```
  x     z
1 a  TRUE
2 b  TRUE
3 c FALSE
4 d  TRUE
```

Copy-on-modify

For optimal use of memory, the copy-on-modify principles for lists also apply to dataframes. Furthermore, the modification of a column leads to the copy of the corresponding column, whereas the modification of a row results in the copy of the entire dataframe.

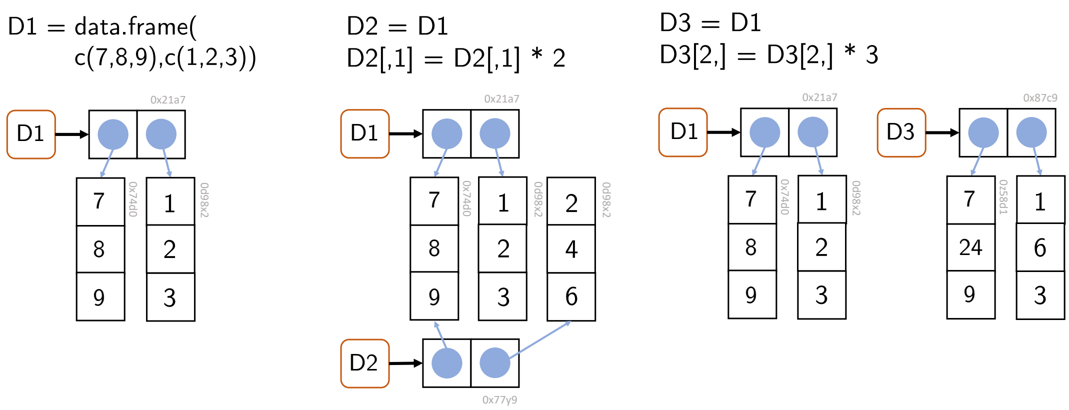

Figure 16.4 Effects of column and row modifications in dataframes. The modification of a column leads to the copy of the corresponding column, and the modification of a row to the copy of the entire dataframe.

16.3 Tibbles

In addition to **R** dataframes, *tibbles* (diminutive for *tables*) have become widespread in the last ten years as part of the `tidyverse` package. Tibbles are a more modern, technically improved form of dataframes and show how **R** has been extended for data science tasks beyond the application of classical linear models. To use tibbles, the `tidyverse` package must be installed and the tibble code must be loaded:

```
install.packages("tidyverse")  # one-time package download and installation
library(tibble)                # session-specific loading of the tibble code
```

By and large, tibbles correspond to dataframes. As with dataframes, tibbles are lists of vector references, where the vectors again have the same length and represent the named columns of a table. Tibbles can be generated with the function `tibble()`. The display in the **R** terminal is optimized compared with dataframes, as the following example shows:

```
library(tibble)                                                        # package
```

```
Warning: Paket 'tibble' wurde unter R Version 4.5.3 erstellt
```

```
T = tibble(x = c("a", "c", "d"), y = c(1,2,3), z = c(TRUE,FALSE,FALSE))  # generation
print(T)                                                               # output
```

```
# A tibble: 3 x 3
  x         y z    
  <chr> <dbl> <lgl>
1 a         1 TRUE 
2 c         2 FALSE
3 d         3 FALSE
```

```
attributes(T)                                                          # attributes
```

```
$class
[1] "tbl_df"     "tbl"        "data.frame"

$row.names
[1] 1 2 3

$names
[1] "x" "y" "z"
```

16.3.1 Tibbles vs. dataframes

Tibbles were introduced in 2016 as part of the `tidyverse` package to compensate for perceived weaknesses of **R** dataframes. These included, among other things, that the function `data.frame()` automatically converted character vectors into the **R** data type `factor`. Since an **R** revision from 2019, however, this is no longer the case, as the following example shows:

```
D = data.frame(x = 1:3, y = c("a", "b", "c"))  # character vector y
str(D)                                         # remains character vector
```

```
'data.frame':   3 obs. of  2 variables:
 $ x: int  1 2 3
 $ y: chr  "a" "b" "c"
```

By default, `data.frame()` still converts inadmissible column names such as a number (`1`) into legal column names (`X1`). If one does not want this, however, one can turn off this behavior by setting the argument `check.names`.


```
names(data.frame(`1` = 1))                    # conversion of nonlegal variable names
```

```
[1] "X1"
```

```
names(data.frame(`1` = 1, check.names = FALSE))  # turning off the default behavior
```

```
[1] "1"
```

As seen in Chapter 14, **R** tends to generate data itself through recycling when data types have unsuitable lengths, which can easily lead to errors. Thus, for example, the function `data.frame()` recycles vectors when generating a dataframe if one of them is an integer multiple of the other:

```
D = data.frame(x = 1:2, y = 3:6)  # length(y) = 2 * length(x)
print(D)                          # output
```

```
  x y
1 1 3
2 2 4
3 1 5
4 2 6
```

By contrast, `tibble()` is more precise here and issues an error in the case above:

```
T = tibble(x = 1:2, y = 3:6)  # tibble issues an error here
```

```
Error in `tibble()`:
! Tibble columns must have compatible sizes.
* Size 2: Existing data.
* Size 4: Column `y`.
i Only values of size one are recycled.
```

Vectors of length 1, however, are also recycled by `tibble()`:

```
T = tibble(x = 1:3, y = 3)  # vectors with length 1
print(T)                    # output
```

```
# A tibble: 3 x 2
      x     y
  <int> <dbl>
1     1     3
2     2     3
3     3     3
```

Dataframes are sometimes inconsistent with respect to the data types they return. As the example below shows, indexing a dataframe column by its name yields a dataframe, whereas indexing the same column by its name in matrix indexing yields a vector:

```
D = data.frame(x = 1:3, y = 4:6)  # dataframe
str(D["x"])                       # indexing with names returns a dataframe
```

```
'data.frame':	3 obs. of  1 variable:
 $ x: int  1 2 3
```

```
str(D[,"x"])                      # matrix indexing with names returns a vector
```

```
 int [1:3] 1 2 3
```

Tibbles are more consistent in this respect:

```
T = tibble(x = 1:3, y = 4:6)  # tibble
str(T["x"])                   # indexing with names yields tibble
```

```
tibble [3 x 1] (S3: tbl_df/tbl/data.frame)
 $ x: int [1:3] 1 2 3
```

```
str(T[,"x"])                  # matrix indexing with names yields tibble
```

```
tibble [3 x 1] (S3: tbl_df/tbl/data.frame)
 $ x: int [1:3] 1 2 3
```

An additional feature of tibbles compared with dataframes is that the function `tibble()` allows column names to be referenced already during the generation of a tibble, which is not possible with dataframes.

```
T = tibble(x = 1, y = 2*x)      # column x is defined and used
D = data.frame(x = 1, y = 2*x)  # data.frame does not allow this
```

```
Error:
! Objekt 'x' nicht gefunden
```

In summary, tibbles can be said to function less flexibly and therefore more precisely than dataframes in some respects. In general, tibbles behave more consistently than dataframes in `tidyverse` contexts. However, as seen above, the differences between dataframes and tibbles are rather subtle, so that efficient handling of dataframes also enables efficient handling of tibbles. Since not all **R** projects use the packages of the `tidyverse`, dataframes will probably remain the standard data structure for tabular data in **R** in the future.

17 Data management

In this section, we want to introduce some general terminology for dealing with research data, discuss common formats for the longer-term storage of data, and finally consider the practical reading and saving of data in the context of **R** directory management.

17.1 The FAIR data ideal

The FAIR data ideal is a principle for the management of research data that takes into account the broad dissemination, usefulness, and transparency of research data in the digital age. *Research data* are generally understood as "electronically represented analog or digital data that arise or are used in the course of scientific projects, for example through observations, experiments, simulations, surveys, questionnaires, source research, recordings of audio and video sequences, digitization of objects, and analyses" (Rat für Informationsinfrastrukturen (2017)). Specifically, these include numerical arrays or character arrays, but also computer software such as **R** scripts, digital tools, workflows, and analysis pipelines.

In addition to these data forms, sometimes also referred to as *primary data*, so-called *metadata* are important for the FAIR data ideal. These are data that represent information about other data. One distinguishes, for example, *descriptive*, *structural*, and *administrative metadata* (Riley (2017), Ulrich et al. (2022)).

- *Descriptive metadata* serve to find and identify a data source. Examples of descriptive metadata include the title of a scientific publication or its Digital Object Identifier (DOI, Rosenblatt (1997)). DOIs are alphanumeric identifiers that uniquely identify digital objects and function as persistent links to objects on the internet.
- *Structural metadata* are metadata about data containers and represent the structural composition of a data source. Examples are the order of the pages of a book or the for-loop encoding of three-dimensional data objects.
- *Administrative metadata*, finally, are data that facilitate the management of a data source. Examples of administrative metadata are provenance, file format, access rights, or other technical information about a data source.

The FAIR ideal for research data management has its origin in discussions within the FORCE11 conference series and was communicated academically in 2016 by Wilkinson et al. (2016). In essence, it states that research data should be prepared and maintained as *Findable*, *Accessible*, *Interoperable*, and *Reusable* for humans and machines. Specifically, the FAIR data ideal formulates the following requirements for the management of research (meta)data:

Findability

- F1. (Meta)data have a persistent globally unique identifier.

- F2. Data are enriched with metadata.
- F3. Metadata are clearly assigned to a dataset.
- F4. (Meta)data are indexed in a searchable resource.

Accessibility

- A1. (Meta)data are retrievable with standardized protocols.
- A1.1. The protocol used is open, free, and usable.
- A1.2. The protocol enables authentication and assignment of rights.
- A2. Metadata remain accessible even if data are no longer available.

Interoperability

- I1. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. (Meta)data use vocabularies that follow the FAIR principles.
- I3. (Meta)data contain qualified references to other (meta)data.

Reusability

- R1. (Meta)data have a variety of accurate and relevant attributes.
- R1.1. (Meta)data contain a clear usage license.
- R1.2. (Meta)data contain detailed provenance information.
- R1.3. (Meta)data meet the standards of the respective disciplinary community.

The demand for fair research data management is now quite widespread, although the FAIR principles can be implemented with varying degrees of bureaucratic effort. It remains to be noted, however, that at present the FAIR principles are essentially an ideal of data management to be aspired to, and by no means all research data are available in FAIR form. In fact, dealing with digital research data remains very unstructured, and the German university system in particular only very slowly understands digital data management as a core task. On a larger, although project-based and thus not sustainable, scale, the initiative for a National Research Data Infrastructure (NFDI) is trying to improve German digital research data management. Unfortunately, it must still be noted that many publicly funded researchers neither want to systematically prepare the research data they collect nor make them available to the public. Here, a cultural change toward greater scientific transparency therefore remains to be hoped for, and open, publicly funded research (Open Research, cf., e.g., Toelch & Ostwald (2018), Miedema (2022), Bertram et al. (2023)) remains an ideal to be aspired to.

17.2 File formats

In general, file formats define the syntax and semantics of data within a file. File formats are bijective mappings of information onto binary long-term storage. In general, one distinguishes *data file formats* from *software file formats*, *textual* from *binary file formats*, and *open* from *proprietary*, that is, copyright-protected, file formats. For binary data file formats, reading data, inspecting them, and manipulating them is generally possible only with special software. Examples of binary file formats are *.docx*, *.pdf*, *.xlsx*, *.jpg*, and *.mp4* files. Binary file formats are often proprietary. In the past, binary file formats were also preferred for research data because of their lower storage requirements. Today, by contrast, textual data file formats such as *.txt*, *.csv*, *.tsv*, and *.json* are favored. In these formats, reading the stored data, inspecting them, and manipulating them with simple general editors such as VS Code is straightforward, and the file formats are generally not proprietary. Figure 17.1 shows the representation of a binary data format and a textual data format in Windows Notepad. In general, open textual data file formats are therefore appropriate in the research context and in the spirit of the FAIR principles. As examples, we want to consider the .csv, .tsv, and .json file formats as textual file formats in more detail here.

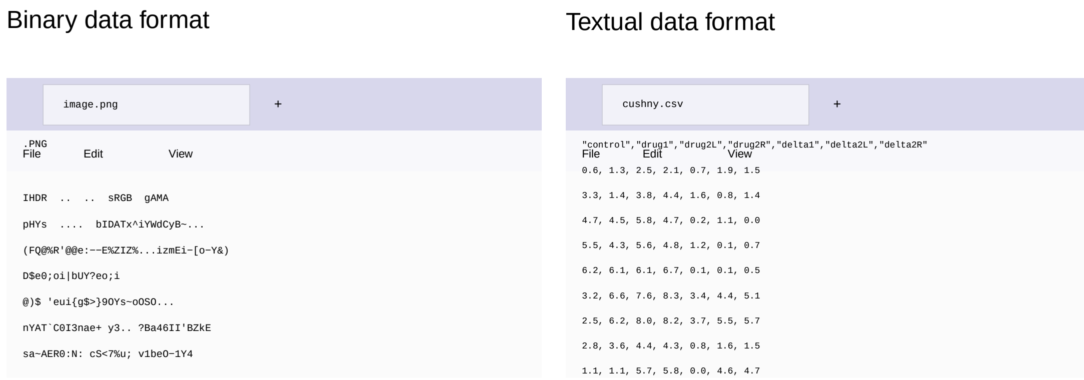

Figure 17.1 Binary and textual file formats as represented in an editor.

17.2.1 .csv and .tsv files

.csv (comma-separated values) and .tsv files (tab-separated values) are important file formats for storing simple, tabularly structured data. In general, .csv and .tsv represent datasets linked row by row. Data fields within a row are separated by commas (.csv) or tab characters (.tsv), and individual datasets are separated from one another by line breaks. The dataset in the first row of a .csv or .tsv file typically contains the definition of the column names and is called the header record. A typical example of a .csv file is shown in Figure 17.1 (right). Here, the rows represent experimental units (here participants), and the columns represent different variables for which a value was determined for the participant.

In the context of data analysis, one distinguishes a *wide-format* and a *long-format* data representation for .csv and .tsv files. In wide-format data representation, all variables of an experimental unit are represented in one row, as in the example above; cf. Table 17.1. For some data analyses, however, it is more helpful to distribute the data of an experimental unit across rows. This leads to the so-called long format, as shown in Table 17.2. In general, the wide format is often more helpful for a first intuitive understanding of a dataset, whereas the long format usually represents data more consistently with the model architecture in model-based analyses.

Table 17.1 Wide format (EU: experimental unit, V1,V2,V3: variables)

EU	V1	V2	V3
1	10.1	67.5	4
2	12.9	51.2	2

EU	V1	V2	V3
3	20.4	70.8	5
4	17.5	60.3	7

Table 17.2 Long format (EU: experimental unit, V1,V2,V3: variables)

Unit	Variable	Measured value
1	V1	10.1
1	V2	67.5
1	V3	4
2	V1	12.9
2	V2	51.2
2	V3	2
3	V1	20.4
3	V2	70.8
3	V3	5
4	V1	17.5
4	V2	60.3
4	V3	7

17.2.2 .json files

.json (JavaScript Object Notation) is a textual data format used to store structured data in key-value form. It has similarities to R lists, especially to named list elements, and is therefore well suited as a format for storing metadata.

A .json file consists of objects that are written in curly braces { } and contain a comma-separated list of properties. Each property is a key-value pair, where the key is always written as a string in quotation marks (" "). The value can in turn be an object, an array, a string, a Boolean, or a number. Figure Figure 17.2 shows the content of a .json file for storing student information.

```
{
  "first_name" : "Maxi",
  "last_name" : "Samplewoman",
  "student_id" : 12345,
  "semester" : 2,
  "program" : "BSc Psychology",
  "modules" :
  {
    "Descriptive statistics" : { "completed" : true,  "grade" : 1.0 },
    "Inferential statistics" : { "completed" : false, "grade" : null }
  }
}
```

Figure 17.2 .json file format.

17.3 Directory management

Because computer directory systems are usually represented in words (e.g., `C:\Users`) rather than numerically, a basis for the automated reading of long-term stored data into memory and, conversely, for saving data from memory in long-term storage is working with so-called *strings* (character strings). Strings are generated in **R** with quotation marks or apostrophes:

```
# quotation marks are the string standard
c("This is a character vector")
```

```
[1] "This is a character vector"
```

```
# apostrophes can help with quotation marks in the string
c('This is a "string"')
```

```
[1] "This is a \"string\""
```

The function `paste()` converts vectors to character and concatenates them element by element.

```
# conversion and concatenation of one-element double vectors
paste(1,2)
```

```
[1] "1 2"
```

```
# concatenation of one-element character strings
paste("This is", "a string")
```

```
[1] "This is a string"
```

The function `paste()` also has a number of other functionalities that can sometimes be helpful:

```
# vector recycling, elementwise links
paste(c("Red", "Yellow"), "flower")
```

```
[1] "Red flower"    "Yellow flower"
```

```
# separator specification
paste(c("Red", "Yellow"), "flower", sep = "-")
```

```
[1] "Red-flower"    "Yellow-flower"
```

```
# concatenation with specified separator
paste(c("Red", "Yellow"), "flower", collapse = ", ")
```

```
[1] "Red flower, Yellow flower"
```

The function `toString()` is a `paste()` variant for numerical vectors:

```
# conversion of a double vector to a formatted string
toString(1:10)
```

```
[1] "1, 2, 3, 4, 5, 6, 7, 8, 9, 10"
```

```
# with the option to restrict output to width characters
toString(1:10, width = 10)
```

```
[1] "1, 2, ...."
```

To load the data of a file in long-term storage into memory, one needs its address in the computer's directory structure. Figure 17.3 shows an excerpt from the directory structure of a typical Windows computer.

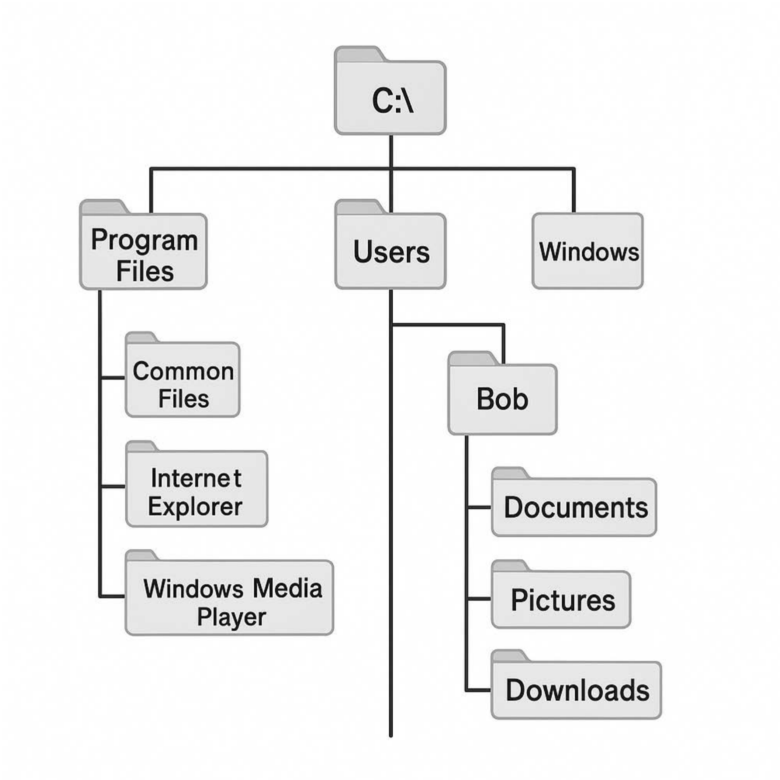

Figure 17.3 Windows directory structure

The addresses of files in the directory structure are called *file paths* (*paths*). A path consists of a list of directory names separated by slashes. Under Windows, left-leaning slashes (`\`, *backslash*) are traditionally used, whereas under Mac and Linux right-leaning slashes (`/`, *slash*) are used. Under Windows, the disk name (e.g., `C:\` or `D:\`) is usually found at the top level of the directory structure, under Mac the user name (`/Users/`), and under Linux the home directory (`/home/`). The following examples are Windows-based.

```
# path that ends in a directory name
C:\Users\dirko\Nextcloud\Home\pdwp-online\dirk-ostwald.github.io\_data

# path that ends in a file name
C:\Users\dirko\Nextcloud\Home\pdwp-online\dirk-ostwald.github.io\_data\cushny.csv
```

Absolute file paths give the address of a directory or file in the overall directory structure of the hard drive and are not very susceptible to file confusions. *Relative file paths*, by contrast, refer to a storage location in relation to the current directory, where `.` and `..` denote the current and the superordinate directory. Relative file paths can certainly lead to file confusions. In general, therefore, the use of adaptively generated absolute paths is recommended.

```
# file name in absolute path form
fname =  "D:\Teaching\Data\cushny.csv"
```

R has a so-called *working directory* from which files are read by default. When an **R** terminal is open, `getwd()` indicates the working directory.

```
getwd()
```

```
[1] "C:/Users/dirko/Nextcloud/Home/pdwp-online/dirk-ostwald.github.io-en/_part_2"
```

`setwd()` changes the working directory. Note that **R** works only with forward slashes `/`. The manual specification of Windows paths therefore requires double backward slashes `\\`

```
setwd("C:\\Users\\dirko\\Nextcloud\\Home\\pdwp-online\\dirk-ostwald.github.io\\_data")
getwd()
```

With the function `file.path()`, operating-system-compliant directory and file paths can be generated:

```
file.path("D:", "Nextcloud", "Home", "Teaching")
```

The function `dirname()` gives the directory that contains a directory or file:

```
getwd()
```

```
[1] "C:/Users/dirko/Nextcloud/Home/pdwp-online/dirk-ostwald.github.io-en/_part_2"
```

```
dirname(getwd())
```

```
[1] "C:/Users/dirko/Nextcloud/Home/pdwp-online/dirk-ostwald.github.io-en"
```

The function `basename()` gives the lowest level of a file or directory path:

```
getwd()
```

```
[1] "C:/Users/dirko/Nextcloud/Home/pdwp-online/dirk-ostwald.github.io-en/_part_2"
```

```
basename(getwd())
```

```
[1] "_part_2"
```

17.4 Data import and data export

For reading simply structured .csv files, the **R** function `read.csv()` is a good choice. This function reads a file and stores its contents in a dataframe.

```
D = read.csv("C:\\Users\\dirko\\Nextcloud\\Home\\pdwp-online\\dirk-ostwald.github.io\\_part_2\\_data\\cushny
print(D)
```

```
   Control drug1 drug2L drug2R delta1 delta2L delta2R
1      0.6   1.3    2.5    2.1    0.7     1.9     1.5
2      3.3   1.4    3.8    4.4    1.6     0.8     1.4
3      4.7   4.5    5.8    4.7    0.2     1.1     0.0
4      5.5   4.3    5.6    4.8    1.2     0.1     0.7
5      6.2   6.1    6.1    6.7    0.1     0.1     0.5
6      3.2   6.6    7.6    8.3    3.4     4.4     5.1
7      2.5   6.2    8.0    8.2    3.7     5.5     5.7
8      2.8   3.6    4.4    4.3    0.8     1.6     1.5
9      1.1   1.1    5.7    5.8    0.0     4.6     4.7
10     2.9   4.9    6.3    6.4    2.0     3.4     3.5
```

The following **R** code defines the absolute path of the file adaptively in the directory structure of the computer with the help of the **R** working directory.

```
wdir  = getwd()                          # working directory
print(wdir)                              # output
ddir  = file.path(wdir, "_part_2/_data") # data directory path
print(ddir)                              # output
fname = "cushny.csv"                     # (base) filename
print(fname)                             # output
fpath = file.path(ddir, fname)           # filepath
print(fpath)                             # output
D     = read.csv(fpath)                  # reading the file
print(D)                                 # output of the dataframe
```

In addition, **R** offers a variety of possibilities for data import using specialized functions:

For .csv and other text files:

- `read.csv()`, `read.csv2()`, `read.delim()`, `read.delim2()` as `read.table()` variants.
- `readLines()` for low-level text file import.
- `fromJSON()` from the `rjson` package for .json files.

For binary files:

- `read.xlsx()` and `read.xlsx2()` from the `xlsx` package for Excel .xlsx files.
- `read.spss()` from the `foreign` package for SPSS .sav files.
- `readMat()` from the `R.matlab` package for MATLAB .mat files.

For databases:

- SQL databases can be queried with the help of the `DBI` and `RSQLite` packages.

18 Control structures

Per se, imperative program code is executed strictly sequentially, command by command. Sometimes, however, one wants to deviate from this purely sequential order of commands and control the execution order of commands more flexibly. Principal tools for this purpose are *conditional control structures* and *iterative control structures*. *Conditional control structures* control program execution by specifying that certain commands are executed only when defined conditions are met. For this purpose, **R** provides `if()` and `switch()` statements. Often, one also wants to repeat (*iterate*) certain sections of code frequently in loops. For this purpose, **R** provides `for()`, `while()`, and `repeat()` loops. In this section, we introduce the basic conditional and iterative control structures in **R**, following the presentation in Cotton (2013), Chapter 8 and Chapter 9, as well as Wickham (2019), Chapter 5.

18.1 Conditional control structures

Conditional control structures allow the condition-dependent execution of commands.

if-statements

The basic conditional control structure is the if-statement. The general syntax for an if-statement in **R** has the following form:

```
if (condition){
  true_actions
}
```

Here, if the variable `condition` has the value `TRUE`, that is, if it is true, the commands denoted by `true_actions` are executed. If the value of the variable `condition` is `FALSE`, by contrast, the commands `true_actions` are not executed, and program execution continues on the next line after the if-statement. For example, in the following code the command `print("x is greater than 0")` is executed, since `x` has the value `1` after the preceding assignment.

```
x = 1
if(x > 0){
  print("x is greater than 0")
}
```

```
[1] "x is greater than 0"
```

The if-statement can be extended by an `else` condition to specify which commands are to be executed in the case that the variable `condition` has the value `FALSE`.

```
if (condition){
  true_actions
} else {
  false_actions
}
```

Here, the commands `true_actions` are executed if the variable `condition` has the value `TRUE`, and the commands `false_actions` are executed if the variable `condition` has the value `FALSE`. For example, in the following code the command `print("y is not greater than 0")` is executed because the condition `y > 0` is not met.

```
y = -1
if(y > 0){
  print("y is greater than 0")
} else{
  print("y is not greater than 0")
}
```

```
[1] "y is not greater than 0"
```

R includes the relation operators listed in Table 18.1 for testing binary logical conditions.

Table 18.1 Relation operators in **R**

Relation operator	Meaning
`==`	Equal
`!=`	Unequal
`<`, `>`	Smaller, greater
`<=`, `>=`	Smaller or equal, greater or equal
`&`	AND
`|`	OR

The relation operators `<`, `<=`, `>`, `>=` are mostly applied only to numerical values. The equality operators `==`,`!=` can be applied to arbitrary data structures to check their equality. The logical links `&` and `|` are mostly applied to logical values. Finally, the function `xor()` implements the exclusive OR. For example, the following tests of equality and relation can be performed:

```
# variable definition
x = 1
y = 2

# test of equality
if(x == y){
  print("x is equal to y")
} else {
  print("x is unequal to y")
}
```

```
[1] "x is unequal to y"
```

```
# test of inequality
if(x < y){
  print("x is smaller than y")
} else {
  print("x is greater than or equal to y")
}
```

```
[1] "x is smaller than y"
```

The if-statement tests can be extended by logical conditions, as the following example is intended to demonstrate:

```
# variable definition
x = 2
y = 2

# logical AND/OR
if(x == y | x < y){
  print("x is smaller than or equal to y")
} else {
  print("x is greater than y")
}
```

```
[1] "x is smaller than or equal to y"
```

```
# logical AND
if(x > y & x != y){
  print("x is greater than y")
} else {
  print("x is smaller than or equal to y")
}
```

```
[1] "x is smaller than or equal to y"
```

For if-statements, it is crucial that the condition corresponds to a single logical value. The following conditions, for example, are erroneous:

```
if("x"){
  print(1)
}
```

```
Error in `if ("x") ...`:
! argument is not interpretable as logical
```

```
if(NA){
  print(1)
}
```

```
Error in `if (NA) ...`:
! missing value where TRUE/FALSE needed
```

```r
if(c(T,F)){
  print(1)
}
```

```
Error in `if (c(T, F)) ...`:
! the condition has length > 1
```

switch-statements

Another conditional control structure is the switch-statement. Switch-statements are motivated by the fact that combined if-else-statements can easily become unclear, as the following example shows:

```r
x = 2
if (x == 1){
  print("action 1")
} else if(x == 2){
  print("action 2")
} else if(x == 3){
  print("action 3")
} else if(x == 4){
  print("action 4")
}
```

```
[1] "action 2"
```

Thus, if one has a case in which one of many actions is to be executed depending on the value of a variable, a switch-statement can be useful for improving code readability. **R** implements switch-statements in the `switch()` function, which works slightly differently depending on the data type of its variable. If the switch variable is of type integer, the ith action is executed, as the following example demonstrates:

```r
x = 2               # switch variable
switch(x,           # x = 2
print("action 1"),  # 1st action
print("action 2"),  # 2nd action
print("action 3"),  # 3rd action
print("action 4"))  # 4th action
```

```
[1] "action 2"
```

If the switch variable is of type character, by contrast, the action with the corresponding name is executed:

```r
x = "a"                 # switch variable
switch(x,               # x = "a"
a = print("action a"),  # a action
b = print("action b"),  # b action
c = print("action c"),  # c action
d = print("action d"))  # d action
```

```
[1] "action a"
```

In general, it can be said of if- and switch-statements that every if-statement can be formulated equivalently as a switch-statement and vice versa. In practice, if-statements are certainly encountered more frequently than switch-statements. In general, for human code comprehension it makes sense not to formulate if-else-statements too complexly. If this is necessary, however, complex if-else-statements can be simplified by switch-statements.

18.2 Iterative control structures

Iterative control structures serve to execute certain commands repeatedly. The basic iterative control structure in **R** is the `for` loop. It has the general form

```
for (item in vector){
  perform_action
}
```

The command `perform_action` is executed once for each `item` in `vector`, and the value of `item` is updated each time. The following example is intended to illustrate this:

```
for (i in 1:3){
  print(i)
}
```

```
[1] 1
[1] 2
[1] 3
```

Here the vector consists of `1:3 = [1,2,3]`, the command executed is `print(i)`, and the `items` are the entries of the vector. However, the command within the `for` loop does not necessarily have to refer to the `items`, as the following example shows:

```
for (i in 1:3){
  print('a')
}
```

```
[1] "a"
[1] "a"
[1] "a"
```

Here, the constant value `a` is output three times. Typically, something is generated and stored within a `for` loop. In doing so, the corresponding storage structure should generally be preallocated, because the corresponding space in memory is then already provided and does not have to be provided during the execution of the `for` loop. The efficiency gain of this strategy may be negligible in most cases, but this procedure also makes it easier in particular to detect errors, for example if there are discrepancies in dimensional expectations or missing entries that are then already allocated as `NaN`. The following example demonstrates the iterative filling of a vector with means.

```
ns    = 3               # number of simulations
X_bar = rep(NaN, ns)    # storage structure

# simulation iterations
for(i in 1:ns){         # iteration indices are typically i,j,k
  X        = rnorm(12)  # realization of 12 Z variables
  X_bar[i] = mean(X)    # mean of the ith realization
  print(X_bar)          # display for demonstration
}
```

```
[1] -0.116743       NaN       NaN
[1] -0.1167430  0.5903326        NaN
[1] -0.11674300  0.59033263 -0.05825925
```

To iterate linearly along a vector with variable entries, the function `seq_along()` is useful. It generates the linear indices `1`,`2`, `3`, ... up to the length of the vector in question, as the following example demonstrates:

```
mu    = c(0,5,50)                # three expected-value parameters
X_bar = rep(NaN, ns)             # storage structure

# simulation iterations
for(i in seq_along(mu)){         # iteration indices are typically i,j,k
  X        = rnorm(12, mu[i], 1) # realization of 12 N(mu,1) variables
  X_bar[i] = mean(X)             # mean of the ith realization
  print(X_bar)                   # display for demonstration
}
```

```
[1] 0.3798827       NaN       NaN
[1] 0.3798827 5.2512809       NaN
[1]  0.3798827  5.2512809 49.7973054
```

`for` loops can also be nested, that is, further `for` loops can be executed within `for` loops. This may take some getting used to. The crucial point here is to realize that for *each* iteration of the outer loop, *all* iterations of the inner loop are run through. It is often helpful first to clarify the general logic of a nested loop structure with the help of `print()` statements, as the following example demonstrates:

```
n_i = 2               # number of iterations of the outer loop
n_j = 3               # number of iterations of the inner loop
for(i in 1:n_i){      # outer loop
  for (j in 1:n_j){   # inner loop
    print(c(i,j))     # display of the iteration indices [i,j]
    }
}
```

```
[1] 1 1
[1] 1 2
[1] 1 3
[1] 2 1
[1] 2 2
[1] 2 3
```

The following example, for instance, generates a 4 x 5 matrix whose elements correspond to the respective column indices. The inner `for` loop with iterator variable `j` serves the column indices, and the outer `for` loop with iterator variable `i` serves the row indices:

```
     [,1] [,2] [,3] [,4] [,5]
[1,]    1    2    3    4    5
[2,]    1    2    3    4    5
[3,]    1    2    3    4    5
[4,]    1    2    3    4    5
```

Analogously, the following example generates a 4 x 5 matrix whose elements correspond to the respective row indices. The inner `for` loop with iterator variable `j` again serves the column indices, and the outer `for` loop with iterator variable `i` serves the row indices:

```
     [,1] [,2] [,3] [,4] [,5]
[1,]    1    1    1    1    1
[2,]    2    2    2    2    2
[3,]    3    3    3    3    3
[4,]    4    4    4    4    4
```

Finally, in the following example, two nested `for` loops and an if-statement are used to generate a 4 x 5 matrix whose elements are equal to the column indices if the respective column index is greater than or equal to the row index, and whose elements are otherwise zero:

```
     [,1] [,2] [,3] [,4] [,5]
[1,]    1    2    3    4    5
[2,]    0    2    3    4    5
[3,]    0    0    3    4    5
[4,]    0    0    0    4    5
```

Part III

Probability theory

Preliminary remarks

Probability theory is a mathematical model for describing and quantitatively reasoning about *random processes* of reality (Figure 18.1). By random processes we mean all phenomena that cannot be predicted by us with absolute certainty, and whose outcome is therefore subject to uncertainty. Obvious and familiar examples of random processes are throwing a die or tossing a coin. However, the concept of a random process, and thus the domain of application of probability theory, should be understood much more broadly. For example, the outcome of an election, tomorrow's weather, the measurement value of an electroencephalography electrode at a particular point in time, or the effect of a psychotherapy intervention on the health status of a patient cannot be predicted with complete certainty and are therefore subject to uncertainty. Once one begins to think about which phenomena of reality are subject to uncertainty, it becomes difficult to name nontrivial phenomena about whose outcome one has complete certainty.

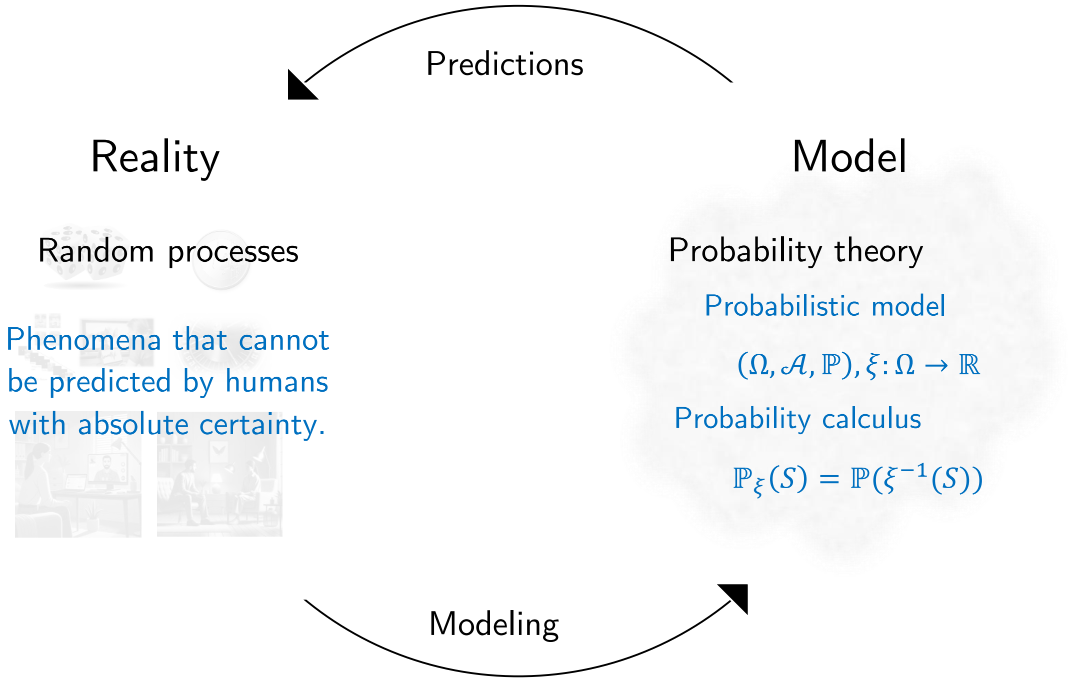

Figure 18.1 *Probability theory as a model of random processes.* The starting point of probability theory is the intention to draw logical-quantitative conclusions about a random process, that is, a phenomenon of reality subject to uncertainty. The representation of central aspects of the random process with the help of probability-theoretic terminology is called *modeling*. The probability-theoretic model itself then guarantees, in the sense of probability calculus, the correctness of logical-quantitative conclusions that can be used to predict aspects of the random process.

As a mathematical model of random processes, probability theory in particular allows reason-based, quantitative reasoning about random processes. This is reflected primarily in so-called *probability calculus*. Quantitative conclusions of probability calculus have, for example, the following form: If I assume that event A occurs with probability p_1 and event B occurs with probability p_2, then event C has probability p_3. The conclusion about the probability of C is logically and mathematically secured in the same sense in which, for example, it is logically and mathematically secured that $1 + 1 = 2$. Whether the assumptions about the probabilities of A and B correspond to the conditions of the random process in reality, however, is not something probability theory makes statements about.

Probability theory itself uses the mathematical theory of sets and functions. At least since Kolmogoroff (1933), an axiomatic approach has prevailed: in probability theory itself, one does not ask what a probability is or to what extent the predictions of probability theory agree with reality, but instead tries to develop a self-consistent formal-mathematical system of ungrounded but intuitively plausible basic assumptions and their consequences. The starting point of this development is the *probability space model* of a random process, which will be introduced in Chapter 19. Indeed, beyond the formal-mathematical system of probability theory, mathematical-philosophical discussions continue to this day about exactly what is to be understood by the term "probability of an event". Roughly speaking, two somewhat opposing interpretations are predominant: on the one hand the so-called *frequentist interpretation* as a prototypical example of *realist interpretations*, and on the other hand the so-called *Bayesian interpretation* as a general form of so-called *evidential interpretations*. For a more detailed overview of different interpretations of the concept of probability, see Table 18.2 after Hájek (2019).

Table 18.2 Interpretations of probability after Hájek (2019). In its essential features, the *frequentist interpretation* corresponds to the *infinite frequentist realist interpretation* sketched in the table; in its essential features, the *Bayesian interpretation* corresponds to the *subjective evidential* and *epistemic evidential interpretations* sketched in the table.

Interpretation	Intuition	Field of application	Problems	References
Finite frequentist	The probability of a coin showing heads corresponds to the relative frequency with which the coin shows heads in a finite number of observations.	-	A single coin toss does not yield a meaningful prediction; probabilities change depending on the observation series and are therefore not uniquely defined.	Venn (1876)
Infinite frequentist	The probability of a coin showing heads corresponds to the relative frequency with which the coin shows heads in infinitely many observations.	Frequentist inference	There are no infinite observation series; for a finite number of coin tosses, the probability of an event is not defined.	Reichenbach (1949) von Mises (1951)

Interpretation	Intuition	Field of application	Problems	References
Propensity	The probability of a coin showing heads corresponds to its inherent physical tendency to land heads up in the gravitational field of the Earth.	Causal inference	The connection to relative frequencies or physical theories is not immediately clear.	Peirce (1957) Popper (1959) Gillies (2000)
Classical	In the absence of further information, the probabilities of a coin showing heads or tails are equal.	Games of chance, textbooks	Restriction to finite outcome spaces	Laplace (1814) Jaynes (1968)
Subjective	The probability of a coin showing heads corresponds to the subjective degree of uncertainty that an observer assigns to the outcome of a coin toss.	Bayesian inference	No principled restriction to axiomatically valid probabilities	De Finetti (1975)
Epistemic	The probability of a coin showing heads corresponds to the subjective degree of uncertainty that a rational observer assigns to the outcome of a coin toss.	Bayesian inference	The connection to relative frequencies or physical theories is not immediately clear	Ramsey (1926) Jeffreys (1939) Savage (1954)

According to the *frequentist interpretation*, the probability of an event is the idealized relative frequency with which an event tends to occur under the same external conditions. For example, the frequentist interpretation of the statement "With a probability of 1/6, the die will show a 2 on the next throw" is the following: "If one were to throw a die infinitely often and determine the relative frequency of the event that the die shows a 2, then this relative frequency would be equal to 1/6." Note that, in this interpretation, one de facto cannot empirically determine the probability of an event, since one cannot throw a die infinitely often. Of course, in this interpretation one can estimate the probability empirically. Estimation procedures themselves, however, are not part of probability theory, but of *inference*.

According to the *Bayesian interpretation*, the probability of an event is the degree of certainty that an observer assigns to the occurrence of the event based on her subjective assessment of the situation. For example, the Bayesian interpretation of the statement "With a probability of 1/6, the die will show a two on the next throw" would then be something like the following: "Based on my own and transmitted experience with throwing a die, I am 16.6% certain that the die will show a two on the next throw."

In models of random processes that can in fact be repeated at least under similar circumstances, such as throwing a die, the difference between the frequentist and Bayesian interpretations is often rather subtle. However, as indicated above, there are many random processes that can be described with probabilities for which, because of their uniqueness, a frequentist interpretation is not appropriate. For example, statements of the form "The probability that the global heat records in 2023 are not attributable to climate change is less than 0.01" (cf. Philip et al. (2020)) make sense only under the Bayesian interpretation, because the weather records of the year 2023 are a unique, non-repeatable event.

Although, then, the interpretation of the concept of probability is by no means unambiguous, the formal definitions and calculation rules for probabilities do not differ. Both frequentist and Bayesian inference have an identical mathematical reference system and common foundation in probability theory. In essence, this fact is not so different from many other forms of mathematical modeling. For example, the derivative of a function can be interpreted as the velocity of an object or as the growth rate of a bacterial population. Although these phenomena differ intuitively very strongly with respect to reality, logical reasoning about both phenomena is identical in the domain of mathematics.

19 Probability spaces

A *probability space* is a very general formal-mathematical model of a random process. The central importance of this model for probability theory and probabilistic inference follows from the fact that the probability space model provides guidance for translating arbitrary random processes about which one wants to reason quantitatively into the formal-mathematical system of probability theory. At the same time, the probability space model and the concepts built on it guarantee that the mechanics of probability calculus lead to logically meaningful quantitative conclusions about random processes. In this chapter, we introduce the concept of a probability space (Section 19.1) and then use probability functions (Section 19.2) to give first examples of modeling random processes (Section 19.3).

19.1 Definition and first properties

We begin with the definition of the probability space model after Kolmogoroff (1933), which we then explain in its individual parts from a frequentist perspective.

Definition 19.1 (Probability space)**.** A *probability space* is a triple $(\Omega, \mathcal{A}, \mathbb{P})$, where

- Ω is an arbitrary nonempty set of *outcomes* ω and is called the *outcome set*,
- $\mathcal{A}$ is a set of subsets of Ω with the properties

  - $\Omega \in \mathcal{A}$,
  - for all $A \in \mathcal{A}$, also $A^c := \Omega \setminus A \in \mathcal{A}$,
  - from $A_1, A_2, ... \in \mathcal{A}$ it follows that also $\cup_{i=1}^{\infty} A_i \in \mathcal{A}$

  holds, is called a *σ-algebra on* Ω, and is called the *event system*,

- $\mathbb{P}$ is a mapping of the form $\mathbb{P} : \mathcal{A} \to [0, 1]$ with the properties

  - $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{A}$ (*non-negativity*),
  - $\mathbb{P}(\Omega) = 1$ (*normalization*),
  - $\mathbb{P}(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$ for pairwise disjoint $A_i \in \mathcal{A}$ (*σ-additivity*)

  holds and is called a *probability measure*.

•

Outcome set and mechanics

We begin with explanations of the concept of the *outcome set* Ω and the implicit mechanics of the probability space model. To make the introduction easier, in what follows we first primarily consider *finite probability spaces*, in which the cardinality of Ω is not infinitely large. Thus let $|\Omega| < \infty$, so that Ω has only finitely many elements. For modeling the throw of a die, for example, one could define $\Omega := \{1, 2, 3, 4, 5, 6\}$.

Behind the formal definition of the probability space model are the following frequentist-influenced assumptions about its mechanics as a model of a random process. We first imagine sequential runs of a random process, for example the repeated throwing of a die. According to the assumptions of the probability space model, in each of these runs exactly one ω from Ω is *realized*, that is, selected as actually occurring. For example, if one throws a die and a two appears, one says that a two was realized. The probability with which an ω from Ω is realized in a run is described by the value $\mathbb{P}(\{\omega\}) \in [0, 1]$. If, for example, $\mathbb{P}(\{\omega\}) = 1$, then this ω is realized in every run of the random process; if $\mathbb{P}(\{\omega\}) = 0$, then this ω is realized in no run of the random process; and if $\mathbb{P}(\{\omega\}) = 1/2$, then ω is realized in about half of the runs of the random process. In the model of throwing a fair die, one usually assumes $\mathbb{P}(\{\omega\}) = 1/6$ for all $\omega \in \Omega$. Here, for example, a four could be realized in the first run, a one in the second run, a five in the third run, then perhaps a four again, and so on.

Events and event system

The concept of an *event* $A \in \mathcal{A}$ is best imagined as a conceptual summary of one or more outcomes. When throwing a die, possible events are, for example, “An even number of pips appears”, that is, $\omega \in \{2, 4, 6\}$, “A number of pips greater than two appears”, that is, $\omega \in \{3, 4, 5, 6\}$, or “A one or a five appears”, that is, $\omega \in \{1, 5\}$. Note that, against the background of the mechanics of the probability space, the event “An even number of pips appears” occurs exactly when, in a run of the random process *throwing a die*, the realized ω is an element of the set $\{2, 4, 6\}$, for example when a four appears. The occurrence of the event “A number of pips greater than two appears” may also be read as “In a run of the random process *throwing a die*, an element of $\{3, 4, 5, 6\}$ is realized”, that is, concretely, either a three, a four, a five, or a six appears. Of course, the outcomes $\omega \in \Omega$ themselves are also possible events, so that, for example, the following interpretations hold: the event “A one appears” corresponds to the realization $\omega \in \{1\}$, and the event “A six appears” corresponds to the realization $\omega \in \{6\}$. If one considers an outcome $\omega \in \Omega$ as an event in this context, it is called an *elementary event* and is written as the one-element set $\{\omega\}$.

Overall, this procedure for describing random events corresponds to the inherent goal of the definition of the probability space. Kolmogoroff (1933) writes that the actual objects of the subsequent considerations, the random events, have been defined as sets. This has the advantage that sets are mathematical objects with which one can work mathematically, so that an aspect of reality, a “random event”, has been translated into the model domain of mathematics.

The sole purpose of the *event system* $\mathcal{A}$ is now to mathematically represent all events that can arise, based on a given outcome set, when an $\omega \in \Omega$ is selected. Thus there should be no events in reality that were not considered in advance in the probability space model of the random process. If this were the case, the model would be deficient, since it could not assign probabilities to certain events occurring in reality. The event system $\mathcal{A}$ should therefore be the complete set of all possible events for a given Ω. The requirement that $\mathcal{A}$ should satisfy the so-called σ-algebra criteria for this purpose is intuitively motivated as follows.

- It should first be ensured that $\omega \in \Omega$ for an arbitrary ω, that is, that some outcome is realized, is one of the possible events. This corresponds to the property $\Omega \in \mathcal{A}$.
- For every event it should also be possible that this event does not occur. This corresponds to the property that from $A \in \mathcal{A}$ it should follow that $A^c := \Omega \setminus A$ is also in $\mathcal{A}$. This also implies in particular that $\emptyset = \Omega \setminus \Omega \in \mathcal{A}$. Thus one event is that no elementary event occurs, although this happens only with probability zero, $\mathbb{P}(\emptyset) = 0$. Thus at least one elementary event always occurs with certainty.
- Finally, the combination of events should always also be an event. In modeling the throw of a die, for example, in addition to the events "An even number appears" and "A number greater than two appears", the event "An even number appears and/or this number is greater than two" should also be an event. In its most general form, this corresponds to the requirement that from $A_1, A_2, ... \in \mathcal{A}$ it should follow that also $\cup_{i=1}^{\infty} A_i \in \mathcal{A}$.

Even if the concepts of the event system and the σ-algebra may seem somewhat abstract, their definition in the practical modeling of random processes is usually not a major challenge, since suitable event systems have long been known both for finite outcome sets and for infinite countable and uncountable outcome sets. Thus, for outcome sets with finite cardinality, the power set of the outcome set always satisfies the requirements of an event system and can always be used to formulate a model of a random process with a finite outcome set. This is the statement of the following theorem.

Theorem 19.1 (Event system for finite outcome set). *Let $\Omega := \{\omega_1, \omega_2, ..., \omega_n\}$ with $n \in \mathbb{N}$ be a finite set. Then the power set $\mathcal{P}(\Omega)$ of Ω is a σ-algebra on Ω and thus a suitable event system in the probability space model.*

∘

Proof. The power set of Ω is the set of all subsets of Ω. We check the σ-algebra properties. First, Ω itself is one of the subsets of Ω, so the first σ-algebra property is satisfied. Now let A be a subset of Ω. Then $A^c = \Omega \setminus A$ is also a subset of Ω, and hence the second σ-algebra property is also satisfied. Finally, consider the union of k subsets $A_1, A_2, ..., A_k \subseteq \Omega$. Then $\cup_{i=1}^{k} A_i$ is the set of $\omega \in \Omega$ for which $\omega \in A_1$ and/or $\omega \in A_2$ ... and/or $\omega \in A_k$. Since for all these ω it holds that $\omega \in \Omega$, also $\cup_{i=1}^{k} A_i$ is a subset of Ω, and therefore the third σ-algebra property is also satisfied. The power set thus satisfies the required properties of an event system. □

For uncountable outcome sets such as the real numbers $\mathbb{R}$ or the n-dimensional real space $\mathbb{R}^n$, the construction of a suitable event system is more complex, so in this respect we refer to the advanced literature for formal developments (cf. Meintrup & Schäffler (2005), Schmidt (2009)). However, with the so-called *Borel σ-algebra*, which goes back to Borel (1898), a set system is known that satisfies the requirements of a σ-algebra on uncountable outcome sets. We denote the Borel σ-algebra on $\mathbb{R}$ by $\mathcal{B}(\mathbb{R})$ and the Borel σ-algebra on $\mathbb{R}^n$ by $\mathcal{B}(\mathbb{R}^n)$. As sets of subsets of $\mathbb{R}$ and $\mathbb{R}^n$, respectively, $\mathcal{B}(\mathbb{R})$ and $\mathcal{B}(\mathbb{R}^n)$ contain all sets whose probability assigned by $\mathbb{P}$ one may be interested in. Intuitively, one may therefore think of the Borel σ-algebras $\mathcal{B}(\mathbb{R})$ and $\mathcal{B}(\mathbb{R}^n)$ as the power sets of $\mathbb{R}$ and $\mathbb{R}^n$, respectively, even though this is formally incorrect. In fact, the Borel σ-algebra contains only subsets generated by countable set operations, not by uncountable ones.

Overall, this leads to the following procedure for selecting event systems depending on the outcome set Ω. If Ω is finite, one chooses as event system $\mathcal{A}$ the power set $\mathcal{P}(\Omega)$ of Ω. If Ω is given by $\mathbb{R}$, one chooses as event system $\mathcal{A}$ the Borel σ-algebra $\mathcal{B}(\mathbb{R})$. Finally, if Ω is given by $\mathbb{R}^n$, one chooses for $\mathcal{A}$ the Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$. In special cases and for very special outcome sets Ω, one may want to deviate from this procedure, but in general the three cases considered cover most applications.

Probability measure $\mathbb{P}$

With Ω and $\mathcal{A}$, which in tuple form $(\Omega, \mathcal{A})$ are also called a *measurable space*, we have so far examined more closely the *structural basis* of a probability space model. Many probability spaces, for example for random processes concerning real numbers, are identical with respect to their measurable space. The *probability measure* $\mathbb{P}$ now represents the probabilistic characteristics of a probability space model and thus forms the *functional basis* of such a model. In what follows, especially after introducing the concepts of random variables and random vectors, we will encounter many different probability measures. At this point, we first want to consider only general properties of probability measures.

With the definition

$$\mathbb{P} : \mathcal{A} \to [0, 1], A \mapsto \mathbb{P}(A) \tag{19.1}$$

it first follows that a probability measure is defined on a set of sets and assigns probabilities, that is, values in the interval $[0, 1]$, to the elements of this set, namely to the sets $A \in \mathcal{A}$. Of course, with $\{\omega\} \in \mathcal{A}$ for all $\omega \in \Omega$, $\mathbb{P}$ also assigns probabilities to the elementary events, but not only to them. We also emphasize that, by definition, the probability $\mathbb{P}(A)$ of an event $A \in \mathcal{A}$ is a number in the interval $[0, 1]$ and not, for example, a percentage or a ratio. In what follows, we will examine more closely the defining properties of *non-negativity*, *normalization*, and *σ-additivity* of $\mathbb{P}$.

The *non-negativity* $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{A}$ is of course implicit in the definition $[0, 1]$ of the target set of $\mathbb{P}$. In fact, the mapping form of $\mathbb{P}$ is an addition we have made to the formulation of Kolmogoroff (1933) in the interest of clarity. Formally, the form of the target set of $\mathbb{P}$ actually follows from the defining properties of non-negativity, normalization, and σ-additivity of $\mathbb{P}$.

The *normalization* $\mathbb{P}(\Omega) = 1$ corresponds to the fact that, in every run of a random process, the realized ω is certainly an element of Ω. Thus in every run of a random process an elementary event occurs and, depending on the nature of the measurable space, many others as well. In the model of throwing a die, for example, the outcome/elementary event realized in a run of the random process is an element of $\Omega := \{1, 2, 3, 4, 5, 6\}$ with probability 1. If the realized outcome is, for example, a one, then in addition to the event “A one appears”, the events “An odd number appears”, “A number less than three appears”, “An odd number less than three appears”, and many others also occur.

The *σ-additivity of the probability measure* $\mathbb{P}$, finally, forms the foundation of *probability calculus*, that is, the basis for calculating with probabilities. The σ-additivity of $\mathbb{P}$ makes it possible to calculate the probabilities of other events from already known event probabilities. Based on a definition of $\Omega, \mathcal{A}$, and $\mathbb{P}$, one can therefore calculate probabilities for all possible events of a probability space model. Whether these probabilities actually have anything to do with real events concerning a random process in reality depends on whether the modeling is reasonably successful or not. However, the calculated probabilities are at least determined rationally, that is, according to the rules of reason, namely logic and mathematics. Overall, the probability model thus allows inferential thinking about random processes, that is, about phenomena subject to uncertainty.

Finally, we want to illustrate probability calculations based on the σ-additivity of $\mathbb{P}$ with two basic examples. The first example states that the probability that the ω realized in a run of a random process is not an element of the outcome set is equal to zero.

Theorem 19.2 (Probability of the impossible event)**.** *Let* $(\Omega, \mathcal{A}, \mathbb{P})$ *be a probability space. Then*

$$\mathbb{P}(\emptyset) = 0. \tag{19.2}$$

∘

Proof. For $i = 1, 2, ...,$ let $A_i := \emptyset$. Then $A_1, A_2, ...$ is a sequence of disjoint events because $\emptyset \cap \emptyset = \emptyset$, and $\cup_{i=1}^{\infty} A_i = \emptyset$. With the σ-additivity of $\mathbb{P}$ it then follows that

$$\mathbb{P}(\emptyset) = \mathbb{P}\left(\cup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}\left(A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}\left(\emptyset\right). \tag{19.3}$$

Thus the infinite addition of the number $\mathbb{P}(\emptyset) \in [0, 1]$ should again yield $\mathbb{P}(\emptyset)$. This is only possible if $\mathbb{P}(\emptyset) = 0$. □

Note that, intuitively, a possible inadequacy of the probability space as a model for random processes in reality can arise here: if, when playing dice, the die falls out of reach under the sofa, then an elementary event $\omega \notin \Omega$ has occurred, even though its modeled probability is equal to zero.

As a second example, we want to show that σ-additivity, which in the definition of the probability space is defined only for the union of infinitely many disjoint events, implies the σ-additivity of finitely many disjoint events as they often occur in applications.

Theorem 19.3 (σ-additivity for finite sequences of disjoint events)**.** *Let* $(\Omega, \mathcal{A}, \mathbb{P})$ *be a probability space and let* $A_1, ..., A_n$ *be a finite sequence of pairwise disjoint events. Then*

$$\mathbb{P}(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i). \tag{19.4}$$

∘

Proof. We consider an infinite sequence of pairwise disjoint events $A_1, A_2, ...,$ where for an $n \in \mathbb{N}$ it is to hold that $A_i := \emptyset$ for $i > n$. Then, with the σ-additivity of $\mathbb{P}$, it first follows that

$$\mathbb{P}\left(\cup_{i=1}^{n} A_i\right) = \mathbb{P}\left(\cup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}\left(A_i\right) = \sum_{i=1}^{n} \mathbb{P}\left(A_i\right) + \sum_{i=n+1}^{\infty} \mathbb{P}\left(A_i\right). \tag{19.5}$$

With $\mathbb{P}\left(A_i\right) = \mathbb{P}(\emptyset) = 0$ for $i = n+1, n+2, ...,$ it follows directly that

$$\mathbb{P}\left(\cup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mathbb{P}\left(A_i\right) + 0 = \sum_{i=1}^{n} \mathbb{P}\left(A_i\right). \tag{19.6}$$

□

To prepare the concept of the joint probability of events, we finally consider one further important property of σ-algebras: in addition to being closed under countable unions, $\mathcal{A}$ is also closed under countable intersections. This is a direct consequence of the closure of $\mathcal{A}$ under complements and countable unions and is recorded in the following theorem.

Theorem 19.4 (Closure of σ-algebras under intersections)**.** *Let $\mathcal{A}$ be a σ-algebra on a set Ω. Then from $A, B \in \mathcal{A}$ it follows that also $A \cap B \in \mathcal{A}$. More generally, for $A_1, A_2, ... \in \mathcal{A}$, also $\cap_{i=1}^{\infty} A_i \in \mathcal{A}$.*

∘

Proof. We show the first statement by way of example. To this end, first note that by the first De Morgan rule and the properties of complements,

$$(A \cap B)^c = A^c \cup B^c \Leftrightarrow ((A \cap B)^c)^c = (A^c \cup B^c)^c \Leftrightarrow A \cap B = (A^c \cup B^c)^c . \tag{19.7}$$

Furthermore, with the defining properties of σ-algebras, for $A, B \in \mathcal{A}$ it holds that also $(A^c \cup B^c)^c \in \mathcal{A}$. For from the closure of $\mathcal{A}$ under complements it first follows that also $A^c \in \mathcal{A}$ and $B^c \in \mathcal{A}$. Further, from the closure of $\mathcal{A}$ under countable unions it also follows that $A^c \cup B^c \in \mathcal{A}$. Finally, with the closure of $\mathcal{A}$ under complements, $(A^c \cup B^c)^c \in \mathcal{A}$. Overall, then, from $A, B \in \mathcal{A}$ it follows that $A \cap B \in \mathcal{A}$. The generalization to countable intersections $A_1, A_2, ... \in \mathcal{A}$ follows analogously. □

According to Theorem 19.4, in addition to the probabilities $\mathbb{P}(A), \mathbb{P}(B)$ and $\mathbb{P}(A \cup B)$ with $A, B \in \mathcal{A}$, the probability $\mathbb{P}(A \cap B)$ is therefore also well-defined.

19.2 Probability functions

In this section, with *probability functions*, we want to learn about a way of specifying probability measures for probability spaces with finite outcome sets. In the sections on random variables and random vectors, we will encounter probability functions again under the name *probability mass functions*. We first define the concept of a probability function as follows.

Definition 19.2 (Probability function)**.** Let Ω be a finite set. Then a function $\pi : \Omega \to [0, 1]$ is called a *probability function* if

$$\sum_{\omega \in \Omega} \pi(\omega) = 1. \tag{19.8}$$

Furthermore, let $\mathbb{P}$ be a probability measure. Then the function defined by

$$\pi : \Omega \to [0, 1], \omega \mapsto \pi(\omega) := \mathbb{P}(\{\omega\}) \tag{19.9}$$

is called the *probability function of* $\mathbb{P}$ *on* Ω.

•

We note that because $\mathbb{P}$ is σ-additive by definition, in particular

$$\mathbb{P}(\Omega) = \mathbb{P}(\cup_{\omega \in \Omega}\{\omega\}) = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = \sum_{\omega \in \Omega} \pi(\omega) = 1. \tag{19.10}$$

The following theorem provides the formal basis for constructing probability measures through probability functions. In particular, it states that for finite Ω, the probabilities of *all* possible events can be calculated from the probabilities of the elementary events $\pi(\omega)$.

Theorem 19.5. *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space with finite outcome set, and let $\pi : \Omega \to [0,1]$ be a probability function. Then there exists a probability measure $\mathbb{P}$ on Ω with π as the probability function of $\mathbb{P}$. This probability measure is defined as*

$$\mathbb{P} : \mathcal{A} \to [0,1], A \mapsto \mathbb{P}(A) := \sum_{\omega \in A} \pi(\omega). \tag{19.11}$$

$\circ$

Proof. We first check the probability measure properties of $\mathbb{P}$. Because $\pi(\omega) \in [0,1]$ for all $\omega \in \Omega$, it is always true that $\sum_{\omega \in A} \pi(\omega) \geq 0$, and hence non-negativity of $\mathbb{P}$ holds. Furthermore, as seen above, normalization of $\mathbb{P}$ follows directly from normalization of π. Now let $A_1, A_2, ... \in \mathcal{A}$. Then

$$\mathbb{P}\left(\cup_{i=1}^{\infty} A_i\right) = \sum_{\omega \in \cup_{i=1}^{\infty} A_i} \pi(\omega) = \sum_{i=1}^{\infty} \sum_{\omega \in A_i} \pi(\omega) = \sum_{i=1}^{\infty} \mathbb{P}(A_i) \tag{19.12}$$

and hence σ-additivity of $\mathbb{P}$. $\square$

Thus, if for a given outcome set Ω one defines a function $\pi : \Omega \to [0,1]$, ensures that its function values $\pi(\omega)$ sum to 1 over all $\omega \in \Omega$, and then interprets the individual function value $\pi(\omega)$ as the probability $\mathbb{P}(\{\omega\})$ of the elementary event $\{\omega\} \in \mathcal{A}$, one has constructed a probability measure. Conversely, if for a finite outcome set Ω one defines a probability $\mathbb{P}(\{\omega\}) \in [0,1]$ for all elementary events $\{\omega\} \in \mathcal{A}$ and ensures that $\mathbb{P}(\Omega) = 1$, then one has of course implicitly also satisfied the requirements of a probability function. In this case, the σ-additivity of $\mathbb{P}$ can be used directly for the pairwise disjoint elementary events to calculate compound events. We illustrate this and the procedure using a probability function in Section 19.3 by means of examples.

19.3 Examples with finite outcome space

From what has been said so far, the following procedure for modeling a random process using a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ can be summarized:

(1) In a first step, one considers a sensible definition of the outcome set Ω, that is, of the outcomes or elementary events that are to be realized in each run of the random process.

(2) In a second step, one then chooses a suitable event system. In the case of a finite outcome set, the power set $\mathcal{P}(\Omega)$ is suitable; in the case of the uncountable outcome set $\Omega := \mathbb{R}$, the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is suitable.

(3) Finally, one defines a probability measure $\mathbb{P}$ that represents the probabilities of the occurrence of all possible events. In the case of a finite outcome set, this is accomplished in particular by defining the probabilities of the elementary events. In what follows, we illustrate this procedure with examples. For the uncountable outcome set $\Omega := \mathbb{R}$, defining $\mathbb{P}$ with the help of *probability density functions* is suitable, as we will see later.

Throwing one die

We model throwing one die. It is certainly sensible to define the outcome set as $\Omega := \{1, 2, 3, 4, 5, 6\}$. However, the definition

$$\Omega := \{\cdot, \cdot\cdot, \cdot\cdot\cdot, \cdot\cdot\cdot\cdot, \cdot\cdot\cdot\cdot\cdot, \cdot\cdot\cdot\cdot\cdot\cdot\} \tag{19.13}$$

would also be possible in an equivalent way.

Because this is a finite outcome set, we choose the power set $\mathcal{P}(\Omega)$ as the σ-algebra $\mathcal{A}$. Then $\mathcal{A}$ automatically contains all possible events. The cardinality of $\mathcal{A} := \mathcal{P}(\Omega)$ is $|\mathcal{P}(\Omega)| = 2^{|\Omega|} = 2^6 = 64$. In Table 19.1 we list six of these 64 events in their verbal description and as a subset A of Ω.

Table 19.1 Selected events in the model of throwing a die.

Description	Set form
Any number of pips appears	$\omega \in A = \Omega$
No number of pips appears	$\omega \in A = \emptyset$
A number of pips greater than 4 appears	$\omega \in A = \{5, 6\}$
An even number of pips appears	$\omega \in A = \{2, 4, 6\}$
A six appears	$\omega \in A = \{6\}$
A one, a three, or a six appears	$\omega \in A = \{1, 3, 6\}$

This completes the definition of the measurable space $(\Omega, \mathcal{A})$ in the modeling of throwing a die.

As described in Section 19.2, a probability measure $\mathbb{P}$ can be defined by specifying $\mathbb{P}(\{\omega\})$ for all $\omega \in \Omega$. For the model of a fair die, one would choose

$$\mathbb{P}(\{\omega\}) := \frac{1}{|\Omega|} := 1/6 \text{ for all } \omega \in \Omega. \tag{19.14}$$

For a model of a biased die that favors throwing a six, one could, for example, define

$$\mathbb{P}(\{\omega\}) := \frac{1}{8} \text{ for } \omega \in \{1, 2, 3, 4, 5\} \text{ and } \mathbb{P}(\{6\}) := \frac{3}{8}. \tag{19.15}$$

In the case of the fair die, for example, the probability of the event "An even number of pips appears" is then obtained with the σ-additivity of $\mathbb{P}$ as

$$\mathbb{P}(\{2, 4, 6\}) = \mathbb{P}(\{2\} \cup \{4\} \cup \{6\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6}. \tag{19.16}$$

In the case of the above model of a biased die, by contrast, the probability of the same event is

$$\mathbb{P}(\{2, 4, 6\}) = \mathbb{P}(\{2\} \cup \{4\} \cup \{6\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{1}{8} + \frac{1}{8} + \frac{3}{8} = \frac{5}{8}. \tag{19.17}$$

The event considered has a higher probability in the model of the biased die than in the model of the fair die, which is intuitively sensible because six is an even number.

Simultaneous throwing of a blue and a red die

We now want to model the simultaneous throwing of a blue and a red die. For this purpose it is sensible to define the outcome set as

$$\Omega := \{(r,b)|r \in \{1,2,3,4,5,6\}, b \in \{1,2,3,4,5,6\}\} \tag{19.18}$$

with cardinality $|\Omega| = 36$, where r is to represent the number of pips on the red die and b the number of pips on the blue die.

Again, choosing the power set of Ω as the σ-algebra is suitable, so we again define $\mathcal{A} := \mathcal{P}(\Omega)$. The number of events possible in this model is $|\mathcal{A}| = 2^{|\Omega|} = 2^{36} = 68,719,476,736$. In Table 19.2 we list six of these events in their verbal description and as a subset A of Ω.

Table 19.2 Selected events in the model of throwing a red and a blue die.

Description	Set form
A three appears on the red die	$\omega \in A = \{(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)\}$
A three appears on the blue die	$\omega \in A = \{(1,3),(2,3),(3,3),(4,3),(5,3),(6,3)\}$
A three appears on both dice	$\omega \in A = \{(3,3)\}$
Doubles appear	$\omega \in A = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$
The sum of the numbers that appear is four	$\omega \in A = \{(1,3),(3,1),(2,2)\}$

The definition of the measurable space $(\Omega, \mathcal{A})$ is therefore complete. A probability measure $\mathbb{P}$ can again be specified by defining $\mathbb{P}(\{\omega\})$ for all $\omega \in \Omega$. For the model of two fair dice, one would choose

$$\mathbb{P}(\{\omega\}) := \frac{1}{|\Omega|} = \frac{1}{36} \text{ for all } \omega \in \Omega. \tag{19.19}$$

Under this probability measure, the probability of the event "The sum of the numbers that appear is four", for example, is obtained with the σ-additivity of $\mathbb{P}$ as

$$\begin{aligned}
\mathbb{P}\left(\{(1,3),(3,1),(2,2)\}\right) &= \mathbb{P}\left(\{(1,3)\} \cup \{(3,1)\} \cup \{(2,2)\}\right) \\
&= \mathbb{P}\left(\{(1,3)\}\right) + \mathbb{P}\left(\{(3,1)\}\right) + \mathbb{P}\left(\{(2,2)\}\right) \\
&= 1/36 + 1/36 + 1/36 \\
&= 1/12.
\end{aligned}$$

Tossing a coin

We model tossing a coin, one side of which shows heads and the other side tails. It is sensible to define the outcome set as $\Omega := \{H, T\}$, where H represents "heads" and T represents "tails". However, any other binary definition of Ω would also be possible, e.g. $\Omega := \{0,1\}, \Omega := \{-1,1\}$, or $\Omega := \{1,2\}$.

The power set $\mathcal{A} := \mathcal{P}(\Omega)$ contains all possible events. In this case, we can easily list the entire set system $\mathcal{A}$, as shown in Table 19.3.

Table 19.3 Event system $\mathcal{A}$ in the model of tossing a coin.

Description	Set form
Neither heads nor tails	$\omega \in A = \emptyset$
Heads appears	$\omega \in A = \{H\}$
Tails appears	$\omega \in A = \{T\}$
Heads or tails appears	$\omega \in A = \{H, T\}$

The definition of the measurable space $(\Omega, \mathcal{A})$ is therefore complete.

A probability measure $\mathbb{P}$ can again be specified by defining $\mathbb{P}(\{\omega\})$ for all $\omega \in \Omega$. The normalization of Ω here implies in particular that

$$\mathbb{P}(\Omega) = 1 \Leftrightarrow \mathbb{P}(\{H\}) + \mathbb{P}(\{T\}) = 1 \Leftrightarrow \mathbb{P}(\{T\}) = 1 - \mathbb{P}(\{H\}). \tag{19.20}$$

Thus, when the probability of the elementary event $\{H\}$ is specified, the probability of the elementary event $\{T\}$ is immediately specified as well, and conversely, of course. For the model of a fair coin, one would choose $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) := 1/2$. In this case, the probabilities of all possible events are

$$\mathbb{P}(\emptyset) = 0, \mathbb{P}(\{H\}) = 1/2, \mathbb{P}(\{T\}) = 1/2 \text{ and } \mathbb{P}(\{H, T\}) = 1. \tag{19.21}$$

Tossing a coin twice

We model tossing a coin twice. Based on the model of a single coin toss, it is sensible to define the outcome set as $\Omega := \{HH, HT, TH, TT\}$. The power set $\mathcal{A} := \mathcal{P}(\Omega)$ again contains all $2^{|\Omega|} = 2^4 = 16$ possible events. In the table below, we list four of them.

Table 19.4 Selected events in the model of tossing a coin twice.

Description	Set form
Heads appears on the first toss	$\omega \in A = \{HH, HT\}$
Heads appears on the second toss	$\omega \in A = \{HH, TH\}$
Heads does not appear	$\omega \in A = \{TT\}$
Tails appears at least once	$\omega \in A = \{HT, TH, TT\}$

The definition of the measurable space $(\Omega, \mathcal{A})$ is therefore complete. A probability measure $\mathbb{P}$ can again be specified by defining $\mathbb{P}(\{\omega\})$ for all $\omega \in \Omega$. For the model of tossing a fair coin twice, one would define

$$\mathbb{P}(\{HH\}) = \mathbb{P}(\{HT\}) = \mathbb{P}(\{TH\}) = \mathbb{P}(\{TT\}) := \frac{1}{4}. \tag{19.22}$$

Value of a BDI-II item

We model the value of a patient for the item “Sadness” of the English version of the Beck Depression Inventory II (Hautzinger et al. (2006)),

(0) I do not feel sad.
(1) I feel sad much of the time.
(2) I am sad all the time.

(3) I am so sad or unhappy that I can't stand it.

The random process under consideration is therefore a patient's response to this item. For modeling, the outcome set $\Omega := \{0, 1, 2, 3\}$ is then suitable. For a first patient, for example, the realization is $3 \in \Omega$, for a second patient the realization is $0 \in \Omega$, for a third patient $1 \in \Omega$, and so on. As the event system $\mathcal{A}$, the power set is again suitable. In the present case, it has cardinality

$$|\mathcal{P}(\Omega)| = 2^{|\Omega|} = 2^4 = 16 \tag{19.23}$$

and can easily be written down completely as shown in Table 19.5.

Table 19.5 Selected events in the model of responding to a BDI-II item.

Given response	Set form
None	$\emptyset$
"I do not feel sad"	$\{0\}$
"I feel sad much of the time"	$\{1\}$
"I am sad all the time"	$\{2\}$
"I am so sad or unhappy that I can't stand it"	$\{3\}$
"I do not feel sad" or "I feel sad much of the time"	$\{0, 1\}$
"I do not feel sad" or "I am sad all the time"	$\{0, 2\}$
"I do not feel sad" or "I am so sad or unhappy that I can't stand it"	$\{0, 3\}$
"I feel sad much of the time" or "I am sad all the time"	$\{1, 2\}$
"I feel sad much of the time" or "I am so sad or unhappy that I can't stand it"	$\{1, 3\}$
"I am sad all the time" or "I am so sad or unhappy that I can't stand it"	$\{2, 3\}$
Not "I am so sad or unhappy that I can't stand it"	$\{0, 1, 2\}$
Not "I am sad all the time"	$\{0, 1, 3\}$
Not "I feel sad much of the time"	$\{0, 2, 3\}$
Not "I do not feel sad"	$\{1, 2, 3\}$
Any response option	$\{0, 1, 2, 3\}$

Now suppose that the probabilities of the item responses are known as

$$\mathbb{P}(\{0\}) := \frac{2}{10}, \quad \mathbb{P}(\{1\}) := \frac{3}{10}, \quad \mathbb{P}(\{2\}) := \frac{4}{10}, \quad \mathbb{P}(\{3\}) := \frac{1}{10}. \tag{19.24}$$

Then these probabilities define, in the sense of a probability function,

$$\pi : \Omega \to [0, 1], \omega \mapsto \pi(\omega) =: \mathbb{P}(\{\omega\}) \tag{19.25}$$

a probability measure, and the definition of the probability space model $(\Omega, \mathcal{A}, \mathbb{P})$ for the scenario under consideration is complete. With the help of the σ-additivity of $\mathbb{P}$, for example, the probability of an item response greater than 1 can now be determined:

$$\mathbb{P}(\{2, 3\}) = \mathbb{P}(\{2\} \cup \{3\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{3\}) = \frac{4}{10} + \frac{1}{10} = \frac{5}{10} = \frac{1}{2} \tag{19.26}$$

Similarly, for example, the probability that a patient gives one of the two item responses "(0) I do not feel sad" or "(3) I am so sad or unhappy that I can't stand it" is

$$\mathbb{P}(\{0, 3\}) = \mathbb{P}(\{0\} \cup \{3\}) = \mathbb{P}(\{0\}) + \mathbb{P}(\{3\}) = \frac{2}{10} + \frac{1}{10} = \frac{3}{10}. \tag{19.27}$$

19.4 Bibliographic remarks

Andrey Kolmogorov's monograph "Grundbegriffe der Wahrscheinlichkeitsrechnung" (Kolmogoroff (1933)) symbolizes the foundation of modern probability calculus. In addition to the axiomatic introduction of the probability space model discussed here, Kolmogoroff (1933) considers many other aspects of probability calculus and thus offers a readable introduction to the whole field of probability theory. Of course, the approach formulated by Kolmogoroff (1933) is only a provisional end product of the long historical development of probability theory. After all, the development of the mathematical modeling of random processes by no means came to an end with Kolmogoroff (1933). Later works in the 20th century concerned in particular the interpretation of the concept of probability (cf. De Finetti (1975)) or introduced generalized quantitative measures of subjective uncertainty (cf. Walley (1991)). A current overview of the interpretation of the concept of probability and its formal foundations is given by Hájek (2019).

Study questions

1. Explain the concept of a random process.
2. State the definition of the concept of a σ-algebra.
3. State the definition of the concept of a probability measure.
4. State the definition of the concept of a probability space.
5. Explain the concept of the outcome set Ω.
6. Explain the tacit mechanics of the probability space model.
7. Explain the concept of an event $A \in \mathcal{A}$.
8. Explain the concept of the event system $\mathcal{A}$.
9. Which σ-algebra is sensibly chosen for a probability space with finite outcome set?
10. Explain the concept of the probability measure $\mathbb{P}$.
11. State the definition of the concept of a probability function.
12. Why is the concept of a probability function helpful when modeling a random process by a probability space with finite outcome set?
13. Explain the modeling of throwing a die using a probability space.
14. Explain the modeling of the simultaneous throwing of a red and a blue die using a probability space.

Study question answers

1. A random process is a phenomenon of reality that cannot be predicted with absolute certainty or is associated with uncertainty for humans. See the introduction to probability theory.
2. See Definition 19.1 regarding $\mathcal{A}$.
3. See Definition 19.1 regarding $\mathbb{P}$.
4. See Definition 19.1.
5. The outcome set contains the elementary events possible in a run of the modeled random process; see the explanations of **Outcome set and mechanics**.
6. See the explanations of **Outcome set and mechanics**.
7. See the explanations of **Events and event system**.
8. See the explanations of **Events and event system**.
9. The power set of the outcome set; it contains all events that are possible in principle; see also Theorem 19.1.
10. See the explanations of **Probability measure** $\mathbb{P}$.
11. See Definition 19.2.
12. By defining the function values of a probability function, a probability measure can be specified; see also Section 19.3.
13. See the explanations of **Throwing one die**.
14. See the explanations of **Simultaneous throwing of a blue and a red die**.

20 Elementary probabilities

In this section, with the concepts of the *joint probability* of two events and the *conditional probability* of an event, we introduce two elementary forms of probabilities. Intuitively, the concept of joint probability refers to the probability of the "simultaneous" occurrence of two events A and B, and the concept of conditional probability refers to the probability of the occurrence of an event A "when one knows about the occurrence of another event B". If it is irrelevant for the probability of an event A whether or not an event B has occurred, then A and B are called *independent events*. Intuitively, independent events model the absence of mutual influences.

20.1 Joint probabilities

The concept of the joint probability of two events is defined as follows.

Definition 20.1 (Joint probability)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $A, B \in \mathcal{A}$. Then

$$\mathbb{P}(A \cap B) \tag{20.1}$$

is called the *joint probability of A and B.*

•

As noted above, $\mathbb{P}(A \cap B)$ corresponds to the probability that the events A and B occur "simultaneously". This is best understood against the background of the mechanics of the probability space model. According to this mechanics, $\mathbb{P}(A \cap B)$ is precisely the probability that, in one run of a random process, an ω is realized for which both $\omega \in A$ and $\omega \in B$ hold.

Example

As a first example of a joint probability of two events, we again consider the probability space model of throwing a fair die. Let A be the event "An even number of pips appears", that is, $A := \{2, 4, 6\}$, and let B be the event "A number of pips greater than three appears", that is, $B := \{4, 5, 6\}$. Set-theoretically, we then have

$$A \cap B = \{2, 4, 6\} \cap \{4, 5, 6\} = \{4, 6\}. \tag{20.2}$$

The interpretation of $A \cap B = \{4, 6\}$ is exactly "An even number of pips appears *and* this number of pips is greater than three". Assuming fairness of the die, that is, for

$\mathbb{P}(\{4\}) = \mathbb{P}(\{6\}) := 1/6$, we can easily calculate the probability of this event using the σ-additivity of $\mathbb{P}$. We obtain

$$\begin{aligned}
\mathbb{P}(A \cap B) &= \mathbb{P}(\{2,4,6\} \cap \{4,5,6\}) \\
&= \mathbb{P}(\{4,6\}) \\
&= \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) \\
&= \frac{1}{6} + \frac{1}{6} \\
&= \frac{1}{3}.
\end{aligned} \tag{20.3}$$

When thinking about joint probabilities, it is of course important not to confuse the joint probability $\mathbb{P}(A \cap B)$ with the probability $\mathbb{P}(A \cup B)$ of the event $A \cup B$. Recall that the union of two sets $\cup$ corresponds to the *inclusive or*, that is, to an *and/or* (cf. Section 1.3 and Section 2.2). The event $A \cup B$ therefore corresponds to the event that event A and/or event B occurs. In particular, $\omega \in A \cup B$ is already satisfied if for the outcome of one run of a random process *only* $\omega \in A$ or *only* $\omega \in B$ holds. Concretely, for the events $A := \{2,4,6\}$ and $B := \{4,5,6\}$ from the die example above, we obtain

$$A \cup B = \{2,4,6\} \cup \{4,5,6\} = \{2,4,5,6\} \tag{20.4}$$

with the interpretation "An even number of pips appears and/or a number of pips greater than three appears". For the corresponding probability we obtain

$$\mathbb{P}(\{2,4,5,6\}) = \frac{2}{3}, \tag{20.5}$$

so in this case apparently $\mathbb{P}(A \cap B) \neq \mathbb{P}(A \cup B)$.

With the help of the following theorem, in this section we finally record a few useful properties for calculating with probabilities, which follow directly from the connection between set operations and the σ-additivity of probability measures. We visualize the corresponding statements in Figure 20.1 using Venn diagrams.

Theorem 20.1 (Further properties of probabilities)**.** *Let* $(\Omega, \mathcal{A}, \mathbb{P})$ *be a probability space and let* $A, B \in \mathcal{A}$ *be events. Then*

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. $A \subset B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$.
3. $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$
4. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

∘

Proof. The second, third, and fourth statements of this theorem are based on elementary set-theoretic statements and the σ-additivity of $\mathbb{P}$. We do not prove these elementary set-theoretic statements here, but instead refer in each case to the intuition conveyed by the Venn diagrams in Figure 20.2.

For 1.: We first note that from $A^c := \Omega \setminus A$ it follows that $A^c \cup A = \Omega$ and that $A^c \cap A = \emptyset$. With normalization and σ-additivity of $\mathbb{P}$ it then follows that

$$\mathbb{P}(\Omega) = 1 \Leftrightarrow \mathbb{P}(A^c \cup A) = 1 \Leftrightarrow \mathbb{P}(A^c) + \mathbb{P}(A) = 1 \Leftrightarrow \mathbb{P}(A^c) = 1 - \mathbb{P}(A). \tag{20.6}$$

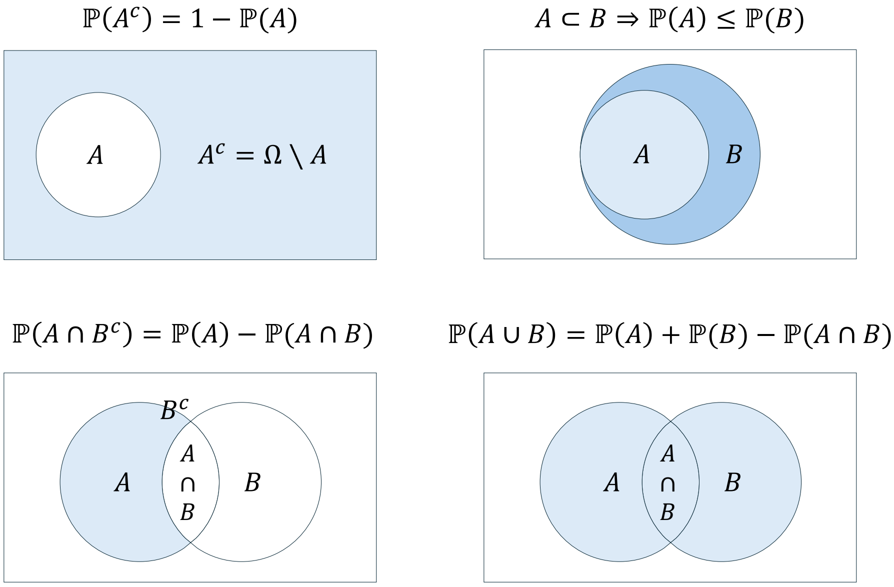

Figure 20.1 Venn diagrams for Theorem 20.1. The light-blue areas correspond to the probabilities of interest.

For 2.: First, we have (cf. Figure A)

$$A \subset B \Rightarrow B = A \cup (B \cap A^c) \text{ with } A \cap (B \cap A^c) = \emptyset. \tag{20.7}$$

With the σ-additivity of $\mathbb{P}$ it follows that

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c). \tag{20.8}$$

With $\mathbb{P}(B \cap A^c) \ge 0$ it follows that $\mathbb{P}(A) \le \mathbb{P}(B)$.

For 3.: First, we have (cf. Figure B)

$$(A \cap B) \cap (A \cap B^c) = \emptyset \text{ and } A = (A \cap B) \cup (A \cap B^c). \tag{20.9}$$

With the σ-additivity of $\mathbb{P}$ it follows that

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \Leftrightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A \cap B^c). \tag{20.10}$$

For 4.: First, we have (cf. Figure C)

$$B \cap (A \cap B^c) = \emptyset \text{ and } A \cup B = B \cup (A \cap B^c). \tag{20.11}$$

With the σ-additivity of $\mathbb{P}$ it follows that

$$\mathbb{P}(A \cup B) = \mathbb{P}(B) + \mathbb{P}(A \cap B^c). \tag{20.12}$$

With 3. it then follows that

$$\mathbb{P}(A \cup B) = \mathbb{P}(B) + \mathbb{P}(A) - \mathbb{P}(A \cap B). \tag{20.13}$$

□

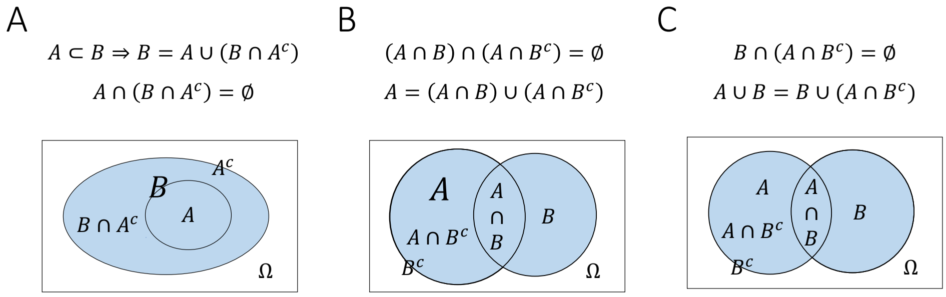

Figure 20.2 Venn diagrams for the proof of Theorem 20.1

20.2 Conditional probabilities

We now turn to the concept of conditional probability.

Definition 20.2 (Conditional probability)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $A, B \in \mathcal{A}$ be events with $\mathbb{P}(B) > 0$. The *conditional probability of event A given event B* is defined as

$$\mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \tag{20.14}$$

•

We note: The conditional probability $\mathbb{P}(A|B)$ of an event A given an event B is the joint probability $\mathbb{P}(A \cap B)$ of the events A and B, scaled by $1/\mathbb{P}(B)$. Thus, if in the formulation of a probabilistic model one specifies the joint probability $\mathbb{P}(A \cap B)$ as well as the probability $\mathbb{P}(B) > 0$ of the event B, then in particular one also specifies the conditional probability $\mathbb{P}(A|B)$ of event A given event B. We further point out that there is no reason to confuse the conditional probabilities

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \text{ and } \mathbb{P}(B|A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \tag{20.15}$$

(cf. Herzog & Ostwald (2013)). In particular, from $\mathbb{P}(A) \neq \mathbb{P}(B)$ it always follows directly that $\mathbb{P}(A|B) \neq \mathbb{P}(B|A)$. Finally, note that a generalization of the conditional probability in Definition 20.2 to the case $\mathbb{P}(B) = 0$ is possible, but technically involved. For this we refer to the advanced literature, e.g. Meintrup & Schäffler (2005) and Schmidt (2009).

Examples

Throwing one die

As a first example of a conditional probability, we again consider the model $(\Omega, \mathcal{A}, \mathbb{P})$ of the fair die. We want to calculate the conditional probability of the event "An even number of pips appears" given the event "A number greater than three appears". We have already seen above that the event "An even number of pips appears" corresponds to the subset $A := \{2, 4, 6\}$ of Ω, and that the event "A number greater than three appears" corresponds to the subset $B := \{4,5,6\}$ of Ω. Furthermore, we have seen that, under the assumption that the modeled die is fair,

$$\mathbb{P}(\{2,4,6\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} \tag{20.16}$$

and that

$$\mathbb{P}(\{4,5,6\}) = \mathbb{P}(\{4\}) + \mathbb{P}(\{5\}) + \mathbb{P}(\{6\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6}. \tag{20.17}$$

Finally, we had also seen that the event $A \cap B$, that is, the event "An even number of pips greater than three appears", has probability

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{2,4,6\} \cap \{4,5,6\}) = \mathbb{P}(\{4,6\}) = \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} \tag{20.18}$$

By the definition of conditional probability, we then directly obtain

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\{4,6\})}{\mathbb{P}(\{4,5,6\})} = \frac{2}{6} \cdot \frac{6}{3} = \frac{2}{3}. \tag{20.19}$$

In this context, an interpretation of the conditional probability $\mathbb{P}(A|B)$ as a decrease in subjective uncertainty or as a gain in subjective information relative to the unconditional probability $\mathbb{P}(A)$ suggests itself: if one knows that a number of pips greater than three has appeared, that is, that the event $\omega \in B$ is present, then the probability that ω is an even number of pips is $2/3$. If, by contrast, one does not know that the event $\omega \in B$ is present (and has no other information about ω), then the probability of an even number of pips appearing is only $1/2$. Conditioning on the presence of an event therefore corresponds to including information and thus to the decrease of uncertainty in probability-theoretic models. This is the basis of Bayesian statistics. A similar interpretation is also suggested by the following, perhaps more everyday, example.

Value of a BDI-II item

We again consider the example of the value of a patient for the item "Sadness" of the English version of the BDI-II. Above, we assigned the probabilities

$$\mathbb{P}(\{0\}) := \frac{2}{10}, \quad \mathbb{P}(\{1\}) := \frac{3}{10}, \quad \mathbb{P}(\{2\}) := \frac{4}{10}, \quad \mathbb{P}(\{3\}) := \frac{1}{10}, \tag{20.20}$$

to the possible item responses (0), (1), (2), (3). We now assume that we know of a patient that she chose a value greater than (1), and ask for the probabilities that the patient chose value (2) or (3). To this end, we first consider the probability of the event "The patient chose a value greater than (1)". This obviously corresponds to the subset $B := \{2,3\}$ of Ω. With σ-additivity, as seen above, the probability of this event is $\mathbb{P}(\{2,3\}) = 1/2$. We therefore consider the events listed in Table 20.1.

Table 20.1 Events when answering a BDI-II item

Description	Set form
The patient chose a value greater than (1)	$\omega \in B := \{2,3\}$
The patient chose response (2)	$\omega \in A_1 := \{2\}$
The patient chose response (3)	$\omega \in A_2 := \{3\}$

We obtain

$$\mathbb{P}(A_1|B) = \frac{\mathbb{P}(A_1 \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\{2\} \cap \{2,3\})}{\mathbb{P}(\{2,3\})} = \frac{\mathbb{P}(\{2\})}{\mathbb{P}(\{2,3\})} = \frac{4}{10} \cdot \frac{2}{1} = \frac{8}{10} \tag{20.21}$$

and

$$\mathbb{P}(A_2|B) = \frac{\mathbb{P}(A_2 \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\{3\} \cap \{2,3\})}{\mathbb{P}(\{2,3\})} = \frac{\mathbb{P}(\{3\})}{\mathbb{P}(\{2,3\})} = \frac{1}{10} \cdot \frac{2}{1} = \frac{2}{10}. \tag{20.22}$$

Note that

$$\mathbb{P}(A_1|B) + \mathbb{P}(A_2|B) = \frac{8}{10} + \frac{2}{10} = 1. \tag{20.23}$$

Based on the definition of the conditional probability of an event under the condition of the occurrence of a fixed event B, one can define a probability measure that gives the probabilities of *all* events of the event system under the condition of the occurrence of a fixed event B. This probability measure is called the *conditional probability given event B*.

Theorem 20.2 (Conditional probability)**.** *For a fixed $B \in \mathcal{A}$ with $\mathbb{P}(B) > 0$, let*

$$\mathbb{P}(\cdot|B) : \mathcal{A} \to [0,1], A \mapsto \mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \tag{20.24}$$

Then $\mathbb{P}(\cdot|B)$ is a probability measure and is called the conditional probability given event B.

∘

Proof. We verify the defining properties of a probability measure.

(1) $\mathbb{P}(A|B) \geq 0$ for all $A \in \mathcal{A}$

By the definition of conditional probability, $\mathbb{P}(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ with $\mathbb{P}(B) > 0$. With the theorem on closure of σ-algebras under intersections and the definition of $\mathbb{P}$, $\mathbb{P}(A \cap B) \geq 0$ holds, and hence the statement follows.

(2) $\mathbb{P}(\Omega|B) = 1$.

We have

$$\mathbb{P}(\Omega|B) = \frac{\mathbb{P}(\Omega \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1. \tag{20.25}$$

(3) $\mathbb{P}\left(\cup_{i=1}^{\infty} A_i|B\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i|B)$ for pairwise disjoint $A_1, A_2, \ldots \in \mathcal{A}$.

With the associative and distributive laws of union and intersection, first

$$\left(\cup_{i=1}^{\infty} A_i\right) \cap B = (A_1 \cup A_2 \cup \cdots) \cap B = (A_1 \cap B) \cup (A_2 \cap B) \cup \cdots = \cup_{i=1}^{\infty} (A_i \cap B). \tag{20.26}$$

Furthermore, with $A_i \cap A_j = \emptyset$ and the associative and distributive laws of union and intersection, the $A_i \cap B$ for $i = 1, 2, \ldots$ are also pairwise disjoint, because

$$(A_i \cap B) \cap (A_j \cap B) = (A_i \cap A_j) \cap (B \cap B) = \emptyset \cap B = \emptyset. \tag{20.27}$$

With the σ-additivity of $\mathbb{P}$ it then follows that

$$
\begin{aligned}
\mathbb{P}\left(\cup_{i=1}^{\infty} A_i | B\right) & = \frac{\mathbb{P}\left(\left(\cup_{i=1}^{\infty} A_i\right) \cap B\right)}{\mathbb{P}(B)} \\
& = \frac{\mathbb{P}\left(\cup_{i=1}^{\infty}\left(A_i \cap B\right)\right)}{\mathbb{P}(B)} \\
& = \frac{\sum_{i=1}^{\infty} \mathbb{P}\left(A_i \cap B\right)}{\mathbb{P}(B)} \\
& = \sum_{i=1}^{\infty} \frac{\mathbb{P}\left(A_i \cap B\right)}{\mathbb{P}(B)} \\
& = \sum_{i=1}^{\infty} \mathbb{P}(A_i | B).
\end{aligned}
\tag{20.28}
$$

□

Note that for the conditional probability $\mathbb{P}(\cdot|B)$, the calculation rules of probability theory apply to the events to the left of the bar. In particular, in contrast to $\mathbb{P}(\cdot \cap B)$, $\mathbb{P}(\cdot|B)$ defines a probability measure for all $A \in \mathcal{A}$. Thus, for example, for $0 < \mathbb{P}(B) < 1$, $\mathbb{P}(\Omega|B) = 1$, but $\mathbb{P}(\Omega \cap B) = \mathbb{P}(B) < 1$. One therefore also says that $\mathbb{P}(\cdot|B)$ is *normalized*, whereas $\mathbb{P}(\cdot \cap B)$ is not.

At the end of this section, we consider three technical consequences of the definition of conditional probability, which we formulate as theorems.

The first theorem concerns the relation between joint and conditional probabilities and reiterates how joint probabilities can be calculated from conditional and total probabilities.

Theorem 20.3 (Joint and conditional probabilities)**.** *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $A, B \in \mathcal{A}$ with $\mathbb{P}(A) > 0$ and $P(B) > 0$. Then*

$$
\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A). \tag{20.29}
$$

∘

Proof. With the definition of the respective conditional probability, it follows directly that

$$
\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \Leftrightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) \tag{20.30}
$$

and

$$
\mathbb{P}(B|A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} \Leftrightarrow \mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A). \tag{20.31}
$$

□

Just as specifying $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A)$ specifies the conditional probability $\mathbb{P}(B|A)$, specifying $\mathbb{P}(A)$ and $\mathbb{P}(B|A)$ therefore specifies the joint probability $\mathbb{P}(A \cap B)$.

The following so-called *law of total probability* states how unconditional, so-called *total probabilities* can be calculated based on joint probabilities.

Theorem 20.4 (Law of total probability)**.** *Let* $(\Omega, \mathcal{A}, \mathbb{P})$ *be a probability space and let* $A_1, ..., A_k$ *with* $\mathbb{P}(A_i) > 0$ *be a partition of* Ω*. Then for every* $B \in \mathcal{A}$*,*

$$\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \cap A_i) = \sum_{i=1}^{k} \mathbb{P}(B|A_i)\mathbb{P}(A_i). \tag{20.32}$$

$\circ$

Proof. For $i = 1, \dots, k$, let $C_i := B \cap A_i$, so that $\cup_{i=1}^k C_i = B$ and $C_i \cap C_j = \emptyset$ for $1 \le i, j \le k, i \neq j$. We illustrate these definitions in Figure 20.3 using a Venn diagram.

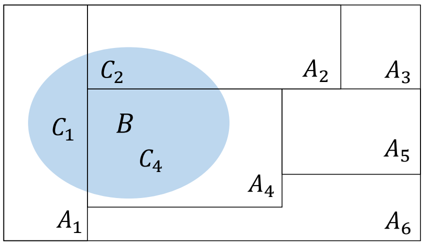

Figure 20.3 Venn diagram for the proof of Theorem 20.4.

Thus, with the definition of conditional probability for $\mathbb{P}(A_i) > 0$, we have

$$\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(C_i) = \sum_{i=1}^{k} \mathbb{P}(B \cap A_i) = \sum_{i=1}^{k} \mathbb{P}(B|A_i)\mathbb{P}(A_i). \tag{20.33}$$

$\square$

Intuitively, $\mathbb{P}(B)$ therefore corresponds to the weighted sum of the conditional probabilities $\mathbb{P}(B|A_i)$, where the weighting factors are precisely the unconditional probabilities $\mathbb{P}(A_i)$ for $i = 1, .., k$. The requirement $\mathbb{P}(A_i) > 0$ for all $i = 1, ..., k$ is necessary here to guarantee the representation of $\mathbb{P}(B \cap A_i)$ using Definition 20.2.

Example

As an example of representing the probability of an event using the law of total probability, we consider the probability space model of the fair die. For the outcome space $\Omega := \{1, 2, 3, 4, 5, 6\}$, we consider the event $B := \{2, 4, 6\}$ ("An even number appears") with probability $\mathbb{P}(B) = \frac{1}{2}$ as well as the partition $A_1 := \{1, 2\}$, $A_2 := \{3\}$, $A_3 := \{4, 5\}$, $A_4 := \{6\}$ of Ω. Then apparently

$$\begin{array}{lllllll}
\mathbb{P}(B \cap A_1) & = & \mathbb{P}(\{2,4,6\} \cap \{1,2\}) & = & \mathbb{P}(\{2\}) & = & \frac{1}{6} \\
\mathbb{P}(B \cap A_2) & = & \mathbb{P}(\{2,4,6\} \cap \{3\}) & = & \mathbb{P}(\emptyset) & = & 0 \\
\mathbb{P}(B \cap A_3) & = & \mathbb{P}(\{2,4,6\} \cap \{4,5\}) & = & \mathbb{P}(\{4\}) & = & \frac{1}{6} \\
\mathbb{P}(B \cap A_4) & = & \mathbb{P}(\{2,4,6\} \cap \{6\}) & = & \mathbb{P}(\{6\}) & = & \frac{1}{6}
\end{array} \tag{20.34}$$

Thus we obtain

$$\begin{aligned}
\sum_{i=1}^{4} \mathbb{P}(B \cap A_i) &= \mathbb{P}(B \cap A_1) + \mathbb{P}(B \cap A_2) + \mathbb{P}(B \cap A_3) + \mathbb{P}(B \cap A_4) \\
&= \frac{1}{6} + 0 + \frac{1}{6} + \frac{1}{6} \\
&= \frac{1}{2}
\end{aligned} \tag{20.35}$$

and hence

$$\mathbb{P}(B) = \sum_{i=1}^{4} \mathbb{P}(B \cap A_i). \tag{20.36}$$

Note that the partition property of A_1, A_2, A_3, A_4 is essential for the validity of the theorem in this example. If, for example, $A_4 = \{4, 6\}$ and thus A_3 and A_4 were not disjoint, then the right-hand side of the law of total probability would yield a value greater than $\mathbb{P}(B)$. If, by contrast, the union of A_1, A_2, A_3, A_4 did not yield Ω, for example because $A_4 = \emptyset$, then conversely a value smaller than $\mathbb{P}(B)$ would result.

Finally, with *Bayes' theorem*, we consider a formula for the alternative calculation of conditional probabilities.

Theorem 20.5 (Bayes' theorem). *Let* $(\Omega, \mathcal{A}, \mathbb{P})$ *be a probability space and let* $A_1, ..., A_k$ *be a partition of* Ω *with* $\mathbb{P}(A_i) > 0$ *for all* $i = 1, ..., k$*. If* $\mathbb{P}(B) > 0$*, then for every* $i = 1, ..., k$*,*

$$\mathbb{P}(A_i|B) = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\sum_{i=1}^{k} \mathbb{P}(B|A_i)\mathbb{P}(A_i)}. \tag{20.37}$$

∘

Proof. With the definition of conditional probability and the law of total probability,

$$\mathbb{P}(A_i|B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\sum_{i=1}^{k} \mathbb{P}(B|A_i)\mathbb{P}(A_i)}. \tag{20.38}$$

□

Note that Bayes' theorem is independent of the frequentist or Bayesian interpretation of probability and merely makes a statement about calculating with conditional probabilities. In frequentist inference, however, Bayes' theorem is used rather rarely. In Bayesian inference, by contrast, Bayes' theorem is central. In this context, $\mathbb{P}(A_i)$ is then often called the *prior probability of event* A_i and $\mathbb{P}(A_i|B)$ the *posterior probability of event* A_i. As explained above, $\mathbb{P}(A_i|B)$ corresponds to the probability of A_i when one knows about the occurrence of B.

In applications, the results formulated here as the *law of total probability* (Theorem 20.4), the *theorem on joint and conditional probabilities* (Theorem 20.3), and *Bayes' theorem* (Theorem 20.5) are often used as the central calculation rules of probability theory. In practice, they are usually called the *summation rule*, *multiplication rule*, and *Bayes rule* (cf. Table 20.2).

Table 20.2 Central calculation rules of probability theory

Calculation rule	Probability space form
Multiplication rule	$\mathbb{P}(A \cap B) = \mathbb{P}(B|A)\,\mathbb{P}(A)$
Summation rule	$\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \cap A_i)$, if $\Omega = \cup_{i=1}^{k} A_i$ and $A_i \cap A_j = \emptyset$
Bayes rule	$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}$, if $\mathbb{P}(B) > 0$

20.3 Independent events

The independence of events serves to model the absence of mutual influences of events. Its definition states that the joint probability of two events should result from the product of the probabilities of the individual events. In this context, one also speaks of the *factorization of the joint probability* of the events. The meaning of this definition becomes clear in light of the concept of conditional probability in Theorem 20.6. We first consider the definition.

Definition 20.3 (Independent events). Two events $A \in \mathcal{A}$ and $B \in \mathcal{A}$ are called *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B). \tag{20.39}$$

A set of events $\{A_i | i \in I\} \subset \mathcal{A}$ with arbitrary index set I is called *independent* if for every finite subset $J \subseteq I$,

$$\mathbb{P}\left(\cap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j). \tag{20.40}$$

•

Note that the independence of certain events can be assumed in the definition of a probabilistic model or can also follow from the definition of a probabilistic model. If two events are not independent, one also says that these events are *dependent*. The meaning of the product property under independence becomes clear, as we show below, in the context of conditional probabilities. First, we give an example that illustrates the relationship between independence and pairwise independence of events.

Example

Independent events are by definition always also pairwise independent. For example, for the set $\{A_i | i \in \{1, 2, 3\}\}$ of events and the possible subsets $J_1 := \{1, 2\}$, $J_2 := \{1, 3\}$, $J_3 := \{2, 3\}$, $J := \{1, 2, 3\}$ of $I = \{1, 2, 3\}$, the defining condition of independence of A_1, A_2, A_3 implies the statements

$$\begin{aligned}
\mathbb{P}(A_1 \cap A_2) &= \mathbb{P}(A_1)\,\mathbb{P}(A_2) \\
\mathbb{P}(A_1 \cap A_3) &= \mathbb{P}(A_1)\,\mathbb{P}(A_3) \\
\mathbb{P}(A_2 \cap A_3) &= \mathbb{P}(A_2)\,\mathbb{P}(A_3) \\
\mathbb{P}(A_1 \cap A_2 \cap A_3) &= \mathbb{P}(A_1)\,\mathbb{P}(A_2)\,\mathbb{P}(A_2)
\end{aligned} \tag{20.41}$$

Conversely, pairwise independence of events does not imply independence of events, as the following example shows (cf. DeGroot & Schervish (2012)). Consider the model of tossing a fair coin twice with outcome space

$$\Omega := \{HH, HT, TH, TT\} \tag{20.42}$$

with elementary event probabilities

$$\mathbb{P}\{HH\} = \mathbb{P}\{HT\} = \mathbb{P}\{TH\} = \mathbb{P}\{TT\} = \frac{1}{4} \tag{20.43}$$

Then the events listed in Table 20.3 are pairwise independent, but not independent.

Table 20.3 Coin-toss example for independence

Verbal description	Set form
Heads appears on the first toss	$A_1 := \{HH, HT\}$
Heads appears on the second toss	$A_2 := \{HH, TH\}$
The same outcome appears on both tosses	$A_3 := \{HH, TT\}$

Here, apparently,

$$\mathbb{P}(A_1) = \mathbb{P}(A_2) = \mathbb{P}(A_2) = \frac{1}{2}, \tag{20.44}$$

and

$$\begin{aligned} \mathbb{P}(A_1 \cap A_2) &= \mathbb{P}(\{HH\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A_1)\,\mathbb{P}(A_2), \\ \mathbb{P}(A_1 \cap A_3) &= \mathbb{P}(\{HH\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A_2)\,\mathbb{P}(A_3), \\ \mathbb{P}(A_2 \cap A_3) &= \mathbb{P}(\{HH\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A_1)\,\mathbb{P}(A_3). \end{aligned} \tag{20.45}$$

The events are therefore pairwise independent. However, it is also true that

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(\{HH\}) = \frac{1}{4} \neq \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A_1)\,\mathbb{P}(A_2)\,\mathbb{P}(A_3) \tag{20.46}$$

and the events A_1, A_2, A_3 are therefore not independent.

Finally, with the relation between independence and conditional probability, we explain the meaning of the product property under independence.

Theorem 20.6 (Conditional probability under independence)**.** *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $A, B \in \mathcal{A}$ be independent events with $\mathbb{P}(B) > 0$. Then*

$$\mathbb{P}(A|B) = \mathbb{P}(A). \tag{20.47}$$

∘

Proof. Under the assumptions of the theorem,

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A). \tag{20.48}$$

□

Given independence of two events A and B, it is therefore irrelevant for the probability of event A whether B also occurs or not; the probability $\mathbb{P}(A)$ remains the same. Independence of events is thus modeled precisely as factorization of the joint probability of A and B so that $\mathbb{P}(A|B) = \mathbb{P}(A)$ follows. From the perspective of modeling subjective uncertainty through probabilities, the independence of two events means that knowing about the presence of one of the two events does not change the probability of the presence of the other event. Conversely, dependence of two events means that knowing about the presence of one of the two events changes the probability of the presence of the other event, that is, either increases or decreases it.

Example

We want to illustrate the relation between independence, conditional probability, and factorization of the joint probability using the model of the fair die. To this end, we consider the event $A := \{2,4,6\}$ ("An even number appears") and the event $B := \{3,4,5,6\}$ ("A number greater than two appears"). Intuitively, the probability that an even number has appeared does not change if one knows that a number greater than two has appeared: if one has the information that a number greater than two has appeared, then three of six cases correspond to the event of an even number appearing; if one has the information, then two of four cases correspond to the event of an even number appearing. The relative proportion of cases corresponding to the event "An even number appears" therefore remains the same in view of the absence or presence of the information "A number greater than two appears".

Formally, we first establish the factorization of the joint probability of the two events, that is, of the event "An even number appears and this number is greater than two", into the product of the probabilities of the two events. Here, with the σ-additivity of $\mathbb{P}$,

$$\begin{aligned}
\mathbb{P}(A \cap B) &= \mathbb{P}\left(\{2,4,6\} \cap \{3,4,5,6\}\right) \\
&= \mathbb{P}\left(\{4,6\}\right) \\
&= \frac{2}{6} \\
&= \frac{1}{2} \cdot \frac{2}{3} \\
&= \mathbb{P}(\{2,4,6\}) \cdot \mathbb{P}(\{3,4,5,6\}) \\
&= \mathbb{P}(A)\mathbb{P}(B)
\end{aligned} \tag{20.49}$$

Thus the joint probability of A and B can be written as the product of the probabilities of A and B. For the conditional probability of A given B, we further have

$$\begin{aligned}
\mathbb{P}(A|B) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \\
&= \frac{\mathbb{P}\left(\{4,6\}\right)}{\mathbb{P}(\{3,4,5,6\})} \\
&= \frac{1}{3} \cdot \frac{3}{2} \\
&= \frac{1}{2} \\
&= \mathbb{P}(\{2,4,6\}) \\
&= \mathbb{P}(A)
\end{aligned} \tag{20.50}$$

Thus in this example both Definition 20.3 and Theorem 20.6 are satisfied.

The property of events of being independent should not be confused with the possibility that events can be disjoint. In fact, the following theorem holds.

Theorem 20.7 (Disjointness and dependence of events)**.** *Let $A \in \mathcal{A}$ and $B \in \mathcal{A}$ be two disjoint events with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. Then A and B are dependent events.*

∘

Proof. First, because of the disjointness of A and B,

$$\mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0 \tag{20.51}$$

Second,

$$\mathbb{P}(A)\mathbb{P}(B) > 0, \tag{20.52}$$

because by assumption both $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$ hold. Thus

$$\mathbb{P}(A \cap B) \neq \mathbb{P}(A)\mathbb{P}(B) \tag{20.53}$$

and by Definition 20.3, A and B are therefore not independent, hence dependent. □

Example

We want to illustrate Theorem 20.7 using the example of the fair die. Let $A = \{2, 4, 6\}$ be the event "An even number appears" and $B = \{1, 3, 5\}$ the event "An odd number appears". Intuitively, the events are dependent, because the information that an even number has appeared reduces the subjective probability that an odd number has appeared to zero. Formally, on the one hand,

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \tag{20.54}$$

but

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{2, 4, 6\} \cap \{1, 3, 5\}) = \mathbb{P}(\emptyset) = 0. \tag{20.55}$$

On the other hand, it is also true that

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\emptyset)}{\mathbb{P}(B)} = \frac{0}{\mathbb{P}(B)} = 0 \neq = \frac{1}{2} = \mathbb{P}(\{2, 4, 6\}) = \mathbb{P}(A). \tag{20.56}$$

20.4 Bibliographic remarks

Many of the concepts introduced in this section are closely interwoven with the historical genesis of probability theory, so no individual references will be given. An introduction to the history of probability theory of the last two centuries is provided by Hald (1990), and an overview of more modern developments is given by Von Plato (1994). Bayes' theorem is generally traced back to Bayes (1763), even though it is not the actual main topic of this work.

Study questions

1. State the definition of the joint probability of two events.
2. Explain the intuitive meaning of the joint probability of two events.
3. State the theorem on further properties of probabilities.
4. State the definition of the conditional probability of an event.
5. State the theorem on conditional probability.
6. State the theorem on joint and conditional probabilities.
7. State the law of total probability.
8. State Bayes' theorem.
9. Reproduce the proof of Bayes' theorem.
10. State the definition of the independence of two events.
11. State the theorem on conditional probability under independence.
12. Reproduce the proof of the theorem on conditional probability under independence.
13. Explain the theorem on conditional probability under independence.

Study question answers

1. See Definition 20.1.
2. See the remarks on Definition 20.1.
3. See Theorem 20.1.
4. See Definition 20.2.
5. See Theorem 20.2.
6. See Theorem 20.3.
7. See Theorem 20.4.
8. See Theorem 20.5.
9. See proof of Theorem 20.5.
10. See Definition 20.3.
11. See Theorem 20.6.
12. See proof of Theorem 20.6.
13. See remarks on Theorem 20.6.

21 Random variables

With the concept of a *random variable*, this chapter introduces the standard probabilistic model for a univariate data point. Central to this model is the possibility of specifying distributions of random variables by means of probability mass functions and probability density functions. Moreover, because random variables are constructed as maps of random outcomes, the study of the distributions of these transformed random outcomes leads directly to a core topic of frequentist inference.

21.1 Definition

We first sketch the construction of a random variable and its distribution using Figure 21.1. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let

$$\xi : \Omega \to \mathcal{X}, \omega \mapsto \xi(\omega) \tag{21.1}$$

be a map. Furthermore, let $\mathcal{S}$ be a σ-algebra on the target set $\mathcal{X}$ of this map. For every $S \in \mathcal{S}$, let the preimage set of S be given by (cf. Definition 4.2)

$$\xi^{-1}(S) := \{\omega \in \Omega | \xi(\omega) \in S\}. \tag{21.2}$$

If now $\xi^{-1}(S) \in \mathcal{A}$ holds for all $S \in \mathcal{S}$, then the map ξ is called *measurable*. Thus, assume that ξ is measurable. Then each $S \in \mathcal{S}$ can be assigned the probability

$$\mathbb{P}_\xi : \mathcal{S} \to [0,1], S \mapsto \mathbb{P}_\xi(S) := \mathbb{P}\left(\xi^{-1}(S)\right) = \mathbb{P}\left(\{\omega \in \Omega | \xi(\omega) \in S\}\right). \tag{21.3}$$

In this context, ξ is called a *random variable* and $\mathbb{P}_\xi$ is called the *image measure* or the *distribution* of ξ. Overall, starting from the probability space $(\Omega, \mathcal{A}, \mathbb{P})$, the random variable ξ has thus been used to construct the probability space $(\mathcal{X}, \mathcal{S}, \mathbb{P}_\xi)$. Formally, a random variable is therefore defined as follows.

Definition 21.1 (Random variable)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $(\mathcal{X}, \mathcal{S})$ be a *measurable space*. A *random variable* is defined as a map $\xi : \Omega \to \mathcal{X}$ with the *measurability property*

$$\{\omega \in \Omega | \xi(\omega) \in S\} \in \mathcal{A} \text{ for all } S \in \mathcal{S}. \tag{21.4}$$

•

According to Definition 21.1, random variables are neither "random" nor "variables", but measurable maps. If one asks about the meaning of randomness for the values $\xi(\omega)$ of random variables, then the implicit frequentist mechanics of the probability space $(\Omega, \mathcal{A}, \mathbb{P})$ again provides the corresponding intuition: in each run of the modeled random process, an ω is realized according to $\mathbb{P}$ and is then mapped deterministically to $\xi(\omega)$. Accordingly, we define the concepts of the *outcome space* and the *realization* of a random variable.

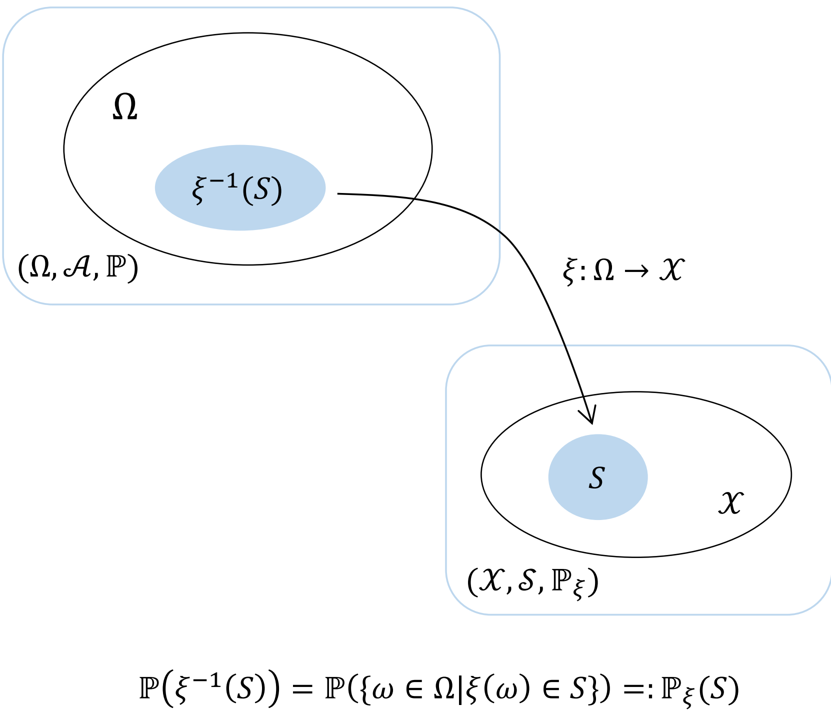

Figure 21.1 Construction of a random variable and a distribution.

Definition 21.2 (Realization of a random variable). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let $(\mathcal{X}, \mathcal{S})$ be a measurable space, and let $\xi : \Omega \to \mathcal{X}$ be a random variable. Then $\mathcal{X}$ is called the *outcome space of the random variable* ξ, and $\xi(\omega) \in \mathcal{X}$ is called a *realization of the random variable* ξ.

•

Example

Building on the example considered in Section 19.3, namely a probability space for modeling the simultaneous throw of a blue and a red die, we consider the sum of the numbers of pips as a first example of a random variable and its distribution. In Section 19.3 we saw that a sensible probability space model for the simultaneous throw of a blue and a red die is given by $(\Omega, \mathcal{A}, \mathbb{P})$ with

- $\Omega := \{(r, b) | r \in \mathbb{N}_6, b \in \mathbb{N}_6\}$,
- $\mathcal{A} := \mathcal{P}(\Omega)$ and
- $\mathbb{P} : \mathcal{A} \to [0, 1]$ with $\mathbb{P}(\{(r, b)\}) = \frac{1}{36}$ for all $(r, b) \in \Omega$.

The evaluation of the sum of the two numbers of pips is then sensibly described by the map

$$\xi : \Omega \to \mathcal{X}, (r, b) \mapsto \xi((r, b)) := r + b, \tag{21.5}$$

where evidently $\mathcal{X} := \{2, 3, ..., 12\}$ must hold. The outcome space of the random variable is therefore again finite, and $\mathcal{S} := \mathcal{P}(\mathcal{X})$ is a sensible σ-algebra on $\mathcal{X}$. With the help of the σ-additivity of $\mathbb{P}$, we can now compute the distribution $\mathbb{P}_\xi$ of ξ for all elementary events $\{x\} \in \mathcal{S}$ and thereby, in particular, also verify the measurability of ξ, as shown in Table 21.1.

Table 21.1 Probability distribution of the sum of pips when throwing two dice

$\mathbb{P}_\xi(\{x\})$	$\mathbb{P}(\xi^{-1}(\{x\}))$	$\mathbb{P}(\{\omega \vert \xi(\omega) \in \{x\}\})$	
$\mathbb{P}_\xi(\{2\})$	$\mathbb{P}\left(\xi^{-1}(\{2\})\right)$	$\mathbb{P}\left(\{(1,1)\}\right)$	$\frac{1}{36}$
$\mathbb{P}_\xi(\{3\})$	$\mathbb{P}\left(\xi^{-1}(\{3\})\right)$	$\mathbb{P}\left(\{(1,2),(2,1)\}\right)$	$\frac{2}{36}$
$\mathbb{P}_\xi(\{4\})$	$\mathbb{P}\left(\xi^{-1}(\{4\})\right)$	$\mathbb{P}\left(\{(1,3),(3,1),(2,2)\}\right)$	$\frac{3}{36}$
$\mathbb{P}_\xi(\{5\})$	$\mathbb{P}\left(\xi^{-1}(\{5\})\right)$	$\mathbb{P}\left(\{(1,4),(4,1),(2,3),(3,2)\}\right)$	$\frac{4}{36}$
$\mathbb{P}_\xi(\{6\})$	$\mathbb{P}\left(\xi^{-1}(\{6\})\right)$	$\mathbb{P}\left(\{(1,5),(5,1),(2,4),(4,2),(3,3)\}\right)$	$\frac{5}{36}$
$\mathbb{P}_\xi(\{7\})$	$\mathbb{P}\left(\xi^{-1}(\{7\})\right)$	$\mathbb{P}\left(\{(1,6),(6,1),(2,5),(5,2),(3,4),(4,3)\}\right)$	$\frac{6}{36}$
$\mathbb{P}_\xi(\{8\})$	$\mathbb{P}\left(\xi^{-1}(\{8\})\right)$	$\mathbb{P}\left(\{(2,6),(6,2),(3,5),(5,3),(4,4)\}\right)$	$\frac{5}{36}$
$\mathbb{P}_\xi(\{9\})$	$\mathbb{P}\left(\xi^{-1}(\{9\})\right)$	$\mathbb{P}\left(\{(3,6),(6,3),(4,5),(5,4)\}\right)$	$\frac{4}{36}$
$\mathbb{P}_\xi(\{10\})$	$\mathbb{P}\left(\xi^{-1}(\{10\})\right)$	$\mathbb{P}\left(\{(4,6),(6,4),(5,5)\}\right)$	$\frac{3}{36}$
$\mathbb{P}_\xi(\{11\})$	$\mathbb{P}\left(\xi^{-1}(\{11\})\right)$	$\mathbb{P}\left(\{(5,6),(6,5)\}\right)$	$\frac{2}{36}$
$\mathbb{P}_\xi(\{12\})$	$\mathbb{P}\left(\xi^{-1}(\{12\})\right)$	$\mathbb{P}\left(\{(6,6)\}\right)$	$\frac{1}{36}$

The probabilities of the elementary events in $\mathcal{S}$ in turn allow us to compute arbitrary event probabilities with respect to the sum of the dice pips (cf. Section 19.2). Overall, based on $(\Omega, \mathcal{A}, \mathbb{P})$ and ξ, we have thus constructed another probability space model $(\mathcal{X}, \mathcal{S}, \mathbb{P}_\xi)$.

The following **R** code demonstrates how the construction of the example considered here can be simulated for one run of a random process using computer-based generation of random outcomes.

```
# probability space model formulation
Omega    = list()                                # outcome space initialization
idx      = 1                                     # outcome index initialization
for(r in 1:6){                                   # outcomes red die
    for(b in 1:6){                               # outcomes blue die
        Omega[[idx]] = c(r,b)                    # \omega \in \Omega
        idx          = idx + 1 }}                # outcome index update
K        = length(Omega)                         # cardinality of \Omega
pi       = rep(1/K,1,K)                          # probability function \pi

# random process run
omega    = Omega[[which(rmultinom(1,1,pi) == 1)]] # selection of \omega according to \mathbb{P}({\omega})

# realization of the random variable
xi_omega = sum(omega)                            # \xi(\omega)
```

```
omega     :  5 5
xi(omega) :  10
```

In the context of measuring randomly selected experimental units, random variables often serve as models of measurement processes. For example, consider the determination of the value of a psychological questionnaire (BDI-II) in randomly selected participants (Figure 21.2). This yields the following interpretation: the outcome space of the underlying probability space (Ω) is meant to represent the set of all eligible participants, and the selection of a participant from this space is the selection of an outcome ω, which is to occur with probability $\mathbb{P}(\{\omega\})$. If a particular property of this participant is then measured in an idealized way, this is a deterministic map to a measurement value $\xi(\omega)$ assigned to this participant in the space of measurement values $\mathcal{X}$. The measurement values themselves then follow a probability distribution that is determined by the underlying distribution and the type of measurement process.

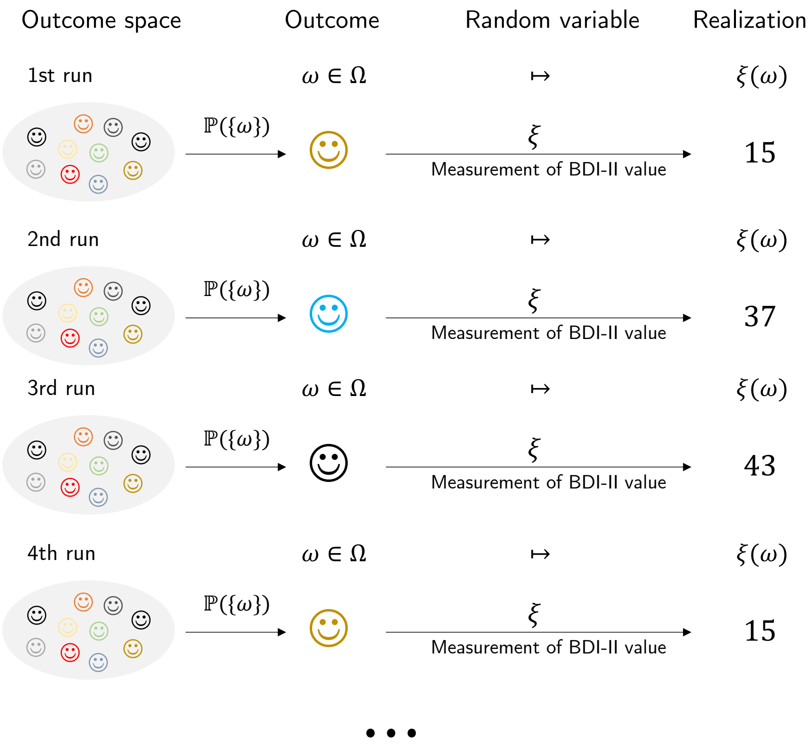

Figure 21.2 Random variable as a model of a measurement process.

Notation

The conventions for denoting probabilities and distributions associated with random variables take some getting used to, so we record them in the following definition.

Definition 21.3 (Notation for random variables)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ and $(\mathcal{X}, \mathcal{S}, \mathbb{P}_\xi)$ be probability spaces and let $\xi : \Omega \to \mathcal{X}$ be a random variable. Then, with $S \in \mathcal{S}$ and $x \in \mathcal{X}$, the following notational conventions hold for events in $\mathcal{A}$:

$$
\begin{aligned}
\{\xi \in S\} &:= \{\omega \in \Omega | \xi(\omega) \in S\} \\
\{\xi = x\} &:= \{\omega \in \Omega | \xi(\omega) = x\} \\
\{\xi \le x\} &:= \{\omega \in \Omega | \xi(\omega) \le x\} \\
\{\xi < x\} &:= \{\omega \in \Omega | \xi(\omega) < x\} \\
\{\xi \ge x\} &:= \{\omega \in \Omega | \xi(\omega) \ge x\} \\
\{\xi > x\} &:= \{\omega \in \Omega | \xi(\omega) > x\}
\end{aligned}
$$

From these conventions follow the following conventions for probabilities of distributions:

$$
\begin{aligned}
\mathbb{P}_{\xi}(\xi \in S) &= \mathbb{P}(\{\xi \in S\}) = \mathbb{P}(\{\omega \in \Omega | \xi(\omega) \in S\}) \\
\mathbb{P}_{\xi}(\xi = x) &= \mathbb{P}(\{\xi = x\}) = \mathbb{P}(\{\omega \in \Omega | \xi(\omega) = x\}) \\
\mathbb{P}_{\xi}(\xi \le x) &= \mathbb{P}(\{\xi \le x\}) = \mathbb{P}(\{\omega \in \Omega | \xi(\omega) \le x\}) \\
\mathbb{P}_{\xi}(\xi < x) &= \mathbb{P}(\{\xi < x\}) = \mathbb{P}(\{\omega \in \Omega | \xi(\omega) < x\}) \\
\mathbb{P}_{\xi}(\xi \ge x) &= \mathbb{P}(\{\xi \ge x\}) = \mathbb{P}(\{\omega \in \Omega | \xi(\omega) \ge x\}) \\
\mathbb{P}_{\xi}(\xi > x) &= \mathbb{P}(\{\xi > x\}) = \mathbb{P}(\{\omega \in \Omega | \xi(\omega) > x\}).
\end{aligned}
$$

Often the subscript on distribution symbols is also omitted, and, for example, $\mathbb{P}_{\xi}(\xi \in S)$ is written simply as $\mathbb{P}(\xi \in S)$ as long as the context does not allow the two probability measures to be confused.

•

The omission of explicit function notation for functions determined uniquely by random variables, as sketched in 1 for $\mathbb{P}_{\xi}$ and $\mathbb{P}$, is a fundamental phenomenon of modern, application-oriented probability theory. In particular, functions of random variables are in most cases identified through their arguments. Thus, from "$\mathbb{P}(\xi \in S)$", an attentive reader should understand that $\mathbb{P}$ refers to the image measure of the random variable ξ, whereas in "$\mathbb{P}(\{\omega \in \Omega | \xi(\omega) \in S\})$" it refers to the probability measure on the measurable space $(\Omega, \mathcal{A})$. In computer-science applications, the identification of functions by means of their arguments, which also appears in the *probability mass functions*, *probability density functions*, and *cumulative distribution functions* discussed in the following sections, is also called *multiple dispatch*.

To conclude this section, we record in Theorem 21.1 that arithmetic operations with real-valued random variables again lead to real-valued random variables.

Theorem 21.1 (Arithmetic of real-valued random variables)**.** *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be the real measurable space, let $\xi_1 : \Omega \to \mathbb{R}$ and $\xi_2 : \Omega \to \mathbb{R}$ be real-valued random variables, and let $c \in \mathbb{R}$ be a constant. Furthermore, let*

$$
\begin{aligned}
\xi_1 + c : \Omega \to \mathbb{R}, \omega \mapsto (\xi_1 + c)(\omega) &:= \xi_1(\omega) + c \text{ for } c \in \mathbb{R} \\
c\xi_1 : \Omega \to \mathbb{R}, \omega \mapsto (c\xi_1)(\omega) &:= c\xi_1(\omega) \text{ for } c \in \mathbb{R} \\
\xi_1 + \xi_2 : \Omega \to \mathbb{R}, \omega \mapsto (\xi_1 + \xi_2)(\omega) &:= \xi_1(\omega) + \xi_2(\omega) \\
\xi_1\xi_2 : \Omega \to \mathbb{R}, \omega \mapsto (\xi_1\xi_2)(\omega) &:= \xi_1(\omega)\xi_2(\omega)
\end{aligned}
\tag{21.6}
$$

be the addition of a constant to a real-valued random variable, the multiplication of a real-valued random variable by a constant, the addition of two real-valued random variables, and the multiplication of two real-valued random variables, respectively. Then $\xi_1 + c$, $c\xi_1$, $\xi_1 + \xi_2$, and $\xi_1\xi_2$ are also real-valued random variables.

∘

For a proof of this theorem, we refer to the advanced literature, for example Hesse (2009). Intuitively, Theorem 21.1 says that adding a random quantity to a constant quantity, multiplying a random quantity by a constant, adding two random quantities, and multiplying two random quantities always again produces random quantities with their own distributions.

21.2 Probability mass functions

In this section, with *probability mass functions (PMFs)*, we introduce a tool for defining distributions of random variables with a *discrete* (more precisely *finite* or *countable*) outcome space. We illustrate the concept using three important examples: Bernoulli and binomial random variables as well as discrete uniform random variables. We first define the concept of a PMF as follows.

Definition 21.4 (Discrete random variable and probability mass function)**.** A random variable ξ is called *discrete* if its outcome space $\mathcal{X}$ is finite or countable and if a function of the form

$$p : \mathcal{X} \to \mathbb{R}_{\geq 0}, x \mapsto p(x) \tag{21.7}$$

exists such that

(1) $\sum_{x \in \mathcal{X}} p(x) = 1$
(2) $\mathbb{P}(\xi = x) = p(x)$ for all $x \in \mathcal{X}$.

Such a function p is called the *probability mass function (PMF)* of ξ.

•

Recall that a set is called *countable* if it can be mapped bijectively to $\mathbb{N}$. In German, PMFs are also called *counting densities*. In English, they are called *probability mass functions (PMFs)*; we follow this terminology here. Without proof, we record that every function $p : \mathcal{X} \to \mathbb{R}_{\geq 0}$ with the normalization property $\sum_{x \in \mathcal{X}} p(x) = 1$ can be interpreted as the PMF of a random variable.

Examples

With *Bernoulli random variables*, *binomial random variables*, and *discrete uniform random variables*, we consider three first examples of defining distributions by means of PMFs.

Definition 21.5 (Bernoulli random variable)**.** Let ξ be a random variable with outcome space $\mathcal{X} = \{0, 1\}$ and PMF

$$p : \mathcal{X} \to [0, 1], x \mapsto p(x) := \mu^x (1 - \mu)^{1-x} \text{ with } \mu \in [0, 1]. \tag{21.8}$$

Then we say that ξ follows a *Bernoulli distribution with parameter* $\mu \in [0, 1]$ and call ξ a *Bernoulli random variable*. We abbreviate this as $\xi \sim \text{Bern}(\mu)$. We denote the PMF of a Bernoulli random variable by

$$\text{Bern}(x; \mu) := \mu^x (1 - \mu)^{1-x}. \tag{21.9}$$

•

Bernoulli random variables can be used for probabilistic modeling whenever the phenomenon under consideration is binary and the possible values of the random variable can be mapped bijectively to $\{0,1\}$. Note that the functional form of the Bernoulli random variable $\text{Bern}(x;\mu)$ makes sense only for $x \in \{0,1\}$ and not, for example, for $x \in \{\text{H},\text{T}\}$. However, if the possible outcomes of a coin toss are mapped to $\{0,1\}$, for example by defining $0 := \text{H}$ and $1 := \text{T}$, then a Bernoulli random variable can certainly serve as a model of a coin toss.

The parameter $\mu \in [0,1]$ of a Bernoulli random variable is the probability that the random variable ξ assumes the value 1; this can be seen from

$$\mathbb{P}(\xi = 1) = p(1) = \mu^1(1-\mu)^{1-1} = \mu. \tag{21.10}$$

We visualize the PMFs of Bernoulli random variables for $\mu := 0.1, \mu := 0.5$, and $\mu := 0.7$ in Figure 21.3.

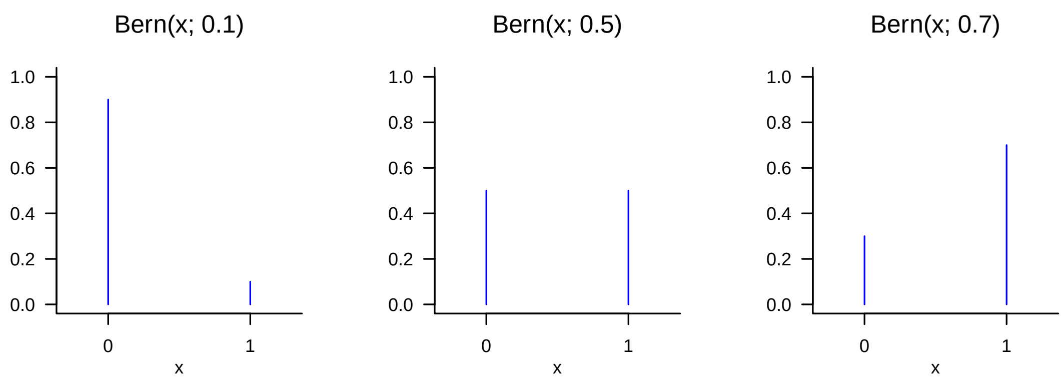

Figure 21.3 PMFs of Bernoulli random variables.

Definition 21.6 (Binomial random variable)**.** Let ξ be a random variable with outcome space $\mathcal{X} := \mathbb{N}_n^0$ and PMF

$$p : \mathcal{X} \to [0,1], x \mapsto p(x) := \binom{n}{x} \mu^x (1-\mu)^{n-x} \text{ for } \mu \in [0,1]. \tag{21.11}$$

Then we say that ξ follows a *binomial distribution with parameters* $n \in \mathbb{N}$ *and* $\mu \in [0,1]$ and call ξ a *binomial random variable*. We abbreviate this as $\xi \sim \text{Bin}(n,\mu)$. We denote the PMF of a binomial random variable by

$$\text{Bin}(x;n,\mu) := \binom{n}{x} \mu^x (1-\mu)^{n-x}. \tag{21.12}$$

•

Without proof, we record that a binomial random variable can be used as a model of the sum of n independent and identically distributed Bernoulli random variables. In particular, $\text{Bin}(x;1,\mu) = \text{Bern}(x;\mu)$ holds. Binomial random variables have the property that with n, one of their parameters determines not only the functional form of their PMF, but also their outcome space $\mathcal{X}$. We visualize the PMFs of binomial random variables for $(n,\mu) := (5,0.1), (n,\mu) := (10,0.5)$, and $(n,\mu) := (15,0.7)$ in Figure 21.4.

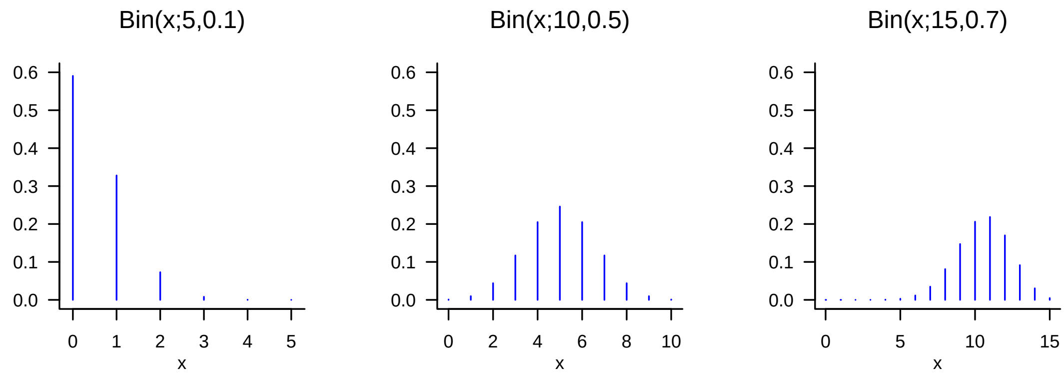

Figure 21.4 PMFs of binomial random variables.

Definition 21.7 (Discrete uniform random variable)**.** Let ξ be a discrete random variable with finite outcome space $\mathcal{X}$ and PMF

$$p : \mathcal{X} \to \mathbb{R}_{\geq 0}, x \mapsto p(x) := \frac{1}{|\mathcal{X}|}. \tag{21.13}$$

Then we say that ξ follows a *discrete uniform distribution* and call ξ a *discrete uniform random variable*. We abbreviate this as $\xi \sim U(|\mathcal{X}|)$. We denote the PMF of a discrete uniform random variable by

$$U(x; |\mathcal{X}|) := \frac{1}{|\mathcal{X}|}. \tag{21.14}$$

•

Discrete uniform random variables can evidently be used for probabilistic modeling whenever the possible discrete outcomes of the modeled phenomenon have the same probability. In the case of discrete uniform random variables, no additional parameter is needed to define their functional form once the outcome space has been specified. Evidently, for $\mathcal{X} := \{0, 1\}$,

$$U(x; |\mathcal{X}|) = \text{Bern}(x; 0.5) = \text{Bin}(x; 1, 0.5). \tag{21.15}$$

We visualize the PMFs of discrete uniform random variables for $\mathcal{X} := \{0, 1\}$, $\mathcal{X} := \{-3, -2, -1, 0, 1\}$, and $\mathcal{X} := \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ in Figure 21.5. Note that Definition 21.7 does not formally require the elements of $\mathcal{X}$ to be sequential, so one can just as well define a discrete uniform random variable with outcome space $\mathcal{X} := \{1, 5, 7\}$ or even with a non-numeric outcome space $\mathcal{X} := \{a, b, x, y\}$.

21.3 Probability density functions

In this section, with *probability density functions (PDFs)*, we introduce a tool for defining distributions of random variables with a *continuous* (more precisely *uncountable*) outcome space. We first illustrate the concept using the most fundamental of all random variables, the normally distributed random variable. With the gamma random variable, the beta

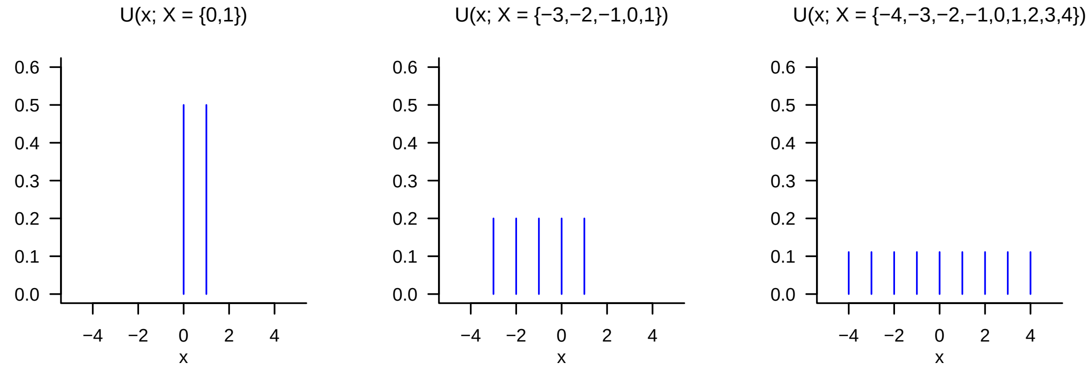

Figure 21.5 PMFs of discrete uniform random variables.

random variable, and the uniform random variable, we then consider three further examples of random variables that are used in many places in the model formulation of both frequentist and Bayesian inference. We define the concept of a PDF as follows.

Definition 21.8 (Continuous random variable and probability density function)**.** A random variable ξ is called *continuous* if $\mathbb{R}$ is the outcome space of ξ and if a function

$$p : \mathbb{R} \to \mathbb{R}_{\geq 0}, x \mapsto p(x) \tag{21.16}$$

exists such that

(1) $\int_{-\infty}^{\infty} p(x)\,dx = 1$ and
(2) $\mathbb{P}(\xi \in [a, b]) = \int_a^b p(x)\,dx$ for all $a, b \in \mathbb{R}$ with $a \leq b$.

Such a function p is called the *probability density function (PDF)* of ξ.

•

Without proof, we record that every function $p : \mathbb{R} \to \mathbb{R}_{\geq 0}$ whose improper integral satisfies $\int_{-\infty}^{\infty} p(x)\,dx = 1$, that is, every normalized such function, can be interpreted as the PDF of a random variable.

When working with PDFs, and in contrast to PMFs, one should always keep the *density property* of a PDF in mind: the values of a PDF are not probabilities but probability densities; probabilities are computed from PDFs by integration according to Definition 21.8 (2). As in the physical sense, the probability mass assigned to a real interval therefore arises only through "multiplication" by the corresponding "interval volume". Here one may also think of the approximation of the definite integral $\int_a^b p(x)\,dx$ by a Riemann sum term (cf. Definition 7.3). Intuitively, we therefore have

$$\text{(probability) mass} = \text{(probability) density} \cdot \text{(set) volume}, \tag{21.17}$$

where volume in the sense of Lebesgue measure refers to the width of the interval $[a, b]$. As in the physical analogy, the probability mass of an interval with no volume is zero,

$$\mathbb{P}(\xi = a) = \int_a^a p(x)\,dx = 0. \tag{21.18}$$

Furthermore, for sufficiently small intervals, PDFs can also assume values greater than 1, even though this cannot be the case for the probability that results from the corresponding integration. Finally, despite these technical details, the following intuition should be emphasized: if one considers the graphical representation of the PDF of a random variable and imagines a partition of $\mathbb{R}$ into equally sized intervals, then the random variable naturally has a higher probability of assuming values in an interval with a higher associated probability density than values in an interval with a relatively lower associated probability density.

Normally distributed random variables

As a first example of defining a continuous random variable by means of a PDF, we now introduce the most important random variable of probabilistic modeling: the normally distributed random variable.

Definition 21.9 (Normally distributed random variable)**.** Let ξ be a random variable with outcome space $\mathbb{R}$ and PDF

$$p : \mathbb{R} \to \mathbb{R}_{>0}, x \mapsto p(x) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{21.19}$$

Then we say that ξ follows a *normal distribution with parameters* $\mu \in \mathbb{R}$ *and* $\sigma^2 > 0$ and call ξ a *normally distributed random variable*. We abbreviate this as $\xi \sim N(\mu, \sigma^2)$. We denote the PDF of a normally distributed random variable by

$$N(x; \mu, \sigma^2) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{21.20}$$

A normally distributed random variable with $\mu = 0$ and $\sigma^2 = 1$ is called a *standard normally distributed random variable.*

•

We visualize the PDFs of normally distributed random variables for $(\mu, \sigma^2) := (0, 1)$, $(\mu, \sigma^2) := (-2.5, 10)$, and $(\mu, \sigma^2) := (3, 0.5)$ in Figure 21.6. This figure makes graphically clear that the PDFs of normally distributed random variables always have exactly one value of highest probability density, namely at the location of the parameter $\mu \in \mathbb{R}$. This follows from the fact that, because of the negative sign of the square of $x - \mu$ and the positivity of σ^2, the argument of the exponential function in the functional form of $N(x; \mu, \sigma^2)$ is always non-positive, and the exponential function on the non-positive real numbers assumes its maximum at $x = \mu$, that is, $x - \mu = 0$. The figure also makes graphically clear that the parameter $\sigma^2 > 0$ encodes the width of the PDF of a normally distributed random variable.

Further examples

With *gamma random variables*, *beta random variables*, and *uniform random variables*, we consider three further examples of defining distributions by means of PDFs.

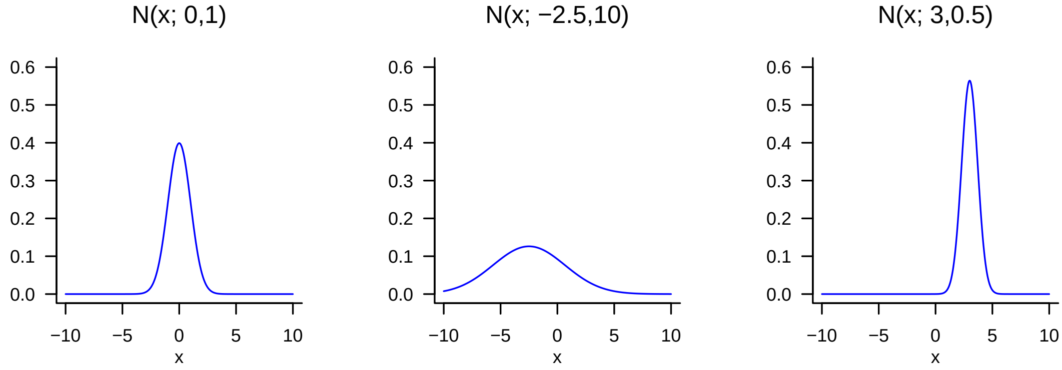

Figure 21.6 PDFs of normally distributed random variables.

Definition 21.10 (Gamma random variable)**.** Let ξ be a random variable with outcome space $\mathcal{X} := \mathbb{R}_{>0}$ and PDF

$$p : \mathbb{R}_{>0} \to \mathbb{R}_{>0}, x \mapsto p(x) := \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp\left(-\frac{x}{\beta}\right), \tag{21.21}$$

where Γ denotes the gamma function. Then we say that ξ follows a *gamma distribution with shape parameter $\alpha > 0$ and scale parameter $\beta > 0$* and call ξ a *gamma-distributed random variable*. We abbreviate this as $\xi \sim G(\alpha, \beta)$. We denote the PDF of a gamma-distributed random variable by

$$G(x; \alpha, \beta) := \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp\left(-\frac{x}{\beta}\right). \tag{21.22}$$

•

The special gamma random variable with PDF $G\left(x; \frac{n}{2}, 2\right)$ is called the *chi-square (χ^2) distribution with n degrees of freedom*. We visualize the PDFs of gamma random variables for $(\alpha, \beta) := (1, 1)$, $(\alpha, \beta) := (2, 2)$, and $(\alpha, \beta) := (5, 1)$ in Figure 21.7.

Definition 21.11 (Beta random variable)**.** Let ξ be a random variable with outcome space $\mathcal{X} := [0, 1]$ and PDF

$$p : \mathcal{X} \to [0, 1], x \mapsto p(x) := \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1 - x)^{\beta-1} \text{ with } \alpha, \beta \in \mathbb{R}_{>0}, \tag{21.23}$$

where Γ denotes the gamma function. Then we say that ξ follows a *beta distribution* with parameters $\alpha > 0$ and $\beta > 0$, and call ξ a *beta random variable*. We abbreviate this as $\xi \sim \text{Beta}(\alpha, \beta)$. We denote the PDF of a beta random variable by

$$\text{Beta}(x; \alpha, \beta) := \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1 - x)^{\beta-1}. \tag{21.24}$$

•

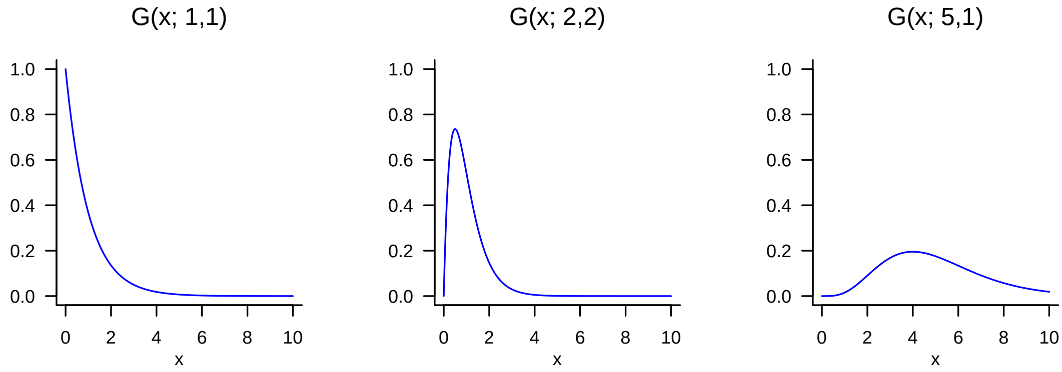

Figure 21.7 PDFs of gamma random variables.

Because the outcome space of a beta random variable is restricted to the interval $\mathcal{X} := [0, 1]$ (more precisely $\mathcal{X} :=]0, 1[$ for $\alpha < 1, \beta < 1$), a beta random variable is useful, among other things, for describing probabilities of probabilities, that is, values between 0 and 1. We visualize the PDFs of beta random variables for $(\alpha, \beta) := (1, 1)$, $(\alpha, \beta) := (3, 2)$, and $(\alpha, \beta) := (10, 5)$ in Figure 21.8.

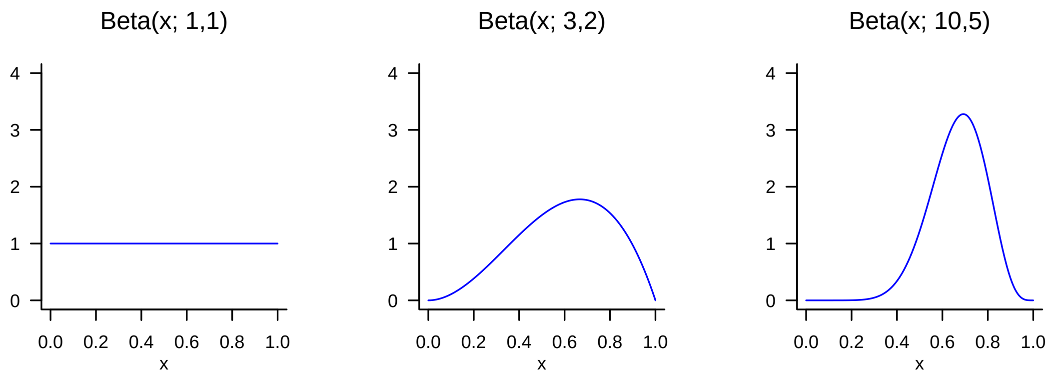

Figure 21.8 PDFs of beta random variables.

With uniform random variables, we finally consider the analogue of discrete uniform random variables for the case of continuous random variables.

Definition 21.12 (Uniform random variable)**.** Let ξ be a continuous random variable with outcome space $\mathbb{R}$ and PDF

$$p : \mathbb{R} \to \mathbb{R}_{\geq 0}, x \mapsto p(x) := \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & x \notin [a, b] \end{cases}. \tag{21.25}$$

Then we say that ξ follows a *uniform distribution with parameters a and b* and call ξ a *uniform random variable*. We abbreviate this as $\xi \sim U(a, b)$. We denote the PDF of a uniform random variable by

$$U(x; a, b) := \frac{1}{b-a}. \tag{21.26}$$

•

We visualize the PDFs of uniform random variables for $(a, b) := (0, 1)$, $(a, b) := (-3, 1)$, and $(a, b) := (-4, 4)$ in Figure 21.9.

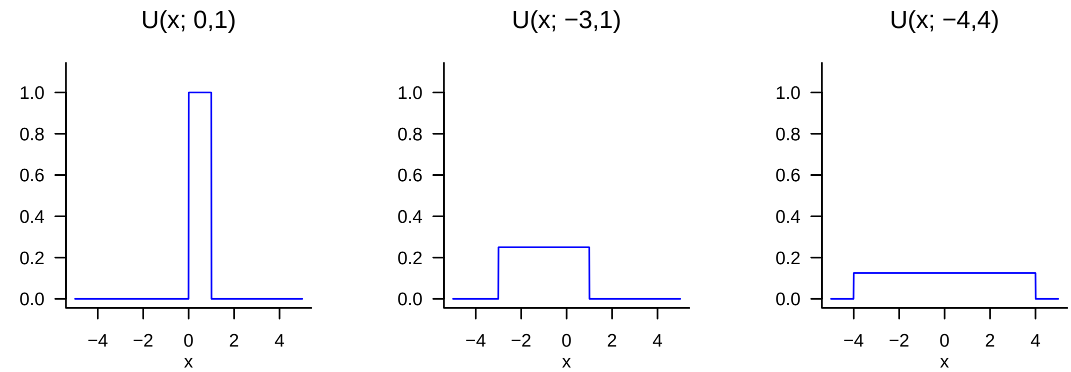

Figure 21.9 PDFs of uniform random variables.

21.4 Cumulative distribution functions

In the final section of this chapter, with *cumulative distribution functions (CDFs)*, we introduce another way of specifying the distributions of discrete or continuous random variables. In general, however, it is more common to do this with PMFs or PDFs. Nevertheless, CDFs are useful in many places because, for both discrete and continuous random variables, they establish a direct connection between values of the random variable and certain probabilities that does not require summation or integration in application. We first consider the general definition of CDFs for both discrete and continuous random variables and then turn to the CDFs of discrete and continuous random variables in detail. For an arbitrary random variable, we define the concept of a CDF as follows.

Definition 21.13 (Cumulative distribution function)**.** The *cumulative distribution function (CDF)* of a random variable ξ is defined as

$$P : \mathbb{R} \to [0, 1], x \mapsto P(x) := \mathbb{P}(\xi \le x). \tag{21.27}$$

•

Note that $P(x)$ is defined for every $x \in \mathbb{R}$, even if for a given $x \in \mathbb{R}$ we have $x \notin \mathcal{X}$. With the help of CDFs, for example, *exceedance probabilities* and *interval probabilities* of random variables can be evaluated directly. This is the content of the following theorems.

Theorem 21.2 (Exceedance probability)**.** *Let ξ be a random variable with outcome space $\mathcal{X}$ and let P be its cumulative distribution function. Then, for the* exceedance probability *$\mathbb{P}(\xi > x)$,*

$$\mathbb{P}(\xi > x) = 1 - P(x) \text{ for all } x \in \mathcal{X}. \tag{21.28}$$

$\circ$

Proof. The events $\{\xi > x\}$ and $\{\xi \le x\}$ are disjoint, and

$$\Omega = \{\omega \in \Omega | \xi(\omega) > x\} \cup \{\omega \in \Omega | \xi(\omega) \le x\} = \{\xi > x\} \cup \{\xi \le x\}. \tag{21.29}$$

With the σ-additivity of $\mathbb{P}$ it follows that

$$\begin{aligned}
\mathbb{P}(\Omega) &= 1 \\
\Leftrightarrow \mathbb{P}(\{\xi > x\} \cup \{\xi \le x\}) &= 1 \\
\Leftrightarrow \mathbb{P}(\{\xi > x\}) + \mathbb{P}(\{\xi \le x\}) &= 1 \\
\Leftrightarrow \mathbb{P}(\{\xi > x\}) &= 1 - \mathbb{P}(\{\xi \le x\}) \\
\Leftrightarrow \mathbb{P}(\{\xi > x\}) &= 1 - P(x).
\end{aligned} \tag{21.30}$$

$\square$

Theorem 21.3 (Interval probabilities)**.** *Let ξ be a random variable with outcome space $\mathcal{X}$ and let P be its CDF. Then, for the* interval probability *$\mathbb{P}(\xi \in]x_1, x_2])$,*

$$\mathbb{P}(\xi \in]x_1, x_2]) = P(x_2) - P(x_1) \text{ for all } x_1, x_2 \in \mathcal{X} \text{ with } x_1 < x_2. \tag{21.31}$$

$\circ$

Proof. We consider the events $\{\xi \le x_1\}$, $\{x_1 < \xi \le x_2\}$, and $\{\xi \le x_2\}$, where

$$\{\xi \le x_1\} \cap \{x_1 < \xi \le x_2\} = \emptyset \text{ and } \{\xi \le x_1\} \cup \{x_1 < \xi \le x_2\} = \{\xi \le x_2\}. \tag{21.32}$$

With the σ-additivity of $\mathbb{P}$ it then follows that

$$\begin{aligned}
\mathbb{P}(\{\xi \le x_1\} \cup \{x_1 < \xi \le x_2\}) &= \mathbb{P}(\{\xi \le x_2\}) \\
\Leftrightarrow \mathbb{P}(\{\xi \le x_1\}) + \mathbb{P}(\{x_1 < \xi \le x_2\}) &= \mathbb{P}(\{\xi \le x_2\}) \\
\Leftrightarrow \mathbb{P}(\{x_1 < \xi \le x_2\}) &= \mathbb{P}(\{\xi \le x_2\}) - \mathbb{P}(\{\xi \le x_1\}) \\
\Leftrightarrow \mathbb{P}(\{x_1 < \xi \le x_2\}) &= P(x_2) - P(x_1) \\
\Leftrightarrow \mathbb{P}(\xi \in]x_1, x_2]) &= P(x_2) - P(x_1).
\end{aligned} \tag{21.33}$$

$\square$

The following theorem states three central properties of CDFs. The third property says that a CDF has no jumps when one approaches limit points from the right. In fact, the properties discussed are precisely the defining properties of CDFs; that is, every function P that satisfies the properties of Theorem 21.4 can be interpreted as the CDF of a random variable. For a proof of this fact, we refer to the advanced literature.

Theorem 21.4 (Properties of cumulative distribution functions)**.** *Let ξ be a random variable and let P be its cumulative distribution function. Then P has the following properties:*

(1) P *is* monotonically increasing, *that is, if $x_1 < x_2$, then $P(x_1) \le P(x_2)$.*
(2) $\lim_{x \to -\infty} P(x) = 0$ *and* $\lim_{x \to \infty} P(x) = 1$.

(3) P is right-continuous*, that is,* $P(x) = \lim_{y \to x, y > x} P(y)$ *for all* $x \in \mathbb{R}$.

∘

Proof. We consider the properties in order.

(1) First, we record that for events $A \subset B$, $\mathbb{P}(A) \le \mathbb{P}(B)$ holds. We then note that for $x_1 < x_2$,

$$\{\xi \le x_1\} = \{\omega \in \Omega | \xi(\omega) \le x_1\} \subset \{\omega \in \Omega | \xi(\omega) \le x_2\} = \{\xi \le x_2\}. \tag{21.34}$$

Thus,

$$\mathbb{P}(\{\xi \le x_1\}) \le \mathbb{P}\{\xi \le x_2\} \Rightarrow P(x_1) \le P(x_2). \tag{21.35}$$

(2) For a proof, we refer to the advanced literature.

(3) We define

$$P\left(x^+\right) = \lim_{y \to x, y > x} P(y). \tag{21.36}$$

Now let $y_1 > y_2 > \cdots$ be such that $\lim_{n\to\infty} y_n = x$. Then

$$\{\xi \le x\} = \cap_{n=1}^{\infty}\{\xi \le y_n\}. \tag{21.37}$$

Thus,

$$P(x) = \mathbb{P}(\{\xi \le x\}) = \mathbb{P}(\cap_{n=1}^{\infty}\{\xi \le y_n\}) = \lim_{n\to\infty} \mathbb{P}(\{\xi \le y_n\}) = P\left(x^+\right), \tag{21.38}$$

where we leave the third equality unjustified.

□

CDFs of discrete random variables

Using Figure 21.10 and Figure 21.11, we want to make the above properties of CDFs of discrete random variables visually clear. One should always keep in mind that the value $P(x)$ of a CDF at the outcome value x of the random variable ξ corresponds to the probability $\mathbb{P}(\xi \le x)$, that is, the probability that the random variable ξ assumes values less than or equal to x. If these probabilities are read off accordingly from the representations of the corresponding PMF, then the functional form of the CDFs follows intuitively in comparison with the corresponding PMFs. Furthermore, the following properties of the CDFs of discrete random variables also become intuitive: if $a < b$ and $\mathbb{P}(a < \xi < b) = 0$, then the CDF of ξ is horizontally constant on $]a, b[$. Furthermore, at every point x with $\mathbb{P}(\xi = x) > 0$, the CDF jumps by the amount $\mathbb{P}(\xi = x)$ and is therefore not left-continuous at that point. In general, the CDF of a discrete random variable with outcome space $\mathbb{N}_0$ is given by

$$P : \mathbb{R} \to [0, 1], x \mapsto P(x) := \sum_{k=0}^{\lfloor x \rfloor} \mathbb{P}(\xi = k), \tag{21.39}$$

where $\lfloor x \rfloor$ denotes the floor function.

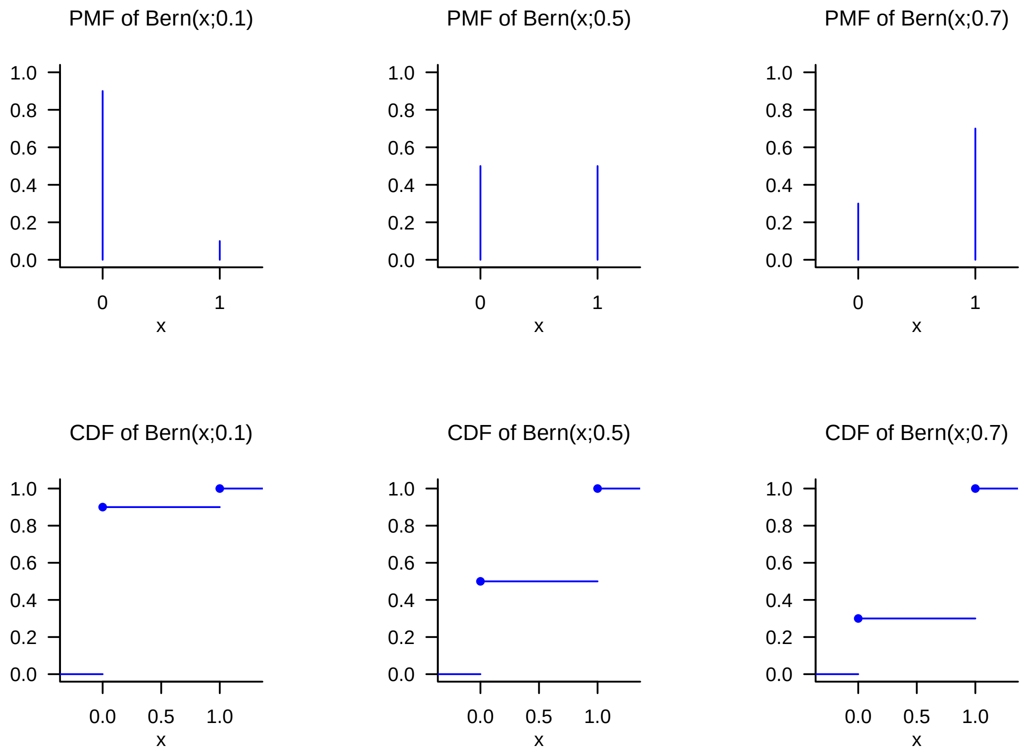

Figure 21.10 PMFs and CDFs of Bernoulli random variables.

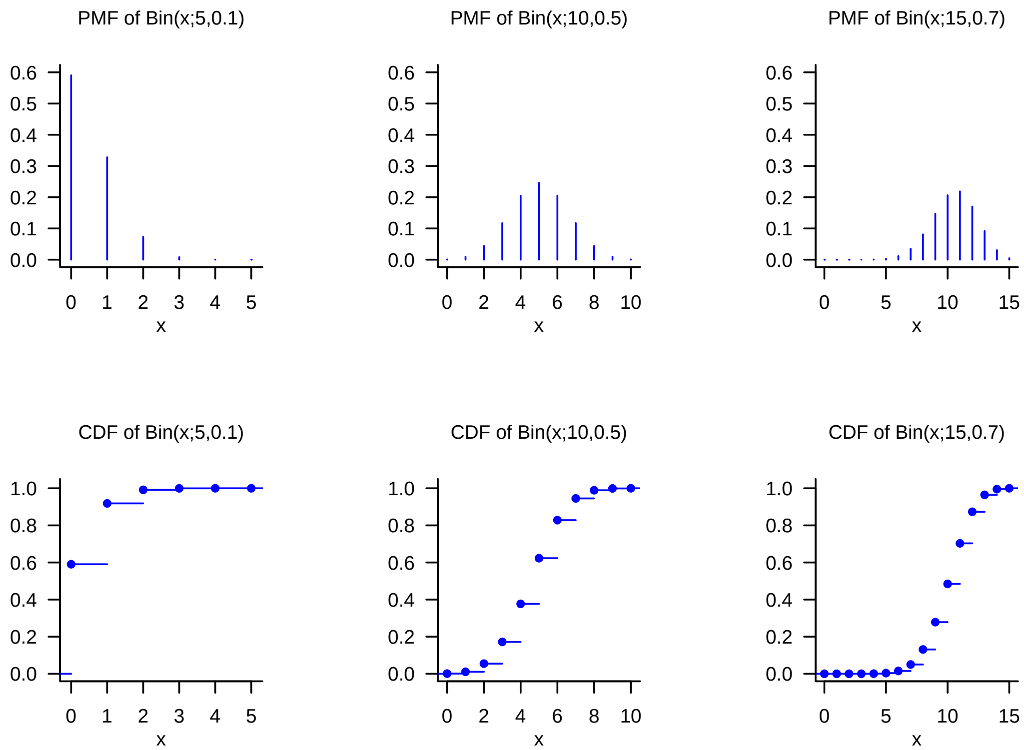

Figure 21.11 PMFs and CDFs of binomial random variables.

CDFs of continuous random variables

The CDFs of continuous random variables are analytically somewhat more accessible than the CDFs of discrete random variables because they have no discontinuities. We first have the following, perhaps somewhat surprising theorem.

Theorem 21.5 (Cumulative distribution functions of continuous random variables)**.** *Let ξ be a continuous random variable with PDF p and CDF P. Then*

$$P(x) = \int_{-\infty}^{x} p(s)\,ds \text{ and } p(x) = P'(x). \tag{21.40}$$

∘

Proof. First, we record that because $\mathbb{P}(\xi = x) = 0$ for all $x \in \mathbb{R}$, the CDF of ξ has no jumps and P is therefore continuous. From the definitions of PDF and CDF, it follows that P has the form of an antiderivative of p. That p is the derivative of P then follows directly from the first fundamental theorem of calculus (Theorem 7.4). □

For continuous random variables, the CDF of the random variable is therefore an antiderivative of the corresponding PDF, and conversely, the PDF is the derivative of the CDF. In the context of the *Radon-Nikodym theorem*, this insight is generalized to general measures (cf. Schmidt (2009)). CDFs of continuous random variables are also often called cumulative density functions.

As an example, consider the CDF of a normally distributed random variable. Let $\xi \sim N(\mu, \sigma^2)$. Then the PDF of ξ is known to be given by

$$p : \mathbb{R} \to \mathbb{R}_{>0}, x \mapsto p(x) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{21.41}$$

For the CDF of ξ, it follows accordingly that

$$P : \mathbb{R} \to]0,1[, x \mapsto P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{x} \exp\left(-\frac{1}{2\sigma^2}(s-\mu)^2\right) ds. \tag{21.42}$$

Interestingly, the defining integral of the CDF of a normally distributed random variable can be computed only numerically, not analytically. We visualize selected PDFs and CDFs of normally distributed random variables in Figure 21.12.

Inverse cumulative distribution function

We conclude this section with the concept of the *inverse cumulative distribution function*, a technical tool that is used especially in confidence intervals and hypothesis tests of frequentist inference to determine critical values. We define the inverse CDF as follows.

Definition 21.14 (Inverse cumulative distribution function)**.** Let ξ be a continuous random variable with CDF P. Then the function

$$P^{-1} :]0,1[\to \mathbb{R}, q \mapsto P^{-1}(q) := \{x \in \mathbb{R} | P(x) = q\} \tag{21.43}$$

is called the *inverse cumulative distribution function of* ξ.

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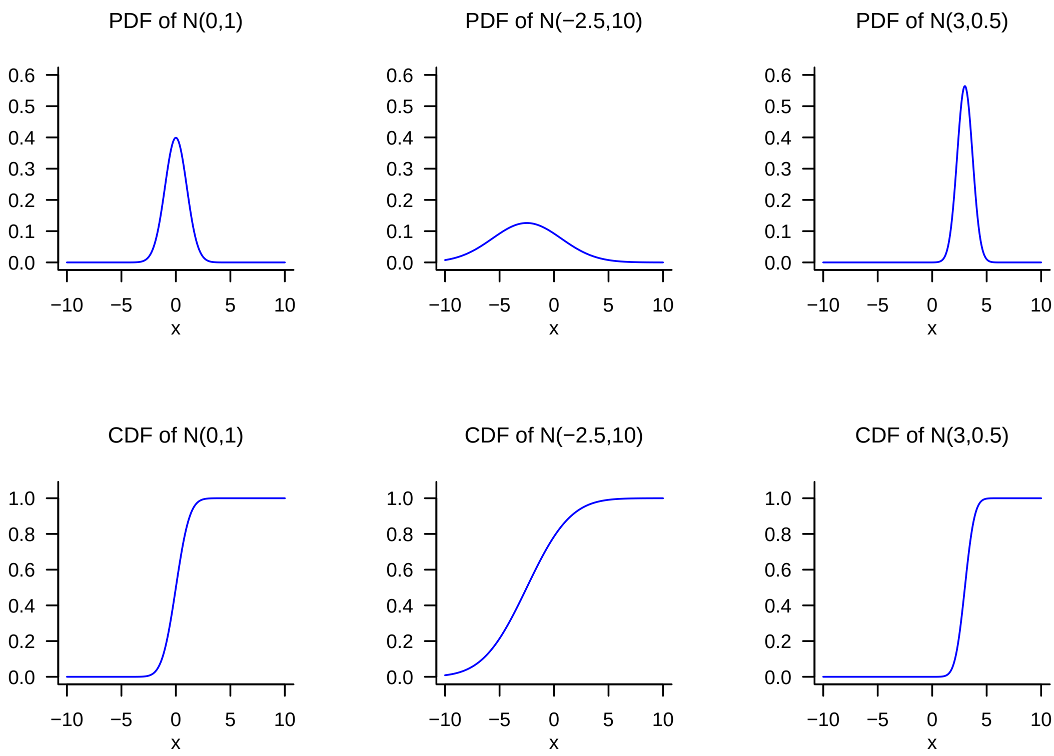


Figure 21.12 PDFs and CDFs of normally distributed random variables.

According to Definition 21.14, the function P^{-1} is evidently the inverse function of P, so that

$$P^{-1}(P(x)) = x. \tag{21.44}$$

Since it is well known that

$$P(x) = q \Leftrightarrow \mathbb{P}(\xi \le x) = q \text{ for } q \in]0,1[, \tag{21.45}$$

$P^{-1}(q)$ is therefore the value x of ξ such that $\mathbb{P}(\xi \le x) = q$. We visualize the CDFs and inverse CDFs of normally distributed random variables in Figure 21.13. In the case of a normally distributed random variable $\xi \sim N(0,1)$, for example,

$$P(1.645) = 0.950 \Leftrightarrow P^{-1}(0.950) = 1.645, \tag{21.46}$$

and

$$P(1.906) = 0.975 \Leftrightarrow P^{-1}(0.975) = 1.960. \tag{21.47}$$

21.5 Random outcomes and random variables

With the *outcomes* ω discussed in Chapter 19 and the values of *random variables* ξ discussed in this chapter, we have now encountered two concepts that can describe uncertainty about a usually numerical value of a random process. For example, the number of pips that a die shows in a single throw can be conceived as the realization of an outcome or

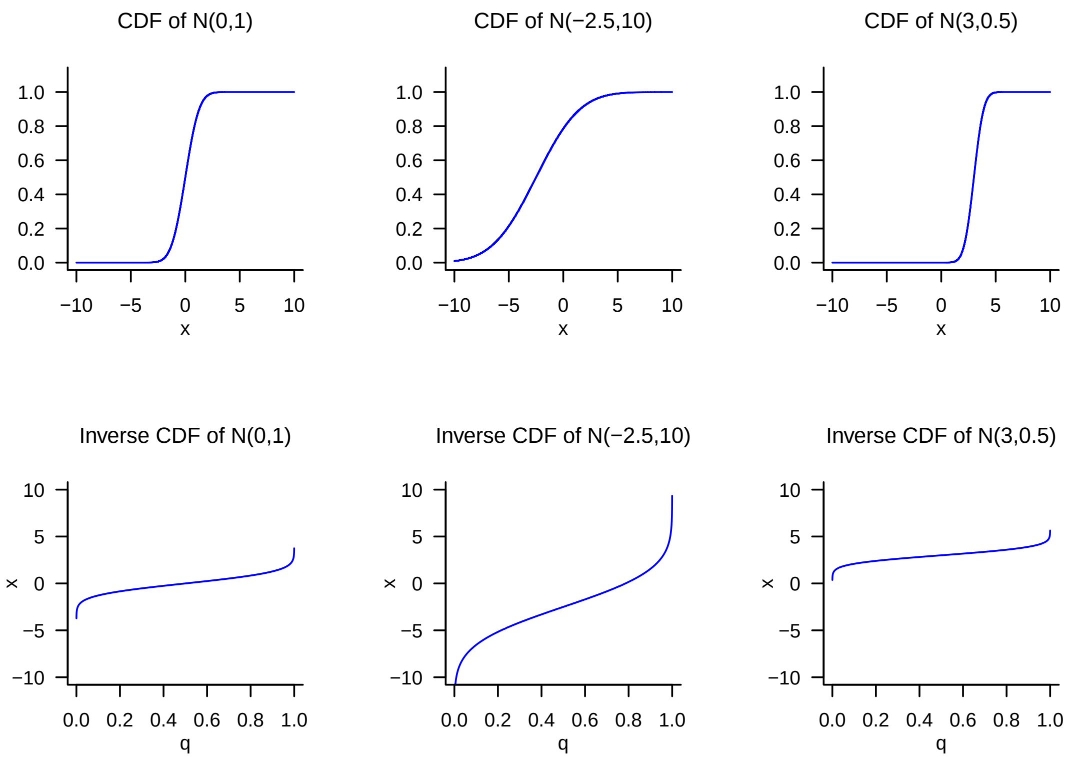


Figure 21.13 CDFs and inverse CDFs of normally distributed random variables.

of a random variable. In fact, there is no standardized answer to the question of whether one should model a random process only with a probability space or with the underlying probability space of a random variable and the probability space induced by both. In applications, the concept of the random variable is usually preferred and corresponding PMFs or PDFs are specified without specifying an underlying probability space or the mapping form of the random variable. The following formulation is typical, for example:

ξ is a normally distributed random variable with expectation parameter μ and variance parameter σ^2.

Implicitly, based on the definition of the normally distributed random variable (cf. Definition 21.9), this statement specifies for ξ the outcome space $\mathcal{X} := \mathbb{R}$, the σ-algebra $\mathcal{S} := \mathcal{B}(\mathbb{R})$, and the distribution $\mathbb{P}_\xi := N(\mu, \sigma^2)$, that is, it considers the "induced" probability space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), N(\mu, \sigma^2))$. However, it remains unclear by which probability space and which exact mapping form $(\mathbb{R}, \mathcal{B}(\mathbb{R}), N(\mu, \sigma^2))$ was induced. This can always be supplemented by assuming that ξ is the identity function. Concretely, one could choose as the underlying probability space here $(\mathbb{R}, \mathcal{B}(\mathbb{R}), N(\mu, \sigma^2))$ and choose ξ as

$$\xi : \mathbb{R} \to \mathbb{R}, \omega \mapsto \xi(\omega) := \omega := x. \tag{21.48}$$

Since ξ then changes nothing about the outcome ω realized randomly in the context of the underlying probability space and merely renames it as x, the distribution of ξ here directly corresponds to the probability measure of the underlying probability space.

In general, one may note that modeling random processes by means of random variables therefore contains the elementary probability space model as a special case, but beyond this it opens up the possibility of formalizing and studying the transformation of probability measures on different measurable spaces by means of random variables that differ from the identity map.

21.6 Bibliographic remarks

The genesis of the concept of the random variable is closely interwoven with the development of probability theory over the last three centuries, so no publication decisive for the concept can be named. The mathematical development of the concept of the normal distribution by Abraham De Moivre (1667-1754), Pierre Simon Laplace (1749-1827), Johann Carl Friedrich Gauss (1777-1855), and many others, its descriptive-statistical counterparts in empirical research of the nineteenth century, and its multivariate generalization in the late nineteenth century are presented in detail in Stigler (1986).

Study questions

1. State the definition of the concept of a random variable.
2. Explain the equation $\mathbb{P}_\xi(\xi = x) = \mathbb{P}(\{\xi = x\})$.
3. Explain the intuitive meaning of $\mathbb{P}(\xi = x)$.
4. State the definition of the concept of a probability mass function.
5. State the definition of the concept of a probability density function.
6. State the definition of the PDF of a normally distributed random variable.
7. State the definition of the concept of a cumulative distribution function.
8. Write the exceedance probability $\mathbb{P}(\xi > x)$ of a random variable using its CDF.
9. Write the interval probability $\mathbb{P}(\xi \in]x_1, x_2]$ of a random variable using its CDF.
10. Write the value $P(x)$ of the CDF of a continuous random variable using its PDF p.
11. Write the value $p(x)$ of the PDF of a continuous random variable using its CDF P.
12. State the definition of the concept of the inverse distribution function.

Study question answers

1. See Definition 21.1.
2. The equation says that the probability value assigned by the distribution $\mathbb{P}_\xi$ of the random variable ξ to the event $\xi = x$ in $\mathcal{X}$ is, by definition, equal to the value assigned by the probability measure $\mathbb{P}$ to the preimage of the event $\xi = x$, that is, to the set $\{\xi = x\} = \{\omega \in \Omega | \xi(\omega) = x\}$ in $\mathcal{A}$. See

1.

1. Intuitively, $\mathbb{P}(\xi = x)$ is the probability that the random variable ξ assumes the value x.
2. See Definition 21.4.
3. See Definition 21.8.
4. See Definition 21.9.
5. See Definition 21.13.
6. See Theorem 21.2.
7. See Theorem 21.3.
8. See Theorem 21.5. We have
$$P(x) = \int_{-\infty}^{x} p(s)\, ds. \tag{21.49}$$
9. See Theorem 21.5. We have
$$p(x) = \frac{d}{dx} P(x) = P'(x). \tag{21.50}$$
10. See Definition 21.14.

22 Random vectors

Random vectors are tuples of random variables. Since every random variable is subject to a probability distribution, a random vector is also subject to a probability distribution. Since a random vector consists of two or more random variables, the distribution of a random vector describes the *joint distribution* of two or more random variables. In this chapter, we first introduce the concept of a random vector and its associated probability distribution, which is often simply called a *multivariate distribution*. We then discuss how the concepts of PMFs, PDFs, and CDFs of random variables can be transferred to random vectors (Section 22.1). With *marginal distributions* and *conditional distributions*, we then introduce concepts that do not arise in the study of random variables. Finally, with the concept of *independent random variables*, we introduce the probabilistic standard model for univariate datasets. This is a special case of the joint distribution of the random variables of a random vector.

22.1 Definition and multivariate distributions

The construction and definition of a random vector is analogous to that of a random variable, with the difference that a random variable is a scalar-valued map, whereas a random vector is a vector-valued map on the outcome space of a probability space.

Definition 22.1 (Random vector)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $(\mathcal{X}, \mathcal{S})$ be an n-dimensional measurable space. An n-dimensional *random vector* is defined as a map

$$\xi : \Omega \to \mathcal{X}, \omega \mapsto \xi(\omega) := \begin{pmatrix} \xi_1(\omega) \\ \vdots \\ \xi_n(\omega) \end{pmatrix} \tag{22.1}$$

with the *measurability property*

$$\{\omega \in \Omega | \xi(\omega) \in S\} \in \mathcal{A} \text{ for all } S \in \mathcal{S}. \tag{22.2}$$

•

The standard example for the outcome space of a random vector is $\mathbb{R}^n$, and the standard example for the σ-algebra defined on it is the n-dimensional Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$. For an explicit and formal introduction to the n-dimensional Borel σ-algebra, we refer to the advanced literature (e.g. Schmidt (2009)). Here we again content ourselves with the still formally incorrect intuition of the n-dimensional Borel σ-algebra as the set of all subsets of $\mathbb{R}^n$. The standard example of an n-dimensional measurable space is therefore $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$.

As with all vector-valued functions, we call the functions ξ_i that constitute the random vector the *component functions* of ξ. If we take the n-dimensional measurable space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ as the basis of the random vector, then these have the form

$$\xi_i : \Omega \to \mathbb{R}, \omega \mapsto \xi_i(\omega). \tag{22.3}$$

Without proof, we record that the random vector ξ is measurable if, for all $i = 1, ..., n$, the functions ξ_i are measurable, and conversely. Thus, the component functions of a random vector are ultimately, by definition, random variables. An n-dimensional random vector is therefore considered a concatenation of n random variables; for $n := 1$, a random vector corresponds to a random variable.

Random vectors are sometimes also called "multivariate random variables". In fact, when studying random vectors, probability-theoretic aspects are initially primary, rather than aspects of geometric vector space theory. The study of vector space structures is, however, quite common in the context of probabilistic standard models such as the general linear model, so here we prefer the term random vector (cf. Christensen (2011)). Nevertheless, as is common in many texts on probability theory, we will often also write random vectors in row form, for example as $\xi := (\xi_1, ..., \xi_n)$.

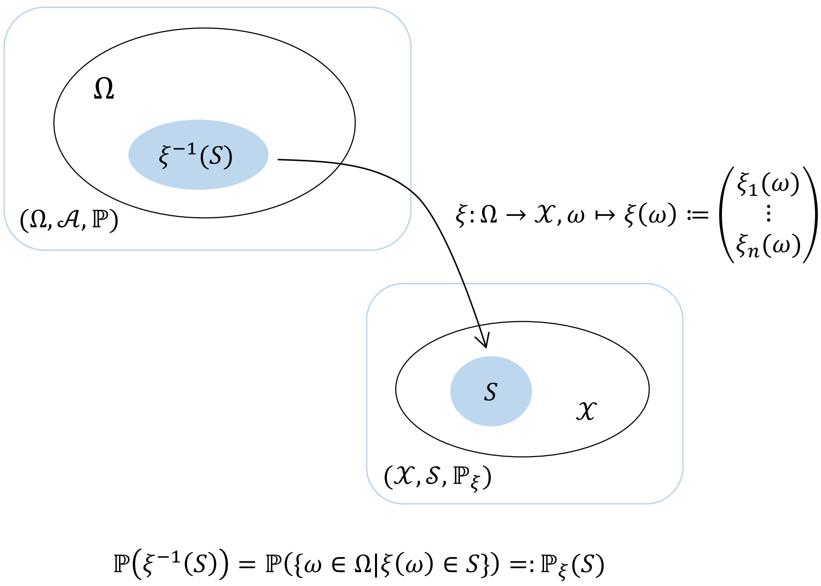


Figure 22.1 Construction of a random vector and a multivariate distribution.

The image measure defined by the construction of a random vector is called the *multivariate distribution of the random vector*, as set out in the following definition (cf. Figure 22.1).

Definition 22.2 (Multivariate distribution)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let $(\mathcal{X}, \mathcal{S})$ be an n-dimensional measurable space, and let

$$\xi : \Omega \to \mathcal{X}, \omega \mapsto \xi(\omega) \tag{22.4}$$

be a random vector. Then the probability measure $\mathbb{P}_\xi$, defined by

$$\mathbb{P}_\xi : \mathcal{S} \to [0, 1], S \mapsto \mathbb{P}_\xi(S) := \mathbb{P}(\xi^{-1}(S)) = \mathbb{P}\left(\{\omega \in \Omega | \xi(\omega) \in S\}\right) \tag{22.5}$$

is called the *multivariate distribution of the random vector* ξ.

•

For simplicity, one often speaks only of *the distribution of the random vector* ξ or of a *multivariate distribution*. The notational conventions for random variables in 1 apply analogously to random vectors. For example, we have

$$\begin{aligned}
\mathbb{P}_{\xi}(\xi \in S) &:= \mathbb{P}\left(\{\xi \in S\}\right) = \mathbb{P}\left(\{\omega \in \Omega | \xi(\omega) \in S\}\right) \\
\mathbb{P}_{\xi}(\xi = x) &:= \mathbb{P}\left(\{\xi = x\}\right) = \mathbb{P}\left(\{\omega \in \Omega | \xi(\omega) = x\}\right) \\
\mathbb{P}_{\xi}(\xi \le x) &:= \mathbb{P}\left(\{\xi \le x\}\right) = \mathbb{P}\left(\{\omega \in \Omega | \xi(\omega) \le x\}\right) \\
\mathbb{P}_{\xi}(x_1 \le \xi \le x_2) &:= \mathbb{P}\left(\{x_1 \le \xi \le x_2\}\right) = \mathbb{P}\left(\{\omega \in \Omega | x_1 \le \xi(\omega) \le x_2\}\right),
\end{aligned} \tag{22.6}$$

where the relational operators $<, \le, >, \ge$ are understood here *componentwise*. For example, $x \le y$ for $x, y \in \mathbb{R}^n$ means that $x_i \le y_i$ holds for all components $x_i, y_i, i = 1, ..., n$. This convention is also followed by the definition of the *multivariate cumulative distribution function* in generalization of Definition 21.13. As with random variables, the qualifying subscripts with respect to a specific random vector are usually omitted in the sense of multiple dispatch. We use this convention already in the following definition of the multivariate cumulative distribution function of a random vector ξ.

Definition 22.3 (Multivariate cumulative distribution functions)**.** Let ξ be a random vector with outcome space $\mathcal{X}$. Then a function of the form

$$P : \mathcal{X} \to [0, 1],\ x \mapsto P(x) := \mathbb{P}(\xi \le x) \tag{22.7}$$

is called a multivariate cumulative distribution function of ξ.

•

Like cumulative distribution functions, multivariate cumulative distribution functions can also be used to define multivariate distributions. More commonly, however, as in the univariate case, multivariate distributions are defined by *multivariate probability mass functions* or *probability density functions*. We generalize the definitions of discrete and continuous random variables and their associated probability mass and probability density functions as follows (cf. Definition 21.4 and Definition 21.8).

Definition 22.4 (Discrete random vector and multivariate probability mass function)**.** A random vector ξ is called *discrete* if its outcome space $\mathcal{X}$ is finite or countable and if a function

$$p : \mathcal{X} \to [0, 1], x \mapsto p(x) \tag{22.8}$$

exists such that

(1) $\sum_{x \in \mathcal{X}} p(x) = 1$
(2) $\mathbb{P}(\xi = x) = p(x)$ for all $x \in \mathcal{X}$.

Such a function p is called the *multivariate probability mass function (PMF)* of ξ.

•

The concept of a multivariate PMF is evidently directly analogous to the concept of a PMF. Like univariate PMFs, multivariate PMFs are non-negative and normalized. For simplicity, one often speaks simply of *the PMF of a random vector* and, when the relevant random vector is clear from the context, omits the ξ subscript in its notation, thus often writing simply p instead of p_ξ.

Example 22.1 (Discrete random vector)**.** To illustrate the concept of the discrete random vector and its PMF, we consider an example. Let $\xi := (\xi_1, \xi_2)$ be a random vector that takes values in $\mathcal{X} := \mathcal{X}_1 \times \mathcal{X}_2$, where $\mathcal{X}_1 := \{1,2,3\}$ and $\mathcal{X}_2 = \{1,2,3,4\}$. Then the outcome space of ξ corresponds to the set of tuples (x_1, x_2) specified in Table 22.1.

Table 22.1 Outcome space

(x_1, x_2)	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$
$x_1 = 1$	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$
$x_1 = 2$	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$
$x_1 = 3$	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$

An exemplary bivariate PMF of the form

$$p : \{1,2,3\} \times \{1,2,3,4\} \to [0,1], (x_1, x_2) \mapsto p(x_1, x_2) \tag{22.9}$$

is then defined by Table 22.2.

Table 22.2 Bivariate PMF $p(x_1, x_2)$

$p(x_1, x_2)$	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$
$x_1 = 1$	0.1	0.0	0.2	0.1
$x_1 = 2$	0.1	0.2	0.0	0.0
$x_1 = 3$	0.0	0.1	0.1	0.1

Note that the function p specified in this way satisfies the normalization and non-negativity requirements of a PMF. In particular, here

$$\sum_{x \in \mathcal{X}} p(x) = \sum_{x_1=1}^{3} \sum_{x_2=1}^{4} p(x_1, x_2) = 1. \tag{22.10}$$

$\triangle$

We define the concept of the continuous random vector and the multivariate probability density function as follows.

Definition 22.5 (Continuous random vector and multivariate probability density function)**.** A random vector ξ is called *continuous* if its outcome space is given by $\mathbb{R}^n$ and if a function

$$p : \mathbb{R}^n \to \mathbb{R}_{\geq 0}, x \mapsto p(x), \tag{22.11}$$

exists such that

(1) $\int_{\mathbb{R}^n} p(x)\, dx = 1$ and
(2) $\mathbb{P}(x_1 \leq \xi \leq x_2) = \int_{x_{1_1}}^{x_{2_1}} \cdots \int_{x_{1_n}}^{x_{2_n}} p(s_1, ..., s_n)\, ds_1 \cdots ds_n$.

Such a function p is called the *multivariate probability density function (PDF) of* ξ.

•

Evidently, the concept of the multivariate PDF of a continuous random vector is analogous to the concept of the PDF of a continuous random variable, and like univariate PDFs, multivariate PDFs are non-negative and normalized. For simplicity, one often also speaks simply of *multivariate PDFs* and omits the subscripts identifying the random vector. As for continuous random variables, for continuous random vectors we have

$$\mathbb{P}(\xi = x) = \mathbb{P}(x \le \xi \le x) = \int_{x_1}^{x_1} \cdots \int_{x_n}^{x_n} p(s_1, ..., s_n)\, ds_1 \cdots ds_n = 0. \tag{22.12}$$

The standard example of a multivariate PDF is the *multivariate normal distribution*, which is discussed in detail in the chapter on normal distributions.

22.2 Marginal distributions

Once the distribution of a random vector has been specified, one may ask which distributions follow from it for the individual components of the random vector, that is, for the random variables that together form the random vector. In the context of a random vector, these are called the *univariate marginal distributions* of the random vector. The following definition is fundamental.

Definition 22.6 (Univariate marginal distribution). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let $(\mathcal{X}, \mathcal{S})$ be an n-dimensional measurable space, let $\xi : \Omega \to \mathcal{X}$ be a random vector, let $\mathbb{P}$ be the distribution of ξ, let $\mathcal{X}_i \subset \mathcal{X}$ be the outcome space of the ith component ξ_i of ξ, and let $\mathcal{S}_i$ be a σ-algebra on ξ_i. Then the distribution defined by

$$\mathbb{P}_{\xi_i} : \mathcal{S}_i \to [0,1], S \mapsto \mathbb{P}\left(\mathcal{X}_1 \times \cdots \times \mathcal{X}_{i-1} \times S \times \mathcal{X}_{i+1} \times \cdots \times \mathcal{X}_n\right) \text{ for } S \in \mathcal{S}_i \tag{22.13}$$

is called the *ith univariate marginal distribution of* ξ.

•

Concretely, for both discrete and continuous random vectors, the PMFs or PDFs of their components can be determined directly from the corresponding multivariate PMF or PDF. This is the statement of the following theorem.

Theorem 22.1 (Marginal probability mass and probability density functions). *(1) Let $\xi = (\xi_1, ..., \xi_n)$ be an n-dimensional discrete random vector with PMF p and component outcome spaces $\mathcal{X}_1, ..., \mathcal{X}_n$. Then the PMF of the ith component ξ_i of ξ is given by*

$$p : \mathcal{X}_i \to [0,1], x_i \mapsto p(x_i) := \sum_{x_1} \cdots \sum_{x_{i-1}} \sum_{x_{i+1}} \cdots \sum_{x_n} p(x_1, ..., x_{i-1}, x_i, x_{i+1}, ..., x_n). \tag{22.14}$$

(2) Let $\xi = (\xi_1, ..., \xi_n)$ be an n-dimensional continuous random vector with PDF p and component outcome space $\mathbb{R}$. Then the PDF of the ith component ξ_i of ξ is given by

$$p : \mathbb{R} \to \mathbb{R}_{\geq 0}, x_i \mapsto p(x_i) := \int_{x_1} \cdots \int_{x_{i-1}} \int_{x_{i+1}} \cdots \int_{x_n} p(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n) \, dx_1 \ldots dx_{i-1} \, dx_{i+1} \ldots dx_n. \tag{22.15}$$

∘

The PMFs of the univariate marginal distributions of discrete random vectors are thus obtained by summing over all values of the random variables complementary to the random variable currently under consideration, and the PDFs of the univariate marginal distributions of continuous random vectors are obtained analogously by integrating over all values of the random variables complementary to the random variable currently under consideration. For a proof of Theorem 22.1, we refer to the advanced literature.

Example 22.2 (Marginal probability mass functions)**.** Continuing Example 22.1, for the PMF defined there for the random vector $\xi := (\xi_1, \xi_2)$, the marginal PMFs are obtained as the PMFs listed on the margins of Table 22.3 using

$$p(x_1) = \sum_{x_2=1}^{4} p(x_1, x_2) \text{ and } p(x_2) = \sum_{x_1=1}^{3} p(x_1, x_2). \tag{22.16}$$

Table 22.3 Bivariate PMF $p(x_1, x_2)$ with marginal PMFs

$p(x_1, x_2)$	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$	$p(x_1)$
$x_1 = 1$	0.1	0.0	0.2	0.1	0.4
$x_1 = 2$	0.1	0.2	0.0	0.0	0.3
$x_1 = 3$	0.0	0.1	0.1	0.1	0.3
$p(x_2)$	0.2	0.3	0.3	0.2	

For the values $p(x_1)$, the corresponding values of $p(x_1, x_2)$ are therefore added row by row, and for the values $p(x_2)$, column by column. Note that the normalization of $p(x_1)$ and $p(x_2)$ follows directly from the normalization of $p(x_1, x_2)$, because the total amount of probability mass does not change:

$$1 = \sum_{x_1=1}^{3} \sum_{x_2=1}^{4} p(x_1, x_2) = \sum_{x_1=1}^{3} p(x_1) = \sum_{x_2=1}^{4} p(x_2). \tag{22.17}$$

An example of realizations using relative frequencies may make the concept of the marginal PMF intuitively clear. Suppose we had $n = 100$ independent realizations of ξ. To estimate the probabilities $p(x_1, x_2)$, we would count the number of realizations of (x_1, x_2) and divide by n. If, for example, we had 12 realizations of $(3, 2)$, we would estimate $p(3, 2) \approx 12/100 = 0.12$. The question of the marginal probability of $x_2 = 2$ would then correspond to the question of how often, among the realizations, there are those for which $x_2 = 2$, irrespective of the value of x_1. This would be precisely the number of realizations of the form $(1, 2), (2, 2)$, and $(3, 2)$. If there were, for example, 0, 22, and 12 of these, respectively, then one would naturally estimate the probability $p(2)$ by

$$\frac{0 + 22 + 12}{100} = \frac{0}{100} + \frac{22}{100} + \frac{12}{100} = 0.00 + 0.22 + 0.12 = 0.34. \tag{22.18}$$

Thus, instead of the probabilities $p(1, 2)$, $p(2, 2)$, and $p(3, 2)$, one adds the corresponding relative frequencies here.

△

Marginal distributions in the case of continuous random vectors are treated using the standard example of the multivariate normal distribution in the chapter on normal distributions.

22.3 Conditional distributions

Once the distribution of a random vector has been specified, one may ask which distribution follows from it for a single component of the random vector if the value of another component is assumed to be known. This leads to the concept of the *conditional distribution*, which arises naturally from the concept of conditional probability (cf. Section 20.2). We first recall that, for a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and two events $A, B \in \mathcal{A}$ with $\mathbb{P}(B) > 0$, the conditional probability of A given B is defined as

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \tag{22.19}$$

Analogously, for two random variables ξ_1, ξ_2 with outcome spaces $\mathcal{X}_1, \mathcal{X}_2$ and measurable sets $S_1 \in \mathcal{X}_1, S_2 \in \mathcal{X}_2$, the conditional distribution of ξ_1 given ξ_2 is defined with the help of the events

$$A := \{\xi_1 \in S_1\} \text{ and } B := \{\xi_2 \in S_2\}. \tag{22.20}$$

For example, the conditional probability that $\xi_1 \in S_1$ given that $\xi_2 \in S_2$, under the assumption that $\mathbb{P}(\{\xi_2 \in S_2\}) > 0$, is

$$\mathbb{P}(\{\xi_1 \in S_1\}|\{\xi_2 \in S_2\}) = \frac{\mathbb{P}(\{\xi_1 \in S_1\} \cap \{\xi_2 \in S_2\})}{\mathbb{P}(\{\xi_2 \in S_2\})}. \tag{22.21}$$

We first consider the definition of conditional distributions of discrete random vectors that consist of only two random variables.

Definition 22.7 (Conditional probability mass function and discrete conditional distribution)**.** Let $\xi := (\xi_1, \xi_2)$ be a discrete random vector with outcome space $\mathcal{X} := \mathcal{X}_1 \times \mathcal{X}_2$, PMF $p(x_1, x_2)$, and marginal PMFs $p(x_1)$ and $p(x_2)$. The conditional PMF of ξ_1 given $\xi_2 = x_2$ is then defined for $p(x_2) > 0$ as

$$p : \mathcal{X}_1 \to [0, 1], x_1 \mapsto p(x_1|x_2) := \frac{p(x_1, x_2)}{p(x_2)}. \tag{22.22}$$

Analogously, for $p(x_1) > 0$, the conditional PMF of ξ_2 given $\xi_1 = x_1$ is defined as

$$p : \mathcal{X}_2 \to [0, 1], x_2 \mapsto p(x_2|x_1) := \frac{p(x_1, x_2)}{p(x_1)}. \tag{22.23}$$

The conditional distributions with PMFs $p(x_1|x_2)$ and $p(x_2|x_1)$ are then called the *discrete conditional distributions of ξ_1 given $\xi_2 = x_2$ and of ξ_2 given $\xi_1 = x_1$*, respectively.

•

In analogy to the definition of the conditional probability of events, we thus have

$$p(x_1|x_2) = \frac{p(x_1, x_2)}{p(x_2)} = \frac{\mathbb{P}(\{\xi_1 = x_1\} \cap \{\xi_2 = x_2\})}{\mathbb{P}(\{\xi_2 = x_2\})}. \tag{22.24}$$

It is crucial to recognize here that conditional distributions are merely normalized joint distributions.

Example 22.3 (Conditional probability mass functions)**.** Continuing the example of a two-dimensional random vector $\xi := (\xi_1, \xi_2)$ considered in Example 22.1 and its marginal distributions determined in Example 22.2, the conditional PMFs for $p(x_1|x_2)$ shown in Table 22.4 and the conditional PMFs for $p(x_2|x_1)$ shown in Table 22.5 are obtained.

Table 22.4 Conditional PMFs $p(x_1|x_2)$

$p(x_1\|x_2)$	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$
$p(1\|x_2)$	$\frac{0.1}{0.2} = 0.5$	$\frac{0.0}{0.3} = 0.0$	$\frac{0.2}{0.3} = 0.6\bar{6}$	$\frac{0.1}{0.2} = 0.5$
$p(2\|x_2)$	$\frac{0.1}{0.2} = 0.5$	$\frac{0.2}{0.3} = 0.6\bar{6}$	$\frac{0.0}{0.3} = 0.0$	$\frac{0.0}{0.2} = 0.0$
$p(3\|x_2)$	$\frac{0.0}{0.2} = 0.0$	$\frac{0.1}{0.3} = 0.3\bar{3}$	$\frac{0.1}{0.3} = 0.3\bar{3}$	$\frac{0.1}{0.2} = 0.5$

Table 22.5 Conditional PMFs $p(x_2|x_1)$

$p(x_2\|x_1)$	$p(1\|x_1)$	$p(2\|x_1)$	$p(3\|x_1)$	$p(4\|x_1)$
$x_1 = 1$	$\frac{0.1}{0.4} = 0.25$	$\frac{0.0}{0.4} = 0.00$	$\frac{0.2}{0.4} = 0.50$	$\frac{0.1}{0.4} = 0.25$
$x_1 = 2$	$\frac{0.1}{0.3} = 0.3\bar{3}$	$\frac{0.2}{0.3} = 0.6\bar{6}$	$\frac{0.0}{0.3} = 0.00$	$\frac{0.0}{0.3} = 0.00$
$x_1 = 3$	$\frac{0.0}{0.3} = 0.00$	$\frac{0.1}{0.3} = 0.3\bar{3}$	$\frac{0.1}{0.3} = 0.3\bar{3}$	$\frac{0.1}{0.3} = 0.3\bar{3}$

Note, first, that

$$\sum_{x_1=1}^{3} p(x_1|x_2) = 1 \text{ for all } x_2 \in \mathcal{X}_2 \text{ and } \sum_{x_2=1}^{4} p(x_2|x_1) = 1 \text{ for all } x_1 \in \mathcal{X}_1, \tag{22.25}$$

so the conditional PMFs are normalized. Second, note the qualitative similarity of the PMFs $p(x_1, x_2)$ and $p(x_1|x_2)$ or $p(x_2|x_1)$, respectively, which arises simply from the fact that, on the one hand, $p(x_1, x_2)$ and $p(x_1|x_2)$ are identical for all $x_2 \in \mathcal{X}_2$ up to the common scaling factor $1/p(x_2)$ and, on the other hand, $p(x_1, x_2)$ and $p(x_2|x_1)$ are identical for all $x_1 \in \mathcal{X}_1$ up to the common scaling factor $1/p(x_1)$.

△

In the case of a continuous random vector, the analogous conditional PDFs are defined as follows.

Definition 22.8 (Conditional probability density function and continuous conditional distribution)**.** Let $\xi := (\xi_1, \xi_2)$ be a continuous random vector with outcome space $\mathbb{R}^2$, PDF $p(x_1, x_2)$, and marginal PDFs $p(x_1)$ and $p(x_2)$. The conditional PDF of ξ_1 given $\xi_2 = x_2$ is then defined for $p(x_2) > 0$ as

$$p : \mathbb{R} \to \mathbb{R}_{\geq 0}, x_1 \mapsto p(x_1|x_2) := \frac{p(x_1, x_2)}{p(x_2)}. \tag{22.26}$$

Analogously, for $p(x_1) > 0$, the conditional PDF of ξ_2 given $\xi_1 = x_1$ is defined as

$$p : \mathbb{R} \to \mathbb{R}_{\geq 0}, x_2 \mapsto p(x_2|x_1) := \frac{p(x_1, x_2)}{p(x_1)}. \tag{22.27}$$

The distributions with PDFs $p(x_1|x_2)$ and $p(x_2|x_1)$ are then called the *continuous conditional distributions of* ξ_1 *given* $\xi_2 = x_2$ *and* ξ_2 *given* $\xi_1 = x_1$, respectively.

•

Note that in the continuous case, although $\mathbb{P}(\xi = x) = 0$, it does not necessarily follow that $p(x) = 0$. Conditional distributions of multivariate normal distributions are discussed in a separate chapter.

22.4 Independent random variables

Similar to the conditional probabilities of events, the concept of independent events can also be transferred to random vectors. We first define the concept of independent random variables.

Definition 22.9 (Independent random variables)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $\xi := (\xi_1, \xi_2)$ be a two-dimensional random vector. The random variables ξ_1, ξ_2 with outcome spaces $\mathcal{X}_1, \mathcal{X}_2$ are called *independent* if, for all $S_1 \subseteq \mathcal{X}_1$ and $S_2 \subseteq \mathcal{X}_2$,

$$\mathbb{P}(\xi_1 \in S_1, \xi_2 \in S_2) = \mathbb{P}(\xi_1 \in S_1)\mathbb{P}(\xi_2 \in S_2). \tag{22.28}$$

•

Definition 22.9 states that the events $\{\xi_1 \in S_1\}$ and $\{\xi_2 \in S_2\}$ are independent. Thus, it also holds that

$$\mathbb{P}(\{\xi_1 \in S_1\}|\{\xi_2 \in S_2\}) = \mathbb{P}(\{\xi_1 \in S_1\}) \tag{22.29}$$

and knowledge of the occurrence of the event $\{\xi_2 \in S_2\}$ does not change the probability of the event $\{\xi_1 \in S_1\}$. The factorization principle for modeling probabilistic independence transfers to PMFs and PDFs of random vectors. This is the statement of the following theorem.

Theorem 22.2 (Independence and factorization)**.**

(1) Let $\xi := (\xi_1, \xi_2)$ be a discrete random vector with outcome space $\mathcal{X}_1 \times \mathcal{X}_2$, PMF $p(x_1, x_2)$, and marginal PMFs $p(x_1), p(x_2)$. Then

$$\xi_1 \text{ and } \xi_2 \text{ are independent random variables} \Leftrightarrow \\ p(x_1, x_2) = p(x_1)p(x_2) \text{ for all } (x_1, x_2) \in \mathcal{X}_1 \times \mathcal{X}_2. \tag{22.30}$$

(2) Let $\xi := (\xi_1, \xi_2)$ be a continuous random vector with outcome space $\mathbb{R}^2$, PDF p, and marginal PDFs p, p. Then

$$\xi_1 \text{ and } \xi_2 \text{ are independent random variables} \Leftrightarrow \\ p(x_1, x_2) = p(x_1)p(x_2) \text{ for all } (x_1, x_2) \in \mathbb{R}^2. \tag{22.31}$$

∘

In general, the independence of two random variables is thus equivalent to the factorization of their joint PMF or PDF. For a proof of Theorem 22.2, we refer to the advanced literature. Theorem 22.2 is fundamental for wide areas of probabilistic modeling.

Example 22.4 (Independent probability mass functions)**.** We again consider the two-dimensional random vector $\xi := (\xi_1, \xi_2)$ from Example 22.1, whose joint and marginal PMFs have the forms specified in Table 22.3. We first ask whether ξ_1 and ξ_2 are independent. This is not the case, since here

$$p(1,1) = 0.10 \neq 0.08 = 0.40 \cdot 0.20 = p(1)p(1). \tag{22.32}$$

If, based on the marginal distributions of ξ, we want to generate a joint distribution in which ξ_1 and ξ_2 are independent, then each entry of the joint distribution $p(\xi_1, \xi_2)$ must be obtained from the corresponding product of the marginal probabilities. The joint distribution of ξ_1 and ξ_2 under this assumption of independence of ξ_1 and ξ_2, with the same marginal distributions as in Table 22.3, is given by the joint PMF in Table 22.6.

Table 22.6 Bivariate PMF under the assumption of independence of ξ_1 and ξ_2

$p(x_1, x_2)$	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$	$p(x_1)$
$x_1 = 1$	0.08	0.12	0.12	0.08	0.40
$x_1 = 2$	0.06	0.09	0.09	0.06	0.30
$x_1 = 3$	0.06	0.09	0.09	0.06	0.30
$p(x_2)$	0.20	0.30	0.30	0.20	

Furthermore, in the case of independence of ξ_1 and ξ_2, the conditional PMFs $p(x_2|x_1)$, for example, are obtained as shown in Table 22.7. In the case of independence of ξ_1 and ξ_2, the distribution of ξ_2 given, or with knowledge of, the value of ξ_1 therefore does not change and corresponds in each case to the marginal distribution of ξ_2. This of course corresponds to the intuition of independence of events in the context of elementary probabilities.

Table 22.7 Conditional PMF under the assumption of independence of ξ_1 and ξ_2

$p(x_2|x_1)$	$p(1|x_1)$	$p(2|x_1)$	$p(3|x_1)$	$p(4|x_1)$
$x_1 = 1$	$\frac{0.08}{0.40} = 0.2$	$\frac{0.12}{0.40} = 0.3$	$\frac{0.12}{0.40} = 0.3$	$\frac{0.08}{0.40} = 0.2$
$x_1 = 2$	$\frac{0.06}{0.30} = 0.2$	$\frac{0.09}{0.30} = 0.3$	$\frac{0.09}{0.30} = 0.3$	$\frac{0.06}{0.30} = 0.2$
$x_1 = 3$	$\frac{0.06}{0.30} = 0.2$	$\frac{0.09}{0.30} = 0.3$	$\frac{0.09}{0.30} = 0.3$	$\frac{0.06}{0.30} = 0.2$

$\triangle$

Finally, we define the concept of independent random variables for more than two random variables.

Definition 22.10 (n independent random variables)**.** Let $\xi := (\xi_1, ..., \xi_n)$ be an n-dimensional random vector with outcome space $\mathcal{X} = \times_{i=1}^n \mathcal{X}_i$. The n random variables $\xi_1, ..., \xi_n$ are called *independent* if, for all $S_i \subseteq \mathcal{X}_i, i = 1, ..., n$,

$$\mathbb{P}(\xi_1 \in S_1, ..., \xi_n \in S_n) = \prod_{i=1}^n \mathbb{P}(\xi_i \in S_i). \tag{22.33}$$

If the random vector has an n-dimensional PMF or PDF p with marginal PMFs or PDFs $p_{\xi_i}, i = 1, ..., n$, then the independence of $\xi_1, ..., \xi_n$ is equivalent to the factorization of the joint PMF or PDF, that is, to

$$p(\xi_1, ..., \xi_n) = \prod_{i=1}^n p(x_i). \tag{22.34}$$

$\bullet$

Thus, Definition 22.10 is a direct generalization of the two-dimensional case. If n random variables are not only independent but also all have the same distribution, they are called *independent and identically distributed (i.i.d.)*.

Definition 22.11 (Independent and identically distributed random variables)**.** n random variables $\xi_1, ..., \xi_n$ are called *independent and identically distributed (i.i.d.)* if

(1) $\xi_1, ..., \xi_n$ are independent random variables, and
(2) the marginal distributions of the ξ_i agree, that is,

$$\mathbb{P}(\xi_i) = \mathbb{P}(\xi_j) \text{ for all } 1 \le i, j \le n. \tag{22.35}$$

If the random variables $\xi_1, ..., \xi_n$ are independent and identically distributed and the ith marginal distribution is $\mathbb{P}$, one also writes

$$\xi_1, ..., \xi_n \sim \mathbb{P}. \tag{22.36}$$

•

In short, one says that $\xi_1, ..., \xi_n$ *are i.i.d.* I.i.d. random variables play an important role in many places in probabilistic modeling. As we will see later, additive error terms in probabilistic models are usually modeled by i.i.d. random variables. Finally, we record that n i.i.d. normally distributed random variables are written as

$$\xi_1, ..., \xi_n \sim N(\mu, \sigma^2). \tag{22.37}$$

In the chapter on normal distributions, we show what the joint distribution of n i.i.d. normally distributed random variables looks like in terms of a random vector.

Study questions

1. State the definition of the concept of a random vector.
2. State the definition of the concept of the multivariate distribution of a random vector.
3. State the definition of the concept of a multivariate PMF.
4. State the definition of the concept of a multivariate PDF.
5. State the definition of the concept of the univariate marginal distribution of a random vector.
6. How does one compute the PMF of the ith component of a discrete random vector?
7. How does one compute the PDF of the ith component of a continuous random vector?
8. State the definition of the concept of independence of two random variables.
9. How can one recognize from the joint PMF or PDF of a two-dimensional random vector $\xi := (\xi_1, \xi_2)$ whether the components of the random vector are independent or not?
10. State the definition of the concept of independence of n random variables.
11. State the definition of the concept of independent and identically distributed random variables.

Study question answers

1. See Definition 22.1.
2. See Definition 22.2.
3. See Definition 22.4.
4. See Definition 22.5.
5. See Definition 22.6.

6. See Theorem 22.1. One obtains the PMF of the ith component of a discrete random vector by summing the values of the PMF of the random vector over all components of the random vector *except* the ith component.
7. See Theorem 22.1. One obtains the PDF of the ith component of a continuous random vector by integrating the values of the PDF of the random vector over all components of the random vector *except* the ith component.
8. See Definition 22.10.
9. One checks whether all values of the joint PMF or PDF $p(x_1, x_2)$ of the random vector can be obtained by multiplying the corresponding marginal PMF or PDF values $p(x_1)$ and $p(x_2)$. If this is the case for all $(x_1, x_2) \in \mathcal{X}_1 \times \mathcal{X}_2$, then ξ_1 and ξ_2 are independent; if this is not the case for all $(x_1, x_2) \in \mathcal{X}_1 \times \mathcal{X}_2$, then ξ_1 and ξ_2 are not independent.
10. See Definition 22.9.
11. See Definition 22.11.

23 Expectations

In this chapter, we introduce the concept of the *expectation* of a random variable. The expectation serves as a single scalar characteristic that summarizes a probability distribution and is often used as a central characteristic quantity of random variables. In this sense, the expectation serves as a measure of the "average realization" of a random variable and forms the basis for further concepts such as the *variance* of a random variable and the *covariance* of two random variables in what follows. We supplement the concept of expectation with its analogues for random vectors and conditional distributions, as well as with its descriptive-statistical equivalent, the so-called *sample mean*.

23.1 Definitions

Definition 23.1 (Expectation of a random variable)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let ξ be a random variable. Then the *expectation of* ξ is defined as

- $\mathbb{E}(\xi) := \sum_{x\in\mathcal{X}} x\,p(x)$ if $\xi : \Omega \to \mathcal{X}$ is discrete with PMF p.
- $\mathbb{E}(\xi) := \int_{-\infty}^{\infty} x\,p(x)\,dx$ if $\xi : \Omega \to \mathbb{R}$ is continuous with PDF p.

•

The expectation is thus a scalar summary of the distribution of a random variable. A definition of the expectation that does not require distinguishing between continuous and discrete random variables is possible, but, with the introduction of the Lebesgue integral, requires some technical effort. We refer to the advanced literature for this (cf. Schmidt (2009), Meintrup & Schäffler (2005)). It is important to recognize that an expectation is determined solely by specifying the outcome space and the distribution of a random variable by means of a PMF or PDF. Nevertheless, the expectation of a random variable corresponds to the value of the random variable expected on average in the long run. This fact is formalized under the concept of the *law of large numbers*. We first consider three examples for Definition 23.1.

Example 23.1 (Expectation of a discrete random variable)**.** Let ξ be a random variable with outcome space $\mathcal{X} := \{-1, 0, 1\}$ and PMF

$$p(-1) := \frac{1}{4}, \quad p(0) := \frac{1}{2}, \quad p(1) := \frac{1}{4}. \tag{23.1}$$

Then

$$\mathbb{E}(\xi) = 0. \tag{23.2}$$

△

Proof. By Definition 23.1, we have

$$
\begin{aligned}
\mathbb{E}(\xi) &= \sum_{x \in \mathcal{X}} x p(x) \\
&= -1 \cdot p(-1) + 0 \cdot p(0) + 1 \cdot p(1) \\
&= -1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \\
&= 0.
\end{aligned} \tag{23.3}
$$

□

Example 23.2 (Expectation of a Bernoulli random variable)**.** Let $\xi \sim \mathrm{Bern}(\mu)$. Then $\mathbb{E}(\xi) = \mu$.

△

Proof. ξ is discrete with $\mathcal{X} = \{0, 1\}$. Thus, by Definition 23.1,

$$
\begin{aligned}
\mathbb{E}(\xi) &= \sum_{x \in \{0,1\}} x \, \mathrm{Bern}(x; \mu) \\
&= 0 \cdot \mu^0 (1-\mu)^{1-0} + 1 \cdot \mu^1 (1-\mu)^{1-1} \\
&= 1 \cdot \mu^1 (1-\mu)^0 \\
&= \mu.
\end{aligned} \tag{23.4}
$$

□

Example 23.3 (Expectation of a normally distributed random variable)**.** Let $\xi \sim N(\mu, \sigma^2)$. Then $\mathbb{E}(\xi) = \mu$.

△

Proof. We omit the proof. □

Intuitively, Definition 23.1 can be explained as shown in Example 23.4.

Example 23.4 (Expectation and mean)**.** We consider a discrete random variable ξ with outcome space $\{1, 2, 3\}$ and PMF defined by

$$
p(1) := \frac{1}{4}, \quad p(2) := \frac{1}{4}, \quad p(3) := \frac{1}{2}. \tag{23.5}
$$

Thus,

$$
\begin{aligned}
\mathbb{E}(\xi) &= \sum_{x \in \{1,2,3\}} x p(x) \\
&= 1 \cdot p(1) + 2 \cdot p(2) + 3 \cdot p(3) \\
&= 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{2} \\
&= \frac{9}{4}.
\end{aligned} \tag{23.6}
$$

Now imagine $n = 100$ independent realizations $x_1, ..., x_{100}$ of the random variable. According to its PMF, we would expect roughly 25 ones, 25 twos, and 50 threes. Suppose, concretely, that we had 27 ones, 22 twos, and 51 threes. The mean of these numbers is then (cf. Chapter 3)

$$\begin{aligned}
\bar{x} &= \frac{1}{100}\sum_{i=1}^{n} x_i \\
&= \frac{1}{100}(27 \cdot 1 + 22 \cdot 2 + 51 \cdot 3) \\
&= \frac{1}{100}(1 \cdot 27 + 2 \cdot 22 + 3 \cdot 51) \\
&= 1 \cdot \frac{27}{100} + 2 \cdot \frac{22}{100} + 3 \cdot \frac{51}{100} \\
&= 1 \cdot 0.27 + 2 \cdot 0.22 + 3 \cdot 0.51 \\
&= 2.24.
\end{aligned} \tag{23.7}$$

If, in this calculation, the factors $0.27 \approx \frac{1}{4}$, $0.22 \approx \frac{1}{4}$, and $0.51 \approx \frac{1}{2}$ are understood as estimates $\hat{p}(1)$, $\hat{p}(2)$, and $\hat{p}(3)$ of the PMF values $p(1)$, $p(2)$, and $p(3)$, then

$$\bar{x} = \sum_{x \in \{1,2,3\}} x\hat{p}(x) \approx \sum_{x \in \{1,2,3\}} xp(x) = \mathbb{E}(\xi) \tag{23.8}$$

makes the formal similarity between forming a mean of realizations of a discrete random variable and the definition of the expectation of such a random variable immediately apparent.

△

In distinction to Definition 23.1, and building on Example 23.4, we define the so-called *sample mean*.

Definition 23.2 (Sample mean)**.** Let $\xi_1, ..., \xi_n$ be random variables. Then $\xi_1, ..., \xi_n$ are also called a *sample*. The *sample mean* of $\xi_1, ..., \xi_n$ is defined as the arithmetic mean

$$\bar{\xi} := \frac{1}{n}\sum_{i=1}^{n} \xi_i. \tag{23.9}$$

•

It is central to recognize that $\mathbb{E}(\xi)$ is a fixed characteristic of a random variable ξ and, in particular, is not itself a random variable. By contrast, $\bar{\xi}$ is a characteristic of a sample $\xi_1, ..., \xi_n$ and, as a function of random variables, is therefore itself a random variable. In Example 23.4, for example, each realization x_i for $i = 1, ..., n$ is modeled as a realization of the random variable $\xi_i := \xi$ for $i = 1, ..., n$. Since $\bar{\xi}$ is a scaled sum of random variables, $\bar{\xi}$ is also itself a random variable. We denote realizations of the random variable $\bar{\xi}$ by $\bar{x}$, as in the following.

Example 23.5 (Sample mean)**.** As an example of the realization of a sample mean, we consider the realizations of the random variables $\xi_1, ..., \xi_{10} \sim N(1, 2)$ shown in Table 23.1.

Table 23.1 Sample realization

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
0.54	1.01	-3.28	0.35	2.75	-0.51	2.32	1.49	0.96	1.25

The sample mean realization is

$$\bar{x} = \frac{1}{10}\sum_{i=1}^{10} x_i = \frac{6.88}{10} = 0.68. \tag{23.10}$$

△

Generalizing Definition 23.1, the following definition applies to the expectation of a function of a random variable.

Definition 23.3 (Expectation of a function of a random variable)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let ξ be a random variable with outcome space $\mathcal{X}$, and let $f : \mathcal{X} \to \mathcal{Z}$ be a function with target set $\mathcal{Z}$. Then the *expectation of the function f of the random variable ξ* is defined as

- $\mathbb{E}(f(\xi)) := \sum_{x\in\mathcal{X}} f(x)\,p(x)$ if $\xi : \Omega \to \mathcal{X}$ is discrete with PMF p,
- $\mathbb{E}(f(\xi)) := \int_{-\infty}^{\infty} f(x)\,p(x)\,dx$ if $\xi : \Omega \to \mathbb{R}$ is continuous with PDF p.

•

The expectation of a random variable follows from Definition 23.3 as the special case in which

$$f : \mathcal{X} \to \mathcal{Z}, x \mapsto f(x) := x. \tag{23.11}$$

In the English-language literature, Definition 23.3 is also known as the "law of the unconscious statistician" and is often used directly to define expectation. The following theorem gives a first example of the expectation of a function of a random variable.

Theorem 23.1 (Expectation under linear-affine transformation of a random variable)**.** *Let ξ be a random variable with outcome space $\mathcal{X}$ and let*

$$f : \mathcal{X} \to \mathcal{Z}, x \mapsto f(x) := ax + b \text{ for } a, b \in \mathbb{R} \tag{23.12}$$

be a linear-affine function. Then

$$\mathbb{E}(f(\xi)) = \mathbb{E}(a\xi + b) = a\mathbb{E}(\xi) + b. \tag{23.13}$$

∘

Proof. The statement of the theorem follows from the linearity properties of sums and integrals. We consider the case of a discrete random variable ξ with finite outcome space $\mathcal{X}$ and PMF p. We have

$$\begin{aligned}
\mathbb{E}(f(\xi)) &= \mathbb{E}(a\xi + b) \\
&= \sum_{x\in\mathcal{X}} (ax + b)p(x) \\
&= \sum_{x\in\mathcal{X}} axp(x) + bp(x) \\
&= a\sum_{x\in\mathcal{X}} xp(x) + b\sum_{x\in\mathcal{X}} p(x) \\
&= a\mathbb{E}(\xi) + b.
\end{aligned} \tag{23.14}$$

□

If one considers a random vector instead of a random variable, the definition of expectation extends as follows.

Definition 23.4 (Expectation of a random vector)**.** Let ξ be an n-dimensional random vector. Then the *expectation of* ξ is defined as the n-dimensional real vector

$$\mathbb{E}(\xi) := \begin{pmatrix} \mathbb{E}(\xi_1) \\ \vdots \\ \mathbb{E}(\xi_n) \end{pmatrix}. \tag{23.15}$$

•

The expectation of a random vector is thus the vector of expectations of its components. Computing the expectation of a random vector is therefore reduced to determining the expectations of the random variables constituting the random vector with respect to their marginal distributions. In analogy to Definition 23.3, one defines the expectation of this transformation for a function of a random vector as follows.

Definition 23.5 (Expectation of a function of a random vector)**.** Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let ξ be a random vector with outcome space $\mathcal{X}$, and let $f : \mathcal{X} \to \mathcal{Z}$ be a function with target set $\mathcal{Z}$. Then the *expectation of the function* f *of the random vector* ξ is defined as

- $\mathbb{E}(f(\xi)) := \sum_{x \in \mathcal{X}} f(x)\, p(x)$ if $\xi : \Omega \to \mathcal{X}$ is discrete with PMF p,
- $\mathbb{E}(f(\xi)) := \int_{-\infty}^{\infty} f(x)\, p(x)\, dx$ if $\xi : \Omega \to \mathbb{R}$ is continuous with PDF p.

•

23.2 Properties

The following statements are particularly relevant when calculating with samples.

Theorem 23.2 (Expectation under linear-affine combination of random variables)**.** *Let* ξ *be an* n*-dimensional random vector with components* $\xi_1, ..., \xi_n$ *and outcome space* $\mathcal{X} := \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$*. Furthermore, let*

$$f : \mathcal{X} \to \mathcal{Z}, x \mapsto f(x) := a_0 + \sum_{i=1}^{n} a_i x_i \text{ for } a_0, ..., a_n \in \mathbb{R}. \tag{23.16}$$

be a linear-affine combination of the components of ξ*. Then*

$$\mathbb{E}(f(\xi)) = \mathbb{E}\left(a_0 + \sum_{i=1}^{n} a_i \xi_i\right) = a_0 + \sum_{i=1}^{n} a_i \mathbb{E}(\xi_i). \tag{23.17}$$

∘

Proof. As shown below, the theorem follows from the linearity properties of sums and integrals, as well as Fubini's theorem for interchanging the order of summation or integration. We restrict ourselves to the discrete case, where, to simplify notation, we write $\sum_{x_i \in \mathcal{X}_i}$ as $\sum_{x_i}$. Then

$$
\begin{aligned}
\mathbb{E}(f(\xi)) &= \mathbb{E}\left(a_0 + \sum_{i=1}^n a_i \xi_i\right) \\
&= \mathbb{E}(a_0 + a_1\xi_1 + \cdots + a_n\xi_n) \\
&= \sum_{x_1} \cdots \sum_{x_n} (a_0 + a_1 x_1 + \cdots + a_n x_n) p(x_1, ..., x_n) \\
&= \sum_{x_1} \cdots \sum_{x_n} a_0 p(x_1, ..., x_n) + a_1 x_1 p(x_1, ..., x_n) + \cdots + a_n x_n p(x_1, ..., x_n) \\
&= \sum_{x_1} \cdots \sum_{x_n} a_0 p(x_1, ..., x_n) + \sum_{x_1} \cdots \sum_{x_n} a_1 x_1 p(x_1, ..., x_n) + \cdots + \sum_{x_1} \cdots \sum_{x_n} a_n x_n p(x_1, ..., x_n) \\
&= a_0 \sum_{x_1} \cdots \sum_{x_n} p(x_1, ..., x_n) + a_1 \sum_{x_1} \cdots \sum_{x_n} x_1 p(x_1, ..., x_n) + \cdots + a_n \sum_{x_n} \cdots \sum_{x_1} x_n p(x_1, ..., x_n) \\
&= a_0 + a_1 \sum_{x_1} x_1 \sum_{x_2} \cdots \sum_{x_n} p(x_1, ..., x_n) + \cdots + a_n \sum_{x_n} x_n \sum_{x_{n-1}} \cdots \sum_{x_1} p(x_1, ..., x_n) \\
&= a_0 + a_1 \sum_{x_1} x_1 p(x_1) + \cdots + a_n \sum_{x_n} x_n p(x_n) \\
&= a_0 + a_1 \mathbb{E}(\xi_1) + \cdots + a_n \mathbb{E}(\xi_n) \\
&= a_0 + \sum_{i=1}^n a_i \mathbb{E}(\xi_i)
\end{aligned}
\tag{23.18}
$$

The result follows analogously for a continuous random vector. □

Theorem 23.3 (Expectation of the product of independent random variables)**.** *Let ξ be an n-dimensional random vector with independent components $\xi_1, ..., \xi_n$ and outcome space $\mathcal{X} := \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$. Furthermore, let*

$$
f : \mathcal{X} \to \mathcal{Z}, x \mapsto f(x) := \prod_{i=1}^n x_i. \tag{23.19}
$$

be the product of the components of ξ. Then

$$
\mathbb{E}(f(\xi)) = \prod_{i=1}^n \mathbb{E}(\xi_i). \tag{23.20}
$$

∘

Proof. As shown below, the theorem follows from the linearity properties of sums and integrals, as well as Fubini's theorem for interchanging the order of summation or integration. We restrict ourselves to the discrete case, where, to simplify notation, we write $\sum_{x_i \in \mathcal{X}_i}$ as $\sum_{x_i}$. Since the components $\xi_1, ..., \xi_n$ are assumed to be independent, the PMF of the random vector first satisfies

$$
p(x_1, ..., x_n) = \prod_{i=1}^n p(x_i). \tag{23.21}
$$

Furthermore,

$$
\begin{aligned}
\mathbb{E}\left(\prod_{i=1}^{n}\xi_i\right) &= \sum_{x_1}\cdots\sum_{x_n}\left(\prod_{i=1}^{n}x_i\right)p(x_1,\dots,x_n) \\
&= \sum_{x_1}\cdots\sum_{x_n}\prod_{i=1}^{n}x_i\prod_{i=1}^{n}p(x_i) \\
&= \sum_{x_1}\cdots\sum_{x_n}\prod_{i=1}^{n}x_ip(x_i) \\
&= \prod_{i=1}^{n}\sum_{x_i}x_ip(x_i) \\
&= \prod_{i=1}^{n}\mathbb{E}(\xi_i).
\end{aligned}
\tag{23.22}
$$

□

23.3 Conditional expectation

If, when forming an expectation, one considers the conditional distribution of a random variable instead of the distribution of a random variable, one is accordingly led to the concept of conditional expectation.

Definition 23.6 (Conditional expectation)**.** Given a random vector $\xi := (\xi_1, \xi_2)$ with outcome space $\mathcal{X} := \mathcal{X}_1 \times \mathcal{X}_2$, PMF or PDF $p(x_1, x_2)$, and conditional PMF or PDF $p(x_1|x_2)$ for all $x_2 \in \mathcal{X}_2$. Then the *conditional expectation* of ξ_1 given $\xi_2 = x_2$ is defined as

- $\mathbb{E}(\xi_1|\xi_2 = x_2) := \sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1|x_2)$ if ξ is a discrete random vector.
- $\mathbb{E}(\xi_1|\xi_2 = x_2) := \int_{\mathcal{X}_1} x_1 p(x_1|x_2)\,dx_1$ if ξ is a continuous random vector.

•

The conditional expectation of a random variable is thus the expectation of a random variable in a conditional distribution. Note that in Definition 23.6 we defined the conditional expectation for a fixed value x_2 of ξ_2. For a fixed value x_2 of ξ_2, $\mathbb{E}(\xi_1|\xi_2 = x_2)$ is therefore a fixed value, and by exchanging the subscripts one obtains correspondingly $\mathbb{E}(\xi_2|\xi_1 = x_1)$.

In general, however, $\mathbb{E}(\xi_1|\xi_2)$ is a random variable because ξ_2 is a random variable and assumes a value $\xi_2 = x_2$ only with a certain probability. We will see later that conditional variances, conditional covariances, and conditional correlations are defined analogously to Definition 23.6. In psychology, the concept of conditional expectation is central because, in classical test theory, the true scores of persons in test measurements are defined as conditional expectations. To illustrate Definition 23.6, we consider an example for a discrete random vector.

Example

For a two-dimensional random vector $\xi := (\xi_1, \xi_2)$ that takes values in $\mathcal{X} := \mathcal{X}_1 \times \mathcal{X}_2$ with $\mathcal{X}_1 := \{1,2,3,4\}$ and $\mathcal{X}_2 = \{1,2,3\}$, let conditional PMFs for $p(x_1|x_2)$ for all $x_1 \in \mathcal{X}_1$ be specified as in Table 23.2.

Table 23.2 Conditional probability distribution of x_1 given x_2

$p(x_1\|x_2)$	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$
$p(x_1 = 1\|x_2)$	$\frac{1}{4}$	$\frac{1}{3}$	0
$p(x_1 = 2\|x_2)$	0	$\frac{2}{3}$	$\frac{1}{3}$
$p(x_1 = 3\|x_2)$	$\frac{1}{2}$	0	$\frac{1}{3}$
$p(x_1 = 4\|x_2)$	$\frac{1}{4}$	0	$\frac{1}{3}$

Then the conditional expectations are

$$
\begin{aligned}
\mathbb{E}(\xi_1|\xi_2 = 1) &= \sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1|x_2 = 1) = 1 \cdot \frac{1}{4} + 2 \cdot 0 + 3 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{11}{4} \\
\mathbb{E}(\xi_1|\xi_2 = 2) &= \sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1|x_2 = 2) = 1 \cdot \frac{1}{3} + 2 \cdot \frac{2}{3} + 3 \cdot 0 + 4 \cdot 0 = \frac{5}{3} \\
\mathbb{E}(\xi_1|\xi_2 = 3) &= \sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1|x_2 = 3) = 1 \cdot 0 + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3} + 4 \cdot \frac{1}{3} = \frac{8}{3}.
\end{aligned}
\tag{23.23}
$$

The following theorem, the so-called *law of iterated expectation*, establishes a connection between the marginal expectation of a random variable and the conditional expectation of this random variable. The theorem is sometimes also called the *law of total expectation*.

Theorem 23.4 (Law of iterated expectation)**.** *Given a random vector $\xi := (\xi_1, \xi_2)$, we have*

$$
\mathbb{E}(\xi_1) = \mathbb{E}\left(\mathbb{E}\left(\xi_1|\xi_2\right)\right). \tag{23.24}
$$

∘

Proof. We restrict ourselves to the proof for a discrete random vector; the proof for a continuous random vector follows analogously. We have

$$
\begin{aligned}
\mathbb{E}(\xi_1) &= \sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1) \\
&= \sum_{x_1 \in \mathcal{X}_1} x_1 \sum_{x_2 \in \mathcal{X}_2} p(x_1, x_2) \\
&= \sum_{x_1 \in \mathcal{X}_1} x_1 \sum_{x_2 \in \mathcal{X}_2} p(x_1|x_2)p(x_2) \\
&= \sum_{x_1 \in \mathcal{X}_1} \sum_{x_2 \in \mathcal{X}_2} x_1 p(x_1|x_2)p(x_2) \\
&= \sum_{x_2 \in \mathcal{X}_2} \sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1|x_2)p(x_2) \\
&= \sum_{x_2 \in \mathcal{X}_2} \mathbb{E}(\xi_1|\xi_2 = x_2)p(x_2) \\
&= \mathbb{E}\left(\mathbb{E}(\xi_1|\xi_2)\right).
\end{aligned}
\tag{23.25}
$$

□

Evidently, the different expectations in Theorem 23.4 refer to expectations with respect to different distributions: the expectation on the left-hand side of the equation denotes the expectation with respect to the marginal distribution of ξ_1, the outer expectation on the right-hand side denotes the expectation with respect to the marginal distribution of ξ_2, and the inner expectation on the right-hand side denotes the conditional expectation of ξ_1 given ξ_2.

Study questions

1. State the definition of the expectation of a random variable.
2. Explain the concept of the expectation of a random variable.
3. Compute the expectation of a Bernoulli random variable.
4. State the definition of the concept of a sample.
5. State the definition of the sample mean.
6. Explain the differences between an expectation and a sample mean.
7. State the definition of the expectation of a function of a random variable.
8. State the theorem on expectation under linear-affine transformation of a random variable.
9. State the definition of the expectation of a random vector.
10. State the definition of the expectation of a function of a random vector.
11. State the theorem on expectation under linear-affine combination of random variables.
12. State the theorem on expectation of the product of independent random variables.

Study question answers

1. See Definition 23.1.
2. See the explanations following Definition 23.1, especially also Example 23.4.
3. See Example 23.2.
4. See Definition 23.2.
5. See Definition 23.2.
6. See the explanations of Example 23.4, Definition 23.2, and Example 23.5.
7. See Definition 23.3.
8. See Theorem 23.1.
9. See Definition 23.4.
10. See Definition 23.5.
11. See Theorem 23.2.
12. See Theorem 23.3.

24 Variances

In this chapter, we introduce the concept of the *variance* of a random variable. Like the expectation, the variance serves as a single scalar characteristic that summarizes a probability distribution and is often used as a central characteristic quantity of random variables. The variance serves as a measure of the "variability" of a random variable. Ultimately, the variance of a random variable is a special covariance, namely the covariance of a random variable with itself, as discussed in the chapter on covariances. Closely related to the concept of variance is the concept of standard deviation, which we introduce in parallel. We supplement these concepts in the present chapter with their descriptive-statistical equivalents, the so-called *sample variance* and *sample standard deviation*, as well as with their occurrence in conditional distributions.

24.1 Definition

Definition 24.1 (Variance and standard deviation)**.** Let ξ be a random variable with finite expectation $\mathbb{E}(\xi)$. Then the *variance of* ξ is defined as

$$\mathbb{V}(\xi) := \mathbb{E}\left((\xi - \mathbb{E}(\xi))^2\right) \tag{24.1}$$

and the *standard deviation of* ξ is defined as

$$\mathbb{S}(\xi) := \sqrt{\mathbb{V}(\xi)}. \tag{24.2}$$

•

Thus, by Definition 23.3, the variance of a random variable ξ with outcome space $\mathcal{X}$ is the expectation of the function

$$f : \mathcal{X} \to \mathcal{Z}, x \mapsto f(x) := (x - \mathbb{E}(\xi))^2 \tag{24.3}$$

and therefore the expected squared deviation of a random variable from its expectation. Squaring the deviation of the random variable from its expectation is necessary because otherwise, by Theorem 23.1, it would always hold that

$$\mathbb{E}(\xi - \mathbb{E}(\xi)) = \mathbb{E}(\xi) - \mathbb{E}(\xi) = 0. \tag{24.4}$$

Intuitively, the variance thus measures the *dispersion* or *variability* of a random variable. In particular, the *Chebyshev inequality*, discussed in more detail in the chapter on inequalities, states that

$$\mathbb{P}(|\xi - \mathbb{E}(\xi)| \geq x) \leq \frac{\mathbb{V}(\xi)}{x^2}. \tag{24.5}$$

With the Chebyshev inequality, for example,

$$\mathbb{P}\left(|\xi - \mathbb{E}(\xi)| \geq 2\sqrt{\mathbb{V}(\xi)}\right) \leq \frac{\mathbb{V}(\xi)}{\left(2\sqrt{\mathbb{V}(\xi)}\right)^2} = \frac{1}{4} \tag{24.6}$$

and

$$\mathbb{P}\left(|\xi - \mathbb{E}(\xi)| \geq 3\sqrt{\mathbb{V}(\xi)}\right) \leq \frac{\mathbb{V}(\xi)}{\left(3\sqrt{\mathbb{V}(\xi)}\right)^2} = \frac{1}{9}. \tag{24.7}$$

For a random variable, it therefore always holds that the probability of an absolute deviation from twice its standard deviation is at most $1/4$, so, from a frequentist perspective, this applies to only about a quarter of its realizations, and the probability of an absolute deviation from three times its standard deviation is at most $1/9$, so, from a frequentist perspective, this applies to only about one tenth of its realizations, each independently of the exact form of the distribution of the random variable. Besides the variance, however, there are many other measures of the variability of random variables. Here we mention the *expected absolute deviation* of a random variable from its expectation,

$$\mathbb{A}(\xi) := \mathbb{E}(|\xi - \mathbb{E}(\xi)|), \tag{24.8}$$

and the so-called *entropy* of a random variable,

$$\mathbb{H}(\xi) := -\mathbb{E}(\ln p(x)). \tag{24.9}$$

We first illustrate the definition of variance in Definition 24.1 using a few examples.

Example 24.1 (Variance of a discrete random variable)**.** Let ξ be a random variable with outcome space $\mathcal{X} := \{-1, 0, 1\}$ and PMF

$$p(-1) = \frac{1}{4}, \quad p(0) = \frac{1}{2}, \quad p(1) = \frac{1}{4}. \tag{24.10}$$

Then

$$\mathbb{V}(\xi) = \frac{1}{2}. \tag{24.11}$$

△

Proof. By Definition 24.1, using the expectation of the same random variable determined in Example 23.1, we obtain

$$\begin{aligned}
\mathbb{V}(\xi) &= \mathbb{E}\left((\xi - \mathbb{E}(\xi))^2\right) \\
&= \mathbb{E}\left((\xi - 0)^2\right) \\
&= \mathbb{E}\left(\xi^2\right) \\
&= \sum_{x \in \mathcal{X}} x^2\, p(x) \\
&= (-1)^2 \cdot p(-1) + 0^2 \cdot p(0) + 1^2 \cdot p(1) \\
&= 1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \\
&= \frac{1}{2}.
\end{aligned} \tag{24.12}$$

□

Example 24.2 (Variance of a Bernoulli random variable)**.** Let $\xi \sim \mathrm{Bern}(\mu)$. Then the variance of ξ is given by

$$\mathbb{V}(\xi) = \mu(1 - \mu). \tag{24.13}$$

△

Proof. ξ is a discrete random variable and $\mathbb{E}(\xi) = \mu$. Thus,

$$
\begin{aligned}
\mathbb{V}(\xi) &= \mathbb{E}\left((\xi-\mu)^2\right) \\
&= \sum_{x\in\{0,1\}} (x-\mu)^2 \mathrm{Bern}(x;\mu) \\
&= (0-\mu)^2\mu^0(1-\mu)^{1-0} + (1-\mu)^2\mu^1(1-\mu)^{1-1} \\
&= \mu^2(1-\mu) + (1-\mu)^2\mu \\
&= \left(\mu^2 + (1-\mu)\mu\right)(1-\mu) \\
&= \left(\mu^2 + \mu - \mu^2\right)(1-\mu) \\
&= \mu(1-\mu).
\end{aligned}
\tag{24.14}
$$

□

Example 24.3 (Variance of a normally distributed random variable)**.** Let $\xi \sim N(\mu, \sigma^2)$. Then the variance of ξ is given by

$$\mathbb{V}(\xi) = \sigma^2. \tag{24.15}$$

△

Proof. We omit the proof. □

Analogously to the concept of the sample mean, one defines the descriptive-statistical equivalents of Definition 24.1 as follows.

Definition 24.2 (Sample variance and sample standard deviation)**.** Let $\xi_1, ..., \xi_n$ be random variables and let $\bar{\xi}$ be their sample mean. The *sample variance of* $\xi_1, ..., \xi_n$ is defined as

$$S^2 := \frac{1}{n-1}\sum_{i=1}^{n}(\xi_i - \bar{\xi})^2 \tag{24.16}$$

and the *sample standard deviation of* $\xi_1, ..., \xi_n$ is defined as

$$S := \sqrt{S^2}. \tag{24.17}$$

•

As in the context of the sample mean, it is central to recognize that $\mathbb{V}(\xi)$ and $\mathbb{S}(\xi)$ are characteristics of a random variable ξ, whereas S^2 and S are characteristics of a sample $\xi_1, ..., \xi_n$. Since S^2 and S are functions of random variables, S^2 and S are also themselves random variables. We denote realizations of S^2 and S by s^2 and s, respectively.

Example 24.4 (Sample variance and sample standard deviation)**.** As an example of the realizations of a sample variance and a sample standard deviation, we again consider the realizations of the random variables $\xi_1, ..., \xi_{10} \sim N(1, 2)$ shown in Table 24.1, with sample mean realization $\bar{x} = 0.68$.

Table 24.1 Sample realization

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
0.54	1.01	-3.28	0.35	2.75	-0.51	2.32	1.49	0.96	1.25

The sample variance realization is then

$$s^2 = \frac{1}{9}\sum_{i=1}^{10}(x_i - \bar{x})^2 = \frac{1}{9}\sum_{i=1}^{10}(x_i - 0.68)^2 = \frac{25.37}{9} = 2.82 \tag{24.18}$$

and the sample standard deviation realization is correspondingly

$$s = \sqrt{s^2} = \sqrt{2.82} = 1.68. \tag{24.19}$$

△

The concepts of variance and standard deviation are evidently equivalent to one another by Definition 24.1, since one is obtained directly from the other by taking the square root or by squaring. Historically, variance is closely interwoven both with the exponential argument of the normal distribution and with the sums of squares of analysis of variance, and is therefore often easier to handle than standard deviation in model building.

Both variance and sample variance measure the variability of a random variable in squared units. An example of this is the calculation of the sample variance in Example 24.4, if one imagines the realizations in Table 24.1 as length measurements in meters (m). According to Definition 24.2, the unit is then squared when the sample variance is calculated,

$$s^2 = \frac{1}{9}\sum_{i=1}^{10}\left(x_i \text{ m} - 0.68 \text{ m}\right)^2 = \frac{1}{9}\cdot 25.37 \text{ m}^2 = 2.82 \text{ m}^2. \tag{24.20}$$

The resulting measure of variability would therefore have the unit of an area (m^2), even though the original data represent lengths in meters (m). Taking the square root corrects the units back to the unit of the data and, as the standard deviation, then provides a measure of variability in the original unit,

$$s = \sqrt{2.82 \text{ m}^2} = \sqrt{2.82}\sqrt{\text{ m}^2} = 1.68 \text{ m}. \tag{24.21}$$

However, in most application contexts one is more interested in the ratios of variabilities of different samples, for which both sample variance and sample standard deviation are equally suitable.

24.2 Properties

The variance has the property of not taking negative values.

Theorem 24.1 (Non-negativity of variance)**.** *Let ξ be a random variable. Then*

$$\mathbb{V}(\xi) \geq 0. \tag{24.22}$$

∘

Proof. We consider the case of a discrete random variable. Then first

$$(x - \mathbb{E}(\xi))^2 \geq 0 \text{ for all } x \in \mathcal{X}. \tag{24.23}$$

Furthermore, for the PMF p of ξ,

$$p(x) \geq 0 \text{ for all } x \in \mathcal{X}. \tag{24.24}$$

Thus,

$$p(x)(x - \mathbb{E}(\xi))^2 \geq 0 \text{ for all } x \in \mathcal{X}. \tag{24.25}$$

But then

$$\mathbb{V}(\xi) = \sum_{x \in \mathcal{X}} p(x)(x - \mathbb{E}(\xi))^2 \geq 0. \tag{24.26}$$

The non-negativity of the variance for continuous random variables is shown analogously. □

Calculating variances is often facilitated by the following theorem, the so-called *variance shift theorem*, especially when the expectation of the squared random variable is easy to determine or known.

Theorem 24.2 (Variance shift theorem)**.** *Let ξ be a random variable. Then*

$$\mathbb{V}(\xi) = \mathbb{E}\left(\xi^2\right) - \mathbb{E}(\xi)^2. \tag{24.27}$$

∘

Proof. With the definition of variance and the linearity of expectation, we have

$$\begin{aligned}
\mathbb{V}(\xi) &= \mathbb{E}\left((\xi - \mathbb{E}(\xi))^2\right) \\
&= \mathbb{E}\left(\xi^2 - 2\xi\mathbb{E}(\xi) + \mathbb{E}(\xi)^2\right) \\
&= \mathbb{E}(\xi^2) - 2\mathbb{E}(\xi)\mathbb{E}(\xi) + \mathbb{E}\left(\mathbb{E}(\xi)^2\right) \\
&= \mathbb{E}(\xi^2) - 2\mathbb{E}(\xi)^2 + \mathbb{E}(\xi)^2 \\
&= \mathbb{E}(\xi^2) - \mathbb{E}(\xi)^2.
\end{aligned} \tag{24.28}$$

□

In analogy to Theorem 23.1, the following result holds for variance.

Theorem 24.3 (Variance and standard deviation under linear-affine transformation of a random variable)**.** *Let ξ be a random variable and let*

$$f : \mathcal{X} \to \mathcal{Z}, x \mapsto f(x) := ax + b \text{ for } a, b \in \mathbb{R} \tag{24.29}$$

be a linear-affine function. Then

$$\mathbb{V}(f(\xi)) = \mathbb{V}(a\xi + b) = a^2\mathbb{V}(\xi) \tag{24.30}$$

and

$$\mathbb{S}(f(\xi)) = \mathbb{S}(a\xi + b) = |a|\mathbb{S}(\xi). \tag{24.31}$$

∘

Proof. We obtain

$$
\begin{aligned}
\mathbb{V}(f(\xi)) &= \mathbb{V}(a\xi + b) \\
&= \mathbb{E}\left((a\xi + b - \mathbb{E}(a\xi + b))^2\right) \\
&= \mathbb{E}\left((a\xi + b - a\mathbb{E}(\xi) - b)^2\right) \\
&= \mathbb{E}\left((a\xi - a\mathbb{E}(\xi))^2\right) \\
&= \mathbb{E}\left(a(\xi - \mathbb{E}(\xi))^2\right) \\
&= \mathbb{E}\left(a^2(\xi - \mathbb{E}(\xi))^2\right) \\
&= a^2\mathbb{E}\left((\xi - \mathbb{E}(\xi))^2\right) \\
&= a^2\mathbb{V}(\xi).
\end{aligned}
\tag{24.32}
$$

Taking square roots while noting that

$$
a \geq 0 \Rightarrow a^2 \geq 0 \Rightarrow \sqrt{a^2} = a \geq 0 \Leftrightarrow \sqrt{a^2} = |a| \text{ for } a \geq 0 \tag{24.33}
$$

and

$$
a < 0 \Rightarrow a^2 > 0 \Rightarrow \sqrt{a^2} = -a > 0 \Leftrightarrow \sqrt{a^2} = |a| \text{ for } a < 0 \tag{24.34}
$$

then shows the result for the standard deviation. □

The variance of the linear combination of two independent random variables can be written as a modified linear combination of the variances of the random variables.

Theorem 24.4 (Variance under linear combination of two independent random variables)**.** *Let ξ_1 and ξ_2 be two independent random variables and let $a_1, a_2 \in \mathbb{R}$. Then*

$$
\mathbb{V}\left(a_1\xi_1 + a_2\xi_2\right) = a_1^2\mathbb{V}(\xi_1) + a_2^2\mathbb{V}(\xi_1). \tag{24.35}
$$

∘

Proof. First, we record that, by Theorem 23.2,

$$
\mathbb{E}\left(a_1\xi_1 + a_2\xi_2\right) = a_1\mathbb{E}(\xi_1) + a_2\mathbb{E}(\xi_2). \tag{24.36}
$$

Furthermore, by Theorem 23.3, we have

$$
\begin{aligned}
\mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2))\right) &= \mathbb{E}\left(\xi_1\xi_2 - \mathbb{E}(\xi_1)\xi_2 - \xi_1\mathbb{E}(\xi_2) + \mathbb{E}(\xi_1)\mathbb{E}(\xi_2)\right) \\
&= \mathbb{E}(\xi_1\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) + \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) \\
&= \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) \\
&= 0.
\end{aligned}
\tag{24.37}
$$

It follows that

$$
\begin{aligned}
\mathbb{V}(a_1\xi_1 + a_2\xi_2) &= \mathbb{E}\left((a_1\xi_1 + a_2\xi_2 - \mathbb{E}\left(a_1\xi_1 + a_2\xi_2\right))^2\right) \\
&= \mathbb{E}\left((a_1\xi_1 + a_2\xi_2 - a_1\mathbb{E}(\xi_1) - a_2\mathbb{E}(\xi_2))^2\right) \\
&= \mathbb{E}\left((a_1\xi_1 - a_1\mathbb{E}(\xi_1) + a_2\xi_2 - a_2\mathbb{E}(\xi_2))^2\right) \\
&= \mathbb{E}\left(a_1(\xi_1 - \mathbb{E}(\xi_1)) + (a_2(\xi_2 - \mathbb{E}(\xi_2)))^2\right) \\
&= \mathbb{E}\left(a_1(\xi_1 - \mathbb{E}(\xi_1)))^2 + 2a_1(\xi_1 - \mathbb{E}(\xi_1))(a_2(\xi_2 - \mathbb{E}(\xi_2)) + (a_2(\xi_2 - \mathbb{E}(\xi_2)))^2\right) \\
&= \mathbb{E}\left(a_1^2(\xi_1 - \mathbb{E}(\xi_1))^2 + 2a_1a_2(\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2)) + a_2^2(\xi_2 - \mathbb{E}(\xi_2))^2\right) \\
&= a_1^2\mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))^2\right) + 2a_1a_2\mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2))\right) + a_2^2\mathbb{E}\left((\xi_2 - \mathbb{E}(\xi_2))^2\right) \\
&= a_1^2\mathbb{V}(\xi_1) + 2a_1a_2\mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2))\right) + a_2^2\mathbb{V}(\xi_2) \\
&= a_1^2\mathbb{V}(\xi_1) + 2a_1a_2 \cdot 0 + a_2^2\mathbb{V}(\xi_2) \\
&= a_1^2\mathbb{V}(\xi_1) + a_2^2\mathbb{V}(\xi_2).
\end{aligned}
\tag{24.38}
$$

□

More general statements about the variance of the sum of non-independent random variables can be made only with the help of the concept of covariance introduced in the chapter on covariances.

24.3 Conditional variance

Definition 24.3 (Conditional variance)**.** Given a random vector $\xi := (\xi_1, \xi_2)$ with outcome space $\mathcal{X} := \mathcal{X}_1 \times \mathcal{X}_2$. Then the *conditional variance of* ξ_1 *given* $\xi_2 = x_2$ is defined as

$$\mathbb{V}(\xi_1|\xi_2 = x_2) = \mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1|\xi_2 = x_2))^2 \,|\xi_2 = x_2\right) \tag{24.39}$$

and the *conditional standard deviation of* ξ_1 *given* $\xi_2 = x_2$ is defined as

$$\mathbb{S}(\xi_1|\xi_2 = x_2) = \sqrt{\mathbb{V}(\xi_1|\xi_2 = x_2)}. \tag{24.40}$$

•

Conditional variance is defined in the sense of conditional expectation; like a conditional expectation, a conditional variance is therefore a random variable. The shift theorem also holds for conditional variance, as stated in the following theorem.

Theorem 24.5 (Shift theorem of conditional variance)**.** *Given a random vector* $\xi := (\xi_1, \xi_2)$*, we have*

$$\mathbb{V}(\xi_1|\xi_2 = x_2) = \mathbb{E}\left(\xi_1^2|\xi_2 = x_2\right) - \mathbb{E}(\xi_1|\xi_2 = x_2)^2 \tag{24.41}$$

∘

Proof. We restrict ourselves to the proof for a discrete random vector. The proof for a continuous random vector follows analogously. With the definition of conditional expectation (Definition 23.6), we have

$$\begin{aligned}
\mathbb{V}(\xi_1|\xi_2 = x_2) &= \mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1|\xi_2 = x_2))^2 \,|\xi_2 = x_2\right) \\
&= \mathbb{E}\left(\xi_1^2 - 2\xi_1\mathbb{E}(\xi_1|\xi_2 = x_2) + \mathbb{E}(\xi_1|\xi_2 = x_2)^2|\xi_2 = x_2\right) \\
&= \sum_{x_1 \in \mathcal{X}_1} p(x_1|x_2)\left(x_1^2 - 2x_1\mathbb{E}(\xi_1|\xi_2 = x_2) + \mathbb{E}(\xi_1|\xi_2 = x_2)^2\right) \\
&= \sum_{x_1 \in \mathcal{X}_1} x_1^2 p(x_1|x_2) - \sum_{x_1 \in \mathcal{X}_1} 2x_1\mathbb{E}(\xi_1|\xi_2 = x_2)p(x_1|x_2) + \sum_{x_1 \in \mathcal{X}_1} \mathbb{E}(\xi_1|\xi_2 = x_2)^2 p(x_1|x_2) \\
&= \mathbb{E}(\xi_1^2|\xi_2 = x_2) - 2\mathbb{E}(\xi_1|\xi_2 = x_2)\sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1|x_2) + \mathbb{E}(\xi_1|\xi_2 = x_2)^2 \sum_{x_1 \in \mathcal{X}_1} p(x_1|x_2) \\
&= \mathbb{E}(\xi_1^2|\xi_2 = x_2) - 2\mathbb{E}(\xi_1|\xi_2 = x_2)\mathbb{E}(\xi_1|\xi_2 = x_2) + \mathbb{E}(\xi_1|\xi_2 = x_2)^2 \cdot 1 \\
&= \mathbb{E}(\xi_1^2|\xi_2 = x_2) - 2\mathbb{E}(\xi_1|\xi_2 = x_2)^2 + \mathbb{E}(\xi_1|\xi_2 = x_2)^2 \\
&= \mathbb{E}(\xi_1^2|\xi_2 = x_2) - \mathbb{E}(\xi_1|\xi_2 = x_2)^2
\end{aligned} \tag{24.42}$$

□

The following theorem, the so-called *law of iterated variance*, establishes a connection between the variance of a random variable, its conditional variance, and the variance of its conditional expectation.

Theorem 24.6 (Law of iterated variance)**.** *Given a random vector* $\xi := (\xi_1, \xi_2)$*, we have*

$$\mathbb{V}(\xi_1) = \mathbb{E}(\mathbb{V}(\xi_1|\xi_2)) + \mathbb{V}(\mathbb{E}(\xi_1|\xi_2)). \tag{24.43}$$

∘

Proof. We restrict ourselves to the proof for a discrete random vector. The proof for a continuous random vector follows analogously. With the variance shift theorem (Theorem 24.2), the law of iterated expectation (Theorem 23.4), and the shift theorem of conditional variance (Theorem 24.5), we have

$$
\begin{aligned}
\mathbb{V}(\xi_1) &= \mathbb{E}(\xi_1^2) - \mathbb{E}(\xi_1)^2 \\
&= \mathbb{E}\left(\mathbb{E}(\xi_1^2|\xi_2)\right) - \mathbb{E}\left(\mathbb{E}(\xi_1|\xi_2)\right)^2 \\
&= \sum_{x_2 \in \mathcal{X}_2} p(x_2)\mathbb{E}(\xi_1^2|\xi_2 = x_2) - \mathbb{E}\left(\mathbb{E}(\xi_1|\xi_2)\right)^2 \\
&= \sum_{x_2 \in \mathcal{X}_2} p(x_2)\left(\mathbb{V}(\xi_1|\xi_2 = x_2) + \mathbb{E}(\xi_1|\xi_2 = x_2)^2\right) - \mathbb{E}\left(\mathbb{E}(\xi_1|\xi_2)\right)^2 \\
&= \sum_{x_2 \in \mathcal{X}_2} p(x_2)\mathbb{V}(\xi_1|\xi_2 = x_2) + \sum_{x_2 \in \mathcal{X}_2} p(x_2)\mathbb{E}(\xi_1|\xi_2 = x_2)^2 - \mathbb{E}\left(\mathbb{E}(\xi_1|\xi_2)\right)^2 \\
&= \mathbb{E}(\mathbb{V}(\xi_1|\xi_2)) + \left(\mathbb{E}\left(\mathbb{E}(\xi_1|\xi_2)^2\right) - \mathbb{E}\left(\mathbb{E}(\xi_1|\xi_2)\right)^2\right) \\
&= \mathbb{E}(\mathbb{V}(\xi_1|\xi_2)) + \mathbb{V}\left(\mathbb{E}(\xi_1|\xi_2)\right)
\end{aligned}
\tag{24.44}
$$

□

Study questions

1. State the definition of the variance and standard deviation of a random variable.
2. Explain the definition of the variance and standard deviation of a random variable.
3. Compute the variance of a Bernoulli random variable.
4. State the definition of sample variance and sample standard deviation.
5. Explain the definitions of sample variance and sample standard deviation.
6. State the theorem on the non-negativity of variance.
7. State the theorem on the variance shift theorem.
8. State the theorem on variance and standard deviation under linear-affine transformation of a random variable.
9. State the theorem on variance under linear combination of two independent random variables.

Study question answers

1. See Definition 24.1.
2. See the explanations of Definition 24.1.
3. See Example 24.2.
4. See Definition 24.2.
5. See the explanations of Definition 24.2 and Example 24.4.
6. See Theorem 24.1.
7. See Theorem 24.2.
8. See Theorem 24.3.
9. See Theorem 24.4.

25 Covariances

With the *covariance* of two random variables and its standardized form, the *correlation*, we introduce in this chapter scalar quantities for the linear-affine dependence of random variables.

25.1 Definitions

Definition 25.1 (Covariance and correlation of two random variables)**.** The *covariance* of two random variables ξ_1 and ξ_2 is defined as

$$\mathbb{C}(\xi_1, \xi_2) := \mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2))\right). \tag{25.1}$$

The *correlation* of two random variables ξ_1 and ξ_2 with $\mathbb{S}(\xi_1) > 0$ and $\mathbb{S}(\xi_2) > 0$ is defined as

$$\rho(\xi_1, \xi_2) := \frac{\mathbb{C}(\xi_1, \xi_2)}{\mathbb{S}(\xi_1)\mathbb{S}(\xi_2)}. \tag{25.2}$$

•

The covariance of a random variable with itself is its variance, since

$$\mathbb{C}(\xi_1, \xi_1) = \mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))(\xi_1 - \mathbb{E}(\xi_1))\right) = \mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))^2\right) = \mathbb{V}(\xi_1). \tag{25.3}$$

The number $\rho(\xi_1, \xi_2)$ is also called the *correlation coefficient of* ξ_1 *and* ξ_2. If $\rho(\xi_1, \xi_2) = 0$, then ξ_1 and ξ_2 are called *uncorrelated*. In the chapter on inequalities, the correlation inequality shows that

$$-1 \leq \rho(\xi_1, \xi_2) \leq 1. \tag{25.4}$$

The definition of covariance in Definition 25.1 implicitly makes clear that the expectations appearing in it are of different kinds. The definition of covariance first assumes that ξ_1 and ξ_2 are the components of a random vector $\xi := (\xi_1, \xi_2)$ with outcome space $\mathcal{X}_1 \times \mathcal{X}_2$, joint distribution $\mathbb{P}(\xi_1, \xi_2)$, and marginal distributions $\mathbb{P}(\xi_1)$ and $\mathbb{P}(\xi_2)$. The expectations $\mathbb{E}(\xi_1)$ and $\mathbb{E}(\xi_2)$ are the expectations of the components ξ_1 and ξ_2 with respect to these marginal distributions. The covariance itself is then the expectation of the function

$$f : \mathcal{X}_1 \times \mathcal{X}_2 \to \mathcal{Z}, (x_1, x_2) \mapsto f(x_1, x_2) := (x_1 - \mathbb{E}(\xi_1))(x_2 - \mathbb{E}(\xi_2)) \tag{25.5}$$

with respect to the joint distribution $\mathbb{P}(\xi_1, \xi_2)$. We illustrate this with the example of a discrete random vector.

Example 25.1 (Covariance of two discrete random variables)**.** Let $\xi := (\xi_1, \xi_2)$ be a discrete random vector with outcome space $\mathcal{X} := \{1,2\} \times \{1,2,3\}$ and the joint PMF $p(x_1, x_2)$ and marginal PMFs $p(x_1)$ and $p(x_2)$ shown in Table 25.1.

Table 25.1 Joint and marginal PMFs of the random vector ξ

$p(x_1,x_2)$	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$	$p(x_1)$
$x_1 = 1$	0.10	0.05	0.15	0.30
$x_1 = 2$	0.60	0.05	0.05	0.70
$p(x_2)$	0.70	0.10	0.20	

Using the definition of the covariance of ξ_1 and ξ_2 and taking Equation 25.5 into account, we have

$$\begin{aligned}\mathbb{C}(\xi_1, \xi_2) &= \mathbb{E}(f(\xi_1, \xi_2)) \\ &= \sum_{x_1=1}^{2}\sum_{x_2=1}^{3} f(x_1, x_2)p(x_1, x_2) \\ &= \sum_{x_1=1}^{2}\sum_{x_2=1}^{3} (x_1 - \mathbb{E}(\xi_1))(x_2 - \mathbb{E}(\xi_2))p(x_1, x_2).\end{aligned} \tag{25.6}$$

Substituting the values from Table 25.1 first gives

$$\mathbb{E}(\xi_1) = \sum_{x_1=1}^{2} x_1 p(x_1) = 1 \cdot 0.3 + 2 \cdot 0.7 = 1.7 \tag{25.7}$$

and

$$\mathbb{E}(\xi_2) = \sum_{x_2=1}^{3} x_2 p(x_2) = 1 \cdot 0.7 + 2 \cdot 0.1 + 3 \cdot 0.2 = 1.5. \tag{25.8}$$

Thus, finally,

$$\begin{aligned}\mathbb{C}(\xi_1, \xi_2) &= \sum_{x_1=1}^{2}\sum_{x_2=1}^{3} (x_1 - 1.7)\,(x_2 - 1.5)\,p(x_1, x_2) \\ &= \sum_{x_1=1}^{2} (x_1 - 1.7)(1 - 1.5)p(x_1, 1) + (x_1 - 1.7)(2 - 1.5)p(x_1, 2) + (x_1 - 1.7)(3 - 1.5)p(x_1, 3) \\ &= \quad (1 - 1.7)(1 - 1.5)p(1, 1) + (1 - 1.7)(2 - 1.5)p(1, 2) + (1 - 1.7)(3 - 1.5)p(1, 3) \\ &\quad\;\; (2 - 1.7)(1 - 1.5)p(2, 1) + (2 - 1.7)(2 - 1.5)p(2, 2) + (2 - 1.7)(3 - 1.5)p(2, 3) \\ &= (-0.7) \cdot (-0.5) \cdot 0.10 + (-0.7) \cdot 0.5 \cdot 0.05 + (-0.7) \cdot 1.5 \cdot 0.15 \\ &\quad + 0.3 \cdot (-0.5) \cdot 0.60 \; + 0.3 \cdot 0.5 \cdot 0.05 \quad + 0.3 \cdot 1.5 \cdot 0.05 \\ &= 0.035 - 0.0175 - 0.1575 - 0.09 + 0.0075 + 0.0225 \\ &= -0.2.\end{aligned} \tag{25.9}$$

The covariance of the random variables ξ_1 and ξ_2 with the distribution specified in the table above is therefore $\mathbb{C}(\xi_1, \xi_2) = -0.2$. Unlike the variance, the covariance can therefore evidently also take negative values.

△

25.2 Properties

We next consider some basic properties of covariance.

Theorem 25.1 (Symmetry of covariance and correlation)**.** *Let ξ_1 and ξ_2 be two random variables. Then*

$$\mathbb{C}(\xi_1,\xi_2) = \mathbb{C}(\xi_2,\xi_1) \text{ and } \rho(\xi_1,\xi_2) = \rho(\xi_2,\xi_1). \tag{25.10}$$

∘

Proof. By the commutativity of multiplication, we first have

$$\mathbb{C}(\xi_1,\xi_2) = \mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2))\right) = \mathbb{E}\left((\xi_2 - \mathbb{E}(\xi_2))(\xi_1 - \mathbb{E}(\xi_1))\right) = \mathbb{C}(\xi_2,\xi_1). \tag{25.11}$$

Furthermore,

$$\rho(\xi_1,\xi_2) = \frac{\mathbb{C}(\xi_1,\xi_2)}{\mathbb{S}(\xi_1)\mathbb{S}(\xi_2)} = \frac{\mathbb{C}(\xi_2,\xi_1)}{\mathbb{S}(\xi_2)\mathbb{S}(\xi_1)} = \rho(\xi_2,\xi_1). \tag{25.12}$$

□

As with the calculation of variances, calculating covariances is sometimes made easier by the following theorem.

Theorem 25.2 (Covariance shift theorem)**.** *Let ξ_1 and ξ_2 be random variables. Then*

$$\mathbb{C}(\xi_1,\xi_2) = \mathbb{E}(\xi_1\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2). \tag{25.13}$$

∘

Proof. By Definition 25.1,

$$\begin{aligned}\mathbb{C}(\xi_1,\xi_2) &= \mathbb{E}\left((\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2))\right)\\ &= \mathbb{E}\left(\xi_1\xi_2 - \xi_1\mathbb{E}(\xi_2) - \mathbb{E}(\xi_1)\xi_2 + \mathbb{E}(\xi_1)\mathbb{E}(\xi_2)\right)\\ &= \mathbb{E}(\xi_1\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) + \mathbb{E}(\xi_1)\mathbb{E}(\xi_2)\\ &= \mathbb{E}(\xi_1\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2).\end{aligned} \tag{25.14}$$

□

Of course, Theorem 25.2 is only truly useful if $\mathbb{E}(\xi_1\xi_2)$ is easy to compute. The variance shift theorem from Theorem 24.2 follows from Theorem 25.2 for the case $\xi_1 = \xi_2 := \xi$ by

$$\mathbb{V}(\xi) = \mathbb{C}(\xi,\xi) = \mathbb{E}(\xi\xi) - \mathbb{E}(\xi)\mathbb{E}(\xi) = \mathbb{E}(\xi^2) - \mathbb{E}(\xi)\mathbb{E}(\xi). \tag{25.15}$$

With regard to the covariance $\mathbb{C}(\xi_1,\xi_2)$, one is also interested in how it behaves under the application of linear-affine transformations to ξ_1 and ξ_2. The following result shows that the covariance of two random variables depends on their scale, whereas the correlation does not. This motivates the frequent use of correlation as a general measure of association that is comparable across different application scenarios.

Theorem 25.3 (Covariance and correlation under linear-affine transformation)**.** *Let ξ_1 and ξ_2 be random variables with joint outcome space $\mathcal{X}_1 \times \mathcal{X}_2$, and let*

$$\begin{aligned}f_1 &: \mathcal{X}_1 \to \mathcal{Z}_1, x_1 \mapsto f_1(x_1) := ax_1 + b\\ f_2 &: \mathcal{X}_2 \to \mathcal{Z}_2, x_2 \mapsto f_2(x_2) := cx_2 + d\end{aligned} \tag{25.16}$$

be two linear-affine functions for $a, b, c, d \in \mathbb{R}$. Then

$$\mathbb{C}(f_1(\xi_1), f_2(\xi_2)) = \mathbb{C}(a\xi_1 + b, c\xi_2 + d) = ac\mathbb{C}(\xi_1,\xi_2) \tag{25.17}$$

and

$$\rho(f_1(\xi_1), f_2(\xi_2)) = \rho(a\xi_1 + b, c\xi_2 + d) = \rho(\xi_1, \xi_2). \tag{25.18}$$

∘

Proof. First,

$$\begin{aligned}
\mathbb{C}(a\xi_1 + b, c\xi_2 + d) &= \mathbb{E}((a\xi_1 + b - \mathbb{E}(a\xi_1 + b))(c\xi_2 + d - \mathbb{E}(c\xi_2 + d))) \\
&= \mathbb{E}((a\xi_1 + b - a\mathbb{E}(\xi_1) - b)(c\xi_2 + d - c\mathbb{E}(\xi_2) - d)) \\
&= \mathbb{E}(a(\xi_1 - \mathbb{E}(\xi_1))c(\xi_2 - \mathbb{E}(\xi_2))) \\
&= \mathbb{E}(ac((\xi_1 - \mathbb{E}(\xi_1))(\xi_2 - \mathbb{E}(\xi_2)))) \\
&= ac\mathbb{C}(\xi_1, \xi_2).
\end{aligned} \tag{25.19}$$

Thus,

$$\begin{aligned}
\rho(a\xi_1 + b, c\xi_2 + d) &= \frac{\mathbb{C}(a\xi_1 + b, c\xi_2 + d)}{\sqrt{\mathbb{V}(a\xi_1 + b)}\sqrt{\mathbb{V}(c\xi_2 + d)}} \\
&= \frac{ac\mathbb{C}(\xi_1, \xi_2)}{\sqrt{a^2\mathbb{V}(\xi_1)}\sqrt{c^2\mathbb{V}(\xi_2)}} \\
&= \frac{ac\mathbb{C}(\xi_1, \xi_2)}{a\mathbb{S}(\xi_1)c\mathbb{S}(\xi_2)} \\
&= \frac{\mathbb{C}(\xi_1, \xi_2)}{\mathbb{S}(\xi_1)\mathbb{S}(\xi_2)} \\
&= \rho(\xi_1, \xi_2).
\end{aligned} \tag{25.20}$$

□

With the concept of covariance, it is possible to make more general statements about the variances of sums and differences of random variables than in Chapter 24, where only the variances of sums of independent random variables were considered. With the following theorem, we first investigate the very general case of the covariance of two random variables that arise as linear combinations of n random variables $\xi_1, ..., \xi_n$ and m random variables $\zeta_1, ..., \zeta_m$, with real constants added. This yields a number of special cases that are important in applications, especially in classical test theory.

Theorem 25.4 (Covariance of linear combinations of random variables)**.** *Let n random variables $\xi_1, ..., \xi_n$ and $n+1$ real constants $a_0, a_1, ..., a_n$ be given, and let m random variables $\zeta_1, ..., \zeta_m$ and $m+1$ real constants $b_0, b_1, ..., b_m$ be given. Then*

$$\mathbb{C}\left(a_0 + \sum_{i=1}^n a_i\xi_i, b_0 + \sum_{j=1}^m b_j\zeta_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \mathbb{C}(\xi_i, \zeta_j). \tag{25.21}$$

∘

Proof.

$$
\begin{aligned}
&\mathbb{C}\left(a_0 + \sum_{i=1}^n a_i\xi_i, b_0 + \sum_{j=1}^m b_j\zeta_j\right) \\
&= \mathbb{E}\left(\left(a_0 + \sum_{i=1}^n a_i\xi_i - \mathbb{E}\left(a_0 + \sum_{i=1}^n a_i\xi_i\right)\right)\left(b_0 + \sum_{j=1}^m b_j\zeta_j - \mathbb{E}\left(b_0 + \sum_{j=1}^m b_j\zeta_j\right)\right)\right) \\
&= \mathbb{E}\left(\left(a_0 + \sum_{i=1}^n a_i\xi_i - a_0 - \mathbb{E}\left(\sum_{i=1}^n a_i\xi_i\right)\right)\left(b_0 + \sum_{j=1}^m b_j\zeta_j - b_0 - \mathbb{E}\left(\sum_{j=1}^m b_j\zeta_j\right)\right)\right) \\
&= \mathbb{E}\left(\left(\sum_{i=1}^n a_i\xi_i - \sum_{i=1}^n a_i\mathbb{E}(\xi_i)\right)\left(\sum_{j=1}^m b_j\zeta_j - \sum_{j=1}^m b_j\mathbb{E}(\zeta_j)\right)\right) \\
&= \mathbb{E}\left(\sum_{i=1}^n a_i\xi_i\sum_{j=1}^m b_j\zeta_j - \sum_{i=1}^n a_i\xi_i\sum_{j=1}^m b_j\mathbb{E}(\zeta_j) - \sum_{i=1}^n a_i\mathbb{E}(\xi_i)\sum_{j=1}^m b_j\zeta_j + \sum_{i=1}^n a_i\mathbb{E}(\xi_i)\sum_{j=1}^m b_j\mathbb{E}(\zeta_j)\right) \\
&= \mathbb{E}\left(\sum_{i=1}^n a_i\xi_i\sum_{j=1}^m b_j\zeta_j\right) - \mathbb{E}\left(\sum_{i=1}^n a_i\xi_i\sum_{j=1}^m b_j\mathbb{E}(\zeta_j)\right) - \mathbb{E}\left(\sum_{i=1}^n a_i\mathbb{E}(\xi_i)\sum_{j=1}^m b_j\zeta_j\right) + \mathbb{E}\left(\sum_{i=1}^n a_i\mathbb{E}(\xi_i)\sum_{j=1}^m b_j\mathbb{E}(\zeta_j)\right) \\
&= \sum_{i=1}^n\sum_{j=1}^m a_ib_j\mathbb{E}(\xi_i\zeta_j) - 2\sum_{i=1}^n\sum_{j=1}^m a_ib_j\mathbb{E}(\xi_i)\mathbb{E}(\zeta_j) + \sum_{i=1}^n\sum_{j=1}^m a_ib_j\mathbb{E}(\xi_i)\mathbb{E}(\zeta_j) \\
&= \sum_{i=1}^n\sum_{j=1}^m a_ib_j\left(\mathbb{E}(\xi_i\zeta_j) - 2\mathbb{E}(\xi_i)\mathbb{E}(\zeta_j) + \mathbb{E}(\xi_i)\mathbb{E}(\zeta_j)\right) \\
&= \sum_{i=1}^n\sum_{j=1}^m a_ib_j\mathbb{E}\left((\xi_i - \mathbb{E}(\xi_i))(\zeta_j - \mathbb{E}(\zeta_j))\right) \\
&= \sum_{i=1}^n\sum_{j=1}^m a_ib_j\mathbb{C}(\xi_i, \zeta_j).
\end{aligned}
$$

□

With the following theorem, we now consider a first special case of Theorem 25.4.

Theorem 25.5 (Covariance under pairwise addition of random variables)**.** *Let $\xi_1, \xi_2, \zeta_1, \zeta_2$ be four random variables. Then*

$$
\mathbb{C}(\xi_1 + \xi_2, \zeta_1 + \zeta_2) = \mathbb{C}(\xi_1, \zeta_1) + \mathbb{C}(\xi_1, \zeta_2) + \mathbb{C}(\xi_2, \zeta_1) + \mathbb{C}(\xi_2, \zeta_2). \tag{25.22}
$$

∘

Proof. Let $n := 2, m := 2, a_0 = b_0 := 0$, and $a_i = b_i := 1$ for $i = 1, 2$. Then, by Theorem 25.4,

$$
\begin{aligned}
\mathbb{C}(\xi_1 + \xi_2, \zeta_1 + \zeta_2) &= \mathbb{C}\left(a_0 + \sum_{i=1}^2 a_i\xi_i, b_0 + \sum_{j=1}^2 b_j\zeta_j\right) \\
&= \sum_{i=1}^2\sum_{j=1}^2 a_ib_j\mathbb{C}(\xi_i, \zeta_j) \\
&= \sum_{i=1}^2\sum_{j=1}^2 \mathbb{C}(\xi_i, \zeta_j) \\
&= \mathbb{C}(\xi_1, \zeta_1) + \mathbb{C}(\xi_1, \zeta_2) + \mathbb{C}(\xi_2, \zeta_1) + \mathbb{C}(\xi_2, \zeta_2).
\end{aligned} \tag{25.23}
$$

□

The following theorem states how, in general, the variance of a linear combination of random variables with an added constant can be determined.

Theorem 25.6 (Variance of a linear combination of random variables)**.** *Let n random variables $\xi_1, ..., \xi_n$ and $n+1$ real constants $a_0, a_1, ..., a_n$ be given. Then*

$$\mathbb{V}\left(a_0 + \sum_{i=1}^{n} a_i \xi_i\right) = \sum_{i=1}^{n} a_i^2 \mathbb{V}(\xi_i) + 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j). \tag{25.24}$$

∘

Proof. With $\mathbb{V}(\xi) = \mathbb{C}(\xi, \xi)$ and Theorem 25.4, we first have

$$\begin{aligned}
\mathbb{V}\left(a_0 + \sum_{i=1}^{n} a_i \xi_i\right) & = \mathbb{C}\left(a_0 + \sum_{i=1}^{n} a_i \xi_i, a_0 + \sum_{i=1}^{n} a_i \xi_i\right) \\
& = \mathbb{C}\left(a_0 + \sum_{i=1}^{n} a_i \xi_i, a_0 + \sum_{j=1}^{n} a_j \xi_j\right) \\
& = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j) \\
& = \sum_{i=1}^{n}\sum_{\substack{j=1 \\ j=i}}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j) + \sum_{i=1}^{n}\sum_{\substack{j=1 \\ j\neq i}}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j) \\
& = \sum_{i=1}^{n} a_i a_i \mathbb{C}(\xi_i, \xi_i) + \sum_{i=1}^{n}\sum_{\substack{j=1 \\ j\neq i}}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j) \\
& = \sum_{i=1}^{n} a_i^2 \mathbb{V}(\xi_i) + \sum_{i=1}^{n}\sum_{\substack{j=1 \\ j\neq i}}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j) \\
& = \sum_{i=1}^{n} a_i^2 \mathbb{V}(\xi_i) + 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j).
\end{aligned} \tag{25.25}$$

In the fourth equality, the double sum was split into those terms for which $i = j$ and those for which $i \neq j$. In the seventh equality, we used that

$$a_i a_j \mathbb{C}(\xi_i, \xi_j) = a_j a_i \mathbb{C}(\xi_j, \xi_i). \tag{25.26}$$

We illustrate the resulting identity

$$\sum_{i=1}^{n}\sum_{\substack{j=1 \\ j\neq i}}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j) = 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} a_i a_j \mathbb{C}(\xi_i, \xi_j), \tag{25.27}$$

for the example $n = 3$ below. Analogously, one may imagine summing over all elements except the diagonal elements of the $i = 1, \ldots, n$ rows and $j = 1, \ldots, n$ columns of a symmetric matrix with entries $a_i a_j \mathbb{C}(\xi_i, \xi_j)$. The left side of the equation above then corresponds to summing row by row over all column entries except the one in the column of the row currently considered, that is, the diagonal entry. The right side corresponds to exploiting the symmetry of the matrix, that is, taking into account that the entries of the matrix above and below the diagonal are identical, so it is enough to add the entries of the upper triangle and double them. For each row $i = 1, ..., n$, only the columns $j = i+1, ..., n$ to the right of the diagonal are considered and their entries are summed. The last row contains only a diagonal element, which is not part of the sum, so the index of the outer sum only runs to $n-1$. Concretely, for $n := 3$,

$$
\begin{aligned}
&\sum_{i=1}^{3}\sum_{\substack{j=1\\ j\neq i}}^{3} a_i a_j \mathbb{C}(\xi_i,\xi_j)\\
&= \sum_{\substack{j=1\\ j\neq 1}}^{3} a_1 a_j \mathbb{C}(\xi_1,\xi_j) + \sum_{\substack{j=1\\ j\neq 2}}^{3} a_2 a_j \mathbb{C}(\xi_2,\xi_j) + \sum_{\substack{j=1\\ j\neq 3}}^{3} a_3 a_j \mathbb{C}(\xi_3,\xi_j)\\
&= a_1a_2\mathbb{C}(\xi_1,\xi_2) + a_1a_3\mathbb{C}(\xi_1,\xi_3) + a_2a_1\mathbb{C}(\xi_2,\xi_1) + a_2a_3\mathbb{C}(\xi_2,\xi_3) + a_3a_1\mathbb{C}(\xi_3,\xi_1) + a_3a_2\mathbb{C}(\xi_3,\xi_2)\\
&= a_1a_2\mathbb{C}(\xi_1,\xi_2) + a_2a_1\mathbb{C}(\xi_2,\xi_1) + a_1a_3\mathbb{C}(\xi_1,\xi_3) + a_3a_1\mathbb{C}(\xi_3,\xi_1) + a_2a_3\mathbb{C}(\xi_2,\xi_3) + a_3a_2\mathbb{C}(\xi_3,\xi_2)\\
&= 2a_1a_2\mathbb{C}(\xi_1,\xi_2) + 2a_1a_3\mathbb{C}(\xi_1,\xi_3) + 2a_2a_3\mathbb{C}(\xi_2,\xi_3)\\
&= 2\left(a_1a_2\mathbb{C}(\xi_1,\xi_2) + a_1a_3\mathbb{C}(\xi_1,\xi_3) + a_2a_3\mathbb{C}(\xi_2,\xi_3)\right)\\
&= 2\left(\sum_{j=2}^{3} a_1 a_j \mathbb{C}(\xi_1,\xi_j) + \sum_{j=3}^{3} a_2 a_j \mathbb{C}(\xi_2,\xi_j)\right)\\
&= 2\left(\sum_{j=1+1}^{3} a_1 a_j \mathbb{C}(\xi_1,\xi_j) + \sum_{j=2+1}^{3} a_2 a_j \mathbb{C}(\xi_2,\xi_j)\right)\\
&= 2\sum_{i=1}^{2}\sum_{j=i+1}^{3} a_i a_j \mathbb{C}(\xi_i,\xi_j).
\end{aligned}
\tag{25.28}
$$

□

Theorem 25.7 (Variances of special linear combinations of random variables)**.**

(1) (Variance under addition of two random variables). Let two random variables ξ and ζ be given. Then

$$\mathbb{V}(\xi+\zeta) = \mathbb{V}(\xi) + \mathbb{V}(\zeta) + 2\mathbb{C}(\xi,\zeta). \tag{25.29}$$

(2) (Variance under subtraction of two random variables). Let two random variables ξ and ζ be given. Then

$$\mathbb{V}(\xi-\zeta) = \mathbb{V}(\xi) + \mathbb{V}(\zeta) - 2\mathbb{C}(\xi,\zeta). \tag{25.30}$$

∘

Proof. Let $n := 2$, $\xi_1 := \xi$, $\xi_2 := \zeta$, $a_0 := 0$, $a_1 := 1$, and $a_2 := 1$. Then, by Theorem 25.6,

$$
\begin{aligned}
\mathbb{V}(\xi+\zeta) &= \mathbb{V}(a_0 + a_1\xi_1 + a_2\xi_2)\\
&= \mathbb{V}\left(a_0 + \sum_{i=1}^{2} a_i\xi_i\right)\\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{i=1}^{2-1}\sum_{j=i+1}^{2} a_i a_j \mathbb{C}(\xi_i,\xi_j)\\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{i=1}^{1}\sum_{j=i+1}^{2} a_i a_j \mathbb{C}(\xi_i,\xi_j)\\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{j=1+1}^{2} a_i a_j \mathbb{C}(\xi_1,\xi_j)\\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{j=2}^{2} a_1 a_j \mathbb{C}(\xi_1,\xi_j)\\
&= a_1^2\mathbb{V}(\xi_1) + a_2^2\mathbb{V}(\xi_2) + 2a_1a_2\mathbb{C}(\xi_1,\xi_2)\\
&= 1^2\cdot\mathbb{V}(\xi) + 1^2\cdot\mathbb{V}(\zeta) + 2\cdot 1\cdot 1\cdot\mathbb{C}(\xi,\zeta)\\
&= \mathbb{V}(\xi) + \mathbb{V}(\zeta) + 2\mathbb{C}(\xi,\zeta).
\end{aligned}
\tag{25.31}
$$

Now let $n := 2$, $\xi_1 := \xi$, $\xi_2 := \zeta$, $a_0 := 0$, $a_1 := 1$, and $a_2 := -1$. Then, by Theorem 25.6,

$$
\begin{aligned}
\mathbb{V}(\xi - \zeta) &= \mathbb{V}(a_0 + a_1\xi_1 + a_2\xi_2) \\
&= \mathbb{V}\left(a_0 + \sum_{i=1}^{2} a_i\xi_i\right) \\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{i=1}^{2-1}\sum_{j=i+1}^{2} a_ia_j\mathbb{C}(\xi_i,\xi_j) \\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{i=1}^{1}\sum_{j=i+1}^{2} a_ia_j\mathbb{C}(\xi_i,\xi_j) \\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{j=1+1}^{2} a_ia_j\mathbb{C}(\xi_1,\xi_j) \\
&= \sum_{i=1}^{2} a_i^2\mathbb{V}(\xi_i) + 2\sum_{j=2}^{2} a_1a_j\mathbb{C}(\xi_1,\xi_j) \\
&= a_1^2\mathbb{V}(\xi_1) + a_2^2\mathbb{V}(\xi_2) + 2a_1a_2\mathbb{C}(\xi_1,\xi_2) \\
&= 1^2\cdot\mathbb{V}(\xi) + (-1)^2\cdot\mathbb{V}(\zeta) + 2\cdot 1\cdot(-1)\cdot\mathbb{C}(\xi,\zeta) \\
&= \mathbb{V}(\xi) + \mathbb{V}(\zeta) - 2\mathbb{C}(\xi,\zeta).
\end{aligned} \tag{25.32}
$$

Alternatively, without using Theorem 25.6 and with Theorem 23.2, we have

$$
\begin{aligned}
&\mathbb{V}(a\xi + b\zeta + c) \\
&= \mathbb{E}\left((a\xi + b\zeta + c - a\mathbb{E}(\xi) - b\mathbb{E}(\zeta) - c)^2\right) \\
&= \mathbb{E}\left((a(\xi - \mathbb{E}(\xi)) + b(\zeta - \mathbb{E}(\zeta)))^2\right) \\
&= \mathbb{E}\left(a^2(\xi - \mathbb{E}(\xi))^2 + 2ab(\xi - \mathbb{E}(\xi))(\zeta - \mathbb{E}(\zeta)) + b^2(\zeta - \mathbb{E}(\zeta))^2\right) \\
&= a^2\mathbb{E}\left((\xi - \mathbb{E}(\xi))^2\right) + b^2\mathbb{E}\left((\zeta - \mathbb{E}(\zeta))^2\right) + 2ab\mathbb{E}\left((\xi - \mathbb{E}(\xi))(\zeta - \mathbb{E}(\zeta))\right) \\
&= a^2\mathbb{V}(\xi) + b^2\mathbb{V}(\zeta) + 2ab\mathbb{C}(\xi,\zeta).
\end{aligned} \tag{25.33}
$$

The special cases then follow directly with $a := b := 1$ and $a := 1, b := -1$, respectively. □

Finally, the following theorem provides a first impression of the relationship between covariance and correlation and the concept of independence of random variables. It turns out that covariance and correlation are sensitive only to certain forms of dependence between random variables and, in particular, that a covariance of zero does not imply independence of the random variables. On the other hand, independence of two random variables always implies that their covariance is zero and that they are therefore uncorrelated. Dependence and independence of random variables are thus much more general concepts for describing the association between random variables than covariance and correlation.

Theorem 25.8 (Correlation and independence of random variables)**.** *Let ξ_1 and ξ_2 be two random variables. If ξ_1 and ξ_2 are independent, then $\mathbb{C}(\xi_1,\xi_2) = 0$, and ξ_1 and ξ_2 are therefore uncorrelated. Conversely, if $\mathbb{C}(\xi_1,\xi_2) = 0$ and ξ_1 and ξ_2 are therefore uncorrelated, then ξ_1 and ξ_2 are not necessarily independent.*

∘

Proof. In a first step, we show that independence of ξ_1 and ξ_2 implies $\mathbb{C}(\xi_1,\xi_2) = 0$. We then provide an example showing that the covariance of dependent random variables ξ_1 and ξ_2 can indeed be zero.

(1) We show that independence of ξ_1 and ξ_2 implies $\mathbb{C}(\xi_1,\xi_2) = 0$. For this purpose, we first note that for independent random variables,

$$\mathbb{E}(\xi_1\xi_2) = \mathbb{E}(\xi_1)\mathbb{E}(\xi_2). \tag{25.34}$$

With the covariance shift theorem, it follows that

$$\mathbb{C}(\xi_1,\xi_2) = \mathbb{E}(\xi_1\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) = \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) = 0. \tag{25.35}$$

With the definition of the correlation coefficient, it then follows that $\rho(\xi_1, \xi_2) = 0$, and thus ξ_1 and ξ_2 are uncorrelated.

(2) We now provide an example showing that the covariance of dependent random variables ξ_1 and ξ_2 can be zero. For this purpose, we consider two discrete random variables ξ_1 and ξ_2 with outcome spaces $\mathcal{X}_1 = \{-1, 0, 1\}$ and $\mathcal{X}_2 = \{0, 1\}$, the marginal PMF of ξ_1 given by $p(x_1) := \frac{1}{3}$ for all $x_1 \in \mathcal{X}_1$, and the definition

$$\xi_2 := \xi_1^2. \tag{25.36}$$

The definition of ξ_2 then implies the conditional PMF shown in Table 25.2.

Table 25.2 Conditional PMF for the proof of Theorem 25.8

$p(x_2\|x_1)$	$x_1 = -1$	$x_1 = 0$	$x_1 = 1$
$x_2 = 0$	0	1	0
$x_2 = 1$	1	0	1

The marginal PMF $p(x_1)$ and the conditional PMF $p(x_2|x_1)$ then imply the joint PMF shown in Table 25.3.

Table 25.3 Joint PMF for the proof of Theorem 25.8

$p(x_1, x_2)$	$x_1 = -1$	$x_1 = 0$	$x_1 = 1$	$p(x_2)$
$x_2 = 0$	0	1/3	0	1/3
$x_2 = 1$	1/3	0	1/3	2/3
$p(x_1)$	1/3	1/3	1/3	

For example, by Table 25.3,

$$p(-1, 0) = 0 \neq \frac{1}{9} = \frac{1}{3} \cdot \frac{1}{3} = p(-1)p(0). \tag{25.37}$$

The joint PMF of ξ_1 and ξ_2 therefore does not factorize, and ξ_1 and ξ_2 are thus not independent. However, with

$$\mathbb{E}(\xi_1) = \sum_{x_1 \in \mathcal{X}_1} x_1 p(x_1) = -1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0 \tag{25.38}$$

and

$$\mathbb{E}(\xi_1 \xi_2) = \mathbb{E}(\xi_1 \xi_1^2) = \mathbb{E}(\xi_1^3) = \sum_{x_1 \in \mathcal{X}_1} x_1^3 p(x_1) = -1^3 \cdot \frac{1}{3} + 0^3 \cdot \frac{1}{3} + 1^3 \cdot \frac{1}{3} = 0, \tag{25.39}$$

as well as the covariance shift theorem, it follows that

$$\mathbb{C}(\xi_1, \xi_2) = \mathbb{E}(\xi_1 \xi_2) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) = \mathbb{E}(\xi_1^3) - \mathbb{E}(\xi_1)\mathbb{E}(\xi_2) = 0 - 0 \cdot \mathbb{E}(\xi_2) = 0. \tag{25.40}$$

The covariance of ξ_1 and ξ_2 is therefore zero, and ξ_1 and ξ_2 are thus uncorrelated, even though ξ_1 and ξ_2 are not independent. □

25.3 Conditional covariance

Like the expectation and the variance, the covariance can also be considered in a conditional joint distribution of two random variables. This leads to the concept of *conditional covariance*.

Definition 25.2 (Conditional covariance)**.** Let a random vector $\xi := (\xi_1, \xi_2, \xi_3)$ with outcome space $\mathcal{X} := \mathcal{X}_1 \times \mathcal{X}_2 \times \mathcal{X}_3$ and PMF or PDF $p(x_1, x_2, x_3)$, and conditional PMF or PDF $p(x_1, x_2|x_3)$ for all $x_3 \in \mathcal{X}_3$, be given. Then the conditional covariance of ξ_1 and ξ_2 given $\xi_3 = x_3$ is defined as

$$\mathbb{C}(\xi_1, \xi_2|\xi_3 = x_3) = \mathbb{E}\left(\left(\xi_1 - \mathbb{E}\left(\xi_1|\xi_3 = x_3\right)\right)\left(\xi_2 - \mathbb{E}\left(\xi_2|\xi_3 = x_3\right)\right)|\xi_3 = x_3\right). \tag{25.41}$$

•

Thus, conditional covariance is defined in the sense of the conditional expectation of the random vector (ξ_1, ξ_2) and, like the conditional expectation and the conditional variance, is generally a random variable. Finally, we note that the covariance shift theorem also holds for conditional covariance.

Theorem 25.9 (Shift theorem of conditional covariance)**.** *Let a random vector* $\xi := (\xi_1, \xi_2, \xi_3)$ *be given. Then*

$$\mathbb{C}(\xi_1, \xi_2|\xi_3) = \mathbb{E}\left(\xi_1\xi_2|\xi_3\right) - \mathbb{E}\left(\xi_1|\xi_3\right)\mathbb{E}\left(\xi_2|\xi_3\right). \tag{25.42}$$

∘

Proof. A proof is obtained by replacing the corresponding expectations in the proof of Theorem 25.2 with the corresponding conditional expectations. □

25.4 Covariance matrices

The multivariate analog of the variance of a random variable is the *covariance matrix of a random vector*. In addition to the variances of the components of the random vector, it also encodes their pairwise covariances and is defined as follows.

Definition 25.3 (Covariance matrix of a random vector)**.** Let ξ be an n-dimensional random vector. Then the *covariance matrix* of ξ is defined as the $n \times n$ matrix

$$\mathbb{C}(\xi) := \mathbb{E}\left((\xi - \mathbb{E}(\xi))(\xi - \mathbb{E}(\xi))^T\right). \tag{25.43}$$

•

The covariance matrix in Definition 25.3 is formally defined analogously to the covariance of two random variables. The following theorem gives a direct reduction of the concept of the covariance matrix of a random vector to the concept of covariance of two random variables known from the univariate context.

Theorem 25.10 (Properties of the covariance matrix)**.** *Let* ξ *be an* m*-dimensional random vector and let* $\mathbb{C}(\xi)$ *be its covariance matrix. Then:*

(1) (Elements) The elements of $\mathbb{C}(\xi)$ *are the covariances of the components of* ξ*,*

$$\mathbb{C}(\xi) = \left(\mathbb{C}(\xi_i, \xi_j)\right)_{1\le i,j\le m}. \tag{25.44}$$

(2) (Covariance matrix shift theorem) We have

$$\mathbb{C}(\xi) = \mathbb{E}\left(\xi\xi^T\right) - \mathbb{E}(\xi)\mathbb{E}(\xi)^T. \tag{25.45}$$

(3) (Linear-affine transformation) For $A \in \mathbb{R}^{n\times m}$ *and* $b \in \mathbb{R}^n$*,*

$$\mathbb{C}(A\xi + b) = A\mathbb{C}(\xi)A^T. \tag{25.46}$$

(4) (Matrix properties) $\mathbb{C}(\xi)$ *is symmetric and positive semidefinite.*

∘

Proof. (1) We have

$$
\begin{aligned}
\mathbb{C}(\xi) &:= \mathbb{E}\left((\xi - \mathbb{E}(\xi))(\xi - \mathbb{E}(\xi))^T\right) \\
&= \mathbb{E}\left(\left(\begin{pmatrix}\xi_1\\ \vdots \\ \xi_n\end{pmatrix} - \begin{pmatrix}\mathbb{E}(\xi_1)\\ \vdots \\ \mathbb{E}(\xi_n)\end{pmatrix}\right)\left(\begin{pmatrix}\xi_1\\ \vdots \\ \xi_n\end{pmatrix} - \begin{pmatrix}\mathbb{E}(\xi_1)\\ \vdots \\ \mathbb{E}(\xi_n)\end{pmatrix}\right)^T\right) \\
&= \mathbb{E}\left(\begin{pmatrix}\xi_1 - \mathbb{E}(\xi_1)\\ \vdots \\ \xi_n - \mathbb{E}(\xi_n)\end{pmatrix}\begin{pmatrix}\xi_1 - \mathbb{E}(\xi_1)\\ \vdots \\ \xi_n - \mathbb{E}(\xi_n)\end{pmatrix}^T\right) \\
&= \mathbb{E}\left(\begin{pmatrix}\xi_1 - \mathbb{E}(\xi_1)\\ \vdots \\ \xi_n - \mathbb{E}(\xi_n)\end{pmatrix}\begin{pmatrix}\xi_1 - \mathbb{E}(\xi_1) & \dots & \xi_n - \mathbb{E}(\xi_n)\end{pmatrix}\right) \\
&= \mathbb{E}\begin{pmatrix}(\xi_1 - \mathbb{E}(\xi_1))(\xi_1 - \mathbb{E}(\xi_1)) & \dots & (\xi_1 - \mathbb{E}(\xi_1))(\xi_n - \mathbb{E}(\xi_n)) \\ \vdots & \ddots & \vdots \\ (\xi_n - \mathbb{E}(\xi_n))(\xi_1 - \mathbb{E}(\xi_1)) & \dots & (\xi_n - \mathbb{E}(\xi_n))(\xi_n - \mathbb{E}(\xi_n))\end{pmatrix} \\
&= \left(\mathbb{E}\left((\xi_i - \mathbb{E}(\xi_i))(\xi_j - \mathbb{E}(\xi_j))\right)\right)_{1\le i,j\le n} \\
&= \left(\mathbb{C}(\xi_i, \xi_j)\right)_{1\le i,j\le n}.
\end{aligned}
\tag{25.47}
$$

(2) With the properties of expectations,

$$
\begin{aligned}
\mathbb{C}(\xi) &= \mathbb{E}\left((\xi - \mathbb{E}(\xi))(\xi - \mathbb{E}(\xi))^T\right) \\
&= \mathbb{E}\left(\xi\xi^T - \xi\mathbb{E}(\xi)^T - \mathbb{E}(\xi)\xi^T + \mathbb{E}(\xi)\mathbb{E}(\xi)^T\right) \\
&= \mathbb{E}\left(\xi\xi^T\right) - \mathbb{E}(\xi)\mathbb{E}(\xi)^T - \mathbb{E}(\xi)\mathbb{E}(\xi)^T + \mathbb{E}(\xi)\mathbb{E}(\xi)^T \\
&= \mathbb{E}\left(\xi\xi^T\right) - \mathbb{E}(\xi)\mathbb{E}(\xi)^T.
\end{aligned}
\tag{25.48}
$$

(3) With the properties of expectations,

$$
\begin{aligned}
\mathbb{C}(A\xi + b) &= \mathbb{E}\left((A\xi + b - \mathbb{E}(A\xi + b))(A\xi + b - \mathbb{E}(A\xi + b))^T\right) \\
&= \mathbb{E}\left((A\xi + b - A\mathbb{E}(\xi) - b)(A\xi + b - A\mathbb{E}(\xi) - b)^T\right) \\
&= \mathbb{E}\left((A(\xi - \mathbb{E}(\xi)))(A(\xi - \mathbb{E}(\xi)))^T\right) \\
&= \mathbb{E}\left(A(\xi - \mathbb{E}(\xi))(\xi - \mathbb{E}(\xi))^T A^T\right) \\
&= A\mathbb{E}\left((\xi - \mathbb{E}(\xi))(\xi - \mathbb{E}(\xi))^T\right) A^T \\
&= A\mathbb{C}(\xi)A^T.
\end{aligned}
\tag{25.49}
$$

(4) The symmetry of $\mathbb{C}(\xi)$ follows from the symmetry of the covariance of random variables,

$$
\mathbb{C}(\xi_i, \xi_j) = \mathbb{C}(\xi_j, \xi_i) \text{ for all } i = 1, \dots, m, j = 1, \dots, m. \tag{25.50}
$$

To show the positive semidefiniteness of $\mathbb{C}(\xi)$, we need to show that $a^T\mathbb{C}(\xi)a \ge 0$ for all $a \in \mathbb{R}^m$ with $a \ne 0_m$. Let $a \in \mathbb{R}^m$ with $a \ne 0$. Then, by statement (3) with $A := a^T \in \mathbb{R}^{1\times m}$,

$$
a^T\mathbb{C}(\xi)a = \mathbb{C}(a^T\xi). \tag{25.51}
$$

Furthermore, by the definition of the covariance matrix,

$$
\mathbb{C}(a^T\xi) = \mathbb{E}\left(\left(a^T\xi - \mathbb{E}(a^T\xi)\right)^2\right) = \mathbb{V}\left(a^T\xi\right). \tag{25.52}
$$

Since, by the properties of variance, the variance of the random variable $a^T\xi$ is always nonnegative, it follows that

$$
a^T\mathbb{C}(\xi)a = \mathbb{V}(a^T\xi) \ge 0, \tag{25.53}
$$

and thus that $\mathbb{C}(\xi)$ is positive semidefinite. □

The diagonal elements of $\mathbb{C}(\xi)$ are the variances of the components of ξ, since

$$
\mathbb{V}(\xi_i) = \mathbb{C}(\xi_i, \xi_i) \text{ for } i = 1, \dots, m. \tag{25.54}
$$

Properties (2) and (3) are essentially analogous to the properties of variance.

The covariance matrix of a random vector ξ is therefore the matrix of the covariances of the components of ξ. Thus, the covariance matrix is also directly given in terms of the concept of covariance of random variables. Since the covariance of a random variable with itself is known to be its variance, the covariance matrix contains the variances of the components of ξ on its diagonal.

The following theorem documents a notation for the covariance matrix of a partitioned random vector in terms of expectations of products of random vectors, which is helpful, for example, in canonical correlation analysis.

Theorem 25.11 (Covariance matrices of random vectors)**.** *Let*

$$\zeta = \begin{pmatrix} \xi \\ \upsilon \end{pmatrix} \text{ with } \mathbb{E}(\zeta) := 0_m \tag{25.55}$$

be an $m_\xi + m_\upsilon$-dimensional random vector and its expectation vector, respectively. Then the $m \times m$ covariance matrix of ζ can be written as

$$\mathbb{C}(\zeta) = \begin{pmatrix} \Sigma_{\xi\xi} & \Sigma_{\xi_1\xi_2} \\ \Sigma_{\upsilon\xi} & \Sigma_{\upsilon\upsilon} \end{pmatrix} \in \mathbb{R}^{m \times m} \tag{25.56}$$

where

$$\begin{aligned} \Sigma_{\xi\xi} &:= \mathbb{E}\left(\xi\xi^T\right) \in \mathbb{R}^{m_\xi \times m_\xi} \\ \Sigma_{\xi_1\xi_2} &:= \mathbb{E}\left(\xi_1\xi_2^T\right) \in \mathbb{R}^{m_\xi \times m_\upsilon} \\ \Sigma_{\upsilon\xi} &:= \mathbb{E}\left(\upsilon\xi^T\right) \in \mathbb{R}^{m_\upsilon \times m_\xi} \\ \Sigma_{\upsilon\upsilon} &:= \mathbb{E}\left(\upsilon\upsilon^T\right) \in \mathbb{R}^{m_\xi \times m_\upsilon}. \end{aligned} \tag{25.57}$$

∘

Proof. By the definition of the covariance matrix of a random vector,

$$\begin{aligned} \mathbb{C}(\zeta) &= \mathbb{E}\left((\zeta - \mathbb{E}(\zeta))(\zeta - \mathbb{E}(\zeta))^T\right) \\ &= \mathbb{E}\left((\zeta - 0_m)(\zeta - 0_m)^T\right) \\ &= \mathbb{E}\left(\zeta\zeta^T\right) \\ &= \mathbb{E}\left(\begin{pmatrix} \xi \\ \upsilon \end{pmatrix} \begin{pmatrix} \xi^T & \upsilon^T \end{pmatrix}\right) \\ &= \mathbb{E}\left(\begin{pmatrix} \xi\xi^T & \xi_1\xi_2^T \\ \upsilon\xi^T & \upsilon\upsilon^T \end{pmatrix}\right) \\ &= \begin{pmatrix} \mathbb{E}\left(\xi\xi^T\right) & \mathbb{E}\left(\xi_1\xi_2^T\right) \\ \mathbb{E}\left(\upsilon\xi^T\right) & \mathbb{E}\left(\upsilon\upsilon^T\right) \end{pmatrix} \\ &= \begin{pmatrix} \Sigma_{\xi\xi} & \Sigma_{\xi_1\xi_2} \\ \Sigma_{\upsilon\xi} & \Sigma_{\upsilon\upsilon} \end{pmatrix}. \end{aligned} \tag{25.58}$$

□

Finally, in some applications one is interested in a normalized, scale-independent representation of the covariances of a random vector. As in the univariate case, the natural approach is to normalize the covariance of two random variables by means of their respective variances in the sense of a correlation. This consideration leads to the concept of the *correlation matrix* of a random vector.

Definition 25.4 (Correlation matrix)**.** Let ξ be an n-dimensional random vector. Then the *correlation matrix* of ξ is defined as the $n \times n$ matrix

$$\mathbb{R}(\xi) := \left(\rho_{ij}\right)_{1\leq i,j\leq n} = \left(\frac{\mathbb{C}(\xi_i, \xi_j)}{\sqrt{\mathbb{V}(\xi_i)}\sqrt{\mathbb{V}(\xi_j)}}\right)_{1\leq i,j\leq n}. \tag{25.59}$$

•

Since the variances of the components of ξ are the diagonal elements of the covariance matrix of ξ, the correlation matrix is of course implicit in the covariance matrix. Furthermore, as always for correlations, the entries ρ_{ij}, $1 \leq i, j \leq n$, of the correlation matrix satisfy

$$\rho_{ij} \in [-1, 1] \text{ for } 1 \leq i, j \leq n \text{ and } \rho_{ii} = 1 \text{ for } 1 \leq i \leq n. \tag{25.60}$$

Study questions

1. State the definition of covariance and correlation of two random variables.
2. Explain the definition of covariance and correlation of two random variables.
3. State the theorem on symmetry of covariance and correlation.
4. State the theorem on covariance and correlation under linear-affine transformation.
5. Explain the theorem on covariance and correlation under linear-affine transformation.
6. State the theorem on variances of special linear combinations of random variables.
7. State the theorem on correlation and independence of random variables.

Study question answers

1. See Definition 25.1.
2. See the explanations of Definition 25.1 and Example 25.1.
3. See Theorem 25.1.
4. See Theorem 25.3.
5. See the explanations of Theorem 25.3.
6. See Theorem 25.7.
7. See Theorem 25.8.

26 Inequalities

The topic of this chapter is inequalities that are frequently used in probability theory to bound probabilities and expectations. We group these inequalities accordingly into *probability inequalities* (*Markov's inequality* and *Chebyshev's inequality*, Section 26.1) and *expectation inequalities* (*Cauchy-Schwarz inequality*, *correlation inequality*, and *Jensen's inequality*, Section 26.2).

26.1 Probability inequalities

Markov's inequality establishes a relation between exceedance probabilities (see Theorem 21.2) and the expectation of a *nonnegative* random variable, that is, a random variable for which $\mathbb{P}(\xi \geq 0) = 1$. In the proof of this inequality, we consider only the case of a continuous random variable.

Theorem 26.1 (Markov's inequality)**.** *Let ξ be a random variable with $\mathbb{P}(\xi \geq 0) = 1$. Then, for all $x > 0$,*

$$\mathbb{P}(\xi \geq x) \leq \frac{\mathbb{E}(\xi)}{x}. \tag{26.1}$$

◦

Proof. We consider the case of a continuous random variable ξ with PDF p. We first note that

$$\mathbb{E}(\xi) = \int_{-\infty}^{\infty} s\, p(s)\, ds = \int_{0}^{\infty} s\, p(s)\, ds = \int_{0}^{x} s\, p(s)\, ds + \int_{x}^{\infty} s\, p(s)\, ds, \tag{26.2}$$

because ξ is nonnegative. It follows that

$$\mathbb{E}(\xi) \geq \int_{x}^{\infty} s\, p(s)\, ds \geq \int_{x}^{\infty} x\, p(s)\, ds = x \int_{x}^{\infty} p(s)\, ds = x\, \mathbb{P}(\xi \geq x). \tag{26.3}$$

Here, the first inequality holds because

$$\int_{0}^{x} s\, p(s)\, ds \geq 0, \tag{26.4}$$

and the second inequality holds because $x \leq \xi$ for $\xi \in [x, \infty[$. Thus,

$$\mathbb{E}(\xi) \geq x\, \mathbb{P}(\xi \geq x) \Leftrightarrow \mathbb{P}(\xi \geq x) \leq \frac{\mathbb{E}(\xi)}{x}. \tag{26.5}$$

□

For example, if for a nonnegative random variable ξ we have $\mathbb{E}(\xi) = 1$, then Markov's inequality implies that

$$\mathbb{P}(\xi \geq 100) \leq 0.01. \tag{26.6}$$

Example 26.1 (Markov's inequality)**.** As an example of Markov's inequality, we consider the case of a gamma random variable $\xi \sim G(\alpha, \beta)$. Gamma random variables are, by definition, nonnegative (see Definition 21.10). For the expectation of a gamma random variable, $\mathbb{E}(\xi) = \alpha\beta$. We specifically consider the case $\alpha := 5$ and $\beta := 2$, so that ξ also corresponds to a χ^2 random variable with degrees-of-freedom parameter $n = 10$. In Figure 26.1 A we show the CDF P of this random variable; in Figure 26.1 B we visualize the quantities $\mathbb{E}(\xi)/x$ and the exceedance probability $\mathbb{P}(\xi \geq x) = 1 - P(x)$ considered in Markov's inequality. Clearly, Markov's inequality holds.

△

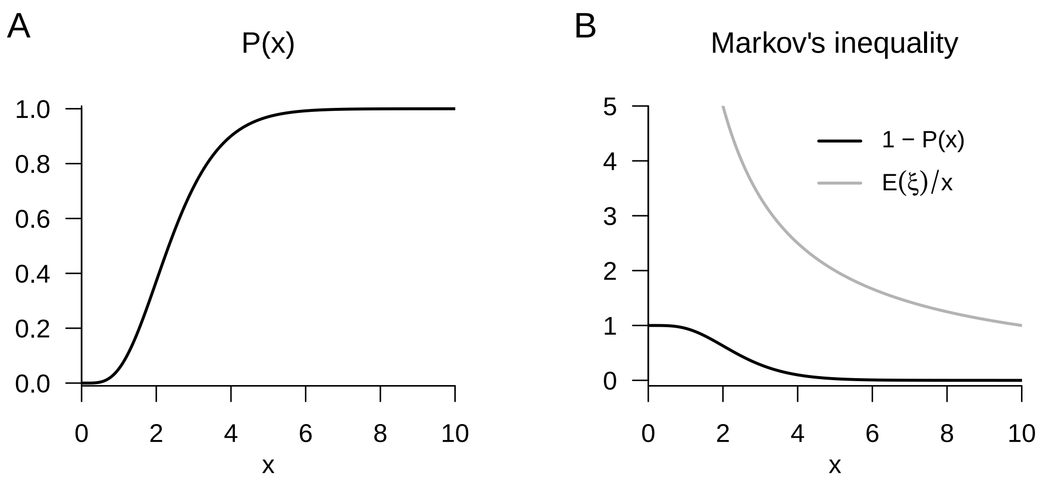


Figure 26.1 Markov's inequality illustrated with a gamma random variable.

Chebyshev's inequality relates the probability that a random variable takes values far away from its expectation to its variance. As explained in Chapter 24, Chebyshev's inequality thus provides a justification for why the quantity formulated in Definition 24.1 can be understood as a measure of the variability of a random variable. The proof of Chebyshev's inequality relies crucially on Markov's inequality.

Theorem 26.2 (Chebyshev's inequality)**.** *Let ξ be a random variable with variance $\mathbb{V}(\xi)$. Then, for all $x \in \mathbb{R}$,*

$$\mathbb{P}(|\xi - \mathbb{E}(\xi)| \geq x) \leq \frac{\mathbb{V}(\xi)}{x^2}. \tag{26.7}$$

∘

Proof. We first note that, for $a, b \in \mathbb{R}$, $a^2 \geq b^2$ implies $|a| \geq b$. To see this, we consider the following four possible cases.

(1) $a^2 \geq b^2$ for $a \geq 0$ and $b \geq 0$. Then

$$a^2 \geq b^2 \Rightarrow \sqrt{a^2} \geq \sqrt{b^2} \Rightarrow a \geq b \Rightarrow |a| \geq b. \tag{26.8}$$

(2) $a^2 \geq b^2$ for $a < 0$ and $b \geq 0$. Then

$$a^2 \geq b^2 \Rightarrow \sqrt{a^2} \geq \sqrt{b^2} \Rightarrow -a \geq b \Rightarrow |a| \geq b. \tag{26.9}$$

(3) $a^2 \geq b^2$ for $a \geq 0$ and $b < 0$. Then

$$a^2 \geq b^2 \Rightarrow \sqrt{a^2} \geq \sqrt{b^2} \Rightarrow a \geq -b \geq b \Rightarrow |a| \geq b. \tag{26.10}$$

(4) $a^2 \geq b^2$ for $a < 0$ and $b < 0$. Then

$$a^2 \geq b^2 \Rightarrow \sqrt{a^2} \geq \sqrt{b^2} \Rightarrow -a \geq -b > b \Rightarrow |a| \geq b. \tag{26.11}$$

Next, define $\upsilon := (\xi - \mathbb{E}(\xi))^2$. Then evidently $\upsilon \geq 0$, and Markov's inequality implies

$$\begin{aligned} \mathbb{P}\left(\upsilon \geq x^2\right) &\leq \frac{\mathbb{E}(\upsilon)}{x^2} \\ \Leftrightarrow \mathbb{P}\left((\xi - \mathbb{E}(\xi))^2 \geq x^2\right) &\leq \frac{\mathbb{E}\left((\xi - \mathbb{E}(\xi))^2\right)}{x^2} \\ \Leftrightarrow \mathbb{P}(|\xi - \mathbb{E}(\xi)| \geq x) &\leq \frac{\mathbb{V}(\xi)}{x^2}. \end{aligned} \tag{26.12}$$

□

26.2 Expectation inequalities

The Cauchy-Schwarz inequality is a central inequality of modern mathematics that is used in various mathematical areas such as analysis, vector space theory, and probability theory (see Steele (2006)). In relation to expectations of random variables, it has the following form.

Theorem 26.3 (Cauchy-Schwarz inequality). *Let ξ and υ be two random variables, and let $\mathbb{E}(\xi\upsilon)$ be finite. Then*

$$\mathbb{E}(\xi\upsilon)^2 \leq \mathbb{E}\left(\xi^2\right)\mathbb{E}\left(\upsilon^2\right). \tag{26.13}$$

∘

For a proof, we refer to the proof of Theorem 4.6.2 in DeGroot & Schervish (2012). Analogously to Theorem 26.3, for example, for vectors $x, y \in \mathbb{R}^n$,

$$\langle x, y\rangle^2 \leq \langle x, x\rangle\langle y, y\rangle. \tag{26.14}$$

In the context of probabilistic data analysis, the Cauchy-Schwarz inequality is especially relevant in the proof of the so-called *correlation inequality*.

Theorem 26.4 (Correlation inequality). *Let ξ and υ be random variables with $\mathbb{V}(\xi) > 0$ and $\mathbb{V}(\upsilon) > 0$. Then*

$$\frac{\mathbb{C}(\xi, \upsilon)^2}{\mathbb{V}(\xi)\mathbb{V}(\upsilon)} \leq 1 \text{ and } -1 \leq \rho(\xi, \upsilon) \leq 1. \tag{26.15}$$

∘

Proof. By the Cauchy-Schwarz inequality for two random variables α and β,

$$\mathbb{E}(\alpha\beta)^2 \leq \mathbb{E}\left(\alpha^2\right)\mathbb{E}\left(\beta^2\right). \tag{26.16}$$

Now define $\alpha := \xi - \mathbb{E}(\xi)$ and $\beta := \upsilon - \mathbb{E}(\upsilon)$. Then the Cauchy-Schwarz inequality states precisely that

$$\mathbb{E}\left((\xi - \mathbb{E}(\xi))(\upsilon - \mathbb{E}(\upsilon))\right)^2 \leq \mathbb{E}\left((\xi - \mathbb{E}(\xi))^2\right)\mathbb{E}\left((\upsilon - \mathbb{E}(\upsilon))^2\right). \tag{26.17}$$

Thus,

$$\mathbb{C}(\xi, \upsilon)^2 \leq \mathbb{V}(\xi)\mathbb{V}(\upsilon) \Leftrightarrow \frac{\mathbb{C}(\xi, \upsilon)^2}{\mathbb{V}(\xi)\mathbb{V}(\upsilon)} \leq 1. \tag{26.18}$$

Furthermore, from the definition of correlation it follows immediately that

$$\rho(\xi, \upsilon)^2 \leq 1. \tag{26.19}$$

But then also

$$|\rho(\xi, \upsilon)| \leq 1 \Leftrightarrow -1 \leq \rho(\xi, \upsilon) \leq 1, \tag{26.20}$$

because

$$\rho(\xi, \upsilon)^2 \leq 1 \Rightarrow \sqrt{\rho(\xi, \upsilon)^2} \leq \sqrt{1} \Rightarrow \quad \rho(\xi, \upsilon) \leq 1 \Rightarrow |\rho(\xi, \upsilon)| \leq 1 \text{ for } \rho(\xi, \upsilon) \geq 0 \tag{26.21}$$

and

$$\rho(\xi, \upsilon)^2 \leq 1 \Rightarrow \sqrt{\rho(\xi, \upsilon)^2} \leq \sqrt{1} \Rightarrow -\rho(\xi, \upsilon) \leq 1 \Rightarrow |\rho(\xi, \upsilon)| \leq 1 \text{ for } \rho(\xi, \upsilon) < 0. \tag{26.22}$$

□

The correlation inequality is sometimes also called the *covariance inequality*. In particular, it states that the correlation of random variables is normalized, that is, it always takes values between -1 and 1 inclusive (see Chapter 25).

Finally, *Jensen's inequality* provides bounds for the expectation of a random variable transformed by a convex or concave function. It is used in the consideration of properties of parameter estimators and, in particular, as a foundation of variational Bayesian inference. Recall that a convex function g is characterized by the fact that the graph of g over an interval $[x_1, x_2]$ always lies below the line connecting the function values $g(x_1)$ and $g(x_2)$, whereas for a concave function g it always lies above the line connecting the function values $g(x_1)$ and $g(x_2)$. We visualize this for a convex function in Figure 26.2.

Theorem 26.5 (Jensen's inequality)**.** *Let ξ be a random variable and let $g : \mathbb{R} \to \mathbb{R}$ be a convex function, that is,*

$$g(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda g(x_1) + (1 - \lambda)g(x_2) \tag{26.23}$$

for all $x_1, x_2 \in \mathbb{R}, \lambda \in [0, 1]$. Then

$$\mathbb{E}(g(\xi)) \geq g(\mathbb{E}(\xi)). \tag{26.24}$$

Analogously, let $g : \mathbb{R} \to \mathbb{R}$ be a concave function, that is,

$$g(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda g(x_1) + (1 - \lambda)g(x_2) \tag{26.25}$$

for all $x_1, x_2 \in \mathbb{R}, \lambda \in [0, 1]$. Then

$$\mathbb{E}(g(\xi)) \leq g(\mathbb{E}(\xi)). \tag{26.26}$$

∘

Proof. Let g be a convex function. Then, for the tangent t of g at $x_0 \in \mathbb{R}$, for all $x \in \mathbb{R}$,

$$g(x) \geq t(x) := g(x_0) + g'(x_0)(x - x_0). \tag{26.27}$$

Now set $x := \xi$ and $x_0 := \mathbb{E}(\xi)$. Then the above inequality gives

$$g(\xi) \geq g(\mathbb{E}(\xi)) + g'(\mathbb{E}(\xi))(\xi - \mathbb{E}(\xi)). \tag{26.28}$$

Taking expectations yields

$$\begin{aligned}
&\mathbb{E}(g(\xi)) \geq \mathbb{E}(g(\mathbb{E}(\xi))) + \mathbb{E}(g'(\mathbb{E}(\xi))(\xi - \mathbb{E}(\xi))) \\
\Leftrightarrow\ &\mathbb{E}(g(\xi)) \geq g(\mathbb{E}(\xi)) + g'(\mathbb{E}(\xi))\mathbb{E}((\xi - \mathbb{E}(\xi))) \\
\Leftrightarrow\ &\mathbb{E}(g(\xi)) \geq g(\mathbb{E}(\xi)) + g'(\mathbb{E}(\xi))(\mathbb{E}(\xi) - \mathbb{E}(\xi)) \\
\Leftrightarrow\ &\mathbb{E}(g(\xi)) \geq g(\mathbb{E}(\xi)).
\end{aligned} \tag{26.29}$$

Now let g be a concave function. Then $-g$ is a convex function. With Jensen's inequality for convex functions, Jensen's inequality for concave functions follows from

$$\begin{aligned}
& \mathbb{E}(-g(\xi)) \geq -g(\mathbb{E}(\xi)) \\
\Leftrightarrow\ & -\mathbb{E}(g(\xi)) \geq -g(\mathbb{E}(\xi)) \\
\Leftrightarrow\ & \mathbb{E}(g(\xi)) \leq g(\mathbb{E}(\xi)).
\end{aligned} \tag{26.30}$$

□

In the context of variational Bayesian inference, it is fundamental that the logarithm is a concave function, and hence for an arbitrary random variable ξ,

$$\mathbb{E}(\ln \xi) \leq \ln \mathbb{E}(\xi). \tag{26.31}$$

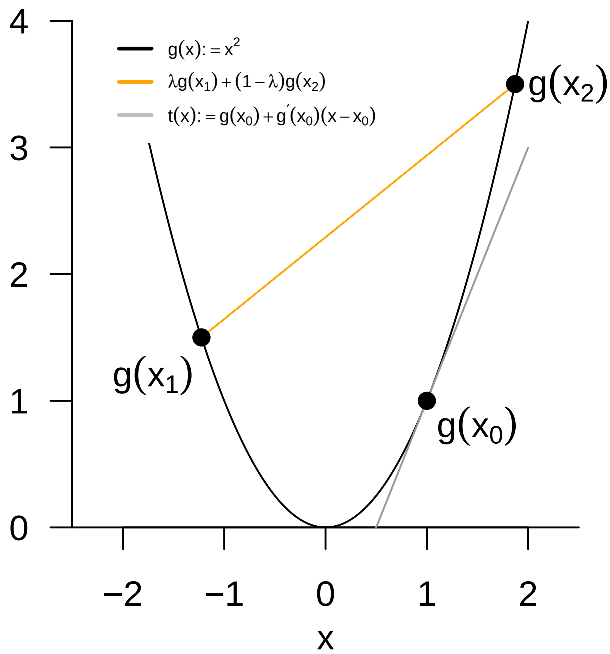


Figure 26.2 Representation of the convex function $g(x) := x^2$ with $x_1 := -\sqrt{1.5}, x_2 := \sqrt{3.5}, \lambda \in [0, 1]$, and $x_0 := 1$.

Study questions

1. State Markov's inequality.
2. State Chebyshev's inequality.
3. State the Cauchy-Schwarz inequality.
4. State the correlation inequality.

Study question answers

1. See Theorem 26.1.
2. See Theorem 26.2.
3. See Theorem 26.3.
4. See Theorem 26.4.

27 Limits

In this chapter, we consider limit statements for sequences of random variables that are fundamental for probabilistic modeling and data analysis. The *laws of large numbers* (Section 27.1) first provide a basic justification for taking means in probabilistic inference. The *central limit theorems* then justify the widespread use of normal distribution assumptions for unknown influences in probabilistic model formulation (Section 27.2). The mathematical depth of these limit statements cannot be exhausted in this introductory treatment, so we have to be content with numerous simplifications. Minimal prior knowledge about sequences of functions and their limit functions is provided by Chapter 5.

27.1 Laws of large numbers

There is a *weak law of large numbers* and a *strong law of large numbers*. Intuitively, both laws state that the sample mean of independent and identically distributed random variables approaches the expectation of the underlying distribution for a large number of random variables. The weak and strong laws of large numbers differ with respect to the forms of *convergence of random variables* used in their formulation. The weak law is based on *convergence in probability*. The strong law is based on *almost sure convergence*. Here we restrict ourselves to the concept of convergence in probability and thus to the weak law of large numbers.

Definition 27.1 (Convergence in probability)**.** A sequence of random variables $\xi_1, \xi_2, ...$ *converges to a random variable* ξ *in probability* if, for every arbitrarily small $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|\xi_n - \xi| < \epsilon) = 1 \Leftrightarrow \lim_{n\to\infty} \mathbb{P}(|\xi_n - \xi| \geq \epsilon) = 0. \tag{27.1}$$

The convergence of $\xi_1, \xi_2, ...$ to ξ in probability is written as

$$\xi_n \xrightarrow[n\to\infty]{\text{P}} \xi. \tag{27.2}$$

•

Thus, $\xi_n \xrightarrow[n\to\infty]{\text{P}} \xi$ means that the probability that ξ_n lies in the random interval

$$]\xi - \epsilon, \xi + \epsilon[\tag{27.3}$$

approaches 1 as n tends to infinity, no matter how small this interval may be. Intuitively, for $n \to \infty$ and a constant random variable $\xi := a$, this means that the distribution of ξ_n concentrates more and more around a as n tends to infinity. Using convergence in probability, the weak law of large numbers is formulated as follows.

Theorem 27.1 (Weak law of large numbers)**.** *Let $\xi_1, ..., \xi_n$ be independent and identically distributed random variables with $\mathbb{E}(\xi_i) = \mu$ for all $i = 1, ..., n$. Furthermore, let*

$$\bar{\xi}_n := \frac{1}{n}\sum_{i=1}^{n} \xi_i \tag{27.4}$$

denote the sample mean of the $\xi_i, i = 1, ..., n$. Then $\bar{\xi}_n$ converges in probability to μ,

$$\bar{\xi}_n \xrightarrow[n\to\infty]{P} \mu. \tag{27.5}$$

∘

Proof. By Theorem 23.2, we first have

$$\mathbb{E}\left(\bar{\xi}_n\right) = \mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n} \xi_i\right) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left(\xi_i\right) = \frac{1}{n} n\mathbb{E}\left(\xi_i\right) = \mathbb{E}\left(\xi_i\right) = \mu. \tag{27.6}$$

The expectation μ of the ith sample variable ξ_i therefore agrees with the expectation of the sample mean $\bar{\xi}_n$. Furthermore, by Theorem 25.6 and Theorem 25.8, independence of the random variables implies

$$\mathbb{V}\left(\bar{\xi}_n\right) = \mathbb{V}\left(\frac{1}{n}\sum_{i=1}^{n} \xi_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathbb{V}\left(\xi_i\right) = \frac{1}{n^2} n\mathbb{V}\left(\xi_i\right) = \frac{\mathbb{V}\left(\xi_i\right)}{n}. \tag{27.7}$$

The variance of the sample mean is therefore obtained by dividing the variance of the ith sample variable by n. By Theorem 26.2,

$$\mathbb{P}(|\bar{\xi}_n - \mathbb{E}\left(\bar{\xi}_n\right)| \geq \epsilon) \leq \frac{\mathbb{V}\left(\bar{\xi}_n\right)}{\epsilon^2} = \frac{\mathbb{V}\left(\xi_i\right)}{n\epsilon^2}. \tag{27.8}$$

For arbitrary $\mathbb{V}\left(\xi_i\right) \geq 0$ and $\epsilon > 0$, and by the nonnegativity of probabilities, we therefore have

$$\lim_{n\to\infty} \mathbb{P}(|\bar{\xi}_n - \mathbb{E}\left(\bar{\xi}_n\right)| \geq \epsilon) \leq \lim_{n\to\infty} \frac{\mathbb{V}\left(\xi_i\right)}{n\epsilon^2} \Leftrightarrow \lim_{n\to\infty} \mathbb{P}(|\bar{\xi}_n - \mu| \geq \epsilon) \leq 0 \Leftrightarrow \lim_{n\to\infty} \mathbb{P}(|\bar{\xi}_n - \mu| \geq \epsilon) = 0, \tag{27.9}$$

which proves the result. □

Intuitively,

$$\bar{\xi}_n \xrightarrow[n\to\infty]{\mathrm{P}} \mu \tag{27.10}$$

means that the probability that the sample mean lies close to the expectation of the underlying distribution approaches 1 as n tends to infinity.

Example 27.1 (Simulation of the weak law of large numbers)**.** To illustrate Theorem 27.1, we consider the case of i.i.d. normally distributed random variables $\xi_1, ..., \xi_n \sim N(0, 1)$. Figure 27.1 A shows realizations of sample means $\bar{\xi}_n$ as a function of n. One can see that, for larger n, more realizations of $\bar{\xi}_n$ lie near the expectation of the $\xi_i, i = 1, ..., n$. Based on these sample-mean realizations, Figure 27.1 B shows estimates of the probability $\mathbb{P}(|\bar{\xi}_n - \mu| \geq \epsilon)$ as functions of n and ϵ. For a large ϵ, a small n is sufficient to make the probability of an absolute deviation of the sample mean from the expectation small. For a smaller ϵ, a larger n is needed. In any case, however, the deviation probabilities decrease as n increases.

△

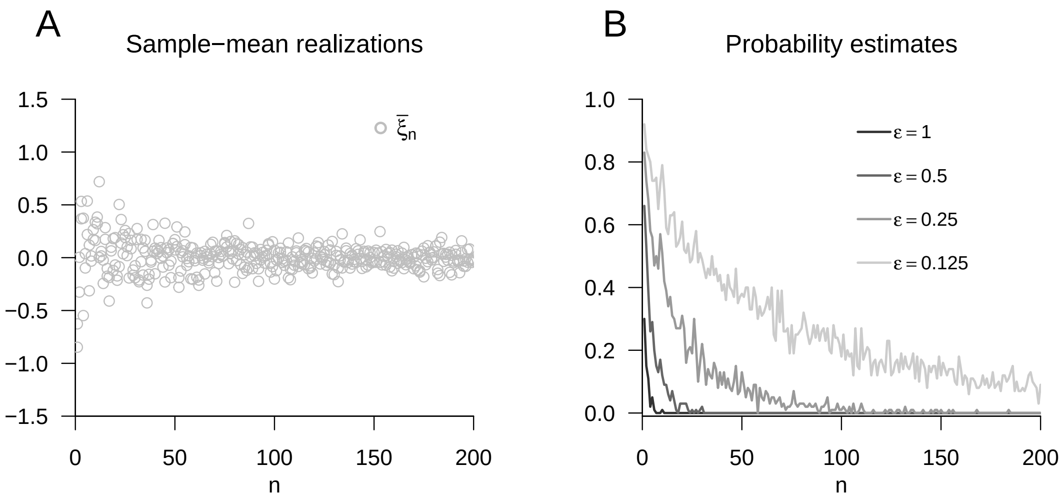


Figure 27.1 Simulation of the weak law of large numbers.

27.2 Central limit theorems

The central limit theorems state, intuitively, that the sum of independent random variables with expectation zero is *asymptotically*, that is, for infinitely many random variables, normally distributed with expectation parameter zero. Thus, if one models an arbitrary measurement quantity y as the sum of a deterministic influence μ and the sum

$$\varepsilon := \sum_{i=1}^{n} \xi_i \tag{27.11}$$

of a large number of independent random variables $\xi_i, i = 1, ..., n$, intended to model unknown influences, then for large n the assumption

$$y = \mu + \varepsilon \text{ with } \varepsilon \sim N(0, \sigma^2) \tag{27.12}$$

is mathematically justified. As we will see later, the assumption in Equation 27.12 underlies a large number of probabilistic models.

Formally, different forms of central limit theorems are distinguished depending on the exact underlying assumptions and their proofs. In the so-called *Lindeberg and Levy form* of the central limit theorem, independent and identically distributed random variables are assumed. In the *Liapunov form*, by contrast, only independent random variables are assumed. In both formulations of the central limit theorem, the considered convergence of random variables is *convergence in distribution*, which we introduce first.

Definition 27.2 (Convergence in distribution)**.** A sequence $\xi_1, \xi_2, ...$ of random variables *converges in distribution to a random variable* ξ if

$$\lim_{n\to\infty} P_{\xi_n}(x) = P_\xi(x) \tag{27.13}$$

for all x at which P_ξ is continuous. The convergence in distribution of $\xi_1, \xi_2, ...$ to ξ is written as

$$\xi_n \xrightarrow[n\to\infty]{\text{D}} \xi. \tag{27.14}$$

If $\xi_n \xrightarrow[n\to\infty]{\text{D}} \xi$, then the distribution of ξ is called the *asymptotic distribution of the sequence* $\xi_1, \xi_2, ...$

•

Convergence in distribution is therefore a statement about the convergence of sequences of functions, specifically of CDFs. Without proof, we note that convergence in probability, considered above, implies convergence in distribution. We now first state the central limit theorem of Lindeberg and Levy. For a proof, we refer to Henze (2024).

Theorem 27.2 (Central limit theorem of Lindeberg and Levy)**.** *Let $\xi_1, ..., \xi_n$ be independent and identically distributed random variables with*

$$\mathbb{E}(\xi_i) := \mu \text{ and } \mathbb{V}(\xi_i) := \sigma^2 > 0 \text{ for all } i = 1, ..., n. \tag{27.15}$$

Furthermore, let ζ_n be the random variable defined as

$$\zeta_n := \sqrt{n}\left(\frac{\bar{\xi}_n - \mu}{\sigma}\right). \tag{27.16}$$

Then, for all $z \in \mathbb{R}$,

$$\lim_{n\to\infty} P_{\zeta_n}(z) = \Phi(z), \tag{27.17}$$

where Φ denotes the cumulative distribution function of the standard normal distribution.

∘

We show later that, for $n \to \infty$, this also implies

$$\sum_{i=1}^{n} \xi_i \sim N(\mu, n\sigma^2) \text{ and } \bar{\xi}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right). \tag{27.18}$$

Example 27.2 (Simulation of the central limit theorem of Lindeberg and Levy)**.** We consider the case of i.i.d. χ^2 random variables $\xi_1, ..., \xi_n \sim \chi^2(3)$. Evidently, the functional form of the $\chi^2(3)$ distribution is quite different from the standard normal distribution; in particular, χ^2 random variables take only nonnegative values with nonzero probability (see Section 21.3). Nevertheless, their standardized sum asymptotically results in a normal distribution, as visualized in Figure 27.2. At the implementation level, we use the fact that for the χ^2 random variables $\xi_i, i = 1, ..., n$ with degrees-of-freedom parameter 3,

$$\mathbb{E}(\xi_i) = 3 \text{ and } \mathbb{V}(\xi_i) = 6. \tag{27.19}$$

The panels in Figure 27.2 A show histogram estimates of the probability density of

$$\zeta_n := \sqrt{n}\left(\frac{\bar{\xi}_n - \mu}{\sigma}\right) \tag{27.20}$$

based on 1000 realizations of ζ_n for $n = 2$ and $n = 200$, together with the PDF of $N(0, 1)$. Evidently, the distribution of the realizations of ζ_2 is still very unlike the standard normal distribution, whereas the distribution of the realizations of ζ_{200} already approaches the standard normal distribution. Figure 27.2 B shows the corresponding estimated CDFs, which are the objects addressed formally by Theorem 27.2.

△

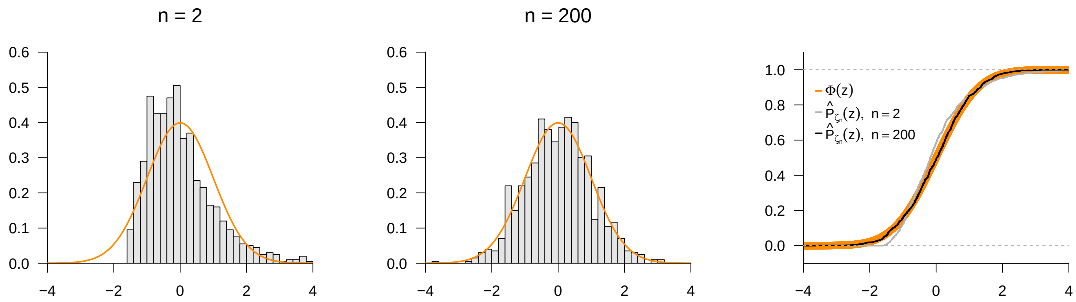


Figure 27.2 Simulation of the central limit theorem of Lindeberg and Levy.

The central limit theorem of Liapounov generalizes Theorem 27.2 to the case of random variables that need not be identically distributed. For a proof, we again refer to Henze (2024).

Theorem 27.3 (Central limit theorem of Liapounov). *Let $\xi_1, ..., \xi_n$ be independent but not necessarily identically distributed random variables with*

$$\mathbb{E}(\xi_i) := \mu_i \text{ and } \mathbb{V}(\xi_i) := \sigma_i^2 > 0 \text{ for all } i = 1, ..., n. \tag{27.21}$$

Furthermore, suppose that $\xi_1, ..., \xi_n$ satisfy the following properties:

$$\mathbb{E}(|\xi_i - \mu_i|^3) < \infty \text{ and } \lim_{n\to\infty} \frac{\sum_{i=1}^n \mathbb{E}\left(|\xi_i - \mu_i|^3\right)}{\left(\sum_{i=1}^n \sigma_i^2\right)^{3/2}} = 0. \tag{27.22}$$

Then, for the random variable ζ_n defined as

$$\zeta_n := \frac{\sum_{i=1}^n \xi_i - \sum_{i=1}^n \mu_i}{\sqrt{\sum_{i=1}^n \sigma_i^2}}, \tag{27.23}$$

and for all $z \in \mathbb{R}$,

$$\lim_{n\to\infty} P_{\zeta_n}(z) = \Phi(z), \tag{27.24}$$

where Φ denotes the CDF of the standard normal distribution.

∘

We show later that, for $n \to \infty$, this also implies

$$\sum_{i=1}^n \xi_i \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right). \tag{27.25}$$

27.3 Bibliographic remarks

On the mathematical-historical genesis of the central limit theorems, see Fischer (2011) and Ulyanov (2024). On the Lindeberg-Levy central limit theorem, see Lindeberg (1922) and Lévy (1937); on the Liapunov central limit theorem, see Liapounoff (1900) and Liapounoff (1901).

Study questions

1. State the weak law of large numbers.
2. Explain the central limit theorem of Lindeberg and Levy.
3. Why are central limit theorems important for probabilistic model building?

Study question answers

1. See Theorem 27.1.
2. The central limit theorem of Lindeberg and Levy states that the standardized sum of independent and identically distributed random variables is asymptotically normally distributed.
3. The central limit theorems justify the frequent assumption of normally distributed unknown influences as disturbance, error, deviation, or uncertainty variables in probabilistic models.

28 Transformations

The transformation of random vectors and the analysis of their resulting distributions is a central topic of probabilistic modeling and, in particular, of frequentist inference. Here, a *transformation* means the application of a function to a random vector, where the function either transforms its components individually or combines them with one another. The central question is the following: "If a random vector ξ has a given distribution, what is the distribution of the random vector υ that results from transforming ξ?" For the cases treated in this chapter, explicit solutions can be given for the distribution of the transformed random vectors. These are among the classical results of frequentist inference and are essential for understanding classical frequentist inference procedures such as confidence intervals and hypothesis tests.

For a first intuition, we restrict ourselves to the case of a single random variable. One can then understand the transformation of a random variable by looking at the transformation of its independent and identically distributed realizations. For example, consider the random variable $\xi \sim N(0,1)$ and its transformation $\upsilon := \xi^2$. If $x_1 = 0.10, x_2 = -0.20, x_3 = 0.80$ are three realizations of three independent copies of ξ, then this corresponds to the realizations $y_1 = x_1^2 = 0.01, y_2 = x_2^2 = 0.04, y_3 = x_3^2 = 0.64$ of υ. In this example, υ takes no negative values; the distribution of υ therefore assigns probability density 0 to negative values. The **R** code below simulates these ideas. Figure 28.1 shows the histogram of the obtained realizations of the random variable ξ considered here, and Figure 28.2 shows the histogram of the squared realizations of ξ, that is, independent and identically distributed realizations of υ.

```
# simulation specification
n       = 1e4                      # number of i.i.d. realizations
mu      = 1                        # expectation parameter of \xi
sigsqr  = 2                        # variance parameter of \xi

# squaring a random variable
x       = rnorm(n, mu, sqrt(sigsqr))   # realizations x_i, i = 1,....,n of \xi
y       = x^2                          # realizations y_i = x_i^2 of \upsilon

# output of the first eight values of the realizations of \xi and \upsilon
print(x[1:8], digits = 2)
```

```
[1]  2.85  2.26  1.62  2.07  0.72 -1.12  2.67  0.68
```

```
print(y[1:8], digits = 2)
```

```
[1] 8.11 5.10 2.63 4.30 0.52 1.25 7.14 0.46
```

The following theorem, which we will not prove, is fundamental for the considerations below.

Theorem 28.1 (Transformation of a random vector)**.** *Let $\xi : \Omega \to \mathcal{X}$ be a random vector, and let $f : \mathcal{X} \to \mathbb{R}^m$ be a multivariate vector-valued function. Then*

$$\upsilon : \Omega \to \mathbb{R}, \omega \mapsto \upsilon(\omega) := (f \circ \xi)(\omega) := f(\xi(\omega)) \tag{28.1}$$

is a random vector.

∘

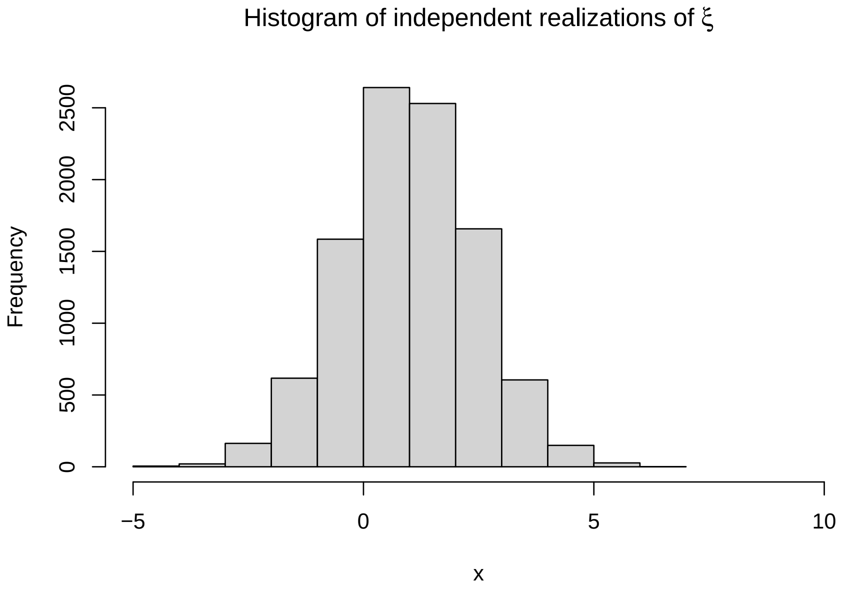


Figure 28.1 Histogram of 10,000 realizations of independent and identically normally distributed random variables.

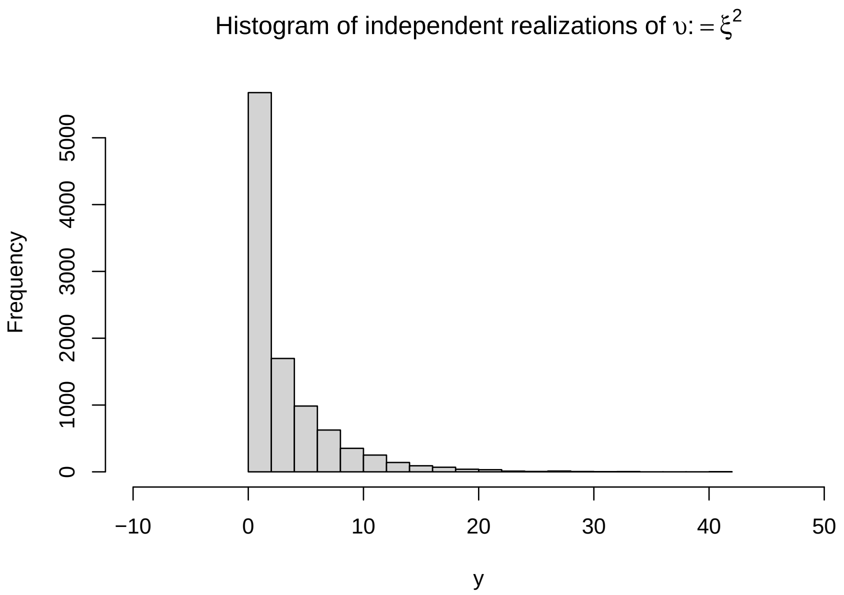


Figure 28.2 Histogram of 10,000 squared realizations of independent and identically normally distributed random variables.

The theorem formalizes the intuition established above that applying a function to a random quantity generally yields another random quantity. A proof of Theorem 28.1 would have to establish the measurability of υ as a consequence of the measurability of ξ. In the following, often $\mathcal{X} := \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}$. In this case we usually simply write $\upsilon := f(\xi)$ and call υ the *transformed random variable*.

In Section 28.1, we consider theorems on the transformation of random variables depending on the type of function f under consideration. In Section 28.2, we consider a theorem on the transformation of random vectors with a bijective transformation function and the determination of distributions under binary combinations of random variables.

28.1 Univariate transformation theorems

Theorem 28.2 gives a formula for computing the PDF p_υ of $\upsilon := f(\xi)$ when ξ is a random variable with PDF p_ξ and f is a bijective function.

Theorem 28.2 (Univariate PDF transformation for bijective mappings). *Let ξ be a random variable with PDF p_ξ for which $\mathbb{P}(]a,b[) = 1$, where a and/or b may be finite or infinite. Furthermore, let*

$$\upsilon := f(\xi), \tag{28.2}$$

where the univariate real-valued function $f :]a,b[\to \mathbb{R}$ is differentiable and bijective on $]a,b[$. Let $f(]a,b[)$ denote the image of $]a,b[$ under f. Finally, let $f^{-1}(y)$ be the value of the inverse function of $f(x)$ for $y \in f(]a,b[)$, and let $f'(x)$ be the derivative of f at x. Then the PDF of υ is given by

$$p_\upsilon : \mathbb{R} \to \mathbb{R}_{\geq 0}, y \mapsto p_\upsilon(y) := \begin{cases} \frac{1}{|f'(f^{-1}(y))|} p_\xi\left(f^{-1}(y)\right) & \text{for } y \in f(]a,b[) \\ 0 & \text{for } y \in \mathbb{R} \setminus f(]a,b[). \end{cases} \tag{28.3}$$

∘

Proof. First note that, because f is a differentiable bijective function on $]a,b[$, f is either strictly increasing or strictly decreasing. Suppose first that f is strictly increasing on $]a,b[$. Then f^{-1} is also increasing for all $y \in f(]a,b[)$, and

$$P_\upsilon(y) = \mathbb{P}(\upsilon \leq y) = \mathbb{P}\left(f(\xi) \leq y\right) = \mathbb{P}\left(f^{-1}(f(\xi)) \leq f^{-1}(y)\right) = \mathbb{P}\left(\xi \leq f^{-1}(y)\right) = P_\xi\left(f^{-1}(y)\right).$$

Thus, P_υ is differentiable at all points y at which both f^{-1} and P_ξ are differentiable. By the chain rule and the inverse function theorem $(f^{-1}(x))' = 1/f'(f^{-1}(x))$, the PDF p_υ is therefore obtained as follows:

$$p_\upsilon(y) = \frac{d}{dy}P_\upsilon(y) = \frac{d}{dy}P_\xi\left(f^{-1}(y)\right) = p_\xi\left(f^{-1}(y)\right)\frac{d}{dy}f^{-1}(y) = \frac{1}{f'\left(f^{-1}(y)\right)}p_\xi\left(f^{-1}(y)\right).$$

Because f^{-1} is strictly increasing, $d/dy(f^{-1}(y))$ is positive, and the theorem applies. Analogously, if f is strictly decreasing on $]a,b[$, then f^{-1} is also decreasing for all $y \in f(]a,b[)$, and

$$P_\upsilon(y) = \mathbb{P}(f(\xi) \leq y) = \mathbb{P}\left(f^{-1}(f(\xi)) \geq f^{-1}(y)\right) = \mathbb{P}\left(\xi \geq f^{-1}(y)\right) = 1 - P_\xi\left(f^{-1}(y)\right).$$

By the chain rule and the inverse function theorem,

$$p_\upsilon(y) = \frac{d}{dy}(1 - P_\upsilon(y)) = -\frac{d}{dy}P_\xi\left(f^{-1}(y)\right) = -p_\xi\left(f^{-1}(y)\right)\frac{d}{dy}f^{-1}(y) = -\frac{1}{f'\left(f^{-1}(y)\right)}p_\xi\left(f^{-1}(y)\right).$$

Because f^{-1} is strictly decreasing, $d/dy(f^{-1}(y))$ is negative, so $-d/dy(f^{-1}(y))$ equals $|d/dy(f^{-1}(y))|$, and the theorem applies. □

An important use case of Theorem 28.2 is Theorem 28.3. Compared with Theorem 28.2, Theorem 28.3 gives a simplified formula for computing the PDF p_υ of $\upsilon := f(\xi)$ when f is a linear-affine function.

Theorem 28.3 (Univariate PDF transformation theorem for linear-affine mappings)**.** *Let ξ be a random variable with PDF p_ξ, and let*

$$\upsilon = f(\xi) \text{ with } f(\xi) := a\xi + b \text{ for } a \neq 0. \tag{28.4}$$

Then the PDF of υ is given by

$$p_\upsilon : \mathbb{R} \to \mathbb{R}_{\geq 0}, y \mapsto p_\upsilon(y) := \frac{1}{|a|} p_\xi \left(\frac{y-b}{a}\right). \tag{28.5}$$

∘

Proof. First note that

$$f^{-1} : \mathbb{R} \to \mathbb{R}, y \mapsto f^{-1}(y) = \frac{y-b}{a} \tag{28.6}$$

because then $f \circ f^{-1} = \mathrm{id}_\mathbb{R}$, as can be seen from

$$f(f^{-1}(x)) = a\left(\frac{x-b}{a}\right) + b = x - b + b = x \text{ for all } x \in \mathbb{R}. \tag{28.7}$$

Furthermore,

$$f' : \mathbb{R} \to \mathbb{R}, x \mapsto f'(x) = \frac{d}{dx}(ax+b) = a. \tag{28.8}$$

Thus, by Theorem 28.2,

$$\begin{aligned} p_\upsilon : \mathbb{R} \to \mathbb{R}_{\geq 0}, y \mapsto p_\upsilon(y) &= \frac{1}{|f'(f^{-1}(y))|} p_\xi\left(f^{-1}(y)\right) \\ &= \frac{1}{|a|} p_\xi\left(\frac{y-b}{a}\right). \end{aligned} \tag{28.9}$$

□

Theorem 28.4 gives a formula for computing the PDF p_υ of $\upsilon := f(\xi)$ when f is bijective at least in pieces.

Theorem 28.4 (Univariate PDF transformation for piecewise bijective mappings)**.** *Let ξ be a random variable with outcome space $\mathcal{X}$ and PDF p_ξ. Furthermore, let*

$$\upsilon = f(\xi), \tag{28.10}$$

where f is such that the outcome space of ξ can be partitioned into a finite number of sets $\mathcal{X}_1, ..., \mathcal{X}_k$, with a corresponding number of sets $\mathcal{Y}_1 := f(\mathcal{X}_1), ..., \mathcal{Y}_k := f(\mathcal{X}_k)$ in the outcome space $\mathcal{Y}$ of υ (where not necessarily $\mathcal{Y}_i \cap \mathcal{Y}_j = \emptyset, 1 \leq i, j \leq k$), such that the mapping f is bijective for all $\mathcal{X}_1, ..., \mathcal{X}_k$, that is, f is a piecewise *bijective mapping. For $i = 1, ..., k$, let f_i^{-1} denote the inverse function of f on $\mathcal{Y}_i$. Finally, assume that the derivatives f_i' exist and are continuous for all $i = 1, ..., k$. Then a PDF of υ is given by*

$$p_\upsilon : \mathcal{Y} \to \mathbb{R}_{\geq 0}, y \mapsto p_\upsilon(y) := \sum_{i=1}^k 1_{\mathcal{Y}_i}(y) \frac{1}{|f_i'(f_i^{-1}(y))|} p_\xi\left(f_i^{-1}(y)\right). \tag{28.11}$$

∘

For a proof, we refer to Casella & Berger (2002).

28.2 Multivariate transformation theorems

Theorem 28.5 gives a formula for computing the PDF p_υ of $\upsilon := f(\xi)$ when ξ is a random vector with PDF p_ξ and f is a bijective multivariate vector-valued function. It is a direct generalization of Theorem 28.2.

Theorem 28.5 (Multivariate PDF transformation for bijective mappings)**.** *Let ξ be an n-dimensional random vector with outcome space $\mathbb{R}^n$ and PDF p_ξ. Furthermore, let*

$$\upsilon := f(\xi), \tag{28.12}$$

where the multivariate vector-valued function $f : \mathbb{R}^n \to \mathbb{R}^n$ is differentiable and bijective. Finally, let

$$J^f(x) = \left(\frac{\partial}{\partial x_j} f_i(x)\right)_{1 \le i \le n, 1 \le j \le n} \in \mathbb{R}^{n \times n} \tag{28.13}$$

be the Jacobian matrix of f at $x \in \mathbb{R}^n$, let $|J^f(x)|$ denote the determinant of $J^f(x)$, and suppose that $|J^f(x)| \neq 0$ for all $x \in \mathbb{R}^n$. Then a PDF of υ is given by

$$p_\upsilon : \mathbb{R}^n \to \mathbb{R}_{\ge 0}, y \mapsto p_\upsilon(y) := \begin{cases} \frac{1}{|J^f(f^{-1}(y))|} p_\xi\left(f^{-1}(y)\right) & \text{for } y \in f(\mathbb{R}^n) \\ 0 & \text{for } y \in \mathbb{R}^n \setminus f(\mathbb{R}^n) \end{cases} \tag{28.14}$$

∘

For a proof, we refer to Billingsley (1995).

The following so-called *convolution theorem* gives a formula for computing the PDF p_υ of $\upsilon := \xi_1 + \xi_2$ when ξ_1 and ξ_2 are two random variables with PDFs p_{ξ_1} and p_{ξ_2}.

Theorem 28.6 (Sum of independent random variables (convolution))**.** *Let ξ_1 and ξ_2 be two continuous independent random variables with PDFs p_{ξ_1} and p_{ξ_2}, respectively. Let $\upsilon := \xi_1 + \xi_2$ be the sum of ξ_1 and ξ_2. Then a PDF of the distribution of υ is given by*

$$p_\upsilon(y) = \int_{-\infty}^{\infty} p_{\xi_1}(y - x_2) p_{\xi_2}(x_2)\, dx_2 = \int_{-\infty}^{\infty} p_{\xi_1}(x_1) p_{\xi_2}(y - x_1)\, dx_1. \tag{28.15}$$

The formula for the PDF p_υ is called the convolution *of p_{ξ_1} and p_{ξ_2}.*

∘

Proof. We use the multivariate PDF transformation theorem for bijective mappings. To this end, first define

$$f : \mathbb{R}^2 \to \mathbb{R}^2, x \mapsto f(x) := \begin{pmatrix} x_1 + x_2 \\ x_2 \end{pmatrix} := \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}. \tag{28.16}$$

The inverse function of f is then given by

$$f^{-1} : \mathbb{R}^2 \to \mathbb{R}^2, z \mapsto f^{-1}(z) := \begin{pmatrix} z_1 - z_2 \\ z_2 \end{pmatrix}, \tag{28.17}$$

because then $f \circ f^{-1} = \mathrm{id}_{\mathbb{R}^2}$, as can be seen from

$$f^{-1}(f(x)) = f^{-1}\begin{pmatrix} x_1 + x_2 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 - x_2 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \tag{28.18}$$

The Jacobian matrix of f is

$$J^f(x) = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \frac{\partial}{\partial x_2} f_1(x) \\ \frac{\partial}{\partial x_1} f_2(x) & \frac{\partial}{\partial x_2} f_2(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1}(x_1 + x_2) & \frac{\partial}{\partial x_2}(x_1 + x_2) \\ \frac{\partial}{\partial x_1} x_2 & \frac{\partial}{\partial x_2} x_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \tag{28.19}$$

and hence the Jacobian determinant is $|J^f(x)| = 1$. Furthermore, independence of ξ_1 and ξ_2 implies

$$p_{\xi_1,\xi_2}(x_1, x_2) = p_{\xi_1}(x_1)p_{\xi_2}(x_2). \tag{28.20}$$

Substituting and integrating with respect to x_2 gives, for $z \in f(\mathbb{R}^2)$,

$$\begin{aligned} p_\zeta(z) &= \frac{1}{|J^f\left(f^{-1}(z)\right)|} p_\xi\left(f^{-1}(z)\right) \\ &= \frac{1}{1} p_{\xi_1,\xi_2}\left(z_1 - z_2, z_2\right) \\ &= p_{\xi_1}(z_1 - z_2)p_{\xi_2}(z_2). \end{aligned} \tag{28.21}$$

Integration over z_2 then gives a PDF for the marginal distribution of ζ_1,

$$p_{\zeta_1}(z_1) = \int_{-\infty}^{\infty} p_{\xi_1}(z_1 - z_2)p_{\xi_2}(z_2)\,dz_2. \tag{28.22}$$

With $\zeta_1 = \xi_1 + \xi_2 = \upsilon$, the first form of the convolution theorem follows as

$$p_\upsilon(y) = \int_{-\infty}^{\infty} p_{\xi_1}(y - x_2)p_{\xi_2}(x_2)\,dx_2. \tag{28.23}$$

□

Study questions

1. Explain the concept of the transformation of a random variable using the example $\upsilon := \xi^2$.

Study question answers

1. See the explanations of Figure 28.1 and Figure 28.2.

29 Normal distributions

The multivariate normal distribution is the multivariate generalization of the univariate normal distribution. The motivation for the widespread normality assumptions in probabilistic modeling is known to lie in the central limit theorem: in probabilistic models, probabilistic terms represent the summation of very many random processes that are not explained by the deterministic components of the respective model, that is, by a formalized scientific theory. According to the central limit theorem, the sum of these unexplained random processes is then precisely normally distributed.

Beyond this fundamental aspect, the normal distribution has many favorable mathematical properties that enable its use in many areas of probabilistic modeling. Applications of multivariate normal distributions are therefore found in the context of the general linear model, the generalizations of the general model to hierarchical linear models or multivariate general linear models, Bayesian inference under normality assumptions, and not least the theory of probabilistic filters, such as the Kalman-Bucy filter.

29.1 Construction of multivariate normal distributions

In this section, we show how a bivariately normally distributed random vector can be constructed by transforming and concatenating two univariately normally distributed random variables. To do so, we first recall the concept of a normally distributed random variable. We repeat the corresponding definition from Chapter 21 here to introduce the terminology for multivariate normal distributions.

Definition 29.1 (Normally distributed random variable)**.** ξ be a random variable with outcome space $\mathbb{R}$ and PDF

$$p : \mathbb{R} \to \mathbb{R}_{>0}, x \mapsto p(x) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{29.1}$$

Then we say that ξ follows a *normal distribution (or Gaussian distribution)* with *expectation parameter* $\mu \in \mathbb{R}$ and *variance parameter* $\sigma^2 > 0$ and call ξ a *normally distributed random variable*. We abbreviate this by $\xi \sim N(\mu, \sigma^2)$. We denote the PDF of a normally distributed random variable by

$$N\left(x; \mu, \sigma^2\right) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{29.2}$$

•

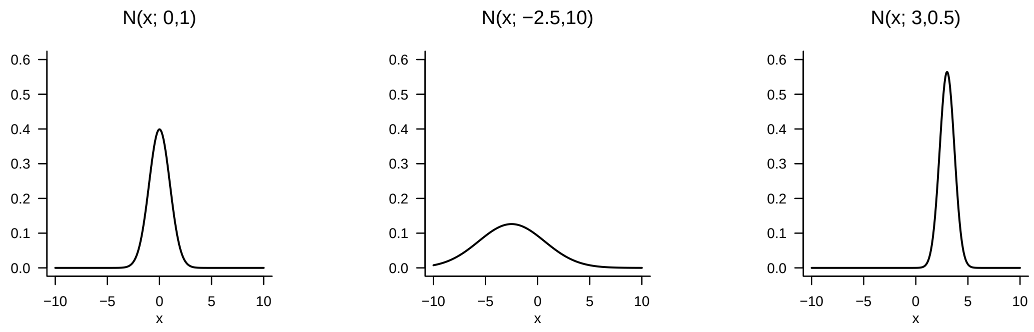


Figure 29.1 Probability density functions of univariate normal distributions.

Visually, the parameter μ of a normally distributed random variable corresponds to the value of highest probability density, and the parameter σ^2 specifies the width of the PDF (Figure 29.1). Further, as is well known, the expectation and the variance of a normally distributed random variable are

$$\mathbb{E}(\xi) = \mu \text{ and } \mathbb{V}(\xi) = \sigma^2. \tag{29.3}$$

Finally, a normally distributed random variable of the form $\xi \sim N(0,1)$ is also called *standard normally distributed.*

The following theorem shows how two independent, univariately standard normally distributed random variables can be combined to construct a bivariately distributed random vector. The distribution of such a random vector is then called a *bivariate normal distribution.*

Theorem 29.1 (Construction of bivariate normal distributions)**.** *Let $\zeta_1 \sim N(0,1)$ and $\zeta_2 \sim N(0,1)$ be two independent standard normally distributed random variables. Further, let $\mu_1, \mu_2 \in \mathbb{R}$, $\sigma_1, \sigma_2 > 0$ and $\rho \in]-1,1[$. Finally, let*

$$\begin{aligned} \xi_1 &:= \sigma_1\zeta_1 + \mu_1 \\ \xi_2 &:= \sigma_2\left(\rho\zeta_1 + (1-\rho^2)^{1/2}\zeta_2\right) + \mu_2. \end{aligned} \tag{29.4}$$

Then the PDF of the random vector $\xi := (\xi_1, \xi_2)^T$, that is, the joint distribution of ξ_1 and ξ_2, has the form

$$p : \mathbb{R}^2 \to \mathbb{R}_{>0},\, x \mapsto p(x) := (2\pi)^{-\frac{n}{2}}|\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right), \tag{29.5}$$

where $n := 2$ and $\mu \in \mathbb{R}^2$ and $\Sigma \in \mathbb{R}^{2\times 2}$ are given by

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \text{ and } \Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_2\sigma_1 & \sigma_2^2 \end{pmatrix}. \tag{29.6}$$

∘

For a proof of the theorem, we refer to DeGroot & Schervish (2012).

Example 29.1 (Construction of a bivariate normal distribution)**.** The following **R** code traces the above theorem using concrete example values for $\mu_1, \mu_2, \sigma_1, \sigma_2$ and ρ and prints the parameters μ and Σ of the resulting bivariate PDF.

```
# parameter definitions
mu_1   = 5.0                                      # \mu_1
mu_2   = 4.0                                      # \mu_2
sig_1  = 1.5                                      # \sigma_1
sig_2  = 1.0                                      # \sigma_2
rho    = 0.9                                      # \rho

# realizations of the standard normally distributed RVs
n      = 100                                      # number of realizations
zeta_1 = rnorm(n)                                 # \zeta_1 \sim N(0,1)
zeta_2 = rnorm(n)                                 # \zeta_1 \sim N(0,1)

# evaluation of realizations of \xi_1 and \xi_2
xi_1   = sig_1*zeta_1 + mu_1                      # realizations of zeta_1
xi_2   = sig_2*(rho*zeta_1 + sqrt(1-rho^2)*zeta_2) + mu_2 # realizations of zeta_2

# parameters of the joint distribution of \xi_1 and \xi_2
mu     = matrix(c(mu_1,                           # \mu \in \mathbb{R}^2
                  mu_2),
                nrow = 2, byrow = TRUE)
Sigma  = matrix(c(sig_1^2        , rho*sig_1*sig_2,       # \Sigma \in \mathbb{R}^{2 x 2}
                rho*sig_1*sig_2, sig_2^2),
                nrow = 2, byrow = TRUE)
print(mu)
```

```
     [,1]
[1,]    5
[2,]    4
```

```
print(Sigma)
```

```
     [,1] [,2]
[1,] 2.25 1.35
[2,] 1.35 1.00
```

The realizations of $\xi = (\xi_1, \xi_2)^T$ generated by the **R** code above, as well as the isocontours of the PDF postulated by Theorem 29.1, are shown in Figure 29.2.

$\triangle$

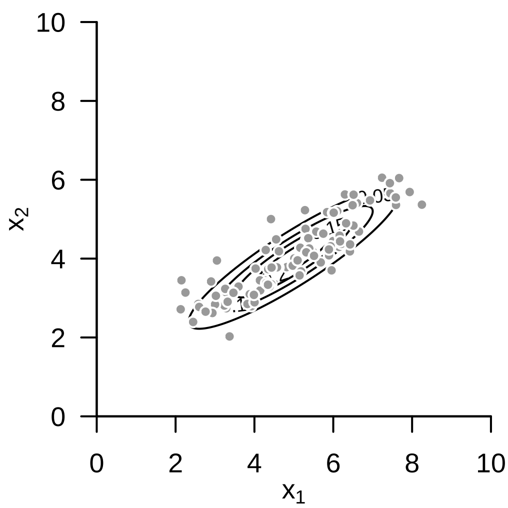


Figure 29.2 Construction of a bivariate normal distribution.

29.2 Definition of multivariate normal distributions

We now formally introduce the multivariate normal distribution and state first properties. We use the following definition for this purpose.

Definition 29.2. ξ be an n-dimensional random vector with outcome space $\mathbb{R}^n$ and PDF

$$p : \mathbb{R}^n \to \mathbb{R}_{>0},\, x \mapsto p(x) := (2\pi)^{-\frac{n}{2}}|\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right). \tag{29.7}$$

Then we say that ξ follows a *multivariate (or n-dimensional) normal distribution* with *expectation parameter* $\mu \in \mathbb{R}^n$ and positive-definite *covariance-matrix parameter* $\Sigma \in \mathbb{R}^{n\times n}$ and call ξ a *(multivariately) normally distributed random vector*. We abbreviate this by $\xi \sim N(\mu, \Sigma)$. We denote the PDF of a multivariately normally distributed random vector by

$$N(x;\mu,\Sigma) := (2\pi)^{-\frac{n}{2}}|\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right). \tag{29.8}$$

•

Example 29.2 (Bivariate normal distributions)**.** Figure 29.3 A, B, and C show the isocontours of the PDFs of bivariately normally distributed random vectors for $\mu = (1,1)^T$ and

$$\Sigma_A := \begin{pmatrix} 0.20 & 0.15 \\ 0.15 & 0.20 \end{pmatrix}, \quad \Sigma_B := \begin{pmatrix} 0.20 & 0.00 \\ 0.00 & 0.20 \end{pmatrix}, \quad \Sigma_C := \begin{pmatrix} 0.20 & -0.15 \\ -0.15 & 0.20 \end{pmatrix} \tag{29.9}$$

respectively. Figure 29.4 A, B, and C each show 200 realizations of the corresponding random vectors.

△

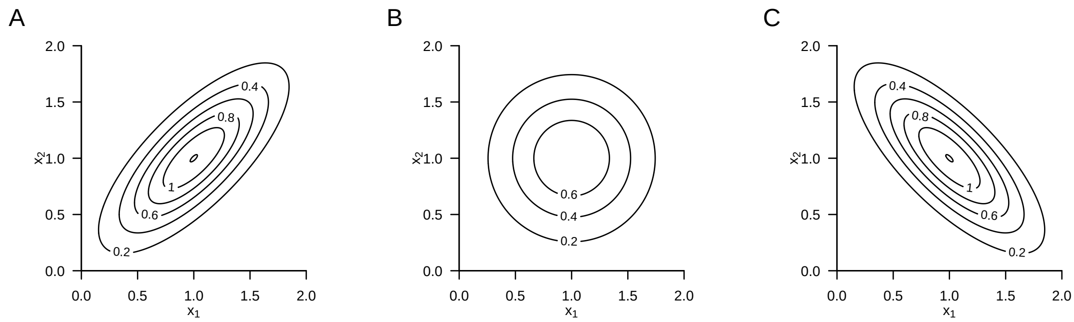


Figure 29.3 Probability density functions of bivariate normal distributions.

Without proof, we note that, as in the case of a univariately normally distributed random variable, the expectation and the covariance matrix of a normally distributed random vector are given by the corresponding parameters.

Theorem 29.2 (Expectation and covariance matrix of normally distributed random vectors)**.** *Let $\xi \sim N(\mu, \Sigma)$ be a multivariately normally distributed random vector with expectation parameter $\mu \in \mathbb{R}^n$ and covariance-matrix parameter $\Sigma \in \mathbb{R}^{n\times n}$ pd. Then*

$$\mathbb{E}(\xi) = \mu \text{ and } \mathbb{C}(\xi) = \Sigma. \tag{29.10}$$

∘

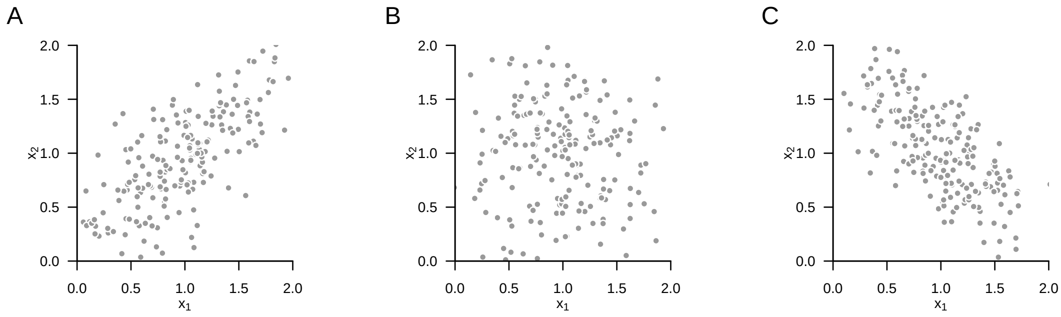


Figure 29.4 Realizations of bivariately normally distributed random vectors.

As in the case of univariately normally distributed random variables, the parameter $\mu \in \mathbb{R}^n$ corresponds to the value of highest probability density of the multivariate normal distribution. Analogously to the variance parameter of a univariately normally distributed random variable, the diagonal elements of $\Sigma \in \mathbb{R}^{n\times n}$ pd specify the width of the PDF with respect to the random-vector components $\xi_1, ..., \xi_n$. In general, in the case of a multivariately normally distributed random vector, the i, jth element of $\Sigma \in \mathbb{R}^{n\times n}$ pd now specifies the covariance of the random-vector components ξ_i and ξ_j.

29.3 Sphericity

The following theorem is central for the basic theory of the general linear model.

Theorem 29.3 (Spherical multivariate normal distribution)**.** *For $i = 1, ..., n$, let $N(x_i; \mu_i, \sigma^2)$ be the PDFs of n independent univariately normally distributed random variables $\xi_1, ..., \xi_n$ with $\mu_1, ..., \mu_n \in \mathbb{R}$ and $\sigma^2 > 0$. Further, let $N(x; \mu, \sigma^2 I_n)$ be the PDF of an n-variate normally distributed random vector ξ with expectation parameter $\mu := (\mu_1, ..., \mu_n)^T \in \mathbb{R}^n$. Then*

$$p_\xi(x) = p_{\xi_1,...,\xi_n}(x_1, ..., x_n) = \prod_{i=1}^n p_{\xi_i}(x_i) \tag{29.11}$$

and, in particular,

$$N\left(x; \mu, \sigma^2 I_n\right) = \prod_{i=1}^n N\left(x_i; \mu_i, \sigma^2\right) \tag{29.12}$$

for all $x = (x_1, ..., x_n)^T \in \mathbb{R}^n$.

∘

Proof. We show the identity of the multivariate PDF $N(x; \mu, \sigma^2 I_n)$ with the product of n univariate

PDFs $N(x_i;\mu_i,\sigma^2 I_n)$, where μ_i is the ith entry of $\mu \in \mathbb{R}^n$. We obtain

$$
\begin{aligned}
N\left(x;\mu,\sigma^2 I_n\right) & = (2\pi)^{-\frac{n}{2}}\left|\sigma^2 I_n\right|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T(\sigma^2 I_n)^{-1}(x-\mu)\right) \\
& = \left(\prod_{i=1}^n 2\pi^{-\frac{1}{2}}\right)(\sigma^2)^{-\frac{n}{2}}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^T(x-\mu)\right) \\
& = \left(\prod_{i=1}^n (2\pi\sigma^2)^{-\frac{1}{2}}\right)\exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i-\mu_i)^2\right) \\
& = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}}\prod_{i=1}^n \exp\left(-\frac{1}{2\sigma^2}(x_i-\mu_i)^2\right) \\
& = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}(x_i-\mu_i)^2\right) \\
& = \prod_{i=1}^n N\left(x_i;\mu_i,\sigma^2\right).
\end{aligned}
\tag{29.13}
$$

□

A covariance-matrix parameter of the form $\Sigma = \sigma^2 I_n$ is also called *spherical*, because the isocontours of the PDF of a normally distributed random vector with such a covariance-matrix parameter form *spheres* (for example, circles for $n = 2$ and balls for $n = 3$). A multivariate normal distribution with a spherical covariance-matrix parameter is correspondingly called a *spherical normal distribution*. Theorem 29.3 states that the PDF of an n-dimensional normally distributed random vector with spherical covariance parameter corresponds to the joint PDF of n independent univariately normally distributed random variables, and conversely. A realization of an n-dimensional normally distributed random vector therefore corresponds to the realizations of n independent univariately normally distributed random variables, and conversely. Note that the identity of the distributions of the $\xi_i, i = 1,...,n$ is not assumed here; in particular, their expectation parameters $\mu_i, i = 1,...,n$ may explicitly differ.

29.4 Marginal, conditional, and joint distributions

Multivariate normal distributions have the property that all other associated distributions are also normal distributions and that their expectation and covariance-matrix parameters can be calculated from the parameters of the respective complementary distribution. Specifically, on the one hand, the univariate and multivariate marginal distributions of multivariate normal distributions are again normal distributions. On the other hand, like all multivariate distributions, multivariate normal distributions can be decomposed multiplicatively into a marginal and a conditional distribution. In particular, however, for multivariate normal distributions these distributions are again (multivariate) normal distributions whose parameters can be calculated from the parameters of the joint distribution, and conversely. We summarize the above observations formally in the following three theorems.

Theorem 29.4 (Marginal normal distributions)**.** *Let $n := k + l$ and let $\xi = (\xi_1,...,\xi_n)^T$ be an n-dimensional normally distributed random vector with expectation parameter*

$$
\mu = \begin{pmatrix}\mu_\upsilon \\ \mu_\zeta\end{pmatrix} \in \mathbb{R}^n, \tag{29.14}
$$

with $\mu_\upsilon \in \mathbb{R}^k$ *and* $\mu_\zeta \in \mathbb{R}^l$ *and covariance-matrix parameter*

$$\Sigma = \begin{pmatrix} \Sigma_{\upsilon\upsilon} & \Sigma_{\upsilon\zeta} \\ \Sigma_{\zeta\upsilon} & \Sigma_{\zeta\zeta} \end{pmatrix} \in \mathbb{R}^{n \times n}, \tag{29.15}$$

with $\Sigma_{\upsilon\upsilon} \in \mathbb{R}^{k\times k}$, $\Sigma_{\upsilon\zeta} \in \mathbb{R}^{k\times l}$, $\Sigma_{\zeta\upsilon} \in \mathbb{R}^{l\times k}$, *and* $\Sigma_{\zeta\zeta} \in \mathbb{R}^{l\times l}$. *Then* $\upsilon := (\xi_1, ..., \xi_k)^T$ *and* $\zeta := (\xi_{k+1}, ..., \xi_n)^T$ *are* k*- and* l*-dimensional normally distributed random vectors, respectively, and*

$$\upsilon \sim N(\mu_\upsilon, \Sigma_{\upsilon\upsilon}) \text{ and } \zeta \sim N(\mu_\zeta, \Sigma_{\zeta\zeta}). \tag{29.16}$$

∘

The marginal distributions of a multivariate normal distribution are therefore also normal distributions, and the parameters of the marginal distributions follow from the parameters of the joint distribution. For proofs of this theorem, we refer to Mardia et al. (1979) and Anderson (2003).

Example 29.3 (Marginal normal distributions)**.** Figure 29.5 visualizes Theorem 29.4 for the case $n := 2, k := 1, l := 1$,

$$\mu := \begin{pmatrix} 1 \\ 2 \end{pmatrix} \in \mathbb{R}^2 \text{ and } \Sigma := \begin{pmatrix} 0.10 & 0.08 \\ 0.08 & 0.15 \end{pmatrix} \in \mathbb{R}^{2\times 2}. \tag{29.17}$$

Figure 29.5 A shows the PDF of the bivariate random vector ξ, and Figure 29.5 B and C show the PDFs of the corresponding marginal random variables υ and ζ.

△

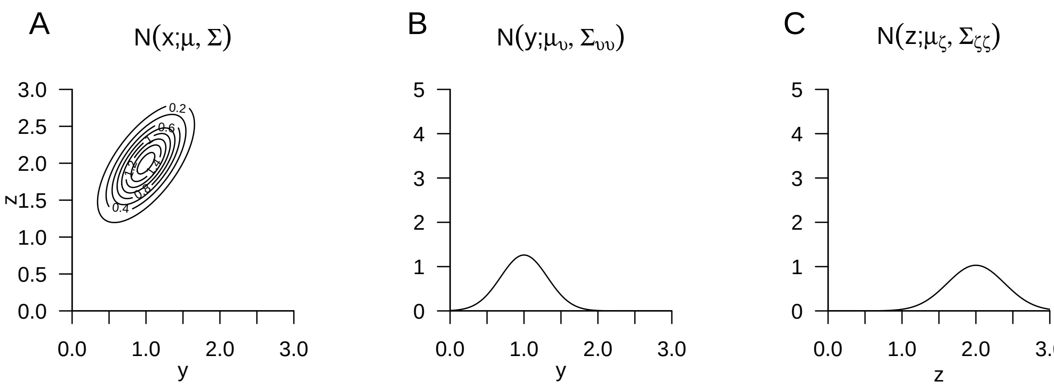


Figure 29.5 Marginal distributions of a bivariate normally distributed random vector.

Using a marginal and a conditional multivariate normal distribution, one can construct a joint multivariate normal distribution whose parameters follow from the parameters of the marginal and conditional distributions. This is the central statement of the following theorem.

Theorem 29.5 (Joint normal distributions). *Let ξ be an m-dimensional normally distributed random vector with PDF*

$$p_\xi : \mathbb{R}^m \to \mathbb{R}_{>0},\ x \mapsto p_\xi(x) := N(x; \mu_\xi, \Sigma_{\xi\xi}) \text{ with } \mu_\xi \in \mathbb{R}^m, \Sigma_{\xi\xi} \in \mathbb{R}^{m\times m}, \tag{29.18}$$

let $A \in \mathbb{R}^{n\times m}$ be a matrix, let $b \in \mathbb{R}^n$ be a vector, and let υ be an n-dimensional conditionally normally distributed random vector with conditional PDF

$$p_{\upsilon|\xi}(\cdot|x) : \mathbb{R}^n \to \mathbb{R}_{>0},\ y \mapsto p_{\upsilon|\xi}(y|x) := N(y; A\xi + b, \Sigma_{\upsilon\upsilon}) \text{ with } \Sigma_{\upsilon\upsilon} \in \mathbb{R}^{n\times n}. \tag{29.19}$$

Then the $m+n$-dimensional random vector $(\xi, \upsilon)^T$ is normally distributed with (joint) PDF

$$p_{\xi,\upsilon} : \mathbb{R}^{m+n} \to \mathbb{R}_{>0},\ \begin{pmatrix} x \\ y \end{pmatrix} \mapsto p_{\xi,\upsilon}\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = N\left(\begin{pmatrix} x \\ y \end{pmatrix}; \mu_{\xi,\upsilon}, \Sigma_{\xi,\upsilon}\right), \tag{29.20}$$

with $\mu_{\xi,\upsilon} \in \mathbb{R}^{m+n}$ and $\Sigma_{\xi,\upsilon} \in \mathbb{R}^{m+n\times m+n}$ and, in particular,

$$\mu_{\xi,\upsilon} = \begin{pmatrix} \mu_\xi \\ A\mu_\xi + b \end{pmatrix} \text{ and } \Sigma_{\xi,\upsilon} = \begin{pmatrix} \Sigma_{\xi\xi} & \Sigma_{\xi\xi}A^T \\ A\Sigma_{\xi\xi} & \Sigma_{\upsilon\upsilon} + A\Sigma_{\xi\xi}A^T \end{pmatrix}. \tag{29.21}$$

∘

The parameters of the joint distribution therefore follow as linear-affine transformations of the parameters of the inducing marginal and conditional distributions.

Example 29.4 (Joint normal distributions). Figure 29.6 visualizes Theorem 29.5 for the case $m := 1, n := 1, \mu_\xi := 1, \Sigma_{\xi\xi} := 0.2, A := 1, b := 1$ and $\Sigma_{\upsilon\upsilon} := 0.1$. Figure 29.6 A shows the PDF of the random variable ξ, Figure 29.6 B shows the PDF of the conditional distribution of the random variable υ given ξ, and Figure 29.6 C finally shows the PDFs of the induced bivariate random vector (ξ, υ).

△

The definition of a multivariate normal distribution further makes it possible to determine the conditional distributions of all components of the corresponding random vector directly using the parameters of the multivariate normal distribution. This is the central statement of the following theorem.

Theorem 29.6 (Conditional normal distributions). *Let (ξ, υ) be an $m+n$-dimensional normally distributed random vector with PDF*

$$p_{\xi,\upsilon} : \mathbb{R}^{m+n} \to \mathbb{R}_{>0},\ \begin{pmatrix} x \\ y \end{pmatrix} \mapsto p_{\xi,\upsilon}\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) := N\left(\begin{pmatrix} x \\ y \end{pmatrix}; \mu_{\xi,\upsilon}, \Sigma_{\xi,\upsilon}\right), \tag{29.22}$$

with

$$\mu_{\xi,\upsilon} = \begin{pmatrix} \mu_\xi \\ \mu_\upsilon \end{pmatrix}, \Sigma_{\xi,\upsilon} = \begin{pmatrix} \Sigma_{\xi\xi} & \Sigma_{\xi\upsilon} \\ \Sigma_{\upsilon\xi} & \Sigma_{\upsilon\upsilon} \end{pmatrix}, \tag{29.23}$$

with $x, \mu_\xi \in \mathbb{R}^m, y, \mu_\upsilon \in \mathbb{R}^n$ and $\Sigma_{\xi\xi} \in \mathbb{R}^{m\times m}, \Sigma_{\xi\upsilon} \in \mathbb{R}^{m\times n}, \Sigma_{\upsilon\upsilon} \in \mathbb{R}^{n\times n}$. Then the conditional distribution of ξ given υ is an m-dimensional normal distribution with conditional PDF

$$p_{\xi|\upsilon}(\cdot|y) : \mathbb{R}^m \to \mathbb{R}_{>0}, x \mapsto p_{\xi|\upsilon}(x|y) := N(x; \mu_{\xi|\upsilon}, \Sigma_{\xi|\upsilon}) \tag{29.24}$$

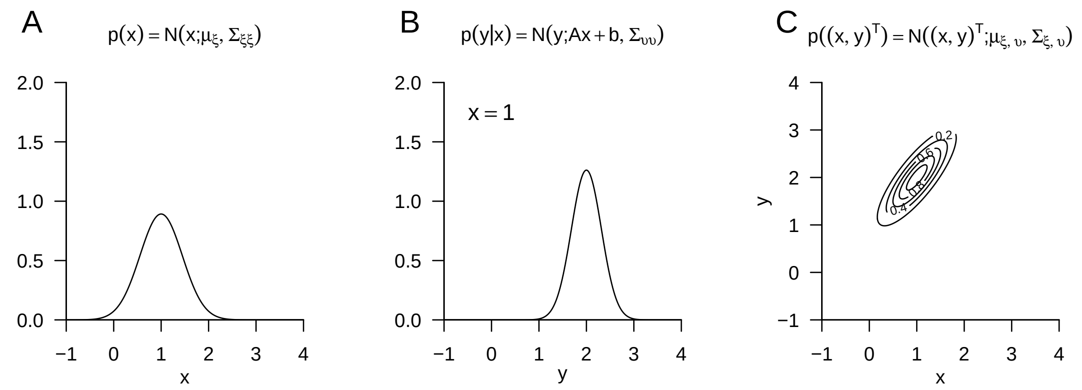


Figure 29.6 Joint distributions of a marginal and a normally distributed random variable conditional on it.

with

$$\mu_{\xi|\upsilon} = \mu_\xi + \Sigma_{\xi\upsilon}\Sigma_{\upsilon\upsilon}^{-1}(y - \mu_\upsilon) \in \mathbb{R}^m \tag{29.25}$$

and

$$\Sigma_{\xi|\upsilon} = \Sigma_{\xi\xi} - \Sigma_{\xi\upsilon}\Sigma_{\upsilon\upsilon}^{-1}\Sigma_{\upsilon\xi} \in \mathbb{R}^{m\times m}. \tag{29.26}$$

∘

Together with Theorem 29.5 and Theorem 29.4, the parameters of conditional and marginal normal distributions can therefore be determined from the parameters of the complementary conditional and marginal normal distributions.

Example 29.5 (Conditional normal distributions)**.** Figure 29.7 visualizes Theorem 29.6 for the case $m := 2, n := 1$,

$$\mu := \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{ and } \Sigma := \begin{pmatrix} 0.12 & 0.09 \\ 0.09 & 0.12 \end{pmatrix}. \tag{29.27}$$

Here Figure 29.7 A shows the PDF of the bivariate random vector $(\xi, \upsilon)^T$, and Figure 29.7 B and C show the PDF of the conditional distribution of the random variable ξ given $\upsilon = 1.5$ and $\upsilon = 2.8$, respectively.

△

29.5 Multivariate transformations

In this section, we collect some results on the distributions of transformed normally distributed random vectors. We omit proofs.

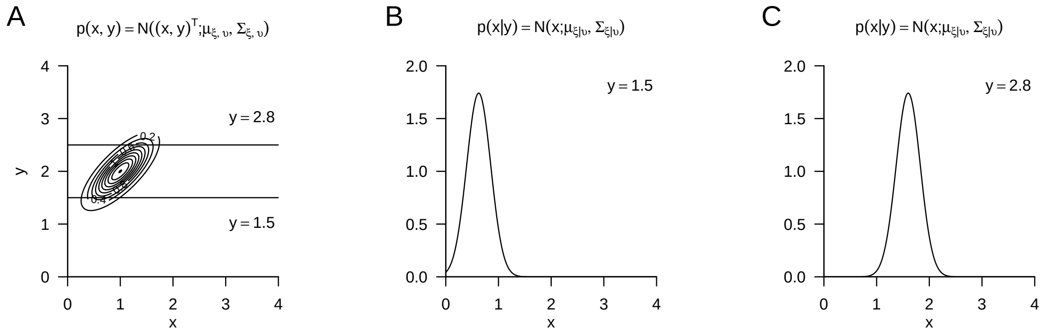


Figure 29.7 Conditional normal distributions.

Theorem 29.7 (Invertible linear transformation of a normally distributed random vector)**.** *Let $\xi \sim N(\mu_\xi, \Sigma_\xi)$ be a normally distributed n-dimensional random vector and let $\zeta := A\xi$ with an invertible matrix $A \in \mathbb{R}^{n\times n}$. Then*

$$\zeta \sim N\left(\mu_\zeta, \Sigma_\zeta\right) \text{ with } \mu_\zeta = A\mu_\xi \text{ and } \Sigma_\zeta = A\Sigma_\xi A^T. \tag{29.28}$$

∘

By Theorem 29.7, the invertible linear transformation of a multivariately normally distributed random vector therefore again yields a multivariately normally distributed random vector, and the parameters of the distribution of this normally distributed random vector follow from the parameters of the distribution of the original random vector and the transformation matrix.

Example 29.6 (Invertible linear transformation of a bivariate normal distribution)**.** As an example, we consider the invertible linear transformation of a bivariately normally distributed random vector ξ. Let

$$\mu_\xi := \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \Sigma_\xi := \begin{pmatrix} 0.20 & 0.15 \\ 0.15 & 0.20 \end{pmatrix} \tag{29.29}$$

be the expectation and covariance-matrix parameters of ξ, respectively, and let

$$A := \begin{pmatrix} -2 & 1 \\ -1 & 2 \end{pmatrix} \tag{29.30}$$

be the transformation matrix. Since $|A| = -3 \neq 0$, A is invertible, and by Theorem 29.7,

$$\zeta \sim N(\mu_\zeta, \Sigma_\zeta) \text{ with } \mu_\zeta = A\mu_\xi = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ and } A\Sigma_\xi A^T = \begin{pmatrix} 0.40 & 0.05 \\ 0.05 & 0.40 \end{pmatrix}. \tag{29.31}$$

Figure 29.8 A shows isocontours of the PDF of ξ and realizations $x^{(i)} \in \mathbb{R}^2$ of ξ for $i = 1, ..., 50$. Figure 29.8 B shows the transformed realizations $z^{(i)} = Ax^{(i)} \in \mathbb{R}^2$ of ζ as well as the isocontours of the PDF of ζ according to Theorem 29.7.

△

The fact that a linearly transformed normally distributed random vector is again normally distributed and that the parameters of the distribution of the transformed random vector can be determined from the parameters of the distribution of the original random vector and the transformation parameters remains true in the case of a not necessarily invertible linear transformation and also in the case of a not necessarily invertible linear-affine transformation. This is the statement of the following central theorem.

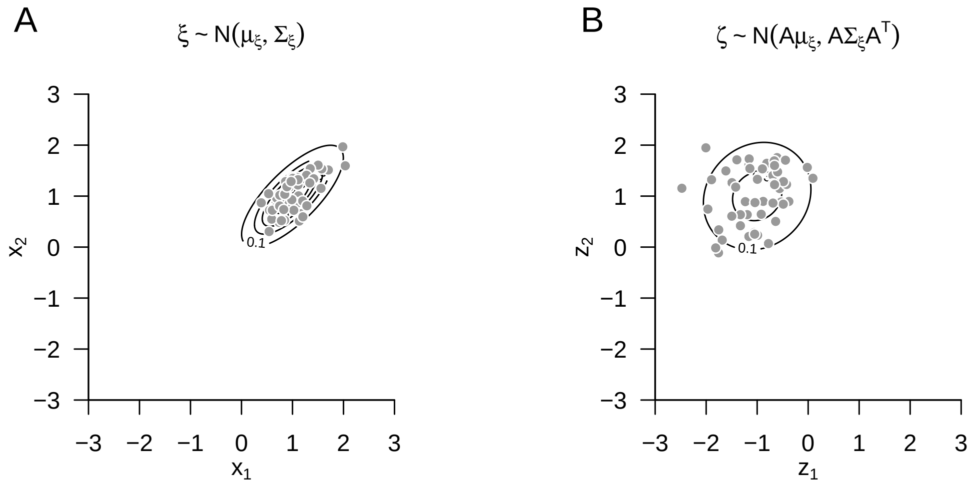


Figure 29.8 Invertible linear transformation of a normally distributed random vector.

Theorem 29.8 (Linear-affine transformation of a normally distributed random vector). *Let $\xi \sim N(\mu_\xi, \Sigma_\xi)$ be a normally distributed n-dimensional random vector and let*

$$\zeta := Ax + b \text{ with } A \in \mathbb{R}^{m \times n} \text{ and } b \in \mathbb{R}^m. \tag{29.32}$$

Then

$$\zeta \sim N(\mu_\zeta, \Sigma_\zeta) \text{ with } \mu_\zeta = A\mu + b \in \mathbb{R}^m \text{ and } \Sigma_\zeta = A\Sigma A^T \in \mathbb{R}^{m \times m}. \tag{29.33}$$

∘

For a proof, we refer to Anderson (2003).

29.6 Transformations of normally distributed random variables

In this section, we discuss six transformations of normally distributed random variables that play central roles in frequentist inference and are applications of the transformation theorems introduced in Chapter 28. Our statements are usually of the general form "If $\xi_i, i = 1, ..., n$ are independent and identically normally distributed random variables and $\upsilon := f(\xi_1, ..., \xi_n)$ is a transformation of these random variables, then the PDF of υ is given by the formula $p_\upsilon := \{\text{formula}\}$, and the distribution of υ is called *distribution name*." Statements of this form are central for frequentist inference because, first, the central limit theorems justify the assumption of additively independent normally distributed disturbance variables, and therefore normally distributed data; second, as we will see in frequentist inference, estimators and statistics are transformations of random variables; and third, quality criteria for parameter estimators, confidence intervals, and hypothesis tests in frequentist inference are characterized and justified by the distributions of the respective estimators and statistics.

Summation transformation

In this section, we consider the resulting distributions under summation and sample-mean formation of independently and identically normally distributed random variables. Specifically, Theorem 29.9 states that the sum of independently normally distributed random variables is again normally distributed and gives the parameters of this distribution, while Theorem 29.10 states that the sample mean of independently normally distributed random variables is again normally distributed and gives the parameters of this distribution.

Theorem 29.9 (Summation transformation)**.** *For $i = 1, ..., n$, let $\xi_i \sim N(\mu_i, \sigma_i^2)$ be independent normally distributed random variables. Then, for the sum $\upsilon := \sum_{i=1}^n \xi_i$,*

$$\upsilon \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right). \tag{29.34}$$

For independent and identically normally distributed random variables $\xi_i \sim N(\mu, \sigma^2)$, it consequently follows that

$$\upsilon \sim N(n\mu, n\sigma^2). \tag{29.35}$$

∘

Proof. Using Theorem 28.6, we sketch that for $\xi_1 \sim N(\mu_1, \sigma_1^2)$, $\xi_2 \sim N(\mu_2, \sigma_2^2)$, and $\upsilon := \xi_1 + \xi_2$, we have $\upsilon \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. For $n > 2$, the theorem then follows by iteration. With the definition of the PDF of the normal distribution, we first obtain

$$\begin{aligned}
p_\upsilon(y) &= \int_{-\infty}^{\infty} p_{\xi_1}(x_1) p_{\xi_2}(y - x_1)\, dx_1 \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left(-\frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2\right) \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{1}{2}\left(\frac{y - x_1 - \mu_2}{\sigma_2}\right)^2\right) dx_1 \\
&= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 - \frac{1}{2}\left(\frac{y - x_1 - \mu_2}{\sigma_2}\right)^2\right) dx_1.
\end{aligned} \tag{29.36}$$

With some algebraic effort, one obtains the identity

$$-\frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 - \frac{1}{2}\left(\frac{y - x_1 - \mu_2}{\sigma_2}\right)^2 = -\frac{(y - \mu_1 - \mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)} - \frac{((\sigma_1^2 + \sigma_2^2)x_1 - \sigma_1^2 y + \mu_2\sigma_1^2 - \mu_1\sigma_2^2)^2}{2\sigma_1^2\sigma_2^2(\sigma_1^2 + \sigma_2^2)}, \tag{29.37}$$

so that we further have

$$\begin{aligned}
p_\upsilon(y) &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(y - \mu_1 - \mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)} - \frac{((\sigma_1^2 + \sigma_2^2)x_1 - \sigma_1^2 y + \mu_2\sigma_1^2 - \mu_1\sigma_2^2)^2}{2\sigma_1^2\sigma_2^2(\sigma_1^2 + \sigma_2^2)}\right) dx_1 \\
&= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(y - \mu_1 - \mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)}\right) \exp\left(-\frac{((\sigma_1^2 + \sigma_2^2)x_1 - \sigma_1^2 y + \mu_2\sigma_1^2 - \mu_1\sigma_2^2)^2}{2\sigma_1^2\sigma_2^2(\sigma_1^2 + \sigma_2^2)}\right) dx_1 \\
&= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(y - \mu_1 - \mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{((\sigma_1^2 + \sigma_2^2)x_1 - \sigma_1^2 y + \mu_2\sigma_1^2 - \mu_1\sigma_2^2)^2}{2\sigma_1^2\sigma_2^2(\sigma_1^2 + \sigma_2^2)}\right) dx_1.
\end{aligned} \tag{29.38}$$

For the remaining integral, one can show by integration through substitution that

$$\int_{-\infty}^{\infty} \exp\left(-\frac{((\sigma_1^2 + \sigma_2^2)x_1 - \sigma_1^2 y + \mu_2\sigma_1^2 - \mu_1\sigma_2^2)^2}{2\sigma_1^2\sigma_2^2(\sigma_1^2 + \sigma_2^2)}\right) dx_1 = \frac{\sqrt{2\pi}\sigma_1\sigma_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}. \tag{29.39}$$

Thus

$$
\begin{aligned}
p_\upsilon(y) &= \frac{1}{2\pi\sigma_1\sigma_2}\frac{\sqrt{2\pi}\sigma_1\sigma_2}{\sqrt{\sigma_1^2+\sigma_2^2}}\exp\left(-\frac{(y-\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right) \\
&= \frac{(2\pi)^{-1}(2\pi)^{2}}{\sqrt{\sigma_1^2+\sigma_2^2}}\exp\left(-\frac{(y-\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right) \\
&= \frac{1}{\sqrt{2\pi}\sqrt{\sigma_1^2+\sigma_2^2}}\exp\left(-\frac{(y-\mu_1-\mu_2)^2}{2(\sigma_1^2+\sigma_2^2)}\right).
\end{aligned} \tag{29.40}
$$

Finally, it follows that

$$
p_\upsilon(y) = \frac{1}{\sqrt{2\pi(\sigma_1^2+\sigma_2^2)}}\exp\left(-\frac{1}{2(\sigma_1^2+\sigma_2^2)}\left(y-(\mu_1+\mu_2)\right)^2\right) = N(y;\mu_1+\mu_2,\sigma_1^2+\sigma_2^2) \tag{29.41}
$$

A simpler approach probably follows after Fourier transformation of the PDF in the sense of the so-called characteristic function of a random variable. In this case, the convolution of the PDFs would correspond to multiplication of the characteristic functions. □

An important application of Theorem 29.9 is the following Theorem 29.10 as well as the generalizations of the central limit theorems mentioned in Equation 27.18 and Equation 27.25. We visualize Theorem 29.9 by example in Figure 29.9.

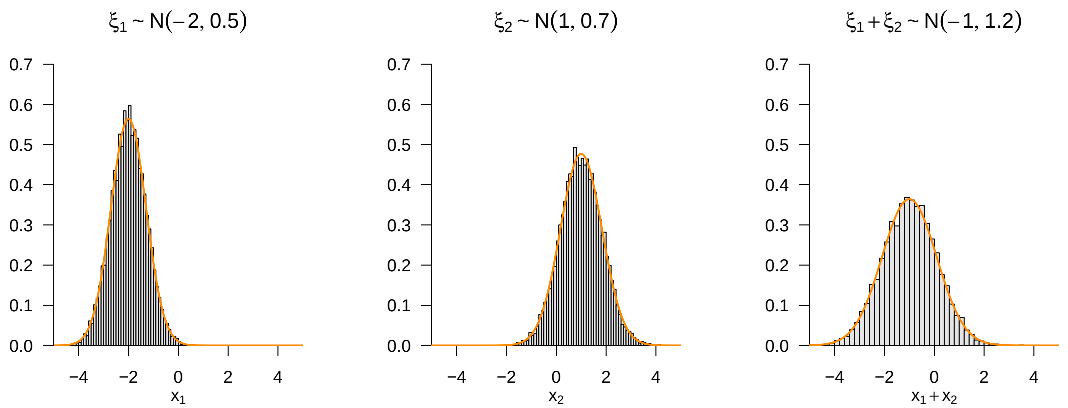


Figure 29.9 Summation of normally distributed random variables.

Sample-mean transformation

Theorem 29.10 (Sample-mean transformation). *For $i = 1, ..., n$, let $\xi_i \sim N(\mu, \sigma^2)$ be independent and identically normally distributed random variables. Then, for the sample mean $\bar{\xi}_n := \frac{1}{n}\sum_{i=1}^n \xi_i$,*

$$
\bar{\xi}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right). \tag{29.42}
$$

∘

Proof. We first note that, by the theorem on the sum of independent normally distributed random variables,

$\bar{\xi}_n = \frac{1}{n}\upsilon$ with $\upsilon := \sum_{i=1}^n \xi_i \sim N(n\mu, n\sigma^2)$. Substitution into Theorem 28.3 then yields

$$
\begin{aligned}
p_{\bar{\xi}_n}(\bar{x}_n) &= \frac{1}{|1/n|} N\left(n\bar{x}_n; n\mu, n\sigma^2\right) \\
&= \frac{n}{\sqrt{2\pi n\sigma^2}} \exp\left(-\frac{1}{2n\sigma^2}\left(n\bar{x}_n - n\mu\right)^2\right) \\
&= \frac{n}{\sqrt{2\pi n\sigma^2}} \exp\left(-\frac{1}{2n\sigma^2}\left(n\bar{x}_n - n\mu\right)^2\right) \\
&= nn^{-\frac{1}{2}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(n\bar{x}_n)^2}{2n\sigma^2} + \frac{2(n\bar{x}_n)(n\mu)}{2n\sigma^2} - \frac{(n\mu)^2}{2n\sigma^2}\right) \\
&= \sqrt{n}\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{n\bar{x}_n^2}{2\sigma^2} + \frac{2n\bar{x}_n\mu}{2\sigma^2} - \frac{n\mu^2}{2\sigma^2}\right) \\
&= \frac{1}{1/\sqrt{n}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\bar{x}_n^2}{2(\sigma^2/n)} + \frac{2\bar{x}_n\mu}{2(\sigma^2/n)} - \frac{\mu^2}{2(\sigma^2/n)}\right) \\
&= \frac{1}{\sqrt{2\pi(\sigma^2/n)}} \exp\left(-\frac{1}{2(\sigma^2/n)}(\bar{x}_n - \mu)^2\right) \\
&= N\left(\bar{x}_n; \mu, \sigma^2/n\right)
\end{aligned}
\tag{29.43}
$$

□

Important applications of Theorem 29.10 are the analysis of expectation estimators in point estimation and the generalization of the central limit theorem after Lindeberg and Levy mentioned in Equation 27.18. We visualize Theorem 29.10 by example in Figure 29.10.

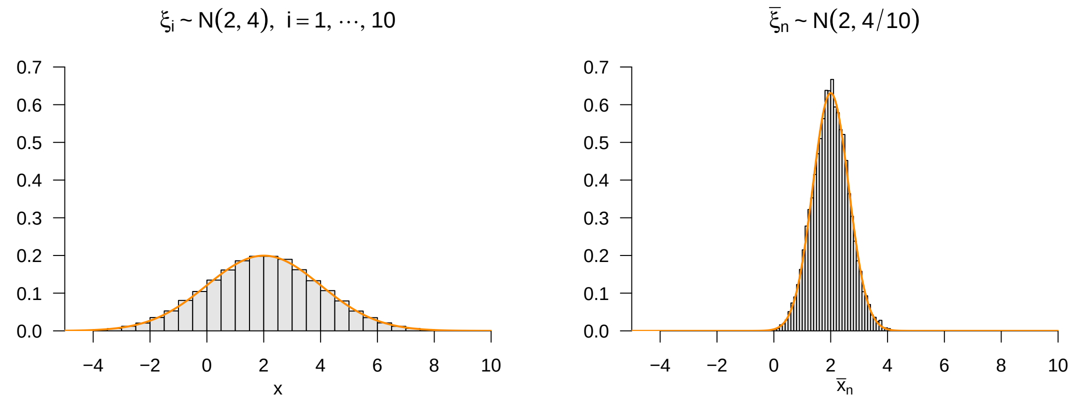


Figure 29.10 Sample-mean formation for normally distributed random variables.

Z-transformation

Theorem 29.11 states that subtracting the expectation parameter and simultaneously dividing by the square root of the variance parameter transforms the distribution of a normally distributed random variable into a standard normal distribution.

Theorem 29.11 (Z-transformation). *Let $\upsilon \sim N(\mu, \sigma^2)$ be a normally distributed random variable. Then the random variable*

$$
Z := \frac{\upsilon - \mu}{\sigma} \tag{29.44}
$$

is a standard normally distributed random variable, that is, $Z \sim N(0, 1)$.

∘

Proof. We use Theorem 28.3. To do so, we first note that the Z-transformation corresponds to a function of the form

$$f(\upsilon) := \frac{\upsilon - \mu}{\sigma} =: Z. \tag{29.45}$$

We further observe that the inverse function of f is given by

$$f^{-1}(Z) := \sigma Z + \mu, \tag{29.46}$$

because for all $z \in \mathbb{R}$ with $z = y - \mu\sigma$,

$$\zeta^{-1}(z) = \zeta^{-1}\left(\frac{y - \mu}{\sigma}\right) = \frac{\sigma(y - \mu)}{\sigma} + \mu = y - \mu + \mu = y. \tag{29.47}$$

Finally, we observe that the derivative f' of f is

$$f'(y) = \frac{d}{dy}\left(\frac{y - \mu}{\sigma}\right) = \frac{d}{dy}\left(\frac{y}{\sigma} - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma}. \tag{29.48}$$

Substitution into Theorem 28.3 then yields

$$\begin{aligned}
p_Z(z) &= \frac{1}{|1/\sigma|} N\left(\sigma z + \mu; \mu, \sigma^2\right) \\
&= \frac{1}{1/\sqrt{\sigma^2}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\left(\sigma z + \mu - \mu\right)^2\right) \\
&= \frac{\sqrt{\sigma^2}}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\sigma^2 z^2\right) \\
&= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) \\
&= N(z; 0, 1)
\end{aligned} \tag{29.49}$$

and hence $Z \sim N(0, 1)$. □

Important applications of Theorem 29.11 include the frequently used standardization of normally distributed random variables in the sense of so-called *Z-values (Z-scores)*, the Z confidence-interval statistic, and the Z test statistic. We visualize Theorem 29.11 by example in Figure 29.11.

χ^2-transformation

With the χ^2-transformation, we now introduce a first transformation of independent and identically normally distributed random variables that does *not* again lead to a normal distribution. Specifically, Theorem 29.12 states that the sum of squared independent standard normally distributed random variables is a χ^2-distributed random variable. To do so, we first recall the concept of a χ^2 random variable as a special case of the gamma random variables considered in Chapter 21 (cf. Definition 21.10).

Definition 29.3 (χ^2 random variable)**.** Let U be a random variable with outcome space $\mathbb{R}_{>0}$ and PDF

$$p : \mathbb{R}_{>0} \to \mathbb{R}_{>0}, u \mapsto p(u) := \frac{1}{\Gamma\left(\frac{n}{2}\right) 2^{\frac{n}{2}}} u^{\frac{n}{2}-1} \exp\left(-\frac{1}{2}u\right), \tag{29.50}$$

where Γ denotes the gamma function. Then we say that U follows a χ^2 distribution with degrees-of-freedom parameter n and call U a χ^2 random variable with degrees-of-freedom

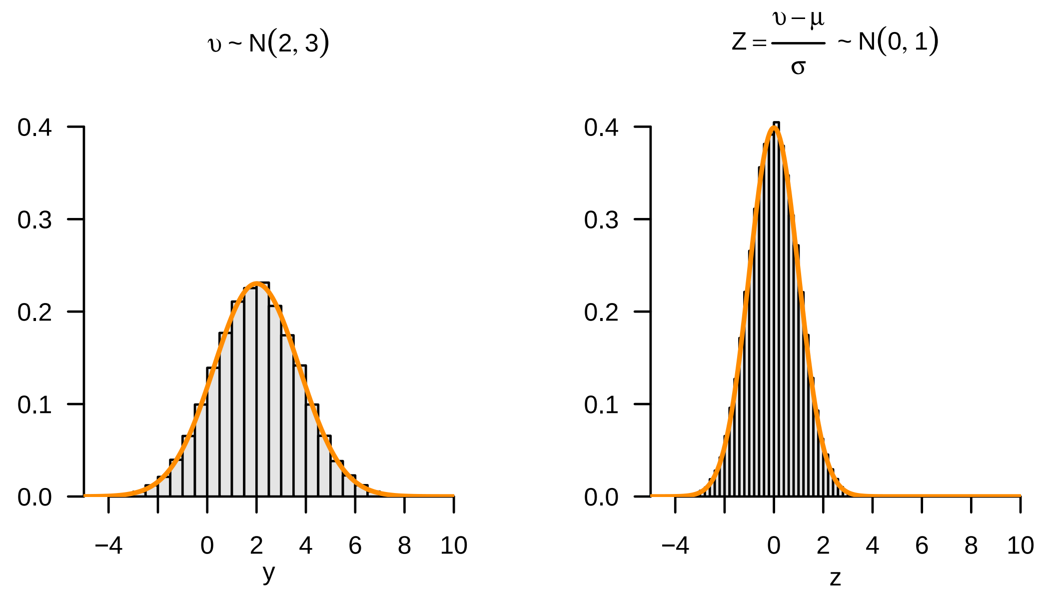


Figure 29.11 Z-transformation of normally distributed random variables.

parameter n. We abbreviate this by $U \sim \chi^2(n)$. We denote the PDF of a χ^2 random variable by

$$\chi^2(u;n) := \frac{1}{\Gamma\left(\frac{n}{2}\right)2^{\frac{n}{2}}} u^{\frac{n}{2}-1} \exp\left(-\frac{1}{2}u\right). \tag{29.51}$$

•

Recall that the PDF of the χ^2 distribution corresponds to the PDF $G\left(u; \frac{n}{2}, 2\right)$ of a gamma distribution. In Figure 29.12, we visualize some PDFs of χ^2 random variables by example. We observe that, as n increases, $\chi^2(u;n)$ broadens and probability mass is shifted to larger values of u.

Theorem 29.12 (χ^2-transformation)**.** *Let $Z_1, ..., Z_n \sim N(0,1)$ be independent and identically standard normally distributed random variables. Then the random variable*

$$U := \sum_{i=1}^{n} Z_i^2 \tag{29.52}$$

is a χ^2-distributed random variable with degrees-of-freedom parameter n, that is, $U \sim \chi^2(n)$. In particular, for $Z \sim N(0,1)$ and $U := Z^2$, we have $U \sim \chi^2(1)$.

∘

Proof. We show the theorem only for the case $n := 1$ using Theorem 28.4. According to that theorem, the PDF of a random variable $U := f(Z)$ that results from the transformation of a random variable Z with PDF p_ζ by a piecewise bijective mapping is given by

$$p_U(u) = \sum_{i=1}^{k} 1_{\mathcal{U}_i} \frac{1}{|f_i'(f_i^{-1}(u))|} p_\zeta\left(f_i^{-1}(u)\right). \tag{29.53}$$

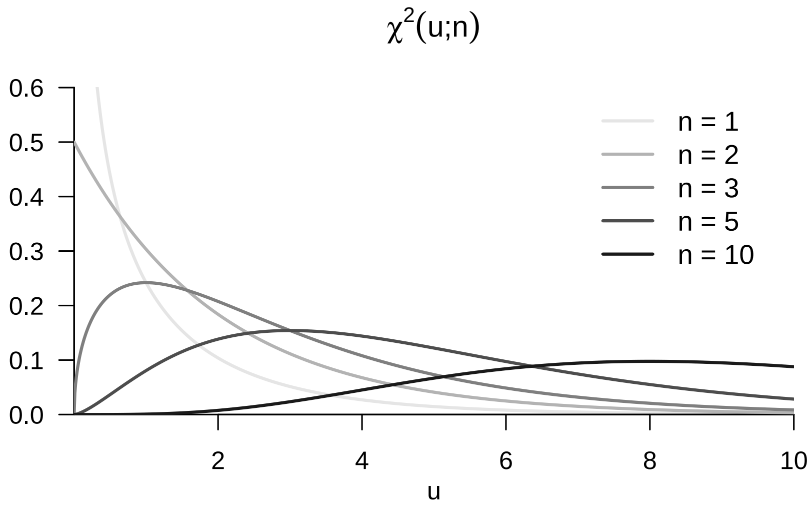


Figure 29.12 PDFs of χ^2 random variables.

We define

$$\mathcal{U}_1 :=]-\infty, 0[, \mathcal{U}_2 :=]0, \infty[, \text{ and } \mathcal{U}_i := \mathbb{R}_{>0} \text{ for } i = 1, 2, \tag{29.54}$$

as well as

$$f_i : \mathcal{Z}_i \to \mathcal{U}_i, x \mapsto f_i(z) := z^2 =: u \text{ for } i = 1, 2. \tag{29.55}$$

The derivative and the inverse function of the f_i are

$$f_i' : \mathcal{Z}_i \to \mathcal{Z}_i, x \mapsto f_i'(z) = 2z \text{ for } i = 1, 2, \tag{29.56}$$

and

$$f_1^{-1} : \mathcal{U}_1 \to \mathcal{U}_1, u \mapsto f_1^{-1}(u) = -\sqrt{u} \text{ and } f_2^{-1} : \mathcal{U}_2 \to \mathcal{U}_2, u \mapsto f_2^{-1}(u) = \sqrt{u}, \tag{29.57}$$

respectively. Substitution then gives

$$\begin{aligned}
p_U(u) &= 1_{\mathcal{U}_1}(u) \frac{1}{|f_1'(f_1^{-1}(u))|} p_\zeta\left(f_1^{-1}(u)\right) + 1_{\mathcal{U}_2}(u) \frac{1}{|f_2'(f_2^{-1}(u))|} p_\zeta\left(f_2^{-1}(u)\right) \\
&= \frac{1}{|2(-\sqrt{u})|} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(-\sqrt{u})^2\right) + \frac{1}{|2(\sqrt{u})|} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(\sqrt{u})^2\right) \\
&= \frac{1}{2\sqrt{u}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u\right) + \frac{1}{2\sqrt{u}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u\right) \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{u}} \exp\left(-\frac{1}{2}u\right).
\end{aligned} \tag{29.58}$$

On the other hand, with $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, the PDF of a χ^2 random variable U with $n = 1$ is given by

$$\frac{1}{\Gamma\left(\frac{1}{2}\right) 2^{\frac{1}{2}}} u^{\frac{1}{2}-1} \exp\left(-\frac{1}{2}u\right) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{u}} \exp\left(-\frac{1}{2}u\right). \tag{29.59}$$

Thus, if $Z \sim N(0, 1)$, then $U := Z^2 \sim \chi^2(1)$. □

Important applications are the U confidence-interval statistic and the t and F random variables introduced below. We visualize Theorem 29.12 by example in Figure 29.13.

T-transformation

The theorem considered in this section goes back to Student (1908) and is the central and style-defining result in the development of frequentist inference in the first half of

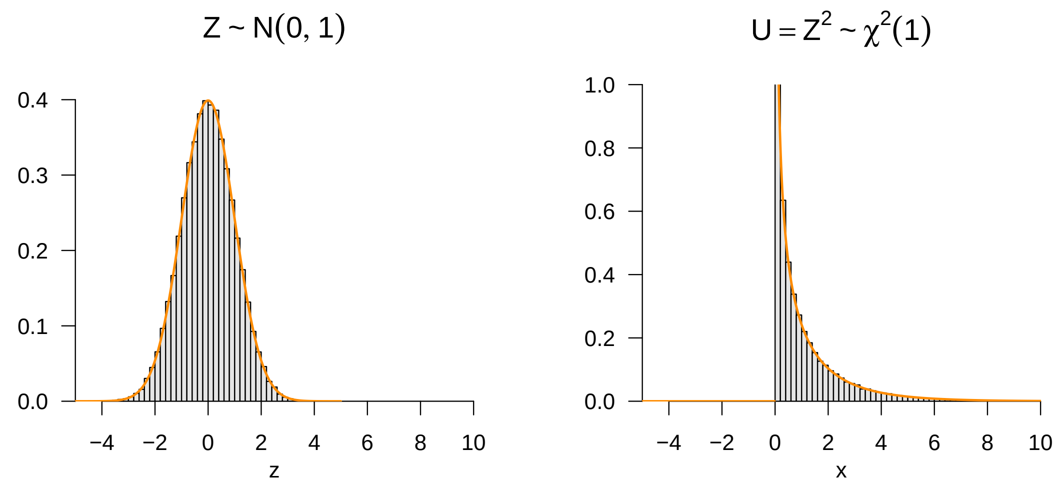


Figure 29.13 χ^2-transformation of normally distributed random variables.

the twentieth century. Hald (2007) and Zabell (2008) provide a historical overview. The central Theorem 29.13 states that the random variable obtained by dividing a standard normally distributed random variable by the square root of a χ^2-distributed random variable divided by an n is a t-distributed random variable. A t-distributed random variable is defined as follows.

Definition 29.4 (t random variable)**.** Let T be a random variable with outcome space $\mathbb{R}$ and PDF

$$p : \mathbb{R} \to \mathbb{R}_{>0}, t \mapsto p(t) := \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)}\left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}, \tag{29.60}$$

where Γ denotes the gamma function. Then we say that T follows a t distribution with degrees-of-freedom parameter n and call T a t random variable with degrees-of-freedom parameter n. We abbreviate this by $T \sim t(n)$. We denote the PDF of a t random variable by

$$T(t; n) := \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)}\left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}. \tag{29.61}$$

•

In Figure 29.14, we visualize some PDFs of t random variables by example. We observe that the t distribution is always symmetric around 0 and that increasing n shifts probability mass from the tails to the center. We note that from about $n = 30$ onward, $T(t; n) \approx N(0, 1)$.

Theorem 29.13 (T-transformation)**.** *Let $Z \sim N(0,1)$ be a standard normally distributed random variable, let $U \sim \chi^2(n)$ be a χ^2 random variable with degrees-of-freedom parameter n, and let Z and U be independent. Then the random variable*

$$T := \frac{Z}{\sqrt{U/n}} \tag{29.62}$$

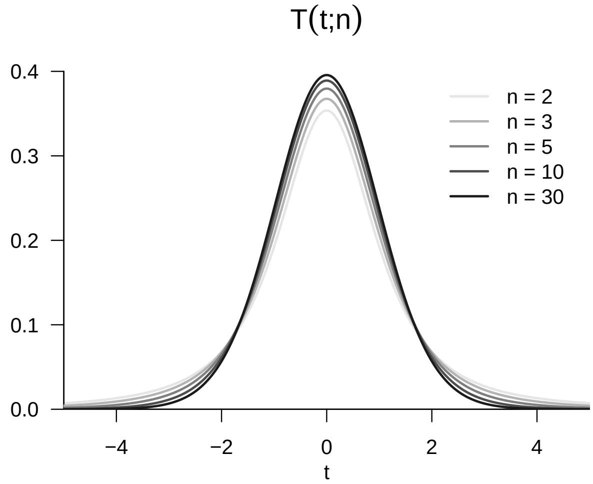


Figure 29.14 PDFs of T random variables.

is a t-distributed random variable with degrees-of-freedom parameter n, that is, $T \sim t(n)$.

∘

Proof. We first note that the two-dimensional PDF of the joint (independent) distribution of Z and U is given by

$$p_{Z,U}(z,u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}} u^{\frac{n}{2}-1} \exp\left(-\frac{1}{2}u\right). \tag{29.63}$$

We then consider the multivariate vector-valued mapping

$$f : \mathbb{R}^2 \to \mathbb{R}^2, (z,u) \mapsto f(z,u) := \left(\frac{z}{\sqrt{u/n}}, u\right) =: (t,w) \tag{29.64}$$

and use the multivariate PDF transformation theorem for bijective mappings to derive the PDF of (t,w). To do so, recall that if ξ is an n-dimensional random vector with PDF p_ξ and $\upsilon := f(\xi)$ for a differentiable and bijective mapping $f : \mathbb{R}^n \to \mathbb{R}^n$, the PDF of the random vector υ is given by

$$p_\upsilon : \mathbb{R}^n \to \mathbb{R}_{\geq 0}, y \mapsto p_\upsilon(y) := \frac{1}{|J^f\left(f^{-1}(y)\right)|} p_\xi\left(f^{-1}(y)\right). \tag{29.65}$$

For the mapping considered here, we first note that

$$f^{-1} : \mathbb{R}^2 \to \mathbb{R}^2, (t,w) \mapsto f^{-1}(t,w) := \left(\sqrt{w/n}t, w\right). \tag{29.66}$$

This follows directly from

$$f^{-1}(f(z,u)) = f^{-1}\left(\frac{z}{\sqrt{u/n}}, u\right) = \left(\frac{\sqrt{u/n}z}{\sqrt{u/n}}, u\right) = (z,u) \text{ for all } (z,u) \in \mathbb{R}^2. \tag{29.67}$$

We then note that the determinant of the Jacobian matrix of f at (z,u) is given by

$$|J^f(z,u)| = \begin{vmatrix} \frac{\partial}{\partial z}\left(\frac{z}{\sqrt{u/n}}\right) & \frac{\partial}{\partial u}\left(\frac{z}{\sqrt{u/n}}\right) \\ \frac{\partial}{\partial z}u & \frac{\partial}{\partial u}u \end{vmatrix} = \left(\frac{v}{n}\right)^{-1/2}, \tag{29.68}$$

so that

$$\frac{1}{|J^f\left(f^{-1}(z,u)\right)|} = \left(\frac{w}{n}\right)^{1/2}. \tag{29.69}$$

Substitution then gives

$$p_{T,W}(t,w) = \left(\frac{w}{n}\right)^{1/2} p_{Z,V}\left(\sqrt{w/n}t, w\right). \tag{29.70}$$

Thus

$$
\begin{aligned}
p_T(t) &= \int_0^\infty p_{T,W}(t,w)\,dw \\
&= \int_0^\infty \left(\frac{w}{n}\right)^{1/2} p_{Z,V}\left(\sqrt{w/n}t, w\right)\,dw \\
&= \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(\sqrt{w/n}t)^2\right) \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}} w^{\frac{n}{2}-1} \exp\left(-\frac{1}{2}w\right)\left(\frac{w}{n}\right)^{1/2} dw \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}n^{\frac{1}{2}}} \int_0^\infty \exp\left(-\frac{1}{2}\frac{w}{n}t^2\right) w^{\frac{n}{2}-1} \exp\left(-\frac{1}{2}w\right) w^{1/2}\,dw \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}n^{\frac{1}{2}}} \int_0^\infty \exp\left(-\frac{1}{2}\frac{w}{n}t^2 - \frac{1}{2}w\right) w^{\frac{n}{2}-1} w^{\frac{1}{2}}\,dw \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}n^{\frac{1}{2}}} \int_0^\infty \exp\left(-\frac{1}{2}\left(\frac{w}{n}t^2 + w\right)\right) w^{\frac{n+1}{2}-1}\,dw \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}n^{\frac{1}{2}}} \int_0^\infty \exp\left(-\frac{1}{2}\left(1 + \frac{t^2}{n}\right)\right) w^{\frac{n+1}{2}-1}\,dw
\end{aligned}
\tag{29.71}
$$

We then note that the integrand on the left-hand side of the equation above corresponds to the kernel of a gamma PDF with parameters $\alpha = \frac{n+1}{2}$ and $\beta = \frac{2}{1+\frac{t^2}{n}}$, as is easily seen:

$$
\begin{aligned}
\Gamma(w;\alpha,\beta) &= \frac{1}{\Gamma(\alpha)\beta^\alpha} w^{\alpha-1} \exp\left(-\frac{w}{\beta}\right) \\
\Rightarrow \Gamma\left(w; \frac{n+1}{2}, \frac{2}{1+\frac{t^2}{n}}\right) &= \frac{1}{\Gamma(\frac{n+1}{2})\left(\frac{2}{1+\frac{t^2}{n}}\right)^{\frac{n+1}{2}}} w^{\frac{n+1}{2}-1} \exp\left(-\frac{w}{\frac{2}{1+\frac{t^2}{n}}}\right) \\
&= \frac{1}{\Gamma(\frac{n+1}{2})\left(\frac{2}{1+\frac{t^2}{n}}\right)^{\frac{n+1}{2}}} \exp\left(-\frac{1}{2}\left(1+\frac{t^2}{n}\right)\right) w^{\frac{n+1}{2}-1}.
\end{aligned}
$$

Thus

$$
p_T(t) = \frac{1}{\sqrt{2\pi}} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}n^{\frac{1}{2}}} \int_0^\infty \Gamma\left(w; \frac{n+1}{2}, \frac{2}{1+\frac{t^2}{n}}\right) dw. \tag{29.72}
$$

Finally, we note that the integral term in the above equation corresponds to the normalization term of a gamma PDF. Hence, finally,

$$
p_T(t) = \frac{1}{\sqrt{2\pi}} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}n^{\frac{1}{2}}} \Gamma\left(\frac{n+1}{2}\right)\left(\frac{2}{1+\frac{t^2}{n}}\right)^{\frac{n+1}{2}}. \tag{29.73}
$$

The distribution of $Z/\sqrt{U/n}$ therefore has the PDF of a t random variable. □

Important applications are the T confidence-interval statistic and the T test statistics of the theory of hypothesis tests in the context of the general linear model. We visualize Theorem 29.13 by example in Figure 29.15.

Noncentral T-transformation

In this section, we consider the case in which the expectation parameter of the numerator variable of the random variable T considered in Theorem 29.13 is different from zero, so that the numerator variable is not distributed according to $N(0,1)$, but according to $N(\mu,1)$ for arbitrary $\mu \in \mathbb{R}$. The resulting random variable T then follows a so-called *noncentral t distribution*. An early detailed discussion of this distribution can be found, for example, in Johnson & Welch (1940). A corresponding noncentral t-distributed random variable is defined as follows (cf. Lehmann (1986)).

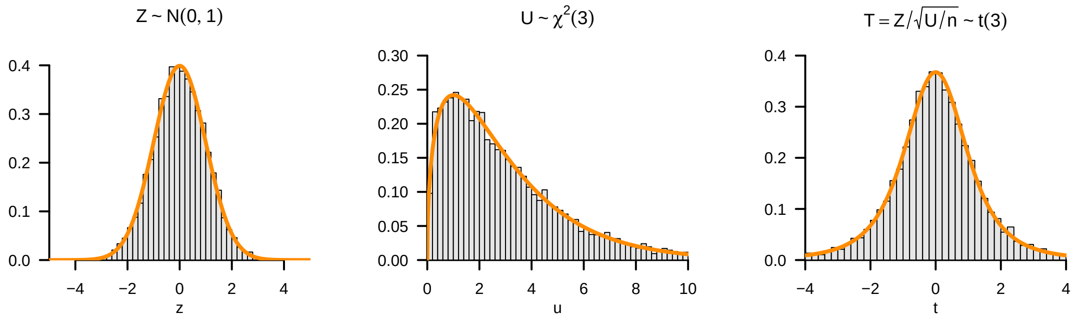


Figure 29.15 T-transformation of normally distributed random variables.

Definition 29.5 (Noncentral t random variable)**.** Let T be a random variable with outcome space $\mathbb{R}$ and PDF

$$p : \mathbb{R} \to \mathbb{R}_{>0}, t \mapsto p(t) := \frac{1}{2^{\frac{n-1}{2}}\Gamma\left(\frac{n}{2}\right)(n\pi)^{\frac{1}{2}}} \times \int_0^\infty \tau^{\frac{n-1}{2}} \exp\left(-\frac{\tau}{2}\right) \exp\left(-\frac{1}{2}\left(t\left(\frac{\tau}{n}\right)^{\frac{1}{2}} - \delta\right)^2\right) d\tau. \quad (29.74)$$

Then we say that T follows a noncentral t distribution with noncentrality parameter δ and degrees-of-freedom parameter n and call T a *noncentral t random variable with noncentrality parameter δ and degrees-of-freedom parameter n.* We abbreviate this by $t(\delta, n)$. We denote the PDF of a noncentral t random variable by $t(T; \delta, n)$. We denote the CDF and inverse CDF of a noncentral t random variable by $\Psi(\cdot; \delta, n)$ and $\Psi^{-1}(\cdot; \delta, n)$, respectively.

•

Without proof, we note that a noncentral t random variable with $\delta = 0$ corresponds to a t random variable, that is,

$$t(T; 0, n) = t(T; n) \quad (29.75)$$

as well as

$$\Psi(T; 0, n) = \Psi(T; n) \text{ and } \Psi^{-1}(T; 0, n) = \Psi^{-1}(T; n). \quad (29.76)$$

In Figure 29.16, we visualize some PDFs of noncentral t random variables by example. We observe that a positive noncentrality parameter δ shifts the distribution to the right and that, with increasing degrees-of-freedom parameter n, the distributions approach correspondingly localized normal distributions with variance parameter 1. Note also the lack of symmetry of the PDFs for small degrees-of-freedom parameters when the positive noncentrality parameter differs from zero.

A noncentral t random variable is the result of a noncentral T-transformation, as stated by the following theorem.

Theorem 29.14 (Noncentral T-transformation)**.** *Let $v \sim N(\mu, 1)$ be a normally distributed random variable, let $U \sim \chi^2(n)$ be a χ^2 random variable with degrees-of-freedom*

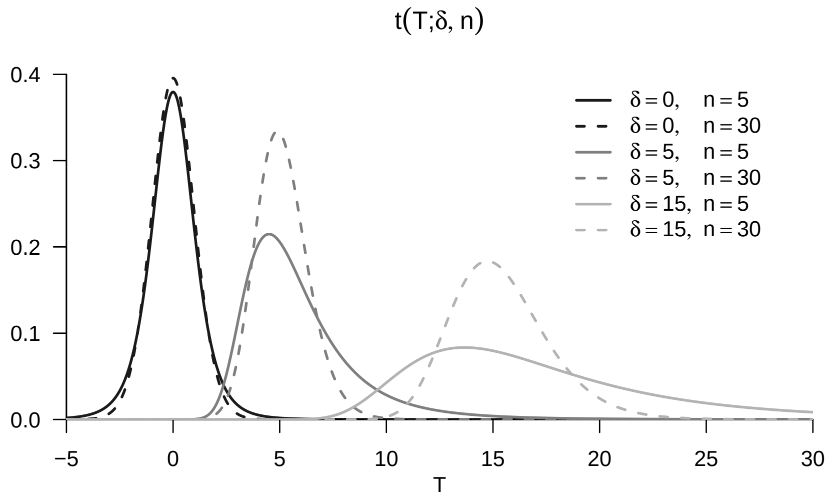


Figure 29.16 PDFs of noncentral T random variables.

parameter n, and let υ and U be independent random variables. Then the random variable

$$T := \frac{\upsilon}{\sqrt{U/n}} \tag{29.77}$$

is a noncentral t random variable with noncentrality parameter μ and degrees-of-freedom parameter n, that is, $T \sim t(\mu, n)$.

∘

We omit a proof. Important applications are the power functions of the T-test variants in the context of the general linear model. We visualize Theorem 29.14 by example in Figure 29.17.

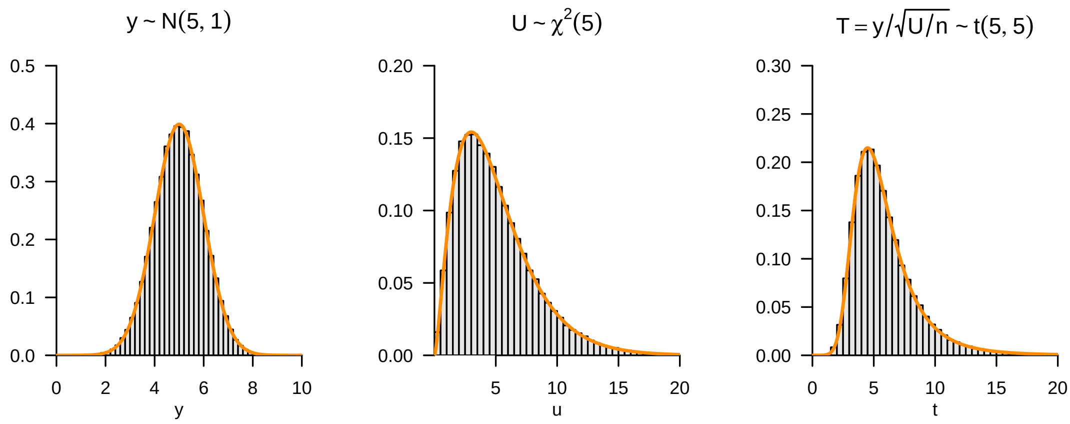


Figure 29.17 Noncentral T-transformation of normally distributed random variables.

F-transformation

The central Theorem 29.15 in this section states that the random variable obtained by dividing two χ^2-distributed random variables, each divided by its degrees-of-freedom parameter, is an F-distributed random variable. An F-distributed random variable is defined as follows.

Definition 29.6 (F random variable). Let F be a random variable with outcome space $\mathbb{R}_{>0}$ and PDF

$$p_F : \mathbb{R} \to \mathbb{R}_{>0}, f \mapsto p_F(f) := \left(\frac{n}{m}\right)^{\frac{n}{2}} \frac{\Gamma\left(\frac{n+m}{2}\right)}{\Gamma\left(\frac{n}{2}\right)\Gamma\left(\frac{m}{2}\right)} \frac{f^{\frac{n}{2}-1}}{\left(1+\frac{n}{m}f\right)^{\frac{n+m}{2}}}, \tag{29.78}$$

where Γ denotes the gamma function. Then we say that F follows an F distribution with degrees-of-freedom parameters n, m and call F an F random variable with degrees-of-freedom parameters n, m. We abbreviate this by $F \sim F(n, m)$. We denote the PDF of an F random variable by

$$F(f; n, m) := \left(\frac{n}{m}\right)^{\frac{n}{2}} \frac{\Gamma\left(\frac{n+m}{2}\right)}{\Gamma\left(\frac{n}{2}\right)\Gamma\left(\frac{m}{2}\right)} \frac{f^{\frac{n}{2}-1}}{\left(1+\frac{n}{m}f\right)^{\frac{n+m}{2}}}. \tag{29.79}$$

•

In Figure 29.18, we visualize some PDFs of F random variables by example. We observe that the shape of the PDFs is initially determined primarily by the degrees-of-freedom parameter n and then secondarily by the degrees-of-freedom parameter m.

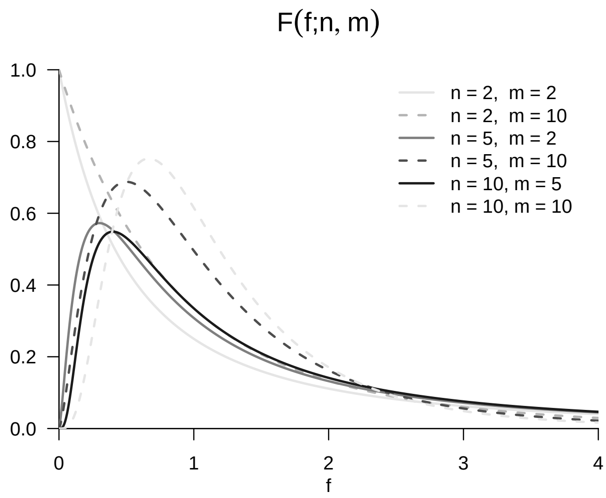


Figure 29.18 PDFs of F-distributed random variables.

Theorem 29.15 (F-transformation). *Let $V \sim \chi^2(n)$ and $W \sim \chi^2(m)$ be two independent χ^2 random variables with degrees-of-freedom parameters n and m, respectively. Then the random variable*

$$F := \frac{V/n}{W/m} \tag{29.80}$$

is an F-distributed random variable with degrees-of-freedom parameters n, m, that is, $F \sim F(n, m)$.

∘

The theorem can be proved by first deriving a transformation theorem for quotients of random variables using Theorem 28.1 and marginalization and then applying this theorem to the PDF of χ^2-distributed random variables. We visualize Theorem 29.15 by example in Figure 29.19. Important applications of Theorem 29.15 are the F statistics considered in the theory of the general linear model.

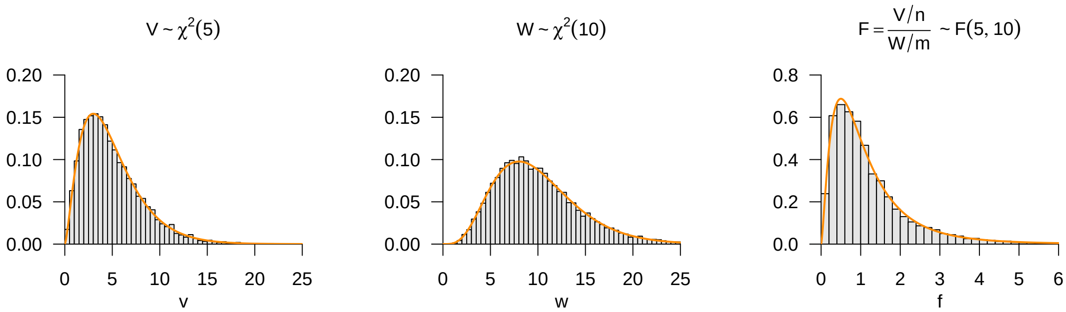


Figure 29.19 F-transformation of normally distributed random variables.

29.7 Bibliographic remarks

The development of the bivariate normal distribution has its origins in the statistical literature around the middle of the nineteenth century, especially in the work of Francis Galton (1822-1911). The mathematical formalization of the bivariate normal distribution probably goes back to Pearson (1896) (Seal (1967)). The original formulation of the multivariate normal distribution is located in Edgeworth (1892). Tong (1990) provides a comprehensive overview of the theory and application of the multivariate normal distribution.

Study questions

1. State the summation transformation theorem.
2. State the sample-mean transformation theorem.
3. State the Z-transformation theorem.
4. State the χ^2-transformation theorem.
5. Describe the PDF of the t distribution as a function of its degrees-of-freedom parameter.
6. State the T-transformation theorem.
7. State the F-transformation theorem.

Study question answers

1. See Theorem 29.9.
2. See Theorem 29.10.
3. See Theorem 29.11.

4. See Theorem 29.12.
5. See Figure 29.14.
6. See Theorem 29.13.
7. See Theorem 29.15.

Part IV

Inference

30 Frequentist inference

30.1 Frequentist inference models

With the following definition, we first introduce some basic terminology for the consideration of frequentist inference models.

Definition 30.1 (Frequentist inference models)**.** A *frequentist inference model* is a tuple

$$\mathcal{M} := (\mathcal{Y}, \mathcal{A}, \{\mathbb{P}_\theta | \theta \in \Theta\}) \tag{30.1}$$

consisting of a *data space* $\mathcal{Y}$, a σ-algebra $\mathcal{A}$ on $\mathcal{Y}$, and a set $\{\mathbb{P}_\theta | \theta \in \Theta\}$ of probability measures on $(\mathcal{Y}, \mathcal{A})$ with at least two elements, indexed by $\theta \in \Theta$. If $\Theta \subset \mathbb{R}^k$, a frequentist inference model is also called a *parametric frequentist inference model*, and Θ is called the *parameter space* of the frequentist inference model. A frequentist inference model $\mathcal{M}$ is called a *discrete model* if $\mathcal{Y}$ is finite or countable and each $\mathbb{P}_\theta$ has a probability mass function p_θ. A frequentist inference model $\mathcal{M}$ is called a *continuous model* if $\mathcal{Y} \subset \mathbb{R}^n$ and each $\mathbb{P}_\theta$ has a probability density function p_θ. If the data space $\mathcal{Y}$ of a frequentist inference model $\mathcal{M}$ is one-dimensional, for example $\mathcal{Y} := \mathbb{R}$, one speaks of a *univariate frequentist inference model.* If the data space $\mathcal{Y}$ of a frequentist inference model $\mathcal{M}$ is multidimensional, for example $\mathcal{Y} := \mathbb{R}^m$ for $m > 1$, one speaks of a *multivariate frequentist inference model.* For a frequentist inference model $\mathcal{M}_0 := (\mathcal{Y}_0, \mathcal{A}_0, \{\mathbb{P}_\theta^0 | \theta \in \Theta\})$, the frequentist inference model $\mathcal{M} := (\mathcal{Y}, \mathcal{A}, \{\mathbb{P}_\theta | \theta \in \Theta\})$, for which $\mathcal{Y}$ is the n-fold Cartesian product of $\mathcal{Y}_0$ with itself, $\mathcal{A}$ is the corresponding product σ-algebra, and $\{\mathbb{P}_\theta | \theta \in \Theta\}$ is the corresponding set of product measures, is called a *frequentist product model* (belonging to $\mathcal{M}_0$).

•

Against the background of a frequentist inference model, the process of data observation is described by a random vector y that takes values in $\mathcal{Y}$ and whose distribution corresponds to one of the principally possible distributions $\mathbb{P}_\theta$. This random vector is called *data*, an *observation*, a *measurement*, or a *sample.* In contrast to the probability space model, in frequentist inference models one therefore explicitly considers two or more probability measures that presumably determine the distribution of y. A realization of y, that is, concretely available data values, is called a *dataset*, an *observed value*, a *measured value*, or a *sample value.* Expectations and (co)variances of y with respect to $\mathbb{P}_\theta$ are usually written as $\mathbb{E}_\theta(y)$, $\mathbb{V}_\theta(y)$, and $\mathbb{C}_\theta(y)$. Frequentist product models model the n-fold independent repetition of a random process. The corresponding set of random vectors $y_1, ..., y_n$ then corresponds to a set of n independent random vectors.

In a concrete data analysis problem based on a parametric frequentist product model, one assumes that the observed values of $y_1, ..., y_n$ were generated by exactly one probability measure $\mathbb{P}_\theta$ with parameter $\theta \in \Theta$. In application, this $\theta \in \Theta$ is then called the *true, but unknown, parameter value.* The true, but unknown, parameter value θ remains unknown even after any form of inference. The general aim of parametric inference procedures is therefore to make, based on an available dataset, a statement that is as valid as possible with respect to the true, but unknown, parameter θ. In this sense, the true, but unknown, parameter value is only indirectly observable. This is sometimes also expressed by saying that the true, but unknown, parameter value is *unobservable* or *latent*, that is, not immediately visible or accessible. In the mathematical analysis of inference procedures, one considers all possible true, but unknown, parameter values and therefore usually dispenses with an explicit notational designation of the true, but unknown, parameter value.

With the univariate normal distribution model and the Bernoulli model, we give two first examples of frequentist inference models.

Example 30.1 (Normal distribution model)**.** The univariate parametric product model

$$\mathcal{M} := (\mathcal{Y}, \mathcal{A}, \{\mathbb{P}_\theta | \theta \in \Theta\}) \tag{30.2}$$

with

$$\mathcal{Y} := \mathbb{R}^n, \mathcal{A} := \mathcal{B}(\mathbb{R}^n), \mathbb{P}_\theta^0 := N(\mu, \sigma^2), \theta := (\mu, \sigma^2), \Theta := \mathbb{R} \times \mathbb{R}_{>0}, \tag{30.3}$$

that is,

$$\{\mathbb{P}_\theta | \theta \in \Theta\} := \left\{ \prod_{i=1}^n N(\mu, \sigma^2) | (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_{>0} \right\}, \tag{30.4}$$

and thus

$$y_1, ..., y_n \sim N(\mu, \sigma^2) \text{ with } (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_{>0} \tag{30.5}$$

is called the *normal distribution model.*

△

The normal distribution model is the basis of many popular statistical procedures that are considered integratively within the framework of the general linear model. Note that the assumption of normally distributed data is motivated by additive normally distributed error terms, as we briefly outlined in Chapter 27 and will deepen at a later point.

Example 30.2 (Bernoulli model)**.** The univariate parametric product model

$$\mathcal{M} := (\mathcal{Y}, \mathcal{A}, \{\mathbb{P}_\theta | \theta \in \Theta\}) \tag{30.6}$$

with

$$\mathcal{Y} := \{0,1\}^n, \mathcal{A} := \mathcal{P}\left(\{0,1\}^n\right), \mathbb{P}_\theta^0 := \text{Bern}(\mu), \theta := \mu, \Theta :=]0,1[, \tag{30.7}$$

that is,

$$\{\mathbb{P}_\theta | \theta \in \Theta\} := \left\{ \prod_{i=1}^n \text{Bern}(\mu) | \mu \in]0,1[\right\}, \tag{30.8}$$

and thus

$$y_1, ..., y_n \sim \text{Bern}(\mu) \text{ with } \mu \in]0,1[, \tag{30.9}$$

is called the *Bernoulli model.*

△

30.2 Statistics and estimators

Against the background of frequentist inference models, we now formalize what is to be understood by the terms *statistic* and *estimator*.

Definition 30.2 (Statistic). Let $\mathcal{M}$ be a frequentist inference model and let $(\Sigma, \mathcal{S})$ be a measurable space. Then a *statistic* is a random vector of the form

$$S : \mathcal{Y} \to \Sigma. \tag{30.10}$$

•

In frequentist inference, both data and statistics are therefore modeled by random vectors (in the univariate case correspondingly by random variables). However, these random vectors differ fundamentally with respect to their intuitive meaning: data represent the outcome of measurement processes under uncertainty, whereas statistics model functions of data constructed by data scientists. In the best case, these provide data-based information from which conclusions can be drawn about the latent data-generating random processes. The fact that statistics are random follows from the fact that they are applied as functions to random data (cf., for example, Theorem 21.1).

Example 30.3 (Examples of statistics). Let $\mathcal{M}$ be the normal distribution model. Then, for example, the following random variables are statistics:

- The *sample mean*
$$\bar{y} : \mathbb{R}^n \to \mathbb{R}, y \mapsto \bar{y}(y) := \frac{1}{n}\sum_{i=1}^{n} y_i, \tag{30.11}$$
- The *sample variance*
$$s^2 : \mathbb{R}^n \to \mathbb{R}_{\geq 0}, y \mapsto s^2(y) := \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}(y))^2, \tag{30.12}$$
- The *sample standard deviation*
$$s : \mathbb{R}^n \to \mathbb{R}_{\geq 0}, y \mapsto s(y) := \sqrt{s^2(y)}, \tag{30.13}$$

△

Often, as in Example 30.3, the nature of statistics as random variables or random vectors remains rather implicit in notation. This does not, however, change the fundamental fact that statistics, as functions of random-dependent values, are themselves random variables or random vectors.

Definition 30.3 (Estimator). Let $\mathcal{M}$ be a frequentist inference model, let $(\Sigma, \mathcal{S})$ be a measurable space, and let $\tau : \Theta \to \Sigma$ be a mapping that assigns a characteristic quantity $\tau(\theta) \in \Sigma$ to every $\theta \in \Theta$. Then a statistic

$$\hat{\tau} : \mathcal{Y} \to \Sigma \tag{30.14}$$

is called an *estimator* for τ.

•

Estimators therefore estimate functions of the parameters of a parametric frequentist inference model. Typical examples of such functions are

- $\tau(\theta) := \theta$ for the estimation of the parameter θ,
- $\tau(\theta) := \theta_i$ with $\theta \in \mathbb{R}^d, d > 1$ for the estimation of a component of the parameter θ,
- $\tau(\theta) := \mathbb{E}_\theta(y_1)$ for the estimation of the expectation,
- $\tau(\theta) := \mathbb{V}_\theta(y_1)$ for the estimation of the variance.

In the case $\tau(\theta) := \theta$, that is, the estimation of parameters, one usually writes $\hat{\theta}$. Note that estimators take numerical values in Σ, in the estimation of parameters, for example, in Θ. They are therefore also called *point estimators*. This is a characteristic of frequentist inference procedures. Within Bayesian inference, estimators can also take generalized forms; for example, probability distributions are also considered as estimators there. Finally, it should be noted that the definition of an estimator makes no statement whatsoever about the validity of estimators. Not every estimator is therefore *per se* a good estimator. In frequentist inference, one therefore additionally defines *estimator quality criteria*, as will be presented in detail in the chapter on point estimation.

Example 30.4 (Estimators in the normal distribution model)**.** Let $\mathcal{M}$ be the normal distribution model. Then the sample mean $\bar{y} : \mathbb{R}^n \to \mathbb{R}$ is an estimator for

$$\tau : \mathbb{R} \times \mathbb{R}_{>0} \to \mathbb{R}, (\mu, \sigma^2) \mapsto \tau(\mu, \sigma^2) := \mu. \tag{30.15}$$

Likewise, $\bar{y}$ is an estimator for

$$\tau : \mathbb{R} \times \mathbb{R}_{>0} \to \mathbb{R}, (\mu, \sigma_2) \mapsto \tau(\mu, \sigma^2) := \mathbb{E}_{\mu,\sigma^2}(y_1). \tag{30.16}$$

Furthermore, the constant function

$$\hat{\tau} : \mathbb{R}^n \to \mathbb{R}, y \mapsto \hat{\tau}(y) := 42 \tag{30.17}$$

is an estimator for

$$\tau : \mathbb{R} \times \mathbb{R}_{>0} \to \mathbb{R}_{>0}, (\mu, \sigma_2) \mapsto \tau(\mu, \sigma^2) := \sigma^2. \tag{30.18}$$

The fact that a function $\hat{\tau} : \mathcal{Y} \to \Sigma$ is an estimator therefore by no means implies that it is a good estimator.

△

30.3 Standard assumptions and standard problem settings

We summarize the concepts considered so far in this chapter once again under the term *data-analytic standard assumptions of frequentist inference* (cf. also Figure 30.1). To this end, let $\mathcal{M}$ be a univariate parametric frequentist product model, and let $y_1, ..., y_n \sim \mathbb{P}_\theta$ be the random variables of the sample, which we may combine, for example, in a random vector $y := (y_1, ..., y_n)$. Of a concretely available dataset, which we may combine, for example, in an n-dimensional real vector, it is then *assumed that it is one of the possible realizations of y on the basis of a distribution $\mathbb{P}_\theta$ with true, but unknown, parameter θ.* From a frequentist perspective, one can repeat the observation of a dataset infinitely often and evaluate estimators or statistics for each data realization, for example the sample mean:

Data realization $y^{(1)} := \left(y_1^{(1)}, ..., y_n^{(1)}\right)$ with $\bar{y}^{(1)} = \frac{1}{n}\sum_{i=1}^n y_i^{(1)}$

Data realization $y^{(2)} := \left(y_1^{(2)}, ..., y_n^{(2)}\right)$ with $\bar{y}^{(2)} = \frac{1}{n}\sum_{i=1}^n y_i^{(2)}$

Data realization $y^{(3)} := \left(y_1^{(3)}, ..., y_n^{(3)}\right)$ with $\bar{y}^{(3)} = \frac{1}{n}\sum_{i=1}^n y_i^{(3)}$

Data realization $y^{(4)} := \left(y_1^{(4)}, ..., y_n^{(4)}\right)$ with $\bar{y}^{(4)} = \frac{1}{n}\sum_{i=1}^n y_i^{(4)}$

Data realization $y^{(5)} := \left(y_1^{(5)}, ..., y_n^{(4)}\right)$ with $\bar{y}^{(5)} = \frac{1}{n}\sum_{i=1}^n y_i^{(5)}$

...

Against this background, frequentist inference usually treats the following standard problem settings:

(1) *Point estimation.* The aim of point estimation is, on the basis of observed data, to give a precise guess for the true, but unknown, parameter value that is as good as possible in the frequentist sense.

(2) *Confidence interval determination.* The aim of confidence interval determination is, based on the assumed data distribution and the observed data, to give by means of interval estimation a guess for the true, but unknown, parameter value that is as certain as possible, even if often imprecise.

(3) *Hypothesis tests.* The aim of frequentist hypothesis testing is, based on the assumed distribution of the data, to decide in a form that is as reliable as possible whether a true, but unknown, parameter value lies in one of two mutually exclusive subsets of the parameter space.

Procedures for solving these problem settings are called *frequentist inference procedures.* To assess the quality of frequentist inference procedures, one usually considers the distributions of estimators and statistics under the assumption of $y = (y_1, ..., y_n) \sim \mathbb{P}_\theta$. One asks, for example, about the distribution of the realizations sketched above, $\bar{y}^{(1)}$, $\bar{y}^{(2)}$, $\bar{y}^{(3)}$, $\bar{y}^{(4)}$, ..., that is, about the distribution of the random variable $\bar{y}_n$. If an inference procedure is considered "good" on the basis of these assumptions, this merely means that the procedure is good "on average when applied frequently". In an individual case, that is, in the normal application case of a single available dataset, the inference procedure may also be "bad". We will deepen this way of thinking especially in the context of point estimation and confidence intervals. Similarly, frequentist inference assesses the *strength of empirical evidence* against the background of the assumed distribution of estimators and statistics in scenarios in which the effect of interest is assumed *not* to exist, so-called *null hypotheses.* We clarify this way of thinking in the context of considering hypothesis tests.

Example 30.5 (Application example)**.** For a first application example of frequentist inference procedures, we consider the evidence-based evaluation of psychotherapy for depression. To this end, let the dataset shown in Table 41.1 be given, consisting of BDI-II score differences (`dBDI`; Beck (1961), Beck et al. (1996)) collected from $n = 12$ patients before and after therapy. The `dBDI` values are intended to reflect the reduction of the patients' BDI-II scores over the period of therapy. High positive values of `dBDI` therefore correspond to a strong decrease in the depressive symptomatology quantified by the BDI-II score, values around zero correspond to no substantial change, and negative values correspond to an increase in the depressive symptomatology quantified by the BDI-II score.

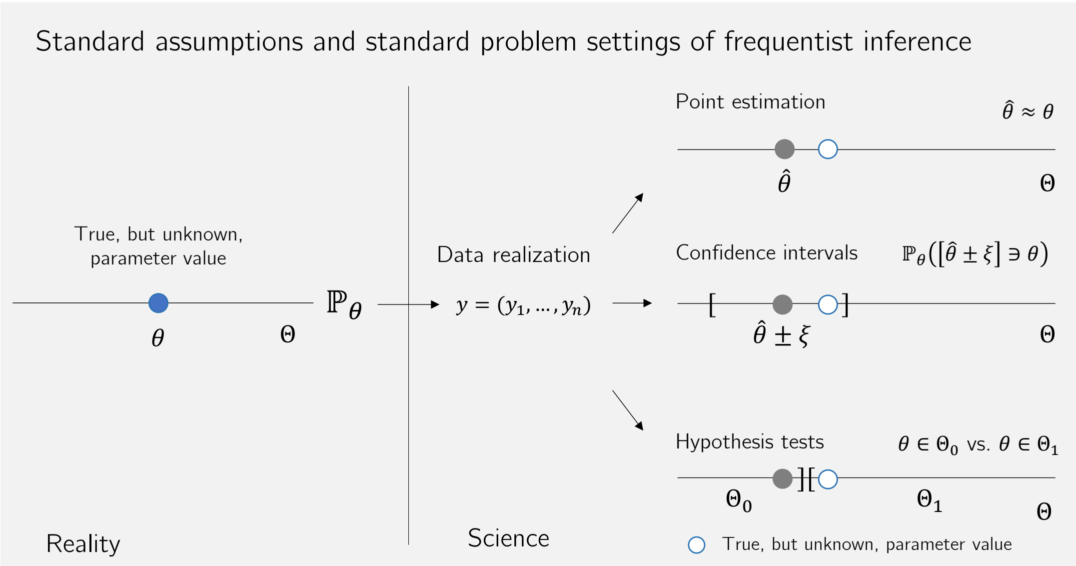


Figure 30.1 Standard assumptions and standard problem settings of frequentist inference. Frequentist inference assumes that in reality there is a true, but unknown, parameter value of the probability measure $\mathbb{P}_\theta$ that serves as the model for collecting a dataset. A concretely available dataset is then one (and in particular *only* one) of the possible realizations of the random vector $y := (y_1, ..., y_n)$ distributed according to $\mathbb{P}_\theta$. On the basis of this realization, the procedures for treating the frequentist standard problem settings of point estimation, confidence interval determination, and hypothesis test evaluation aim to make statements that are as valid as possible with respect to the true, but unknown, parameter value; these will be discussed in the chapters on point estimation, confidence intervals, and hypothesis tests. The actual true, but unknown, parameter value always remains unknown even after completion of an inference procedure.

Table 30.1 Pre-post therapy BDI-II reduction scores of n = 12 patients

dBDI
-1
3
-2
9
3
-2
4
5
5
1
9
4

For each of the $n := 12$ `dBDI` values, we now use the model

$$y_i := \mu + \varepsilon_i \text{ with } \varepsilon_i \sim N(0, \sigma^2) \text{ i.i.d. for } i = 1, ..., n \tag{30.19}$$

as a basis. Thus, the `dBDI` value of the ith patient is explained with the help of a BDI-II score reduction $\mu \in \mathbb{R}$ that is identical across the group of patients and a patient-specific normally distributed BDI-II score reduction deviation ε_i, and it is assumed that these reduction deviations do not influence one another between patients. Intuitively, it is therefore assumed that the therapy has an effect that leads to the same BDI-II score reduction μ in all patients, and that the differences in the observed `dBDI` values can be explained by a large number of further random processes that are normally distributed and centered in the aggregate. Alternatively, one may understand these deviations as realizations of the uncertainty with which the model in Equation 30.19 is associated.

It then follows directly from Equation 30.19 that

$$y_1, ..., y_n \sim N(\mu, \sigma^2), \tag{30.20}$$

because for $i = 1, ..., n$ and with

$$y_i = f(\varepsilon_i) \text{ with } f : \mathbb{R} \to \mathbb{R}, e_i \mapsto f(e_i) := e_i + \mu. \tag{30.21}$$

it holds for the probability density functions of the y_i that

$$\begin{aligned}
p_{y_i}(y_i) &= \frac{1}{|1|} p_{\varepsilon_i}\left(\frac{y_i - \mu}{1}\right) \\
&= N\left(y_i - \mu; 0, \sigma^2\right) \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu - 0)^2\right) \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu)^2\right) \\
&= N(y_i; \mu, \sigma^2)
\end{aligned} \tag{30.22}$$

The standard problem settings of frequentist inference then lead, in this application scenario, to the following questions, which we will take up again in the chapters on point estimation, confidence interval determination, and hypothesis test evaluation, respectively:

(1) What are sensible guesses for the true, but unknown, parameter values μ and σ^2, that is, the true, but unknown, expectation of the BDI-II score reduction and its true, but unknown, variance? How good, then, is the therapy in this quantitative sense, if we try to take into account the patient-dependent deviations, and how large is the uncertainty inherent in the data generation?

(2) How can a maximally certain estimate of the true, but unknown, expectation of the BDI-II score reduction be achieved in the sense of interval estimation? How imprecise must such an estimate be in order to be as reliable as possible?

(3) Do we sensibly decide in favor of one of the hypotheses that the therapy is not effective ($\mu = 0$), or that it is effective, for example, in a positive ($\mu > 0$) or also in a negative sense ($\mu < 0$)? And if we were to decide in favor of one of these hypotheses, with what error probability would we do so? How high, then, is the uncertainty inherent in such a decision?

$\triangle$

Study questions

1. State the definition of a parametric frequentist inference model.
2. What is the difference between a univariate and a multivariate product model?
3. State the definition of the normal distribution model.
4. State the definition of the Bernoulli model.
5. State the definition of the term statistic.
6. State the definition of the term estimator.
7. Explain the standard problem settings of frequentist inference.
8. Explain the standard assumptions of frequentist inference.

Study question answers

1. See Definition 30.1.
2. See Definition 30.1.
3. See Example 30.1.
4. See Example 30.2.
5. See Definition 30.2.
6. See Definition 30.3.
7. See Section 30.3.
8. See Section 30.3.

31 Point estimation

In this chapter, in the sense of the terminology introduced in Chapter 30, we always assume a parametric product model

$$\mathcal{M} := \{\mathcal{Y}, \mathcal{A}, \{\mathbb{P}_\theta | \theta \in \Theta\}\} \tag{31.1}$$

with n-dimensional sample space (e.g. $\mathcal{Y} := \mathbb{R}^n$), d-dimensional parameter space $\Theta \subset \mathbb{R}^d$, and given PMF or PDF p_θ for all $\theta \in \Theta$. $y := (y_1, ..., y_n)$ denotes the sample of independent and identically distributed random variables belonging to $\mathcal{M}$; throughout, therefore,

$$y_1, ..., y_n \sim \mathbb{P}_\theta. \tag{31.2}$$

The nature and aim of the *point estimation* treated here is to give, based on the sample, a guess that is as good as possible for a characteristic quantity of the distribution $\mathbb{P}_\theta$ of a sample variable. The guess has the same mathematical nature as the corresponding characteristic quantity, for example a scalar value for a scalar parameter. This is not the only possibility for estimation; with confidence intervals we will encounter one way of estimating scalar values by intervals. The characteristic quantities of $\mathbb{P}_\theta$ to be estimated are often simply the true, but unknown, parameters themselves. We treat this case in detail in Section 31.1. However, many basic results of frequentist point estimation also remain valid when the quantities to be estimated are not the parameters themselves, but functions of them. Examples are the estimation of the expectation, variance, or standard deviation of $\mathbb{P}_\theta$. We begin, however, with *parameter estimation*. To estimate the true, but unknown, parameter of an inference model, frequentist inference uses so-called *parameter point estimators*.

Definition 31.1. $\mathcal{M} := (\mathcal{Y}, \mathcal{A}, \{\mathbb{P}_\theta | \theta \in \Theta\})$ be a frequentist inference model, let $(\Theta, \mathcal{S})$ be a measurable space, and let $\hat{\theta} : \mathcal{Y} \to \Theta$ be a mapping. Then $\hat{\theta}$ is called a *parameter point estimator* for θ.

•

Parameter point estimators are usually also simply called *parameter estimators*. In the sense of Definition 30.3, parameter point estimators are estimators with $\tau := \mathrm{id}_\Theta$. Parameter point estimators are therefore functions of data and take numerical values in the parameter space. As functions of random variables, parameter estimators are of course also random variables. Often, notation does not distinguish between $\hat{\theta}$ as a random variable and $\hat{\theta}(y)$ as the value of this random variable.

Definition 31.1 evidently does not specify how a parameter point estimator is to be constructed or to what extent it may then be a sensible estimator. In what follows, with *maximum likelihood estimation*, we first discuss a general principle that allows parameter estimators to be determined for a given frequentist inference model and that, as we will see later, guarantees certain desirable properties (Section 31.4). These properties generally concern its qualitative distributional behavior for a fixed sample size or in the limit of an infinitely large sample size. We introduce these properties in Section 31.2 and Section 31.3.

31.1 Maximum likelihood estimation

The basic idea of maximum likelihood estimation is to choose, as a guess for a true, but unknown, parameter value, that parameter value for which the probability of the observed data is maximal. For this, it is first necessary to consider the probability of observed data in a frequentist inference model as a function of the parameter in question. This is enabled and formalized by the *likelihood function* and its logarithm, the *log-likelihood function*. We define these terms here for parametric product models.

Definition 31.2 (Likelihood function and log-likelihood function)**.** Let $\mathcal{M}$ be a parametric product model with PMF or PDF p_θ. Then the *likelihood function* is defined as

$$L : \Theta \to [0, \infty[, \theta \mapsto L(\theta) := \prod_{i=1}^{n} p_\theta(y_i) \tag{31.3}$$

and the *log-likelihood function* is defined as

$$\ell_n : \Theta \to \mathbb{R}, \theta \mapsto \ell(\theta) := \ln L(\theta). \tag{31.4}$$

•

The likelihood function is thus a function of the parameter, and its function values are the values of the joint PMF or PDF of observed data values $y_1, ..., y_n$. In general, there is no reason to assume that a likelihood function integrates to 1 over the parameter space; the likelihood function is therefore generally not a PMF or PDF. The log-likelihood function is simply the logarithm of the likelihood function. A parameter estimator obtained according to the principle of maximum likelihood estimation is now intended to maximize the likelihood function or the log-likelihood function. This leads to the following definition of a *maximum likelihood estimator*.

Definition 31.3 (Maximum likelihood estimator)**.** Let $\mathcal{M}$ be a parametric product model with parameter $\theta \in \Theta$. A *maximum likelihood estimator* of θ is defined as

$$\hat{\theta}^{\text{ML}} : \mathcal{Y} \to \Theta, y \mapsto \hat{\theta}^{\text{ML}}(y) := \text{argmax}_{\theta\in\Theta} L(\theta) = \text{argmax}_{\theta\in\Theta} \ell(\theta) \tag{31.5}$$

•

Note in Definition 31.3 that a maximum point of the log-likelihood function corresponds to the maximum point of the likelihood function, because the logarithm function is monotonically increasing. Working with the log-likelihood function is often easier than working directly with the likelihood function, for example when exponential functions occur in the PMF or PDF of the model. Furthermore, note that Definition 31.2 implies

$$\hat{\theta}^{\text{ML}}(y) = \text{argmax}_{\theta\in\Theta} \prod_{i=1}^{n} p_\theta(y_i) = \text{argmax}_{\theta\in\Theta} \sum_{i=1}^{n} \ln p_\theta(y_i) \tag{31.6}$$

which makes the dependence of a maximum likelihood estimator on the data clear.

With Definition 31.3, maximum likelihood estimation is therefore the problem of determining extrema of a function. For these extrema, differential calculus provides, as is well known, necessary and sufficient conditions (cf. Section 6.2). In its application to obtaining maximum likelihood estimators, one usually contents oneself, due to the functional form of the functions considered, with the fulfillment of the necessary condition. Depending on the nature of the log-likelihood function, methods of analytic or numerical optimization then suggest themselves. In the following classical examples we use an analytic approach. We proceed in the following steps:

(1) Formulation of the log-likelihood function.
(2) Determination of the first derivative of the log-likelihood function and setting it to zero.
(3) Solving for potential maximum points.

In Theorem 31.1 we show that the maximum likelihood estimator for the parameter of the Bernoulli model from Example 30.2 is given by the corresponding sample mean, and in Theorem 31.2 we show that the maximum likelihood estimators for the expectation and variance parameters of the normal distribution model from Example 30.1 are given by the sample mean and a modified sample variance, respectively.

Examples

Theorem 31.1 (Maximum likelihood estimator of the Bernoulli model)**.** *Let $\mathcal{M}$ be the Bernoulli model, that is, $y_1, ..., y_n \sim Bern(\mu)$. Then*

$$\hat{\mu}^{ML} : \{0,1\}^n \to [0,1], y \mapsto \hat{\mu}^{ML}(y) := \frac{1}{n}\sum_{i=1}^n y_i \tag{31.7}$$

is a maximum likelihood estimator of μ.

∘

Proof. We first formulate the log-likelihood function. For the likelihood function,

$$L :]0,1[\to]0,1[, \mu \mapsto L(\mu) := \prod_{i=1}^n \mu^{y_i}(1-\mu)^{1-y_i} = \mu^{\sum_{i=1}^n y_i}(1-\mu)^{n-\sum_{i=1}^n y_i}. \tag{31.8}$$

Taking logarithms gives

$$\ell :]0,1[\to \mathbb{R}, \mu \mapsto \ell(\mu) = \ln\mu \sum_{i=1}^n y_i + \ln(1-\mu)\left(n - \sum_{i=1}^n y_i\right). \tag{31.9}$$

We then evaluate the derivative of the log-likelihood function. We have

$$\begin{aligned}
\frac{d}{d\mu}\ell(\mu) &= \frac{d}{d\mu}\left(\ln\mu\sum_{i=1}^n y_i + \ln(1-\mu)\left(n - \sum_{i=1}^n y_i\right)\right) \\
&= \frac{d}{d\mu}\ln\mu\sum_{i=1}^n y_i + \frac{d}{d\mu}\ln(1-\mu)\left(n - \sum_{i=1}^n y_i\right) \\
&= \frac{1}{\mu}\sum_{i=1}^n y_i - \frac{1}{1-\mu}\left(n - \sum_{i=1}^n y_i\right).
\end{aligned} \tag{31.10}$$

Setting this to zero then gives the following *maximum likelihood equation* as a necessary condition for a maximum likelihood estimator in the Bernoulli model:

$$\frac{1}{\hat{\mu}^{\mathrm{ML}}}\sum_{i=1}^{n} y_i - \frac{1}{1-\hat{\mu}^{\mathrm{ML}}}\left(n - \sum_{i=1}^{n} y_i\right) = 0. \tag{31.11}$$

Solving the maximum likelihood equation for $\hat{\mu}^{\mathrm{ML}}$ gives

$$\begin{aligned}
\frac{1}{\hat{\mu}^{\mathrm{ML}}}\sum_{i=1}^{n} y_i - \frac{1}{1-\hat{\mu}^{\mathrm{ML}}}\left(n - \sum_{i=1}^{n} y_i\right) &= 0 \\
\Leftrightarrow \hat{\mu}^{\mathrm{ML}}(1-\hat{\mu}^{\mathrm{ML}})\left(\frac{1}{\hat{\mu}^{\mathrm{ML}}}\sum_{i=1}^{n} y_i - \frac{1}{1-\hat{\mu}^{\mathrm{ML}}}\left(n - \sum_{i=1}^{n} y_i\right)\right) &= 0 \\
\Leftrightarrow \sum_{i=1}^{n} y_i - \hat{\mu}^{\mathrm{ML}}\sum_{i=1}^{n} y_i - n\hat{\mu}^{\mathrm{ML}} + \hat{\mu}^{\mathrm{ML}}\sum_{i=1}^{n} y_i &= 0 \\
\Leftrightarrow n\hat{\mu}^{\mathrm{ML}} &= \sum_{i=1}^{n} y_i \\
\Leftrightarrow \hat{\mu}^{\mathrm{ML}} &= \frac{1}{n}\sum_{i=1}^{n} y_i .
\end{aligned} \tag{31.12}$$

Thus, $\hat{\mu}^{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} y_i$ is a candidate for a maximum likelihood estimator of μ. This could be verified by considering the second derivative of ℓ, but we refrain from doing so here. □

Theorem 31.2 (Maximum likelihood estimators of the normal distribution model)**.** *Let $\mathcal{M}$ be the normal distribution model, that is, $y_1, ..., y_n \sim N(\mu, \sigma^2)$. Then*

$$\hat{\mu}^{ML} : \mathbb{R}^n \to \mathbb{R}, y \mapsto \hat{\mu}^{ML}(y) := \frac{1}{n}\sum_{i=1}^{n} y_i \tag{31.13}$$

and

$$\hat{\sigma}^{2^{ML}} : \mathbb{R}^n \to \mathbb{R}_{\geq 0}, y \mapsto \hat{\sigma}^{2^{ML}}(y) := \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\mu}^{ML}\right)^2 . \tag{31.14}$$

are maximum likelihood estimators for μ and σ^2, respectively.

∘

Proof. We first formulate the log-likelihood function. For the likelihood function we obtain

$$\begin{aligned}
L : \mathbb{R} \times \mathbb{R}_{>0} \to \mathbb{R}_{>0}, (\mu, \sigma^2) \mapsto L(\mu, \sigma^2) &:= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu)^2\right) \\
&= \left(2\pi\sigma^2\right)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2\right).
\end{aligned} \tag{31.15}$$

Taking logarithms gives

$$\ell : \mathbb{R} \times \mathbb{R}_{>0} \to \mathbb{R}, (\mu, \sigma^2) \mapsto \ell_n(\mu, \sigma^2) = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln \sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2. \tag{31.16}$$

Evaluation of the partial derivatives of the log-likelihood function gives

$$\frac{\partial}{\partial \mu}\ell(\mu, \sigma^2) = -\frac{\partial}{\partial \mu}\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2 = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}\frac{\partial}{\partial \mu}(y_i - \mu)^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i - \mu) \tag{31.17}$$

and

$$\frac{\partial}{\partial \sigma^2}\ell(\mu, \sigma^2) = -\frac{n}{2}\frac{\partial}{\partial \sigma^2}\ln \sigma^2 - \frac{\partial}{\partial \sigma^2}\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(y_i - \mu)^2. \tag{31.18}$$

The system of maximum likelihood equations as an expression of the necessary conditions for extrema of the log-likelihood function therefore has the form

$$\sum_{i=1}^{n}(y_i - \hat{\mu}^{\mathrm{ML}}) = 0 \text{ and } -\frac{n}{2\hat{\sigma}^{2\mathrm{ML}}} + \frac{1}{2\hat{\sigma}^{4\mathrm{ML}}}\sum_{i=1}^{n}(y_i - \mu)^2 = 0. \tag{31.19}$$

Solving the system of maximum likelihood equations first gives

$$\sum_{i=1}^{n}(y_i - \hat{\mu}^{\mathrm{ML}}) = 0 \Leftrightarrow \sum_{i=1}^{n} y_i = n\hat{\mu}^{\mathrm{ML}} \Leftrightarrow \hat{\mu}^{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} y_i. \tag{31.20}$$

Thus,

$$\hat{\mu}^{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} y_i \tag{31.21}$$

is a potential maximum likelihood estimator of μ. Substituting this estimator into the second maximum likelihood equation then gives

$$\begin{aligned} -\frac{n}{2\hat{\sigma}^{2\mathrm{ML}}} + \frac{1}{2\hat{\sigma}^{4\mathrm{ML}}}\sum_{i=1}^{n}(y_i - \hat{\mu}^{\mathrm{ML}})^2 &= 0 \\ \Leftrightarrow -n\hat{\sigma}^{2\mathrm{ML}} + \sum_{i=1}^{n}(y_i - \hat{\mu}^{\mathrm{ML}})^2 &= 0 \\ \Leftrightarrow \hat{\sigma}^{2\mathrm{ML}} &= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\mu}^{\mathrm{ML}})^2. \end{aligned} \tag{31.22}$$

Thus,

$$\hat{\sigma}^{2\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\mu}^{\mathrm{ML}}\right)^2 \tag{31.23}$$

is a potential maximum likelihood estimator of σ^2. Both potential maximum likelihood estimators can be verified by considering the second derivative of ℓ, but we refrain from doing so here. □

Note in Theorem 31.2 that $\hat{\mu}^{\mathrm{ML}}$ is identical to the sample mean $\bar{y}$, but $\hat{\sigma}^{2\mathrm{ML}}$ does not coincide with the sample variance S^2. In contrast to the sample variance, the maximum likelihood estimator of σ^2 contains the multiplicative factor $\frac{1}{n}$, not, as in the sample variance, the multiplicative factor $\frac{1}{n-1}$. We will return to this difference in the context of estimator properties.

Example 31.1 (Application example)**.** To conclude this section, we consider Theorem 31.2 in the context of the application example from Example 30.5. There we based the observed `dBDI` values on the normal distribution model

$$y_1, ..., y_n \sim N(\mu, \sigma^2) \tag{31.24}$$

The maximum likelihood estimators for the parameters of this model can then be evaluated, using Theorem 31.2, with the **R** sample mean and sample variance functions `mean()` and `var()`, and taking into account the identity

$$\frac{n-1}{n}s^2 = \frac{n-1}{n}\cdot\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\mu}^{\mathrm{ML}}\right)^2 = \hat{\sigma}^{2\mathrm{ML}} \tag{31.25}$$

as in the following **R** code.

```
D          = read.csv("./_data/bdi-ii-dataset.csv")       # read dataset
y          = D$dBDI                                        # select data
mu_hat     = mean(y)                                       # maximum likelihood estimate of expectation parameter
n          = length(y)                                     # number of data points
sigsqr_hat = ((n-1)/n)*var(y)                              # maximum likelihood estimate of variance parameter
cat("mu_hat     :", mu_hat,"\nsigsqr_hat :", sigsqr_hat)   # output
```

```
mu_hat     : 3.166667
sigsqr_hat : 12.63889
```

Based on the principle of maximum likelihood estimation and the available $n = 12$ data points,

$$\hat{\mu}^{\text{ML}} = 3.17 \text{ and } \hat{\sigma}^{2^{\text{ML}}} = 12.6 \tag{31.26}$$

are therefore guesses for the true, but unknown, parameters of the model.

31.2 Estimator properties in finite samples

In general, frequentist estimator properties concern the distribution of estimators as a function of the distribution of the data on which they are based. Because data are random in frequentist inference, estimators are random as well. In particular, observed data values are interpreted as realizations of random variables. Estimators as functions of random variables are therefore also random variables, even though, of course, they take only one concrete value when a concrete dataset is available. We distinguish between *estimator properties in finite samples* and *asymptotic estimator properties*. The former are the content of this section and concern the properties of an estimator for a fixed sample size n; the latter are the content of Section 31.3 and concern the properties of an estimator in the limiting case $n \to \infty$ of large sample sizes.

Let (Σ, S) first be a measurable space and $\hat{\tau} : \mathcal{Y} \to \Sigma$ an estimator of $\tau : \Theta \to \Sigma$ (cf. Definition 30.3). In what follows, in addition to parameter estimators of the form

$$\tau : \Theta \to \Sigma, \tau(\theta) := \theta \tag{31.27}$$

we will repeatedly consider estimators that, in parametric product models, estimate only functions of the parameters, such as the expectation, variance, or standard deviation of the sample variables. Since, by assumption, the distributions of the sample variables $y_1, ..., y_n$ are identical, these are estimators of the form

$$\tau : \Theta \to \Sigma,\ \theta \mapsto \tau(\theta) \text{ with } \tau(\theta) := \mathbb{E}_\theta(y_1), \tau(\theta) := \mathbb{V}_\theta(y_1) \text{ and } \tau(\theta) := \mathbb{S}_\theta(y_1). \tag{31.28}$$

In this section we specifically illuminate four aspects of estimator properties in finite samples. In Section 31.2.1 we first deal with the *unbiasedness* of an estimator. An estimator is called *unbiased* if its expectation is identical to the true, but unknown, value $\tau(\theta)$ for all $\theta \in \Theta$. In Section 31.2.2 we introduce, with the *variance* and the *standard error* of an estimator, two measures of the frequentist variability of estimators. With the *mean squared error* of an estimator $\hat{\tau}$ as the expectation of the squared deviation of $\hat{\tau}(y)$ from $\tau(\theta)$, in Section 31.2.3 we then introduce an estimator property that allows the accuracy and variability of an estimator to be related in the sense of a so-called *bias-variance trade-off*. Finally, the *Cramér-Rao inequality* discussed in Section 31.2.4 gives a lower bound for the variance of unbiased estimators. An unbiased estimator with variance equal to the lower bound given by the Cramér-Rao inequality has the smallest possible variance of all unbiased estimators and is in this sense an optimal estimator.

31.2.1 Unbiasedness

The concept of unbiasedness of an estimator arises in the context of the *error* and the *bias* of an estimator as follows.

Definition 31.4 (Error, bias, and unbiasedness)**.** Let $y = (y_1, ..., y_n)$ be a sample of a frequentist inference model and let $\hat{\tau}$ be an estimator for τ.

- The *error* of $\hat{\tau}$ is defined as
$$\hat{\tau}(y) - \tau(\theta). \tag{31.29}$$
- The *bias* of $\hat{\tau}$ is defined as
$$\mathrm{B}(\hat{\tau}) := \mathbb{E}_\theta(\hat{\tau}(y)) - \tau(\theta). \tag{31.30}$$
- The estimator $\hat{\tau}$ is called *unbiased* if
$$\mathrm{B}(\hat{\tau}) = 0 \Leftrightarrow \mathbb{E}_\theta(\hat{\tau}(y)) = \tau(\theta) \text{ for all } \theta \in \Theta \text{ and all } n \in \mathbb{N}. \tag{31.31}$$
Otherwise, $\hat{\tau}$ is called *biased*.

•

Note that in Definition 31.4 the error of an estimator depends on the specific realization of the sample y. The bias, in contrast, is the expected error over sample realizations and is therefore independent of a specific realization. For the special case of a parameter point estimator, Definition 31.4 says that it is unbiased if

$$\mathbb{E}_\theta(\hat{\theta}) = \theta. \tag{31.32}$$

As first examples of unbiased estimators, the following theorem considers the sample mean and the sample variance as estimators of the expectation and the variance of a sample variable.

Theorem 31.3 (Unbiasedness of sample mean and sample variance)**.** *Let $y := (y_1, ..., y_n)$ be the sample of a parametric product model. Then:*

(1) The sample mean
$$\bar{y} := \frac{1}{n}\sum_{i=1}^n y_i \tag{31.33}$$
is an unbiased estimator of the expectation $\mathbb{E}_\theta(y_1)$.

(2) The sample variance
$$S^2 := \frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \tag{31.34}$$
is an unbiased estimator of the variance $\mathbb{V}_\theta(y_1)$.

∘

Proof. (1) The unbiasedness of the sample mean follows with Theorem 23.2 from

$$\mathbb{E}_\theta(\bar{y}) = \mathbb{E}_\theta\left(\frac{1}{n}\sum_{i=1}^n y_i\right) = \frac{1}{n}\sum_{i=1}^n \mathbb{E}_\theta(y_i) = \frac{1}{n}\sum_{i=1}^n \mathbb{E}_\theta(y_1) = \frac{1}{n}n\mathbb{E}_\theta(y_1) = \mathbb{E}_\theta(y_1). \tag{31.35}$$

(2) To show the unbiasedness of the sample variance, we first note that with Theorem 25.6 and independence of the random variables considered,

$$\mathbb{V}_\theta(\bar{y}) = \mathbb{V}_\theta\left(\frac{1}{n}\sum_{i=1}^n y_i\right) = \frac{1}{n^2}\sum_{i=1}^n \mathbb{V}_\theta(y_i) = \frac{1}{n^2}\sum_{i=1}^n \mathbb{V}_\theta(y_1) = \frac{1}{n^2}n\mathbb{V}_\theta(y_1) = \frac{\mathbb{V}_\theta(y_1)}{n}. \tag{31.36}$$

Furthermore, for the term of summed squared deviations in the sample variance,

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))^2 - n(\bar{y} - \mathbb{E}_\theta(y_1))^2, \tag{31.37}$$

because

$$\begin{aligned}
\sum_{i=1}^{n}(y_i - \bar{y})^2 &= \sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1) - \bar{y} + \mathbb{E}_\theta(y_1))^2 \\
&= \sum_{i=1}^{n}((y_i - \mathbb{E}_\theta(y_1)) - (\bar{y} - \mathbb{E}_\theta(y_1)))^2 \\
&= \sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))^2 - 2(\bar{y} - \mathbb{E}_\theta(y_1))\left(\sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))\right) + \sum_{i=1}^{n}(\bar{y} - \mathbb{E}_\theta(y_1))^2 \\
&= \sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))^2 - 2(\bar{y} - \mathbb{E}_\theta(y_1))\left(\sum_{i=1}^{n}y_i - n\mathbb{E}_\theta(y_1)\right) + n(\bar{y} - \mathbb{E}_\theta(y_1))^2 \\
&= \sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))^2 - 2(\bar{y} - \mathbb{E}_\theta(y_1))\left(n\left(\frac{1}{n}\sum_{i=1}^{n}y_i\right) - n\mathbb{E}_\theta(y_1)\right) + n(\bar{y} - \mathbb{E}_\theta(y_1))^2 \\
&= \sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))^2 - 2n(\bar{y} - \mathbb{E}_\theta(y_1))^2 + n(\bar{y} - \mathbb{E}_\theta(y_1))^2 \\
&= \sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))^2 - n(\bar{y} - \mathbb{E}_\theta(y_1))^2.
\end{aligned} \tag{31.38}$$

Together, this gives

$$\begin{aligned}
\mathbb{E}_\theta\left((n-1)S^2\right) &= \mathbb{E}_\theta\left(\sum_{i=1}^{n}(y_i - \bar{y})^2\right) && (31.39) \\
&= \mathbb{E}_\theta\left(\sum_{i=1}^{n}(y_i - \mathbb{E}_\theta(y_1))^2 - n(\bar{y} - \mathbb{E}_\theta(y_1))^2\right) && (31.40) \\
&= \sum_{i=1}^{n}\mathbb{E}_\theta\left((y_i - \mathbb{E}_\theta(y_1))^2\right) - n\mathbb{E}_\theta\left((\bar{y} - \mathbb{E}_\theta(y_1))^2\right) && (31.41) \\
&= n\mathbb{V}_\theta(y_1) - n\mathbb{V}_\theta(\bar{y}) && (31.42) \\
&= n\mathbb{V}_\theta(y_1) - n\frac{\mathbb{V}_\theta(y_1)}{n} && (31.43) \\
&= n\mathbb{V}_\theta(y_1) - \mathbb{V}_\theta(y_1) && (31.44) \\
&= (n-1)\mathbb{V}_\theta(y_1). && (31.45)
\end{aligned}$$

Finally,

$$\mathbb{E}_\theta\left(S^2\right) = \mathbb{E}_\theta\left(\frac{1}{n-1}(n-1)S^2\right) = \frac{1}{n-1}\mathbb{E}_\theta\left((n-1)S^2\right) = \frac{1}{n-1}(n-1)\mathbb{V}_\theta(y_1) = \mathbb{V}_\theta(y_1) \tag{31.46}$$

and thus the unbiasedness of the sample variance as an estimator of the variance. □

Of course, in Theorem 31.3, due to the identical distribution of the sample variables of a parametric product model, the sample mean and the sample variance are also unbiased estimators of the expectation and variance of any sample variable y_i with $1 \le i \le n$. Note that in the proof of the unbiasedness of the sample variance, the denominator $n-1$ in the definition of the sample variance plays a decisive role.

Although the sample variance is an unbiased estimator of the variance of a sample variable of a parametric product model, this does not hold for the sample standard deviation as an estimator of the standard deviation. This is the content of the following theorem.

Theorem 31.4 (Bias of the sample standard deviation)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model. Then the sample standard deviation*

$$S := \sqrt{S^2} \tag{31.47}$$

is a biased estimator of the standard deviation $\mathbb{S}_\theta(y_1)$.

∘

Proof. We first note that $\sqrt{\cdot}$ is a strictly concave function and $\sigma^2 > 0$. With Jensen's inequality $\mathbb{E}(f(\xi)) < f(\mathbb{E}(\xi))$ for strictly concave functions (cf. Theorem 26.5), it follows that

$$\mathbb{E}_\theta(S) = \mathbb{E}_\theta\left(\sqrt{S^2}\right) < \sqrt{\mathbb{E}_\theta(S^2)} = \sqrt{\mathbb{V}_\theta(y_1)} = \mathbb{S}_\theta(y_1). \tag{31.48}$$

□

In general, nonlinear transformations of unbiased estimators often lead to biased estimators, but we will not deepen this point further here.

Example 31.2 (Simulation example)**.** The following **R** code illustrates the concepts of unbiasedness and bias of the sample mean, sample variance, and sample standard deviation using the example of a parametric product model with sample distribution

$$y_1, ..., y_{12} \sim N(1.7, 2) \tag{31.49}$$

The expectations of the estimators are estimated by their sample means across many realizations of $y_1, ..., y_{12}$ as a function of the number of realizations (simulations).

```
# model formulation
set.seed(0)                          # random number generator
mu      = 1.7                         # true, but unknown, expectation parameter
sigsqr  = 2                           # true, but unknown, variance parameter
n       = 12                          # sample size n
nsim    = 5e4                         # number of simulations
y_bar   = rep(NaN,nsim)               # sample mean array
s_sqr   = rep(NaN,nsim)               # sample variance array
s       = rep(NaN,nsim)               # sample standard deviation array

# simulation iterations
for(sim in 1:nsim){

    # sample realization of y_1,...,y_{12}
    y          = rnorm(n,mu,sqrt(sigsqr))

    # expectation, variance, and standard deviation estimators
    y_bar[sim] = mean(y)              # sample mean
    s_sqr[sim] = var(y)               # sample variance
    s[sim]     = sd(y)                # sample standard deviation
}

# expectation estimation
E_hat_y_bar = cumsum(y_bar)/(1:nsim)  # \mathbb{E}(\bar{y}) estimates
E_hat_s_sqr = cumsum(s_sqr)/(1:nsim)  # \mathbb{E}(S^2) estimates
E_hat_s     = cumsum(s)    /(1:nsim)  # \mathbb{E}(S) estimates
```

Figure 31.1 visualizes the results of the above simulation. Shown are estimates of the expectations of the sample mean, sample variance, and sample standard deviation as a function of the number of realizations of the sample variables $y_1, ..., y_{12}$, as well as the true, but unknown, values of the expectation, variance, and standard deviation of $y_1, ..., y_{12}$. It is noticeable that these estimates are more variable for a small number of realizations. From an estimate based on about 10,000 realizations of $y_1, ..., y_{12}$ onward, the sample means of $\bar{y}$ and S^2 correspond to their true, but unknown, values in accordance with their unbiasedness. The sample standard deviation, in contrast, in accordance with its bias, still shows a consistently too low estimate of the true, but unknown, standard deviation even as the number of realizations of $y_1, ..., y_{12}$ continues to increase.

△

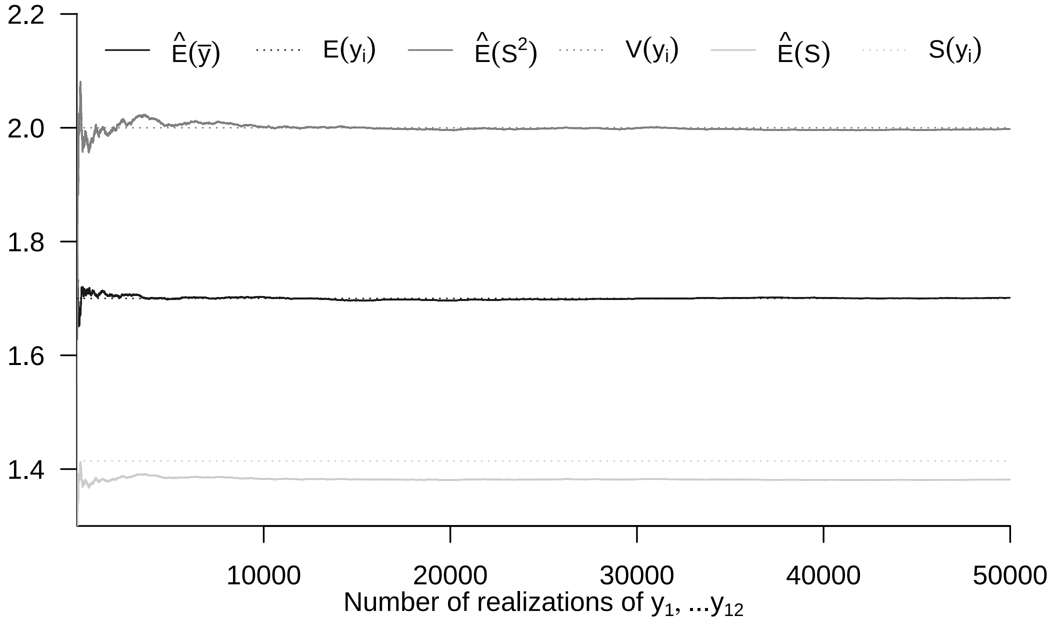


Figure 31.1 Simulation of the unbiasedness of the sample mean and sample variance as estimators of the expectation and variance under normally distributed sample variables, and simulation of the bias of the sample standard deviation as an estimator of the standard deviation under normally distributed sample variables.

31.2.2 Variance and standard error

In the previous section we considered the expectation of an estimator. In this section we consider its variance and its standard deviation. We use the following definition.

Definition 31.5 (Variance and standard error)**.** Let $y = (y_1, ..., y_n)$ be the sample of a frequentist inference model and let $\hat{\tau}$ be an estimator of τ.

- The *variance* of $\hat{\tau}$ is defined as

$$\mathbb{V}_\theta(\hat{\tau}) := \mathbb{E}_\theta\left((\hat{\tau}(y) - \mathbb{E}_\theta(\hat{\tau}(y)))^2\right). \tag{31.50}$$

- The *standard error* of $\hat{\tau}$ is defined as

$$\mathrm{SE}(\hat{\tau}) := \sqrt{\mathbb{V}_\theta(\hat{\tau})}. \tag{31.51}$$

•

The variance of an estimator $\hat{\tau}$ is thus defined as the variance of the random variable $\hat{\tau}(y)$. The standard error of an estimator $\hat{\tau}$ is defined as the standard deviation of $\hat{\tau}(y)$. As a first example of a standard error, we consider the *standard error of the sample mean*.

Theorem 31.5 (Standard error of the sample mean)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model. Then the standard error of the sample mean is given by*

$$SE(\bar{y}) = \frac{\mathbb{S}_\theta(y_1)}{\sqrt{n}}. \tag{31.52}$$

.

∘

Proof. With the variance of the sample mean, we obtain

$$\mathrm{SE}(\bar{y}) = \sqrt{\mathbb{V}_\theta(\bar{y})} = \sqrt{\frac{\mathbb{V}_\theta(y_1)}{n}} = \frac{\mathbb{S}_\theta(y_1)}{\sqrt{n}}. \tag{31.53}$$

□

The standard error of the mean describes the variability of the sample mean. Since the standard deviation $\mathbb{S}_\theta(y_1)$ is unknown, the standard error $\mathrm{SE}(\bar{y})$ is also unknown and can therefore only be estimated. With the sample standard deviation as a biased estimator of the standard deviation $\mathbb{S}_\theta(y_1)$, an also biased estimator for the standard error of the sample mean is

$$\hat{\mathrm{SE}}(\bar{y}) = \frac{S}{\sqrt{n}}. \tag{31.54}$$

As a second example, we consider the standard error of the maximum likelihood estimator for the parameter of a Bernoulli model.

Theorem 31.6 (Standard error of the maximum likelihood estimator of the Bernoulli model parameter)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a Bernoulli model and let $\hat{\mu}^{ML}$ be the maximum likelihood estimator for the Bernoulli model parameter μ. Then the standard error of $\hat{\mu}^{ML}$ is given by*

$$SE(\hat{\mu}^{ML}) = \sqrt{\frac{\mu(1-\mu)}{n}}. \tag{31.55}$$

∘

Proof. We have

$$\mathrm{SE}\left(\hat{\mu}^{\mathrm{ML}}\right) = \sqrt{\mathbb{V}_\mu\left(\hat{\mu}^{\mathrm{ML}}\right)} = \sqrt{\mathbb{V}_\mu\left(\frac{1}{n}\sum_{i=1}^n y_i\right)} = \sqrt{\frac{1}{n^2}\sum_{i=1}^n \mathbb{V}_\mu(y_i)} = \sqrt{\frac{n\mu(1-\mu)}{n^2}} = \sqrt{\frac{\mu(1-\mu)}{n}}, \tag{31.56}$$

where the third equality follows from the independence of the y_i and the fourth equality from the variance

$$\mathbb{V}_\mu(y_1) = \mathbb{V}_\mu(y_i) = \mu(1-\mu) \tag{31.57}$$

of the sample variables. □

As in the case of the standard error of the sample mean, the standard error of the maximum likelihood estimator of the Bernoulli model parameter is a true, but unknown, value. An estimator for $\mathrm{SE}\left(\hat{\mu}^{\mathrm{ML}}\right)$ is obtained with the maximum likelihood estimator for the Bernoulli model parameter as

$$\hat{\mathrm{SE}}\left(\hat{\mu}^{\mathrm{ML}}\right) = \sqrt{\frac{\hat{\mu}^{\mathrm{ML}}(1-\hat{\mu}^{\mathrm{ML}})}{n}}. \tag{31.58}$$

Example 31.3 (Simulation example)**.** The following **R** code simulates the distribution of the maximum likelihood estimator for the parameter of a Bernoulli model with true, but unknown, parameter value $\mu := 0.4$ for the sample sizes $n = 20$, $n = 100$, and $n = 200$. Figure 31.2 visualizes the resulting distributions using histograms. The variability of the estimates, that is, the width of the histogram distributions, evidently depends on the sample size. Larger sample sizes result in lower variability of the estimator. We will deepen this idea in Section 31.3.

```
# model formulation
mu        = 0.4                                                      # true, but unknown, parameter value
n_all     = c(20,100,200)                                            # sample sizes n
ns        = 1e4                                                      # number of simulations
mu_hat    = matrix(rep(NaN, length(n_all)*ns), nrow = length(n_all)) # maximum likelihood estimator array

# sample size iterations
for(i in seq_along(n_all)){

    # simulation iterations
    for(s in 1:ns){
        y             = rbinom(n_all[i],1,mu)                        # sample realization of y_1,...,y_n
        mu_hat[i,s]   = mean(y)                                      # sample mean
    }
}
```

△

△

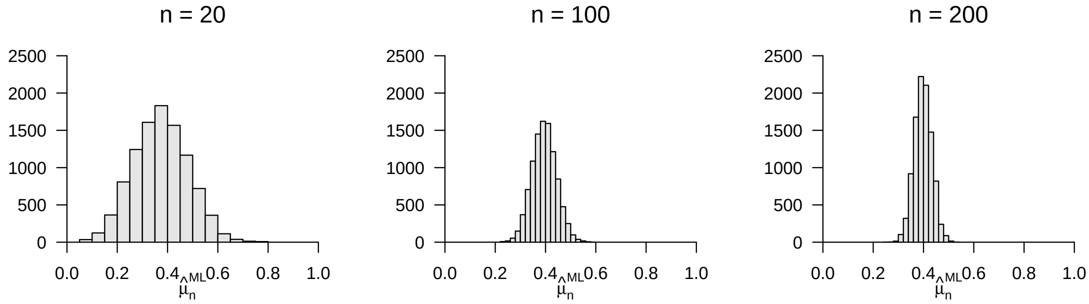


Figure 31.2 Simulation of the distribution of the maximum likelihood estimator of a Bernoulli model. The variability of the estimator evidently depends on the sample size n.

31.2.3 Mean squared error

With the unbiasedness and variance of an estimator, we have encountered in the previous two sections two independent criteria for the quality of estimators. The *mean squared error* of an estimator introduced in this section enables an integrated consideration of the accuracy (unbiasedness) and variability (variance) of an estimator in the sense of its so-called *bias-variance decomposition*. We first define the mean squared error of an estimator as follows.

Definition 31.6 (Mean squared error)**.** Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model and $\hat{\tau}$ an estimator for τ. Then the *mean squared error* of $\hat{\tau}$ is defined as

$$\mathrm{MSE}(\hat{\tau}) := \mathbb{E}_\theta\left((\hat{\tau}(y) - \tau(\theta))^2\right). \tag{31.59}$$

•

The mean squared error of $\hat{\tau}$ is thus the expected squared deviation of $\hat{\tau}(y)$ from $\tau(\theta)$. Note that, by contrast, the variance of $\hat{\tau}$ is the expected squared deviation of $\hat{\tau}$ from $\mathbb{E}_\theta(\hat{\tau}(y))$. As seen in Section 31.2.1, $\mathbb{E}_\theta(\hat{\tau}(y))$ may coincide with $\tau(\theta)$, so an estimator may be unbiased, but need not be. If one uses the mean squared error as a quality criterion for an estimator, for example by trying to construct an estimator with the smallest possible mean squared error, then one may accept possible slight deviations from unbiasedness in favor of a small estimator variance. For the mean squared error, the following theorem holds.

Theorem 31.7 (Decomposition of the mean squared error)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model, let $\hat{\tau}$ be an estimator for τ, and let MSE$(\hat{\tau})$ be the mean squared error of $\hat{\tau}$. Then*

$$MSE(\hat{\tau}) = B(\hat{\tau})^2 + \mathbb{V}_\theta(\hat{\tau}). \tag{31.60}$$

∘

Proof. To simplify notation, let $\tau := \tau(\theta)$, $\hat{\tau} := \hat{\tau}(y)$, and $\bar{\tau}_n := \mathbb{E}_\theta(\hat{\tau}(y))$. Then:

$$
\begin{aligned}
\mathbb{E}_\theta\left((\hat{\tau}-\tau)^2\right) &= \mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n+\bar{\tau}_n-\tau)^2\right) \\
&= \mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n)^2+2(\hat{\tau}-\bar{\tau}_n)(\bar{\tau}_n-\tau)+(\bar{\tau}_n-\tau)^2\right) \\
&= \mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n)^2\right)+2\mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n)(\bar{\tau}_n-\tau)\right)+\mathbb{E}_\theta\left((\bar{\tau}_n-\tau)^2\right) \\
&= \mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n)^2\right)+2\mathbb{E}_\theta\left(\hat{\tau}\bar{\tau}_n-\hat{\tau}\tau-\bar{\tau}_n\bar{\tau}_n+\bar{\tau}_n\tau\right)+\mathbb{E}_\theta\left((\bar{\tau}_n-\tau)^2\right) \\
&= \mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n)^2\right)+2\left(\bar{\tau}_n\bar{\tau}_n-\bar{\tau}_n\tau\right)+\mathbb{E}_\theta\left((\bar{\tau}_n-\tau)^2\right) \\
&= \mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n)^2\right)+0+\mathbb{E}_\theta\left((\bar{\tau}_n-\tau)^2\right) \\
&= \mathbb{E}_\theta\left((\bar{\tau}_n-\tau)^2\right)+\mathbb{E}_\theta\left((\hat{\tau}-\bar{\tau}_n)^2\right) \\
&= \mathbb{E}_\theta\left((\mathbb{E}_\theta(\hat{\tau})-\tau)^2\right)+\mathbb{E}_\theta\left((\hat{\tau}-\mathbb{E}_\theta(\hat{\tau}))^2\right) \\
&= (\mathbb{E}_\theta(\hat{\tau})-\tau)^2+\mathbb{V}_\theta(\hat{\tau}) \\
&= \mathrm{B}(\hat{\tau})^2+\mathbb{V}_\theta(\hat{\tau}).
\end{aligned}
\tag{31.61}
$$

□

31.2.4 Cramér-Rao inequality

If several unbiased estimators are available, the estimator with the smallest variance most reliably serves its purpose. But because sample realizations of frequentist inference models are generally variable, the variability of unbiased estimators cannot be arbitrarily small either. The *Cramér-Rao inequality* gives a lower bound for the variance of unbiased estimators. An unbiased estimator with variance equal to this lower bound therefore has the smallest possible variance of all unbiased estimators and is, in this sense, an optimal estimator.

The Cramér-Rao inequality is based on the concept of the so-called *Fisher information*, which in turn is based on the concept of the *score function* of a frequentist inference model. We therefore first introduce these two concepts before formulating and proving the Cramér-Rao inequality.

The results presented generally hold only under a series of mathematical assumptions, the so-called *Fisher regularity conditions*. For a frequentist inference model with PMF or PDF p_θ and parameter space Θ, these consist in assuming that (1) Θ is an open set, so the true, but unknown, parameter value cannot lie on a boundary of the parameter space, (2) the subset of Θ on which p_θ takes nonzero values does not depend on θ, (3) the model itself is identifiable, that is, PMFs or PDFs with different parameter values are different functions and therefore imply different sample distributions, (4) the likelihood function of the model is twice continuously differentiable, and (5) for the likelihood function integration and differentiation may be interchanged. We thus assume the Fisher regularity conditions to be fulfilled and consider only models with one-dimensional parameter spaces $\Theta \subseteq \mathbb{R}$. We first define the terms *score function* and *Fisher information* as follows.

Definition 31.7 (Score function and Fisher information)**.** Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model with one-dimensional parameter $\theta \in \Theta \subseteq \mathbb{R}$, and let ℓ be the associated log-likelihood function. Then:

- The first derivative of ℓ is called the *score function of the sample* and is denoted by

$$S(\theta) := \frac{d}{d\theta}\ell(\theta) \tag{31.62}$$

  For $n = 1$ we write $S(\theta) := S_1(\theta)$ and call $S(\theta)$ the *score function of a random variable.*

- The negative second derivative of ℓ is called the *Fisher information of the sample* and is denoted by
$$I(\theta) := -\frac{d^2}{d\theta^2}\ell(\theta) \tag{31.63}$$
For $n = 1$ we write $I(\theta) := I_1(\theta)$ and call $I(\theta)$ the *Fisher information of a random variable.*

•

Because likelihood and log-likelihood functions depend on the realization of a sample, they are random functions against the background of a frequentist inference model. Since Fisher information, as a function of the log-likelihood function, is therefore also a random variable, one must distinguish between the *observed* and the *expected* values of the Fisher information.

Definition 31.8 (Observed and expected Fisher information)**.** Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model with one-dimensional parameter $\theta \in \Theta \subseteq \mathbb{R}$, let ℓ be the associated log-likelihood function, and let $\hat{\theta}^{\text{ML}}$ be a maximum likelihood estimator of θ. Then:

- The *observed Fisher information of the sample* is defined as
$$I\left(\hat{\theta}^{\text{ML}}\right) := -\frac{d^2}{d\theta^2}\ell\left(\hat{\theta}^{\text{ML}}\right), \tag{31.64}$$
that is, the observed Fisher information of the sample is the Fisher information at the point of the maximum likelihood estimator $\hat{\theta}^{\text{ML}}$.
- The *expected Fisher information of the sample* is defined as
$$J(\theta) := \mathbb{E}_\theta(I(\theta)). \tag{31.65}$$
For $n = 1$ we write $J(\theta) := J_1(\theta)$ and call $J(\theta)$ the *expected Fisher information of a random variable.*

•

Before illustrating these concepts using the Bernoulli model (Theorem 31.10) and the normal distribution model (Theorem 31.12 and Theorem 31.11), we introduce, with the *additivity of Fisher information* in parametric product models (Theorem 31.8) and the expectation and variance of the score function (Theorem 31.9), important properties of these concepts that simplify the following discussion.

Theorem 31.8 (Additivity of Fisher information)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model with parameter $\theta \in \Theta \subseteq \mathbb{R}$, let ℓ be the associated log-likelihood function, and let $I(\theta)$ and $J(\theta)$ be the Fisher information and the expected Fisher information of the sample, respectively. Then*
$$I(\theta) = nI_1(\theta) \text{ and } J(\theta) = nJ_1(\theta). \tag{31.66}$$

∘

Proof. We show the result for the expected Fisher information; the result for the observed Fisher information then holds implicitly. With the linearity of derivatives and expectations,

$$
\begin{aligned}
J(\theta) &= \mathbb{E}_\theta\left(-\frac{d^2}{d\theta^2}\ell(\theta)\right) \\
&= \mathbb{E}_\theta\left(-\frac{d^2}{d\theta^2}\ln\left(\prod_{i=1}^n p_\theta(y_i)\right)\right) \\
&= \mathbb{E}_\theta\left(-\frac{d^2}{d\theta^2}\sum_{i=1}^n \ln p_\theta(y_i)\right) \\
&= \mathbb{E}_\theta\left(-\frac{d^2}{d\theta^2}\sum_{i=1}^n \ln p_\theta(y_1)\right) \\
&= \mathbb{E}_\theta\left(-\frac{d^2}{d\theta^2}\ell_1(\theta)n\right) \\
&= n\mathbb{E}_\theta\left(-\frac{d^2}{d\theta^2}\ell_1(\theta))\right) \\
&= nJ_1(\theta).
\end{aligned}
\tag{31.67}
$$

□

According to Theorem 31.8, to compute the observed or expected Fisher information of a sample in parametric product models, it is sufficient to compute the observed or expected Fisher information of one of the random variables in the sample. Further simplifications in determining Fisher information and justifying the Cramér-Rao inequality follow from the identities formulated in the following theorem.

Theorem 31.9 (Expectation and variance of the score function)**.** *The expectation of the score function of a random variable is*

$$\mathbb{E}_\theta(S(\theta)) = 0 \tag{31.68}$$

and the variance of the score function of a random variable is

$$\mathbb{V}_\theta(S(\theta)) = J(\theta). \tag{31.69}$$

∘

Proof. We consider only the case in which p_θ is a PDF and first show that $\mathbb{E}_\theta(S(\theta)) = 0$.

$$
\begin{aligned}
\mathbb{E}_\theta(S(\theta)) &= \int S(\theta)p_\theta(x)\,dx \\
&= \int \frac{d}{d\theta}\ell(\theta)p_\theta(x)\,dx \\
&= \int \frac{d}{d\theta}\ln L(\theta)p_\theta(x)\,dx \\
&= \int \frac{1}{L(\theta)}\frac{d}{d\theta}L(\theta)p_\theta(x)\,dx \\
&= \int \frac{1}{p_\theta(x)}\frac{d}{d\theta}L(\theta)p_\theta(x)\,dx \\
&= \int \frac{d}{d\theta}L(\theta)\,dx \\
&= \frac{d}{d\theta}\int p_\theta(x)\,dx \\
&= \frac{d}{d\theta}1 \\
&= 0.
\end{aligned}
\tag{31.70}
$$

With the definition of variance, it follows immediately that $\mathbb{V}_\theta(S(\theta)) = \mathbb{E}_\theta(S(\theta)^2)$. Next we show that $J(\theta) = \mathbb{E}_\theta(S(\theta)^2)$ and hence $\mathbb{V}_\theta(S(\theta)) = J(\theta)$.

$$
\begin{aligned}
J(\theta) &= \mathbb{E}_\theta\left(-\frac{d^2}{d\theta^2}\ln L(\theta)\right) \\
&= \mathbb{E}_\theta\left(-\frac{d}{d\theta}\frac{\frac{d}{d\theta}L(\theta)}{L(\theta)}\right) \\
&= \mathbb{E}_\theta\left(-\frac{\frac{d^2}{d\theta^2}L(\theta)L(\theta) - \frac{d}{d\theta}L(\theta)\frac{d}{d\theta}L(\theta)}{L(\theta)L(\theta)}\right) \\
&= -\mathbb{E}_\theta\left(\frac{\frac{d^2}{d\theta^2}L(\theta)}{L(\theta)}\right) + \mathbb{E}_\theta\left(\frac{\left(\frac{d}{d\theta}L(\theta)\right)^2}{(L(\theta))^2}\right) \\
&= -\int \frac{\frac{d^2}{d\theta^2}L(\theta)}{L(\theta)} p_\theta(x)\,dx + \int \frac{\left(\frac{d}{d\theta}L(\theta)\right)^2}{(L(\theta))^2} p_\theta(x)\,dx \\
&= -\frac{d^2}{d\theta^2}\int p_\theta(x)\,dx + \int \left(\frac{1}{L(\theta)}\frac{d}{d\theta}L(\theta)\right)^2 p_\theta(x)\,dx \\
&= -\frac{d^2}{d\theta^2}1 + \int \left(\frac{d}{d\theta}\ln L(\theta)\right)^2 p_\theta(x)\,dx \\
&= \mathbb{E}_\theta\left(S(\theta)^2\right).
\end{aligned}
\tag{31.71}
$$

□

The expectation of the derivative of the log-likelihood function is thus always zero, and the expected Fisher information is always equal to the variance of the score function. We now compute the score function and the different forms of Fisher information concretely for the familiar frequentist inference models. The following theorem first summarizes the results for the Bernoulli model.

Theorem 31.10 (Score function and Fisher information of the Bernoulli model)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a Bernoulli model with parameter $\mu \in]0,1[$. Then:*

- *The score function of the sample is*

$$
S :]0,1[\to \mathbb{R}, \mu \mapsto S(\mu) := \frac{1}{\mu}\sum_{i=1}^n y_i - \frac{1}{1-\mu}\left(n - \sum_{i=1}^n y_i\right). \tag{31.72}
$$

- *The Fisher information of the sample is*

$$
I :]0,1[\to \mathbb{R}, \mu \mapsto I(\mu) := I(\mu) = \frac{ny}{\mu^2} + \frac{n(1-y)}{1-\mu}^2. \tag{31.73}
$$

- *The observed Fisher information of the sample is*

$$
I :]0,1[\to \mathbb{R}, \hat{\mu}^{ML} \mapsto I\left(\hat{\mu}^{ML}\right) := \frac{ny}{\hat{\mu}_n^{ML^2}} + \frac{n(1-y)}{1-\hat{\mu}^{ML}}. \tag{31.74}
$$

- *The expected Fisher information of the sample is*

$$
J :]0,1[\to \mathbb{R}, \mu \mapsto J(\mu) := \frac{n}{\mu(1-\mu)}. \tag{31.75}
$$

∘

Proof. The score function was already derived in the context of maximum likelihood estimation of μ. We consider the Fisher information of a single Bernoulli random variable y.

$$
\begin{aligned}
I(\mu) &:= -\frac{d^2}{d\mu^2}\ell_1(\mu) \\
&= -\frac{d^2}{d\mu^2}\ln p_\mu(y) \\
&= -\frac{d^2}{d\mu^2}\left(y\ln\mu + (1-y)\ln(1-\mu)\right) \\
&= -\frac{\partial}{\partial\mu}\left(\frac{\partial}{\partial\mu}\left(y\ln\mu + (1-y)\ln(1-\mu)\right)\right) \\
&= -\frac{\partial}{\partial\mu}\left(\frac{y}{\mu} + \frac{(1-y)}{1-\mu}\right) \\
&= -\left(-\frac{y}{\mu^2} - \frac{(1-y)}{1-\mu}^2\right) \\
&= \frac{y}{\mu^2} + \frac{(1-y)}{1-\mu}^2.
\end{aligned}
\tag{31.76}
$$

Thus the expected Fisher information of the random variable y is

$$
\begin{aligned}
J(\mu) &= \mathbb{E}_\mu(I(\mu)) \\
&= \mathbb{E}_\mu\left(\frac{y}{\mu^2} + \frac{(1-y)}{1-\mu}^2\right) \\
&= \frac{\mathbb{E}_\mu(y)}{\mu^2} + \frac{(1-\mathbb{E}_\mu(y))}{1-\mu}^2 \\
&= \frac{\mu}{\mu^2} + \frac{(1-\mu)}{1-\mu}^2 \\
&= \frac{1}{\mu(1-\mu)}.
\end{aligned}
\tag{31.77}
$$

With the additivity property of Fisher information and the definition of observed Fisher information, it follows immediately that

$$
I(\mu) = \frac{ny}{\mu^2} + \frac{n(1-y)}{1-\mu}^2 \text{ and } J(\mu) = \frac{n}{\mu(1-\mu)}.
\tag{31.78}
$$

□

We consider the score function and Fisher information of the normal distribution model only under the additional assumption of a known variance parameter (Theorem 31.11) or a known expectation parameter (Theorem 31.12).

Theorem 31.11 (Score function and Fisher information of the normal distribution model with known variance parameter)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a normal distribution model and assume that the variance parameter σ^2 is known. Then:*

- *The score function of the sample is*

$$
S : \mathbb{R} \to \mathbb{R}, \mu \mapsto S(\mu) := \frac{1}{\sigma^2}\sum_{i=1}^n (y_i - \mu).
\tag{31.79}
$$

- *The Fisher information of the sample is*

$$
I : \mathbb{R} \to \mathbb{R}, \mu \mapsto I(\mu) := \frac{n}{\sigma^2}.
\tag{31.80}
$$

- *The observed Fisher information of the sample is*

$$I(\hat{\mu}_n^{ML}) = \frac{n}{\sigma^2}. \tag{31.81}$$

- *The expected Fisher information of the sample is*

$$J : \mathbb{R} \to \mathbb{R}, \mu \mapsto J(\mu) := \frac{n}{\sigma^2}. \tag{31.82}$$

∘

Proof. Recall that the log-likelihood function of a normal distribution model with known variance parameter σ^2 is given by

$$\ell : \mathbb{R} \to \mathbb{R}, \mu \mapsto \ell(\mu) := -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln \sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2 \tag{31.83}$$

Thus the score function is

$$S(\mu) = \frac{\partial}{\partial \mu}\ell(\mu) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i - \mu) \tag{31.84}$$

The Fisher information of the sample is

$$I(\mu) = -\frac{d^2}{d\mu^2}\ell(\mu) = -\frac{\partial}{\partial \mu}S(\mu) = -\frac{1}{\sigma^2}\frac{\partial}{\partial \mu}\left(\sum_{i=1}^{n} y_i - n\mu\right) = \frac{n}{\sigma^2}. \tag{31.85}$$

The observed Fisher information is the Fisher information at the point of the maximum likelihood estimator $\hat{\mu}_n^{\text{ML}}$. The expected Fisher information finally is

$$J(\mu) = \mathbb{E}_\mu(I(\mu)) = \mathbb{E}_\mu\left(\frac{n}{\sigma^2}\right) = \frac{n}{\sigma^2}. \tag{31.86}$$

□

Theorem 31.12 (Score function and Fisher information of the normal distribution model with known expectation parameter)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a normal distribution model and assume that the expectation parameter μ is known. Then:*

- *The score function of the sample is given by*

$$S : \mathbb{R}_{>0} \to \mathbb{R}, \sigma^2 \mapsto S(\sigma^2) := -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(y_i - \mu)^2 \tag{31.87}$$

- *The Fisher information of the sample is given by*

$$I : \mathbb{R}_{>0} \to \mathbb{R}, \sigma^2 \mapsto I(\sigma^2) := \frac{1}{\sigma^6}\sum_{i=1}^{n}(y_i - \mu)^2 - \frac{n}{2\sigma^4} \tag{31.88}$$

- *The observed Fisher information of the sample is given by*

$$I(\hat{\sigma}_n^{2\,ML}) = \frac{n}{2\hat{\sigma}_{ML}^4} \tag{31.89}$$

- *The expected Fisher information of the sample is given by*

$$J : \mathbb{R}_{>0} \to \mathbb{R}, \sigma^2 \mapsto J(\sigma^2) := \frac{n}{2\sigma^4}. \tag{31.90}$$

∘

Proof. Recall that the log-likelihood function of the sample of a normal distribution model with known expectation parameter μ is given by

$$\ell : \mathbb{R}_{>0} \to \mathbb{R}, \sigma^2 \mapsto \ell(\sigma^2) := -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln \sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n (y_i - \mu)^2. \tag{31.91}$$

The score function is therefore

$$S(\sigma^2) = \frac{\partial}{\partial \sigma^2}\ell(\sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^n (y_i - \mu)^2. \tag{31.92}$$

The Fisher information of the sample is

$$\begin{aligned}
I(\sigma^2) &= -\frac{\partial}{\partial \sigma^2} S(\sigma^2) \\
&= -\frac{\partial}{\partial \sigma^2}\left(\frac{n}{2\sigma^4} - \frac{1}{\sigma^6}\sum_{i=1}^n (y_i - \mu)^2\right) \\
&= \frac{1}{\sigma^6}\sum_{i=1}^n (y_i - \mu)^2 - \frac{n}{2\sigma^4}.
\end{aligned} \tag{31.93}$$

The observed Fisher information is the Fisher information at the point of the maximum likelihood estimator $\hat{\sigma}^{2\,\mathrm{ML}}$,

$$\begin{aligned}
I(\hat{\sigma}_n^{2\,\mathrm{ML}}) &= \frac{\sum_{i=1}^n (y_i - \mu)^2}{\left(\hat{\sigma}_n^{2\,\mathrm{ML}}\right)^3} - \frac{n}{2\left(\hat{\sigma}_n^{2\,\mathrm{ML}}\right)^2} \\
&= \frac{\sum_{i=1}^n (y_i - \mu)^2}{\frac{1}{n^3}\left(\sum_{i=1}^n (y_i - \mu)^2\right)^3} - \frac{n}{2\left(\hat{\sigma}_n^{2\,\mathrm{ML}}\right)^2} \\
&= \frac{1}{\frac{1}{n^3}\left(\sum_{i=1}^n (y_i - \mu)^2\right)^2} - \frac{n}{2\left(\hat{\sigma}_n^{2\,\mathrm{ML}}\right)^2} \\
&= \frac{n}{\left(\hat{\sigma}_n^{2\,\mathrm{ML}}\right)^2} - \frac{n}{2\left(\hat{\sigma}_n^{2\,\mathrm{ML}}\right)^2} \\
&= \frac{n}{2\left(\hat{\sigma}_n^{2\,\mathrm{ML}}\right)^2} \\
&= \frac{n}{2\hat{\sigma}_n^{4\,\mathrm{ML}}}.
\end{aligned} \tag{31.94}$$

The expected Fisher information finally is

$$\begin{aligned}
J(\sigma^2) &= \mathbb{E}_{\sigma^2}(I(\sigma^2)) \\
&= \mathbb{E}_{\sigma^2}\left(\frac{1}{\sigma^6}\sum_{i=1}^n (y_i - \mu)^2 - \frac{n}{2\sigma^4}\right) \\
&= \frac{1}{\sigma^6}\sum_{i=1}^n \mathbb{E}_{\sigma^2}\left((y_i - \mu)^2\right) - \frac{n}{2\sigma^4} \\
&= \frac{1}{\sigma^6}\sum_{i=1}^n \sigma^2 - \frac{n}{2\sigma^4} \\
&= \frac{n\sigma^2}{\sigma^6} - \frac{n}{2\sigma^4} \\
&= \frac{n}{\sigma^4} - \frac{n}{2\sigma^4} \\
&= \frac{n}{2\sigma^4}.
\end{aligned} \tag{31.95}$$

□

With the properties of the score function discussed above, we can now formulate and prove the Cramér-Rao inequality.

Theorem 31.13 (Cramér-Rao inequality). *Let a frequentist inference model with one-dimensional parameter $\theta \in \Theta \subseteq \mathbb{R}$, PMF or PDF p_θ, be given, and let $\hat{\tau}$ be an unbiased estimator of $\tau(\theta)$. Then*

$$\mathbb{V}_\theta(\hat{\tau}) \geq \frac{\left(\frac{d}{d\theta}\tau(\theta)\right)^2}{J(\theta)}. \tag{31.96}$$

In particular, for a parameter estimator with $\tau(\theta) := \theta$ and thus

$$\hat{\tau} = \hat{\theta} \text{ and } \left(\frac{d}{d\theta}\tau(\theta)\right)^2 = 1, \tag{31.97}$$

we have

$$\mathbb{V}_\theta(\hat{\theta}) \geq \frac{1}{J(\theta)}. \tag{31.98}$$

The right-hand side of the above inequalities is called the Cramér-Rao bound*.*

∘

Proof. We first note that for the random variables $S(\theta)$ and $\hat{\tau}$, with the correlation inequality (Theorem 26.4) and the identity of $\mathbb{V}_\theta(S(\theta))$ and $J(\theta)$ (Theorem 31.9),

$$\frac{\mathbb{C}_\theta(S(\theta), \hat{\tau})^2}{\mathbb{V}_\theta(S(\theta))\mathbb{V}_\theta(\hat{\tau})} \leq 1 \Leftrightarrow \mathbb{V}_\theta(\hat{\tau}) \geq \frac{\mathbb{C}_\theta(S(\theta), \hat{\tau})^2}{J(\theta)}. \tag{31.99}$$

With the covariance shift theorem (Theorem 25.2), the fact that the expectation of the score function is always zero (Theorem 31.9), and the assumed unbiasedness of $\hat{\tau}$, we first obtain

$$\begin{aligned}
\mathbb{C}_\theta(S(\theta), \hat{\tau}) &= \mathbb{E}_\theta(S(\theta)\hat{\tau}) - \mathbb{E}_\theta(S(\theta))\mathbb{E}_\theta(\hat{\tau}) \\
&= \mathbb{E}_\theta(S(\theta)\hat{\tau}) \\
&= \int S(\theta)\,\hat{\tau}\,p_\theta(x)\,dx \\
&= \int \frac{d}{d\theta}\ln L(\theta)\,\hat{\tau}\,p_\theta(x)\,dx \\
&= \int \frac{\frac{d}{d\theta}L(\theta)}{L(\theta)}\,\hat{\tau}\,p_\theta(x)\,dx \\
&= \int \frac{\frac{d}{d\theta}L(\theta)}{p_\theta(x)}\,\hat{\tau}\,p_\theta(x)\,dx \\
&= \int \frac{d}{d\theta}L(\theta)\,\hat{\tau}\,dx \\
&= \frac{d}{d\theta}\int L(\theta)\,\hat{\tau}\,dx \\
&= \frac{d}{d\theta}\int \hat{\tau}\,p_\theta(x)\,dx \\
&= \frac{d}{d\theta}\mathbb{E}_\theta(\hat{\tau}) \\
&= \frac{d}{d\theta}\tau(\theta).
\end{aligned} \tag{31.100}$$

Thus it follows directly that

$$\mathbb{V}_\theta(\hat{\tau}) \geq \frac{\left(\frac{d}{d\theta}\tau(\theta)\right)^2}{J(\theta)}. \tag{31.101}$$

□

For parameter estimators, in particular, the variance of an unbiased parameter estimator $\hat{\theta}$ is therefore always greater than or equal to the reciprocal expected Fisher information $J(\theta)$. If even

$$\mathbb{V}_\theta(\hat{\theta}) = \frac{1}{J(\theta)} \tag{31.102}$$

holds, then the variance of the parameter estimator is minimal and the estimator is thus shown to be an optimal estimator in the sense of the Cramér-Rao inequality. We return to this idea in Section 31.4.

31.3 Asymptotic estimator properties

In this section we give a brief introduction to *asymptotic statistics* (Vaart (1998)). Asymptotic statistics is the area of frequentist inference that deals with the behavior of statistics and estimators at large sample sizes. The methods of asymptotic statistics are used, on the one hand, to study qualitative estimator properties, as here, and, on the other hand, to approximate estimator properties at large sample sizes. Since sample sizes can nowadays be quite large, the methods of asymptotic statistics are therefore well motivated for application and can be used in many ways there.

Continuing Section 31.2, in this section we illuminate four asymptotic estimator properties. To emphasize that in this section the properties of an estimator $\hat{\tau}$ depend on the sample size n, we write $\hat{\tau}_n$ for an estimator in this section. In Section 31.3.1 we consider the *asymptotic unbiasedness* of an estimator. An estimator $\hat{\tau}_n$ for τ is called *asymptotically unbiased* if the expectation of $\hat{\tau}_n$ for large sample sizes $n \to \infty$ is identical to the true, but unknown, value $\tau(\theta)$. In Section 31.3.2 we introduce the concept of *consistency* of an estimator. Intuitively, an estimator $\hat{\tau}_n$ for τ is called *consistent* if, for large sample sizes $n \to \infty$, the probability that $\hat{\tau}_n(y)$ deviates from the true, but unknown, value $\tau(\theta)$ becomes arbitrarily small. For large sample sizes, the distributions of estimators often result in normal distributions. In Section 31.3.3 we introduce a corresponding formalization with the concept of an *asymptotic normal distribution*. An estimator $\hat{\tau}_n$ for τ is then called *asymptotically normally distributed* if, for large sample sizes $n \to \infty$, the distribution of $\hat{\tau}_n$ is given by a normal distribution. Finally, in Section 31.3.4, with *asymptotic efficiency*, we consider an optimality criterion for estimators at sample sizes tending to infinity. An estimator $\hat{\tau}_n$ for τ is then called *asymptotically efficient* if, for large sample sizes $n \to \infty$, the distribution of $\hat{\tau}_n$ is given by a normal distribution with expectation parameter $\tau(\theta)$ and variance parameter equal to the Cramér-Rao bound.

31.3.1 Asymptotic unbiasedness

Asymptotic unbiasedness of an estimator refines the concept of an unbiased estimator as follows.

Definition 31.9 (Asymptotic unbiasedness)**.** Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model and let $\hat{\tau}_n$ be an estimator for τ. $\hat{\tau}_n$ is called *asymptotically unbiased* if

$$\lim_{n\to\infty} \mathbb{E}_\theta(\hat{\tau}_n(y)) = \tau(\theta) \text{ for all } \theta \in \Theta. \tag{31.103}$$

•

Asymptotically unbiased estimators are therefore unbiased only when the sample size tends to infinity. Unbiased estimators are always also asymptotically unbiased, because their unbiasedness is independent of the sample size. As an example of an estimator that is only asymptotically unbiased, we consider the maximum likelihood estimator of the variance parameter of the normal distribution model.

Theorem 31.14 (Asymptotic unbiasedness of the variance parameter estimator)**.** *Let $y = (y_1, ..., y_n)$ be the sample of a normal distribution model with variance parameter σ^2. Then the maximum likelihood estimator of σ^2,*

$$\hat{\sigma}_n^{2^{ML}} := \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y}_n)^2, \tag{31.104}$$

is not unbiased, but asymptotically unbiased.

∘

Proof. With the unbiasedness of the sample variance we first obtain (cf. Theorem 31.3)

$$\mathbb{E}_{\mu,\sigma^2}\left(\hat{\sigma}_n^{2^{\mathrm{ML}}}\right) = \mathbb{E}_{\mu,\sigma^2}\left(\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y}_n)^2\right) = \frac{1}{n}\mathbb{E}_{\mu,\sigma^2}\left(\sum_{i=1}^{n}(y_i - \bar{y}_n)^2\right) = \frac{n-1}{n}\sigma^2. \tag{31.105}$$

Thus

$$\mathbb{E}_{\mu,\sigma^2}\left(\hat{\sigma}_n^{2^{\mathrm{ML}}}\right) \neq \sigma^2 \tag{31.106}$$

and $\hat{\sigma}_n^{2^{\mathrm{ML}}}$ is not an unbiased estimator of σ^2. However,

$$\frac{n-1}{n} \to 1 \text{ for } n \to \infty, \tag{31.107}$$

so that

$$\lim_{n\to\infty}\mathbb{E}_{\mu,\sigma^2}\left(\hat{\sigma}_n^{2^{\mathrm{ML}}}\right) = \lim_{n\to\infty}\frac{n-1}{n}\sigma^2 = \sigma^2\lim_{n\to\infty}\frac{n-1}{n} = \sigma^2 \tag{31.108}$$

and the maximum likelihood estimator of σ^2 is therefore asymptotically unbiased. □

Example 31.4 (Simulation example)**.** The following **R** code simulates the behavior of the maximum likelihood estimator for the variance parameter of a normal distribution model with true, but unknown, parameters $\mu = 1$ and $\sigma^2 = 2$ as a function of the sample size.

```
# model formulation
mu        = 1                                           # true, but unknown, expectation parameter
sigsqr    = 2                                           # true, but unknown, variance parameter
n         = seq(1,100, by = 2)                          # sample sizes
ns        = 1e3                                         # number of simulations per sample size
sigsqr_ml = matrix(rep(NaN, length(n)*ns),ncol = length(n)) # \hat{\sigma^2}^{ML} array

# simulation
for(i in seq_along(n)){                                 # sample size iterations
    for(s in 1:ns){                                     # sample realization iterations
        y               = rnorm(n[i], mu, sqrt(sigsqr)) # sample realization
        sigsqr_ml[s,i]  = ((n[i]-1)/n[i])*var(y)        # \hat{\sigma^2}^{ML}
    }
}
E_sigsqr_ml = colMeans(sigsqr_ml)                       # estimator expectation estimate
```

We visualize the expectation of the estimator estimated on the basis of the above simulations as a function of sample size in Figure 31.3. For small sample sizes, the estimator clearly underestimates the true, but unknown, variance parameter; for larger sample sizes, it does not.

△

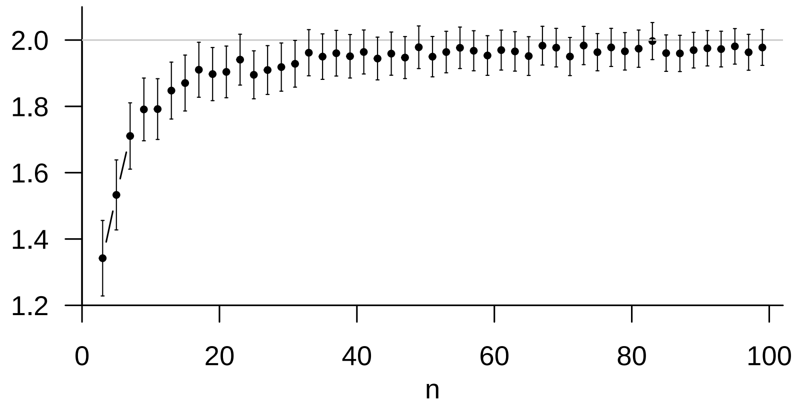


Figure 31.3 Simulation of the expectation of the maximum likelihood estimator for the variance parameter of a normal distribution model. For small sample sizes, the maximum likelihood estimator systematically underestimates the true, but unknown, variance parameter; for larger sample sizes, the estimator is approximately unbiased.

31.3.2 Consistency

The concept of consistency of an estimator generalizes the weak law of large numbers from sample means and expectations (cf. Theorem 27.1) to arbitrary estimators and true, but unknown, parameter values.

Definition 31.10 (Consistency)**.** Let $y := (y_1, ..., y_n)$ be the sample of a parametric product model and let $\hat{\tau}_n$ be an estimator of τ. Then a sequence of estimators $\hat{\tau}_1, \hat{\tau}_2, ...$ is called a *consistent sequence of estimators* if, for every arbitrarily small $\epsilon > 0$ and every $\theta \in \Theta$,

$$\lim_{n\to\infty} \mathbb{P}_\theta \left(|\hat{\tau}_n(y) - \tau(\theta)| \geq \epsilon\right) = 0. \tag{31.109}$$

If $\hat{\tau}_1, \hat{\tau}_2, ...$ is a consistent sequence of estimators, then $\hat{\tau}_n$ is called a *consistent estimator*.

•

The intuition behind Definition 31.10 corresponds to the law of large numbers. For sample sizes with $n \to \infty$, the probability that $\hat{\tau}_n(y)$ is arbitrarily close to $\tau(\theta)$, or equivalently that $\hat{\tau}_n(y)$ deviates far from $\tau(\theta)$, becomes small. However, these properties must hold for all possible true, but unknown, parameter values. As will be shown below, the sample mean is a consistent estimator for the expectation parameter of a normal distribution model.

Example 31.5. Analogously to Section 27.1, we demonstrate the meaning of consistency by simulation for $\mu = 1$ and $\sigma^2 = 2$ using the following **R** code.

```
# model formulation
mu      = 1                                  # true, but unknown, value of \mu
sigsqr  = 2                                  # true, but unknown, value of \sigma^2
n       = seq(1,1e3,by = 10)                 # sample size n
eps     = c(0.15, 0.10, 0.05)                # \epsilon values
ne      = length(eps)                        # number of \epsilon values
nn      = length(n)                          # number of sample sizes
ns      = 1000                               # number of simulations
E       = array(rep(NaN,nn*ne*ns),dim = c(nn,ne,ns))  # event indicator array

# simulation
for(e in seq_along(eps)){                    # \epsilon iterations
    for(i in seq_along(n)){                  # n iterations
        for(s in 1:ns){                      # simulation iterations

            # sample realizations
            y   = rnorm(n[i], mu, sqrt(sigsqr))
            if(abs(mean(y) - mu) >= eps[e]){         # |y_bar - \mu)| \ge \epsilon
                E[i,e,s] = 1
            } else {                                  # |y_bar - \mu)| < \epsilon
                E[i,e,s] = 0
            }
        }
    }
}

# estimate of \mathbb{P}(|\hat{\tau}_n(y)-\tau(\theta)| \ge \epsilon)
P_hat       = apply(E, c(1,2), mean)
```

We visualize the estimate of the probability $\mathbb{P}(|\hat{\tau}_n(y) - \tau(\theta)| \geq \epsilon)$ for this example in Figure 31.4 as a function of sample size and criterion value ϵ. For larger values of ϵ, small sample sizes suffice for a small probability of deviation of the estimator from the true, but unknown, parameter value; for smaller values of ϵ, larger sample sizes are necessary.

△

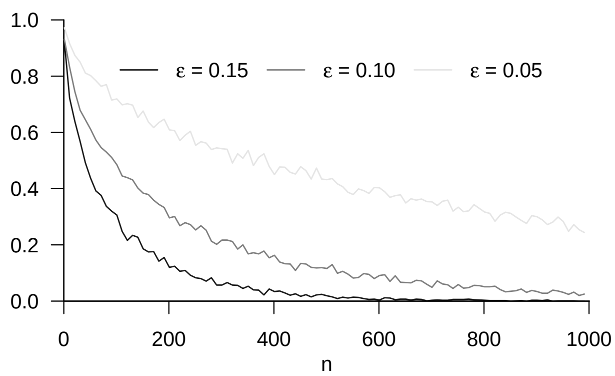


Figure 31.4 Simulation of the consistency of the sample mean as an estimator for the expectation parameter of the normal distribution model.

The criteria for consistency of estimators stated in Theorem 31.15 and Theorem 31.16 simplify the proof of consistency of an estimator with the help of the mean squared error (Definition 31.6).

Theorem 31.15 (Mean squared error criterion for consistency). *Let $y := (y_1, ..., y_n)$ be the sample of a parametric product model and let $\hat{\tau}_n$ be an estimator of τ. If*

$$\lim_{n\to\infty} MSE(\hat{\tau}_n) = 0, \tag{31.110}$$

then $\hat{\tau}_n$ is a consistent estimator.

∘

Proof. With Chebyshev's inequality,

$$\mathbb{P}_\theta\left(|\hat{\tau}_n(y) - \tau(\theta)| \geq \epsilon\right) \leq \frac{\mathbb{E}_\theta\left((\hat{\tau}_n(y) - \tau(\theta))^2\right)}{\epsilon^2} \tag{31.111}$$

Taking limits gives

$$\lim_{n\to\infty} \mathbb{P}_\theta\left(|\hat{\tau}_n(y) - \tau(\theta)| \geq \epsilon\right) \leq \frac{1}{\epsilon^2} \lim_{n\to\infty} \mathbb{E}_\theta\left((\hat{\tau}_n(y) - \tau(\theta))^2\right). \tag{31.112}$$

If therefore $\lim_{n\to\infty} \mathbb{E}_\theta\left((\hat{\tau}_n(y) - \tau(\theta))^2\right) = 0$, then, with $\mathbb{P}_\theta(|\hat{\tau}_n(y) - \tau(\theta)| \geq \epsilon) \geq 0$,

$$\lim_{n\to\infty} \mathbb{P}_\theta\left(|\hat{\tau}_n(y) - \tau(\theta)| \geq \epsilon\right) = 0. \tag{31.113}$$

Thus $\hat{\tau}_n$ is a consistent estimator. □

Theorem 31.16 (Bias-variance criterion for consistency)**.** *Let $y := (y_1, ..., y_n)$ be the sample of a parametric product model and let $\hat{\tau}_n$ be an estimator of τ. If*

$$\lim_{n\to\infty} B(\hat{\tau}_n) = 0 \text{ and } \lim_{n\to\infty} \mathbb{V}_\theta(\hat{\tau}_n) = 0 \tag{31.114}$$

hold, then $\hat{\tau}_n$ is a consistent estimator.

∘

Proof. If $n \to \infty$, then $\mathrm{B}(\hat{\tau}_n) \to 0$, and thus also $\mathrm{B}(\hat{\tau}_n)^2 \to 0$. If for $n \to \infty$ both $\mathrm{B}(\hat{\tau}_n)^2 \to 0$ and $\mathbb{V}_\theta(\hat{\tau}_n) \to 0$, then also $\lim_{n\to\infty} \mathrm{MSE}(\hat{\theta}_n) = 0$. Hence, by Theorem 31.15, $\hat{\tau}_n$ is consistent. □

As an application of Theorem 31.16, the following theorem proves consistency of the sample mean as an estimator of the expectation under normality.

Theorem 31.17 (Consistency of the expectation estimator under normality)**.** *Let y be the sample of a normal distribution model. Then $\bar{y}_n$ is a consistent estimator of $\mathbb{E}(y_1)$.*

∘

Proof. With the unbiasedness of the sample mean as an estimator of the expectation, we first have

$$\lim_{n\to\infty} \mathrm{B}(\bar{y}_n) = 0. \tag{31.115}$$

Furthermore, with the variance of the sample mean,

$$\lim_{n\to\infty} \mathbb{V}_\theta(\bar{y}_n) = \lim_{n\to\infty} \frac{1}{n}\mathbb{V}(y_1) = 0. \tag{31.116}$$

By the bias-variance criterion, the consistency of $\bar{y}_n$ as an estimator of $\mathbb{E}(y_1)$ follows. □

Note that Theorem 31.17 is of course already implied by the weak law of large numbers (cf. Theorem 27.1).

31.3.3 Asymptotic normality

In some cases, the frequentist distribution of an estimator approaches a normal distribution at large sample sizes. Such an estimator is called *asymptotically normally distributed.*

Definition 31.11 (Asymptotically normally distributed estimator)**.** Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model and let $\hat{\theta}_n$ be a parameter estimator for θ. Furthermore, let

$$\tilde{\theta} \sim N(\mu, \sigma^2) \tag{31.117}$$

be a normally distributed random variable with expectation parameter μ and variance parameter σ^2. If $\hat{\theta}_n$ converges in distribution to $\tilde{\theta}$, that is, if for the CDFs P_n and P of $\hat{\theta}_n$ and $\tilde{\theta}$, respectively,

$$\lim_{n\to\infty} P_n(\hat{\theta}_n) = P(\tilde{\theta}), \tag{31.118}$$

then $\hat{\theta}_n$ is called *asymptotically normally distributed* and we write

$$\hat{\theta}_n \overset{a}{\sim} N(\mu, \sigma^2). \tag{31.119}$$

•

As an example of an asymptotically normally distributed estimator, in Section 31.3.4 we consider the maximum likelihood estimator of the Bernoulli model parameter. It is remarkable that this estimator is asymptotically normally distributed, since the sample variables of the Bernoulli model take only the values zero and one.

31.3.4 Asymptotic efficiency

In some cases, in addition to asymptotic normality of an estimator, statements about the expectation parameter and variance parameter of this asymptotic normal distribution can also be justified by the form of the frequentist model. The concept of an *efficient estimator* formulates such a special case.

Definition 31.12. Let $y = (y_1, ..., y_n)$ be the sample of a parametric product model and let $\hat{\theta}_n$ be a parameter estimator for θ. Furthermore, let $J(\theta)$ be the expected Fisher information of the sample y. If

$$\hat{\theta}_n \overset{a}{\sim} N\left(\theta, J(\theta)^{-1}\right), \tag{31.120}$$

then $\hat{\theta}_n$ is called *asymptotically efficient.*

•

An asymptotically efficient estimator is evidently always also asymptotically normally distributed and unbiased. The variance of the asymptotic distribution of an estimator is also called the *asymptotic variance.* For an asymptotically efficient estimator, the asymptotic variance is identical to the Cramér-Rao bound and is therefore minimal in the set of unbiased asymptotically normally distributed estimators.

Example 31.6. As an example of an asymptotically efficient estimator, we consider the maximum likelihood estimator of the Bernoulli model parameter. The following **R** code simulates the frequentist distribution of this estimator and determines the PDF of its asymptotic distribution.

```
# model formulation
mu          = 0.4                                                     # true, but unknown, parameter value
n_all       = c(1e1,5e1,1e2)                                          # sample size
ns          = 1e4                                                     # number of sample realizations
mu_hat      = matrix(rep(NaN, length(n_all)*ns), nrow = length(n_all))  # maximum likelihood estimator array
mu_hat_r    = 1e3                                                     # maximum likelihood estimator-space resolution
mu_hat_y    = seq(0,1,len = mu_hat_r)                                 # maximum likelihood estimator space
mu_hat_p    = matrix(rep(NaN, length(n_all)*mu_hat_r), nrow = length(n_all)) # maximum likelihood PDF array

# sample size iterations
for(i in seq_along(n_all)){

    # simulation iterations
    for(s in 1:ns){
        y           = rbinom(n_all[i],1,mu)                           # sample realization
        mu_hat[i,s] = mean(y)                                         # maximum likelihood estimator realization
    }
    mu_hat_p[i,] = dnorm(mu_hat_y, mu, sqrt(mu*(1-mu)/n_all[i]))      # PDF of asymptotic distribution
}
```

△

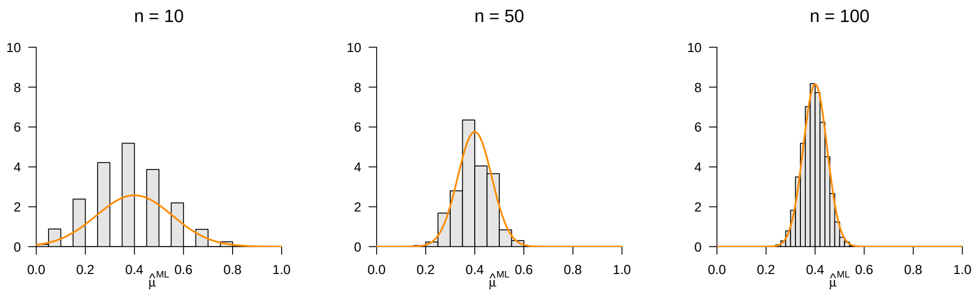


Figure 31.5 Simulation of the asymptotic efficiency of the maximum likelihood parameter estimator for the parameter of a Bernoulli model. With increasing sample size, the distribution of the simulated estimator values shown by a histogram becomes similar to the normal distribution formulated in Definition 31.12.

31.4 Properties of maximum likelihood estimators

The maximum likelihood principle for obtaining estimators for parameters of frequentist inference models is justified by the following theorem on the properties of maximum likelihood estimators.

Theorem 31.18 (Properties of maximum likelihood estimators)**.** *Let y be the sample of a parametric product model and let $\hat{\theta}_n^{ML}$ be a maximum likelihood estimator for θ. Then* $\hat{\theta}_n^{ML}$

(1) is not necessarily unbiased, but
(2) is asymptotically unbiased,
(3) is consistent,
(4) is asymptotically normally distributed, and
(5) is asymptotically efficient.

∘

For a proof of Theorem 31.18 we refer to Held & Sabanés Bové (2014), Section 3.4. Thus, if one uses the maximum likelihood principle to obtain an estimator, the estimator obtained is guaranteed to satisfy the estimator quality criteria summarized in Theorem 31.18.

31.5 Bibliographic remarks

The results presented in this chapter go back in a very substantial way to Fisher (1922). Aldrich (1997) provides historical context.

Study questions

1. State the definition of a parameter point estimator.
2. Explain the concept of a parameter point estimator.
3. State the definition of the likelihood function and the log-likelihood function.
4. State the definition of a maximum likelihood estimator.
5. Explain the procedure for obtaining maximum likelihood estimators.
6. State the theorem on the maximum likelihood estimator of the Bernoulli model parameter.
7. State the theorem on the maximum likelihood estimators of the normal distribution model parameters.
8. State the definition of unbiasedness of an estimator.
9. Explain the concept of unbiasedness of an estimator.
10. State the definition of the variance and standard error of an estimator.
11. State the definition of asymptotic unbiasedness of an estimator.
12. Explain the concept of asymptotic unbiasedness of an estimator.
13. State the definition of consistency of an estimator.
14. Explain the concept of consistency of an estimator.
15. State the first three statements of the theorem on the properties of maximum likelihood estimators.

Study question answers

1. See Definition 31.1.
2. A parameter point estimator gives, based on a sample, a guess for the corresponding true, but unknown, parameter.
3. See Definition 31.2.
4. See Definition 31.3.
5. To obtain a maximum likelihood estimator, one first formulates the log-likelihood function and then determines the zeros of its derivative as potential maximum points. In classical examples one usually uses analytic optimization for this; in applications to more complex models, numerical optimization is usually used.
6. See Theorem 31.1.
7. See Theorem 31.2.
8. See Definition 31.4.
9. An estimator is called unbiased if its expectation is identical to the true, but unknown, value it estimates.
10. See Definition 31.5.
11. See Definition 31.9.
12. An estimator is called asymptotically unbiased if, for sample sizes tending to infinity, its expectation is identical to the true, but unknown, value it estimates.
13. See Definition 31.10.
14. An estimator is called consistent if, for large sample sizes, the probability that the estimator value deviates from the true, but unknown, value becomes arbitrarily small.
15. See Theorem 31.18, statements (1), (2), and (3).

32 Confidence intervals

Confidence intervals are interval estimates of true, but unknown, parameters that are constructed so that they are correct in most cases. The gain in certainty about the accuracy of the estimate, compared with point estimation, comes at the cost of a loss in precision. In point estimation, the estimate of a true, but unknown, parameter value, such as $\theta = 2$, is very precise, for example $\hat{\theta} = 2.14$. However, against the background of the vanishing probability that a continuous random variable takes exactly *one* real value, this estimate is almost certainly wrong. Estimating a true, but unknown, parameter by an interval, such as $[1.94, 2.34]$, is coarser. Such an estimate, however, can be constructed so that it is correct with a large desired probability. To express this intuition formally, we first focus in this chapter on one-dimensional parameter spaces. Thus, we consider $\Theta \subseteq \mathbb{R}$ and hence indeed only confidence *intervals*, that is, subsets of $\mathbb{R}$. A generalization of the concepts introduced here to higher-dimensional parameter spaces in the sense of confidence *sets* is, however, possible without major difficulty.

32.1 Definition

Definition 32.1 (δ confidence interval)**.** Let y be the sample of a frequentist inference model with true, but unknown, parameter $\theta \in \Theta$, let $\delta \in]0,1[$, and let $G_u(y)$ and $G_o(y)$ be given. Then an interval of the form

$$\kappa(y) := [G_u(y), G_o(y)], \tag{32.1}$$

such that

$$\mathbb{P}_\theta\left(\kappa(y) \ni \theta\right) = \mathbb{P}_\theta\left(G_u(y) \le \theta \le G_o(y)\right) = \delta \text{ for all } \theta \in \Theta \tag{32.2}$$

holds is called a δ *confidence interval for* θ. δ is the coverage probability of $\kappa(y)$ for θ and is usually called the *confidence level*. The statistics $G_u(y)$ and $G_o(y)$ are called the *lower* and the *upper bound* of the confidence interval, respectively.

•

Note that, as in all frequentist inference models, the parameter θ is a true, but unknown, value here and therefore, in particular, fixed and not random. Because the upper and lower bounds of a confidence interval are random variables as functions of the random sample, however, the confidence interval defined by them is a random interval. The somewhat unusual notation $\kappa(y) \ni \theta$ simply means $\theta \in \kappa(y)$. Since $\kappa(y)$ is the random entity in the expression $\mathbb{P}_\theta\left(\kappa(y) \ni \theta\right)$, as just described, $\kappa(y)$ is conventionally written on the left. In this respect, think for example of an expression such as $\mathbb{P}(\xi = x)$. According to Definition 32.1, a δ confidence interval covers the true, but unknown, parameter θ with probability δ. A high coverage probability of $\delta := 0.95$ is often chosen. In this case, one speaks of a 95% *confidence interval*.

Intuitively, one may interpret δ confidence intervals in two ways. In the first case, one assumes repeated independent realizations of samples with an identical true, but unknown, parameter θ. Thus, if one repeatedly realizes data "under the same circumstances", a δ confidence interval covers this true, but unknown, parameter in $\delta \cdot 100\%$ of the realized cases in the long run. Alternatively, however, according to Definition 32.1, this frequentist probability of covering the true, but unknown, parameter also holds for any arbitrary true, but unknown, parameter value $\theta_i, i = 1, 2, ...$ Thus, even if one considers different true, but unknown, parameter values $\theta_1, \theta_2, ...$ and, in each case, records a sample realization that is independent of the other realizations, the corresponding δ confidence intervals cover these true, but unknown, parameters in $\delta \cdot 100\%$ of the cases in the long run. Intuitively, one therefore does not need to repeat "one study", that is, the investigation of one true, but unknown, parameter value, "infinitely often" under the same circumstances in order to benefit from the coverage probability of a confidence interval. Rather, it is sufficient to determine confidence intervals according to Definition 32.1 in "different studies", that is, investigations of different true, but unknown, parameters. In this case, too, their coverage probability for the true, but unknown, parameters is guaranteed.

To construct δ confidence intervals for given frequentist inference models by explicitly specifying the statistics $G_u(y)$ and $G_o(y)$, one proceeds as follows. First, one defines the inference model and thereby fixes the distribution of the sample y. In a second step, one defines a statistic, that is, a function of the sample, that serves as the basis for $G_u(y)$ and $G_o(y)$, and analyzes its distribution, which is based on the sampling distribution. Once the corresponding distribution has been found, it can be used to guarantee the coverage probability of the true, but unknown, parameter by an appropriately defined confidence interval. We trace this procedure in the development and constructive proofs of the following examples.

32.2 Examples of confidence intervals

Confidence interval for the expectation parameter of the normal distribution model

We consider the construction of a δ confidence interval for the expectation parameter of the normal distribution model. To this end, we first define the following confidence-interval statistic.

Definition 32.2 (T confidence-interval statistic)**.** Given the normal distribution model

$$y_1, ..., y_n \sim N\left(\mu, \sigma^2\right) \tag{32.3}$$

the statistic defined by the sample mean and the sample standard deviation

$$\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i \text{ and } S := \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}, \tag{32.4}$$

namely

$$T := \sqrt{n}\frac{\bar{y} - \mu}{S} \tag{32.5}$$

is called the T *confidence-interval statistic.*

•

For the distribution of the T confidence-interval statistic, the following theorem holds.

Theorem 32.1 (Distribution of the T confidence-interval statistic)**.** *The T confidence-interval statistic is a t-distributed random variable with parameter $n-1$; that is,*

$$T \sim t(n-1) \tag{32.6}$$

∘

Note that, according to Definition 32.2, the T confidence-interval statistic is a function of the sample, whereas its distribution according to Theorem 32.1 is independent of the true, but unknown, parameters of the sampling distribution. This is also called the *pivot property* of the T confidence-interval statistic. For the following developments, recall that we denote the PDF of a t-distributed random variable by t, the CDF of a t-distributed random variable by Ψ, and the inverse CDF of a t-distributed random variable by Ψ^{-1}. The following **R** code first simulates the distribution of the T confidence-interval statistic.

```
# model specification
mu       = 10                                        # true, but unknown, expectation parameter
sigsqr   = 4                                         # true, but unknown, variance parameter
n        = 12                                        # sample size
ns       = 1e4                                       # number of sample realizations
res      = 1e3                                       # outcome-space resolution

# analytic definitions and results
yx       = seq(3,17,len = res)                       # y_i space
ssqrx    = seq(0,20,len = res)                       # S^2 space
tx       = seq(-4,4,len = res)                       # T space
p_y_i    = dnorm(yx,mu,sqrt(sigsqr))                 # y_i PDF
p_y_bar  = dnorm(yx,mu,sqrt(sigsqr/n))               # y_bar PDF
p_sqr    = dchisq(ssqrx,n-1)                         # S^2 PDF
p_t      = dt(tx,n-1)                                # T PDF

# simulation
y_i      = rep(NaN,ns)                               # y_i array
y_bar    = rep(NaN,ns)                               # \bar{y} array
S        = rep(NaN,ns)                               # S array
TKS      = rep(NaN,ns)                               # T confidence-interval statistic array
for(s in 1:ns){                                      # simulation iterations
  y         = rnorm(n,mu,sqrt(sigsqr))               # sample realization
  y_i[s]    = y[1]                                   # sample realization y_i with i = 1
  y_bar[s]  = mean(y)                                # sample mean realization
  S[s]      = sd(y)                                  # sample standard-deviation realization
  TKS[s]    = sqrt(n)*((y_bar[s]-mu)/S[s])           # T confidence-interval statistic realization
}
```

In Figure 32.1, we visualize the distribution of the T confidence-interval statistic as a result of its underlying distributions of the sample variables, the sample mean, and the sample variance (cf. Theorem 29.13). Using the distribution of the T confidence-interval statistic, we can now prove the following theorem on the confidence interval for the expectation parameter of the normal distribution model.

Theorem 32.2 (Confidence interval for the expectation parameter of the normal distribution model)**.** *Given the normal distribution model*

$$y_1, ..., y_n \sim N(\mu, \sigma^2) \tag{32.7}$$

with true, but unknown, parameters μ and σ^2, let $\delta \in]0,1[$ and let

$$t_\delta := \Psi^{-1}\left(\frac{1+\delta}{2}; n-1\right). \tag{32.8}$$

where Ψ^{-1} is the inverse CDF of a t-distributed random variable. Then, for the interval

$$\kappa(y) := \left[\bar{y} - \frac{S}{\sqrt{n}}t_\delta, \bar{y} + \frac{S}{\sqrt{n}}t_\delta\right], \tag{32.9}$$

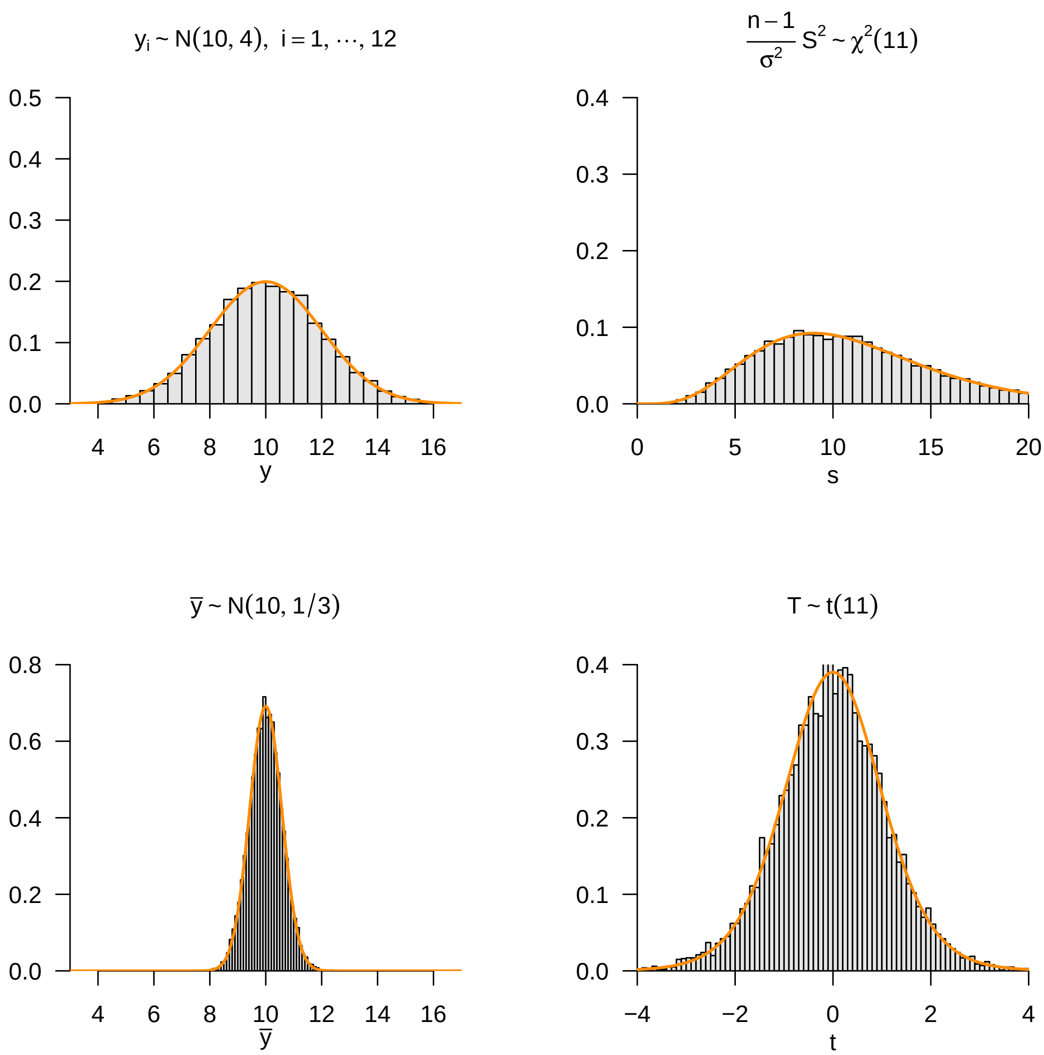


Figure 32.1 Simulation of the distribution of the T confidence-interval statistic and the underlying distributions of the sample variable, the sample mean, and the sample variance.

with the sample mean and the sample standard deviation

$$\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i \text{ and } S := \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}, \tag{32.10}$$

respectively, it holds that

$$\mathbb{P}_\mu(\kappa(y) \ni \mu) = \delta. \tag{32.11}$$

∘

Proof. For $\delta \in]0,1[$, first define

$$t_1 := \Psi^{-1}\left(\frac{1-\delta}{2}; n-1\right) \text{ and } t_2 := \Psi^{-1}\left(\frac{1+\delta}{2}; n-1\right) \tag{32.12}$$

Then

$$\frac{1+\delta}{2} - \frac{1-\delta}{2} = \delta \tag{32.13}$$

and, by the symmetry of the PDF of the t distribution, also

$$t_1 = -t_2. \tag{32.14}$$

By definition, using Definition 32.2 and Theorem 32.1, it follows that

$$\mathbb{P}_\mu\left(-t_\delta \le T \le t_\delta\right) = \delta. \tag{32.15}$$

Consequently, we obtain directly

$$\begin{aligned}
\delta &= \mathbb{P}_\mu\left(-t_\delta \le T \le t_\delta\right) \\
&= \mathbb{P}_\mu\left(-t_\delta \le \frac{\sqrt{n}}{S}(\bar{y} - \mu) \le t_\delta\right) \\
&= \mathbb{P}_\mu\left(-\frac{S}{\sqrt{n}}t_\delta \le \bar{y} - \mu \le \frac{S}{\sqrt{n}}t_\delta\right) \\
&= \mathbb{P}_\mu\left(-\bar{y} - \frac{S}{\sqrt{n}}t_\delta \le -\mu \le -\bar{y} + \frac{S}{\sqrt{n}}t_\delta\right) \\
&= \mathbb{P}_\mu\left(\bar{y} + \frac{S}{\sqrt{n}}t_\delta \ge \mu \ge \bar{y} - \frac{S}{\sqrt{n}}t_\delta\right) \\
&= \mathbb{P}_\mu\left(\bar{y} - \frac{S}{\sqrt{n}}t_\delta \le \mu \le \bar{y} + \frac{S}{\sqrt{n}}t_\delta\right) \\
&= \mathbb{P}_\mu\left(\left[\bar{y} - \frac{S}{\sqrt{n}}t_\delta, \bar{y} + \frac{S}{\sqrt{n}}t_\delta\right] \ni \mu\right). \\
&= \mathbb{P}_\mu\left(\kappa(y) \ni \mu\right).
\end{aligned} \tag{32.16}$$

□

The decisive step for guaranteeing the coverage probability δ of the true, but unknown, expectation parameter by the confidence interval defined in Theorem 32.2 is the definition of

$$t_\delta := \Psi^{-1}\left(\frac{1+\delta}{2}; n-1\right). \tag{32.17}$$

As traced in the proof of Theorem 32.2, the coverage probability of the confidence interval for the true, but unknown, expectation parameter is equivalent to the fact that, when this t_δ is chosen, the T confidence-interval statistic has probability δ of taking a value in the interval $[-t_\delta, t_\delta]$. We visualize the choice of t_δ for the case $\delta := 0.95$ and $n := 5$ in Figure 32.2. In this case,

$$-t_\delta = \Psi^{-1}(0.025; 4) = -2.57 \text{ and } t_\delta = \Psi^{-1}(0.975; 4) = 2.57. \tag{32.18}$$

Figure 32.2 A shows this choice from the perspective of the PDF of the T confidence-interval statistic. The probability mass enclosed by $-t_\delta$ and t_δ is δ by construction, so T takes a value between $-t_\delta$ and t_δ with probability δ. Figure 32.2 B shows the corresponding perspective of the CDF of the T confidence-interval statistic. Based on the specification of $\frac{1-\delta}{2}$ and $\frac{1+\delta}{2}$, the corresponding values for $-t_\delta$ and t_δ are determined using the inverse CDF Ψ^{-1}. Note that the centrality of the probability mass given here is not implicit in Definition 32.1, but results from the properties of the distribution of the T confidence-interval statistic, in particular its symmetry around 0.

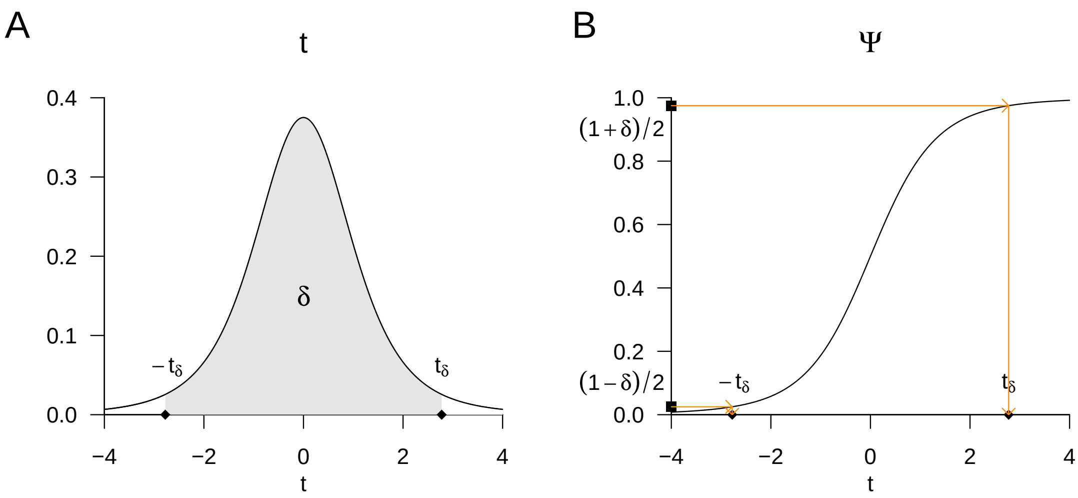


Figure 32.2 Guaranteeing the coverage probability of the confidence interval for the expectation parameter of the normal distribution model for $\delta := 0.95$ and $n := 5$ from the perspective of the PDF (A) and the CDF (B) of the distribution of the T confidence-interval statistic.

Finally, we want to demonstrate the coverage probability of the confidence interval given by Theorem 32.2 using a simulation. We consider only the first interpretation of a confidence interval with a constant true, but unknown, parameter. In this sense, the following **R** code determines the corresponding confidence interval for each sample realization.

```
# model specification
set.seed(1)                                         # random-number generator
mu      = 2                                         # true, but unknown, expectation parameter
sigsqr  = 1                                         # true, but unknown, variance parameter
sigma   = sqrt(sigsqr)                              # true, but unknown, standard-deviation parameter
n       = 12                                        # sample size
delta   = 0.95                                      # confidence condition
t_delta = qt((1+delta)/2,n-1)                       # \Psi^-1((\delta + 1)/2, n-1)

# sample realizations
ns      = 1e2                                       # number of sample realizations
y_bar   = rep(NaN,ns)                               # sample-mean array
S       = rep(NaN,ns)                               # standard-deviation array
kappa   = matrix(rep(NaN,2*ns), ncol = 2)           # confidence-interval array
for(i in 1:ns){
   y           = rnorm(n,mu,sigma)                  # sample realization
   y_bar[i]    = mean(y)                            # sample mean
   S[i]        = sd(y)                              # sample standard deviation
   kappa[i,1] = y_bar[i] - (S[i]/sqrt(n))*t_delta  # lower confidence-interval bound
   kappa[i,2] = y_bar[i] + (S[i]/sqrt(n))*t_delta  # upper confidence-interval bound
}
```

We visualize the results of this simulation in Figure 32.3.

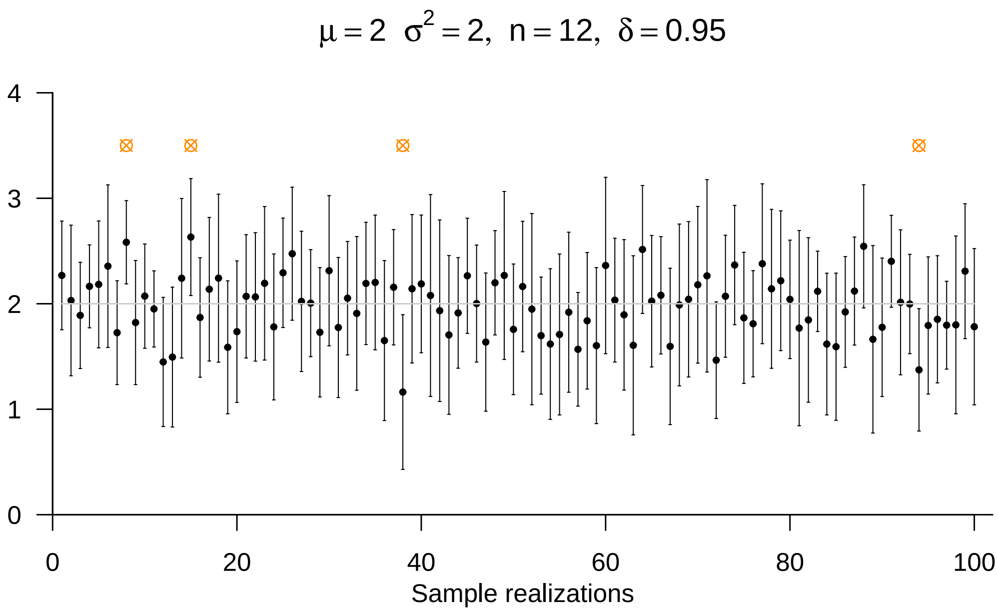


Figure 32.3 Simulation of the coverage probability of the confidence interval for the expectation parameter of the normal distribution model with constant true, but unknown, expectation parameter $\mu := 2$ for $\sigma^2 := 2, n := 12$ and a desired coverage probability of $\delta := 0.95$. The figure shows the confidence interval and the corresponding expectation-parameter estimator for each sample realization. In the present simulation, the confidence intervals cover the identical true, but unknown, expectation parameter $\mu := 2$, shown as a gray line, in 96 out of 100 cases. The sample realizations for which this is not the case are marked by an orange circle.

Confidence interval for the variance parameter of the normal distribution model

We consider the construction of a δ confidence interval for the variance parameter of the normal distribution model. To this end, we first define the following confidence-interval statistic.

Definition 32.3 (U confidence-interval statistic)**.** Given the normal distribution model

$$y_1, ..., y_n \sim N(\mu, \sigma^2) \tag{32.19}$$

the statistic defined by the sample variance

$$\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 \tag{32.20}$$

namely

$$U := \frac{n-1}{\sigma^2}S^2 \tag{32.21}$$

is called the U *confidence-interval statistic.*

•

For the distribution of the U confidence-interval statistic, the following theorem holds.

Theorem 32.3 (Distribution of the U confidence-interval statistic)**.** *The U confidence-interval statistic is a χ^2-distributed random variable with parameter $n-1$; that is,*

$$U \sim \chi^2(n-1) \tag{32.22}$$

∘

For a proof of Theorem 32.3, we refer to Casella & Berger (2002). Like the T confidence-interval statistic, the U confidence-interval statistic has the pivot property because it is a function of the sample, but its distribution according to Theorem 32.3 does not depend on the true, but unknown, distribution parameters of the sample. For the following developments, recall that we denote the PDF of a χ^2-distributed random variable by χ^2, the CDF of a χ^2-distributed random variable by Ξ, and the inverse CDF of a χ^2-distributed random variable by Ξ^{-1}. The following **R** code first simulates the distribution of the U confidence-interval statistic.

```
# model specification
mu      = 10                              # true expectation parameter
sigsqr  = 4                               # true known variance parameter
n       = 12                              # sample size
ns      = 1e4                             # number of sample realizations
res     = 1e3                             # outcome-space resolution

# analytic definitions and results
yx      = seq(3,17,len = res)             # y_i space
ux      = seq(0,30,len = res)             # U space
p_y_i   = dnorm(yx,mu,sqrt(sigsqr))       # y_i PDF
p_y_bar = dnorm(yx,mu,sqrt(sigsqr/n))     # y_bar PDF
p_u     = dchisq(ux,n-1)                  # U PDF

# simulation
y_i     = rep(NaN,ns)                     # y_i array
y_bar   = rep(NaN,ns)                     # \bar{y} array
S_sqr   = rep(NaN,ns)                     # S^2 array
UKS     = rep(NaN,ns)                     # U confidence-interval statistic array
```

```
for(s in 1:ns){                                   # simulation iterations
  y          = rnorm(n,mu,sqrt(sigsqr))           # sample realization
  y_i[s]     = y[1]                               # sample realization y_i with i = 1
  y_bar[s]   = mean(y)                            # sample-mean realization
  S_sqr[s]   = var(y)                             # sample-variance realization
  UKS[s]     = ((n-1)/sigsqr)*S_sqr[s]            # U confidence-interval statistic realization
}
```

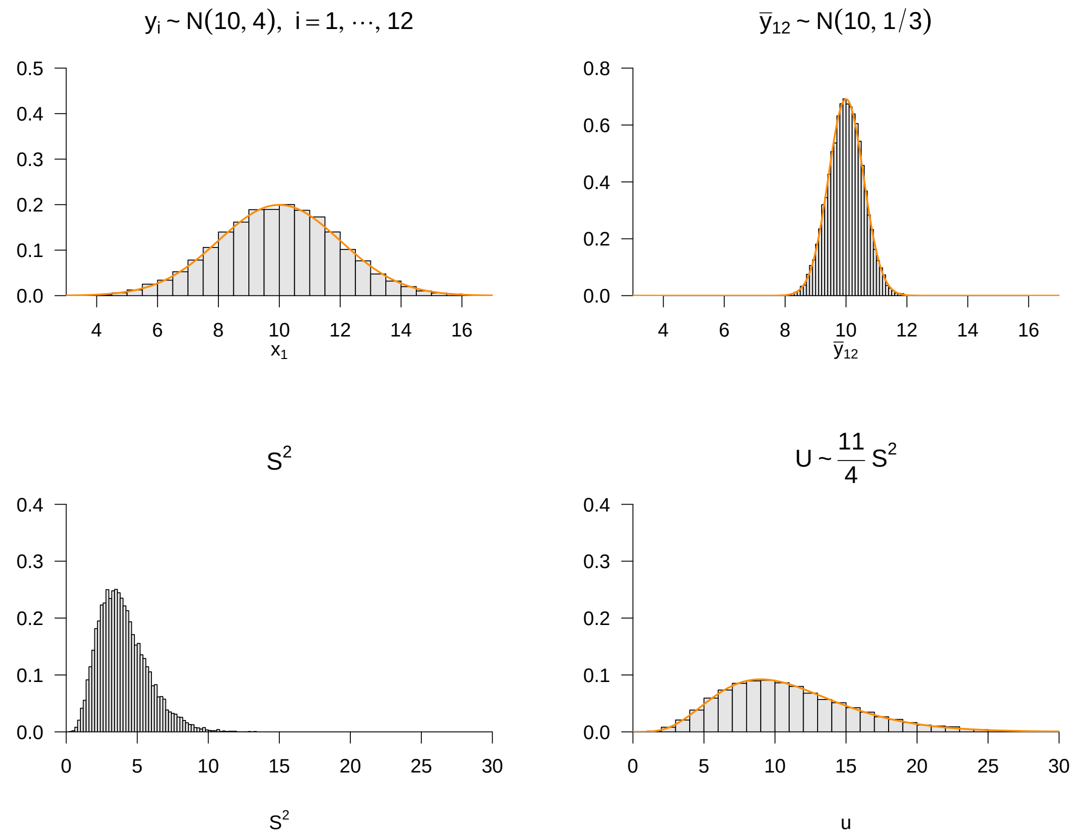


Figure 32.4 Simulation of the distribution of the U confidence-interval statistic and the underlying distributions of the sample variable, the sample mean, and the sample variance.

Using the distribution of the U confidence-interval statistic, we can now prove the following theorem on the confidence interval for the variance parameter of the normal distribution model.

Theorem 32.4 (Confidence interval for the variance parameter of the normal distribution model)**.** *Given the normal distribution model*

$$y_1, ..., y_n \sim N(\mu, \sigma^2) \tag{32.23}$$

with true, but unknown, parameters μ *and* σ^2*, let* $\delta \in]0, 1[$ *and let*

$$u_\delta := \Xi^{-1}\left(\frac{1-\delta}{2}; n-1\right) \text{ and } u'_\delta := \Xi^{-1}\left(\frac{1+\delta}{2}; n-1\right) \tag{32.24}$$

where Ξ^{-1} *is the inverse CDF of a* χ^2*-distributed random variable. Then, for the interval*

$$\kappa(y) := \left[\frac{(n-1)S^2}{u'_\delta}, \frac{(n-1)S^2}{u_\delta}\right]. \tag{32.25}$$

with the sample variance

$$S^2 := \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2, \tag{32.26}$$

it holds that

$$\mathbb{P}_{\sigma^2}(\kappa(y) \ni \sigma^2) = \delta. \tag{32.27}$$

∘

Proof. By definition, using Definition 32.3 and Theorem 32.3, we have

$$\mathbb{P}_{\sigma^2}\left(u_\delta \le U \le u'_\delta\right) = \delta. \tag{32.28}$$

Consequently, it follows directly that

$$\begin{aligned}
\delta &= \mathbb{P}_{\sigma^2}\left(u_\delta \le U \le u'_\delta\right) \\
&= \mathbb{P}_{\sigma^2}\left(u_\delta \le \frac{n-1}{\sigma^2}S^2 \le u'_\delta\right) \\
&= \mathbb{P}_{\sigma^2}\left(u_\delta^{-1} \ge \frac{\sigma^2}{(n-1)S^2} \ge {u'_\delta}^{-1}\right) \\
&= \mathbb{P}_{\sigma^2}\left(\frac{(n-1)S^2}{u_\delta} \ge \sigma^2 \ge \frac{(n-1)S^2}{u'_\delta}\right) \\
&= \mathbb{P}_{\sigma^2}\left(\frac{(n-1)S^2}{u'_\delta} \le \sigma^2 \le \frac{(n-1)S^2}{u_\delta}\right) \\
&= \mathbb{P}_{\sigma^2}\left(\left[\frac{(n-1)S^2}{u'_\delta}, \frac{(n-1)S^2}{u_\delta}\right] \ni \sigma^2\right).
\end{aligned} \tag{32.29}$$

□

As in the case of Theorem 32.2, the decisive step for guaranteeing the coverage probability δ of the true, but unknown, variance parameter by the confidence interval defined in Theorem 32.4 is the definition of

$$u_\delta := \Xi^{-1}\left(\frac{1-\delta}{2}; n-1\right) \text{ and } u'_\delta := \Xi^{-1}\left(\frac{1+\delta}{2}; n-1\right). \tag{32.30}$$

As traced in the proof of Theorem 32.4, the coverage probability of the confidence interval for the true, but unknown, variance parameter is equivalent to the fact that, when precisely these values of u_δ and u'_δ are chosen, the U confidence-interval statistic has probability δ of taking a value in the interval $[u_\delta, u'_\delta]$. We visualize the choice of u_δ and u'_δ for the case $\delta := 0.95$ and $n := 10$ in Figure 32.5. In this case,

$$u_\delta := \Xi^{-1}(0.025; 9) = 2.70 \text{ and } u'_\delta := \Xi^{-1}(0.975; 9) = 19.0. \tag{32.31}$$

Figure 32.5 A shows this choice from the perspective of the PDF of the U confidence-interval statistic. The probability mass enclosed by u_δ and u'_δ is δ by construction, so U takes a value between u_δ and u'_δ with probability δ. Figure 32.5 B shows the corresponding perspective of the CDF of the U confidence-interval statistic. Based on the specification of $\frac{1-\delta}{2}$ and $\frac{1+\delta}{2}$, the corresponding values for u_δ and u'_δ are determined using the inverse CDF Ξ^{-1}. Note that in this case the probability mass is located rather arbitrarily with respect to the mode of the distribution of the U confidence-interval statistic. Accordingly, there are more advanced procedures for locating the coverage probability of a confidence-interval statistic in such a way that, for example, it occupies a maximal interval in its outcome space or satisfies a symmetry property around the expectation. We do not pursue these approaches here.

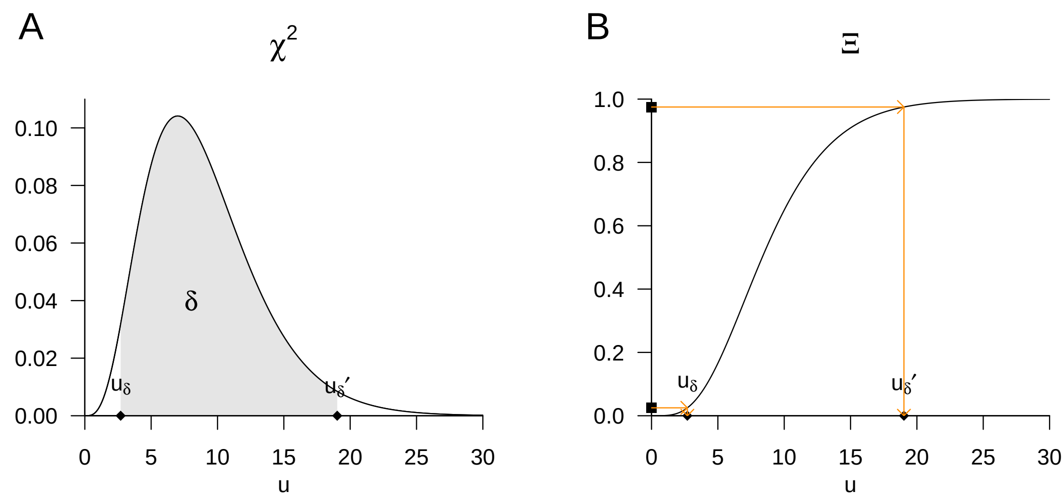


Figure 32.5 Guaranteeing the coverage probability of the confidence interval for the variance parameter of the normal distribution model for $\delta := 0.95$ and $n := 10$ from the perspective of the PDF (A) and the CDF (B) of the distribution of the U confidence-interval statistic.

Finally, we want to demonstrate the coverage probability of the confidence interval given by Theorem 32.4 using a simulation. To this end, we first consider the first interpretation of a confidence interval given in Section 32.1, with the same true, but unknown, parameter throughout. In this sense, the following **R** code determines the corresponding confidence interval for each sample realization.

```
# model specification
set.seed(1)                                   # random-number generator
mu          = 2                               # true, but unknown, expectation parameter
sigsqr      = 2                               # true, but unknown, variance parameter
n           = 12                              # sample size
delta       = 0.95                            # confidence condition
u_delta     = qchisq((1-delta)/2, n - 1)      # \Xi^2((1-\delta)/2; n - 1)
u_delta_p   = qchisq((1+delta)/2, n - 1)      # \Xi^2((1+\delta)/2; n - 1)

# sample realizations
ns          = 1e2                             # number of simulations
y_bar       = rep(NaN,ns)                     # sample-mean array
S2          = rep(NaN,ns)                     # sample-variance array
kappa       = matrix(rep(NaN,2*ns), ncol = 2) # confidence-interval array
for(i in 1:ns){                               # simulation iterations
   y            = rnorm(n,mu,sqrt(sigsqr))    # sample realization
   S2[i]        = var(y)                      # sample variance
   kappa[i,1]   = (n-1)*S2[i]/u_delta_p       # lower confidence-interval bound
   kappa[i,2]   = (n-1)*S2[i]/u_delta         # upper confidence-interval bound
}
```

Example 32.1 (Application example)**.** To conclude this section, we consider the evaluation of confidence intervals for the expectation and variance parameters under normality in the context of Example 30.5. To this end, we first evaluate the unbiased point estimators of μ and σ^2, that is, the sample mean and the sample variance of the dataset, using the following **R** code.

```
D           = read.csv("./_data/bdi-ii-dataset.csv")        # read dataset
y           = D$dBDI                                        # select data
mu_hat      = round(mean(y),2)                              # sample mean
sigsqr_hat  = round(var(y),2)                               # sample variance
cat("mu_hat     :", mu_hat,"\nsigsqr_hat :", sigsqr_hat)    # output
```

```
mu_hat     : 3.17
sigsqr_hat : 13.79
```

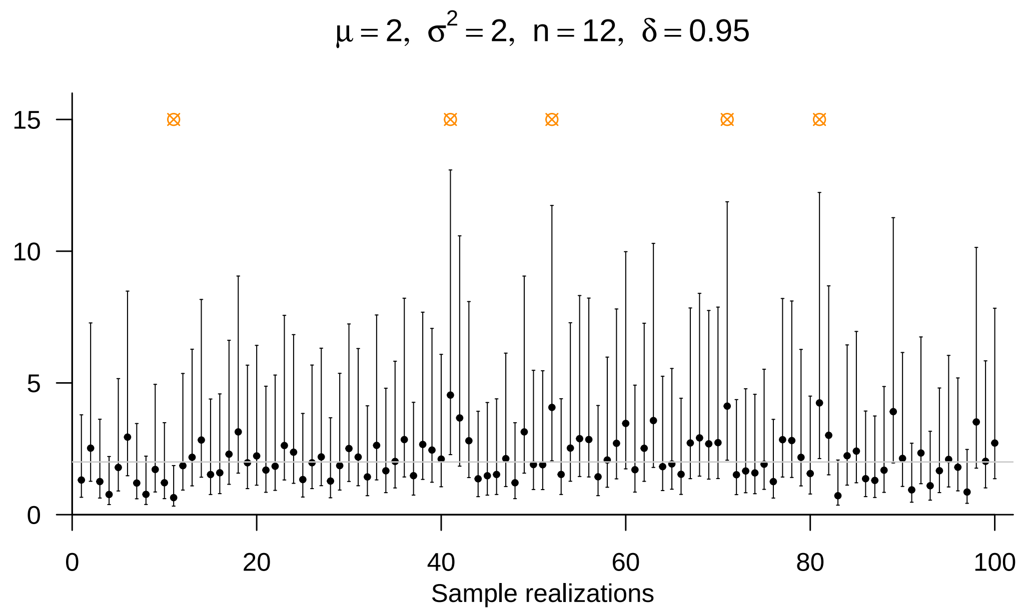


Figure 32.6 Simulation of the coverage probability of the confidence interval for the variance parameter of the normal distribution model with constant true, but unknown, variance parameter $\sigma^2 := 2$ for $\mu := 2, n := 12$ and a desired coverage probability of $\delta := 0.95$. The figure shows the confidence interval and the corresponding variance-parameter estimator for each sample realization. In the present simulation, the confidence intervals cover the identical true, but unknown, variance parameter $\sigma^2 := 2$, shown as a gray line, in 95 out of 100 cases. The sample realizations for which this is not the case are marked by an orange circle. Note that the confidence intervals are not arranged symmetrically around the variance-parameter estimator.

Based on these estimators and the available $n = 12$ data points,

$$\hat{\mu} = 3.17 \text{ and } \hat{\sigma}^2 = 13.8 \tag{32.32}$$

are therefore reasonable guesses for μ and σ^2. In addition to these point estimates, which are very precise but match the true, but unknown, parameters with probability 0, we also determine the 95% confidence-interval estimates for μ and σ^2. The following **R** code determines the 95% confidence interval for the expectation parameter.

```
# confidence interval for the expectation parameter
delta    = 0.95                                    # confidence level
n        = length(y)                               # number of data points
t_delta  = qt((1+delta)/2,n-1)                     # \psi^-1((\delta+1)/2,n-1)
y_bar    = mean(y)                                 # sample mean
s        = sd(y)                                   # sample standard deviation
mu_hat   = y_bar                                   # expectation-parameter estimator
kappa_u  = y_bar - (s/sqrt(n))*t_delta             # lower confidence-interval bound
kappa_o  = y_bar + (s/sqrt(n))*t_delta             # upper confidence-interval bound
cat("kappa_u:", kappa_u, "\nkappa_o:", kappa_o)    # output
```

```
kappa_u: 0.8074098
kappa_o: 5.525923
```

The 0.95 confidence interval for the expectation parameter is therefore

$$\kappa(y) = [0.80, 5.52]. \tag{32.33}$$

In the long-run average, a confidence interval calculated in this way covers the true, but unknown, expectation parameter in 95 out of 100 cases. In this sense, the true, but unknown, treatment effect is therefore very likely to lie in an interval between 0.80 and 5.52 BDI-II score pre-post differences.

The following **R** code determines the 95% confidence interval for the variance parameter.

```
# confidence interval for the variance parameter
delta       =  0.95                                # confidence level
n           = length(y)                            # number of data points
u_delta_u   = qchisq((1-delta)/2, n - 1)           # \Xi^2((1-\delta)/2; n - 1)
u_delta_o   = qchisq((1+delta)/2, n - 1)           # \Xi^2((1+\delta)/2; n - 1)
s2          = var(y)                               # sample variance
sigsqr_hat  = s2                                   # variance-parameter estimator
kappa_u     = (n-1)*s2/u_delta_o                   # lower confidence-interval bound
kappa_o        = (n-1)*s2/u_delta_u                # upper confidence-interval bound
cat("kappa_u:", kappa_u, "\nkappa_o:", kappa_o)    # output
```

```
kappa_u: 6.919084
kappa_o: 39.74756
```

The 0.95 confidence interval for the variance parameter is therefore

$$\kappa(y) = [6.91, 39.74]. \tag{32.34}$$

In the long-run average, a confidence interval calculated in this way covers the true, but unknown, variance parameter in 95 out of 100 cases. In this sense, the true, but unknown, treatment-effect dispersion is therefore very likely to lie in an interval between 6.91 and 39.74 squared BDI-II score pre-post differences.

$\triangle$

32.3 Bibliographic remarks

The results presented in this chapter go back in a very substantial way to Neyman (1937).

Study questions

1. State the definition of the concept of a δ confidence interval.
2. Explain the two interpretations of a δ confidence interval.
3. Explain the typical steps for constructing a δ confidence interval.
4. State the theorem on the confidence interval for the expectation parameter of the normal distribution model.
5. State the theorem on the confidence interval for the variance parameter of the normal distribution model.

Study question answers

1. See Definition 32.1.
2. See the discussion following Definition 32.1.
3. See the last paragraph of the discussion following Definition 32.1.
4. See Theorem 32.2.
5. See Theorem 32.4.

33 Hypothesis tests

The basic logic of frequentist hypothesis tests can be sketched using the normal distribution model. One assumes that an observed dataset is a realization of a sample $y_1, ..., y_n \sim N(\mu, \sigma^2)$ and computes, based on the dataset, a *test statistic*, for example the sample mean standardized by the sample standard deviation and sample size, $\sqrt{n}\frac{\bar{y}}{s}$.

One then asks how likely it would be to observe the observed or a more extreme value of the test statistic under the assumption of a *null model*. Intuitively, a null model is a probability distribution model in which no "interesting effect" is present, for example $\mu = 0$ in the normal distribution model. The concept of probability is, of course, to be understood frequentistically, that is, as an idealized relative frequency one would obtain if many sample realizations of the null model were generated. Depending on the underlying frequentist inference model and the test statistic under consideration, it may be possible to determine this probability exactly.

If the probability of observing the observed or a more extreme value of the test statistic under the null model is large, one intuitively concludes that "it is quite plausible that the null model generated the data". In scientific jargon, this is called a "statistically non-significant result". If the probability of observing the observed or a more extreme value of the test statistic under the null model is small, one intuitively concludes that "it is not very plausible that the null model generated the data". In this case, scientific jargon speaks of a "statistically significant result".

As always in frequentist statistics, after carrying out such a procedure one does not know whether, in the present case, the null model or another model actually generated the data. One only knows how often one would be right or wrong on average with this procedure if all assumptions held and the procedure were repeated very often.

In the following sections, we formalize these intuitive ideas. It is important to distinguish between hypotheses in the sense of frequentist inference and the general concept of a scientific hypothesis. Stating a scientific hypothesis by no means implies the necessity of a frequentist hypothesis test. It merely means that, in quantitative work, uncertainty should be quantified and communicated in a meaningful way using observed data relevant to the hypothesis. Frequentist hypothesis tests are only one, albeit very popular, way of doing this. It should already be noted here that "null-hypothesis significance testing", as we present it below, is scientifically controversial (see, for example, Amrhein & Greenland (2018) and McShane et al. (2019)).

33.1 Test hypotheses and tests

In the context of a frequentist hypothesis test, the concept of a frequentist inference model (cf. Definition 30.1) is first extended by so-called *test hypotheses* to form a *test scenario*. We use the following definition.

Definition 33.1 (Test hypotheses and test scenario)**.** Given a frequentist inference model with sample y, outcome space $\mathcal{Y}$, and parameter space Θ, let $\{\Theta_0, \Theta_1\}$ be a partition of the parameter space such that

$$\Theta = \Theta_0 \cup \Theta_1 \text{ and } \Theta_0 \cap \Theta_1 = \emptyset. \tag{33.1}$$

Then a *test hypothesis* is a statement about the true, but unknown, parameter θ with respect to the subsets Θ_0 and Θ_1 of the parameter space. Specifically,

- $\theta \in \Theta_0$ is called the *null hypothesis* and
- $\theta \in \Theta_1$ is called the *alternative hypothesis*.

For simplicity, Θ_0 and Θ_1 themselves are also referred to as the null hypothesis and alternative hypothesis, respectively. The unit consisting of a frequentist inference model and test hypotheses is called a *test scenario*.

•

Depending on the structure of Θ_0 and Θ_1, one first distinguishes between *simple* and *composite* test hypotheses.

Definition 33.2 (Simple and composite test hypotheses)**.** For the test hypotheses Θ_i with $i = 0, 1$:

- If Θ_i contains only a single element, then Θ_i is called *simple*.
- If Θ_i contains more than one element, then Θ_i is called *composite*.

•

Because, by assumption, the true, but unknown, parameter θ determines the distribution $\mathbb{P}_\theta$ of the sample, a simple test hypothesis fixes the sampling distribution to exactly one distribution. A composite test hypothesis, by contrast, corresponds to a set of possible sampling distributions. An example of a simple test hypothesis in a test scenario with parameter space $\Theta := \mathbb{R}$ is

$$\Theta_0 := \{0\}. \tag{33.2}$$

The corresponding composite alternative hypothesis is then

$$\Theta_1 = \mathbb{R} \setminus \{0\}. \tag{33.3}$$

The null hypothesis, that is, the statement "$\theta \in \Theta_0$", corresponds to the statement "$\theta = 0$", because Θ_0 contains only the single element 0.

If the parameter space of a test scenario is one-dimensional, one further distinguishes between *one-sided* and *two-sided* test hypotheses.

Definition 33.3 (One-sided and two-sided test hypotheses)**.** Given a test scenario with one-dimensional parameter space $\Theta := \mathbb{R}$ and $\theta_0 \in \Theta$, composite null hypotheses of the form $\Theta_0 :=]-\infty, \theta_0]$ or $\Theta_0 := [\theta_0, \infty[$ are called *one-sided null hypotheses* and are also written as $H_0 : \theta \leq \theta_0$ and $H_0 : \theta \geq \theta_0$, respectively. The corresponding alternative hypotheses have the form $\Theta_1 :=]\theta_0, \infty[$ and $\Theta_1 :=]-\infty, \theta_0[$, also written as $H_1 : \theta > \theta_0$ and $H_1 : \theta < \theta_0$, respectively. For a simple null hypothesis of the form $\Theta_0 := \{\theta_0\}$, also written as $H_0 : \theta = \theta_0$, the alternative hypothesis $\Theta_1 := \Theta \setminus \{\theta_0\}$, also written as $H_1 : \theta \neq \theta_0$, is called a *two-sided alternative hypothesis*.

•

Against the background of a test scenario, we now define the concept of a *test*.

Definition 33.4 (Test)**.** Given a test scenario, a *test* is a map ϕ from the sample outcome space $\mathcal{Y}$ to the set $\{0,1\}$, that is,

$$\phi : \mathcal{Y} \to \{0,1\}, y \mapsto \phi(y), \tag{33.4}$$

where

- $\phi(y) = 0$ represents not rejecting the null hypothesis and
- $\phi(y) = 1$ represents rejecting the null hypothesis.

•

The concept of a test is not trivial because tests, like estimators and confidence intervals, are themselves random variables as functions of the sample variables. Strictly speaking, tests are therefore defined on random-vector spaces. For simplicity, Definition 33.4 considers a concrete realization $y \in \mathcal{Y}$ of the sample y, which is mapped by ϕ to the set $\{0,1\}$. Against this background, the function value $\phi(y)$ of ϕ is a realization of the random variable $\phi(y)$.

In applications, one is often interested in tests that have a particular structure. We formalize this structure under the term *standard tests*.

Definition 33.5 (Standard test)**.** Given a test scenario, a *standard test* ϕ is defined as the composition of a *test statistic*

$$\gamma : \mathcal{Y} \to \Gamma \tag{33.5}$$

and a *decision rule*

$$\delta : \Gamma \to \{0,1\} \tag{33.6}$$

and can therefore be written as

$$\phi := \delta \circ \gamma : \mathcal{Y} \to \{0,1\}. \tag{33.7}$$

•

As above, note that in Definition 33.5 the test statistic is a function of the sample variables, that is, of random variables, which we have defined here as a function of the values of these random variables in $\mathcal{Y}$. Likewise, the decision rule is a function of the random test statistic, which we have also written as a function of the values of this random variable with outcome space Γ. In a test scenario, test statistic and decision rule are therefore random variables. Accordingly, if y is a realization of the sample y, $\gamma(y) \in \Gamma$ is a realization of $\gamma(y)$ and $(\delta \circ \gamma)(y)$ is a realization of $(\delta \circ \gamma)(y)$.

The distribution-theoretic properties of a test result from the distribution-theoretic properties of the corresponding frequentist inference model and hence, in particular, from the distribution of the sample variables. Important bridges between these two levels are the concepts of the *critical region* and the *rejection region* of a test.

Definition 33.6 (Critical region of a test)**.** Given a test scenario and a test ϕ, the set

$$K := \{y \in \mathcal{Y} | \phi(y) = 1\} \subset \mathcal{Y}, \tag{33.8}$$

that is, the subset of the outcome space $\mathcal{Y}$ of the sample y for which the test takes the value 1, is called the *critical region* of the test.

•

Against the background of Definition 33.6, the random events $\{y \in K\}$ and $\{\phi(y) = 1\}$, that is, the event that the sample takes a value in the critical region of the test and the event that the test takes the value 1, are equivalent and thus have the same probability. Thus, asking for the probability that a test takes the value 1, that is, that the null hypothesis is rejected, is exactly the same as asking for the probability that the sample takes a value in the critical region of the test. Since the distribution of the sample is assumed to be known, this probability of rejecting the null hypothesis can be determined from it. If one has a standard test, the same consideration also applies directly to the test statistic placed between sample and test. This leads to the following definition.

Definition 33.7 (Rejection region of a standard test)**.** Given a test scenario and a standard test ϕ with test statistic γ, the set

$$A := \{\gamma(y) \in \Gamma | \phi(y) = 1\} \subset \Gamma, \tag{33.9}$$

that is, the subset of the outcome space Γ of the test statistic for which the test takes the value 1, is called the *rejection region* of the test.

•

As noted for the critical region, the events $\{\phi(y) = 1\}$ and $\{\gamma(y) \in A\}$ are equivalent and therefore have the same probability. Overall, for a standard test, Definition 33.6 and Definition 33.7 imply

$$\{y \in K\} \Leftrightarrow \{\gamma(y) \in A\} \Leftrightarrow \{\phi(y) = 1\} \tag{33.10}$$

and

$$\mathbb{P}_\theta\left(\{y \in K\}\right) = \mathbb{P}_\theta\left(\{\gamma(y) \in A\}\right) = \mathbb{P}_\theta\left(\{\phi(y) = 1\}\right). \tag{33.11}$$

The subscript θ indicates that the corresponding distributions are determined by the parameter of the sampling distribution.

In applications, the general form of the decision rule of a standard test given in Definition 33.5 is usually based on whether an observed test statistic with outcome space $\Gamma := \mathbb{R}$ exceeds or falls below a particular so-called *critical value* $k \in \mathbb{R}$. This leads to the concepts of *one-sided* and *two-sided critical-value based tests.*

Definition 33.8 (Critical-value based tests)**.** A *critical-value based test* is a standard test whose decision rule δ depends on a critical value k of a test statistic with outcome space $\mathbb{R}$. Specifically,

- a *one-sided critical-value based test* is of the form

$$\phi : \mathcal{Y} \to \{0, 1\}, y \mapsto \phi(y) := 1_{\{\gamma(y) \ge k\}} = \begin{cases} 1 & \gamma(y) \ge k \\ 0 & \gamma(y) < k \end{cases}, \tag{33.12}$$

- a *two-sided critical-value based test* is of the form

$$\phi : \mathcal{Y} \to \{0,1\}, y \mapsto \phi(y) := 1_{\{|\gamma(y)| \geq k\}} = \begin{cases} 1 & |\gamma(y)| \geq k \\ 0 & |\gamma(y)| < k \end{cases}. \tag{33.13}$$

•

With the definition of critical-value based tests, the practical implementation of a hypothesis test is now outlined. As always in frequentist inference, one first assumes a frequentist inference model for the observed data, that is, one assumes that the data are a realization of a sample. Based on this realization, one computes a test statistic and finally compares it with a critical value to either not reject or reject the null hypothesis. In the next section, we ask how, against the background of null and alternative hypotheses, the critical value of a critical-value based test can be determined so that good test decisions are made in the sense of frequentist probability.

33.2 Test quality criteria and test construction

The fact that, in a test scenario, the true, but unknown, parameter may lie in the null-hypothesis or alternative-hypothesis region, while the null hypothesis can either be rejected or not rejected based on the value of the test, implies that a test decision can be correct or incorrect. The following definition first clarifies the terminology.

Definition 33.9 (Correct test decisions and test errors)**.** Given a test scenario and a test, there are two forms of *correct test decision*: not rejecting the null hypothesis, $\phi(y) = 0$, when the null hypothesis $\theta \in \Theta_0$ holds, and rejecting the null hypothesis, $\phi(y) = 1$, when the alternative hypothesis $\theta \in \Theta_1$ holds. Likewise, there are two types of *test error*. Rejecting the null hypothesis, $\phi(y) = 1$, when the null hypothesis $\theta \in \Theta_0$ holds is called a *type I error*. Not rejecting the null hypothesis, $\phi(y) = 0$, when the alternative hypothesis $\theta \in \Theta_1$ holds is called a *type II error*.

•

	Null hypothesis true	Alternative hypothesis true
Do not reject H0	Correct decision	Type II error
Reject H0	Type I error	Correct decision

Figure 33.1 Correct test decisions and type I and type II errors.

Figure 33.1 gives an overview of the possible correct test decisions and test errors when carrying out a test. Naturally, one would prefer to make a correct test decision. The decisive tool for constructing tests that are as good as possible against the frequentist background of the test scenario is the so-called *power function*.

Definition 33.10 (Power function)**.** Given a test scenario and a test ϕ, the *power function* of ϕ is defined as

$$q_\phi : \Theta \to [0,1], \theta \mapsto q_\phi(\theta) := \mathbb{P}_\theta(\phi(y) = 1). \tag{33.14}$$

For $\theta \in \Theta_1$, q_ϕ is also called the *sensitivity function* or *power function*.

•

Note that $\mathbb{P}_\theta$ in Definition 33.10 denotes the distribution of the random variable $\phi(y)$ under the assumption that the distribution of y is determined by θ. For each $\theta \in \Theta$, q_ϕ therefore gives the probability that the null hypothesis is rejected by the test ϕ. Using the concepts of the critical region (cf. Definition 33.6) and the rejection region (cf. Definition 33.7), these probabilities satisfy

$$\mathbb{P}_\theta(\phi(y) = 1) = \mathbb{P}_\theta(\gamma(y) \in A) = \mathbb{P}_\theta(y \in K). \tag{33.15}$$

The power function is specific to a given test. If the test changes, for example because a different critical value is chosen in a critical-value based test, the above probabilities and therefore the power function change.

Using the power function, test construction proceeds according to the following considerations. Ideally, one would have a test ϕ with

$$q_\phi(\theta) = \mathbb{P}_\theta(\phi(y) = 1) = 0 \text{ for } \theta \in \Theta_0 \text{ and } q_\phi(\theta) = \mathbb{P}_\theta(\phi(y) = 1) = 1 \text{ for } \theta \in \Theta_1. \tag{33.16}$$

The test decision of such a test would be correct with probability 1, because such a test rejects the null hypothesis with probability 0 when it holds and rejects it with probability 1 when it does not hold. More generally, small values of q_ϕ for $\theta \in \Theta_0$, that is, small probabilities of rejecting the null hypothesis when it holds, and large values of q_ϕ for $\theta \in \Theta_1$, that is, large probabilities of rejecting the null hypothesis when it does not hold, are favorable for minimizing test errors. However, dependencies generally exist between the values of the power function for $\theta \in \Theta_0$ and $\theta \in \Theta_1$, as the following examples illustrate.

Example (A) Let ϕ_a be a test defined by

$$\phi_a : \mathcal{Y} \to \{0,1\}, y \mapsto \phi_a(y) := 0. \tag{33.17}$$

Thus, ϕ_a is a test that never rejects the null hypothesis, regardless of the observed data. For ϕ_a,

$$q_{\phi_a}(\theta) = \mathbb{P}_\theta(\phi(y) = 1) = 0 \text{ for } \theta \in \Theta_0. \tag{33.18}$$

However, for ϕ_a it also automatically holds that

$$q_{\phi_a}(\theta) = \mathbb{P}_\theta(\phi(y) = 1) = 0 \text{ for } \theta \in \Theta_1. \tag{33.19}$$

Thus, ϕ_a has minimal sensitivity for detecting the fact that the alternative hypothesis holds.

Example (B) Conversely, let ϕ_b be a test defined by

$$\phi_b : \mathcal{Y} \to \{0,1\}, y \mapsto \phi_b(y) := 1. \tag{33.20}$$

Thus, ϕ_b is a test that always rejects the null hypothesis, regardless of the observed data. For ϕ_b,

$$q_{\phi_b}(\theta) = \mathbb{P}_\theta(\phi(y) = 1) = 1 \text{ for } \theta \in \Theta_1. \tag{33.21}$$

Thus, ϕ_b is maximally sensitive to the alternative hypothesis being true. However, for ϕ_b it also automatically holds that

$$q_{\phi_b}(\theta) = \mathbb{P}_\theta(\phi(y) = 1) = 1 \text{ for } \theta \in \Theta_0, \tag{33.22}$$

and ϕ_b therefore always rejects the null hypothesis when it holds and, in this sense, generates many false-positive results.

Against the background of these examples, the goal of test construction must be to find an appropriate balance between small values of the power function when the null hypothesis holds and large values when the alternative hypothesis holds. The most popular way to achieve this is to choose, in a first step, a small value $\alpha_0 \in [0,1]$ and ensure that

$$q_\phi(\theta) \le \alpha_0 \text{ for all } \theta \in \Theta_0, \tag{33.23}$$

that is, that the probability of rejecting the null hypothesis when it holds, and hence the probability of a type I error, is at most α_0. Conventional values for such an α_0 are $\alpha_0 := 0.001$ and $\alpha_0 := 0.05$. Among all tests, and, when sample sizes are optimized, among all frequentist inference models, that satisfy (33.23), one then seeks, in a second step, a test for which $q_\phi(\theta)$ is as large as possible for $\theta \in \Theta_1$. This two-step approach is not the only possible one; one could, for example, minimize a linear combination of type I and type II errors simultaneously. However, the sketched two-step approach is the most popular in applications, and we therefore restrict ourselves to it in what follows. Inequality (33.23) motivates the definitions of a *level-α_0 test*, the *significance level* α_0, and the *test size* α.

Definition 33.11 (Level-α_0 test, significance level α_0, and test size α)**.** Given a test scenario, a test ϕ, its power function q_ϕ, and $\alpha_0 \in [0,1]$, ϕ is called a *level-α_0 test* if

$$q_\phi(\theta) \le \alpha_0 \text{ for all } \theta \in \Theta_0. \tag{33.24}$$

If ϕ is a level-α_0 test, the value α_0 is also called the *significance level* of the test. Furthermore, the number

$$\alpha := \max_{\theta \in \Theta_0} q_\phi(\theta) \tag{33.25}$$

is called the *test size* of ϕ.

•

According to Definition 33.11, the test size α is the maximal probability of a type I error, and a test is a level-α_0 test if and only if this maximal probability is less than or equal to the significance level α_0. In applications, it is important to keep the subtle differences between the probability of a type I error, the test size, and the significance level of a test clear. Against the background of the difference between simple and composite null hypotheses (cf. Definition 33.2), one must first distinguish the concepts of type I error probability and test size. For a simple null hypothesis Θ_0, the test size is always equal to the probability of a type I error, because

$$\alpha := \max_{\theta\in\Theta_0} q_\phi(\theta) = \max_{\theta\in\{\theta_0\}} q_\phi(\theta) = q_\phi(\theta_0) = \mathbb{P}_{\theta_0}(\phi(y) = 1). \tag{33.26}$$

For a composite null hypothesis Θ_0, there are different probabilities of a type I error depending on the value of $\theta \in \Theta_0$. The largest of these probabilities is the test size

$$\alpha := \max_{\theta\in\Theta_0} q_\phi(\theta) = \max_{\theta\in\Theta_0} \mathbb{P}_\theta(\phi(y) = 1). \tag{33.27}$$

Likewise, the terms significance level α_0 and test size α should be clearly separated. A significance level is a freely chosen upper bound on the maximal probability of a type I error. The actual maximal probability of a type I error may be identical to it, as in most examples discussed in Section 33.3, but it need not be, for example in multiple testing scenarios with non-independent sample variables. Accordingly, a test is called *exact* if its test size is identical to its significance level, that is, if

$$\alpha = \alpha_0. \tag{33.28}$$

A test whose test size is smaller than its significance level, that is,

$$\alpha < \alpha_0, \tag{33.29}$$

is called *conservative*. Finally, a test whose test size is larger than its significance level,

$$\alpha > \alpha_0 \tag{33.30}$$

and which therefore cannot be a level-α_0 test, is called *liberal*.

p-value

A defining characteristic of a test is its binary range. The result of a test is either $\phi(y) = 0$, the null hypothesis is not rejected, or $\phi(y) = 1$, the null hypothesis is rejected. As the final result of a data analysis, the information contained in a dataset is thus maximally compressed. In particular, reporting only the test result suppresses interesting information about the signal-to-noise ratio of the dataset under consideration. For example, it is possible that the null hypothesis is rejected in the context of a critical-value based test because the test statistic exceeded the critical value by only a few decimal places. Conversely, it is equally possible that the null hypothesis is rejected because the test statistic took a value that was a multiple of the critical value. In both cases, the test result $\phi(y) = 1$ would be identical. In addition to pure test-size control and the reporting of the binary test result, it has therefore become customary for critical-value based tests to consider, based on the observed value of the test statistic, all values of the significance level α_0 for which a level-α_0 test would have returned $\phi(y) = 1$, that is, for which the null hypothesis would have been rejected. This consideration leads to the following general definition of the so-called p-value, where p stands for *probability*.

Definition 33.12 (p-value)**.** Let ϕ be a test. The *p-value* is the smallest significance level α_0 at which the null hypothesis is rejected based on a given value of the test statistic.

•

Especially in simple application examples, such as the one-sample t-test scenario considered in Section 33.3.1, p-values then reflect the answer to the intuitive question of how likely it would be, in the frequentist sense, to observe the observed or a more extreme value of the test statistic under the assumption of a null model. Reporting p-values is very popular in many areas of basic science, but it is also controversial (cf. Wasserstein et al. (2019)). P-values should not be overinterpreted. Based on what has been said, there is no reason to draw the following fallacies, but we mention them as a precaution: p-values do not quantify the probability that the null hypothesis is true, one cannot conclude from $p < 0.05$ that the alternative hypothesis holds, and one cannot conclude from $p \geq 0.05$ that the null hypothesis holds.

Like the value of a test statistic and of a test, p-values quantify the signal-to-noise ratio observed in a given dataset. Nothing less, but also nothing more.

Notes on the choice of null and alternative hypothesis

We conclude this section with some remarks on carrying out hypothesis tests. Against the background of the theory sketched above, the first question in a concrete application is how null and alternative hypotheses should be assigned to the objects of scientific interest, that is, to the scientific hypotheses.

For example, if one wants to carry out a test to decide, in the sense of frequentist inference, whether a particular psychotherapy procedure was effective in a clinical study, the question arises whether the absence of a treatment effect should be formulated as the null or as the alternative hypothesis.

The two-step procedure described above for test construction, in which the test size is first fixed by choosing a significance level and only in a second step the probability of rejecting the null hypothesis when the alternative hypothesis holds is made as large as possible, implies a clear asymmetry in the treatment of null and alternative hypotheses. In this procedure, type I errors are treated as more serious than type II errors.

This suggests a possible strategy for specifying null and alternative hypotheses: the null hypothesis is the scientific hypothesis whose associated test decision is the one for which one especially wants to avoid an error, or whose error probability one primarily wants to control. In science, it is common practice to regard the false confirmation of one's favored theory, for example the erroneous confirmation that a newly developed psychotherapy procedure is more effective than another, as a more serious error than its false rejection.

Accordingly, the false confirmation of one's own theory should be a type I error, whereas its false rejection should be classified as a type II error. For the false confirmation of one's own theory to indeed be a type I error, that is, the rejection of the null hypothesis even though it holds, one's own theory must be formulated as the alternative hypothesis. The erroneous rejection of the alternative hypothesis is then a type II error.

Intuitively, this yields the following assignment:

$$
\begin{aligned}
\text{Non-favored scientific hypothesis} &\rightarrow \text{Null hypothesis} \\
\text{Favored scientific hypothesis} &\rightarrow \text{Alternative hypothesis}
\end{aligned}
$$

Hypothesis tests in decision contexts and basic science

Another question is whether one should carry out a hypothesis test at all to evaluate scientific hypotheses. Superficially, hypothesis tests first provide simple binary statements of the form *"Given the evidence, the hypothesis is to be rejected or accepted"*. Such statements can be useful in concrete decision contexts if a decision actually has to be made. However, frequentist hypothesis tests, as we have seen, are formulated without an explicit decision-utility function, so potential decision costs are not explicitly included in decision making. For this purpose, there are several accessible theories that allow good long-run decisions under uncertainty, see for example Pratt et al. (1995), Puterman (1994), or Kochenderfer et al. (2022).

If one turns away from practically relevant decision contexts to basic science, whose essence is precisely not to establish final truths but to quantify and communicate the extent of uncertainty about the current state of theory, the binarity of the hypothesis-test decision appears, at best, unnecessary and, at worst, grossly misleading. Questions in basic science should therefore, in principle, not be formulated as decision problems.

Despite the widespread view that Bayesian approaches such as positive predictive values or Bayes factors offer advantages here, this is not the case as long as the uncertainty associated with a particular model preference is not communicated clearly. Nevertheless, frequentist hypothesis testing remains very popular even in the basic-science community, sometimes merely under the guise of demands for basic studies with "higher power". To retain access to the psychological and scientific literature, it is therefore still unavoidable to engage with hypothesis testing, even though it is not particularly meaningful from a basic-science perspective.

33.3 Test examples

33.3.1 One-sample t-test

The application scenario of a one-sample t-test is characterized by considering n univariate data points of a sample, or group, of randomized experimental units, which are assumed to be realizations of n independent and identically normally distributed random variables. For the identical univariate normal distributions $N(\mu, \sigma^2)$ of these random variables, both the expectation parameter μ and the variance parameter σ^2 are assumed to be unknown. Finally, it is assumed that there is interest in an inferential comparison of the unknown expectation parameter μ with a specified value μ_0 in the sense of a hypothesis test.

There are, however, at least four scenarios that may be of interest. A first case is the scenario of a simple null hypothesis and a simple alternative hypothesis,

$$H_0 : \mu = \mu_0 \text{ and } H_1 : \mu = \mu_1. \tag{33.31}$$

This case is well understood theoretically and forms the basis of the so-called *Neyman-Pearson lemma* (Neyman & Pearson (1933)). Its practical relevance is rather limited, however, because the alternative hypothesis assumes an exact specification of the expectation parameter. A second case is the scenario of a simple null hypothesis and a composite alternative hypothesis,

$$H_0 : \mu = \mu_0 \text{ and } H_1 : \mu \neq \mu_0. \tag{33.32}$$

In this case, one also speaks of an *undirected hypothesis* and usually uses a two-sided test. Intuitively, this corresponds to the undirected question of inferential evidence for a difference. This is the case we consider in detail below. Finally, there are at least two scenarios with composite null and alternative hypotheses, for example of the form

$$H_0 : \mu \leq \mu_0 \text{ and } H_1 : \mu > \mu_0 \text{ or } H_0 : \mu \geq \mu_0 \text{ and } H_1 : \mu < \mu_0. \tag{33.33}$$

In this case, one also speaks of *directed* hypotheses and usually uses one-sided tests. We do not consider these cases below.

Frequentist inference model

Definition 33.13 (Frequentist inference model of the one-sample t-test)**.** The frequentist inference model of the one-sample t-test is the normal distribution model

$$y_1, ..., y_n \sim N(\mu, \sigma^2) \text{ with } (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_{>0}. \tag{33.34}$$

•

Recall that, from a generative perspective, the normal distribution model corresponds to

$$y_i = \mu + \varepsilon_i \text{ with } \varepsilon_i \sim N(0, \sigma^2) \text{ for } i = 1, ..., n \tag{33.35}$$

(cf. Example 30.5). As seen in Example 30.5, the assumption of independent and identically normally distributed random variables as the basis for modeling the observation of n data points is equivalent to the assumption that each random variable y_i modeling a data point is the sum of a fixed, true, but unknown, value μ that is constant across random variables and a random-variable- or data-point-specific deviation term ε_i. As seen, μ models the actual effect of interest assumed in the scientific application context. In contrast, ε_i models the aspect of data variability that cannot be explained by μ but, in the sense of the central limit theorem, results from the summation of infinitely many disturbance influences and therefore remains as uncertainty about the explanation of the data variability by μ.

Test hypotheses

We consider the case of the one-sample t-test with a simple null hypothesis and a composite alternative hypothesis.

Definition 33.14 (Simple null hypothesis and composite alternative hypothesis of the one-sample t-test)**.** Given the frequentist inference model of the one-sample t-test

$$y_1, ..., y_n \sim N(\mu, \sigma^2) \text{ with } (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_{>0} \tag{33.36}$$

and let $\Theta := \mathbb{R}$ be the parameter subspace of the parameter of interest μ. Then, for the *null-hypothesis parameter value* $\mu_0 \in \mathbb{R}$, the *simple null hypothesis* and the *composite alternative hypothesis* of the one-sample t-test are

$$\Theta_0 := \{\mu_0\} \Leftrightarrow H_0 : \mu = \mu_0 \text{ and } \Theta_1 := \mathbb{R} \setminus \{\mu_0\} \Leftrightarrow H_1 : \mu \neq \mu_0. \tag{33.37}$$

•

The simple null hypothesis and the composite alternative hypothesis are parameterized by the value $\mu_0 \in \mathbb{R}$. Different hypothesis scenarios therefore arise depending on the choice of μ_0. If, for example, $\mu_0 := 0$ is chosen, the null hypothesis $\Theta_0 := \{0\}$ states that the true, but unknown, parameter μ is equal to 0, and the alternative hypothesis $\Theta_1 := \mathbb{R} \setminus \{0\}$ states that the true, but unknown, parameter μ is not equal to 0. If, by contrast, $\mu_0 := 2$ is chosen, the null hypothesis $\Theta_0 := \{2\}$ states that the true, but unknown, parameter μ is equal to 2, and the alternative hypothesis $\Theta_1 := \mathbb{R} \setminus \{2\}$ states that the true, but unknown, parameter μ is not equal to 2. In the application context, μ_0 is therefore a freely chosen, and hence known, parameter of the one-sample t-test, whereas μ is, and remains, true but unknown.

Definition of the test statistic

With the *one-sample t-test statistic*, we now define a test statistic that can serve as the basis of a critical-value based test and whose absolute value can indicate a deviation from the null hypothesis.

Definition 33.15 (One-sample t-test statistic)**.** Given the test scenario of a one-sample t-test with sample $y_1, ..., y_n$, sample mean $\bar{y}$, sample standard deviation S, and null-hypothesis parameter μ_0, the *one-sample t-test statistic* is defined as

$$T := \sqrt{n}\frac{\bar{y} - \mu_0}{S}. \tag{33.38}$$

•

The one-sample t-test statistic is evidently very similar to the T confidence-interval statistic (cf. Definition 32.2). Note, however, that in the case of the one-sample t-test statistic the null-hypothesis parameter μ_0 need not be identical to the true, but unknown, parameter value μ appearing in the T confidence-interval statistic.

Because the one-sample t-test statistic is central to the one-sample t-test, it is useful to be aware of its intuitive mechanics. In the numerator of the fraction, the difference between the sample mean $\bar{y}$ and the parameter μ_0 assumed under the null hypothesis appears first. As already seen, the sample mean is an unbiased estimator of the expectation parameter μ of the sample variables. The difference $\bar{y} - \mu_0$ thus represents an estimate of the deviation of the true, but unknown, expectation parameter from the null-hypothesis parameter and provides evidence, in absolute value, for a deviation from the null hypothesis. Roughly speaking, the numerator $\bar{y} - \mu_0$ is therefore a measure of the "signal" contained in the sample, understood as deviation from the null hypothesis or as "systematic variability".

The denominator of the one-sample t-test statistic allows this signal to be expressed in units of the sample standard deviation S. For example, if $\bar{y} - \mu_0 = 2$ and $S = 1$, the deviation of the sample mean from the null-hypothesis parameter is two standard deviations. If, by contrast, $S = 2$, the deviation corresponds to one standard deviation. Moreover, the denominator S is a measure of the observed data variability and an estimator of the standard deviation σ of the error terms in the generative form of the one-sample t-test model. Roughly speaking, the denominator of the one-sample t-test statistic therefore represents the "noise" contained in the data or their "unsystematic variability". Overall, the fraction $\frac{\bar{y}-\mu_0}{S}$ can therefore be interpreted as an estimate of the signal-to-noise ratio of the data.

Finally, in the one-sample t-test statistic, this ratio is weighted by the square root of the sample size, $\sqrt{n}$. Intuitively, this weighting reflects the fact that a given signal-to-noise ratio can be assigned more validity if it is based on a larger number of observations than if it is based on only a few data points.

Overall, the one-sample t-test statistic thus provides a scalar summary of the evidence contained in the data against the null hypothesis, taking into account both data variability and sample size.

Distribution of the test statistic

For the distribution of the one-sample t-test statistic, the following theorem holds.

Theorem 33.1 (Distribution of the one-sample t-test statistic). *Given the test scenario of a one-sample t-test with sample $y_1, ..., y_n$, sample mean $\bar{y}$, sample standard deviation S, null-hypothesis parameter μ_0, and one-sample t-test statistic*

$$T := \sqrt{n}\frac{\bar{y}-\mu_0}{S}. \tag{33.39}$$

Then T is a noncentral t random variable with noncentrality parameter

$$d = \sqrt{n}\frac{\mu-\mu_0}{\sigma} \tag{33.40}$$

and degrees-of-freedom parameter $n-1$, that is, $T \sim t(d, n-1)$.

∘

If the null hypothesis holds, the null-hypothesis parameter μ_0 is identical to the true, but unknown, expectation parameter μ, and the noncentrality parameter of the distribution of the one-sample t-test statistic takes the value $d = 0$. If the null hypothesis of the one-sample t-test scenario holds, the one-sample t-test statistic is therefore a t-distributed random variable with degrees-of-freedom parameter $n-1$. In Figure 33.2, we visualize the distribution of the one-sample t-test statistic by example for a one-sample t-test scenario with $n = 12$, true, but unknown, parameters $\mu = 3$ and $\sigma^2 = 2$, and null-hypothesis parameter $\mu_0 = 0$. The parameters of this distribution are

$$d = \sqrt{n}\frac{\mu-\mu_0}{\sigma} = \sqrt{12}\frac{3-0}{\sqrt{2}} \approx 7.34 \tag{33.41}$$

and

$$n-1 = 11. \tag{33.42}$$

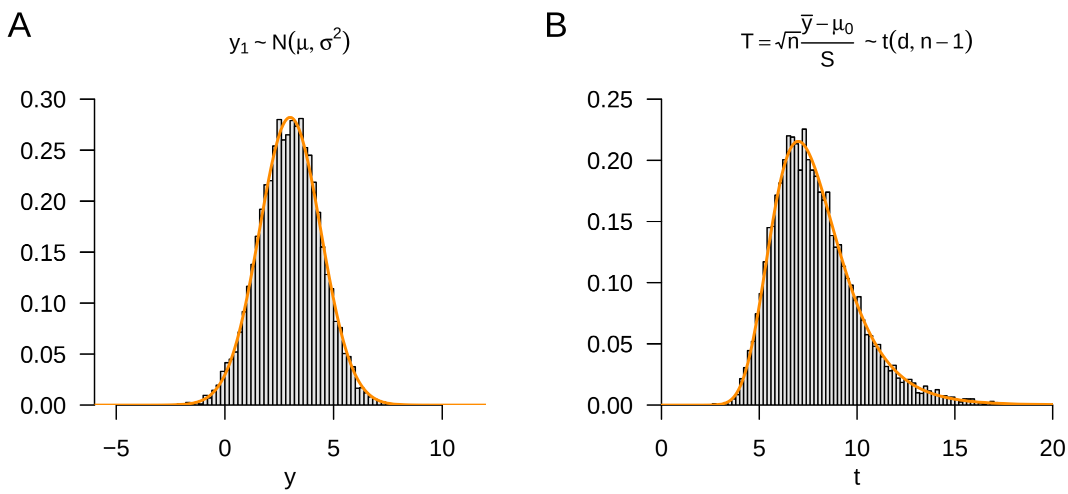


Figure 33.2 Distribution of the one-sample t-test statistic for $n = 12, \mu = 3, \sigma^2 = 2$ and $\mu_0 = 0$. (A) Distribution of the sample variable. (B) Distribution of the one-sample t-test statistic.

Test definition

We can now define the two-sided one-sample t-test with a simple null hypothesis and composite alternative hypothesis and analyze its power function.

Definition 33.16 (Two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis)**.** Given the frequentist inference model of the one-sample t-test with simple null hypothesis and composite alternative hypothesis, and let T denote the one-sample t-test statistic. Then the *two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis* is defined as the two-sided critical-value based test

$$\phi : \mathcal{Y} \to \{0, 1\}, y \mapsto \phi(y) := 1_{\{|T| \geq k\}} = \begin{cases} 1 & |T| > k \\ 0 & |T| \leq k \end{cases}. \tag{33.43}$$

•

The two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis therefore takes the value 0 if the absolute value of the one-sample t-test statistic is smaller than the critical value, and it takes the value 1 if the absolute value of the one-sample t-test statistic is equal to or larger than the critical value.

Power function

For controlling the test size by choosing a critical value and for determining the power function of this test, the following theorem is central.

Theorem 33.2 (Power function of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis)**.** *Let ϕ be the two-sided one-sample t-test* *with simple null hypothesis and composite alternative hypothesis. Then the power function of ϕ is*

$$q_\phi : \mathbb{R} \to [0,1], \mu \mapsto q_\phi(\mu) := 1 - \Psi(k; d_\mu, n-1) + \Psi(-k; d_\mu, n-1), \tag{33.44}$$

where $\Psi(\cdot; d_\mu, n-1)$ denotes the CDF of the noncentral t distribution with noncentrality parameter

$$d_\mu := \sqrt{n}\frac{\mu - \mu_0}{\sigma} \tag{33.45}$$

and degrees-of-freedom parameter $n-1$.

∘

Proof. The power function of the test under consideration in the present test scenario is defined as

$$q_\phi : \mathbb{R} \to [0,1], \mu \mapsto q_\phi(\mu) := \mathbb{P}_\mu(\phi(y) = 1). \tag{33.46}$$

Because the probability of $\phi(y) = 1$ is equal to the probability that the associated test statistic lies in the rejection region of the test, we first need the distribution of the test statistic. We have already seen above that the one-sample t-test statistic T is distributed according to a noncentral t distribution $t(d_\mu, n-1)$ with noncentrality parameter d_μ. The rejection region of the two-sided one-sample t-test is

$$A =]-\infty, -k] \cup]k, \infty[. \tag{33.47}$$

With this rejection region, we obtain

$$\begin{aligned}
q_\phi(\mu) &= \mathbb{P}_\mu(\phi(y) = 1) \\
&= \mathbb{P}_\mu\left(T \in]-\infty, -k] \cup]k, \infty[\right) \\
&= \mathbb{P}_\mu\left(T \in]-\infty, -k]\right) + \mathbb{P}_\mu\left(T \in]k, \infty[\right) \\
&= \mathbb{P}_\mu(T \le -k) + \mathbb{P}_\mu(T > k) \\
&= \mathbb{P}_\mu(T \le -k) + (1 - \mathbb{P}_\mu(T \le k)) \\
&= 1 - \mathbb{P}_\mu(T \le k) + \mathbb{P}_\mu(T \le -k) \\
&= 1 - \Psi(k; d_\mu, n-1) + \Psi(-k; d_\mu, n-1),
\end{aligned} \tag{33.48}$$

where $\Psi(\cdot; d_\mu, n-1)$ denotes the CDF of the noncentral t distribution with noncentrality parameter d_μ and degrees-of-freedom parameter $n-1$. □

In Figure 33.3, we visualize the power function of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis from Theorem 33.2 for $\sigma^2 = 9$ and $\mu_0 = 4$ as a function of the critical value k. First, note that the power function as a function of μ covers both the scenario in which the null hypothesis holds, $\mu = \mu_0$, and the scenario in which the alternative hypothesis holds, $\mu \neq \mu_0$. Second, note that the value of the power function, that is, the probability that the test takes the value 1, increases for both positive and negative deviations of the true, but unknown, expectation parameter μ from the null-hypothesis parameter μ_0. This is due to the fact that the test decision is based on the absolute value of the test statistic. Finally, the exact shape and location of the power function depend on the choice of the critical value k. If this critical value is chosen larger, a larger absolute value of the test statistic is necessary for the test to take the value 1, and the probability of doing so is smaller for otherwise constant parameters than for smaller critical values.

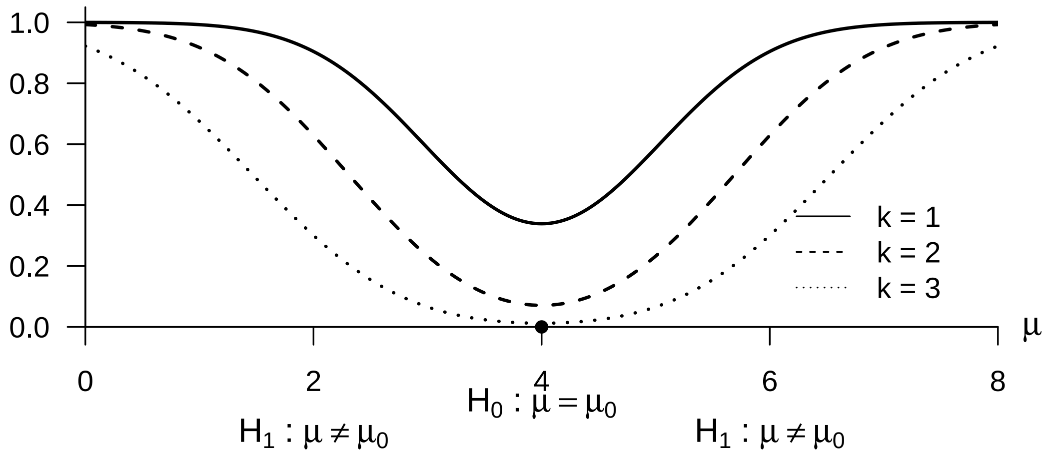


Figure 33.3 Power function of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis for $\sigma^2 = 9, \mu_0 = 4, n = 12$ and $k = 1, 2, 3$.

Test-size control

The values of the power function at $\mu = \mu_0$ in Figure 33.3 give a visual impression of how the critical value controls the test size. The exact determination of the critical value for a desired test size is the content of the following theorem.

Theorem 33.3 (Test-size control for the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis)**.** *Let ϕ be the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis. Then ϕ is a level-α_0 test with test size α_0 if the critical value is defined by*

$$k_{\alpha_0} := \Psi^{-1}\left(1 - \frac{\alpha_0}{2}; n - 1\right), \tag{33.49}$$

where $\Psi^{-1}(\cdot; n-1)$ denotes the inverse CDF of the t distribution with degrees-of-freedom parameter $n - 1$.

∘

Proof. For the test under consideration to be a level-α_0 test, we must have $q_\phi(\mu) \le \alpha_0$ for all $\mu \in \{\mu_0\}$, here $q_\phi(\mu_0) \le \alpha_0$. Furthermore, the test size of the test under consideration is $\alpha = \max_{\mu \in \{\mu_0\}} q_\phi(\mu)$, here $\alpha = q_\phi(\mu_0)$. We therefore have to show that choosing k_{α_0} guarantees that ϕ is a level-α_0 test with test size α_0. First, for $\mu = \mu_0$,

$$\begin{aligned} q_\phi(\mu_0) &= 1 - \Psi(k; d_{\mu_0}, n-1) + \Psi(-k; d_{\mu_0}, n-1) \\ &= 1 - \Psi(k; 0, n-1) + \Psi(-k; 0, n-1) \\ &= 1 - \Psi(k; n-1) + \Psi(-k; n-1), \end{aligned} \tag{33.50}$$

where $\Psi(\cdot; d, n-1)$ and $\Psi(\cdot; n-1)$ denote the CDFs of the noncentral t distribution with noncentrality parameter d and degrees-of-freedom parameter $n - 1$, and of the t distribution with degrees-of-freedom parameter $n-1$, respectively. Now let $k := k_{\alpha_0}$. Then

$$
\begin{aligned}
q_\phi(\mu_0) &= 1 - \Psi(k_{\alpha_0}; n-1) + \Psi(-k_{\alpha_0}; n-1) \\
&= 1 - \Psi(k_{\alpha_0}; n-1) + (1 - \Psi(k_{\alpha_0}; n-1)) \\
&= 2(1 - \Psi(k_{\alpha_0}; n-1)) \\
&= 2\left(1 - \Psi\left(\Psi^{-1}\left(1 - \frac{\alpha_0}{2}, n-1\right), n-1\right)\right) \\
&= 2\left(1 - 1 + \frac{\alpha_0}{2}\right) \\
&= \alpha_0,
\end{aligned}
\tag{33.51}
$$

where the second equality follows from the symmetry of the t distribution. It follows directly that, with $k = k_{\alpha_0}$, $q_\phi(\mu_0) \le \alpha_0$ and therefore the test under consideration is a level-α_0 test. It also follows directly that the test size of the test under consideration equals α_0 when $k = k_{\alpha_0}$. □

According to Theorem 33.3, the test considered here is exact; its test size is identical to the significance level. In Figure 33.4, we visualize the choice of the critical value k_{α_0} in a two-sided one-sample t-test scenario with simple null hypothesis and composite alternative hypothesis for $\alpha_0 := 0.05$ and $n = 12$.

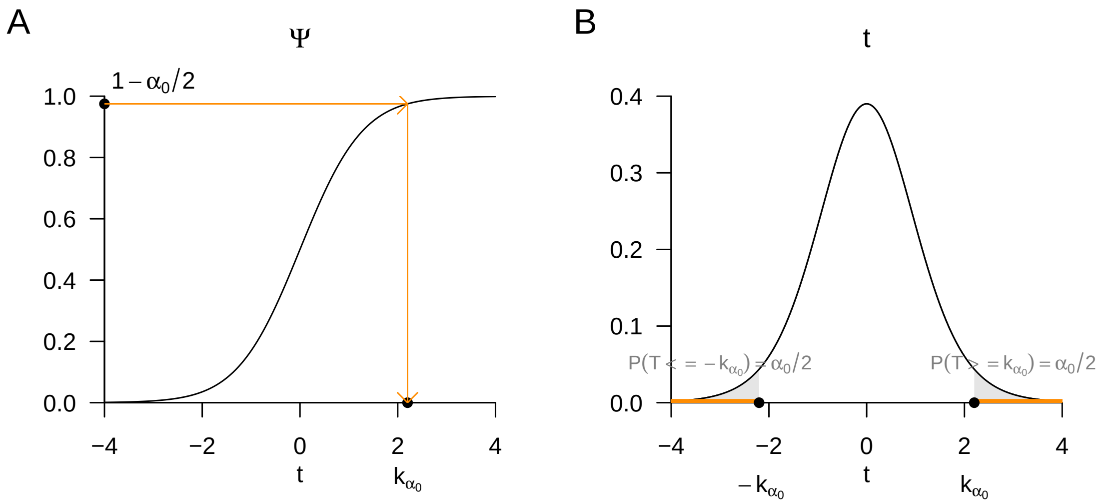


Figure 33.4 Determination of the critical value k_{α_0} for the two-sided one-sample t-test with simple null hypothesis for $n = 12$ and $\alpha_0 := 0.05$ to control the test size. (A) For a chosen α_0, the critical value of the test under consideration is given by the value of the inverse CDF of the t distribution with degrees-of-freedom parameter $n-1$ at $1 - \frac{\alpha_0}{2}$. The figure shows this value as $k_{0.05} = 2.2$ using the CDF of the t distribution with degrees-of-freedom parameter $n-1$. (B) The figure shows the rejection region resulting from the choice of k_{α_0} as gray areas under the PDF of the t distribution with degrees-of-freedom parameter $n-1$. The symmetric distribution of the subsets of the rejection region in the tails of the PDF results from defining the test as a function of the absolute value of the one-sample t-test statistic, and hence from the two-sided character of the test considered here.

The following **R** code demonstrates the determination of the critical value of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis using the inverse CDF of the t distribution, which is implemented in **R** as the function `qt()`. In addition, the code simulates 10^6 sample realizations for the test scenario under the null hypothesis and evaluates the test. The estimated probability that the test takes the value 1 when the null hypothesis holds agrees very well with the desired value $\alpha_0 = 0.05$.

```
# model parameters
n         = 12                                          # number of data points
mu        = 0                                           # true, but unknown, expectation parameter
sigsqr    = 2                                           # true, but unknown, variance parameter

# test parameters
mu_0      = 0                                           # null-hypothesis parameter, here \mu = \mu_0
alpha_0   = 0.05                                        # significance level
k_alpha_0 = qt(1-alpha_0/2,n-1)                         # critical value

# simulation of test-size control
set.seed(1)                                             # random-number generator seed
nsim      = 1e6                                         # number of simulations
phi       = rep(NaN,nsim)                               # test-decision array
for(j in 1:nsim){                                       # simulation iterations
    y      = rnorm(n, mu, sqrt(sigsqr))                 # y_i \sim N(\mu,\Sigma), i = 1,...,n
    y_bar  = mean(y)                                    # sample mean
    s      = sd(y)                                      # sample standard deviation
    Tee    = sqrt(n)*((y_bar - mu_0)/s)                 # one-sample t-test statistic
    if(abs(Tee) > k_alpha_0){                           # test 1_{\vert t \vert >= k_alpha_0}
        phi[j] = 1                                      # reject the null hypothesis
    } else {
        phi[j] = 0                                      # do not reject the null hypothesis
    }
}
```

```
Critical value           = 2.2
Estimated test size alpha = 0.05
```

p-value

The p-value associated with a given value of the test statistic of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis follows from the following theorem.

Theorem 33.4 (p-value of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis). *Given the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis, and let t be a value of the one-sample t-test statistic. Then*

$$\textit{p-value} = 2(1 - \Psi(|t|; n-1)), \tag{33.52}$$

where $\Psi(\cdot; n-1)$ denotes the CDF of the t distribution with degrees-of-freedom parameter $n-1$.

∘

Proof. According to Definition 33.12, the p-value is the smallest significance level α_0 at which the null hypothesis is rejected for the test under consideration based on the value of t. In the present case, the null hypothesis is rejected for every α_0 with

$$|t| \geq \Psi^{-1}\left(1 - \frac{\alpha_0}{2}; n-1\right) \tag{33.53}$$

(cf. Theorem 33.3). For these α_0,

$$\alpha_0 \geq 2(1 - \Psi(|t|; n-1)), \tag{33.54}$$

because

$$\begin{aligned}
& |t| \geq \Psi^{-1}\left(1 - \frac{\alpha_0}{2}; n-1\right) \\
\Leftrightarrow\ & \Psi(|t|; n-1) \geq \Psi\left(\Psi^{-1}\left(1 - \frac{\alpha_0}{2}; n-1\right); n-1\right) \\
\Leftrightarrow\ & \Psi(|t|; n-1) \geq 1 - \frac{\alpha_0}{2} \\
\Leftrightarrow\ & \mathbb{P}(T \leq |t|) \geq 1 - \frac{\alpha_0}{2} \\
\Leftrightarrow\ & \frac{\alpha_0}{2} \geq 1 - \mathbb{P}(T \leq |t|) \\
\Leftrightarrow\ & \alpha_0 \geq 2(1 - \Psi(|t|; n-1)).
\end{aligned} \tag{33.55}$$

The smallest $\alpha_0 \in [0,1]$ with

$$\alpha_0 \geq 2\mathbb{P}(T \geq |t|) \tag{33.56}$$

is therefore

$$\alpha_0 = 2(1 - \Psi(|t|; n-1)). \tag{33.57}$$

□

In Figure 33.5, we visualize the determination of p-values for $t = 2.26$ and $t = 3.81$, which are $\mathrm{p} = 0.045$ and $\mathrm{p} = 0.003$, respectively. Note that, for example, the p-value for $t = -2.26$ is also $\mathrm{p} = 0.045$.

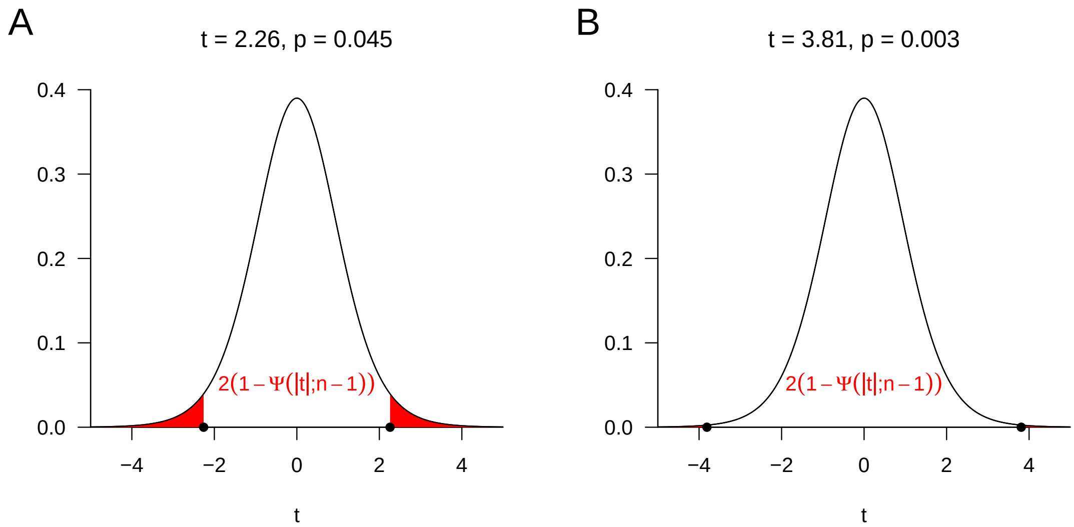


Figure 33.5 p-values for two possible values of the one-sample t-test statistic at $n = 12$. The regions in the outcome space of the one-sample t-test statistic over which the red areas are drawn mark the regions in which $|T| \geq |t|$. The summed probabilities assigned to them, that is, the corresponding red areas under the PDFs of the t distributions, equal the p-value.

Power function

The power function of a test is the power function of a test restricted to the region of the parameter space corresponding to the alternative hypothesis. Changes in the value of the power function of a test, often simply referred to as the *power of the test*, therefore initially arise from changes in the value of the true, but unknown, parameter in the alternative-hypothesis region. However, it has become customary to consider the probability that a test takes the value 1, that is, that the null hypothesis is rejected, not only as a function of the true, but unknown, parameter, but also as a function of further parameters of the test scenario that are relevant in practical applications. The sample size n is of primary interest here. In practical *power analyses*, one usually asks which sample size is associated with a certain probability of rejecting the null hypothesis under an assumed value of the true, but unknown, parameter in the alternative-hypothesis region. Based on the asymmetric treatment of type I and type II error probabilities (cf. Section 33.2), one first fixes a significance level α_0 to control the test size. For the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis discussed here, we therefore consider the power function

$$q_\phi : \mathbb{R} \to [0,1], \mu \mapsto q_\phi(\mu) := 1 - \Psi(k_{\alpha_0}; d_\mu, n-1) + \Psi(-k_{\alpha_0}; d_\mu, n-1) \tag{33.58}$$

under controlled test size, that is, for

$$k_{\alpha_0} := \Psi^{-1}\left(1 - \frac{\alpha_0}{2}; n-1\right) \tag{33.59}$$

with fixed α_0 as a function of the noncentrality parameter d and the sample size n. In particular, the noncentrality parameter d depends on the ratio of the true, but unknown, parameters μ and σ^2, that is, the true, but unknown, signal-to-noise ratio of the test scenario, and k_{α_0} depends on n. These considerations lead to the following definition.

Definition 33.17 (Power function of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis)**.** Given the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis, the *power function* of the test is

$$\pi : \mathbb{R} \times \mathbb{N} \to [0,1], (d,n) \mapsto \pi(d,n) := 1 - \Psi(k_{\alpha_0}; d, n-1) + \Psi(-k_{\alpha_0}; d, n-1). \tag{33.60}$$

•

The following **R** code demonstrates the evaluation of the power function of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis in a scenario with $\alpha_0 := 0.05$ using the **R** implementations of the CDF and inverse CDF of the noncentral t distribution, `pt()` and `qt()`.

```
alpha_0     = 0.05                                                   # significance level
d_min       = -5                                                     # minimal noncentrality parameter
d_max       =  5                                                     # maximal noncentrality parameter
d_res       = 50                                                     # noncentrality-parameter resolution
d           = seq(d_min, d_max, len = d_res)                         # noncentrality-parameter space
n_min       = 1                                                      # minimal sample size
n_max       = 30                                                     # maximal sample size
n_res       = 50                                                     # sample-size resolution
n           = seq(n_min,n_max, len = n_res)                          # sample-size space
pi          = matrix(rep(NaN, d_res*n_res), nrow = d_res)            # power-function array
for(i in 1:d_res){                                                   # noncentrality-parameter iterations
  for(j in 1:n_res){                                                 # sample-size iterations
    k_alpha_0 = qt(1 - alpha_0/2, n[j]-1)                            # critical value
    pi[i,j]   = 1-pt(k_alpha_0, n[j]-1, d[i])+pt(-k_alpha_0, n[j]-1, d[i])  # evaluation of the power function
  }
}
```

We visualize the dependence of the power function of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis on the noncentrality parameter and sample size in Figure 33.6 and Figure 33.7. In general, the power function of the test under consideration increases monotonically with positive or negative deviation of the noncentrality parameter from the null-hypothesis scenario $d = 0$ and with increasing sample size n. Depending on the choice of the significance level, this increase is steeper or less steep.

Practical implementation

Against the background of the theory of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis discussed so far, the routine implementation of the test is as follows. One assumes that a given univariate dataset is a realization of the frequentist inference model $y_1, ..., y_n \sim N(\mu, \sigma^2)$ of the one-sample t-test with true, but unknown, parameters μ and $\sigma^2 > 0$. One further assumes that one must decide whether, for a chosen null-hypothesis parameter μ_0, the null hypothesis $H_0 : \mu = \mu_0$ or

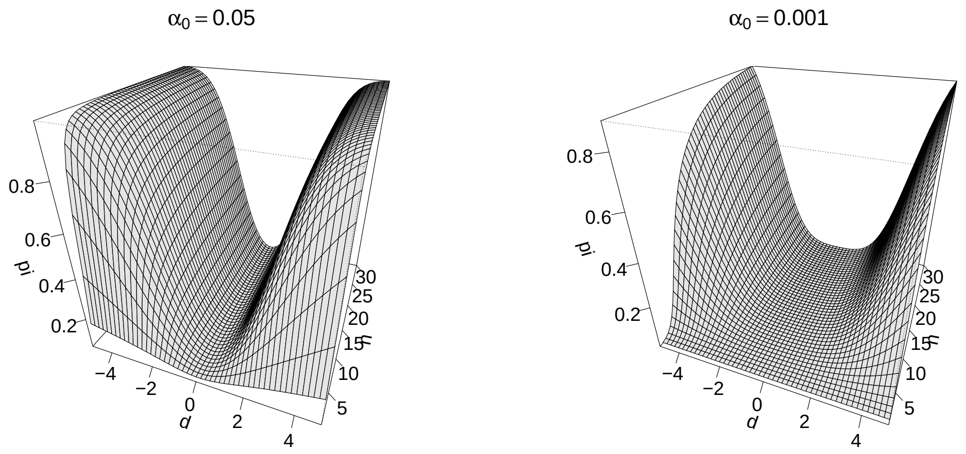


Figure 33.6 Power functions of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis. (A) Dependence of the power function π on the noncentrality parameter d and sample size n for $\alpha_0 := 0.05$. (B) Dependence of the power function π on the noncentrality parameter d and sample size n for $\alpha_0 := 0.001$.

the alternative hypothesis $H_1 : \mu \neq \mu_0$ is more plausible. To control the test size over many repetitions of this testing procedure, one chooses a significance level α_0 and determines the corresponding critical value k_{α_0}, so that, for example, at sample size $n = 12$ and $\alpha_0 := 0.05$, a critical value of $k_{0.05} = 2.20$ is chosen. Using the sample size n, the null-hypothesis parameter μ_0, the sample mean $\bar{y}$, and the sample standard deviation s, one then computes the value of the one-sample t-test statistic

$$t := \sqrt{n}\left(\frac{\bar{y} - \mu_0}{s}\right). \tag{33.61}$$

If the value t determined in this way for the present dataset is greater than k_{α_0} or less than $-k_{\alpha_0}$, one rejects the null hypothesis. Otherwise, one does not reject the null hypothesis. The theory developed here then guarantees that, in the long-run average, one falsely rejects the null hypothesis in at most $\alpha_0 \cdot 100$ out of 100 cases. Furthermore, based on the present value of the one-sample t-test statistic, one determines the associated p-value by

$$\text{p-value} = 2(1 - \Psi(|t|; n - 1)). \tag{33.62}$$

The following **R** code demonstrates this procedure under the assumption of a given data vector `y` of length `n`.

```
n            = length(y)                                # sample size
mu_0         = 0                                        # null-hypothesis parameter
alpha_0      = 0.05                                     # significance level
k_alpha_0    = qt(1-alpha_0/2,n-1)                      # critical value
Tee          = sqrt(n)*((mean(y) - mu_0)/sd(y))         # one-sample t-test statistic
if(abs(Tee) > k_alpha_0){phi = 1} else {phi = 0}        # test evaluation
p            = 2*(1 - pt(abs(Tee),n-1))                # p-value evaluation
```

If one wants to carry out a power analysis to optimize the sample size, then, in principle, the power function of the test increases as sample size increases. From the perspective of test power, a larger sample size is therefore always better than a smaller sample size. However, possible costs of increasing the sample size, such as possible risks for study

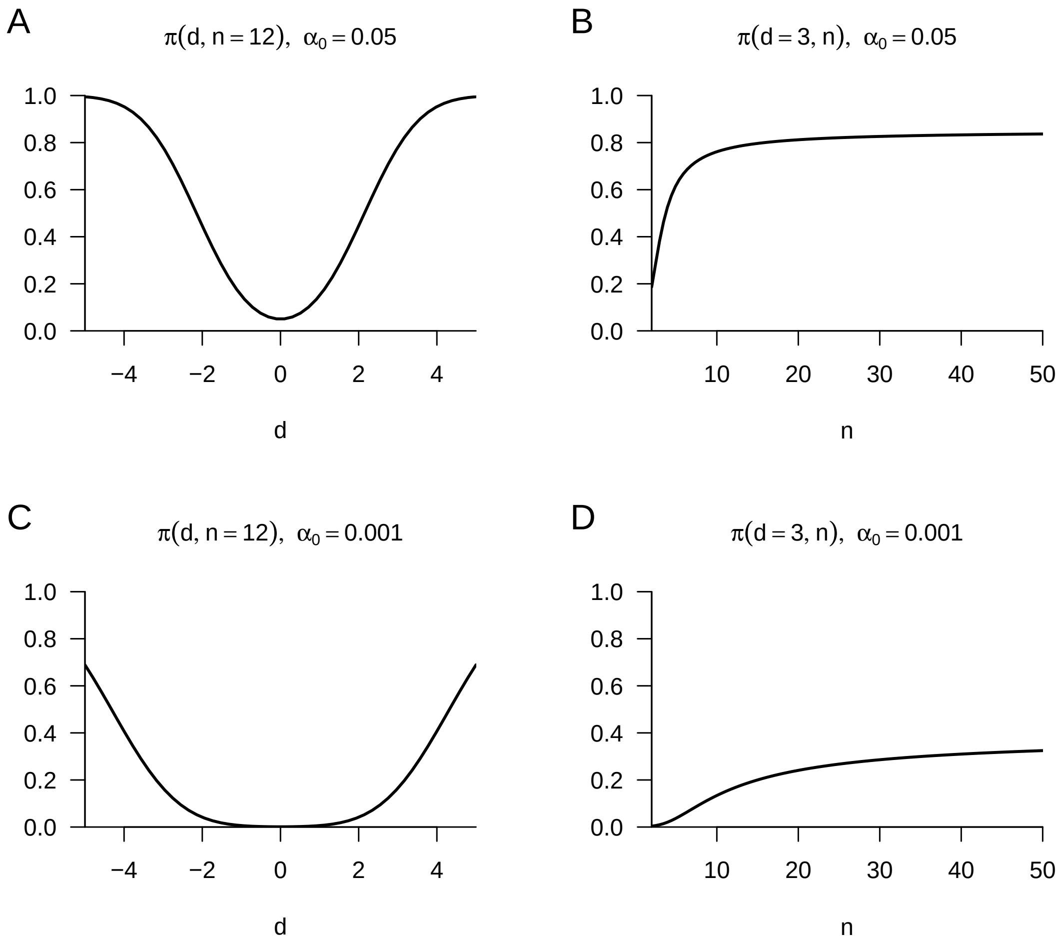


Figure 33.7 Slices of the power functions of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis. (A) Dependence of the power function π on the noncentrality parameter d at constant sample size $n = 12$ and $\alpha_0 := 0.05$. (B) Dependence of the power function π on sample size n at noncentrality parameter $d = 3$ and $\alpha_0 := 0.05$. (C) Dependence of the power function π on the noncentrality parameter d at constant sample size $n = 12$ and $\alpha_0 := 0.001$. (D) Dependence of the power function π on sample size n at noncentrality parameter $d = 3$ and $\alpha_0 := 0.001$.

participants, are not taken into account. Furthermore, the value that the power function takes at a chosen sample size always depends on the true, but unknown, parameter values μ and σ, which enter the value of the noncentrality parameter d. If one already knew these values very precisely in a given application context, one would probably not carry out a study. In study planning, the following procedure is therefore generally favored. First, one chooses a significance level α_0 to control the test size and evaluates the power function. One then chooses a noncentrality value d^* that one would like to detect with power at least β, where a conventional value is $\beta = 0.8$. One then evaluates the sample size required for a power-function value

$$\pi(d = d^*, n) = \beta. \tag{33.63}$$

Because of the monotonicity of the power function of the two-sided one-sample t-test with simple null hypothesis and composite alternative hypothesis in the range of nonnegative noncentrality parameters, this guarantees that the power of the test for noncentrality parameters greater than d^* is greater than or equal to β. The following **R** code implements this procedure for optimizing the sample size, and Figure 33.8 visualizes it.

```
# power-function based sample-size optimization
alpha_0   = 0.05                                                 # significance level
beta      = 0.8                                                  # desired power-function value
d_stern   = 3                                                    # fixed noncentrality parameter
n_min     = 2                                                    # minimal considered sample size
n_max     = 20                                                   # maximal considered sample size
n_res     = 1e2                                                  # sample-size-space resolution
n         = seq(n_min,n_max, len = n_res)                        # sample-size space
k_alpha_0 = qt(1-alpha_0/2, n-1)                                 # critical values as function of sample size
pi_n      = 1-pt(k_alpha_0, n-1, d_stern)+pt(-k_alpha_0, n-1, d_stern)  # power function at fixed noncentrality parameter
i         = 1                                                    # index initialization
n_min     = NaN                                                  # minimal n initialization
while(pi_n[i] < beta){                                           # while \pi(d*,n) < \beta
    n_min = n[i]                                                 # record minimal required n
    i     = i + 1}                                               # and increase index
cat("Minimal required n =", ceiling(n_min))                      # output
```

```
Minimal required n = 16
```

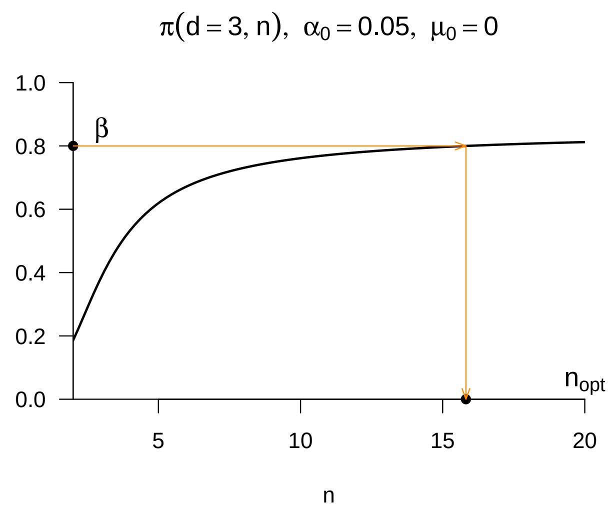


Figure 33.8 Practical implementation of a power-function based sample-size optimization. At a chosen significance level α_0 and fixed assumed noncentrality parameter d^*, one evaluates the sample size for which the test power function has a value of β or larger.

Application example

Finally, we demonstrate the procedure sketched above using Example 30.5. Substantively, in this case the simple null hypothesis $H_0 : \mu = 0$ corresponds to the hypothesis that the therapy has no effect on BDI-II reduction scores. The composite alternative hypothesis $H_1 : \mu \neq 0$ corresponds to the hypothesis that the therapy has a systematic nonzero effect on BDI-II reduction scores. The **R** code below applies the procedure demonstrated above to the pre-post therapy BDI-II reduction-score dataset of $n = 12$ patients and, in addition, evaluates the 95% confidence interval for the expectation parameter.

```
D         = read.csv("./_data/bdi-ii-dataset.csv")        # read dataset
y         = D$dBDI                                        # select data
n         = length(y)                                     # sample size
mu_hat    = mean(y)                                       # expectation-parameter estimator
delta     = 0.95                                          # confidence level
t_delta   = qt((1+delta)/2,n-1)                           # \Psi^-1((\delta + 1)/2, n-1)
G_u       = mean(y) - (sd(y)/sqrt(n))*t_delta             # lower confidence-interval bound
G_o       = mean(y) + (sd(y)/sqrt(n))*t_delta             # upper confidence-interval bound
mu_0      = 0                                             # null-hypothesis parameter, here \mu = \mu_0
alpha_0   = 0.05                                          # significance level
k_alpha_0 = qt(1-alpha_0/2,n-1)                           # critical value
Tee       = sqrt(n)*((mean(y) - mu_0)/sd(y))              # t-test statistic
if(abs(Tee) > k_alpha_0){phi = 1} else {phi = 0}          # test 1_{\vert t \vert >= k_alpha_0}
p         = 2*(1 - pt(abs(Tee),n-1))                      # p-value
```

```
Parameter estimate      = 3.17
95% confidence interval = 0.81 5.53
Significance level     = 0.05
Critical value         = 2.2
Test statistic         = 2.95
Test value             = 1
p-value                = 0.01
```

The same analysis can also be carried out with the **R** function `t.test()`; the syntax for using it and the formatting of the results it determines are shown below.

```
t.test(y)
```

```

	One Sample t-test

data:  y
t = 2.9542, df = 11, p-value = 0.01311
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.8074098 5.5259235
sample estimates:
mean of x 
 3.166667 
```

In the present case, one would reject the null hypothesis at a significance level of $\alpha_0 = 0.05$. Whether the null hypothesis actually holds in the present case, however, remains unknown, just like the true, but unknown, expectation parameter. In the long-run average, based on the assumptions described above, one falsely rejects the null hypothesis in only 5 out of 100 cases.

33.4 Confidence intervals and hypothesis tests

In this section, we examine the extent to which confidence intervals and hypothesis tests can be regarded as equivalent. We start from the scenario of a confidence interval.

Theorem 33.5 (Duality of confidence intervals and hypothesis tests)**.** *Let y be the sample of a frequentist inference model with outcome space $\mathcal{Y}$ and parameter space Θ. Furthermore, for $\delta \in]0,1[$, let $[G_u(y), G_o(y)]$ be a δ confidence interval for the true, but unknown, parameter $\theta \in \Theta$. Then the hypothesis test defined by*

$$\phi_\theta : \mathcal{Y} \to \{0,1\}, y \mapsto \phi(y) := \begin{cases} 0, & [G_u(y), G_o(y)] \ni \theta_0 \\ 1, & [G_u(y), G_o(y)] \not\ni \theta_0 \end{cases} \tag{33.64}$$

is a hypothesis test with significance level $\alpha_0 = 1 - \delta$ for the hypotheses

$$\Theta_0 := \{\theta_0\} \text{ and } \Theta_1 := \Theta \setminus \{\theta_0\}. \tag{33.65}$$

∘

Proof. Because of the simple null hypothesis and hence $\alpha_0 = \alpha$, it follows that

$$\alpha_0 = \alpha = \mathbb{P}_{\theta_0}(\phi(y) = 1) = \mathbb{P}_{\theta_0}([G_u(y), G_o(y)] \not\ni \theta_0) = 1 - \mathbb{P}_{\theta_0}([G_u(y), G_o(y)] \ni \theta_0) = 1 - \delta. \tag{33.66}$$

□

Theorem 33.5 thus states that one can use a δ confidence interval to construct a hypothesis test with significance level $\alpha_0 = 1-\delta$ with simple null hypothesis and composite alternative hypothesis. In this test, the null hypothesis $\theta = \theta_0$ is rejected whenever the confidence interval does not cover the null-hypothesis parameter θ_0. With the following theorem, we concretize Theorem 33.5 for the confidence interval for the expectation parameter of the normal distribution model considered in Section 32.2 and the one-sample t-test considered in Section 33.3.1.

Theorem 33.6 (Duality of expectation confidence interval and one-sample t-test)**.** *Given the normal distribution model, let*

$$\kappa := \left[\bar{y} - \frac{S}{\sqrt{n}} t_\delta, \bar{y} + \frac{S}{\sqrt{n}} t_\delta\right]. \tag{33.67}$$

be the δ confidence interval for the expectation parameter defined by

$$t_\delta := \Psi^{-1}\left(\frac{1+\delta}{2}; n-1\right) \tag{33.68}$$

in Theorem 32.2. Then, by Theorem 33.5, the test

$$\phi : \mathcal{Y} \to \{0,1\}, y \mapsto \phi(y) := \begin{cases} 0, & \left[\bar{y} - \frac{S}{\sqrt{n}} t_\delta, \bar{y} + \frac{S}{\sqrt{n}} t_\delta\right] \ni \mu_0 \\ 1, & \left[\bar{y} - \frac{S}{\sqrt{n}} t_\delta, \bar{y} + \frac{S}{\sqrt{n}} t_\delta\right] \not\ni \mu_0 \end{cases} \tag{33.69}$$

is a test of the simple null hypothesis $H_0 : \mu = \mu_0$ and the composite alternative hypothesis $H_1 : \mu \neq \mu_0$ with significance level $\alpha_0 = 1 - \delta$.

∘

Proof. It holds that

$$\mathbb{P}_{\mu_0}(\phi(y)=1) = 1 - \mathbb{P}_{\mu_0}(\phi(y)=0) = 1 - \mathbb{P}_{\mu_0}\left(\left[\bar{y} - \frac{S}{\sqrt{n}}t_\delta, \bar{y} + \frac{S}{\sqrt{n}}t_\delta\right] \ni \mu_0\right) = 1 - \delta. \quad (33.70)$$

□

The following **R** code simulates this confidence-interval based hypothesis test under the null hypothesis and reports estimates of the confidence level and the significance level over 100 realizations of a sample with sample size $n = 12$ and true, but unknown, parameters $\mu = 2$ and $\sigma^2 = 1$.

```
n       = 12                                        # sample size
mu      = 2                                         # true, but unknown, expectation parameter
sigsqr  = 1                                         # true, but unknown, variance parameter
delta   = 0.95                                      # confidence level
t_delta = qt((1+delta)/2, n-1)                      # \Psi^{-1}((\delta + 1)/2, n-1)
mu_0    = mu                                        # null-hypothesis parameter when H_0 holds
set.seed(1)                                         # random-number generator seed
ns      = 1e2                                       # number of simulations
y_bar   = rep(NaN,ns)                               # sample-mean array
s       = rep(NaN,ns)                               # sample-standard-deviation array
kappa   = matrix(rep(NaN,2*ns), ncol = 2)           # confidence-interval array
kfn     = rep(NaN,ns)                               # coverage-indicator array
phi     = rep(NaN,ns)                               # test array
for(i in 1:ns){                                     # simulation iterations
  y          = rnorm(n,mu_0,sqrt(sigsqr))           # sample realization
  y_bar[i]   = mean(y)                              # sample mean
  s[i]       = sd(y)                                # sample standard deviation
  kappa[i,1] = y_bar[i] - (s[i]/sqrt(n))*t_delta    # lower confidence-interval bound
  kappa[i,2] = y_bar[i] + (s[i]/sqrt(n))*t_delta    # upper confidence-interval bound
  if(kappa[i,1] <= mu_0 & mu_0 <= kappa[i,2]){
      kfn[i] = 1} else{kfn[i] = 0}                  # coverage-indicator evaluation
  if(kappa[i,1] <= mu_0 & mu_0 <= kappa[i,2]){
      phi[i] = 0} else{phi[i] = 1}}                 # test evaluation
```

```
cat(   "Estimated confidence level =", round(mean(kfn),2),    # output
     "\nEstimated test size        =", round(mean(phi),2))
```

```
Estimated confidence level = 0.96
Estimated test size        = 0.04
```

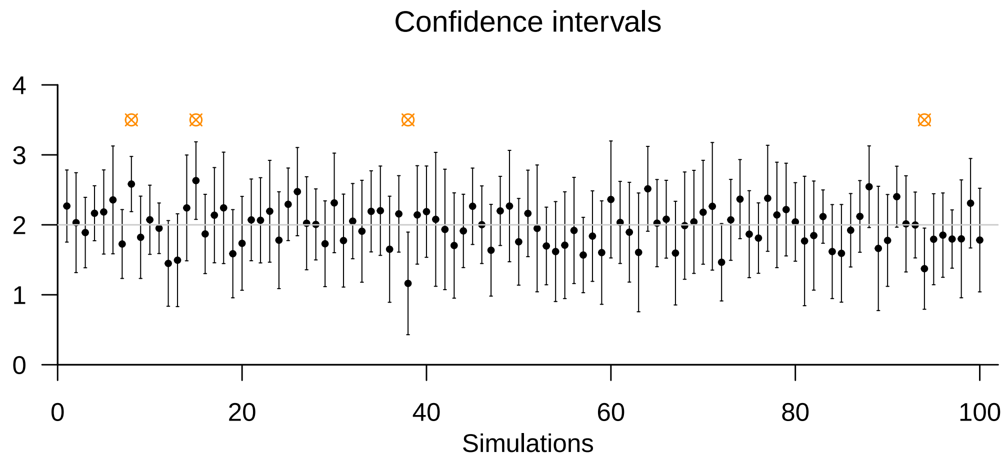
We visualize the results of this simulation in Figure 33.9.

33.5 Bibliographic remarks

The theory of hypothesis tests presented here goes back essentially to Neyman & Pearson (1928) and Neyman & Pearson (1933). Gigerenzer (2004) and Lehmann (2011) provide historical classifications of the genesis of the concept of hypothesis testing.

Study questions

1. Explain the basic logic of frequentist hypothesis tests.
2. State the definitions of test hypotheses and a test scenario.
3. State the definitions of simple and composite test hypotheses.
4. State the definitions of one-sided and two-sided test hypotheses.
5. State the definition of the concept of a test.
6. State the definition of the concept of a standard test.
7. State the definition of the concept of a critical region.
8. State the definition of the concept of a rejection region.
9. State the definition of the concept of a critical-value based test.



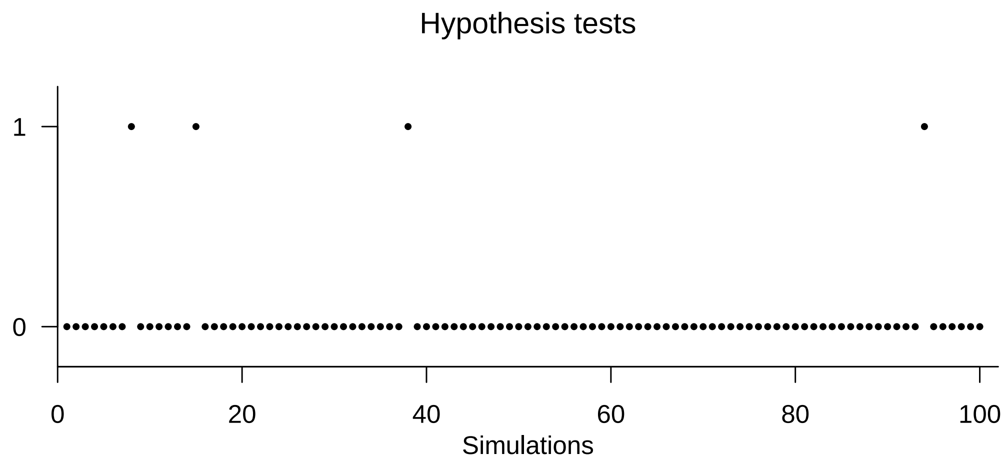


Figure 33.9 Duality of confidence interval and hypothesis test by example of the normal distribution model. If the δ confidence interval for the expectation parameter is used to reject the null hypothesis whenever it does not cover the null-hypothesis parameter, the result is a hypothesis test with significance level $\alpha_0 = 1 - \delta$.

10. State the definitions of correct test decisions and test errors.
11. State the definition of the concept of a power function.
12. Explain the importance of the power function for constructing hypothesis tests.
13. State the definition of the concept of a level-α_0 test.
14. State the definition of the concept of a significance level α_0.
15. State the definition of the concept of a test size α.
16. State the definition of the concept of a p-value.

Study question answers

1. See the introductory remarks of this chapter.
2. See Definition 33.1.
3. See Definition 33.2.
4. See Definition 33.3.
5. See Definition 33.4.
6. See Definition 33.5.
7. See Definition 33.6.
8. See Definition 33.7.
9. See Definition 33.8.
10. See Definition 33.9.
11. See Definition 33.10.
12. See the explanations following Definition 33.10.
13. See Definition 33.11.
14. See Definition 33.11.
15. See Definition 33.11.
16. See Definition 33.12.

Part V

General linear model

34 Regression

The fundamental goal of regression analyses is to model relationships between independent and dependent variables. A central topic in this context is fitting functions to observed datasets. With the concepts of the fitting line in the framework of the method of least squares and of simple linear regression, we want to approach these central topics of probabilistic data modeling step by step in this section. The concepts of fitting line and simple linear regression differ in one central aspect: for the fitting line, independent and dependent variables are not modeled as random variables; in the framework of simple linear regression, the dependent variable then takes the form of a random variable. Finally, in the context of correlation, both dependent and independent variables are modeled as random variables.

To illustrate the concepts of this section, we consider an example dataset in which the number of psychotherapy sessions as independent variable x is related to symptom reduction in a group of $n = 20$ patients as dependent variable y (Figure 34.1). Visual inspection of this dataset suggests that more therapy sessions imply more symptom reduction. The goal of the method of least squares and of simple linear regression is to put this intuitive functional relationship between independent and dependent variable on a quantitative basis.

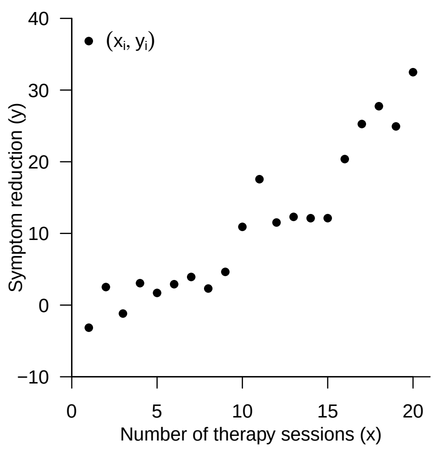


Figure 34.1 Example dataset.

34.1 Method of least squares

We first define the concept of a *fitting line*.

Definition 34.1 (Fitting line). For $\beta := (\beta_0, \beta_1)^T \in \mathbb{R}^2$, the linear-affine function

$$f_\beta : \mathbb{R} \to \mathbb{R}, x \mapsto f_\beta(x) := \beta_0 + \beta_1 x \tag{34.1}$$

for which, for a dataset $\{(x_1, y_1), \dots, (x_n, y_n)\} \subset \mathbb{R}^2$, the function

$$q : \mathbb{R}^2 \to \mathbb{R}_{\geq 0}, \beta \mapsto q(\beta) := \sum_{i=1}^{n} \left(y_i - f_\beta(x_i)\right)^2 = \sum_{i=1}^{n} \left(y_i - (\beta_0 + \beta_1 x_i)\right)^2 \tag{34.2}$$

of the squared vertical deviations of the y_i from the function values $f_\beta(x_i)$ attains its minimum is called the fitting line for the dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$.

•

The fitting line is thus a linear-affine function of the form

$$f_\beta : \mathbb{R} \to \mathbb{R}, x \mapsto f_\beta(x) := \beta_0 + \beta_1 x \tag{34.3}$$

Figure 34.2 shows three linear-affine functions parameterized by different values of β_0 and β_1, together with the value set of the example dataset.

As with all linear-affine functions, for f_β the value of β_0 corresponds to the value that f_β takes for $x = 0$,

$$f_\beta(0) = \beta_0 + \beta_1 \cdot 0 = \beta_0 \tag{34.4}$$

and hence graphically to the intersection of the function graph with the y-axis. Because β_0 therefore corresponds to the offset of the function graph from $y = 0$ at $x = 0$, β_0 is also often called the *offset parameter*. Analogously, as with all linear-affine functions, the value of β_1 corresponds to the function-value difference per unit argument difference. For example, for $\beta_0 = 5$ and $\beta_1 = 0.5$ we have

$$\begin{aligned} f_\beta(2) - f_\beta(1) &= (5 + 0.5 \cdot 2) - (5 + 0.5 \cdot 1) = 1 - 0.5 = 0.5 \\ f_\beta(9) - f_\beta(8) &= (5 + 0.5 \cdot 9) - (5 + 0.5 \cdot 8) = 9.5 - 8 = 0.5 \end{aligned} \tag{34.5}$$

Thus, an argument difference of 1 results in a function-value difference of 0.5. β_1 therefore encodes the strength of the change in function values per unit argument difference and hence the slope of the graph of the linear-affine function. Accordingly, β_1 is called the *slope parameter*.

By definition, however, the fitting line is not an arbitrary linear-affine function of the form f_β, but precisely the one that, for a given dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$, minimizes the sum of squared vertical deviations

$$q(\beta) := \sum_{i=1}^{n} \left(y_i - (\beta_0 + \beta_1 x_i)\right)^2 \tag{34.6}$$

For a fixed dataset of (x_i, y_i) pairs, the value of this sum depends on the values of β_0 and β_1 and can therefore be minimized by choosing suitable values of β_0 and β_1.

Because a sum of squared deviations between data points and values of the fitting line is minimized here, one also often speaks somewhat imprecisely of the *method of least squares*. Figure 34.3 shows the vertical deviations between y_i and $\beta_0 + \beta_1 x_i$ for $i = 1, \dots, n$ in the example dataset as orange lines, and the sum of their squares $q(\beta)$ in the title. For the

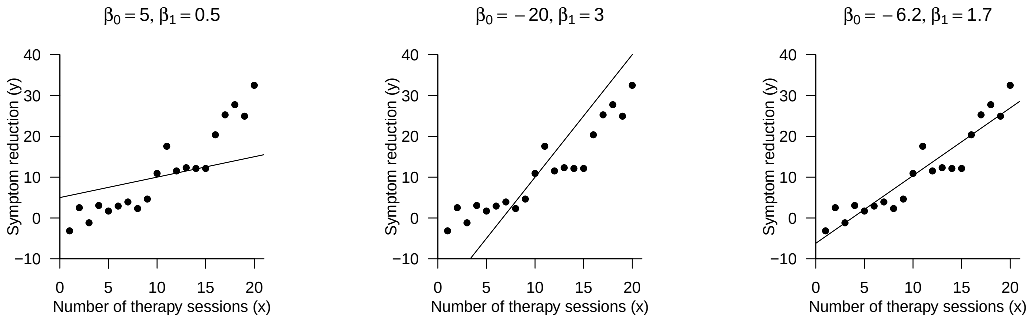


Figure 34.2 Linear-affine functions with different parameter values against the background of the example dataset.

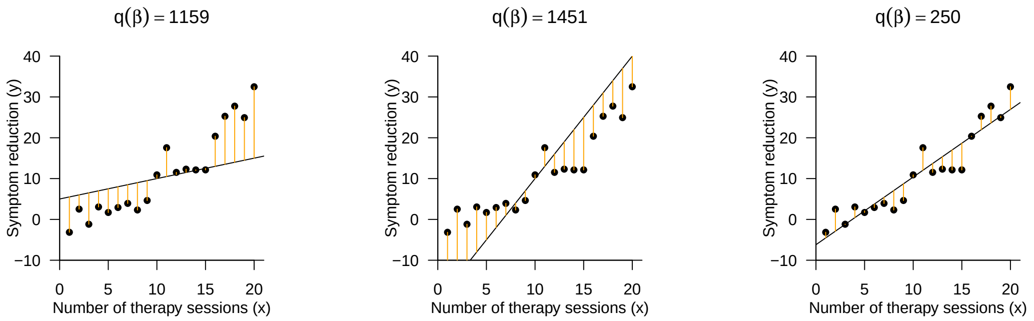


Figure 34.3 Vertical deviations and sums of squares for different parameter values.

parameter values $\beta_0 = -6.2$ and $\beta_1 = 1.7$ (cf. Figure 34.2), this sum attains its smallest value.

Concrete formulas for determining the parameter values of the fitting line are provided by Theorem 34.1.

Theorem 34.1 (Fitting line)**.** *For a dataset* $\{(x_1, y_1), \ldots, (x_n, y_n)\} \subset \mathbb{R}^2$*, the fitting line has the form*

$$f_\beta : \mathbb{R} \to \mathbb{R}, x \mapsto f_\beta(x) := \hat{\beta}_0 + \hat{\beta}_1 x \tag{34.7}$$

where, with the sample covariance c_{xy} *of the* (x_i, y_i) *values, the sample variance* s_x^2 *of the* x_i *values, and the sample means* $\bar{x}$ *and* $\bar{y}$ *of the* x_i *and* y_i *values, respectively, it holds that*

$$\hat{\beta}_1 = \frac{c_{xy}}{s_x^2} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \tag{34.8}$$

∘

Proof. We consider the sum of squared vertical deviations of the y_i from the function values $f(x_i)$ as a function of β_0 and β_1, and determine values $\hat{\beta}_0$ and $\hat{\beta}_1$ for which this function attains its minimum, that is, for which the sum of squared vertical deviations of the y_i from the function values $f(x_i)$ becomes minimal. We consider the function

$$q : \mathbb{R}^2 \to \mathbb{R}, (\beta_0, \beta_1) \mapsto q(\beta_0, \beta_1) := \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x))^2 \tag{34.9}$$

To determine the minimum of this function, we first compute the partial derivatives with respect to β_0 and β_1 and set them equal to 0. We first obtain

$$\begin{aligned}
\frac{\partial}{\partial \beta_0} q(\beta_0, \beta_1) &= \frac{\partial}{\partial \beta_0} \left(\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2 \right) \\
&= \sum_{i=1}^n \frac{\partial}{\partial \beta_0} (y_i - (\beta_0 + \beta_1 x_i))^2 \\
&= \sum_{i=1}^n 2 (y_i - (\beta_0 + \beta_1 x_i)) \frac{\partial}{\partial \beta_0} (y_i - \beta_0 - \beta_1 x_i) \\
&= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i).
\end{aligned} \tag{34.10}$$

Furthermore, we obtain

$$\begin{aligned}
\frac{\partial}{\partial \beta_1} q(\beta_0, \beta_1) &= \frac{\partial}{\partial \beta_1} \left(\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2 \right) \\
&= \sum_{i=1}^n \frac{\partial}{\partial \beta_1} (y_i - (\beta_0 + \beta_1 x_i))^2 \\
&= \sum_{i=1}^n 2 (y_i - (\beta_0 + \beta_1 x_i)) \frac{\partial}{\partial \beta_1} (y_i - \beta_0 - \beta_1 x_i) \\
&= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i.
\end{aligned} \tag{34.11}$$

Setting both partial derivatives equal to zero then yields

$$\begin{aligned}
&\frac{\partial}{\partial \beta_0} q(\beta_0, \beta_1) = 0 \text{ and } \frac{\partial}{\partial \beta_1} q(\beta_0, \beta_1) = 0 \\
\Leftrightarrow\ & -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \text{ and } -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0 \\
\Leftrightarrow\ & \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \text{ and } \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0
\end{aligned} \tag{34.12}$$

and further

$$
\begin{aligned}
\sum_{i=1}^n y_i - \sum_{i=1}^n \beta_0 - \beta_1 \sum_{i=1}^n x_i = 0 \text{ and } \sum_{i=1}^n y_i x_i - \sum_{i=1}^n \beta_0 x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0 \\
\Leftrightarrow \beta_0 n + \beta_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i \text{ and } \beta_0 \sum_{i=1}^n x_i + \beta_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i x_i
\end{aligned}
\tag{34.13}
$$

The resulting system of equations

$$
\begin{aligned}
\beta_0 n + \beta_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i \\
\beta_0 \sum_{i=1}^n x_i + \beta_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i
\end{aligned}
\tag{34.14}
$$

is called the system of normal equations and describes the necessary condition for a minimum of q. Solving this system of equations for β_0 and β_1 then yields the values $\hat{\beta}_0$ and $\hat{\beta}_1$ of the theorem. To see this, we first note that the first equation of the system of normal equations implies

$$
n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i \Leftrightarrow \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y} \Leftrightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}
\tag{34.15}
$$

Substituting the form of $\hat{\beta}_0$ into the second equation of the system of normal equations first yields

$$
\begin{aligned}
\hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i \\
\Leftrightarrow \left(\bar{y} - \hat{\beta}_1 \bar{x}\right) \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i \\
\Leftrightarrow \bar{y} \sum_{i=1}^n x_i - \hat{\beta}_1 \bar{x} \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i \\
\Leftrightarrow -\hat{\beta}_1 \bar{x} \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i - \bar{y} \sum_{i=1}^n x_i \\
\Leftrightarrow \hat{\beta}_1 \left(\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i \right) &= \sum_{i=1}^n y_i x_i - \bar{y} \sum_{i=1}^n x_i .
\end{aligned}
\tag{34.16}
$$

We now first note that

$$
\begin{aligned}
\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \bar{x} \sum_{i=1}^n x_i \\
&= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + n \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \bar{x} \\
&= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + n\bar{x}^2 \\
&= \sum_{i=1}^n \left(x_i^2 - 2\bar{x} x_i + \bar{x}^2\right) \\
&= \sum_{i=1}^n \left(x_i - \bar{x}\right)^2
\end{aligned}
\tag{34.17}
$$

Furthermore, we first note that

$$
\begin{aligned}
\sum_{i=1}^n y_i x_i - \bar{y}\sum_{i=1}^n x_i &= \sum_{i=1}^n y_i x_i - \bar{y}\sum_{i=1}^n x_i - n\bar{y}\bar{x} + n\bar{y}\bar{x} \\
&= \sum_{i=1}^n y_i x_i - \bar{y}\sum_{i=1}^n x_i - \sum_{i=1}^n y_i\bar{x} + \sum_{i=1}^n \bar{y}\bar{x} \\
&= \sum_{i=1}^n y_i x_i - \sum_{i=1}^n y_i\bar{x} - \sum_{i=1}^n \bar{y}x_i + \sum_{i=1}^n \bar{y}\bar{x} \\
&= \sum_{i=1}^n \left(y_i x_i - y_i\bar{x} - \bar{y}x_i + \bar{y}\bar{x}\right) \\
&= \sum_{i=1}^n \left(y_i - \bar{y}\right)\left(x_i - \bar{x}\right).
\end{aligned}
\tag{34.18}
$$

Continuing from (1.16), we then obtain

$$
\begin{aligned}
\hat{\beta}_1\left(\sum_{i=1}^n x_i^2 - \bar{x}\sum_{i=1}^n x_i\right) &= \sum_{i=1}^n y_i x_i - \bar{y}\sum_{i=1}^n x_i \\
\Leftrightarrow \hat{\beta}_1\left(\sum_{i=1}^n \left(x_i - \bar{x}\right)^2\right) &= \sum_{i=1}^n \left(y_i - \bar{y}\right)\left(x_i - \bar{x}\right) \\
\Leftrightarrow \hat{\beta}_1 &= \frac{\sum_{i=1}^n \left(y_i - \bar{y}\right)\left(x_i - \bar{x}\right)}{\sum_{i=1}^n \left(x_i - \bar{x}\right)^2} \\
\Leftrightarrow \hat{\beta}_1 &= \frac{c_{xy}}{s_x^2}.
\end{aligned}
\tag{34.19}
$$

□

Theorem 34.1 states that, for a given dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$, the parameter values that minimize the sum of squared vertical deviations for a linear-affine function can be computed using the sample means of the x_i and y_i values, the sample variance of the x_i values, and the sample covariance of the x_i and y_i values. The terminology is oriented toward the concepts of descriptive statistics; in particular, the x_i values are often not understood as realizations of random variables, but the term sample is nevertheless used. From an applied perspective, according to Theorem 34.1, the parameters of the fitting line can thus be determined using the familiar functions for evaluating descriptive statistics. The following **R** code demonstrates this.

```
# read example dataset
D           = read.csv("./_data/501-regression.csv")

# sample statistics
x_bar       = mean(D$x_i)                                          # sample mean of x_i values
y_bar       = mean(D$y_i)                                          # sample mean of y_i values
s2x         = var(D$x_i)                                           # sample variance of x_i values
cxy         = cov(D$x_i, D$y_i)                                    # sample covariance of (x_i,y_i) values

# fitting-line parameters
beta_1_hat  = cxy/s2x                                              # \hat{\beta}_1, slope parameter
beta_0_hat  = y_bar - beta_1_hat*x_bar                             # \hat{\beta}_0, offset parameter

# output
cat("beta_0_hat:", beta_0_hat,
    "\nbeta_1_hat:", beta_1_hat)
```

```
beta_0_hat: -6.194704
beta_1_hat: 1.657055
```

A typical visualization of the fitting line of a dataset as in Figure 34.4 is implemented by the following **R** code.

```
# data values
plot(
D$x_i,
D$y_i,
pch         = 16,
xlab        = "Number of therapy sessions (x)",
ylab        = "Symptom reduction (y)",
xlim        = c(0,21),
ylim        = c(-10, 40),
main        = TeX("$\\hat{\\beta}_0 =  -6.19, \\hat{\\beta}_1 = 1.66$"))

# fitting line
abline(
coef        = c(beta_0_hat, beta_1_hat),
lty         = 1,
col         = "black")

# legend
legend(
"topleft",
c(TeX("$(x_i,y_i)$"), TeX("$f(x) = \\hat{\\beta}_0 + \\hat{\\beta}_1x$")),
lty        = c(0,1),
pch        = c(16, NA),
bty        = "n")
```

The idea of minimizing, for a given dataset of (x_i, y_i) pairs, the sum of squared vertical deviations between a function of the x_i values and the y_i values, and thus fitting a function as well as possible to a set of values, is not restricted to linear-affine functions. The following definition generalizes the definition of the fitting line to polynomial functions of arbitrary degree.

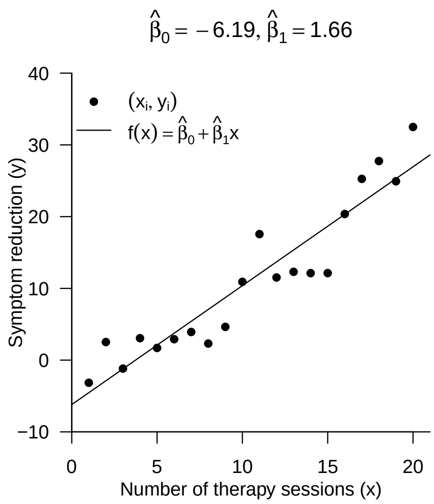


Figure 34.4 Fitting line for the example dataset.

Definition 34.2 (Fitting polynomial). For $\beta := (\beta_0, \dots, \beta_k)^T \in \mathbb{R}^{k+1}$, the polynomial function of degree k

$$f_\beta : \mathbb{R} \to \mathbb{R}, x \mapsto f_\beta(x) := \sum_{i=0}^{k} \beta_i x^i \tag{34.20}$$

for which, for a dataset $\{(x_1, y_1), \dots, (x_n, y_n)\} \subset \mathbb{R}^2$, the function

$$q : \mathbb{R}^{k+1} \to \mathbb{R}_{\geq 0}, \beta \mapsto q(\beta) := \sum_{i=1}^{n} \left(y_i - f_\beta(x_i)\right)^2 = \sum_{i=1}^{n} \left(y_i - \sum_{i=0}^{k} \beta_i x^i\right)^2 \tag{34.21}$$

of the squared vertical deviations of the y_i from the function values $f_\beta(x_i)$ attains its minimum is called the *fitting polynomial of degree k for the dataset* $\{(x_1, y_1), \dots, (x_n, y_n)\}$.

•

The fitting line is therefore the fitting polynomial of first degree. We do not want to deepen the concept of the fitting polynomial here; in particular, we will determine the parameter values $\hat{\beta}_0, ..., \hat{\beta}_k$ for which the function q attains its minimum in general at a later point in the framework of the theory of the general linear model. In Figure 34.5, we visualize the fitting polynomials of first to fourth degree for the example dataset; the value of the function q at the minimum is shown in the title in each case.

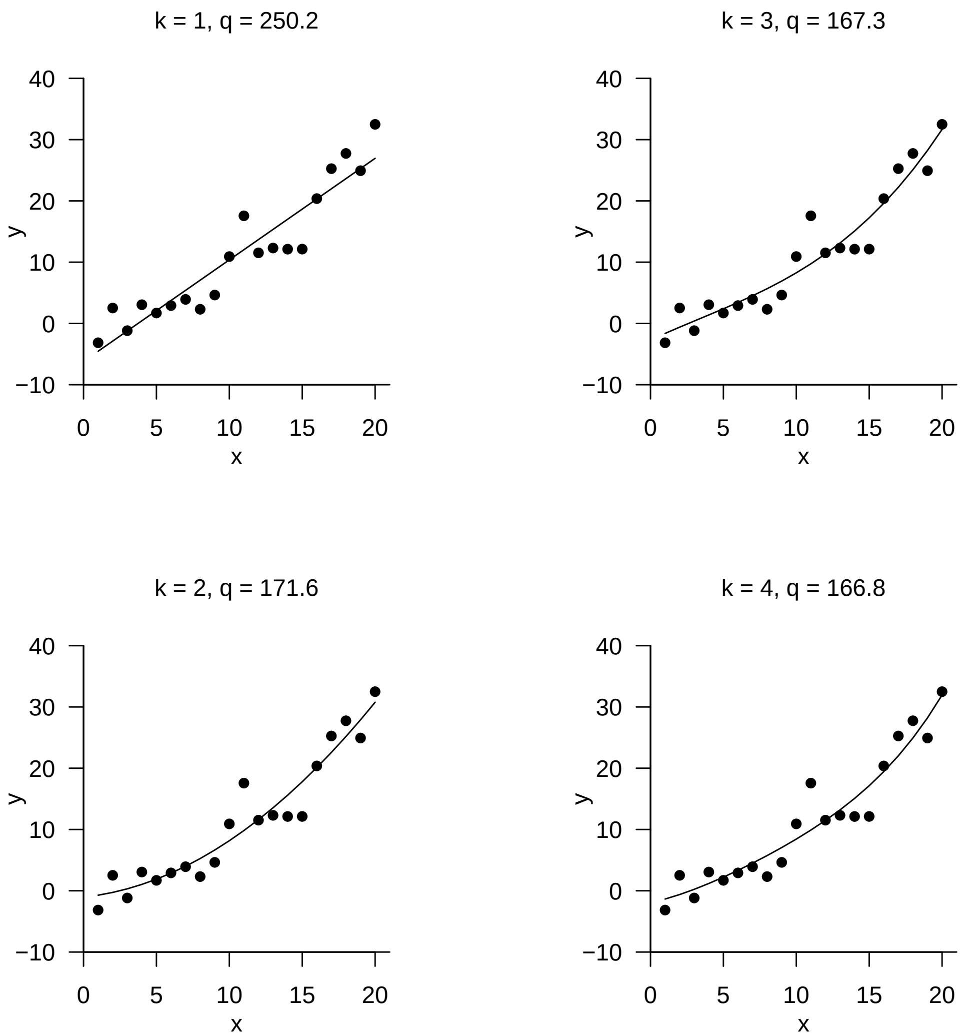


Figure 34.5 Fitting polynomials of first to fourth degree for the example dataset.

34.2 Simple linear regression

A fitting line allows statements about unobserved values of the dependent variable. However, a fitting line allows only implicit statements about the uncertainty associated with fitting a linear-affine function to a dataset. In simple linear regression, the idea of a fitting line is extended by a probabilistic component. In the sense of frequentist inference, simple linear regression thus makes it possible in particular to state confidence intervals for the fitting-line parameters and to carry out hypothesis tests concerning the fitting-line parameters. Here, we first want to consider only the model of simple linear regression and the maximum likelihood estimation of the fitting-line parameters based on it.

We then treat the assessment of the uncertainty associated with this estimation as well as parameter-centered hypothesis tests at a later point, first in the context of the general linear model. We begin with the following definition.

Definition 34.3 (Simple linear regression model)**.** Let

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \text{ with } \varepsilon_i \sim N\left(0, \sigma^2\right) \text{ i.i.d. for } i = 1, \dots, n \tag{34.22}$$

where

- y_i are observable random variables that model values of a dependent variable,
- $x_i \in \mathbb{R}$ are fixed predictor values or regressor values that model values of an independent variable,
- $\beta_0, \beta_1 \in \mathbb{R}$ are true, but unknown, offset and slope parameter values, and
- ε_i are independent and identically normally distributed non-observable random variables with true, but unknown, variance parameter $\sigma^2 > 0$ that model error or disturbance variables.

Then Equation 34.22 is called the *simple linear regression model.*

•

In contrast to the fitting line, random variables occur explicitly in the simple linear regression model. Specifically, the simple linear regression model defines how n observable (dependent) random variables y_i are generated using the values x_i of an independent variable, the parameter values β_0 and β_1, and addition of the normally distributed error variables ε_i. The model has three parameters, the offset parameter β_0, the slope parameter β_1, and the variance parameter σ^2 of the normally distributed error variables. Addition of the fixed values β_0 and $\beta_1 x_i$ to the normally distributed random variable ε_i implies a normal distribution of y_i. This is the statement of the following theorem.

Theorem 34.2 (Data distribution of simple linear regression)**.** *The simple linear regression model*

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \textit{ with } \varepsilon_i \sim N\left(0, \sigma^2\right) \textit{ i.i.d. for } i = 1, \dots, n \tag{34.23}$$

can equivalently be written, with $\mu_i := \beta_0 + \beta_1 x_i$, in the form

$$y_i \sim N\left(\mu_i, \sigma^2\right) \textit{ independently for } i = 1, \dots, n \tag{34.24}$$

∘

Proof. We show the equivalence for one i; we show the independence of the y_i at a later point in the framework of the general linear model. The equivalence of the two model forms for one i follows directly from the transformation of normally distributed random variables by linear-affine functions. Specifically, in the present case, for $\varepsilon_i \sim N(0, \sigma^2)$, we have

$$y_i = f(\varepsilon_i) \text{ with } f : \mathbb{R} \to \mathbb{R}, \varepsilon_i \mapsto f(\varepsilon_i) := \varepsilon_i + (\beta_0 + \beta_1 x_i) \tag{34.25}$$

By the PDF transformation theorem for linear-affine mappings, it then follows that

$$\begin{aligned}
p_{y_i}(y_i) &= \frac{1}{|1|} p_{\varepsilon_i}\left(\frac{y_i - \beta_0 - \beta_1 x_i}{1}\right) \\
&= N\left(y_i - \beta_0 - \beta_1 x_i; 0, \sigma^2\right) \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\left(y_i - \beta_0 - \beta_1 x_i - 0\right)^2\right) \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\left(y_i - (\beta_0 + \beta_1 x_i)\right)^2\right) \\
&= N\left(y_i; \beta_0 + \beta_1 x_i, \sigma^2\right).
\end{aligned} \tag{34.26}$$

Defining $\mu_i := \beta_0 + \beta_1 x_i$ then yields the statement of the theorem. □

Theorem 34.2 states in particular that the data variables y_i are univariate normally distributed random variables whose expectation parameters each depend on the value of the independent variable x_i. Figure 34.6 visualizes the model and a realization of simple linear regression for true, but unknown, parameter values $\beta_0 := 0$, $\beta_1 := 1$, and $\sigma^2 := 1$.

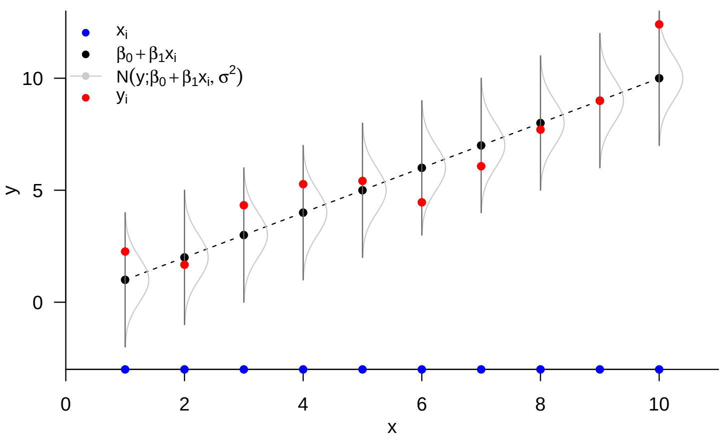


Figure 34.6 Simple linear regression model for $\beta_0 := 0$, $\beta_1 := 1$, and $\sigma^2 := 1$.

Because the simple linear regression model is a parametric frequentist model, estimators for the model parameters can be obtained using the maximum likelihood principle. In particular, it turns out that the maximum likelihood estimators of the offset and slope parameters are identical to the values of the fitting-line parameters. This is one of the statements of the following theorem. For notational clarity, we omit $^{\text{ML}}$ superscripts for the estimators here.

Theorem 34.3 (Maximum likelihood estimation of simple linear regression)**.**

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \textit{ with } \varepsilon_i \sim N\left(0, \sigma^2\right) \textit{ i.i.d. for } i = 1, \ldots, n \tag{34.27}$$

is the simple linear regression model. Then maximum likelihood estimators of the model parameters β_0, β_1, and σ^2 are given by

$$\hat{\beta}_1 := \frac{c_{xy}}{s_x^2}, \quad \hat{\beta}_0 := \bar{y} - \hat{\beta}_1 \bar{x} \quad \text{and } \hat{\sigma}^2 := \frac{1}{n}\sum_{i=1}^n \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2 \tag{34.28}$$

∘

Proof. We first show that the fitting-line parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ equal the corresponding maximum likelihood estimators. To this end, we first note that, because of the independence of $y_1, ..., y_n$, the likelihood function of the simple linear regression model with respect to β_0 and β_1 has the form

$$\begin{aligned} L : \mathbb{R}^2 \to \mathbb{R}_{>0}, (\beta_0, \beta_1) \mapsto L(\beta_0, \beta_1) : &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - (\beta_0 + \beta_1 x_i))^2\right) \\ &= \left(2\pi\sigma^2\right)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2\right) \end{aligned} \tag{34.29}$$

Because for the exponential function, for $a < b \le 0$, it holds that $\exp(a) < \exp(b)$, the exponential term of this likelihood function is maximized when the non-negative term

$$q := \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2 \tag{34.30}$$

becomes minimal and hence $-q$ becomes maximal. In the proof of the form of the fitting line, however, we have already shown that the term (1.30) is minimized for

$$\hat{\beta}_1 := \frac{c_{xy}}{s_x^2} \text{ and } \hat{\beta}_0 := \bar{y} - \hat{\beta}_1 \bar{x} \tag{34.31}$$

and that therefore $\hat{\beta}_1$ and $\hat{\beta}_0$ maximize the likelihood function.

In a second step, we now consider the likelihood function of the simple linear regression model with respect to σ^2 at the values of $\hat{\beta}_0$ and $\hat{\beta}_1$. We obtain the likelihood function

$$L : \mathbb{R}_{>0} \to \mathbb{R}_{>0}, \sigma^2 \mapsto L(\sigma^2) = \left(2\pi\sigma^2\right)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2\right) \tag{34.32}$$

and the corresponding log-likelihood function

$$\ell : \mathbb{R}_{>0} \to \mathbb{R}, \sigma^2 \mapsto \ell(\sigma^2) = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2 \tag{34.33}$$

Analogously to the derivation of the maximum likelihood estimator for σ^2 in the normal distribution model, taking into account

$$\hat{\mu} = \hat{\beta}_0 + \hat{\beta}_1 x_i \tag{34.34}$$

we then obtain here

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2 \tag{34.35}$$

□

In application, maximum likelihood estimation of the parameters of simple linear regression is thus essentially identical to determining the fitting-line parameters, as the following **R** code for estimating the parameters for the example dataset demonstrates.

```
# read example dataset
fname       = "./_data/501-regression.csv"
D           = read.table(fname, sep = ",", header = TRUE)

# sample statistics
n           = length(D$y_i)                                           # number of data points
x_bar       = mean(D$x_i)                                             # sample mean of x_i values
y_bar       = mean(D$y_i)                                             # sample mean of y_i values
s2x         = var(D$x_i)                                              # sample variance of x_i values
cxy         = cov(D$x_i, D$y_i)                                       # sample covariance of (x_i,y_i) values

# parameter estimators
beta_1_hat  = cxy/s2x                                                 # \hat{\beta}_1, slope parameter
beta_0_hat  = y_bar - beta_1_hat*x_bar                                # \hat{\beta}_0, offset parameter
sigsqr_hat  = (1/n)*sum((D$y_i-(beta_0_hat+beta_1_hat*D$x_i))^2)      # variance parameter

# output
cat("beta_0_hat:"  , beta_0_hat,
    "\nbeta_1_hat:", beta_1_hat,
    "\nsigsqr_hat:", sigsqr_hat)
```

```
beta_0_hat: -6.194704
beta_1_hat: 1.657055
sigsqr_hat: 12.50892
```

34.3 Bibliographic remarks

The idea of minimizing a sum of squared deviations when fitting a polynomial function to observed values goes back to the work of Legendre (1805) and Gauss (1809) in the context of determining planetary orbits. A historical classification is provided by Stigler (1981). The concept of regression goes back to Galton (1886). Stigler (1986) provides a detailed historical overview.

35 Correlation

The concepts of regression and correlation are closely intertwined and ultimately serve to quantify linear-affine dependencies between independent and dependent variables. A central topic of this section is therefore the equivalences between the two concepts. In contrast to regression, in correlation both the dependent and the independent variable are modeled as random variables; the probabilistic complexity of correlation is therefore somewhat higher than that of regression and of the general linear model. To approach the relationships between regression and correlation, we first recall the concept and mechanics of sample correlation in Section 35.1. In Section 35.2, we use the R^2 statistic to consider the relationship between sample correlation and fitting line and give a first example of the idea of a variance decomposition. Finally, in Section 35.3, we discuss first aspects for understanding sample correlation in a probabilistic context. As an application example, we use throughout the example dataset on the relationship between psychotherapy duration and symptom reduction introduced in Chapter 34.

35.1 Basics

We first recall the concept of the correlation of two random variables.

Definition 35.1 (Correlation)**.** The correlation of two random variables ξ and υ is defined as

$$\rho(\xi, \upsilon) := \frac{\mathbb{C}(\xi, \upsilon)}{\mathbb{S}(\xi)\mathbb{S}(\upsilon)} \tag{35.1}$$

where $\mathbb{C}(\xi, \upsilon)$ denotes the covariance of ξ and υ, and $\mathbb{S}(\xi)$ and $\mathbb{S}(\upsilon)$ denote the standard deviations of ξ and υ, respectively.

•

The number $\rho(\xi, \upsilon)$ is also called the correlation coefficient of ξ and υ. We have already seen that $-1 \le \rho(\xi, \upsilon) \le 1$ holds and that ξ and υ are called uncorrelated when $\rho(\xi, \upsilon) = 0$. Furthermore, we have already seen that independence of the random variables ξ and υ always implies uncorrelatedness of ξ and υ, but that, in general, uncorrelatedness of ξ and υ does not imply independence of ξ and υ.

If realizations of two random variables are available as a bivariate dataset, then the realization-specific sample correlation can be determined according to the following definition.

Definition 35.2 (Sample correlation)**.** Let $\{(x_1, y_1), ..., (x_n, y_n)\} \subset \mathbb{R}^2$ be a dataset. Furthermore, let:

- The sample means of the x_i and y_i be defined as

$$\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i \text{ and } \bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i \tag{35.2}$$

- The sample standard deviations of the x_i and y_i be defined as

$$s_x := \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2} \text{ and } s_y := \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}. \tag{35.3}$$

- The sample covariance of $(x_1, y_1), \ldots, (x_n, y_n)$ be defined as

$$c_{xy} := \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right) \tag{35.4}$$

Then the *sample correlation* of $(x_1, y_1), \ldots, (x_n, y_n)$ is defined as

$$r_{xy} := \frac{c_{xy}}{s_x s_y} \tag{35.5}$$

and is also called *Pearson's sample correlation coefficient.*

•

The following **R** code evaluates the sample correlation of the example dataset.

```
# load example dataset
fname = "./_data/502-correlation.csv"                                  # file path
D     = read.table(fname, sep = ",", header = TRUE)                    # load as dataframe
x_i   = D$x_i                                                          # x_i values
y_i   = D$y_i                                                          # y_i values
n     = length(x_i)                                                    # n

# "manual" computation of sample correlation
x_bar = (1/n)*sum(x_i)                                                 # \bar{x}
y_bar = (1/n)*sum(y_i)                                                 # \bar{y}
s_x   = sqrt(1/(n-1)*sum((x_i - x_bar)^2))                             # s_x
s_y   = sqrt(1/(n-1)*sum((y_i - y_bar)^2))                             # s_y
c_xy  = 1/(n-1) * sum((x_i - x_bar) * (y_i - y_bar))                   # c_{xy}
r_xy  = c_xy/(s_x * s_y)                                               # r_{xy}
print(r_xy)                                                            # output
```

```
[1] 0.9378162
```

```
# automatic computation with cor()
r_xy  = cor(x_i,y_i)                                                   # r_{xy}
print(r_xy)                                                            # output
```

```
[1] 0.9378162
```

In the example dataset, the number of therapy sessions and symptom reduction are therefore highly correlated with $r_{xy} = 0.93$. In general, absolute values of r_{xy} larger than about 0.70 are called high correlation, absolute values of r_{xy} between about 0.30 and 0.70 are called medium correlation, and absolute values of r_{xy} between 0.00 and 0.30 are called low correlation. A low correlation between two variables does not necessarily mean, however, that this correlation is irrelevant (think of health risk factors), just as a high correlation between two variables need not be trivial (think of the correlation between body height and shoe size).

Because the sample correlation is merely the sample covariance c_{xy} normalized to the interval $[-1, 1]$, the magnitude of the sample correlation and, in particular, its sign are

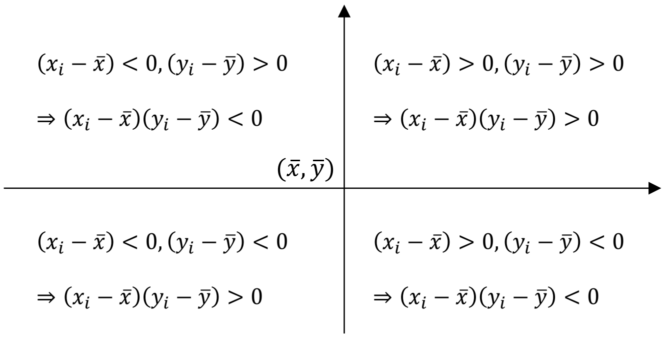


Figure 35.1 Mechanics of the sample-correlation sum terms.

decisively determined by the values of the sample covariance sum terms $(x_i - \bar{x})(y_i - \bar{y})$. What is decisive is how often, across the data pairs (x_i, y_i), the x_i and y_i deviate from their respective sample means in the same or in opposite directions. This is shown schematically in Figure 35.1. When deviations from the respective means occur frequently in the same direction, both in the positive and in the negative direction, the product $(x_i - \bar{x})(y_i - \bar{y})$ is positive and thus contributes to a positive sample covariance. When deviations from the respective means occur frequently in opposite directions, the product $(x_i - \bar{x})(y_i - \bar{y})$ is often negative and thus contributes to a negative sample covariance. If deviations of the x_i and y_i in the same direction and in opposite directions both occur often, positive and negative contributions to the sample covariance sum tend to cancel, and the result is a low sample covariance or a sample correlation coefficient near zero. Figure 35.2 shows bivariate datasets of $n = 30$ data points each together with their respective sample correlation coefficients.

The advantage of the sample correlation coefficient over the sample covariance as an association measure is that the absolute value of the sample correlation coefficient remains unchanged under linear-affine transformation of the underlying set of values, whereas the sample covariance changes its value depending on the chosen scale. One therefore also says that the sample correlation coefficient is scale-independent. This is the central statement of the following theorem.

Theorem 35.1 (Sample correlation under linear-affine transformation)**.** *For a dataset* $\{(x_i, y_i)\}_{i=1,...n} \subset \mathbb{R}^2$*, let* $\{(\tilde{x}_i, \tilde{y}_i)\}_{i=1,...n} \subset \mathbb{R}^2$ *be the linearly-affinely transformed dataset with*

$$(\tilde{x}_i, \tilde{y}_i) = \left(a_x x_i + b_x, a_y y_i + b_y\right), a_x, a_y \neq 0. \tag{35.6}$$

Then

$$\left|r_{\tilde{x}\tilde{y}}\right| = \left|r_{xy}\right|. \tag{35.7}$$

∘

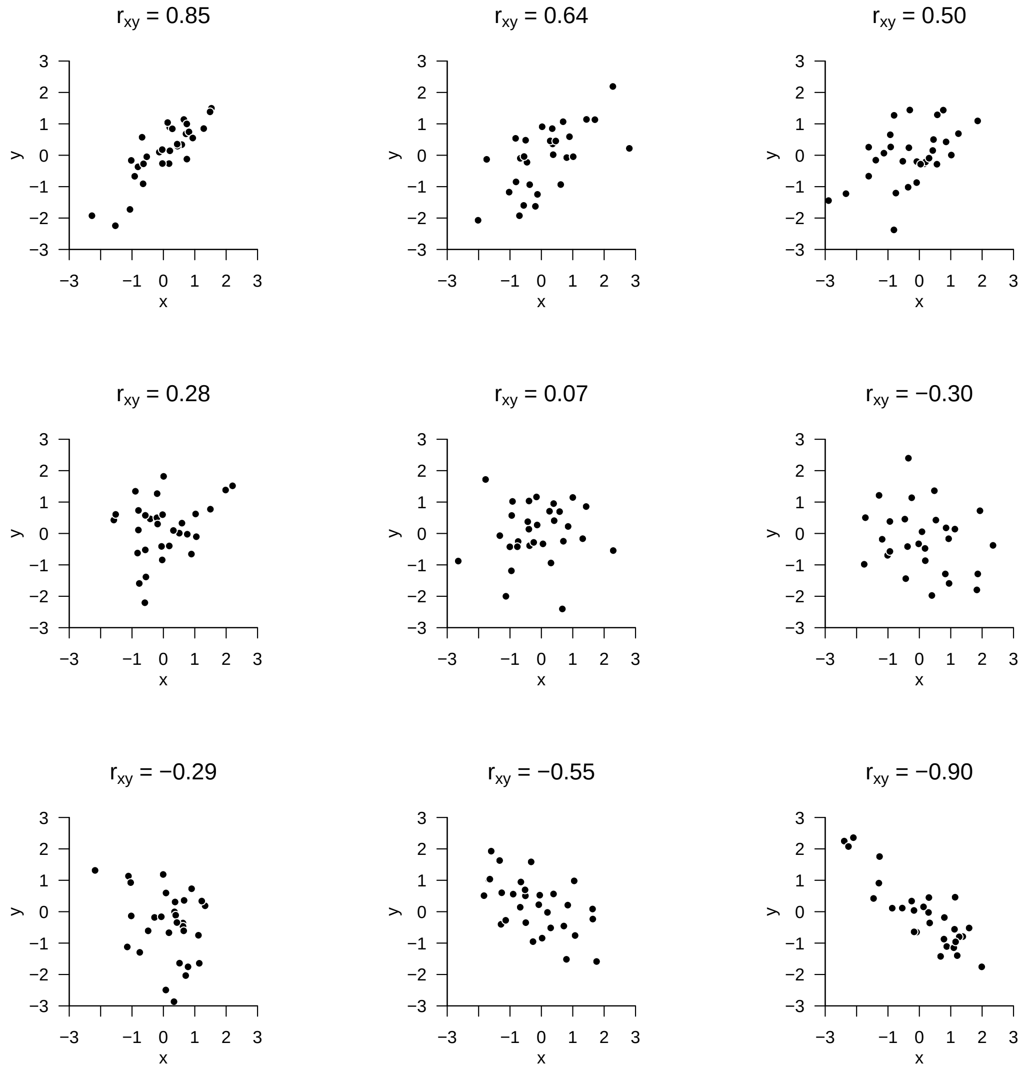


Figure 35.2 Examples of bivariate datasets and their sample correlations.

Proof.

$$
\begin{aligned}
r_{\tilde{x}\tilde{y}} &:= \frac{\frac{1}{n-1}\sum_{i=1}^{n}\left(\tilde{x}_i-\bar{\tilde{x}}\right)\left(\tilde{y}_i-\bar{\tilde{y}}\right)}{\sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n}\tilde{x}_i-\bar{\tilde{x}}\right)^2}\sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n}\tilde{y}_i-\bar{\tilde{y}}\right)^2}} \\
&= \frac{\sum_{i=1}^{n}\left(a_x x_i+b_x-(a_x\bar{x}+b_x)\right)\left(a_y y_i+b_y-(a_y\bar{y}+b_y)\right)}{\sqrt{\sum_{i=1}^{n}\left(a_x x_i+b_x-(a_x\bar{x}+b_x)\right)^2}\sqrt{\sum_{i=1}^{n}\left(a_y y_i+b_y-(a_y\bar{y}+b_y)\right)^2}} \\
&= \frac{a_x a_y\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{a_x^2\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{a_y^2\sum_{i=1}^{n}(y_i-\bar{y})^2}} \\
&= \frac{a_x a_y}{|a_x|\,|a_y|}\frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}} \\
&= \frac{a_x a_y}{|a_x|\,|a_y|}\frac{c_{xy}}{s_x s_y} \\
&= \frac{a_x a_y}{|a_x|\,|a_y|}r_{xy}.
\end{aligned}
\tag{35.8}
$$

Therefore, by considering all possible sign cases, it follows that

$$
\left|r_{\tilde{x}\tilde{y}}\right| = \left|r_{xy}\right|. \tag{35.9}
$$

□

35.2 Correlation and coefficient of determination

The so-called coefficient of determination R^2 is a popular, frequently reported statistic for describing the strength of association between the values of an independent and a dependent variable. Numerically, R^2 is merely the squared sample correlation coefficient. If the sample correlation is, for example, $r_{xy} = 0.5$, then $\mathrm{R}^2 = 0.5^2 = 0.25$; if, by contrast, the sample correlation is $r_{xy} = -0.5$, then analogously $\mathrm{R}^2 = (-0.5)^2 = 0.25$. These examples show that R^2 contains less information about the raw data than r_{xy} because the sign, and hence the direction of the association, is lost. In itself, reporting R^2 instead of r_{xy} as a descriptive statistic for describing the strength of association between the values of an independent and a dependent variable has no advantage. Nevertheless, we want to look at R^2 somewhat more closely here, because a deeper understanding of R^2 allows, on the one hand, entry into the concept of variance decompositions and, on the other hand, further clarifies the relationships between the concepts of fitting line and sample correlation. To this end, we first extend the description of the fitting line from Section 34.1 by the concepts of explained values and residuals.

Definition 35.3 (Explained values and residuals of a fitting line)**.** Given a dataset $\{(x_1, y_1), \ldots, (x_n, y_n)\} \subset \mathbb{R}^2$ and the fitting line belonging to this dataset

$$
f_{\hat{\beta}} : \mathbb{R} \to \mathbb{R}, x \mapsto f_{\hat{\beta}}(x) := \hat{\beta}_0 + \hat{\beta}_1 x \tag{35.10}
$$

then, for $i = 1, \ldots, n$,

$$
\hat{y}_i := \hat{\beta}_0 + \hat{\beta}_1 x_i \tag{35.11}
$$

are called the *explained values* by the fitting line, and

$$
\hat{\varepsilon}_i := y_i - \hat{y}_i \tag{35.12}
$$

are called the *residuals* of the fitting line.

•

Formulated somewhat more generally, the *explained values* are thus the data predictions of the model based on the estimated parameter values, whereas the residuals denote the differences between the estimated data predictions and the observed data values. Figure 35.3 illustrates these concepts using the fitting line of the example dataset.

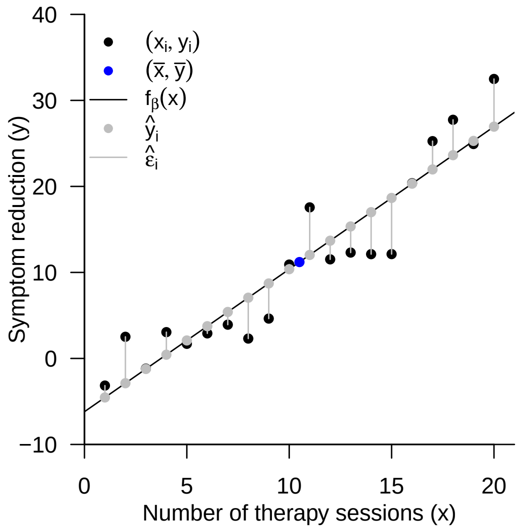


Figure 35.3 Fitting line, explained values, and residuals for the example dataset.

Using the concepts of explained values and residuals, the following sum-of-squares decomposition can now be stated when a fitting line of a dataset is available.

Theorem 35.2 (Sum-of-squares decomposition for a fitting line). *For a dataset* $\{(x_1, y_1), \ldots, (x_n, y_n)\} \subset \mathbb{R}^2$ *and its associated fitting line* $f_{\hat{\beta}}$, *let, for*

$$\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i \text{ and } \hat{y}_i := \hat{\beta}_0 + \hat{\beta}_1 x_i, \text{ for } i = 1, \ldots, n \tag{35.13}$$

the sample mean of the y_i *values and the values explained by the fitting line be given, respectively. Furthermore, let*

- *the total sum of squares be defined as*

$$SQT := \sum_{i=1}^{n} (y_i - \bar{y})^2 \tag{35.14}$$

- *the explained sum of squares be defined as*

$$SQE := \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 \tag{35.15}$$

- *the residual sum of squares be defined as*

$$SQR := \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{35.16}$$

Then

$$SQT = SQE + SQR \tag{35.17}$$

∘

Proof.

$$
\begin{aligned}
\text{SQT} &= \sum_{i=1}^{n} (y_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} ((y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}))^2 \\
&= \sum_{i=1}^{n} \left((y_i - \hat{y}_i)^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) + (\hat{y}_i - \bar{y})^2\right) \\
&= \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 + 2\sum_{i=1}^{n} (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) + \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\
&= \text{SQE} + 2\sum_{i=1}^{n} (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) + \text{SQR} \\
&= \text{SQE} + \text{SQR}
\end{aligned}
\tag{35.18}
$$

The last equality follows from

$$\bar{\hat{y}} := \frac{1}{n}\sum_{i=1}^{n} \hat{y}_i = \frac{1}{n}\sum_{i=1}^{n} \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x} = \bar{y} \tag{35.19}$$

and therefore also

$$\bar{\hat{y}} = \bar{y} \Leftrightarrow \frac{1}{n}\sum_{i=1}^{n} \hat{y}_i = \frac{1}{n}\sum_{i=1}^{n} y_i \Leftrightarrow \sum_{i=1}^{n} \hat{y}_i = \sum_{i=1}^{n} y_i \Leftrightarrow \bar{y}\sum_{i=1}^{n} \hat{y}_i = \bar{y}\sum_{i=1}^{n} y_i \tag{35.20}$$

as well as

$$\bar{\hat{y}} = \bar{y} \Leftrightarrow \frac{1}{n}\sum_{i=1}^{n} \hat{y}_i = \frac{1}{n}\sum_{i=1}^{n} y_i \Leftrightarrow \sum_{i=1}^{n} y_i = \sum_{i=1}^{n} \hat{y}_i \Leftrightarrow \sum_{i=1}^{n} y_i \hat{y}_i = \sum_{i=1}^{n} \hat{y}_i \hat{y}_i \tag{35.21}$$

from

$$
\begin{aligned}
\sum_{i=1}^{n} (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) &= \sum_{i=1}^{n} (y_i \hat{y}_i - y_i \bar{y} - \hat{y}_i \hat{y}_i + \hat{y}_i \bar{y}) \\
&= \sum_{i=1}^{n} y_i \hat{y}_i - \sum_{i=1}^{n} y_i \bar{y} - \sum_{i=1}^{n} \hat{y}_i \hat{y}_i + \sum_{i=1}^{n} \hat{y}_i \bar{y} \\
&= \sum_{i=1}^{n} y_i \hat{y}_i - \sum_{i=1}^{n} \hat{y}_i \hat{y}_i + \bar{y}\sum_{i=1}^{n} \hat{y}_i - \bar{y}\sum_{i=1}^{n} y_i \\
&= 0 + 0 \\
&= 0
\end{aligned}
\tag{35.22}
$$

□

The concepts introduced in Theorem 35.2 can be intuitively explained as follows:

- SQT represents the total variability of the y_i values around their mean $\bar{y}$.
- SQE represents the variability of the explained values $\hat{y}_i$ around their mean. Large values of SQE therefore represent a large absolute slope of the y_i with the x_i, and small values of SQE represent a small absolute slope of the y_i with the x_i. SQE is thus a measure of the strength of the linear association between the x_i and y_i values.

- SQR is the sum of squared residuals, because

$$\mathrm{SQR} := \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 := \sum_{i=1}^{n} \hat{\varepsilon}_i^2 \tag{35.23}$$

Large values of SQR therefore represent large deviations of the explained from the observed y_i values, and small values of SQR represent small deviations of the explained from the observed y_i values. SQR is thus a measure of the quality of the description of the dataset by the fitting line.

The central statement of Theorem 35.2 is that the total scatter of the y_i values around their mean $\bar{y}$ is composed precisely of the sum of the strength of the linear association between the x_i and y_i values (that is, the "deterministic influence" of the x_i on the y_i) and the deviations from this linear association (that is, the "noise"). Although SQT is formally not a variance measure, in this context one often speaks of a variance decomposition into explained variance and residual variance. This motif is a central aspect of the general linear model and will be taken up again in later chapters. For the moment, Theorem 35.2 allows the following definition of the coefficient of determination R^2.

Definition 35.4 (Coefficient of determination R^2)**.** For a dataset $\{(x_1, y_1), \dots, (x_n, y_n)\} \subset \mathbb{R}^2$ and its associated fitting line $f_{\hat{\beta}}$, as well as the corresponding explained sum of squares SQE and total sum of squares SQT,

$$\mathrm{R}^2 := \frac{\mathrm{SQE}}{\mathrm{SQT}} \tag{35.24}$$

is called the *coefficient of determination*.

•

The following theorem now gives the relationship mentioned above between the coefficient of determination and the sample correlation.

Theorem 35.3 (Sample correlation and coefficient of determination)**.** *For a dataset $\{(x_1, y_1), \dots, (x_n, y_n)\} \subset \mathbb{R}^2$, let R^2 be the coefficient of determination and let r_{xy} be the sample correlation. Then*

$$\mathrm{R}^2 = r_{xy}^2 \tag{35.25}$$

∘

Proof. We first note that, with

$$\bar{\hat{y}} := \frac{1}{n}\sum_{i=1}^{n} \hat{y}_i = \frac{1}{n}\sum_{i=1}^{n} \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x} = \bar{y} \tag{35.26}$$

it follows that

$$\begin{aligned}
\mathrm{SQE} &= \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} (\hat{y}_i - \bar{\hat{y}})^2 \\
&= \sum_{i=1}^{n} \left(\hat{\beta}_0 + \hat{\beta}_1 x_i - \hat{\beta}_0 - \hat{\beta}_1 \bar{x}\right)^2 \\
&= \sum_{i=1}^{n} \left(\hat{\beta}_1 (x_i - \bar{x})\right)^2 \\
&= \hat{\beta}_1^2 \sum_{i=1}^{n} (x_i - \bar{x})^2
\end{aligned} \tag{35.27}$$

This then yields

$$
\begin{aligned}
\mathrm{R}^2 &= \frac{\mathrm{SQE}}{\mathrm{SQT}} \\
&= \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\
&= \hat{\beta}_1^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\
&= \frac{c_{xy}^2}{s_x^4} \frac{\frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})^2}{\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2} \\
&= \frac{c_{xy}^2}{s_x^4} \frac{s_x^2}{s_y^2} \\
&= \frac{c_{xy}^2}{s_x^2 s_y^2} \\
&= \left(\frac{c_{xy}}{s_x s_y}\right)^2 \\
&= r_{xy}^2 .
\end{aligned}
\tag{35.28}
$$

□

Note that, with $-1 \leq r_{xy} \leq 1$, Theorem 35.3 directly implies that $0 \leq \mathrm{R}^2 \leq 1$. According to Definition 35.4, $\mathrm{R}^2 = 0$ holds exactly when $\mathrm{SQE} = 0$. Thus, $\mathrm{R}^2 = 0$ means that the data variability explained by the fitting line is zero and therefore describes the case of the worst possible data explanation by the fitting line. On the other hand, $\mathrm{R}^2 = 1$ holds exactly when $\mathrm{SQE} = \mathrm{SQT}$. Thus, $\mathrm{R}^2 = 1$ means that the total scatter equals the scatter explained by the fitting line and describes the case in which all data variability can be explained by the fitting line. One therefore also often says, somewhat imprecisely, that R^2 represents the variance explained by the fitting line as a proportion of the total variance of the data. In addition to the respective sample correlation, Figure 35.2 also lists the coefficient of determination for each of the bivariate example datasets.

35.3 Correlation and linear-affine dependence

The fact that stochastic independence implies uncorrelatedness, but conversely the uncorrelatedness of two random variables does not imply their stochastic independence, suggests that the correlation of two random variables measures only certain forms of dependence between variables. Figure 35.4 illustrates this using three simulation examples.

Figure 35.4 A shows a realization of the model

$$
y_i = x_i + \varepsilon_i \text{ with } \varepsilon_i \sim N(0,1) \text{ for } i = 1, ..., n. \tag{35.29}
$$

The sample correlation of the dataset $\{(x_i, y_i)\}_{i=1}^n$ is $r_{xy} = 0.90$ here. There is a clear dependence of the value of the y_i realization y_i on the value x_i: the higher the value of x_i, the higher the expectation for the value of y_i. Figure 35.4 B shows a realization of the model

$$
y_i = x_i^2 + \varepsilon_i \text{ with } \varepsilon_i \sim N(0,1) \text{ for } i = 1, ..., n. \tag{35.30}
$$

The sample correlation of the dataset $\{(x_i, y_i)\}_{i=1}^n$ is $r_{xy} = -0.01$ here. The sample correlation is therefore minimal. However, here too there is a clear dependence of the value of the y_i realization y_i on the value x_i: the higher or lower the value of x_i, the higher

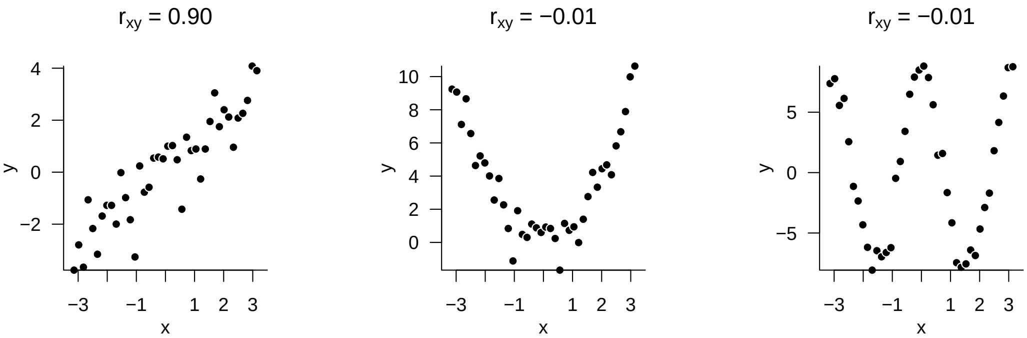


Figure 35.4 Correlation and dependence.

the expectation for the value of y_i; there is a quadratic relationship. A similar picture emerges when considering Figure 35.4 C, a realization of the model

$$y_i = 8\cos(2x_i) + \varepsilon_i \text{ with } \varepsilon_i \sim N(0,1) \text{ for } i = 1, ..., n. \tag{35.31}$$

Here too, a dependence of the value of the y_i realization y_i on the value x_i is apparent, in this case in the sense of a cyclic dependence, but the sample correlation is again minimal with $r_{xy} = -0.01$. These simulation examples therefore intuitively demonstrate that sample correlation can be vanishingly small even when clear dependencies between two variables exist. As a general measure of the dependence of two variables, correlation is therefore unsuitable. This circumstance is based on the fact that correlation is only a measure of the linear-affine association of two random variables, but not of their stochastic dependence. We formalize and specify this statement in the following theorem.

Theorem 35.4 (Correlation and linear-affine dependence)**.** *Let x and y be two random variables with positive variance. Then there is a linear-affine dependence of the form*

$$y = \beta_0 + \beta_1 x \textit{ with } \beta_0, \beta_1 \in \mathbb{R} \tag{35.32}$$

between x and y if and only if

$$\rho(x,y) = 1 \textit{ or } \rho(x,y) = -1 \tag{35.33}$$

holds.

∘

Proof. We restrict ourselves to proving the statement that $y = \beta_0 + \beta_1 x$ implies $\rho(x,y) = \pm 1$. To this end, we first note that, by the theorems on the properties of expectation and variance,

$$\mathbb{E}(y) = \beta_0 + \beta_1 \mathbb{E}(x) \text{ and } \mathbb{V}(y) = \beta_1^2 \mathbb{V}(x) \tag{35.34}$$

hold. Because $\mathbb{V}(x) > 0$ and $\mathbb{V}(y) > 0$, it follows that $\beta_1 \neq 0$. Thus,

$$\beta_1 > 0 \Rightarrow \mathbb{S}(y) = \beta_1 \mathbb{S}(x) > 0 \text{ and } \beta_1 < 0 \Rightarrow \mathbb{S}(y) = -\beta_1 \mathbb{S}(x) > 0. \tag{35.35}$$

Furthermore,

$$\begin{aligned} y - \mathbb{E}(y) &= \beta_0 + \beta_1 x - \mathbb{E}(y) \\ &= \beta_0 + \beta_1 x - \beta_0 - \beta_1 \mathbb{E}(x) \\ &= \beta_1 x - \beta_1 \mathbb{E}(x) \\ &= \beta_1 (x - \mathbb{E}(x)). \end{aligned} \tag{35.36}$$

For the covariance of x and y, we therefore obtain

$$\begin{aligned}\mathbb{C}(x,y) &= \mathbb{E}((y-\mathbb{E}(y))(x-\mathbb{E}(x))) \\ &= \mathbb{E}\left(\beta_1(x-\mathbb{E}(x))(x-\mathbb{E}(x))\right) \\ &= \beta_1\mathbb{E}\left((x-\mathbb{E}(x))^2\right) \\ &= \beta_1\mathbb{V}(x).\end{aligned} \tag{35.37}$$

Thus, for the correlation of x and y,

$$\rho(x,y) = \frac{\mathbb{C}(x,y)}{\mathbb{S}(x)\mathbb{S}(y)} = \pm\frac{\beta_1\mathbb{V}(x)}{\mathbb{S}(x)\beta_1\mathbb{S}(x)} = \pm\frac{\beta_1\mathbb{V}(x)}{\beta_1\mathbb{V}(x)} = \pm 1 \tag{35.38}$$

□

The correlation of two random variables is therefore maximal exactly when a linear-affine relationship exists between the two random variables. Here, linear-affine dependence of y on x also always implies linear-affine dependence of x on y, because

$$y = \beta_0 + \beta_1 x \Leftrightarrow -\beta_0 + y = \beta_1 x \Leftrightarrow x = -\frac{\beta_0}{\beta_1} + \frac{1}{\beta_1}y \Leftrightarrow x = \tilde{\beta}_0 + \tilde{\beta}_1 y \tag{35.39}$$

with

$$\tilde{\beta}_0 = -\frac{\beta_0}{\beta_1} \text{ and } \tilde{\beta}_1 = \frac{1}{\beta_1}. \tag{35.40}$$

35.4 Bibliographic remarks

The concept of correlation appears, though based on earlier work for example by Bravais (1844), first in Galton (1890) (cf. Stigler (1986)), and is further developed among others through the work of Pearson (1895), Pearson (1896), Pearson (1900), and Pearson (1901) in the context of multivariate normal distributions. An early study on the relationship between correlation and causality is Wright (1921).

36 Model formulation

36.1 General theory

We define the general linear model (GLM) as follows.

Definition 36.1 (General linear model)**.** Let

$$y = X\beta + \varepsilon \tag{36.1}$$

where

- y is an n-dimensional observable random vector, called the *data*,
- $X \in \mathbb{R}^{n\times p}$ for $n > p$ and $\text{rank}(X) = p$ is a matrix, called the *design matrix*,
- $\beta \in \mathbb{R}^p$ is an unknown parameter vector, called the *beta-parameter vector*,
- ε is an n-dimensional unobservable random vector, called the *random error*, for which it is assumed that, with an unknown variance parameter $\sigma^2 > 0$,

$$\varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{36.2}$$

Then Equation 36.1 is called the *general linear model (GLM)*.

•

In Equation 36.1, we refer to $X\beta \in \mathbb{R}^n$ as the *deterministic aspect* of the GLM and to ε as the *probabilistic aspect* of the GLM. The GLM thus postulates that data arise by adding a deterministic aspect $X\beta \in \mathbb{R}^n$ and a multivariate normally distributed probabilistic aspect ε. Note that $X\beta \in \mathbb{R}^n$ is an n-dimensional vector and ε is an n-dimensional random vector. The resulting vector y is a random vector because it results from adding the random vector ε to the vector $X\beta \in \mathbb{R}^n$. The GLM is thus a probabilistic model in which datasets represented by y are modeled. From a generative perspective, in the GLM a dataset $y \in \mathbb{R}^n$ arises as a realization of y by adding the deterministic model aspect and an unobservable realization $e \in \mathbb{R}^n$ of ε,

$$y = X\beta + e \tag{36.3}$$

with $y \in \mathbb{R}^n, X\beta \in \mathbb{R}^n$, and $e \in \mathbb{R}^n$.

The total number of parameters of the GLM is $p+1$, consisting of p scalar beta parameters and one variance parameter σ^2. The beta-parameter vector $\beta \in \mathbb{R}^p$ is also called a weight vector or effect vector. Its entries weight the entries of the columns of the design matrix $X \in \mathbb{R}^{n\times p}$ in generating the deterministic model aspect $X\beta \in \mathbb{R}^n$. The columns of the design matrix are named differently in different contexts; common terms are, for example, predictors, regressors, or covariates. In general, the columns of the design matrix model independent variables, and the data vector models dependent variables.

Note that the covariance-matrix parameter of ε is assumed to be spherical. It follows directly that $\varepsilon_1, \dots, \varepsilon_n$ are independent normally distributed random variables with identical variance parameter. Because the expectation parameter of ε is additionally assumed to be $0_n \in \mathbb{R}^n$, the $\varepsilon_1, \dots, \varepsilon_n$ are also identically normally distributed random variables. If $x_{ij} \in \mathbb{R}$ denotes the ijth element of the design matrix $X \in \mathbb{R}^{n \times p}$, then, for each component $y_i, i = 1, \dots, n$ of y, Equation 36.1 implies that

$$y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{ip}\beta_p + \varepsilon_i \text{ with } \varepsilon_1, \dots, \varepsilon_n \sim N\left(0, \sigma^2\right). \tag{36.4}$$

We record the distribution of the data vector y implied by Equation 36.1 in the following theorem.

Theorem 36.1 (Data distribution of the general linear model)**.** *Let*

$$y = X\beta + \varepsilon \textit{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{36.5}$$

be the GLM. Then

$$y \sim N\left(\mu, \sigma^2 I_n\right) \textit{ with } \mu := X\beta \in \mathbb{R}^n \tag{36.6}$$

∘

Proof. By the theorem on linear-affine transformations of multivariate normal distributions, for

$$\varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \text{ and } y := I_n\varepsilon + X\beta, \tag{36.7}$$

we have

$$y \sim N\left(I_n 0_n + X\beta, I_n\left(\sigma^2 I_n\right) I_n^T\right) = N\left(X\beta, \sigma^2 I_n\right). \tag{36.8}$$

With the definition $\mu := X\beta \in \mathbb{R}^n$, the statement of the theorem follows directly. □

In the GLM, the data y are therefore an n-dimensional normally distributed random vector with expectation parameter $\mu = X\beta \in \mathbb{R}^n$ and covariance-matrix parameter $\sigma^2 I_n \in \mathbb{R}^{n \times n}$. The GLM is thus a multivariate normal distribution whose expectation parameter is parameterized by means of a design matrix and a beta-parameter vector. Moreover, the components $y_1, \dots, y_n$ of y, that is, the random variables that model scalar data points, are independent normally distributed random variables of the form

$$y_i \sim N\left((X\beta)_i, \sigma^2\right) \text{ for } i = 1, \dots, n. \tag{36.9}$$

Because, however, $(X\beta)_i \neq (X\beta)_j$ generally holds for $i \neq j$, the $y_i, i = 1, \dots, n$ are generally not identically distributed. The scenario of independent and identically normally distributed random variables can, of course, be formulated as a special case of the GLM.

Example (1) Independent and identically normally distributed random variables

We consider the scenario of n independent and identically normally distributed random variables with expectation parameter $\mu \in \mathbb{R}$ and variance parameter σ^2,

$$y_1, \dots, y_n \sim N\left(\mu, \sigma^2\right). \tag{36.10}$$

Then Equation 36.10 is equivalent to

$$y_i = \mu + \varepsilon_i, \varepsilon_i \sim N\left(0, \sigma^2\right) \text{ for } i = 1, \dots, n \text{ with independent } \varepsilon_i. \tag{36.11}$$

In matrix notation, this is in turn equivalent to the GLM special case

$$y \sim N\left(X\beta, \sigma^2 I_n\right) \text{ with } X := 1_n \in \mathbb{R}^{n\times 1}, \beta := \mu \in \mathbb{R}^1, \sigma^2 > 0. \tag{36.12}$$

In **R**, realizations of the GLM can easily be obtained using a random-number generator for multivariate normal distributions by specifying the corresponding expectation and covariance parameters. The following **R** code shows how n independent and identically normally distributed scalar data points can be realized in the sense of the GLM. Note that n scalar data points correspond to one realization of the GLM.

```
# model formulation
library(MASS)                                                   # multivariate normal distribution
set.seed(0)                                                     # random number generator seed
n      = 12                                                     # number of data points
p      = 1                                                      # number of beta parameters
X      = matrix(rep(1,n), nrow = n)                             # design matrix
I_n    = diag(n)                                                # n x n identity matrix
beta   = 2                                                      # true, but unknown, beta parameter
sigsqr = 1                                                      # true, but unknown, variance parameter

# data realization
y      = mvrnorm(1, X %*% beta, sigsqr*I_n)                     # one realization of an n-dimensional random vector
```

```
Realizations:  1.2 2.76 4.4 1.99 1.71 1.07 0.46 2.41 3.27 3.33 1.67 3.26
```

Example (2) Simple linear regression

We consider the generative model of simple linear regression

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \varepsilon_i \sim N\left(0, \sigma^2\right) \text{ for } i = 1, \dots, n, \tag{36.13}$$

We have already seen that this model is equivalent to the normal-distribution model of regression

$$y_i \sim N\left(\mu_i, \sigma^2\right) \text{ with } \mu_i := \beta_0 + \beta_1 x_i \text{ for } i = 1, \dots, n. \tag{36.14}$$

In matrix notation, this is in turn equivalent to the GLM special case

$$y \sim N\left(X\beta, \sigma^2 I_n\right) \text{ with } X := \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \in \mathbb{R}^{n\times 2}, \beta := \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \in \mathbb{R}^2, \sigma^2 > 0. \tag{36.15}$$

R code for simulating realizations of a simple linear regression therefore has a very similar structure to the **R** code above for simulating realizations of n independent and identically normally distributed random variables.

```
# model formulation
library(MASS)                                                   # multivariate normal distribution
set.seed(0)                                                     # random number generator seed
n      = 10                                                     # number of data points
p      = 2                                                      # number of beta parameters
x      = 1:n                                                    # predictor values
X      = matrix(c(rep(1,n),x), nrow = n)                        # design matrix
I_n    = diag(n)                                                # n x n identity matrix
beta   = matrix(c(0,1), nrow = p)                               # true, but unknown, beta parameter
sigsqr = 1                                                      # true, but unknown, variance parameter

# data realization
y      = mvrnorm(1, X %*% beta, sigsqr*I_n)                     # one realization of an n-dimensional random vector
```

```
Realizations:  3.4 1.99 2.71 3.07 3.46 6.41 8.27 9.33 8.67 11.26
```

We visualize the above realization in Figure 36.1.

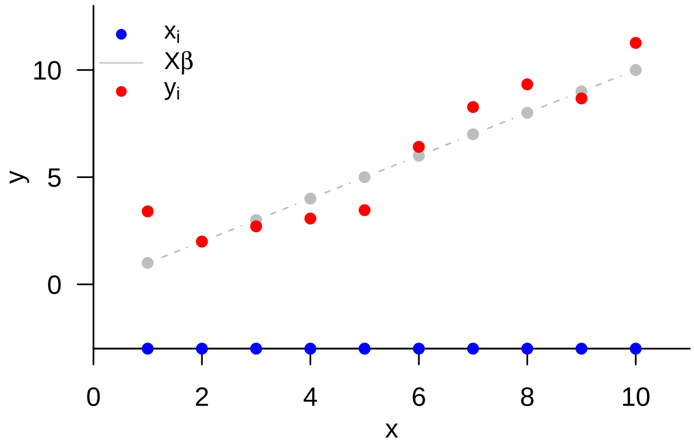


Figure 36.1 Realization of a simple linear regression.

36.2 Identifiability and estimability

Above, we saw that the data distribution of the GLM is given by

$$y \sim N\left(\mu, \sigma^2 I_n\right) \text{ with } \mu := X\beta \in \mathbb{R}^n \text{ for } X \in \mathbb{R}^{n \times p}, \beta \in \mathbb{R}^p. \tag{36.16}$$

The GLM is therefore a multivariate normal distribution with a special expectation-parameterization. To introduce the concepts of identifiability and estimability in the context of GLMs, it is helpful first to generalize the concept of a *parameterization of a multivariate normal distribution* somewhat.

Definition 36.2 (Beta parameterization)**.** Let $N\left(\mu, \sigma^2 I_n\right)$ be a multivariate normal distribution with spherical covariance-matrix parameter. We then call a multivariate, vector-valued function of the form

$$f : \mathbb{R}^p \to \mathbb{R}^n, \beta \mapsto f(\beta) =: \mu \tag{36.17}$$

a beta parameterization of μ.

•

The GLM is evidently based on the design-matrix-dependent beta parameterization

$$f : \mathbb{R}^p \to \mathbb{R}^n, \beta \mapsto f(\beta) := X\beta. \tag{36.18}$$

Using the concept of beta parameterization, we can now formulate the concept of *beta-parameter identifiability*:

Definition 36.3 (Beta-parameter identifiability)**.** Let $N(\mu, \sigma^2 I_n)$ be a multivariate normal distribution with spherical covariance-matrix parameter, and let f be a beta parameterization of μ. β is called identifiable if and only if, for arbitrary $\beta_1, \beta_2 \in \mathbb{R}^p$, it holds that $f(\beta_1) = f(\beta_2)$ implies $\beta_1 = \beta_2$.

•

The beta parameters of general linear models whose design matrix has full rank are identifiable. This is the statement of the following theorem:

Theorem 36.2 (Beta-parameter identifiability under full design-matrix rank)**.** *Let $y \sim N(X\beta, \sigma^2 I_n)$ be the data distribution of a GLM with rank$(X) = p$. Then β is identifiable.*

∘

Proof. For $X \in \mathbb{R}^{n \times p}$, $\operatorname{rank}(X) = p$ implies that $(X^T X) \in \mathbb{R}^{p \times p}$ is an invertible matrix. It then follows, for arbitrary $\beta_1, \beta_2 \in \mathbb{R}^p$, that

$$\begin{aligned} f(\beta_1) = f(\beta_2) &\Leftrightarrow X\beta_1 = X\beta_2 \\ &\Leftrightarrow X^T X\beta_1 = X^T X\beta_2 \\ &\Leftrightarrow (X^T X)^{-1} X^T X\beta_1 = (X^T X)^{-1} X^T X\beta_2 \\ &\Leftrightarrow \beta_1 = \beta_2. \end{aligned} \tag{36.19}$$

□

For the analysis of estimable functions, we also need the concept of identifiability of vector-valued functions of the beta parameters. We define:

Definition 36.4 (Identifiability of functions of the beta parameters)**.** Let $N(\mu, \sigma^2 I_n)$ be a multivariate normal distribution with spherical covariance-matrix parameter, and let f be a beta parameterization of μ. Furthermore, let

$$g : \mathbb{R}^p \to \mathbb{R}^k, \beta \mapsto g(\beta) \tag{36.20}$$

be a function of the beta-parameter vector. g is called identifiable if and only if, for arbitrary $f(\beta_1), f(\beta_2) \in \mathbb{R}^n$, $f(\beta_1) = f(\beta_2)$ implies $g(\beta_1) = g(\beta_2)$.

•

Estimable functions are linear functions of β that are identifiable. In general, the following theorem holds:

Theorem 36.3 (Identifiable beta-parameter functions)**.** *Let $N(\mu, \sigma^2 I_n)$ be a multivariate normal distribution with spherical covariance-matrix parameter, and let f be a beta parameterization of μ. Furthermore, let*

$$g : \mathbb{R}^p \to \mathbb{R}^k, \beta \mapsto g(\beta) \tag{36.21}$$

be a function of the beta-parameter vector. The function g is identifiable if and only if g is a function of f, that is, if there exists a function ϕ such that

$$g = \phi \circ f \tag{36.22}$$

∘

Proof. We show the statement only in one direction. $\Rightarrow$ We assume that there exists a function ϕ such that

$$g = \phi \circ f. \tag{36.23}$$

Then the fact that a function assigns exactly one function value to an argument implies that

$$f(\beta_1) = f(\beta_2) \Leftrightarrow \phi(f(\beta_1)) = \phi(f(\beta_2)) \Leftrightarrow g(\beta_1) = g(\beta_2) \tag{36.24}$$

Thus g is identifiable, because $f(\beta_1) = f(\beta_2)$ implies $g(\beta_1) = g(\beta_2)$. □

As mentioned above, estimable functions are linear functions of β that are identifiable. The classical definition of an estimable function is the following.

Definition 36.5 (Estimable function)**.** Let $N(X\beta, \sigma^2 I_n)$ be a GLM. Then a linear function

$$g : \mathbb{R}^p \to \mathbb{R}^k, \beta \mapsto g(\beta) := C^T\beta \tag{36.25}$$

with $C \in \mathbb{R}^{p\times k}$ is called estimable if there exists a matrix $P \in \mathbb{R}^{n\times k}$ such that

$$C^T\beta = P^T X\beta. \tag{36.26}$$

•

This definition can be understood as follows: by the theorem on identifiable functions, an identifiable linear function of β must be a function of the form

$$g(\beta) = (\phi \circ f)(\beta) \tag{36.27}$$

Because, furthermore, for a GLM $f(\beta) = X\beta$ and g is a linear function, and can therefore be written using a matrix C^T, ϕ must also be a linear function. Hence ϕ can also be written as

$$\phi(f(\beta)) = \phi(X\beta) = P^T X\beta \tag{36.28}$$

with a suitable matrix $P \in \mathbb{R}^{n\times k}$.

36.3 Design spectrum

The choice of design matrix and beta parameters opens up great freedom for implementing a wide range of expectation-parameter scenarios of the GLM data distribution. In principle, all special designs lie on a continuum between the following two extremes:

(1) The expectations of all data variables are identical, that is,

$$y_i \sim N(\mu, \sigma^2) \text{ i.i.d. for } i = 1, \dots, n \tag{36.29}$$

that is,

$$y = X\beta + \varepsilon \text{ with } X := 1_n \in \mathbb{R}^{n\times 1}, \beta := \mu \in \mathbb{R}, \varepsilon \sim N(0_n, \sigma^2 I_n). \tag{36.30}$$

(2) The expectations of all data variables are pairwise different, that is,

$$y_i \sim N(\mu_i, \sigma^2) \text{ independently for } i = 1, \dots, n, \tag{36.31}$$

that is,

$$y = X\beta + \varepsilon \text{ with } X := I_n \in \mathbb{R}^{n\times n}, \beta := (\mu_1, \dots, \mu_n)^T \in \mathbb{R}^n, \varepsilon \sim N(0_n, \sigma^2 I_n). \tag{36.32}$$

In scenario (1), all data variability is attributed to the random-error term; in scenario (2), by contrast, all data variability is attributed to the expectation parameter. Both extreme scenarios are not scientifically fruitful, because they do not represent any theory-guided systematic dependence between independent and dependent variables. All GLM designs considered in the following chapters therefore lie between these two extreme scenarios and represent different forms of systematic dependence between independent and dependent variables. In particular, one distinguishes

(1) *Factorial designs*, in which the design matrix essentially contains only 1s and 0s and sometimes -1s. In this case, the beta parameters represent group expectations, and we will see that the beta-parameter estimators result in representations of group sample means. One sometimes says that factorial designs serve the investigation of *difference hypotheses*. Examples of factorial designs are different designs for implementing T*-tests* and *analyses of variance*.
(2) *Parametric designs*, in which the design matrix consists of columns with continuous real values. Especially in this context, the columns of the design matrix are often called *regressors* or *predictors*. In this case, the beta parameters represent partial slope parameters, and we will see that the corresponding beta-parameter estimators arise as normalized regressor-data covariances. One sometimes says that parametric designs serve the investigation of *association hypotheses*. Examples of parametric designs are different designs for implementing *simple linear regression* and, in particular, *multiple linear regression*.
(3) *Factorial-parametric designs*, in which the columns of the design matrix represent both factorial and parametric predictors. In this context, the parametric regressors are often considered *covariates*. Mixed factorial-parametric designs are the central characteristic of *analysis of covariance*, which aims at a controlled investigation of difference hypotheses in the presence of further possible dependencies between independent and dependent variables, or at the controlled investigation of association hypotheses in the presence of further possible group differences.

36.4 Bibliographic remarks

The general linear model has a long history, whose modern incarnation is usually traced back to the work of Legendre (1805) and Gauss (1809). Matrix-based formulations of multiple regression can be found at least as early as Aitken (1936) and Scheffé (1959). The general linear model entered the canon of psychological methods at least with Cohen (1968). Seal (1967) gives a detailed overview of the history of the general linear model in the nineteenth and first half of the twentieth century. The discussion of identifiability and estimability given in this section is based on the presentation in Christensen (2011).

37 Parameter estimation

In this section, we consider frequentist point estimation of the beta-parameter vector and the variance parameter in the GLM. As example applications, we consider the scenario of n independent and identically normally distributed random variables and the scenario of simple linear regression. We conclude with the documentation of the frequentist parameter-estimator distributions of the GLM.

37.1 Beta-parameter estimation

We summarize frequentist point estimation of the beta-parameter vector in the following theorem.

Theorem 37.1 (Beta-parameter estimator). *Let*

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{37.1}$$

be the general linear model, and let

$$\hat{\beta} := \left(X^T X\right)^{-1} X^T y. \tag{37.2}$$

Then:

(1) $\hat{\beta}$ is the least-squares estimator of $\beta \in \mathbb{R}^p$; that is, for an arbitrary fixed value $y \in \mathbb{R}^n$ of y,

$$\hat{\beta} = \operatorname{argmin}_{\tilde{\beta}}(y - X\tilde{\beta})^T(y - X\tilde{\beta}). \tag{37.3}$$

(2) $\hat{\beta}$ is an unbiased maximum-likelihood estimator of $\beta \in \mathbb{R}^p$.

∘

Proof. (1) In a first step, we show that $\hat{\beta}$ is a least-squares estimator; that is, for an arbitrary fixed value $y \in \mathbb{R}^n$ of y, $\hat{\beta}$ minimizes the sum of squared deviations

$$(y - X\tilde{\beta})^T(y - X\tilde{\beta}) = \sum_{i=1}^n \left(y_i - (X\tilde{\beta})_i\right)^2 \tag{37.4}$$

(the notation $\tilde{\beta}$ for the minimization argument is used here only to distinguish it from the true, but unknown, parameter value $\beta \in \mathbb{R}^p$ and has no further meaning). To this end, we first note that

$$\hat{\beta} = \left(X^T X\right)^{-1} X^T y \Leftrightarrow X^T X\hat{\beta} = X^T y \Leftrightarrow X^T y - X^T X\hat{\beta} = 0_p \Leftrightarrow X^T(y - X\hat{\beta}) = 0_p. \tag{37.5}$$

Furthermore, it also follows that

$$X^T(y - X\hat{\beta}) = 0_p \Leftrightarrow \left(X^T(y - X\hat{\beta})\right)^T = 0_p^T \Leftrightarrow (y - X\hat{\beta})^T X = 0_p^T. \tag{37.6}$$

We also note without proof that, for every matrix $X \in \mathbb{R}^{n\times p}$,

$$z^T X^T X z \geq 0 \text{ for all } z \in \mathbb{R}^p. \tag{37.7}$$

For fixed y and arbitrary $\tilde{\beta}$, we now consider the sum of squared deviations

$$(y - X\tilde{\beta})^T(y - X\tilde{\beta}). \tag{37.8}$$

This gives

$$\begin{aligned}
&(y - X\tilde{\beta})^T(y - X\tilde{\beta}) \\
&= (y - X\hat{\beta} + X\hat{\beta} - X\tilde{\beta})^T(y - X\hat{\beta} + X\hat{\beta} - X\tilde{\beta}) \\
&= ((y - X\hat{\beta}) + X(\hat{\beta} - \tilde{\beta}))^T((y - X\hat{\beta}) + X(\hat{\beta} - \tilde{\beta})) \\
&= (y - X\hat{\beta})^T(y - X\hat{\beta}) + (y - X\hat{\beta})^T X(\hat{\beta} - \tilde{\beta}) + (\hat{\beta} - \tilde{\beta})^T X^T(y - X\hat{\beta}) + (\hat{\beta} - \tilde{\beta})^T X^T X(\hat{\beta} - \tilde{\beta}) \\
&= (y - X\hat{\beta})^T(y - X\hat{\beta}) + 0_p^T(\hat{\beta} - \tilde{\beta}) + (\hat{\beta} - \tilde{\beta})^T 0_p + (\hat{\beta} - \tilde{\beta})^T X^T X(\hat{\beta} - \tilde{\beta}) \\
&= (y - X\hat{\beta})^T(y - X\hat{\beta}) + (\hat{\beta} - \tilde{\beta})^T X^T X(\hat{\beta} - \tilde{\beta}).
\end{aligned} \tag{37.9}$$

On the right-hand side of the above equation, only the second term depends on $\tilde{\beta}$. Since this term satisfies

$$(\hat{\beta} - \tilde{\beta})^T X^T X(\hat{\beta} - \tilde{\beta}) \geq 0, \tag{37.10}$$

it attains its minimum value 0 exactly when

$$(\hat{\beta} - \tilde{\beta}) = 0_p \Leftrightarrow \tilde{\beta} = \hat{\beta}. \tag{37.11}$$

Thus,

$$\hat{\beta} = \operatorname{argmin}_{\tilde{\beta}}(y - X\tilde{\beta})^T(y - X\tilde{\beta}). \tag{37.12}$$

(2) To show that $\hat{\beta}$ is a maximum-likelihood estimator, we consider, for an arbitrary value $y \in \mathbb{R}^n$ of y and fixed $\sigma^2 > 0$, the log-likelihood function

$$\ell : \mathbb{R}^p \to \mathbb{R}, \tilde{\beta} \mapsto \ln p_{\tilde{\beta}}(y) = \ln N\left(y; X\tilde{\beta}, \sigma^2 I_n\right), \tag{37.13}$$

where

$$\begin{aligned}
\ln N\left(y; X\tilde{\beta}, \sigma^2 I_n\right) &= \ln\left((2\pi)^{-\frac{n}{2}}\left|\sigma^2 I_n\right|^{-\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}(y - X\tilde{\beta})^T(y - X\tilde{\beta})\right)\right) \\
&= -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln\left|\sigma^2 I_n\right| - \frac{1}{2\sigma^2}(y - X\tilde{\beta})^T(y - X\tilde{\beta}).
\end{aligned} \tag{37.14}$$

Only the term $-\frac{1}{2\sigma^2}(y - X\tilde{\beta})^T(y - X\tilde{\beta})$ depends on $\tilde{\beta}$. Because $(y - X\tilde{\beta})^T(y - X\tilde{\beta}) \geq 0$, this term is maximized, due to the negative sign, when $(y - X\tilde{\beta})^T(y - X\tilde{\beta})$ is minimized. As shown above, this is exactly the case for $\tilde{\beta} = \hat{\beta}$. Finally, the unbiasedness of $\hat{\beta}$ follows from

$$\mathbb{E}(\hat{\beta}) = \mathbb{E}\left(\left(X^T X\right)^{-1} X^T y\right) = \left(X^T X\right)^{-1} X^T \mathbb{E}(y) = \left(X^T X\right)^{-1} X^T X\beta = \beta. \tag{37.15}$$

□

With

$$\hat{\beta} = \left(X^T X\right)^{-1} X^T y, \tag{37.16}$$

Theorem 37.1 provides a formula for concretely estimating β based on the design matrix and a realization $y \in \mathbb{R}^n$ of y. As a random vector, $\hat{\beta}$ is an unbiased estimator of β, and as a maximum-likelihood estimator it is, in particular, also consistent, asymptotically normally distributed, and asymptotically efficient. We will see later that $\hat{\beta}$ is even normally distributed. In addition to these properties, $\hat{\beta}$ has further good properties. For example, $\hat{\beta}$ has the smallest variance in the class of linear unbiased estimators of β. This property is the core statement of the *Gauss-Markov theorem*, which we do not discuss in detail here.

Using the beta-parameter estimator, we can define the concepts of explained data, the residual vector, and residuals, which we will need in many places.

Definition 37.1 (Explained data, residual vector, and residuals)**.** Let

$$y = X\beta + \varepsilon \,\text{with}\, \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{37.17}$$

be the general linear model, and let

$$\hat{\beta} := \left(X^T X\right)^{-1} X^T y \tag{37.18}$$

be the beta-parameter estimator. Then the random vector

$$\hat{y} := X\hat{\beta} = X\left(X^T X\right)^{-1} X^T y \tag{37.19}$$

is called the *explained data*, the random vector

$$\hat{\varepsilon} := y - \hat{y} = y - X\hat{\beta} \tag{37.20}$$

is called the *residual vector*, and, for $i = 1, \dots, n$, the components of this random vector

$$\hat{\varepsilon}_i := y_i - \hat{y}_i = y_i - (X\hat{\beta})_i \tag{37.21}$$

are called the *residuals*.

•

37.2 Variance-parameter estimation

We summarize frequentist point estimation of the variance parameter in the following theorem, which we do not prove here.

Theorem 37.2 (Variance-parameter estimator)**.** *Let*

$$y = X\beta + \varepsilon \;\textit{with}\; \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{37.22}$$

be the general linear model. Then

$$\hat{\sigma}^2 := \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n - p} \tag{37.23}$$

is an unbiased estimator of $\sigma^2 > 0$.

∘

With

$$\hat{\sigma}^2 = \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{n - p}, \tag{37.24}$$

Theorem 37.2 provides a formula for estimating σ^2 based on the design matrix, the beta-parameter estimator, and a realization $y \in \mathbb{R}^n$ of y. Evidently, by Theorem 37.2,

$$\hat{\sigma}^2 = \frac{1}{n - p} \sum_{i=1}^{n} \left(y_i - (X\hat{\beta})_i\right)^2. \tag{37.25}$$

Thus $\hat{\sigma}^2$ is estimated by the sum of squared residuals, that is, by a sum of squared deviations. For a proof of Theorem 37.2, we refer, for example, to S. R. Searle (1971), S. R. Searle & Gruber (2017), or Rencher & Schaalje (2008). From a probabilistic perspective, $\hat{\sigma}^2$ is not a maximum-likelihood estimator, but a restricted maximum-likelihood estimator of σ^2 (cf. Harville (1977), Foulley (1993), Starke & Ostwald (2017)). From a geometric perspective, $\hat{\sigma}^2$ is a least-squares estimator (cf. Christensen (2011)).

37.3 Independent identically normally distributed random variables

As a first application of Theorem 37.1 and Theorem 37.2, we analyze the scenario of n independent and identically normally distributed random variables with expectation parameter $\mu \in \mathbb{R}$ and variance parameter σ^2,

$$y_i \sim N\left(\mu, \sigma^2\right) \text{ for } i = 1, \ldots, n. \tag{37.26}$$

Writing this model in its design-matrix form (cf. Equation 36.10), we have, as shown below,

$$\hat{\beta} = \frac{1}{n}\sum_{i=1}^{n} y_i =: \bar{y} \text{ and } \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2 =: s_y^2. \tag{37.27}$$

In this case, the beta-parameter estimator is therefore identical to the sample mean $\bar{y}$ of the $y_1, \ldots, y_n$, and the variance-parameter estimator is identical to the sample variance s_y^2 of the $y_1, \ldots, y_n$.

Equation 37.27 follows as follows. On the one hand,

$$\begin{aligned}
\hat{\beta} &= \left(X^T X\right)^{-1} X^T y \\
&= \left(1_n^T 1_n\right)^{-1} 1_n^T y \\
&= \left(\begin{pmatrix} 1 & \cdots & 1 \end{pmatrix}\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}\right)^{-1}\begin{pmatrix} 1 & \cdots & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \\
&= n^{-1}\sum_{i=1}^{n} y_i \\
&= \frac{1}{n}\sum_{i=1}^{n} y_i \\
&=: \bar{y}.
\end{aligned} \tag{37.28}$$

On the other hand,

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n-1}(y - X\hat{\beta})^T(y - X\hat{\beta}) \\
&= \frac{1}{n-1}\left(y - 1_n\bar{y}\right)^T\left(y - 1_n\bar{y}\right) \\
&= \frac{1}{n-1}\left(\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} - \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}\bar{y}\right)^T\left(\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} - \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}\bar{y}\right) \\
&= \frac{1}{n-1}\begin{pmatrix} y_1 - \bar{y} & \cdots & y_n - \bar{y} \end{pmatrix}\begin{pmatrix} y_1 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{pmatrix} \\
&= \frac{1}{n-1}\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2 \\
&= s_y^2.
\end{aligned} \tag{37.29}$$

We demonstrate parameter estimation in this scenario in the following **R** code.

```
# model formulation
library(MASS)                                              # multivariate normal distribution
n          = 12                                            # number of data points
p          = 1                                             # number of beta parameters
X          = matrix(rep(1,n), nrow = n)                    # design matrix
I_n        = diag(n)                                       # n x n identity matrix
beta       = 2                                             # true, but unknown, beta parameter
sigsqr     = 1                                             # true, but unknown, variance parameter

# data realization
```

```
y          =  mvrnorm(1, X %*% beta, sigsqr*I_n)                             # one realization of an n-dimensional random vector

# parameter estimation
beta_hat   = solve(t(X) %*% X) %*% t(X) %*% y                               # beta-parameter estimator
eps_hat    = y - X %*% beta_hat                                              # residual vector
sigsqr_hat = (t(eps_hat) %*% eps_hat) /(n-p)                                 # variance-parameter estimator
```

```
beta        :  2
hat{beta}   :  2.029446
sigsqr      :  1
hat{sigsqr}:  1.65161
```

The following **R** code simulates the frequentist meaning of estimator unbiasedness in this scenario.

```
# model formulation
library(MASS)                                                                # multivariate normal distribution
n          = 12                                                              # number of data points
p          = 1                                                               # number of beta parameters
X          = matrix(rep(1,n), nrow = n)                                      # design matrix
I_n        = diag(n)                                                         # n x n identity matrix
beta       = 2                                                               # true, but unknown, beta parameter
sigsqr     = 1                                                               # true, but unknown, variance parameter

# frequentist simulation
nsim       = 1e4                                                             # number of data realizations
beta_hat   = rep(NaN,nsim)                                                   # \hat{\beta} realization array
sigsqr_hat = rep(NaN,nsim)                                                   # \hat{sigsqr} realization array
for(i in 1:nsim){                                                            # simulation iterations
  y             = mvrnorm(1, X %*% beta, sigsqr*I_n)                         # data realization
  beta_hat[i]   = solve(t(X) %*% X) %*% t(X) %*% y                          # beta-parameter estimator
  eps_hat       = y - X %*% beta_hat[i]                                      # residual vector
  sigsqr_hat[i] = (t(eps_hat) %*% eps_hat) /(n-p)                            # variance-parameter estimator
}
```

```
True, but unknown, beta parameter                  :  2
Estimated expectation of the beta-parameter estimator    :  1.998326
True, but unknown, variance parameter               :  1
Estimated expectation of the variance-parameter estimator :  0.9958223
```

37.4 Simple linear regression

As a second application of Theorem 37.1 and Theorem 37.2, we analyze the scenario of simple linear regression

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \text{ with } \varepsilon_i \sim N\left(0, \sigma^2\right) \text{ for } i = 1, \dots, n. \tag{37.30}$$

Based on the design-matrix form Equation 36.15 of this model, we obtain, as shown below,

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \bar{y} - \frac{c_{xy}}{s_x^2}\bar{x} \\ \frac{c_{xy}}{s_x^2} \end{pmatrix} \text{ and } \hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^n \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2 \tag{37.31}$$

where $\bar{x}$ and $\bar{y}$ denote the sample means of the $x_1, \dots, x_n$ and $y_1, \dots, y_n$, c_{xy} denotes the sample covariance of the $x_1, \dots, x_n$ and $y_1, \dots, y_n$, and s_x^2 denotes the sample variance of the $x_1, \dots, x_n$. As in Chapter 1, the terms sample covariance and sample variance with respect to the $x_1, \dots, x_n$ are meant here only formally, because no assumption is made that the $x_1, \dots, x_n$ are realizations of random variables.

We therefore note that, for a parametric design-matrix column, the corresponding beta-parameter estimator is obtained from the sample covariance of the respective column with the data divided by the sample variance of the corresponding column, and thus corresponds to a standardized sample covariance. A comparison with the parameters of the fitting line in Chapter 34 further shows the identity of the beta-parameter-estimator components $\hat{\beta}_0$ and $\hat{\beta}_1$ with the parameters derived there under the criterion of minimizing squared vertical deviations. This is not surprising, because both $\hat{\beta}$ and the parameters of the fitting line, for a fixed value $y \in \mathbb{R}^n$ of y, minimize the value

$$q(\tilde{\beta}) = \sum_{i=1}^{n} \left(y_i - \left(\tilde{\beta}_0 + \tilde{\beta}_1 x_i\right)\right)^2 = (y - X\tilde{\beta})^T (y - X\tilde{\beta}) \tag{37.32}$$

with respect to $\tilde{\beta}$.

To derive the form of the beta-parameter estimator in Equation 37.31, we first note that

$$\begin{aligned}
\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) &= \sum_{i=1}^{n} (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x}\bar{y}) \\
&= \sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \bar{y} - \sum_{i=1}^{n} \bar{x} y_i + \sum_{i=1}^{n} \bar{x}\bar{y} \\
&= \sum_{i=1}^{n} x_i y_i - \bar{y} \sum_{i=1}^{n} x_i - \bar{x} \sum_{i=1}^{n} y_i + n\bar{x}\bar{y} \\
&= \sum_{i=1}^{n} x_i y_i - \bar{y} n \bar{x} - \bar{x} n \bar{y} + n\bar{x}\bar{y} \\
&= \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y} - n\bar{x}\bar{y} + n\bar{x}\bar{y} \\
&= \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}.
\end{aligned} \tag{37.33}$$

Furthermore, we note that

$$\begin{aligned}
\sum_{i=1}^{n} (x_i - \bar{x})^2 &= \sum_{i=1}^{n} (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\
&= \sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} 2x_i\bar{x} + \sum_{i=1}^{n} \bar{x}^2 \\
&= \sum_{i=1}^{n} x_i^2 - 2\bar{x} \sum_{i=1}^{n} x_i + n\bar{x}^2 \\
&= \sum_{i=1}^{n} x_i^2 - 2\bar{x} n \bar{x} + n\bar{x}^2 \\
&= \sum_{i=1}^{n} x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\
&= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2.
\end{aligned} \tag{37.34}$$

From the definition of $\hat{\beta}$, we obtain

$$\begin{aligned}
\hat{\beta} &= (X^T X)^{-1} X^T y \\
&= \left(\begin{pmatrix} 1 & \cdots & 1 \\ x_1 & \cdots & x_n \end{pmatrix} \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \right)^{-1} \begin{pmatrix} 1 & \cdots & 1 \\ x_1 & \cdots & x_n \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \\
&= \begin{pmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{pmatrix} \\
&= \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_{i=1}^{n} x_i^2 \end{pmatrix}^{-1} \begin{pmatrix} n\bar{y} \\ \sum_{i=1}^{n} x_i y_i \end{pmatrix}.
\end{aligned} \tag{37.35}$$

The inverse of $X^T X$ is given by

$$\frac{1}{s_{xx}} \begin{pmatrix} \frac{s_{xx}}{n} + \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix}, \tag{37.36}$$

where $s_{xx} := \sum_{i=1}^n (x_i - \bar{x})^2$, because

$$
\begin{aligned}
&\frac{1}{s_{xx}} \begin{pmatrix} \frac{s_{xx}}{n} + \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix} \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_{i=1}^n x_i^2 \end{pmatrix} \\
&= \frac{1}{s_{xx}} \begin{pmatrix} \frac{ns_{xx}}{n} + n\bar{x}^2 - n\bar{x}^2 & \frac{s_{xx}n\bar{x}}{n} + n\bar{x}^2\bar{x} - \bar{x}\sum_{i=1}^n x_i^2 \\ -\bar{x}n + n\bar{x} & -n\bar{x}^2 + \sum_{i=1}^n x_i^2 \end{pmatrix} \\
&= \frac{1}{s_{xx}} \begin{pmatrix} s_{xx} & s_{xx}\bar{x} - \bar{x}\left(\sum_{i=1}^n x_i^2 - n\bar{x}^2\right) \\ 0 & \sum_{i=1}^n x_i^2 - n\bar{x}^2 \end{pmatrix} \\
&= \frac{1}{s_{xx}} \begin{pmatrix} s_{xx} & s_{xx}\bar{x} - \bar{x}s_{xx} \\ 0 & s_{xx} \end{pmatrix} \\
&= \frac{1}{s_{xx}} \begin{pmatrix} s_{xx} & 0 \\ 0 & s_{xx} \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.
\end{aligned} \tag{37.37}
$$

Thus,

$$
\begin{aligned}
\hat{\beta} = \begin{pmatrix} \frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} & -\frac{\bar{x}}{s_{xx}} \\ -\frac{\bar{x}}{s_{xx}} & \frac{1}{s_{xx}} \end{pmatrix} \begin{pmatrix} n\bar{y} \\ \sum_{i=1}^n x_i y_i \end{pmatrix} &= \begin{pmatrix} \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}\right) n\bar{y} - \frac{\bar{x}\sum_{i=1}^n x_i y_i}{s_{xx}} \\ \frac{\sum_{i=1}^n x_i y_i}{s_{xx}} - \frac{n\bar{x}\bar{y}}{s_{xx}} \end{pmatrix} \\
&= \begin{pmatrix} \frac{n\bar{y}}{n} + \frac{\bar{x}^2 n\bar{y}}{s_{xx}} - \frac{\bar{x}\sum_{i=1}^n x_i y_i}{s_{xx}} \\ \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{s_{xx}} \end{pmatrix} \\
&= \begin{pmatrix} \bar{y} + \frac{\bar{x}n\bar{x}\bar{y} - \bar{x}\sum_{i=1}^n x_i y_i}{s_{xx}} \\ \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{s_{xx}} \end{pmatrix} \\
&= \begin{pmatrix} \bar{y} - \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{s_{xx}}\bar{x} \\ \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{s_{xx}} \end{pmatrix} \\
&= \begin{pmatrix} \bar{y} - \frac{c_{xy}}{s_x^2}\bar{x} \\ \frac{c_{xy}}{s_x^2} \end{pmatrix}.
\end{aligned} \tag{37.38}
$$

We demonstrate parameter estimation in this scenario in the following **R** code. Note the extensive agreement with the implementation of parameter estimation in the scenario of independent and identically normally distributed random variables; only the design matrix and the dimension of the beta parameter change.

```
# model formulation
library(MASS)                                               # multivariate normal distribution
n          = 10                                             # number of data points
p          = 2                                              # number of beta parameters
x          = 1:n                                            # predictor values
X          = matrix(c(rep(1,n),x), nrow = n)                # design matrix
I_n        = diag(n)                                        # n x n identity matrix
beta       = matrix(c(0,1), nrow = p)                       # true, but unknown, beta parameter
sigsqr     = 1                                              # true, but unknown, variance parameter

# data realization
y          = mvrnorm(1, X %*% beta, sigsqr*I_n)             # one realization of an n-dimensional random vector

# parameter estimation
beta_hat   = solve(t(X) %*% X) %*% t(X) %*% y               # beta-parameter estimator
eps_hat    = y - X %*% beta_hat                             # residual vector
sigsqr_hat = (t(eps_hat) %*% eps_hat) /(n-p)                # variance-parameter estimator
```

```
beta       :  0 1
hat{beta}  :  0.3507292 0.9623078
sigsqr     :  1
hat{sigsqr}:  1.085769
```

Analogously to the scenario of independent and identically normally distributed random variables, the frequentist meaning of estimator unbiasedness can also be simulated here.

```
# model formulation
library(MASS)                                                          # multivariate normal distribution
n          = 10                                                        # number of data points
p          = 2                                                         # number of beta parameters
x          = 1:n                                                       # predictor values
X          = matrix(c(rep(1,n),x), nrow = n)                           # design matrix
I_n        = diag(n)                                                   # n x n identity matrix
beta       = matrix(c(0,1), nrow = p)                                  # true, but unknown, beta parameter
sigsqr     = 1                                                         # true, but unknown, variance parameter

# frequentist simulation
nsim       = 1e4                                                       # number of realizations of the n-dimensional random vector
beta_hat   = matrix(rep(NaN,p*nsim), nrow = p)                         # \hat{\beta} realization array
sigsqr_hat = rep(NaN,nsim)                                             # \hat{sigsqr} realization array
for(i in 1:nsim){                                                      # simulation iterations
  y             = mvrnorm(1, X %*% beta, sigsqr*I_n)                   # data realization
  beta_hat[,i]  = solve(t(X) %*% X) %*% t(X) %*% y                     # beta-parameter estimator
  eps_hat       = y - X %*% beta_hat[,i]                               # residual vector
  sigsqr_hat[i] = (t(eps_hat) %*% eps_hat) /(n-p)                      # variance-parameter estimator
}
```

```
True, but unknown, beta parameter                         :  0 1
Estimated expectation of the beta-parameter estimator     :  -0.00782099 1.000713
True, but unknown, variance parameter                     :  1
Estimated expectation of the variance-parameter estimator :  0.9944547
```

37.5 Frequentist estimator distributions

We document the frequentist distribution of the beta-parameter estimator in the following theorem.

Theorem 37.3 (Frequentist distribution of the beta-parameter estimator)**.** *Let*

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{37.39}$$

be the GLM. Furthermore, let

$$\hat{\beta} := \left(X^T X\right)^{-1} X^T y \tag{37.40}$$

be the beta-parameter estimator. Then

$$\hat{\beta} \sim N\left(\beta, \sigma^2 \left(X^T X\right)^{-1}\right). \tag{37.41}$$

∘

Proof. The theorem follows directly from the theorem on linear transformations of multivariate normal distributions. Specifically, here

$$\hat{\beta} \sim N\left(\left(X^T X\right)^{-1} X^T X\beta, \left(X^T X\right)^{-1} X^T \left(\sigma^2 I_n\right)\left(\left(X^T X\right)^{-1} X^T\right)^T\right). \tag{37.42}$$

The expectation parameter then simplifies to

$$\left(X^T X\right)^{-1} X^T X\beta = \beta. \tag{37.43}$$

The covariance-matrix parameter simplifies as follows:

$$\begin{aligned}\left(X^T X\right)^{-1} X^T \left(\sigma^2 I_n\right)\left(\left(X^T X\right)^{-1} X^T\right)^T &= \left(X^T X\right)^{-1} X^T \left(\sigma^2 I_n\right) X \left(X^T X\right)^{-1} \\ &= \sigma^2 \left(X^T X\right)^{-1} X^T X \left(X^T X\right)^{-1} \\ &= \sigma^2 \left(X^T X\right)^{-1}.\end{aligned} \tag{37.44}$$

Here, the first equality follows from the fact that both $X^T X$ and its inverse $\left(X^T X\right)^{-1}$ are symmetric matrices. Thus, it follows immediately that

$$\hat{\beta} \sim N\left(\beta, \sigma^2 \left(X^T X\right)^{-1}\right). \tag{37.45}$$

□

With Theorem 6.3, it follows in particular for the expectation and covariance matrix of the beta-parameter estimator that

$$\mathbb{E}(\hat{\beta}) = \beta \text{ and } \mathbb{C}(\hat{\beta}) = \sigma^2 \left(X^T X\right)^{-1}. \tag{37.46}$$

As diagonal elements of $\mathbb{C}(\hat{\beta})$, the variances of the beta-parameter-estimator components therefore depend both on the variance parameter of the error variables and on the design matrix. In particular, for fixed, true, but unknown $\sigma^2 > 0$, the design matrix can therefore be chosen such that the variance of the beta-parameter-estimator components is minimized.

We document the frequentist distribution of the variance-parameter estimator in the following theorem, which we do not prove here.

Theorem 37.4 (Frequentist distribution of the variance-parameter estimator)**.** *Let*

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{37.47}$$

be the GLM. Furthermore, let

$$\hat{\sigma}^2 = \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-p} \tag{37.48}$$

be the variance-parameter estimator. Then

$$\frac{n-p}{\sigma^2}\hat{\sigma}^2 \sim \chi^2(n-p). \tag{37.49}$$

∘

Because $(n-p)\hat{\sigma}^2$ is a sum of normally distributed random variables, the χ^2 distribution is at least plausible in light of the χ^2 transformation for normally distributed random variables. However, $\hat{\sigma}^2$ itself is not χ^2 distributed, but only its version scaled by multiplication with $\frac{n-p}{\sigma^2}$. We want to illustrate the frequentist estimator distributions from Theorem 37.3 and Theorem 37.4 using the two standard examples.

Example (1) Independent and identically normally distributed random variables

Let

$$y \sim N\left(X\beta, \sigma^2 I_n\right) \text{ with } X := 1_n \in \mathbb{R}^{n\times 1}, \beta := \mu \in \mathbb{R} \text{ and } \sigma^2 > 0 \tag{37.50}$$

be the GLM scenario of independent and identically normally distributed random variables with known variance. We have already seen that, in this case, $\hat{\beta}$ is identical to the sample mean $\bar{y}$. Theorem 37.3 then implies, with

$$\left(X^T X\right)^{-1} = \left(1_n^T 1_n\right)^{-1} = \frac{1}{n}, \tag{37.51}$$

that

$$\bar{y} \sim N\left(\mu, \frac{\sigma^2}{n}\right). \tag{37.52}$$

The sample mean of n independent and identically normally distributed random variables with expectation parameter μ and variance parameter σ^2 is therefore normally distributed with expectation parameter μ and variance parameter σ^2/n. We have already seen this fact in the context of transformations of normal distributions under the term mean transformation.

Example (2) Simple linear regression

Let

$$y \sim N\left(X\beta, \sigma^2 I_n\right) \text{ with } \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \in \mathbb{R}^{n\times 2}, \beta \in \mathbb{R}^2 \text{ and } \sigma^2 > 0 \tag{37.53}$$

be the scenario of simple linear regression. We have already seen that

$$\sigma^2 \left(X^T X\right)^{-1} = \frac{\sigma^2}{s_{xx}} \begin{pmatrix} \frac{s_{xx}}{n} + \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix} \text{ with } s_{xx} := \sum_{i=1}^{n} \left(x_i - \bar{x}\right)^2. \tag{37.54}$$

The variance of the intercept-parameter estimator thus depends both on the sum of squared differences of the values of the independent variable from their sample mean and on the sample mean of the values of the independent variable itself. The variance of the slope-parameter estimator, by contrast, depends only on the sum of squared differences of the independent variable from its sample mean. Finally, the covariance of intercept- and slope-parameter estimators depends on the mean of the values of the independent variable. The following **R** code simulates the frequentist distributions of beta- and variance-parameter estimators in the scenario of simple linear regression.

```
# model formulation
library(MASS)                                                               # multivariate normal distribution
n        = 10                                                               # number of data points
p        = 2                                                                # number of beta parameters
x        = 1:n                                                              # predictor values
X        = matrix(c(rep(1,n),x), nrow = n)                                  # design matrix
I_n      = diag(n)                                                          # n x n identity matrix
beta     = matrix(c(0,1), nrow = p)                                         # true, but unknown, beta parameter
sigsqr   = .5                                                               # true, but unknown, variance parameter

# frequentist simulation
nsim     = 10                                                               # number of realizations of the n-dimensional random vector
y        = matrix(rep(NaN,n*nsim), nrow = n)                                # y realization array
beta_hat = matrix(rep(NaN,p*nsim), nrow = p)                                # \hat{\beta} realization array
for(i in 1:nsim){
  y[,i]        = mvrnorm(1, X %*% beta, sigsqr*I_n)                         # one realization of an n-dimensional random vector
  beta_hat[,i] = solve(t(X) %*% X) %*% t(X) %*% y[,i]                       # \hat{\beta} = (X^T X)^{-1}X^Ty
}
```

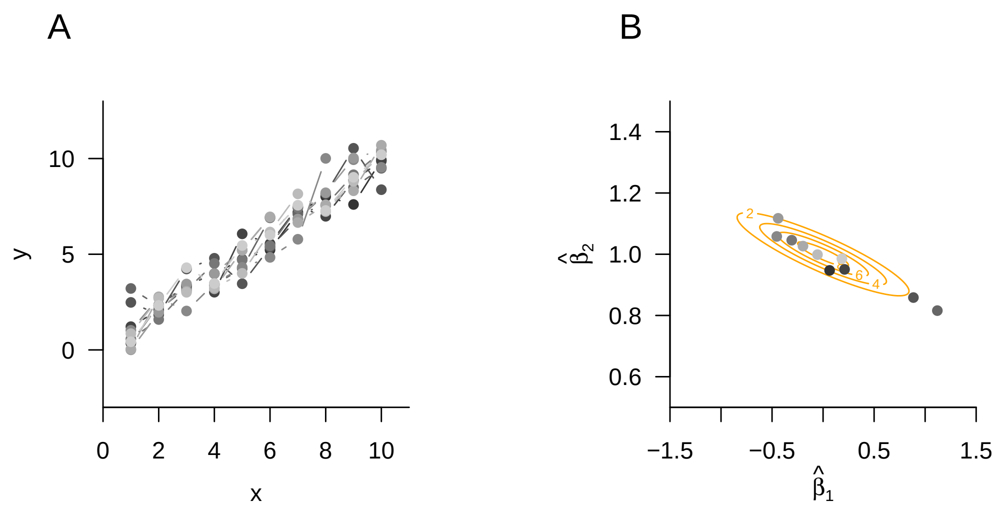


Figure 37.1 Frequentist beta-parameter-estimator distribution in simple linear regression.

Figure 37.1 shows 10 realizations of the model of simple linear regression, and Figure 37.1 B shows the corresponding beta-parameter-estimator realizations as well as the analytical distribution of the beta-parameter estimator.

37.6 Bibliographic remarks

Plackett (1949) gives a historical overview of the development of beta-parameter estimation and, in particular, of the Gauss-Markov theorem. The problem of variance-parameter estimation in the GLM in the sense of the restricted-maximum-likelihood method first appears in Patterson & Thompson (1971) (cf. Harville (1977)), Verbyla (1990), and remains, in generalized GLMs, a topic of current research (cf. Lindholm & Wahl (2020)).

38 T-statistics

In this section, we introduce T-statistics as measures for evaluating beta-parameter estimators in the GLM. T-statistics quantify the estimated effects of the beta-parameter estimator relative to the residual variability estimated by the variance-parameter estimator. The value of a T-statistic can therefore initially be understood simply as a signal-to-noise ratio.

T-statistics also allow the evaluation of linear combinations of the components of the beta-parameter estimator in terms of frequentist confidence intervals and hypothesis tests. Here, we first consider only the functional form of T-statistics and their frequentist distribution for the purpose of determining confidence intervals. The use of T-test statistics in one-sample and two-sample T-tests is the topic of Chapter 40.

38.1 Definition and examples

Against the background of the GLM and its parameter point estimators, we define the T-statistic as follows.

Definition 38.1 (T-statistic)**.** Let

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{38.1}$$

be the GLM. Furthermore, let

$$\hat{\beta} := \left(X^T X\right)^{-1} X^T y \text{ and } \hat{\sigma}^2 := \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{n-p} \tag{38.2}$$

be the beta-parameter and variance-parameter estimators, respectively. Then, for a contrast-weight vector $c \in \mathbb{R}^p$ and a parameter $\beta_0 \in \mathbb{R}^p$, the T-statistic is defined as

$$T := \frac{c^T \hat{\beta} - c^T \beta_0}{\sqrt{\hat{\sigma}^2 c^T \left(X^T X\right)^{-1} c}}. \tag{38.3}$$

•

Suitable choices of the contrast-weight vector $c \in \mathbb{R}^p$ and the parameter $\beta_0 \in \mathbb{R}^p$ allow a wide range of uses of the T-statistic. Let us first consider the contrast-weight vector. Evidently, the contrast-weight vector serves to transform the random vector $\hat{\beta} \in \mathbb{R}^p$ into the random variable $c^T \hat{\beta}$ and thereby ensures that the T-statistic is scalar. Furthermore, choosing p-dimensional unit vectors for the contrast-weight vector allows the selection of individual components of the beta parameter for evaluation using the T-statistic. Finally, a general choice of the contrast-weight vector allows the evaluation of arbitrary linear combinations of the beta-parameter components, such as differences of individual components. As examples, for $\hat{\beta} \in \mathbb{R}^2$ the following possibilities for choosing $c \in \mathbb{R}^2$ with respect to the scalar product $c^T\hat{\beta}$ are:

$$c := \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow c^T\hat{\beta} = \hat{\beta}_1, \quad c := \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow c^T\hat{\beta} = \hat{\beta}_2, \quad c := \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow c^T\hat{\beta} = \hat{\beta}_1 - \hat{\beta}_2. \quad (38.4)$$

The choice of the parameter $\beta_0 \in \mathbb{R}^p$ opens up different ways of using the T-statistic. If, for example, one chooses $\beta_0 := 0_p$, then the T-statistic is a descriptive statistic that makes it possible to relate estimated regressor effects, that is, components or linear combinations of $\hat{\beta}$, as a signal-to-noise ratio to the residual data variability quantified by $\hat{\sigma}^2$. The denominator of the T-statistic ensures that, in particular, the appropriate (co)standard deviation of the corresponding beta-parameter component combination serves as the reference quantity, because $\hat{\sigma}^2 \left(X^TX\right)^{-1}$ is, as is well known, the covariance of the beta-parameter estimator (cf. Theorem 37.3).

If, by contrast, one chooses β_0 as β, that is, the true, but unknown, beta-parameter value, then the T-statistic opens up the possibility of determining confidence intervals for individual components of the beta-parameter vector. We deepen this aspect of the T-statistic in Section 38.2. Finally, if β_0 is declared as the element of a null hypothesis in the context of a test scenario, then the T-statistic opens up hypothesis-test-based inference about beta-parameter components and their linear combinations. We discuss applications of this type in detail in Chapter 40.

The use of the T-statistic for frequentist inference in terms of confidence intervals and hypothesis tests is, of course, based on the frequentist distribution of the T-statistic against the background of the GLM. This is the central content of the following theorem, whose proof we omit.

Theorem 38.1 (Frequentist distribution of the T-statistic)**.** *Let*

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \quad (38.5)$$

be the GLM. Furthermore, let

$$\hat{\beta} := \left(X^TX\right)^{-1} X^Ty \text{ and } \hat{\sigma}^2 := \frac{(y - X\hat{\beta})^T(y - X\hat{\beta})}{n-p} \quad (38.6)$$

be the beta-parameter and variance-parameter estimators, respectively. Finally, for a contrast-weight vector $c \in \mathbb{R}^p$ and a parameter $\beta_0 \in \mathbb{R}^p$, let

$$T := \frac{c^T\hat{\beta} - c^T\beta_0}{\sqrt{\hat{\sigma}^2 c^T \left(X^TX\right)^{-1} c}} \quad (38.7)$$

be the T-statistic. Then

$$T \sim t(\delta, n-p) \text{ with } \delta := \frac{c^T\beta - c^T\beta_0}{\sqrt{\sigma^2 c^T \left(X^TX\right)^{-1} c}}. \quad (38.8)$$

∘

In general, the T-statistic is therefore noncentrally t-distributed. If, as in the determination of confidence intervals (cf. Section 38.2), $\beta_0 := \beta$, or if, in a test scenario, $\beta := \beta_0$ holds when the null hypothesis is true (cf. Chapter 40), then the T-statistic is even t-distributed, each time with degrees-of-freedom parameter $n-p$. If, by contrast, the null hypothesis is not true in a test scenario, then the distribution of the T-statistic from Theorem 38.1 can be used to derive the power function and thus to determine the power of the test (cf. Chapter Chapter 40). We will deepen these aspects in the appropriate place. Here, we first want to illustrate the T-statistic and its distribution only using the examples of independent identically normally distributed random variables and simple linear regression.

Example (1) Independent and identically normally distributed random variables

Let

$$y \sim N\left(X\beta, \sigma^2 I_n\right) \text{ with } X := 1_n \in \mathbb{R}^{n\times 1}, \beta := \mu \in \mathbb{R} \text{ and } \sigma^2 > 0 \tag{38.9}$$

be the GLM scenario of independent and identically normally distributed random variables, and let $c := 1$ and $\beta_0 := \mu_0 \in \mathbb{R}$. Then, for the T-statistic,

$$T = \frac{c^T\hat{\beta} - c^T\mu_0}{\sqrt{\hat{\sigma}^2 c^T\left(X^TX\right)^{-1}c}} = \frac{1^T\bar{y} - 1^T\mu_0}{\sqrt{s_y^2 1^T\left(1_n^T 1_n\right)^{-1}1}} = \sqrt{n}\left(\frac{\bar{y}-\mu_0}{s_y}\right). \tag{38.10}$$

This evidently corresponds to the familiar one-sample T-test statistic (cf. Chapter 33). Like that statistic, the T-statistic considered here takes large absolute values, with the respective complementary terms held constant, for a large absolute difference $\bar{y}-\mu_0$ (often called an effect), for small values of s_y^2 (that is, low data variability), and for a large value of n (that is, a large sample size). The following **R** code simulates the frequentist distribution of this T-statistic for the cases $\beta = \beta_0$ and $\beta \neq \beta_0$.

```
# libraries
library(MASS)                                                                  # multivariate normal distribution

# model formulation
n           = 12                                                               # number of data points
p           = 1                                                                # number of beta parameters
X           = matrix(c(rep(1,n)), nrow = n)                                    # design matrix
I_n         = diag(n)                                                          # identity matrix
beta        = c(0,1)                                                           # true, but unknown, beta parameters
nscn        = length(beta)                                                     # number of true, but unknown, hypothesis scenarios
sigsqr      = 1                                                                # true, but unknown, variance parameter
c           = 1                                                                # contrast vector of interest
beta_0      = 0                                                                # null-hypothesis beta parameter

# frequentist simulation
nsim        = 1e4                                                              # number of simulations
delta       = rep(NaN, nscn)                                                   # noncentrality-parameter array
Tee         = matrix(rep(NaN, nscn*nsim), ncol = nscn)                         # T-test-statistic realization array
for(s in 1:nscn){                                                              # hypothesis scenarios
  delta[s]     = ((t(c) %*% beta[s] - t(c) %*% beta_0)/                        # noncentrality parameter
                 sqrt(sigsqr*t(c)%*%solve(t(X)%*%X)%*%c))
  for(i in 1:nsim){                                                            # simulation iterations
    y           = mvrnorm(1, X %*% beta[s], sigsqr*I_n)                        # y
    beta_hat    = solve(t(X) %*% X) %*% t(X) %*% y                             # \hat{\beta}
    eps_hat     = y - X %*% beta_hat                                           # \hat{\eps}
    sigsqr_hat  = (t(eps_hat) %*% eps_hat)/(n-p)                               # \hat{\sigma}^2
    Tee[i,s]    = ((t(c) %*% beta_hat - t(c) %*% beta_0)/                      # T
                  sqrt(sigsqr_hat*t(c)%*%solve(t(X)%*%X)%*%c))
  }
}
```

Figure 38.1 A and B show the resulting simulated and analytical distributions of the T-statistic.

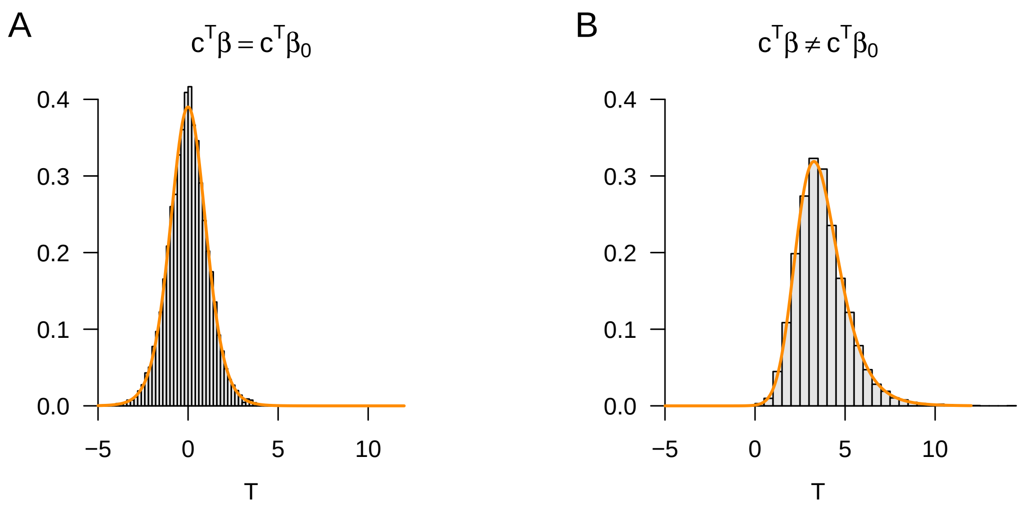


Figure 38.1 Distributions of the T-statistic for independent and identically normally distributed random variables.

Example (2) Simple linear regression

In this example, we do not want to discuss the specific form of the T-statistic but rather demonstrate, by means of a simulation, how the principle of the T-statistic appears in the context of simple linear regression. To this end, we consider the familiar example model of simple linear regression (cf. Chapter 36), here with the true, but unknown, parameter values $\beta_A := (1, 0)$ and $\beta_B := (1, 1)$. Furthermore, we consider the contrast-weight vector $c := (0, 1)$, so that the T-statistic can be used to evaluate the slope parameter of the simple linear regression. Finally, in both cases we consider the parameter $\beta_0 := (0, 0)^T$, so that in the case of β_A, $c^T\beta = c^T\beta_0$ holds, and in the case of β_B, $c^T\beta \neq c^T\beta_0$ holds. The following **R** code implements the outlined scenarios; Figure 38.2 A and B show the resulting simulated and analytical distributions of the T-statistic.

```
# model formulation
library(MASS)                                                          # multivariate normal distribution
n          = 10                                                        # number of data points
p          = 2                                                         # number of beta parameters
x          = 1:n                                                       # predictor values
X          = matrix(c(rep(1,n),x), ncol = p)                           # design matrix
I_n        = diag(n)                                                   # identity matrix
beta       = matrix(c(1,0,1,1), nrow = 2)                              # true, but unknown, beta parameters
nscn       = ncol(beta)                                                # number of true, but unknown, hypothesis scenarios
sigsqr     = 1                                                         # true, but unknown, variance parameter
c          = matrix(c(0,1), nrow = 2)                                  # contrast vector of interest
beta_0     = matrix(c(0,0), nrow = 2)                                  # null-hypothesis beta parameter

# frequentist simulation
nsim       = 1e4                                                       # number of simulations
delta      = rep(NaN, nscn)                                            # noncentrality-parameter array
Tee        = matrix(rep(NaN, nscn*nsim), ncol = nscn)                  # T-test-statistic realization array
for(s in 1:nscn){                                                      # hypothesis scenarios
  delta[s]    = ((t(c) %*% beta[,s] - t(c) %*% beta_0)/                # noncentrality parameter
                sqrt(sigsqr*t(c)%*%solve(t(X)%*%X)%*%c))
  for(i in 1:nsim){                                                    # simulation iterations
    y          = mvrnorm(1, X %*% beta[,s], sigsqr*I_n)                # y
    beta_hat   = solve(t(X) %*% X) %*% t(X) %*% y                      # \hat{\beta}
    eps_hat    = y - X %*% beta_hat                                    # \hat{\eps}
    sigsqr_hat = (t(eps_hat) %*% eps_hat)/(n-p)                        # \hat{\sigma}^2
    Tee[i,s]   = ((t(c) %*% beta_hat - t(c) %*% beta_0)/               # T
                  sqrt(sigsqr_hat*t(c)%*%solve(t(X)%*%X)%*%c))
  }
}
```

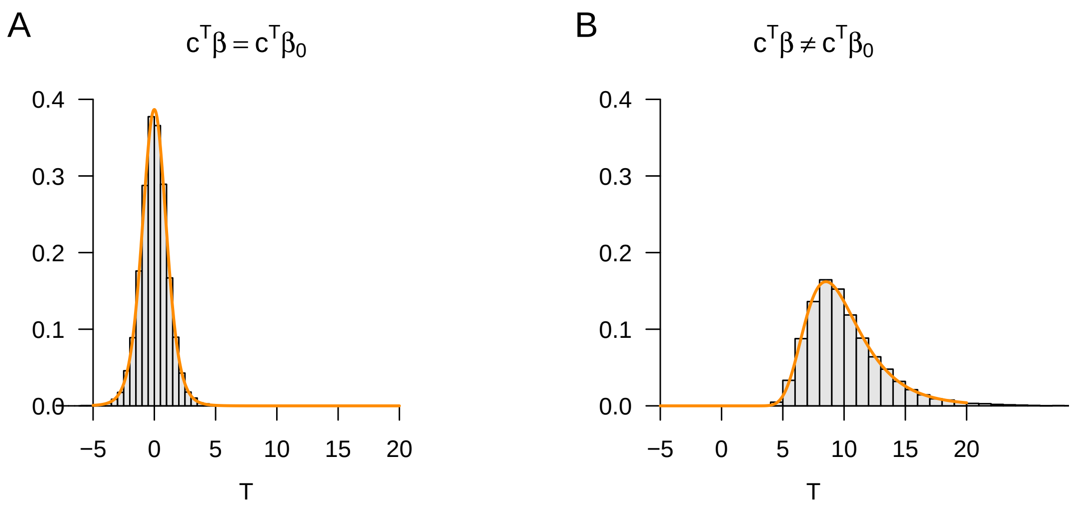


Figure 38.2 Distributions of the T-statistic in simple linear regression.

38.2 Confidence intervals for beta-parameter components

Using the T-statistic, confidence intervals for the components of the beta-parameter vector can be determined. The following theorem is the central statement of this section.

Theorem 38.2 (Confidence intervals for beta-parameter components)**.** *Let*

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{38.11}$$

be the GLM, let

$$\hat{\beta} := \left(X^T X\right)^{-1} X^T y \text{ and } \hat{\sigma}^2 := \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{n-p} \tag{38.12}$$

be the beta-parameter and variance-parameter estimators, respectively, and, for $\delta \in]0,1[$, let

$$t_\delta := \Psi^{-1}\left(\frac{1+\delta}{2}; n-p\right). \tag{38.13}$$

Finally, for $j = 1, \ldots, p$, let

$$\lambda_j := \left(\left(X^T X\right)^{-1}\right)_{jj} \text{ be the jth diagonal element of } \left(X^T X\right)^{-1}. \tag{38.14}$$

Then, for $j = 1, \ldots, p$,

$$\kappa_j := \left[\hat{\beta}_j - \hat{\sigma}\sqrt{\lambda_j} t_\delta, \hat{\beta}_j + \hat{\sigma}\sqrt{\lambda_j} t_\delta\right] \tag{38.15}$$

is a δ confidence interval for the jth component β_j of the beta parameter $\beta = \left(\beta_1, \ldots, \beta_p\right)^T$.

∘

Proof. We have to show that

$$\mathbb{P}\left(\kappa_j \ni \beta_j\right) = \delta. \tag{38.16}$$

To this end, we first note that, for all $j = 1, \dots, p$, when choosing $\beta_0 = \beta$ and $c := e_j$, Theorem 38.1 implies

$$T = \frac{e_j^T\hat{\beta} - e_j^T\beta}{\sqrt{\hat{\sigma}^2 e_j^T\left(X^TX\right)^{-1}e_j}} = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2\left(\left(X^TX\right)^{-1}\right)_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\lambda_j}} =: T_j \tag{38.17}$$

and

$$\frac{e_j^T\beta - e_j^T\beta}{\sqrt{\sigma^2 e_j^T\left(X^TX\right)^{-1}e_j}} = 0. \tag{38.18}$$

Thus $T_j \sim t(n-p)$. Furthermore, we recall (cf. Chapter 32) that, by definition of t_δ,

$$\mathbb{P}\left(-t_\delta \le T_j \le t_\delta\right) = \delta. \tag{38.19}$$

From the definition of a δ confidence interval, it follows that

$$\begin{aligned}
\delta &= \mathbb{P}\left(-t_\delta \le T_j \le t_\delta\right)\\
&= \mathbb{P}\left(-t_\delta \le \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\lambda_j}} \le t_\delta\right)\\
&= \mathbb{P}\left(-t_\delta\hat{\sigma}\sqrt{\lambda_j} \le \hat{\beta}_j - \beta_j \le t_\delta\hat{\sigma}\sqrt{\lambda_j}\right)\\
&= \mathbb{P}\left(-\hat{\beta}_j - t_\delta\hat{\sigma}\sqrt{\lambda_j} \le -\beta_j \le -\hat{\beta}_j + t_\delta\hat{\sigma}\sqrt{\lambda_j}\right)\\
&= \mathbb{P}\left(\hat{\beta}_j + t_\delta\hat{\sigma}\sqrt{\lambda_j} \ge \beta_j \ge \hat{\beta}_j - t_\delta\hat{\sigma}\sqrt{\lambda_j}\right)\\
&= \mathbb{P}\left(\hat{\beta}_j - t_\delta\hat{\sigma}\sqrt{\lambda_j} \le \beta_j \le \hat{\beta}_j + t_\delta\hat{\sigma}\sqrt{\lambda_j}\right)\\
&= \mathbb{P}\left(\left[\hat{\beta}_j - \hat{\sigma}\sqrt{\lambda_j}t_\delta, \hat{\beta}_j + \hat{\sigma}\sqrt{\lambda_j}t_\delta\right] \ni \beta_j\right)\\
&= \mathbb{P}\left(\kappa_j \ni \beta_j\right)
\end{aligned} \tag{38.20}$$

and the claim is proved. □

Example (1) Independent and identically normally distributed random variables

As usual, we first consider the GLM form of the scenario of independent and identically normally distributed random variables

$$y \sim N\left(X\beta, \sigma^2 I_n\right) \text{ with } X := 1_n \in \mathbb{R}^n, \beta := \mu \in \mathbb{R}, \sigma^2 > 0. \tag{38.21}$$

Then, as already seen,

$$\hat{\beta} = \frac{1}{n}\sum_{i=1}^n y_i =: \bar{y}, \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^n \left(y_i - \bar{y}\right)^2 =: s_y^2 \text{ and } \lambda_1 = \left(1_n^T 1_n\right)^{-1} = \frac{1}{n}. \tag{38.22}$$

By Theorem 38.2, it follows that

$$\kappa := \left[\bar{y} - \frac{s_y}{\sqrt{n}}t_\delta, \bar{y} + \frac{s_y}{\sqrt{n}}t_\delta\right] \tag{38.23}$$

is a δ confidence interval for β, and this is evidently identical to the familiar δ confidence interval for the expectation parameter of the normal distribution.

Example (2) Simple linear regression

In this example, we do not want to discuss the specific form of the confidence intervals for the intercept and slope parameters, but merely use the following simulation to recall the frequentist meaning of a δ confidence interval: realizations of δ confidence intervals cover the true, but unknown, parameter value with frequentist probability δ. Based on the following **R** code, Figure 38.3 shows that, in the concrete simulation with $\delta = 0.95$, this is the case in 94 of 100 cases for the intercept parameter and in 93 of 100 cases for the slope parameter.

```r
# model formulation
library(MASS)                                                                  # multivariate normal distribution
set.seed(0)                                                                    # random number generator seed
ns          = 1e2                                                              # number of simulations
n           = 10                                                               # number of data points
p           = 2                                                                # number of beta parameters
x           = 1:n                                                              # predictor values
X           = matrix(c(rep(1,n),x), ncol = p)                                  # design matrix
I_n         = diag(n)                                                          # identity matrix
beta        = matrix(c(1,2), nrow = 2)                                         # true, but unknown, beta parameters
sigsqr      = 1                                                                # true, but unknown, variance parameter
delta       = 0.95                                                             # confidence level
t_delta     = qt((1+delta)/2,n-p)                                              # \Psi^{-1}((1+\delta)/2,n-p)
lambda      = diag(solve(t(X) %*% X))                                          # \lambda_j values

# simulation
kappa       = array(rep(NaN, ns*p*p), dim=c(ns,2,2))                           # confidence-interval array
beta_hat    = matrix(rep(NaN,p*ns), nrow = p)                                  # beta-parameter estimator
for(i in 1:ns){                                                                # iteration over realizations
  y              = mvrnorm(1, X %*% beta, sigsqr*I_n)                          # data realization
  beta_hat[,i]   = solve(t(X) %*% X) %*% t(X) %*% y                            # \hat{\beta}
  eps_hat        = y - X %*% beta_hat[,i]                                      # \hat{\varepsilon}
  sigsqr_hat     = (t(eps_hat) %*% eps_hat)/(n-p)                              # \hat{\sigma}^2
  for(j in 1:p){                                                               # iteration over beta-array components
    kappa[i,1,j] = beta_hat[j,i]-sqrt(sigsqr_hat*lambda[j])*t_delta            # lower CI bound
    kappa[i,2,j] = beta_hat[j,i]+sqrt(sigsqr_hat*lambda[j])*t_delta            # upper CI bound
  }
}
```

38.3 Bibliographic remarks

Box (1981) and Zabell (2008) give a historical overview of the development of the T-statistic and its distribution in the context of the work of Student (1908) and Fisher (1925c), Fisher (1925b), and Fisher (1925a). The theory of confidence intervals goes back to Neyman (1935) and Neyman (1937).

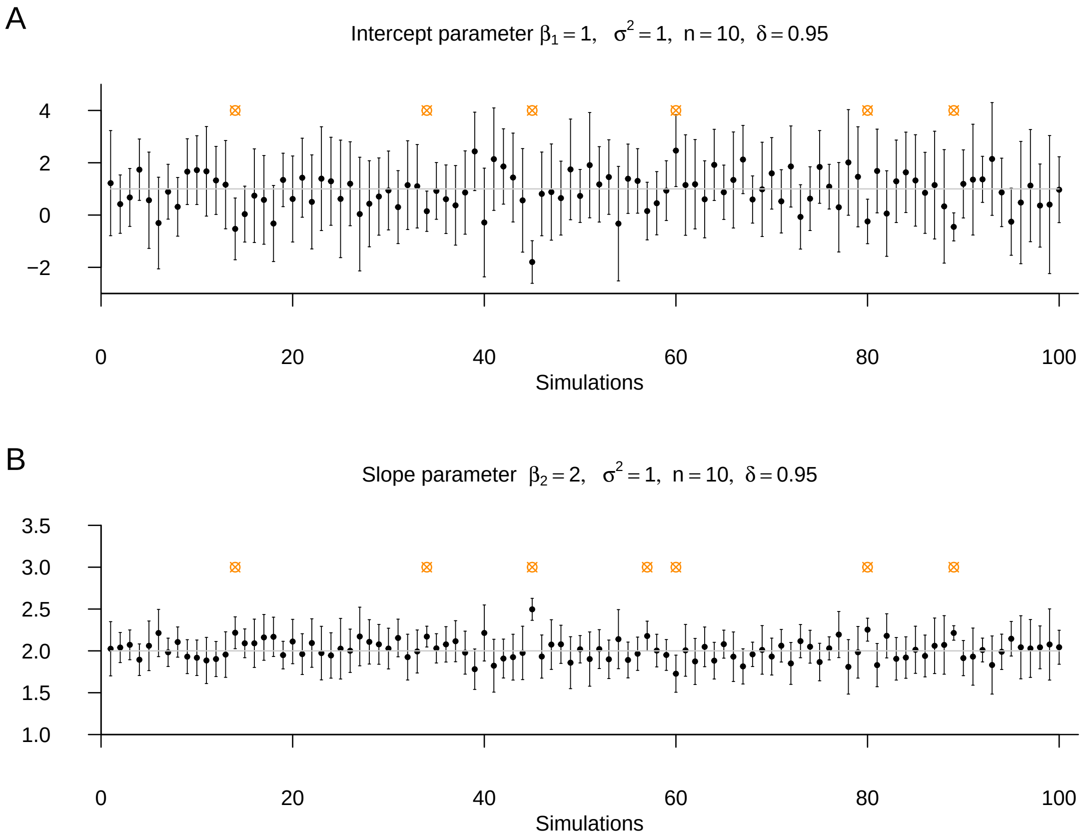


Figure 38.3 Confidence intervals for beta-parameter components in simple linear regression.

39 F-statistics

In this section, we introduce F-statistics against the background of model comparisons based on likelihood-ratio statistics. Using the (maximized or marginal) likelihood of a dataset under a given probabilistic model as a model comparison criterion is a widely used procedure in probabilistic data analysis. In contrast to T-statistics, the goal of computing F-statistics can therefore be not only to evaluate linear combinations of beta-parameter estimates probabilistically, but also to evaluate a model's fit to a dataset as a whole. The model-comparison capacity of F-statistics is, however, somewhat limited because the F-statistic refers only to GLMs and, in particular, to nested GLMs in which one model is part of another model. F-statistics usually form the basis for hypothesis tests in the context of analysis-of-variance procedures. The use of F-statistics is not, however, restricted per se to analyses of variance, but can also be appropriate in parametric GLM designs. In the following, we first introduce the concept of the likelihood-ratio statistic and then consider the definition of the F-statistic against this background. We close with the frequentist distribution of the F-statistic.

39.1 Likelihood-ratio statistics

We define the concept of the likelihood-ratio statistic as follows.

Definition 39.1 (Likelihood-ratio statistic)**.** Let two frequentist inference models

$$\mathcal{M}_0 := \left(\mathcal{Y}, \mathcal{A}, \left\{\mathbb{P}^0_{\theta_0} \mid \theta_0 \in \Theta_0\right\}\right) \text{ and } \mathcal{M}_1 := \left(\mathcal{Y}, \mathcal{A}, \left\{\mathbb{P}^1_{\theta_1} \mid \theta_1 \in \Theta_1\right\}\right) \tag{39.1}$$

be given, with identical data space, identical σ-algebra, and potentially distinct sets of probability measures and parameter spaces. Furthermore, let y be a random vector with data space $\mathcal{Y}$. Finally, let L_0^y and L_1^y be the likelihood functions of $\mathcal{M}_0$ and $\mathcal{M}_1$, respectively, where the superscript ^{y} is meant to recall the data dependence of the likelihood function. Then

$$\Lambda := \frac{\max_{\theta_0 \in \Theta_0} L_0^y(\theta_0)}{\max_{\theta_1 \in \Theta_1} L_1^y(\theta_1)} \tag{39.2}$$

is called the *likelihood-ratio statistic*.

•

A likelihood-ratio statistic relates the probability mass or density of an observed dataset $y \in \mathcal{Y}$ under two frequentist inference models after optimization of the respective model parameters. A high value of the likelihood-ratio statistic corresponds to a higher probability mass or density of the observed dataset $y \in \mathcal{Y}$ under $\mathcal{M}_0$ than under $\mathcal{M}_1$, and vice versa.

Considering the probability masses or densities of observed data after model estimation under different models is a general procedure for comparing models. Ultimately, this procedure makes it possible to compare different scientific theories about the genesis of observable data quantitatively and to quantify the associated uncertainty. Model comparisons are a central topic in Bayesian inference, which generalizes the logic of likelihood-ratio statistics to general probabilistic models, for example under the terms Bayes factors or Bayesian information criterion. However, as seen here, model comparisons are also possible and useful in frequentist inference; model comparison is therefore not a unique feature of Bayesian inference relative to frequentist inference.

With the *reduced model* and the *full model*, we consider in the following two special forms of $\mathcal{M}_0$ and $\mathcal{M}_1$, respectively, in the context of the GLM.

Definition 39.2 (Full and reduced model)**.** For $p > 1$ with $p = p_0 + p_1$, let

$$X := \begin{pmatrix} X_0 & X_1 \end{pmatrix} \in \mathbb{R}^{n \times p} \text{ with } X_0 \in \mathbb{R}^{n \times p_0} \text{ and } X_1 \in \mathbb{R}^{n \times p_1} \tag{39.3}$$

and

$$\beta := \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \in \mathbb{R}^p \text{ with } \beta_0 \in \mathbb{R}^{p_0} \text{ and } \beta_1 \in \mathbb{R}^{p_1} \tag{39.4}$$

be partitions of an $n \times p$ design matrix and a p-dimensional beta-parameter vector. Then we call

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{39.5}$$

the *full model* and

$$y = X_0\beta_0 + \varepsilon_0 \text{ with } \varepsilon_0 \sim N\left(0_n, \sigma^2 I_n\right) \tag{39.6}$$

the *reduced model*, and we speak of a *partition of a (full) model.*

•

One also says that the reduced model is *nested* in the full model. The likelihood-ratio statistic for comparing a full and a reduced model has a simple form. This is the central aspect of the following theorem.

Theorem 39.1 (Likelihood-ratio statistic of full and reduced model)**.** *For $p = p_0 + p_1, p > 1$, let a partition of a full GLM be given, and let $\hat{\sigma}^2$ and $\hat{\sigma}_0^2$ be the maximum-likelihood estimators of the variance parameter under the full and reduced model, respectively. Furthermore, let the two parametric statistical models $\mathcal{M}_0$ and $\mathcal{M}_1$ in the definition of the likelihood-ratio statistic be given by the reduced model and the full model. Then*

$$\Lambda = \left(\frac{\hat{\sigma}^2}{\hat{\sigma}_0^2}\right)^{\frac{n}{2}}. \tag{39.7}$$

∘

Proof. We first recall that the maximum-likelihood estimators of the variance parameter are given by

$$\hat{\sigma}^2 = \frac{1}{n}(y - X\hat{\beta})^T(y - X\hat{\beta}) \text{ and } \hat{\sigma}_0^2 = \frac{1}{n}\left(y - X_0\hat{\beta}_0\right)^T\left(y - X_0\hat{\beta}_0\right), \tag{39.8}$$

respectively, where $\hat{\beta}$ and $\hat{\beta}_0$ denote the maximum-likelihood estimators of the beta parameters under the full and reduced model, respectively. Furthermore, we note that for the likelihood function of the full model at the maximum-likelihood estimators,

$$
\begin{aligned}
L_1^y\left(\hat{\beta}, \hat{\sigma}^2\right) &= (2\pi)^{-\frac{n}{2}} \left(\hat{\sigma}^2\right)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\hat{\sigma}^2}(y - X\hat{\beta})^T(y - X\hat{\beta})\right) \\
&= (2\pi)^{-\frac{n}{2}} \left(\hat{\sigma}^2\right)^{-\frac{n}{2}} \exp\left(-\frac{n}{2}\frac{(y - X\hat{\beta})^T(y - X\hat{\beta})}{(y - X\hat{\beta})^T(y - X\hat{\beta})}\right) \\
&= (2\pi)^{-\frac{n}{2}} \left(\hat{\sigma}^2\right)^{-\frac{n}{2}} e^{-\frac{n}{2}},
\end{aligned}
\tag{39.9}
$$

and, analogously, that for the likelihood function of the reduced model at the maximum-likelihood estimators,

$$
L_0^y\left(\hat{\beta}_0, \hat{\sigma}_0^2\right) = (2\pi)^{-\frac{n}{2}} \left(\hat{\sigma}_0^2\right)^{-\frac{n}{2}} e^{-\frac{n}{2}}. \tag{39.10}
$$

This yields

$$
\Lambda = \frac{\max_{\theta_0 \in \Theta_0} L_0^y(\theta_0)}{\max_{\theta_1 \in \Theta_1} L_1^y(\theta_1)} = \frac{L_0^y\left(\hat{\beta}_0, \hat{\sigma}_0^2\right)}{L_1^y\left(\hat{\beta}, \hat{\sigma}^2\right)} = \frac{(2\pi)^{-\frac{n}{2}} \left(\hat{\sigma}_0^2\right)^{-\frac{n}{2}} e^{-\frac{n}{2}}}{(2\pi)^{-\frac{n}{2}} \left(\hat{\sigma}^2\right)^{-\frac{n}{2}} e^{-\frac{n}{2}}} = \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}^2}\right)^{-\frac{n}{2}} = \left(\frac{\hat{\sigma}^2}{\hat{\sigma}_0^2}\right)^{\frac{n}{2}}. \tag{39.11}
$$

□

39.2 Definition and distribution

We now define the F-statistic against the background of a full and a reduced model.

Definition 39.3 (F-statistic)**.** For $X \in \mathbb{R}^{n \times p}, \beta \in \mathbb{R}^p$ and $\sigma^2 > 0$, let a GLM of the form

$$
y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{39.12}
$$

be given with the partition

$$
X = \begin{pmatrix} X_0 & X_1 \end{pmatrix}, X_0 \in \mathbb{R}^{n \times p_0}, X_1 \in \mathbb{R}^{n \times p_1}, \text{ and } \beta := \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \beta_0 \in \mathbb{R}^{p_0}, \beta_1 \in \mathbb{R}^{p_1}, \tag{39.13}
$$

with $p = p_0 + p_1$. Furthermore, with

$$
\hat{\beta}_0 := \left(X_0^T X_0\right)^{-1} X_0^T y \text{ and } \hat{\beta} := \left(X^T X\right)^{-1} X^T y, \tag{39.14}
$$

let the residual vectors

$$
\hat{\varepsilon}_0 := y - X_0\hat{\beta}_0 \text{ and } \hat{\varepsilon} := y - X\hat{\beta} \tag{39.15}
$$

be defined. Then the *F-statistic* is defined as

$$
F := \frac{\left(\hat{\varepsilon}_0^T \hat{\varepsilon}_0 - \hat{\varepsilon}^T \hat{\varepsilon}\right)/p_1}{\hat{\varepsilon}^T \hat{\varepsilon}/(n-p)}. \tag{39.16}
$$

•

The numerator of the F-statistic,

$$
\frac{\hat{\varepsilon}_0^T \hat{\varepsilon}_0 - \hat{\varepsilon}^T \hat{\varepsilon}}{p_1}, \tag{39.17}
$$

measures the extent to which the p_1 regressors in X_1 reduce the residual sum of squares, relative to the number of these regressors. This means that, for the same magnitude of residual-sum-of-squares reduction (and the same denominator), a larger value of F results when this reduction is achieved by fewer additional regressors, that is, when p_1 is small, and vice versa. In terms of the number of columns of X and the corresponding components of β, the F-statistic therefore favors less "complex" models.

For the denominator of the F-statistic,

$$\frac{\hat{\varepsilon}^T\hat{\varepsilon}}{n-p} = \hat{\sigma}^2, \tag{39.18}$$

where $\hat{\sigma}^2$ is here the estimator of σ^2 estimated on the basis of the full model. Note that this estimator is not identical to the maximum-likelihood estimator of the variance parameter considered above, whose denominator is n, because of the denominator $n-p$. If the data are in fact generated under the reduced model, the full model can represent this by $\hat{\beta}_2 \approx 0_{p_1}$ and achieves a similar estimate of σ^2 as the reduced model. If the data are de facto generated under the full model, then $\hat{\varepsilon}^T\hat{\varepsilon}/(n-p)$ is a better estimator of σ^2 than $\hat{\varepsilon}_0^T\hat{\varepsilon}_0/(n-p)$, because in the latter the data variability not explained by the p_0 regressors in X_0 would be reflected in the estimate of σ^2. The denominator of the F-statistic is therefore the more meaningful estimator of σ^2 in both cases.

Taken together, the F-statistic thus measures the residual-sum-of-squares reduction due to the p_1 regressors in X_1 relative to the p_0 regressors in X_0 per unit of data variability (σ^2) and per regressor (p_1).

Example (1) Simple linear regression

As an example, the **R** code below evaluates the F-statistic in the context of the following partition of the simple linear regression model,

$$X = \begin{pmatrix} X_0 & X_1 \end{pmatrix}, X_0 := 1_n, X_1 := (x_1, \dots, x_n)^T, \beta := \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \tag{39.19}$$

once for the case $\beta = (1, 0)^T$, that is, the reduced model is the true, but unknown, data-generating model, and once for the case $\beta = (1, 0.5)^T$, that is, the full model is the true, but unknown, data-generating model. In the sense of the above discussion, the first case yields an F-statistic close to zero, whereas the second case yields a high F-statistic.

```
# model formulation
library(MASS)                                                  # multivariate normal distribution
set.seed(0)                                                    # random number generator seed
nmod   = 2                                                     # number of models
n      = 10                                                    # number of data points
p      = 2                                                     # number of beta parameters
p_0    = 1                                                     # number of beta parameters in the reduced model
p_1    = 1                                                     # number of additional beta parameters in the full model
p      = p_0 + p_1                                             # number of beta parameters in the full model
x      = 1:n                                                   # predictor values
X      = matrix(c(rep(1,n),x), nrow = n)                       # design matrix of the full model
X_0    = X[,1]                                                 # design matrix of the reduced model
I_n    = diag(n)                                               # n x n identity matrix
beta   = matrix(c(1,0,1,.5), nrow = 2)                         # true, but unknown, beta parameters
nscn   = ncol(beta)                                            # number of true, but unknown, hypothesis scenarios
sigsqr = 1                                                     # true, but unknown, variance parameter

# model simulation and evaluation
Eff    = matrix(rep(NaN, nscn), nrow = nscn)                   # F-statistic realization array
for(s in 1:nscn){                                              # scenario iterations
  y             = mvrnorm(1, X %*%beta[,s], sigsqr*I_n)        # data realization
  beta_hat_0    = solve(t(X_0)%*%X_0)%*%t(X_0)%*%y             # beta-parameter estimator, reduced model
  beta_hat      = solve(t(X)  %*%X  )%*%t(X)  %*%y             # beta-parameter estimator, full model
  eps_0_hat     = y-X_0%*%beta_hat_0                           # residual vector, reduced model
  eps_hat       = y-X%*%beta_hat                               # residual vector, full model
  eps_0_eps_0_hat = t(eps_0_hat) %*% eps_0_hat                 # RSS, reduced model
  eps_eps_hat   = t(eps_hat)   %*% eps_hat                     # RSS, full model
  Eff[s]        = (((eps_0_eps_0_hat-eps_eps_hat)/p_1)/        # F-statistic
                    (eps_eps_hat/(n-p)))}
```

```
F-statistic for beta_1  = 0_{p_1}: 0.03223958
F-statistic for beta_1 != 0_{p_1}: 47.8873
```

The likelihood-ratio statistic and the F-statistic of full and reduced models can be transformed into one another. This is the central statement of the following theorem.

Theorem 39.2 (F-statistic and likelihood-ratio statistic)**.** *Let a partition of a GLM into a full and a reduced model be given, and let F and Λ be the corresponding F- and likelihood-ratio statistics. Then*

$$F = \frac{n-p}{p_1}\left(\Lambda^{-\frac{2}{n}} - 1\right). \tag{39.20}$$

∘

Proof. We first recall that the maximum-likelihood estimators of the variance parameter are given by

$$\hat{\sigma}^2 = \frac{1}{n}(y - X\hat{\beta})^T(y - X\hat{\beta}) = \frac{\hat{\varepsilon}^T\hat{\varepsilon}}{n} \text{ and } \hat{\sigma}_0^2 = \frac{1}{n}\left(y - X_0\hat{\beta}_0\right)^T\left(y - X_0\hat{\beta}_0\right) = \frac{\hat{\varepsilon}_0^T\hat{\varepsilon}_0}{n} \tag{39.21}$$

respectively. With the definition of the F-statistic and the form of the likelihood-ratio statistic for comparing reduced and full models, we obtain

$$\begin{aligned} F &= \frac{\left(\hat{\varepsilon}_0^T\hat{\varepsilon}_0 - \hat{\varepsilon}^T\hat{\varepsilon}\right)/p_1}{\hat{\varepsilon}^T\hat{\varepsilon}/(n-p)} \\ &= \frac{n\left(\hat{\sigma}_0^2 - \hat{\sigma}^2\right)/p_1}{n\hat{\sigma}^2/(n-p)} \\ &= \frac{n-p}{p_1}\frac{\hat{\sigma}_0^2 - \hat{\sigma}^2}{\hat{\sigma}^2} \\ &= \frac{n-p}{p_1}\left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}^2} - \frac{\hat{\sigma}^2}{\hat{\sigma}^2}\right) \\ &= \frac{n-p}{p_1}\left(\Lambda^{-\frac{2}{n}} - 1\right). \end{aligned} \tag{39.22}$$

□

Thus, between the F-statistic and the likelihood-ratio statistic there is a nonlinear, reciprocal relationship, which we visualize for $n = 12, p = 2$ and $p_1 = 1$ in Figure 39.1. Note that for $\Lambda = 1$, $F = 0$. A value of $F = 0$ therefore implies that the reduced model has the same plausibility as the full model in light of an observed dataset.

We document the frequentist distribution of the F-statistic in the following theorem, whose proof we omit.

Theorem 39.3 (F-statistic)**.** *For $X \in \mathbb{R}^{n\times p}, \beta \in \mathbb{R}^p$ and $\sigma^2 > 0$, let a GLM of the form*

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{39.23}$$

be given with the partition

$$X = \begin{pmatrix} X_0 & X_1 \end{pmatrix}, X_0 \in \mathbb{R}^{n\times p_0}, X_1 \in \mathbb{R}^{n\times p_1}, \text{ and } \beta := \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \beta_0 \in \mathbb{R}^{p_0}, \beta_1 \in \mathbb{R}^{p_1}, \tag{39.24}$$

with $p = p_0 + p_1$. Finally, let

$$c := \begin{pmatrix} 0_{p_0} \\ 1_{p_1} \end{pmatrix} \in \mathbb{R}^p \tag{39.25}$$

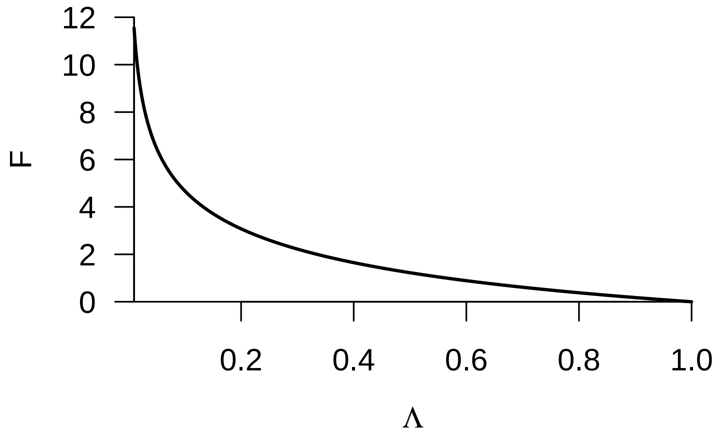


Figure 39.1 Relationship between the F-statistic and the likelihood-ratio statistic.

be a vector. Then

$$F \sim f\left(\delta, p_1, n-p\right) \;\; \textit{with}\; \delta := \frac{c^T\beta\left(c^T\left(X^TX\right)^{-1}c\right)^{-1}c^T\beta}{\sigma^2}. \tag{39.26}$$

∘

Note that the F-statistic is a function of the parameter estimator, whereas δ is a function of the true, but unknown, parameters. Like the distribution of the T-statistic, the distribution of the F-statistic can be used for the evaluation of frequentist confidence intervals and hypothesis tests. We illustrate the latter aspect in particular in Chapter 41 and **?@sec-two-way-analysis-of-variance**.

Example (1) Simple linear regression

As an example, using the **R** code below we evaluate the distribution of the F-statistic in the context of the partition of the simple linear regression model, again once for the case $\beta = (1,0)^T$, that is, the reduced model is the true, but unknown, data-generating model, and once for the case $\beta = (1,0.5)^T$, that is, the full model is the true, but unknown, data-generating model. Figure 39.2 A and B visualize the resulting distributions, respectively.

```
# model formulation
library(MASS)                                          # multivariate normal distribution
set.seed(0)                                            # random number generator seed
nmod   = 2                                             # number of models
n      = 10                                            # number of data points
p_0    = 1                                             # number of beta parameters in the reduced model
p_1    = 1                                             # number of additive beta parameters in the full model
p      = p_0 + p_1                                     # number of beta parameters in the full model
x      = 1:n                                           # predictor values
X      = matrix(c(rep(1,n),x), nrow = n)               # design matrix of the full model
X_0    = X[,1]                                         # design matrix of the reduced model
I_n    = diag(n)                                       # n x n identity matrix
beta   = matrix(c(1,0,1,.5), nrow = 2)                 # true, but unknown, beta parameters
nscn   = ncol(beta)                                    # number of true, but unknown, hypothesis scenarios
```

```
sigsqr = 1                                                                # true, but unknown, variance parameter
c      = matrix((c(0,1)), nrow = 2)                                       # vector

# frequentist simulation
nsim   = 1e4                                                              # number of realizations of the n-dimensional random vector
delta  = rep(NaN,nscn)                                                    # noncentrality-parameter array
Eff    = matrix(rep(NaN, nscn*nsim), nrow = nscn)                         # F-statistic realization array
for(s in 1:nscn){                                                         # scenario iterations
  delta[s] = (t(t(c)%*%beta[,s])%*%                                       # noncentrality parameter
              solve(t(c)%*%solve(t(X)%*%X)%*%c) %*%
              (t(c)%*%beta[,s])/sigsqr)
  for(i in 1:nsim){                                                       # simulation iterations
    y                = mvrnorm(1, X %*%beta[,s], sigsqr*I_n)              # data realization
    beta_hat_0       = solve(t(X_0)%*%X_0)%*%t(X_0)%*%y                   # beta-parameter estimator, reduced model
    beta_hat         = solve(t(X)  %*%X  )%*%t(X)  %*%y                   # beta-parameter estimator, full model
    eps_0_hat        = y-X_0%*%beta_hat_0                                 # residual vector, reduced model
    eps_hat          = y-X%*%beta_hat                                     # residual vector, full model
    eps_0_eps_0_hat = t(eps_0_hat) %*% eps_0_hat                          # RSS, reduced model
    eps_eps_hat      = t(eps_hat)   %*% eps_hat                           # RSS, full model
    Eff[s,i]         = (((eps_0_eps_0_hat-eps_eps_hat)/p_1)/              # F-statistic
                         (eps_eps_hat/(n-p)))}}
```

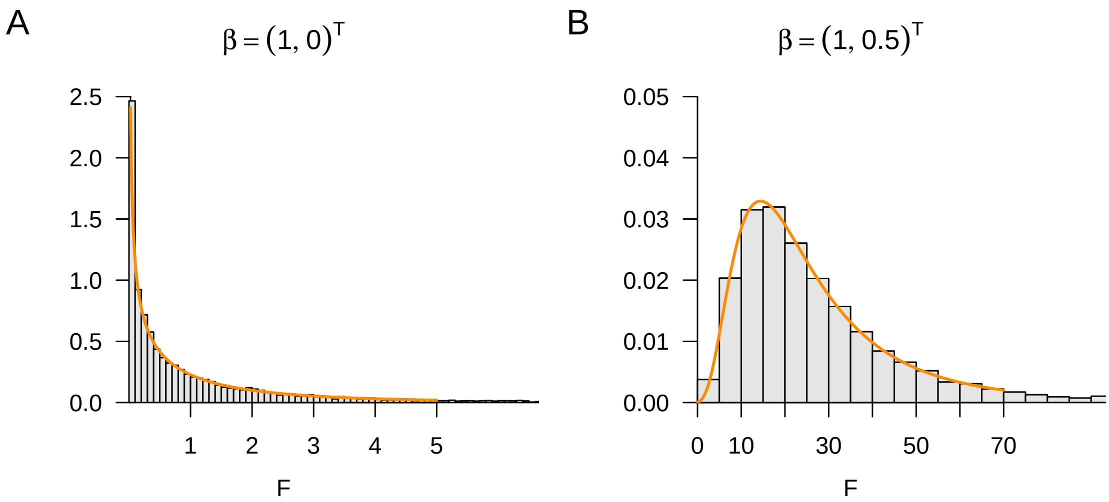


Figure 39.2 Exemplary F-statistic distributions under reduced and full true, but unknown, data-generating models in simple linear regression.

39.3 Bibliographic remarks

The popularity of F-statistics, especially in the context of analysis of variance, is generally traced back to Fisher (1925a). Seal (1967) gives a historical overview. Likelihood-ratio statistics are considered in particular by Neyman & Pearson (1928) and Wilks (1938). Lehmann (2011) gives an integrated historical overview of both approaches. The equivalence of likelihood-ratio statistic and F-statistic discussed here is based on the presentation in Seber & Lee (2003).

40 T-tests

40.1 One-sample T-tests

Application scenario

The application scenario of a one-sample T-test is characterized, as is well known, by considering n univariate data points from one sample (group) of randomized experimental units, which are assumed to be realizations of n independent and identically normally distributed random variables. With respect to the identical univariate normal distributions $N(\mu, \sigma^2)$ of these random variables, both the expectation parameter μ and the variance parameter σ^2 are assumed to be unknown. Finally, one assumes an interest in an inferential comparison of the unknown expectation parameter μ with a specified value μ_0, for example $\mu_0 := 0$.

Application example

For a concrete application example, we consider the analysis of pre-post intervention BDI difference scores for one group of $n = 12$ patients, as shown in Table 40.1. The first two columns of this table list the patient-specific BDI values before (PreBDI) and after (PosBDI) the psychotherapeutic intervention; the third column, dBDI, shows the corresponding PreBDI-PosBDI difference scores. A positive value here corresponds to an improvement in depressive symptoms, and a negative value to a worsening of depressive symptoms.

Table 40.1 Pre- and post-intervention BDI values

PreBDI	PosBDI	dBDI
29	28	1
32	26	6
33	29	4
28	26	2
31	30	1
32	27	5
33	27	6
30	27	3
29	27	2
30	28	2
33	29	4
31	25	6

When applying a one-sample T-test to the dBDI data of this dataset, we thus assume that the `dBDI` data are realizations of $n = 12$ independently normally distributed random variables $y_i \sim N(\mu, \sigma^2)$. We further assume that we are interested in quantifying our uncertainty in the inferential comparison of the true, but unknown, expectation parameter μ with a comparison value μ_0 in terms of a hypothesis test.

Independently of this inferential procedure, we first consider the descriptive statistics of the dBDI data, as shown in Table 40.2. In particular, it is noticeable that the sample mean is relatively small compared with the standard deviation. On average, the PreBDI and PosBDI values therefore differ in a positive direction, which implies a reduction in depressive symptoms, but the data also vary considerably across patients, as is already apparent from Table 40.1.

Table 40.2 Descriptive statistics of the pre-post BDI difference scores

	n	Max	Min	Median	Mean	Var	Std
F2F	12	6	1	3.5	3.5	3.73	1.93

Model formulation

We now define the one-sample T-test model as follows.

Definition 40.1 (One-sample T-test model)**.** For $i = 1, \dots, n$, let y_i be random variables that model the n data points of a one-sample T-test scenario. Then the one-sample T-test model has the structural form

$$y_i = \mu + \varepsilon_i \text{ with } \varepsilon_i \sim N\left(0, \sigma^2\right) \text{ i.i.d. for } i = 1, \dots, n \text{ with } \mu \in \mathbb{R} \text{ and } \sigma^2 > 0, \tag{40.1}$$

the data-distribution form

$$y_i \sim N\left(\mu, \sigma^2\right) \text{ i.i.d. for } i = 1, \dots, n \text{ with } \mu \in \mathbb{R} \text{ and } \sigma^2 > 0, \tag{40.2}$$

and, for the data vector $y = (y_1, \dots, y_n)^T$, the design-matrix form

$$y = X\beta + \varepsilon \text{ with } X := 1_n \in \mathbb{R}^{n \times 1}, \beta := \mu \in \mathbb{R}, \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \text{ and } \sigma^2 > 0. \tag{40.3}$$

•

The one-sample T-test model is evidently identical to the model of independent and identically normally distributed random variables (cf. Chapter 36). The equivalence of the structural, data-distribution, and design-matrix forms of the one-sample T-test model was already discussed in detail in Chapter 36. Accordingly, simulating data based on the one-sample T-test model has the same form as simulating independent and identically normally distributed random variables. The **R** code below demonstrates this.

```
# model formulation
library(MASS)                                                    # multivariate normal distribution
n      = 12                                                      # number of data points
p      = 1                                                       # number of beta parameters
X      = matrix(rep(1,n), nrow = n)                              # design matrix
I_n    = diag(n)                                                 # n x n identity matrix
beta   = 5                                                       # true, but unknown, beta parameter
sigsqr = 14                                                      # true, but unknown, variance parameter

# data realization
y      = mvrnorm(1, X %*% beta, sigsqr*I_n)                      # one realization of an n-dimensional random vector
```

Model estimation

Because the form of the one-sample T-test model is identical to the scenario of independent and identically normally distributed random variables, this also applies to the corresponding beta- and variance-parameter estimators. We thus obtain the following theorem, which was already proved in Chapter 37.

Theorem 40.1 (Parameter estimators in the one-sample T-test model). *Let the design-matrix form of the one-sample T-test model be given. Then, for the beta-parameter estimator,*

$$\hat{\beta} = \frac{1}{n}\sum_{i=1}^{n} y_i =: \bar{y} \tag{40.4}$$

and for the variance-parameter estimator,

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n} (y_i - \bar{y})^2 =: s_y^2. \tag{40.5}$$

Here, $\bar{y}$ and s_y^2 again denote the sample mean and sample variance of the random variables $y_1, \dots, y_n$.

∘

40.2 Model evaluation

Based on Definition 38.1, we now formulate the T-test statistic for the one-sample T-test scenario and state its frequentist distribution.

Theorem 40.2 (T-test statistic of the one-sample T-test). *Let the design-matrix form of the one-sample T-test model be given. Then, for the T-test statistic with*

$$c := 1 \text{ and } c^T\beta_0 =: \mu_0, \tag{40.6}$$

we obtain

$$T = \sqrt{n}\left(\frac{\bar{y} - \mu_0}{s_y}\right), \tag{40.7}$$

and

$$T \sim t(\delta, n-1) \text{ with } \delta = \sqrt{n}\left(\frac{\mu - \mu_0}{\sigma}\right). \tag{40.8}$$

∘

Proof. By Theorem 38.1,

$$T = \frac{c^T\hat{\beta} - c^T\beta_0}{\sqrt{\hat{\sigma}^2 c^T \left(X^T X\right)^{-1} c}} = \frac{1^T\bar{y} - 1^T\mu_0}{\sqrt{s_y^2 1^T \left(1_n^T 1_n\right)^{-1} 1}} = \sqrt{n}\left(\frac{\bar{y} - \mu_0}{s_y}\right). \tag{40.9}$$

Furthermore, by the same theorem,

$$\delta = \frac{c^T\beta - c^T\beta_0}{\sqrt{\sigma^2 c^T \left(X^T X\right)^{-1} c}} = \frac{1^T\mu - 1^T\mu_0}{\sqrt{\sigma^2 1^T \left(1_n^T 1_n\right)^{-1} 1}} = \sqrt{n}\left(\frac{\mu - \mu_0}{\sigma}\right). \tag{40.10}$$

□

The forms of the T-test statistic and its distribution in the one-sample T-test special case of the GLM are therefore naturally identical to the corresponding forms in the GLM-free context. The corresponding theory for confidence intervals and for controlling the size of one-sample T-tests, as well as the use of the power function for evaluating test power, therefore follows analogously.

Application example

The following **R** code demonstrates the evaluation of a 95% confidence interval for the expectation parameter μ and the implementation of a two-sided one-sample T-test with simple null hypothesis $\Theta_0 := \{0\}$ and significance level $\alpha_0 := 0.05$ for the application example outlined above.

```
# data analysis
D           = read.csv("./_data/507-t-tests.csv")                          # load dataset
y           = D$dBDI[D$COND == "F2F"]                                      # pre-post difference scores
n           = length(y)                                                    # number of data points
p           = 1                                                            # number of beta parameters
c           = 1                                                            # contrast-weight vector
mu_0        = 0                                                            # null-hypothesis parameter
delta       = 0.95                                                         # confidence level
alpha_0     = 0.05                                                         # significance level
X           = matrix(rep(1,n), nrow = n)                                   # one-sample T-test design matrix
beta_hat    = solve(t(X)%*%X)%*%t(X)%*%y                                   # beta-parameter estimator
eps_hat     = y - X %*% beta_hat                                           # residual vector
sigsqr_hat  = (t(eps_hat) %*% eps_hat)/(n-p)                               # variance-parameter estimator
t_delta     = qt((1+delta)/2,n-1)                                          # \Psi^{-1}((1+\delta)/2, n-1)
lambda      = diag(solve(t(X) %*% X))                                      # \lambda_j values
kappa_u     = beta_hat - sqrt(sigsqr_hat*lambda)*t_delta                   # lower confidence-interval bound
kappa_o     = beta_hat + sqrt(sigsqr_hat*lambda)*t_delta                   # upper confidence-interval bound
t_num       = t(c) %*% beta_hat - mu_0                                     # numerator of the one-sample T-test statistic
t_den       = sqrt(sigsqr_hat %*% t(c) %*% solve(t(X) %*% X) %*% c)        # denominator of the one-sample T-test statistic
t           = t_num/t_den                                                  # value of the one-sample T-test statistic
pval        = 2*(1 - pt(abs(t), n-1))                                      # p-value for a two-sided one-sample T-test
k_alpha_0   = qt(1-alpha_0/2, n-1)                                         # critical value
if(abs(t) > k_alpha_0){phi = 1} else {phi = 0}                             # one-sample T-test
```

```
Beta-parameter estimator      :  3.5
95% confidence interval       :  2.27 4.73
Variance-parameter estimator  :  3.73
alpha_0                       :  0.05
Critical value                :  2.2
One-sample T-test statistic   :  6.28
phi                           :  1
p-value                       :  0
```

In this case, the null hypothesis is rejected at a critical value of $k_{0.05} = 2.20$ and a value of the T-statistic of $T = 6.28$. The 95% confidence interval for the true, but unknown, expectation parameter is $[2.27, 4.73]$ and therefore does not cover the null-hypothesis parameter value $\mu_0 = 0$.

40.3 Two-sample T-tests

Application scenario

The application scenario of a two-sample T-test for independent samples is characterized, as is well known, by considering a total of n univariate data points from two samples (groups) of randomized experimental units. It is assumed in particular that the n_1 univariate data points of the first group are realizations of n_1 independent and identically normally distributed random variables with expectation parameter μ_1 and variance parameter σ^2, while the n_2 univariate data points of the second group are assumed to be realizations of n_2 independent and identically normally distributed random variables with expectation parameter μ_2 and variance parameter σ^2. Thus, in particular, the true, but unknown, expectation parameters of both groups of random variables may differ, whereas the variance parameters of both groups are assumed to be identical. Finally, one assumes an interest in the inferential comparison of the unknown expectation parameters μ_1 and μ_2, for example their equality $\mu_1 = \mu_2$ or inequality $\mu_1 \neq \mu_2$.

Application example

For a concrete application example, we consider the analysis of pre-post intervention BDI difference scores from two groups of 12 patients each in different therapy settings, as shown in Table 40.3. The first column of the table (`COND`) lists the patient-specific therapy setting (F2F: face-to-face, ONL: online). The second column (`dBDI`) lists the corresponding patient-specific pre-post intervention BDI difference scores. Positive values again correspond to a decrease in depressive symptoms, and negative values to an increase in depressive symptoms.

Table 40.3 Pre-post BDI difference scores for two samples

COND	dBDI
F2F	1
F2F	6
F2F	4
F2F	2
F2F	1
F2F	5
F2F	6
F2F	3
F2F	2
F2F	2
F2F	4
F2F	6
ONL	9
ONL	2
ONL	10
ONL	-1
ONL	4
ONL	11
ONL	10
ONL	15
ONL	4
ONL	5
ONL	10
ONL	10

When applying a two-sample T-test to the dBDI data, we assume that the 12 data points of the F2F therapy group are realizations of $n_1 = 12$ independent and identically normally distributed random variables $y_{1j} \sim N(\mu_1, \sigma^2)$ with $j = 1, \ldots, n_1$ and that the 12 data points of the ONL therapy group are realizations of $n_2 = 12$ independent and identically normally distributed random variables $y_{2j} \sim N(\mu_2, \sigma^2)$ with $j = 1, \ldots, n_2$.

Independently of the inferential procedure described below, we again first consider the descriptive statistics of the therapy-setting-specific dBDI values. These are listed in Table 40.4.

Table 40.4 Descriptive statistics of the pre-post BDI difference scores in different therapy settings

	n	Max	Min	Median	Mean	Var	Std
F2F	12	6	1	3.5	3.50	3.73	1.93
ONL	12	15	-1	9.5	7.42	20.81	4.56

Model formulation

Using the index i for the groups and the index j for the experimental units in each group, we define the two-sample T-test model as follows.

Definition 40.2 (Two-sample T-test model)**.** For $i = 1, 2$ and $j = 1, \dots, n_i$, let y_{ij} be random variables that model the $n = n_1 + n_2$ data points of a two-sample T-test scenario. Then the two-sample T-test model has the structural form

$$y_{ij} = \mu_i + \varepsilon_{ij} \text{ with } \varepsilon_{ij} \sim N\left(0, \sigma^2\right) \text{ i.i.d. with } \mu_i \in \mathbb{R} \text{ and } \sigma^2 > 0, \tag{40.11}$$

the data-distribution form

$$y_{ij} \sim N\left(\mu_i, \sigma^2\right) \text{ i.i.d. with } \mu_i \in \mathbb{R} \text{ and } \sigma^2 > 0, \tag{40.12}$$

and, for the n-dimensional data vector defined as

$$y := \left(y_{11}, \cdots, y_{1n_1}, y_{21}, \cdots, y_{2n_2}\right)^T, \tag{40.13}$$

the design-matrix form

$$y = X\beta + \varepsilon \tag{40.14}$$

with

$$X := \begin{pmatrix} 1_{n_1} & 0_{n_1} \\ 0_{n_2} & 1_{n_2} \end{pmatrix} \in \mathbb{R}^{n \times 2}, \beta := \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \in \mathbb{R}^2, \varepsilon \sim N\left(0_n, \sigma^2 I_n\right), \sigma^2 > 0. \tag{40.15}$$

•

The definition of the two-sample T-test model in design-matrix form chosen here is not the only possible one, but it is the one under which the equivalence to the two-sample T-test model in the GLM-free context is most clearly visible. In Chapter 41, we will encounter an alternative parameterization of the two-sample T-test model as well. As in the one-sample T-test scenario, the equivalence of the model forms formulated in Definition 40.2 follows from the results in Chapter 36. Apart from the definition of the design matrix and beta-parameter vector, simulating data based on the two-sample T-test model is identical to the simulations of GLM special cases already known, as the following **R** code demonstrates.

```
# model formulation
library(MASS)                                     # multivariate normal distribution
n_1    = 12                                       # number of data points in group 1
n_2    = 12                                       # number of data points in group 2
n      = n_1 + n_2                                # total number of data points
p      = 2                                        # number of beta parameters
X      = matrix(c(rep(1,n_1), rep(0,n_1),         # design matrix
                  rep(0,n_2), rep(1,n_2)),
                  nrow  = n)
I_n    = diag(n)                                  # n x n identity matrix
beta   = matrix(c(1,2), nrow = p)                 # true, but unknown, beta parameter
sigsqr = 10                                       # true, but unknown, variance parameter

# data realization
y      = mvrnorm(1, X %*% beta, sigsqr*I_n)  # one realization of an n-dimensional random vector
```

Model estimation

The two beta-parameter components of the two-sample T-test model in design-matrix form are, unsurprisingly, estimated by the corresponding group sample means. For the variance-parameter estimator, the so-called *pooled sample variance* results. These are the two core statements of the following theorem.

Theorem 40.3 (Parameter estimation in the two-sample T-test model)**.** *Let the design-matrix form of the two-sample T-test model be given. Then, for the beta-parameter estimator,*

$$\hat{\beta} = \begin{pmatrix} \frac{1}{n_1}\sum_{j=1}^{n_1} y_{1j} \\ \frac{1}{n_2}\sum_{j=1}^{n_2} y_{2j} \end{pmatrix} =: \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \end{pmatrix} \tag{40.16}$$

and, for the variance-parameter estimator,

$$\hat{\sigma}^2 = \frac{\sum_{j=1}^{n_1} \left(y_{1j} - \bar{y}_1\right)^2 + \sum_{j=1}^{n_2} \left(y_{2j} - \bar{y}_2\right)^2}{n_1 + n_2 - 2} =: s_{12}^2. \tag{40.17}$$

∘

Proof. For $i = 1, 2$, let $y_i := \left(y_{i1}, \dots, y_{in_i}\right)^T$. Then, for the beta-parameter estimator,

$$\begin{aligned}
\hat{\beta} &= \left(X^T X\right)^{-1} X^T y \\
&= \left(\begin{pmatrix} 1_{n_1} & 0_{n_2} \\ 0_{n_1} & 1_{n_2} \end{pmatrix} \begin{pmatrix} 1_{n_1} & 0_{n_1} \\ 0_{n_2} & 1_{n_2} \end{pmatrix}\right)^{-1} \begin{pmatrix} 1_{n_1} & 0_{n_2} \\ 0_{n_1} & 1_{n_2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \\
&= \begin{pmatrix} n_1 & 0 \\ 0 & n_2 \end{pmatrix}^{-1} \begin{pmatrix} \sum_{j=1}^{n_1} y_{1j} \\ \sum_{j=1}^{n_2} y_{2j} \end{pmatrix} \\
&= \begin{pmatrix} n_1^{-1} & 0 \\ 0 & n_2^{-1} \end{pmatrix} \begin{pmatrix} \sum_{j=1}^{n_1} y_{1j} \\ \sum_{j=1}^{n_2} y_{2j} \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{n_1}\sum_{j=1}^{n_1} y_{1j} \\ \frac{1}{n_2}\sum_{j=1}^{n_2} y_{2j} \end{pmatrix} \\
&=: \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \end{pmatrix}.
\end{aligned} \tag{40.18}$$

Similarly, for the variance-parameter estimator with $n = n_1 + n_2$ and $p = 2$,

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{n - p} \\
&= \frac{1}{n_1 + n_2 - 2} \left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} - \begin{pmatrix} 1_{n_1} & 0_{n_1} \\ 0_{n_2} & 1_{n_2} \end{pmatrix} \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \end{pmatrix}\right)^T \left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} - \begin{pmatrix} 1_{n_1} & 0_{n_1} \\ 0_{n_2} & 1_{n_2} \end{pmatrix} \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \end{pmatrix}\right) \\
&= \frac{1}{n_1 + n_2 - 2} \begin{pmatrix} y_{11} - \bar{y}_1 \\ \vdots \\ y_{1n_1} - \bar{y}_1 \\ y_{21} - \bar{y}_2 \\ \vdots \\ y_{2n_2} - \bar{y}_2 \end{pmatrix}^T \begin{pmatrix} y_{11} - \bar{y}_1 \\ \vdots \\ y_{1n_1} - \bar{y}_1 \\ y_{21} - \bar{y}_2 \\ \vdots \\ y_{2n_2} - \bar{y}_2 \end{pmatrix} \\
&= \frac{\sum_{j=1}^{n_1} \left(y_{1j} - \bar{y}_1\right)^2 + \sum_{j=1}^{n_2} \left(y_{2j} - \bar{y}_2\right)^2}{n_1 + n_2 - 2} \\
&=: s_{12}^2.
\end{aligned} \tag{40.19}$$

□

Note that the sample variance s_y^2 of the components of y generally differs from the pooled sample variance s_{12}^2. This is due not least to the fact that the sample variance is based on the overall sample mean $\bar{y}$, whereas the pooled sample variance is based on the group-specific sample means $\bar{y}_1$ and $\bar{y}_2$. We do not want to deepen the concept of pooled sample variance further here.

Model evaluation

Based on Theorem 38.1, we now formulate the T-test statistic for the two-sample T-test model defined in design-matrix form in Definition 40.2 and state its frequentist distribution.

Theorem 40.4 (T-test statistic of the two-sample T-test)**.** *Let the design-matrix form of the two-sample T-test be given. Then, for the T-test statistic with*

$$c := (1, -1)^T \text{ and } c^T\beta_0 =: \mu_0, \tag{40.20}$$

we obtain

$$T = \sqrt{\frac{n_1 n_2}{n_1 + n_2}} \left(\frac{\bar{y}_1 - \bar{y}_2 - \mu_0}{s_{12}}\right), \tag{40.21}$$

and

$$T \sim t\left(\delta, n_1 + n_2 - 2\right) \text{ with } \delta = \sqrt{\frac{n_1 n_2}{n_1 + n_2}} \left(\frac{\mu_1 - \mu_2 - \mu_0}{\sigma}\right). \tag{40.22}$$

∘

Proof. By Theorem 38.1, for the numerators of T and δ,

$$c^T\hat{\beta} - c^T\beta_0 = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \end{pmatrix} - \mu_0 = \bar{y}_1 - \bar{y}_2 - \mu_0 \tag{40.23}$$

and

$$c^T\beta - c^T\beta_0 = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} - \mu_0 = \mu_1 - \mu_2 - \mu_0, \tag{40.24}$$

respectively. Furthermore, for the denominators of T and δ,

$$c^T\left(X^TX\right)^{-1} c = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} n_1^{-1} & 0 \\ 0 & n_2^{-1} \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} n_1^{-1} & -n_2^{-1} \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{n_1} + \frac{1}{n_2}. \tag{40.25}$$

In addition,

$$\left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{-\frac{1}{2}} = \left(\frac{n_2}{n_1 n_2} + \frac{n_1}{n_1 n_2}\right)^{-\frac{1}{2}} = \left(\frac{n_1 + n_2}{n_1 n_2}\right)^{-\frac{1}{2}} = \left(\frac{n_1 n_2}{n_1 + n_2}\right)^{\frac{1}{2}}. \tag{40.26}$$

Taken together, it follows directly that

$$T = \sqrt{\frac{n_1 n_2}{n_1 + n_2}} \left(\frac{\bar{y}_1 - \bar{y}_2 - \mu_0}{s_{12}}\right) \text{ and } \delta = \sqrt{\frac{n_1 n_2}{n_1 + n_2}} \left(\frac{\mu_1 - \mu_2 - \mu_0}{\sigma}\right). \tag{40.27}$$

□

The forms of the T-test statistic and its distribution in the two-sample T-test model in design-matrix form are therefore again naturally identical to the corresponding forms in the GLM-free context. The corresponding theory for controlling the size of two-sample T-tests and the use of the power function for evaluating test power therefore follows analogously.

Application example

The following **R** code demonstrates the evaluation of 95% confidence intervals for the expectation parameters μ_1 and μ_2 and the implementation of a two-sided two-sample T-test with null hypothesis

$$\Theta_0 := \left\{ \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \in \mathbb{R}^2 \mid \mu_1 = \mu_2 \right\} \tag{40.28}$$

and significance level $\alpha_0 := 0.05$ for the application example outlined above.

```
# data loading
D           = read.csv("./_data/507-t-tests.csv")              # dataframe
y_1         = D$dBDI[D$COND == "F2F"]                            # BDI difference scores in the F2F group
y_2         = D$dBDI[D$COND == "ONL"]                            # BDI difference scores in the ONL group

# model formulation
n_1         = length(y_1)                                        # number of data points in group 1 (F2F)
n_2         = length(y_2)                                        # number of data points in group 2 (ONL)
n           = n_1 + n_2                                          # total number of data points
y           = matrix(c(y_1, y_2), nrow = n)                      # data vector
p           = 2                                                  # number of beta parameters
X           = matrix(c(rep(1,n_1), rep(0,n_2),                   # two-sample T-test design matrix
                       rep(0,n_1), rep(1,n_2)),
              nrow = n)

# parameter estimation
beta_hat    = solve(t(X) %*% X) %*% t(X) %*% y                   # beta-parameter estimator
eps_hat     = y - X %*% beta_hat                                 # residual vector
sigsqr_hat  = (t(eps_hat) %*% eps_hat) /(n-p)                    # variance-parameter estimator

# confidence interval
delta       = 0.95                                               # confidence level
t_delta     = qt((1+delta)/2,n-p)                                # \Psi^{-1}((1+\delta)/2,n-p)
lambda      = diag(solve(t(X) %*% X))                            # \lambda_j values
kappa       = matrix(rep(NaN,p*2), nrow = p)                     # \beta_j confidence-interval array
for(j in 1:p){                                                   # iteration over \beta_j
  kappa[j,1] = beta_hat[j]-sqrt(sigsqr_hat*lambda[j])*t_delta    # lower CI bound
  kappa[j,2] = beta_hat[j]+sqrt(sigsqr_hat*lambda[j])*t_delta}   # upper CI bound

# hypothesis test
c           = matrix(c(1,-1), nrow = 2)                          # contrast-weight vector
mu_0        = 0                                                  # null hypothesis H_0
alpha_0     = 0.05                                               # significance level
k_alpha_0   = qt(1 - (alpha_0/2), n-p)                           # critical value
t_num       = t(c) %*% beta_hat - mu_0                           # T-test statistic numerator
t_den       = sqrt(sigsqr_hat*t(c) %*% solve(t(X) %*% X)%*%c)    # T-test statistic denominator
t           = t_num/t_den                                        # T-test statistic
if(abs(t) >= k_alpha_0){phi = 1} else {phi = 0}                  # test 1_{|T(X) >= k_alpha_0|}
pval        = 2*(1-pt(abs(t), n_1+n_2-2))                        # p-value
```

```
Beta-parameter estimator       :  3.5 7.42
95% confidence intervals      :  1.4 5.32 5.6 9.51
Variance-parameter estimator  :  12.27
alpha_0                       :  0.05
Critical value                :  2.07
Two-sample T-test statistic   :  -2.74
phi                           :  1
p-value                       :  0.01
```

In this case, the null hypothesis is rejected at a critical value of $k_{0.05} = 2.07$ and a value of the T-statistic of $T = -2.74$. Inferentially, there is therefore evidence that the true, but unknown, expectation parameter in the F2F therapy setting differs from the true, but unknown, expectation parameter in the ONL therapy setting. The 95% confidence intervals for the true, but unknown, expectation parameters μ_1 and μ_2 are $[1.40, 5.60]$ and $[5.32, 9.51]$, respectively.

40.4 Bibliographic remarks

Although the frequentist literature of the first half of the twentieth century is permeated by the equivalence of regression and analysis-of-variance linear models, it is difficult to name a definite source that would have priority with respect to describing T-tests as special cases of the GLM. We therefore refer here more generally to Fisher (1925c) and Fisher (1935).

41 One-way analysis of variance

41.1 Application scenario

The application scenario of one-way analysis of variance is characterized by n univariate data points from two or more groups of randomized experimental units that differ with respect to the levels of an experimental factor. If the number of data points in each group is the same, the design is also called a balanced one-way analysis-of-variance design. The data points of the ith group, or of the ith factor level, are assumed to be realizations of n_i independent and identically normally distributed random variables whose true, but unknown, expectation parameters may differ across groups and whose true, but unknown, variance parameter is identical across groups. In these basic assumptions, the one-way analysis-of-variance scenario is therefore a direct generalization of the one-sample and two-sample T-test scenarios to, potentially, more than two groups. Conversely, one-sample and two-sample T-test scenarios can of course also be viewed as one-way analysis-of-variance scenarios in which the experimental factor has only one or two levels, respectively. Finally, as in the case of T-test scenarios, one usually assumes that there is interest in an inferential comparison of the true, but unknown, factor-level-specific expectation parameters.

41.2 Application example

As a concrete application example, we consider the analysis of pre-post intervention BDI difference values from three groups of 12 patients each, who underwent different therapy settings, face-to-face and online, or a waitlist control condition, as illustrated in Table 41.1. The first column of the table (`COND`) lists the patient-specific therapy setting (`F2F`: face-to-face, `ONL`: online, `WLC`: waitlist control) for selected patients in each study group. The second column of the table (`dBDI`) lists the corresponding patient-specific pre-post intervention BDI difference values. Positive values again correspond to a decrease in depressive symptoms, negative values to an increase in depressive symptoms.

Table 41.1 Exemplary pre-post intervention BDI difference values of the example data set

	COND	dBDI
1	F2F	9
2	F2F	7
3	F2F	10
4	F2F	11
13	ONL	2
14	ONL	7
15	ONL	9
16	ONL	9
17	ONL	8
25	WLC	-1
26	WLC	2

Table 41.1 Exemplary pre-post intervention BDI difference values of the example data set

	COND	dBDI
27	WLC	-3
28	WLC	-1
29	WLC	0

Figure 41.1 shows a visualization of the complete data set. The bars represent the group-specific sample means, the corresponding error bars represent the group-specific sample standard deviations. The point clouds represent the group-specific data points. In the F2F and ONL groups, the change in the BDI value is more pronounced than in the WLC group.

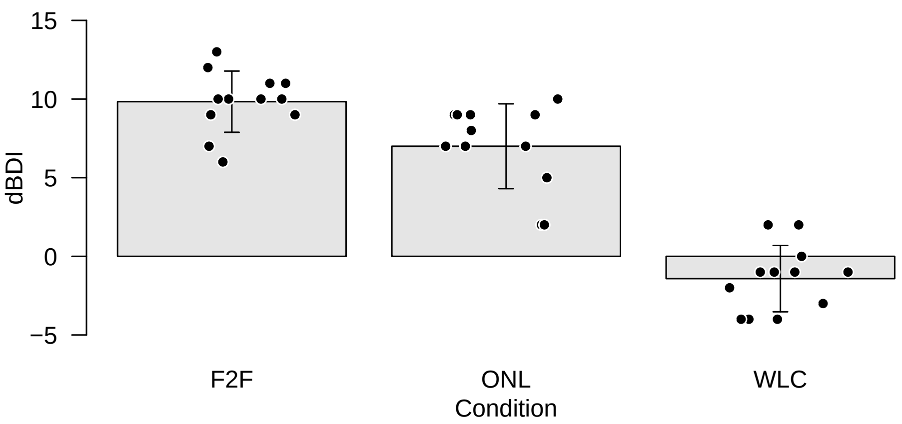


Figure 41.1 Data display of the application example for one-way analysis of variance.

In Table 41.2, we summarize the descriptive statistics of the example data set, broken down by therapy conditions.

Table 41.2 Descriptive statistics of the pre-post BDI difference values

	n	Max	Min	Median	Mean	Var	Std
F2F	12	13	6	10.0	9.83	3.79	1.95
ONL	12	10	2	7.5	7.00	7.27	2.70
WLC	12	2	-4	-1.0	-1.42	4.45	2.11

41.3 Model formulation

We first define the one-way ANOVA model in expectation-parameter representation. We use the index i to index the experimental groups and the index j to index the experimental units within the groups.

Definition 41.1 (One-way ANOVA model in expectation-parameter representation)**.** For $i = 1, \dots, p$ and $j = 1, \dots, n_i$, let y_{ij} be random variables that model the $n := \sum_{i=1}^{p} n_i$ data points of a one-way ANOVA scenario. Then the one-way ANOVA model in expectation-parameter representation has the structural form

$$y_{ij} = \mu_i + \varepsilon_{ij} \text{ with } \varepsilon_{ij} \sim N\left(0, \sigma^2\right) \text{ i.i.d. with } \mu_i \in \mathbb{R}, \sigma^2 > 0, \tag{41.1}$$

the data distribution form

$$y_{ij} \sim N\left(\mu_i, \sigma^2\right) \text{ independent with } \mu_i \in \mathbb{R}, \sigma^2 > 0, \tag{41.2}$$

and, for the n-dimensional data vector defined as

$$y := \left(y_{11}, \dots, y_{1n_1}, y_{21}, \dots, y_{2n_2}, \dots, y_{p1}, \dots, y_{pn_p}\right)^T, \tag{41.3}$$

the design-matrix form

$$y = X\beta + \varepsilon \tag{41.4}$$

with

$$X := \begin{pmatrix} 1_{n_1} & 0_{n_1} & \cdots & 0_{n_1} \\ 0_{n_2} & 1_{n_2} & \cdots & 0_{n_2} \\ \vdots & \vdots & \ddots & \vdots \\ 0_{n_p} & 0_{n_p} & \cdots & 1_{n_p} \end{pmatrix}, \beta := \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} \in \mathbb{R}^p, \varepsilon \sim N\left(0_n, \sigma^2 I_n\right), \sigma^2 > 0. \tag{41.5}$$

For $n_i := m$ for all $i = 1, \dots, p$, the model is called balanced.

•

The comparison with the definition of the two-sample T-test model in Definition 40.2 shows that this formulation of the one-way ANOVA model is the direct generalization of the two-sample T-test model from $p = 2$ to arbitrary $p \in \mathbb{N}$.

Motivation of the effect representation

The one-way ANOVA model in expectation-parameter representation is a valid model on the basis of which both parameter estimation and parameter and model inference for the one-way ANOVA scenario can be developed; see Georgii (2009). For consistency with models of multi-factor analysis of variance, however, a reparameterization of the beta-parameter vector is useful. The core idea of this reparameterization is to model the expectation parameter of the ith group as the sum of a group-unspecific expectation parameter $\mu_0 \in \mathbb{R}$ and a group-specific effect parameter $\alpha_i \in \mathbb{R}$,

$$\mu_i := \mu_0 + \alpha_i \text{ for } i = 1, \dots, p. \tag{41.6}$$

Here α_i models the difference between the ith expectation parameter μ_i and the group-unspecific expectation parameter μ_0,

$$\alpha_i = \mu_i - \mu_0 \text{ for } i = 1, \dots, p. \tag{41.7}$$

In this form, however, the reparameterization has one decisive disadvantage: p expectation parameters $\mu_i, i = 1, \dots, p$ are represented by the $p + 1$ parameters μ_0 and $\alpha_i, i = 1, \dots, p$. This representation is generally not unique. For example, the expectation parameters $\mu_1 =$

$3, \mu_2 = 5, \mu_3 = 6$ can be represented both by the group-unspecific expectation parameter $\mu_0 = 0$ and the group-specific effect parameters $\alpha_1 = 3, \alpha_2 = 5, \alpha_3 = 6$, and by the group-unspecific expectation parameter $\mu_0 = 1$ and the group-specific effect parameters $\alpha_1 = 2, \alpha_2 = 4, \alpha_3 = 5$. In this context one also says that the one-way ANOVA model in the form of Equation 41.6 is overparameterized.

From a data-analytic perspective, the overparameterization of an analysis-of-variance model has the disadvantage that $p + 1$ beta-parameter estimates would have to be determined from p estimated expectation parameters, which, as seen above, cannot be done uniquely. To avoid these problems in the effect-parameter representation of the one-way ANOVA model and to transfer this representation consistently to multi-factor analysis-of-variance models, it is natural to introduce the side condition

$$\alpha_1 := 0. \tag{41.8}$$

Thus one effect parameter is assumed to be identically zero from the outset. The group-specific expectation parameters are then

$$\begin{aligned} \mu_1 &:= \mu_0 \\ \mu_i &:= \mu_0 + \alpha_i \text{ for } i = 2, \dots, p. \end{aligned} \tag{41.9}$$

The first group is then called the reference group, and the α_i model the difference between the expectation parameter of the ith group and the expectation parameter of the first group:

$$\alpha_i = \mu_i - \mu_0 = \mu_i - \mu_1 \text{ for } i = 1, \dots, p. \tag{41.10}$$

Under the side condition $\alpha_1 := 0$, μ_0 is therefore no longer a group-unspecific expectation parameter, but identical to the expectation parameter of the first group. Which actual experimental group is defined as the "first group" is irrelevant from a data-analytic perspective. What is data-analytically decisive, however, is that the corresponding expectation-parameter estimator $\hat{\mu}_0$ is correctly understood as the expectation-parameter estimator of the reference group and that the $\hat{\alpha}_i$ for $i = 2, \dots, p$ are correctly understood as estimated expectation-parameter differences between the expectation parameter of the reference group and the expectation parameter of the ith group.

We formalize the above in the following theorem.

Theorem 41.1 (One-way ANOVA model in effect representation with reference group)**.** *Let the one-way ANOVA model in expectation-parameter representation be given. Then the random variables that model the data points of the one-way ANOVA scenario can equivalently be written in the structural form*

$$\begin{aligned} y_{1j} &= \mu_0 + \varepsilon_{1j} \quad \textit{with } \varepsilon_{1j} \sim N\left(0, \sigma^2\right) \textit{ i.i.d. for } j = 1, \dots, n_1 \\ y_{ij} &= \mu_0 + \alpha_i + \varepsilon_{ij} \textit{ with } \varepsilon_{ij} \sim N\left(0, \sigma^2\right) \textit{ i.i.d. for } i = 2, \dots, p, j = 1, \dots, n_i \end{aligned} \tag{41.11}$$

with

$$\alpha_i := \mu_i - \mu_1 \textit{ for } i = 2, \dots, p \tag{41.12}$$

and in the corresponding data distribution form

$$\begin{aligned} &y_{1j} \sim N\left(\mu_0, \sigma^2\right) \quad \textit{i.i.d. for } j = 1, \dots, n_1 \textit{ with } \mu_0 \in \mathbb{R}, \sigma^2 > 0 \\ &y_{ij} \sim N\left(\mu_0 + \alpha_i, \sigma^2\right) \textit{ independent for } i = 2, \dots, p, j = 1, \dots, n_i \textit{ with } \alpha_i \in \mathbb{R}, \sigma^2 > 0. \end{aligned} \tag{41.13}$$

∘

Proof. By definition,

$$\mu_i = \mu_0 + \mu_i - \mu_0. \tag{41.14}$$

The parameterizations with μ_i and with $\mu_0 + \mu_i - \mu_0$ are thus the same and therefore equivalent. It then also follows that

$$\mu_i = \mu_0 + (\mu_i - \mu_0) =: \mu_0 + \alpha_i \text{ for } i = 1, \dots, p. \tag{41.15}$$

With $\alpha_1 := 0$, we have $\mu_1 = \mu_0$ and $\mu_i = \mu_0 + \alpha_i$ for $i = 2, \dots, p$, as claimed in the theorem. □

Based on Theorem 41.1, we now define the one-way ANOVA model in effect representation.

Definition 41.2 (One-way ANOVA model in effect representation with reference group)**.** For $i = 1, \dots, p$ and $j = 1, \dots, n_i$, let y_{ij} be random variables that model the $n := \sum_{i=1}^p n_i$ data points of a one-way ANOVA scenario. Then the one-way ANOVA model in effect representation has the structural form

$$\begin{aligned} y_{1j} &= \mu_0 + \varepsilon_{1j} \quad \text{with } \varepsilon_{1j} \sim N(0, \sigma^2) \text{ i.i.d. for } j = 1, \dots, n_1 \\ y_{ij} &= \mu_0 + \alpha_i + \varepsilon_{ij} \text{ with } \varepsilon_{ij} \sim N(0, \sigma^2) \text{ i.i.d. for } i = 2, \dots, p, j = 1, \dots, n_i \end{aligned} \tag{41.16}$$

with

$$\alpha_i := \mu_i - \mu_1 \text{ for } i = 2, \dots, p, \tag{41.17}$$

the data distribution form

$$\begin{aligned} & y_{1j} \sim N(\mu_0, \sigma^2) \quad \text{i.i.d. for } j = 1, \dots, n_1 \text{ with } \mu_0 \in \mathbb{R}, \sigma^2 > 0 \\ & y_{ij} \sim N(\mu_0 + \alpha_i, \sigma^2) \text{ independent for } i = 2, \dots, p, j = 1, \dots, n_i \text{ with } \alpha_i \in \mathbb{R}, \sigma^2 > 0, \end{aligned} \tag{41.18}$$

and the design-matrix form

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N(0_n, \sigma^2 I_n) \tag{41.19}$$

and

$$n := \sum_{i=1}^p n_i, y := \begin{pmatrix} y_{11} \\ \vdots \\ y_{1n_1} \\ y_{21} \\ \vdots \\ y_{2n_2} \\ \vdots \\ y_{p1} \\ \vdots \\ y_{pn_p} \end{pmatrix} \in \mathbb{R}^n, X := \begin{pmatrix} 1_{n_1} & 0_{n_1} & \cdots & 0_{n_1} \\ 1_{n_2} & 1_{n_2} & \cdots & 0_{n_2} \\ \vdots & \vdots & \ddots & \vdots \\ 1_{n_p} & 0_{n_p} & \cdots & 1_{n_p} \end{pmatrix} \in \mathbb{R}^{n \times p}, \beta := \begin{pmatrix} \mu_0 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{pmatrix} \in \mathbb{R}^p \text{ and } \sigma^2 > 0. \tag{41.20}$$

•

Example

To illustrate the differences and commonalities between the expectation-parameter representation and the effect-parameter representation of the one-way ANOVA model in their design-matrix forms, we consider an example scenario with $n_i := 4$ and $p = 3$ for

$i = 1, \dots, p$, and hence $n = 12$. For the expectation-parameter representation we then have

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_{12}, \sigma^2 I_{12}\right) \tag{41.21}$$

with

$$y := \begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \\ y_{31} \\ y_{32} \\ y_{33} \\ y_{41} \\ y_{42} \\ y_{43} \end{pmatrix}, X := \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{12 \times 3}, \beta := \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix} \in \mathbb{R}^3 \text{ and } \sigma^2 > 0. \tag{41.22}$$

For the effect-parameter representation, in contrast, we have

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_{12}, \sigma^2 I_{12}\right) \tag{41.23}$$

with

$$y := \begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \\ y_{31} \\ y_{32} \\ y_{33} \\ y_{41} \\ y_{42} \\ y_{43} \end{pmatrix}, X := \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{12 \times 3}, \beta := \begin{pmatrix} \mu_0 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \in \mathbb{R}^3 \text{ and } \sigma^2 > 0. \tag{41.24}$$

The following **R** code demonstrates the realization of data in a one-way ANOVA scenario with precisely this effect-parameter representation.

```
# model formulation
library(MASS)                                                        # multivariate normal distribution
m      = 4                                                           # number of data points in the ith group
p      = 3                                                           # number of groups
n      = p*m                                                         # total number of data points
Xt     = cbind(                                                      # design matrix
         matrix(1,nrow = n, ncol = 1),
         kronecker(diag(p), matrix(1,nrow = m,ncol = 1)))
X      = Xt[,-2]
I_n    = diag(n)                                                     # n x n identity matrix
beta   = matrix(c(10,-3,-12), nrow = p)                              # beta = (mu_0,alpha_2,alpha_3)
sigsqr = 14                                                          # sigma^2
y      = mvrnorm(1, X %*% beta, sigsqr*I_n)                          # one realization of an n-dimensional random variable
print(X)
```

```
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    1    0    0
[3,]    1    0    0
[4,]    1    0    0
```



```
 [5,]    1    1    0
 [6,]    1    1    0
 [7,]    1    1    0
 [8,]    1    1    0
 [9,]    1    0    1
[10,]    1    0    1
[11,]    1    0    1
[12,]    1    0    1
```

41.4 Model estimation

We now consider beta-parameter estimation in the effect-parameter representation of the one-way ANOVA model with reference group. In accordance with the interpretation of the beta-parameter components, μ_0 is estimated by the sample mean of the reference group and $\alpha_2, \ldots, \alpha_p$ are estimated by the differences between the respective group sample mean and the reference group sample mean. We do not further discuss the estimation of the variance parameter here; as in the two-sample T-test model, it results in a pooled sample variance.

Theorem 41.2 (Beta-parameter estimation in the one-way ANOVA model)**.** *Let the design-matrix form of the one-way ANOVA model in effect representation with reference group be given. Then the beta-parameter estimator is*

$$\hat{\beta} := \begin{pmatrix} \hat{\mu}_0 \\ \hat{\alpha}_2 \\ \vdots \\ \hat{\alpha}_p \end{pmatrix} = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 - \bar{y}_1 \\ \vdots \\ \bar{y}_p - \bar{y}_1 \end{pmatrix}, \tag{41.25}$$

where

$$\bar{y}_i := \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} \tag{41.26}$$

denotes the sample mean of the ith group.

∘

Proof. We first note that

$$X^T X = \begin{pmatrix} 1 & \cdots & 1 & 1 & \cdots & 1 & \cdots & 1 & \cdots & 1 \\ 0 & \cdots & 0 & 1 & \cdots & 1 & \cdots & 0 & \cdots & 0 \\ & \vdots & & & \vdots & & \vdots & & \vdots & \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 1 & 0 & & 0 \\ 1 & 1 & & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 1 & 1 & & 0 \\ & & & \\ \vdots & \vdots & \cdots & \vdots \\ 1 & 0 & & 1 \\ \vdots & \vdots & \cdots & \vdots \\ 1 & 0 & & 1 \end{pmatrix} \tag{41.27}$$

$$= \begin{pmatrix} n & n_2 & n_3 & \cdots & n_p \\ n_2 & n_2 & 0 & \cdots & 0 \\ n_3 & 0 & n_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n_p & 0 & 0 & \cdots & n_p \end{pmatrix}.$$

The inverse of X^TX is

$$(X^TX)^{-1} = \begin{pmatrix} \frac{1}{n_1} & -\frac{1}{n_1} & \cdots & -\frac{1}{n_1} \\ -\frac{1}{n_1} & \frac{n_1+n_2}{n_1n_2} & \cdots & \frac{1}{n_1} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{1}{n_1} & \frac{1}{n_1} & \cdots & \frac{n_1+n_p}{n_1n_p} \end{pmatrix}. \tag{41.28}$$

For example, for $p = 3$,

$$X^TX = \begin{pmatrix} n & n_2 & n_3 \\ n_2 & n_2 & 0 \\ n_3 & 0 & n_3 \end{pmatrix} \text{ and } (X^TX)^{-1} = \begin{pmatrix} \frac{1}{n_1} & -\frac{1}{n_1} & -\frac{1}{n_1} \\ -\frac{1}{n_1} & \frac{n_1+n_2}{n_1n_2} & \frac{1}{n_1} \\ -\frac{1}{n_1} & \frac{1}{n_1} & \frac{n_1+n_3}{n_1n_3} \end{pmatrix}. \tag{41.29}$$

We further note that

$$X^Ty = \begin{pmatrix} \sum_{i=1}^{p}\sum_{j=1}^{n_i} y_{ij} \\ \sum_{j=1}^{n_2} y_{2j} \\ \vdots \\ \sum_{j=1}^{n_p} y_{pj} \end{pmatrix}. \tag{41.30}$$

Therefore,

$$\hat{\beta} = (X^TX)^{-1}X^Ty. \tag{41.31}$$

For the first component of $\hat{\beta}$, this yields

$$\hat{\beta}_0 = \frac{1}{n_1}\sum_{j=1}^{n_1} y_{1j} = \bar{y}_1. \tag{41.32}$$

For the second component of $\hat{\beta}$, and analogously for all further components, this yields

$$\begin{aligned} \hat{\beta}_2 &= \frac{1}{n_2}\sum_{j=1}^{n_2} y_{2j} - \frac{1}{n_1}\sum_{j=1}^{n_1} y_{1j} \\ &= \bar{y}_2 - \bar{y}_1. \end{aligned} \tag{41.33}$$

□

The following **R** code implements parameter estimation of the one-way ANOVA model for the example data set. In addition to the estimates for the reference group and effect parameters obtained with the beta-parameter estimator, the code also evaluates the estimators based on sample means and sample-mean differences from Theorem 41.2. The code further evaluates the variance-parameter estimator, the pooled sample variance, and the total sample variance, which differ clearly.

```
# data management
D           = read.csv("./_data/508-one-way-analysis-of-variance.csv")          # data frame
y           = D$dBDI                                                             # data vector
y_1         = D$dBDI[D$COND == "F2F"]                                            # F2F data
y_2         = D$dBDI[D$COND == "ONL"]                                            # ONL data
y_3         = D$dBDI[D$COND == "WLC"]                                            # WLC data

# model formulation
p          = 3                                                                   # three groups
m          = length(y_1)                                                         # balanced design
n          = p*m                                                                 # data vector dimension
Xt         = cbind(                                                              # design matrix
             matrix(1, nrow = n, ncol = 1),
             kronecker(diag(p), matrix(1, nrow = m, ncol = 1)))
X          = Xt[,-2]

# model estimation
beta_hat   = solve(t(X) %*% X) %*% t(X) %*% y                                    # beta-parameter estimator
eps_hat    = y - X %*% beta_hat                                                  # residual vector
sigsqr_hat = (t(eps_hat) %*% eps_hat) /(n-p)                                     # variance-parameter estimator
s_sqr_123  = ((m-1)*var(y_1) +                                                   # pooled sample variance
              (m-1)*var(y_2) +
              (m-1)*var(y_3))/(m+m+m-p)
```

```
hat{beta}                                     :  9.83 -2.83 -11.25
bar{y}_1,bar{y}_2,bar{y}_3                    :  9.83 7 -1.42
bar{y}_1,bar{y}_2-bar{y}_1,bar{y}_3-bar{y}_1 :  9.83 -2.83 -11.25
hat{sigsqr}                                   :  5.17
s_123^2                                       :  5.17
s_y^2                                         :  28.35
```

41.5 Model evaluation

In principle, all parameter estimates in a one-way ANOVA model are of interest and can be evaluated with T-statistics in the sense of confidence intervals or hypothesis tests. Traditionally, however, one-way ANOVA scenarios often focus on evaluating the null hypothesis that the true, but unknown, expectation parameters of all groups are identical. Against the background of the one-way ANOVA effect representation with reference group, this corresponds to the null hypothesis that the effect parameters are equal to zero, formally

$$\Theta_0 := \left\{ \begin{pmatrix} \mu_0 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{pmatrix} \in \mathbb{R}^p \mid \alpha_i = 0 \text{ for } i = 2, \dots, p \right\} = \mathbb{R} \times \left\{ 0_{p-1} \right\} \tag{41.34}$$

and

$$\Theta_1 := \left\{ \begin{pmatrix} \mu_0 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{pmatrix} \in \mathbb{R}^p \mid \alpha_i \neq 0 \text{ for at least one } i = 2, \dots, p \right\} = \mathbb{R}^p \backslash \Theta_0. \tag{41.35}$$

An F-statistic is generally used to evaluate the null hypothesis. In the following, we first develop this F-statistic by means of a sum-of-squares decomposition of data variability in a one-way ANOVA scenario. In this context, we also introduce η^2, eta-squared, as an effect-size measure analogous to the coefficient of determination R^2 known from Chapter 35. Equipped with the special form of the F-statistic for the one-way ANOVA model, we then discuss the traditional test of the one-way ANOVA null hypothesis.

41.5.1 Sum-of-squares decomposition and the coefficient of determination η^2

The variability of the data in a one-way ANOVA scenario can be written in terms of a sum-of-squares decomposition as shown in the following theorem.

Theorem 41.3 (Sum-of-squares decomposition for one-way analysis of variance)**.** *For $i = 1, \dots, p$ and $j = 1, \dots, n_i$, let y_{ij} be the jth data variable in the ith group of a one-way ANOVA scenario. Furthermore, with $n := \sum_{i=1}^{p} n_i$, let*

$$\bar{y} := \frac{1}{n} \sum_{i=1}^{p} \sum_{j=1}^{n_i} y_{ij} \textit{ and } \bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} \tag{41.36}$$

denote the overall sample mean and the ith sample mean, respectively. Finally, let

- *the total sum of squares be defined as*

$$SQT := \sum_{i=1}^{p} \sum_{j=1}^{n_i} \left(y_{ij} - \bar{y} \right)^2, \tag{41.37}$$

- *the between sum of squares be defined as*

$$SQB := \sum_{i=1}^{p} n_i \left(\bar{y}_i - \bar{y}\right)^2, \tag{41.38}$$

and the within sum of squares be defined as

$$SQW := \sum_{i=1}^{p}\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_i\right)^2. \tag{41.39}$$

Then

$$SQT = SQB + SQW. \tag{41.40}$$

∘

Proof. It holds that

$$\begin{aligned}
\mathrm{SQT} &= \sum_{i=1}^{p}\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}\right)^2 \\
&= \sum_{i=1}^{p}\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_i + \bar{y}_i - \bar{y}\right)^2 \\
&= \sum_{i=1}^{p}\sum_{j=1}^{n_i} \left(\left(y_{ij} - \bar{y}_i\right) + \left(\bar{y}_i - \bar{y}\right)\right)^2 \\
&= \sum_{i=1}^{p}\left(\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_i\right)^2 + 2\left(\bar{y}_i - \bar{y}\right)\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_i\right) + n_i\left(\bar{y}_i - \bar{y}\right)^2\right).
\end{aligned} \tag{41.41}$$

Because

$$\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_i\right) = 0 \tag{41.42}$$

for each group i, it follows that

$$\begin{aligned}
\mathrm{SQT} &= \sum_{i=1}^{p}\left(\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_i\right)^2 + n_i\left(\bar{y}_i - \bar{y}\right)^2\right) \\
&= \sum_{i=1}^{p}\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_i\right)^2 + \sum_{i=1}^{p} n_i\left(\bar{y}_i - \bar{y}\right)^2 \\
&= \mathrm{SQW} + \mathrm{SQB}.
\end{aligned} \tag{41.43}$$

Hence

$$\mathrm{SQT} = \mathrm{SQB} + \mathrm{SQW}. \tag{41.44}$$

□

The sum-of-squares decomposition in the one-way ANOVA scenario is obviously analogous to the sum-of-squares decomposition for a fitting line; see Section 34.1. Correspondingly, the within sum of squares in one-way ANOVA is also often called the residual sum of squares. The terms in Theorem 41.3 can be understood intuitively as follows:

- SQT represents the total variability of the data y_{ij} around the overall sample mean $\bar{y}$.

- SQB represents the variability of the group sample means around the overall sample mean, weighted by the respective group size n_i. Large values of SQB therefore imply a strong dependence of the group sample means on the factor level i under consideration, whereas small values of SQB imply a weak dependence of the group sample means on the factor level under consideration. In this sense, SQB represents the data variability explained by considering the respective factor level and is analogous to the explained sum of squares, SQE, in the sum-of-squares decomposition for a fitting line.
- SQW represents the data variability summed over factor levels that remains after explaining the data variability in the ith group by its respective sample mean. SQW is therefore analogous to the residual sum of squares in the sum-of-squares decomposition for a fitting line.

Taken together, SQB quantifies the strength of the differences between factor levels. The effect-size measure η^2 and the F-statistic of one-way ANOVA now put SQB into different ratios: η^2 compares SQB with SQT and thus considers the proportion of variability between factor levels in the total variability of the data. The F-statistic, in contrast, compares SQB with SQW and therefore puts the influence of factor levels in relation to the unexplained data variability after subtracting this influence. We first consider the effect-size measure η^2 by means of the following definition.

Definition 41.3 (Effect-size measure η^2)**.** For a one-way ANOVA scenario, let the between sum of squares SQB and the total sum of squares SQT be defined as in Theorem 41.3. Then the effect-size measure η^2 is defined as

$$\eta^2 := \frac{\mathrm{SQB}}{\mathrm{SQT}}. \tag{41.45}$$

•

The comparison with Definition 35.4 shows that η^2 is defined analogously to the coefficient of determination R^2. As described above, η^2 gives the proportion of data variability between factor levels in the total data variability. Finally, with

$$\mathrm{SQT} = \mathrm{SQB} + \mathrm{SQW}, \tag{41.46}$$

it follows immediately, analogously to R^2, that for $\mathrm{SQT} \neq 0$, $\eta^2 \in [0, 1]$, because on the one hand

$$\mathrm{SQB} = 0 \Rightarrow \mathrm{SQT} = \mathrm{SQW} \text{ and } \eta^2 = 0 \tag{41.47}$$

and on the other hand

$$\mathrm{SQW} = 0 \Rightarrow \mathrm{SQT} = \mathrm{SQB} \text{ and } \eta^2 = 1. \tag{41.48}$$

41.5.2 F-test statistic

We now show that, for the design-matrix form of the effect representation with reference group of the one-way ANOVA model, the F-statistic for the partition $p_0 := 1$ and $p_1 := p - 1$; see Chapter 39, can be written as a ratio of scaled versions of SQB and SQW. Here $p_0 := 1$ in particular implies that the reduced one-way ANOVA model under consideration has design matrix $X_0 := 1_n$ and beta-parameter $\beta := \mu_0$, and thus in particular has no effect parameters. The relevant scaling factors relate SQB and SQW to the number of effect parameters and to the difference between the total number of data points and the number of beta parameters, respectively. The following theorem holds.

Theorem 41.4 (F-statistic of one-way analysis of variance)**.** *Let*

$$y = X\beta + \varepsilon \text{ with } \varepsilon \sim N\left(0_n, \sigma^2 I_n\right) \tag{41.49}$$

be the design-matrix form of the effect representation with reference group of the one-way ANOVA model. Furthermore, let this model be partitioned in the sense of Definition 39.2 with $p_0 := 1$ and $p_1 := p - 1$. Finally, let

- *the mean between sum of squares be defined as*

$$MSB := \frac{SQB}{p-1}, \tag{41.50}$$

- *and the mean within sum of squares be defined as*

$$MSW := \frac{SQW}{n-p}, \tag{41.51}$$

where $p-1$ is also called the between degrees of freedom and $n-p$ the within degrees of freedom. Then, with the definition of the F-statistic (Definition 39.3),

$$F = \frac{MSB}{MSW}. \tag{41.52}$$

∘

Proof. We first note that, for the beta-parameter estimator of the reduced model,

$$\hat{\beta}_0 = \left(X_0^T X_0\right)^{-1} X_0^T y = \left(1_n^T 1_n\right)^{-1} 1_n^T y = \frac{1}{n}\sum_{i=1}^{p}\sum_{j=1}^{n_i} y_{ij} = \bar{y}. \tag{41.53}$$

Furthermore,

$$\hat{\varepsilon}_0^T \hat{\varepsilon}_0 = \left(y - X_0\hat{\beta}_0\right)^T \left(y - X_0\hat{\beta}_0\right) = \left(y - 1_n\bar{y}\right)^T \left(y - 1_n\bar{y}\right) = \sum_{i=1}^{p}\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}\right)^2 = \mathrm{SQT}. \tag{41.54}$$

The beta-parameter estimator of the full model is, as seen above,

$$\hat{\beta} = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 - \bar{y}_1 \\ \vdots \\ \bar{y}_p - \bar{y}_1 \end{pmatrix}. \tag{41.55}$$

It follows that

$$\hat{\varepsilon}^T \hat{\varepsilon} = \sum_{i=1}^{p}\sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_i\right)^2 = \mathrm{SQW}. \tag{41.56}$$

With the theorem on the sum-of-squares decomposition for one-way analysis of variance,

$$\mathrm{SQT} = \mathrm{SQB} + \mathrm{SQW} \Leftrightarrow \mathrm{SQB} = \mathrm{SQT} - \mathrm{SQW}, \tag{41.57}$$

it follows immediately that

$$\mathrm{SQB} = \mathrm{SQT} - \mathrm{SQW} = \hat{\varepsilon}_0^T \hat{\varepsilon}_0 - \hat{\varepsilon}^T \hat{\varepsilon}. \tag{41.58}$$

Therefore,

$$\frac{\mathrm{MSB}}{\mathrm{MSW}} = \frac{\frac{\mathrm{SQB}}{p-1}}{\frac{\mathrm{SQW}}{n-p}} = \frac{\frac{\hat{\varepsilon}_0^T \hat{\varepsilon}_0 - \hat{\varepsilon}^T \hat{\varepsilon}}{p-1}}{\frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-p}} = F. \tag{41.59}$$

□

Given the very similar definitions of the effect-size measure η^2 and the F-statistic of one-way ANOVA, the following result is natural.

Theorem 41.5 (Effect-size measure η^2 and F-test statistic)**.** *For a one-way ANOVA scenario with p groups and total number of data points n, let the effect-size measure η^2 and the F-statistic of one-way analysis of variance be given. Then*

$$\eta^2 = \frac{F(p-1)}{F(p-1)+(n-p)}. \tag{41.60}$$

∘

Proof. We first note that

$$F = \frac{\mathrm{SQB}}{\mathrm{SQW}} \cdot \frac{n-p}{p-1} \Leftrightarrow \mathrm{SQB} = \frac{p-1}{n-p} \cdot \mathrm{SQW} \cdot F. \tag{41.61}$$

It follows that

$$\begin{aligned}
\eta^2 &= \frac{\mathrm{SQB}}{\mathrm{SQT}} \\
&= \frac{\mathrm{SQB}}{\mathrm{SQB}+\mathrm{SQW}} \\
&= \frac{\frac{p-1}{n-p} \cdot \mathrm{SQW} \cdot F}{\frac{p-1}{n-p} \cdot \mathrm{SQW} \cdot F + \mathrm{SQW}} \\
&= \frac{F(p-1)}{F(p-1)+(n-p)}.
\end{aligned} \tag{41.62}$$

□

Intuitively, the relation between η^2 and the F-statistic is analogous to the relation between Cohen's d and the T-statistic in one-sample and two-sample T-tests. In particular, the simultaneous reporting of η^2 and the F-statistic is redundant when group sizes are known.

41.5.3 F-test of one-way analysis of variance

We now discuss the use of the F-statistic for carrying out a test of the null hypothesis (Equation 41.34). Recall that this null hypothesis intuitively states that the expectation parameters are identical across all levels of the factor, or that all effect parameters are equal to zero. Rejecting this null hypothesis therefore implies that there is inferential evidence that at least one effect parameter differs from zero. However, rejecting the null hypothesis of the F-test of one-way analysis of variance does not state which effect parameter this is. We define the F-test of one-way analysis of variance as follows.

Definition 41.4 (F-test of one-way analysis of variance)**.** Let the one-way ANOVA model in effect-parameter representation with reference group be given, together with the null and alternative hypotheses

$$\Theta_0 := \left\{ \begin{pmatrix} \mu_0 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{pmatrix} \in \mathbb{R}^p \mid \alpha_i = 0 \text{ for } i = 2, \dots, p \right\} \text{ and } \Theta_1 := \mathbb{R}^p \backslash \Theta_0. \tag{41.63}$$

Furthermore, let the F-test statistic be defined by

$$F = \frac{\mathrm{MSB}}{\mathrm{MSW}}, \tag{41.64}$$

with the mean between sum of squares MSB and the mean within sum of squares defined as in Theorem 41.4. Then the F-test of one-way analysis of variance is defined as the critical-value-based test

$$\phi(y) := 1_{\{F \geq k\}} := \begin{cases} 1 & F \geq k \\ 0 & F < k. \end{cases} \tag{41.65}$$

•

Control of the type-I error probability is based on the f distribution of the F-statistic. This is the core statement of the following theorem.

Theorem 41.6 (Test-size control of the F-test of one-way analysis of variance)**.** *Let ϕ be the F-test of one-way analysis of variance. Then ϕ is a level-α_0 test with test size α_0 if the critical value is defined by*

$$k_{\alpha_0} := \varphi^{-1}\left(1 - \alpha_0; p-1, n-p\right), \tag{41.66}$$

where $\varphi^{-1}(\cdot; p-1, n-p)$ is the inverse CDF of the f distribution with degrees-of-freedom parameters $p-1$ and $n-p$.

∘

Proof. The power function of the test under consideration in the present test scenario is defined as

$$q : \mathbb{R} \to [0,1], \beta \mapsto q_\phi(\beta) := \mathbb{P}_\beta(\phi = 1). \tag{41.67}$$

We saw in Chapter 39 that the F-statistic for $p_1 = p - 1$ follows a noncentral f distribution,

$$F \sim f(\delta, p-1, n-p). \tag{41.68}$$

Furthermore, the rejection region of the test considered here is $[k, \infty[$. Thus the functional form of the power function is

$$\begin{aligned} \mathbb{P}_\beta(\phi = 1) &= \mathbb{P}_\beta(F \in [k, \infty[) \\ &= \mathbb{P}_\beta(F \geq k) \\ &= 1 - \mathbb{P}_\beta(F \leq k) \\ &= 1 - \varphi(k; \delta, p-1, n-p), \end{aligned} \tag{41.69}$$

where $\varphi(k; \delta, p-1, n-p)$ denotes the value of the CDF of the noncentral f distribution at k, with noncentrality parameter δ and degrees-of-freedom parameters $p-1$ and $n-p$; see Chapter 39. For the test under consideration to be a level-α_0 test, it must hold that

$$q_\phi(\beta) \leq \alpha_0 \text{ for all } \beta \in \Theta_0 \text{ with } \Theta_0 = \mathbb{R} \times \left\{0_{p-1}\right\}. \tag{41.70}$$

Using the form of the noncentrality parameter from Theorem 39.3,

$$\delta = \frac{1}{\sigma^2} c^T \beta \left(c^T \left(X^T X\right)^{-1} c\right)^{-1} c^T \beta, \tag{41.71}$$

and with $\beta \in \Theta_0$,

$$c = \begin{pmatrix} 0 \\ 1_{p-1} \end{pmatrix} \in \mathbb{R}^p \text{ and } \beta = \begin{pmatrix} \mu_0 \\ 0_{p-1} \end{pmatrix} \in \mathbb{R}^p, \tag{41.72}$$

we obtain $\delta = 0$ and hence

$$q_\phi(\beta) = 1 - \varphi(k; p-1, n-p) \text{ for all } \beta \in \Theta_0. \tag{41.73}$$

Here $\varphi(k; p-1, n-p)$ denotes the value of the CDF of the f distribution at k with degrees-of-freedom parameters $p-1$ and $n-p$. The test size of the test under consideration is

$$\alpha = \max_{\beta \in \Theta_0} q_\phi(\beta) = 1 - \varphi(k; p-1, n-p), \tag{41.74}$$

because $q_\phi(\beta)$ does not depend on μ_0 for $\beta \in \Theta_0$. It remains only to show that choosing k_{α_0} as in the theorem guarantees that ϕ has test size α_0. Let $k := k_{\alpha_0}$. Then, for all $\beta \in \Theta_0$,

$$q_\phi(\beta) = 1 - \varphi\left(\varphi^{-1}\left(1-\alpha_0; p-1, n-p\right); p-1, n-p\right) = 1 - (1-\alpha_0) = \alpha_0, \tag{41.75}$$

and the claim is shown. □

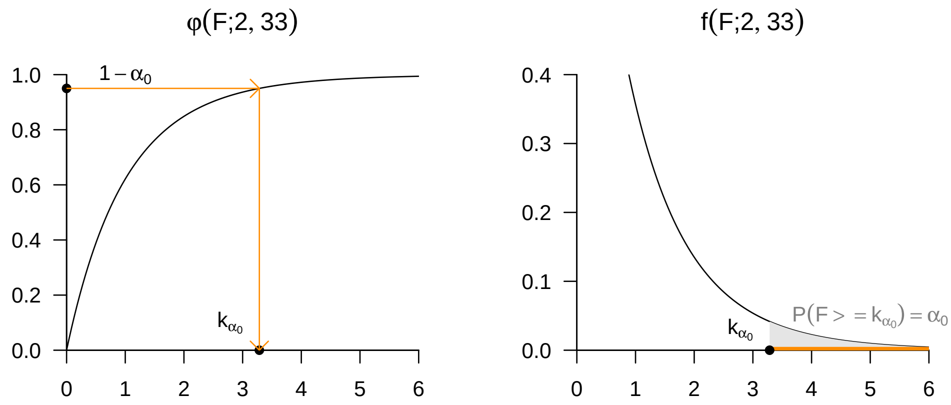


Figure 41.2 Exemplary critical value and rejection region for an F-test of one-way analysis of variance.

In Figure 41.2, we visualize the choice of the critical value k_{α_0} for controlling the test size by means of the CDF of the f distribution as well as the resulting rejection region for $\alpha_0 := 0.05$, a factor with three levels $p = 3$, and a balanced design with group size $m = 12$, hence $n = 3 \cdot 12 = 36$. In this case, the null hypothesis would be rejected for a value of the F-statistic greater than 3.28.

p-value

Let f be an observed value of the F-statistic in the context of one-way analysis of variance. Then the p-value belonging to f is obtained from the following theorem.

Theorem 41.7 (p-value of the F-statistic in one-way analysis of variance)**.** *Given the scenario of an F-test in one-way analysis of variance specified in Definition 41.4, the p-value associated with an observed value f of the F-statistic is*

$$p\text{-}value = \mathbb{P}(F \geq f) = 1 - \varphi(f; p-1, n-p). \tag{41.76}$$

∘

Proof. By definition, the p-value is the smallest significance level α_0 at which one would reject the null hypothesis based on an observed value of the test statistic. For $F = f$, H_0 would be rejected for every α_0 with $f \geq \varphi^{-1}\left(1-\alpha_0; p-1, n-p\right)$. For such an α_0,

$$\alpha_0 \geq \mathbb{P}(F \geq f), \tag{41.77}$$

because

$$
\begin{aligned}
& f \geq \varphi^{-1}\left(1-\alpha_0 ; p-1, n-p\right) \\
\Leftrightarrow\ & \varphi(f ; p-1, n-p) \geq \varphi\left(\varphi^{-1}\left(1-\alpha_0 ; p-1, n-p\right), p-1, n-p\right) \\
\Leftrightarrow\ & \varphi(f ; p-1, n-p) \geq 1-\alpha_0 \\
\Leftrightarrow\ & \mathbb{P}(F \leq f) \geq 1-\alpha_0 \\
\Leftrightarrow\ & \alpha_0 \geq 1-\mathbb{P}(F \leq f) \\
\Leftrightarrow\ & \alpha_0 \geq \mathbb{P}(F \geq f).
\end{aligned} \tag{41.78}
$$

The smallest $\alpha_0 \in [0,1]$ with $\alpha_0 \geq \mathbb{P}(F \geq f)$ is then $\alpha_0 = \mathbb{P}(F \geq f)$, and hence

$$
\text{p-value} = \mathbb{P}(F \geq f) = 1 - \varphi(f; p-1, n-p). \tag{41.79}
$$

□

Practical procedure

The results of this section imply the following procedure for evaluating a one-way ANOVA model with an F-test. First assume that a given data set consisting of p group data sets

$$
y_{11}, \ldots, y_{1n_1}, y_{21}, \ldots, y_{2n_2}, \ldots, y_{p1}, \ldots, y_{pn_p} \tag{41.80}
$$

contains realizations of

$$
y_{1j} \sim N\left(\mu_0, \sigma^2\right) \tag{41.81}
$$

and

$$
y_{ij} \sim N\left(\mu_0 + \alpha_i, \sigma^2\right) \tag{41.82}
$$

for $i = 2, \ldots, p$, with true, but unknown, parameters $\mu_0, \alpha_i, i = 2, \ldots, p$ and $\sigma^2 > 0$. Further assume that one wants to decide whether the null hypothesis of identical true, but unknown, expectation parameters across factor levels, or of true, but unknown, effect parameters identical to zero, is more plausible or less plausible; see Equation 41.34. To this end, one first chooses a significance level α_0 and determines the corresponding degrees-of-freedom-dependent critical value k_{α_0}. For example, for $\alpha_0 := 0.05, p = 3, m = 12, i = 1,2,3$ and hence $n = 36$, we have $k_{\alpha_0} = \varphi^{-1}(1-0.05; 2, 33) \approx 3.28$. One then uses the given data set to compute MSB and MSW and thereby determines the realized value f of the F-statistic. If f is greater than or equal to k_{α_0}, the null hypothesis is rejected; otherwise it is not rejected. Overall, the theory developed in this section then guarantees that, on average, one falsely rejects the null hypothesis in at most $\alpha_0 \cdot 100$ out of 100 cases. Finally, one may determine the p-value associated with the value f as $1 - \varphi(f; p-1, n-p)$ and report it in the documentation of the analysis.

41.5.4 Application example

The following **R** code implements the practical procedure above for the example data set.

```
# data management
D           = read.csv("./_data/508-one-way-analysis-of-variance.csv")          # data set
y           = D$dBDI                                                             # data vector
n           = length(y)                                                          # total data size
p           = 3                                                                  # number of groups
m           = n/p                                                                # number of data points per group

# model formulation
Xt          = cbind(                                                              # design matrix of full model
              matrix(1, nrow = n, ncol = 1),
              kronecker(diag(p),
                        matrix(1, nrow = m, ncol = 1)))
X           = Xt[,-2]
X_0         = X[,1]                                                              # design matrix of reduced model
```

```
# F-test statistic evaluation
beta_hat    = solve(t(X) %*% X) %*% t(X) %*% y                # beta-parameter estimator of full model
beta_hat_0  = solve(t(X_0) %*% X_0) %*% t(X_0) %*% y          # beta-parameter estimator of reduced model
eps_hat     = y - X %*% beta_hat                              # residual vector of full model
eps_hat_0   = y - X_0 %*% beta_hat_0                          # residual vector of reduced model
SQT         = t(eps_hat_0) %*% eps_hat_0                      # sum of squares total
SQW         = t(eps_hat)   %*% eps_hat                        # sum of squares within
SQB         = SQT - SQW                                       # sum of squares between
DFB         = p - 1                                           # between degrees of freedom
DFW         = n - p                                           # within degrees of freedom
MSB         = SQB/DFB                                         # mean sum of squares between
MSW         = SQW/DFW                                         # mean sum of squares within
Eff         = MSB/MSW                                         # F-test statistic
pW          = 1 - pf(Eff, p-1, n-p)                           # p-value

# F-test evaluation
alpha_0     = 0.05                                            # significance level
k_alpha_0   = qf(1 - alpha_0, p-1,n-p)                        # critical value
if(Eff > k_alpha_0){phi = 1} else {phi = 0}                   # test value
```

```
DFB : 2
DFW : 33
SQB : 821.72
SQW : 170.58
MSB : 410.86
MSW : 5.17
F   : 79.48
p   : 0
phi : 1
```

In the present case, the null hypothesis

$$\Theta_0 := \left\{ \begin{pmatrix} \mu_0 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \in \mathbb{R}^3 \mid \alpha_i = 0 \text{ for } i = 2, 3 \right\} \tag{41.83}$$

would be rejected.

The following **R** code demonstrates carrying out and documenting the same analysis with the **R** function `aov()`.

```
D       = read.csv("./_data/508-one-way-analysis-of-variance.csv")     # data
res.aov = aov(D$dBDI ~ D$COND, data = D)                               # model formulation and model estimation
summary(res.aov)                                                       # model evaluation
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)    
D$COND       2  821.7   410.9   79.48 2.41e-13 ***
Residuals   33  170.6     5.2                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

41.6 Literature notes

The popularity of analysis-of-variance procedures is generally traced back to Fisher (1925a) and Fisher (1935). Everitt & Howell (2005) and Stigler (1986) provide a brief and a detailed historical overview, respectively.

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